# Optimal Transport for Machine Learners

*Compact teaching notes*

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# 1 Optimal Matching between Point Clouds

## 1.1 Monge Problem for Discrete Points

### Assignment problem.

**Definition 1.1** (Optimal assignment problem)**.** For a cost matrix $\mathrm{C} \in \mathbb{R}^{n\times n}$ and the set $\mathrm{Perm}(n)$ of bijections of $[\![n]\!] = \{1, \dots, n\}$, the optimal assignment value is

$$\min_{\sigma\in \mathrm{Perm}(n)} \frac{1}{n}\sum_{i=1}^{n} \mathrm{C}_{i,\sigma(i)}. \tag{1.1}$$

A minimizer is an optimal assignment, or optimal permutation.

### Convex costs on the line.

**Proposition 1.2** (Monotone matching on the line)**.** *Assume that $x_1, \dots, x_n$ are pairwise distinct and that $y_1, \dots, y_n$ are pairwise distinct. If $\mathrm{C}_{i,j} = h(x_i-y_j)$ for a strictly convex function $h : \mathbb{R} \to \mathbb{R}$, then the unique optimizer is the order-preserving permutation, characterized by $(x_i - x_{i'})(y_{\sigma(i)} - y_{\sigma(i')}) > 0 \qquad$ for every $i \neq i'$. In particular, the result applies to $h(s) = |s|^p$ for $p > 1$.*

*Proof.* A permutation that does not preserve order contains an inversion: after relabeling, $x < x'$ are matched to $y' > y$. Set $d = y' - y > 0$ and $D(s) = \frac{h(s)-h(s-d)}{d}$. Strict convexity of $h$ makes $D$ strictly increasing. Therefore $h(x - y') + h(x' - y) - h(x - y) - h(x' - y') = d\big(D(x' - y) - D(x - y)\big) > 0$. Swapping the two inverted targets strictly lowers the cost. Repeating this exchange eliminates every inversion; the only order-preserving bijection between the two sorted clouds pairs equal ranks. □

If $h$ is convex but not strictly convex, the same exchange inequality is non-strict. The equal-rank matching remains optimal, but other, non-monotone optimizers can coexist. To construct a monotone optimizer, choose sorting permutations $\sigma_X, \sigma_Y$ such that $x_{\sigma_X(1)} \leqslant x_{\sigma_X(2)} \leqslant \dots \quad$ and $\quad y_{\sigma_Y(1)} \leqslant y_{\sigma_Y(2)} \leqslant \dots$ and match $x_{\sigma_X(k)}$ with $y_{\sigma_Y(k)}$. Thus $\sigma = \sigma_Y \circ \sigma_X^{-1}$ is optimal.

**Concave costs on the line.** $p_1 < q_1 < p_2 < q_2 < \dots < p_N < q_N$, with the opposite orientation handled by exchanging the roles of $p$ and $q$, Delon, Salomon and Sobolevski define the local indicators

$$I_k^p(i) = c(p_i, q_{i+k}) + \sum_{r=0}^{k-1} c(p_{i+r+1}, q_{i+r}) - \sum_{r=0}^{k} c(p_{i+r}, q_{i+r}),$$
$$I_k^q(i) = c(p_{i+k+1}, q_i) + \sum_{r=1}^{k} c(p_{i+r}, q_{i+r}) - \sum_{r=0}^{k} c(p_{i+r+1}, q_{i+r}),$$

For arbitrary real masses, the stratification argument in the same work gives a larger $O(n^3)$ worst-case bound. Very concave costs also motivate the simpler greedy rule that repeatedly matches the closest red–blue pair. These methods are not generic linear-programming solvers: they exploit the order and concavity specific to the line.

### Histogram equalization.

### Flat directions for the linear cost.

**Optimal transport on the circle.** $d_{\mathbb{S}^1}(x, y) := \min_{k\in\mathbb{Z}} |x - y + k|, \qquad c_p(x, y) := d_{\mathbb{S}^1}(x, y)^p, \qquad p > 1.$

**Proposition 1.3** (Discrete circle transport by a cut)**.** *Assume that the $2n$ points $x_1, \dots, x_n, y_1, \dots, y_n$ are pairwise distinct. Let $x_{(1)}, \dots, x_{(n)}$ and $y_{(1)}, \dots, y_{(n)}$ be fixed cyclic orderings, with indices understood modulo $n$. For $p > 1$, every optimal assignment for the cost $c_p$ is a cyclic shift $x_{(k)} \longmapsto y_{(k+s)}, \qquad k \in [\![n]\!], \quad s \in \{0, \dots, n-1\}$.*

$$E_s := \frac{1}{n}\sum_{k=1}^{n} d_{\mathbb{S}^1}\big(x_{(k)}, y_{(k+s)}\big)^p$$

*over the $n$ shifts. For each optimal assignment there is a cut $\theta \in \mathbb{S}^1 \setminus (\{x_i\}_i \cup \{y_j\}_j)$, crossed by none of its shortest transport arcs, such that the lifted assignment on $(\theta, \theta + 1)$ is the equal-rank matching on the line.*

*Proof.* Fix an optimal assignment $\sigma$ and let $\gamma_i$ be an open shortest arc from $x_i$ to $y_{\sigma(i)}$, with either orientation fixed in the antipodal case. The path-uncrossing lemma of Delon, Rabin and Gousseau states that if two arcs of an optimal assignment intersect, then they have the same orientation and neither is strictly contained in the other. Its proof is a two-edge exchange: opposite orientations or strict containment would, by monotonicity and strict convexity of $r \mapsto r^p$, produce a strictly cheaper pairing of the four endpoints.

Suppose that the arcs cover the circle. Since each $\gamma_i$ is open, every source point lies in another transport arc. Order the sources cyclically. The path-uncrossing lemma then propagates one common orientation: either $x_{(k+1)} \in \gamma_{(k)}$ for every $k$, or $x_{(k-1)} \in \gamma_{(k)}$ for every $k$. In the first case, define a new assignment by giving $x_{(k+1)}$ the target previously assigned to $x_{(k)}$; every transport arc becomes strictly shorter. The opposite orientation gives the analogous backward reassignment. Both contradict optimality. Hence some $\theta$ lies outside every transport arc.

Cut the circle at such a $\theta$ and lift it to $(\theta, \theta + 1)$. Every matched geodesic avoids the cut, so its circular length equals the ordinary distance between the corresponding lifts. Proposition 1.2 now implies that the lifted matching preserves order. Returning to the circle, an order-preserving bijection between two cyclically ordered $n$-point sets is a cyclic shift. Minimizing over these $n$ shifts therefore recovers the optimum. □

### Two-dimensional geometry.

**Proposition 1.4** (No proper crossings in Euclidean-distance matchings)**.** *Let $x_1, \dots, x_n, y_1, \dots, y_n \in \mathbb{R}^2$ and let $c(x, y) = \|x - y\|$. If $\sigma$ is optimal, then two matched segments cannot cross properly: their relative interiors cannot meet at a point where their supporting lines are distinct.*

*Proof.* Suppose that $[x_i, y_{\sigma(i)}]$ and $[x_j, y_{\sigma(j)}]$ cross properly at $z$. The triangle inequality along the two reconnected paths gives $\|x_i - y_{\sigma(j)}\| \leqslant \|x_i - z\| + \|z - y_{\sigma(j)}\|, \qquad \|x_j - y_{\sigma(i)}\| \leqslant \|x_j - z\| + \|z - y_{\sigma(i)}\|$. The sum of the two right-hand sides is exactly the cost of the two original segments. At least one inequality is strict because a proper crossing is non-collinear. Swapping $\sigma(i)$ and $\sigma(j)$ therefore strictly decreases the assignment cost, contradicting optimality. Collinear overlaps are excluded from the statement precisely because the exchange can then have equal cost. $\square$

$C_n = \frac{1}{n+1}\binom{2n}{n} \sim \frac{4^n}{\sqrt{\pi} n^{3/2}}$.

### 1.2 Hungarian Algorithm

**Hungarian primal-dual method.**

$$\max_{(\mathrm{f},\mathrm{g}) \in \mathbb{R}^n \times \mathbb{R}^n} \sum_{i=1}^{n} \mathrm{f}_i + \sum_{j=1}^{n} \mathrm{g}_j \qquad \text{subject to} \qquad \mathrm{f}_i + \mathrm{g}_j \leqslant \mathrm{C}_{i,j} \quad \forall i, j. \tag{1.2}$$

**Proposition 1.5** (Dual certificate for an assignment)**.** *Let* $(\mathrm{f},\mathrm{g})$ *be feasible for* (1.2)*. Then every permutation* $\tau \in \mathrm{Perm}(n)$ *satisfies*

$$\sum_i \mathrm{C}_{i,\tau(i)} \geqslant \sum_i \mathrm{f}_i + \sum_j \mathrm{g}_j. \tag{1.3}$$

*If a permutation* $\sigma$ *uses only tight constraints,*

$$\mathrm{f}_i + \mathrm{g}_{\sigma(i)} = \mathrm{C}_{i,\sigma(i)} \qquad \textit{for every } i, \tag{1.4}$$

*then* $\sigma$ *is an optimal assignment and* $(\mathrm{f},\mathrm{g})$ *is dual optimal. Equivalently, for a feasible pair, equality in* (1.3) *at* $\tau = \sigma$ *holds if and only if every edge* $(i, \sigma(i))$ *is tight.*

*Proof.* Dual feasibility gives $\mathrm{C}_{i,\tau(i)} \geqslant \mathrm{f}_i + \mathrm{g}_{\tau(i)}$ for every $i$. Summing and using that $\tau$ is a permutation yields $\sum_i \mathrm{C}_{i,\tau(i)} \geqslant \sum_i \mathrm{f}_i + \sum_i \mathrm{g}_{\tau(i)} = \sum_i \mathrm{f}_i + \sum_j \mathrm{g}_j$, which proves (1.3). If (1.4) holds, the assignment cost of $\sigma$ equals the dual value of $(\mathrm{f},\mathrm{g})$. The lower bound shows that no permutation can have smaller cost. Applied to $\sigma$ and any other feasible dual pair, it also shows that no feasible dual value can exceed this cost. Thus $\sigma$ and $(\mathrm{f},\mathrm{g})$ are respectively primal and dual optimal. Finally, the difference between the two sides of (1.3) at $\tau = \sigma$ is the sum of the nonnegative slacks $\mathrm{C}_{i,\sigma(i)} - \mathrm{f}_i - \mathrm{g}_{\sigma(i)}$. It vanishes exactly when every one of them vanishes. $\square$

$$s_{i,j} := \mathrm{C}_{i,j} - \mathrm{f}_i - \mathrm{g}_j \geqslant 0. \tag{1.5}$$

$E(\mathrm{f},\mathrm{g}) := \{(i,j)\ ;\ s_{i,j} = 0\} = \{(i,j)\ ;\ \mathrm{f}_i + \mathrm{g}_j = \mathrm{C}_{i,j}\}$. The invariant $M \subset E(\mathrm{f},\mathrm{g})$ means that every currently matched pair is already certified by equality in the dual constraint. If $M$ is perfect, its associated permutation $\sigma$ satisfies (1.4); Proposition 1.5 then certifies that $\sigma$ is optimal.

To increase the matching, choose an unmatched source $i_0$ as a root. The algorithm grows a tree in the bipartite equality graph whose reached source and target vertices are denoted by $S$ and $T$.

$$\delta = \min_{i \in S,\ j \notin T} \big(\mathrm{C}_{i,j} - \mathrm{f}_i - \mathrm{g}_j\big), \qquad \mathrm{f}_i \leftarrow \mathrm{f}_i + \delta\ (i \in S), \qquad \mathrm{g}_j \leftarrow \mathrm{g}_j - \delta\ (j \in T). \tag{1.6}$$

Here $\delta > 0$ because no such edge is currently tight. The update leaves the slack unchanged on $S \times T$, decreases it by $\delta$ on $S \times T^c$, and increases it by $\delta$ on $S^c \times T$.

For an efficient implementation, one does not recompute all the minima in (1.6). For each unreached target, store

$$\ell_j = \min_{i \in S} s_{i,j} \tag{1.7}$$

and a parent source $p(j) \in S$ attaining this minimum. When a source $i'$ enters $S$, each unreached target is updated by comparing $\ell_j$ with the single new slack $s_{i',j}$.

**Proposition 1.6** (Correctness and complexity of the Hungarian primal-dual method)**.** *For every cost matrix* $\mathrm{C} \in \mathbb{R}^{n \times n}$*, Algorithm* **??** *terminates after* $n$ *augmentation phases. It returns a permutation* $\sigma \in \mathrm{Perm}(n)$ *and feasible potentials* $(\mathrm{f},\mathrm{g})$ *such that* $\mathrm{f}_i + \mathrm{g}_{\sigma(i)} = \mathrm{C}_{i,\sigma(i)} \quad \forall i$*. With the maintained slacks* (1.7)*, the algorithm uses* $O(n^3)$ *arithmetic and comparison operations and* $O(n^2)$ *storage, including the cost matrix.*

*Proof.* Initialization is dual feasible because $\mathrm{f}_i = \min_j \mathrm{C}_{i,j}$ and $\mathrm{g} = 0$, and the empty matching is contained in the equality graph. At the start of a phase, the initialization of $\ell_j$ and $p(j)$ gives $\ell_j = \min_{i \in S} s_{i,j} = s_{p(j),j} \quad (j \notin T)$. This identity remains true when a matched source $i'$ enters $S$, because the algorithm compares every $\ell_j$ with the one new candidate $s_{i',j}$.

Consider an inner iteration before a free target is reached. Every target in $T$ is matched to a source in $S$, and every source in $S$ except the unmatched root is reached through its matched target. Hence $|S| = |T| + 1$, so $T \neq [\![n]\!]$. Dual feasibility gives $\ell_j \geqslant 0$, and therefore $\delta = \min_{j \notin T} \ell_j = \min_{i \in S,\, j \notin T} s_{i,j} \geqslant 0$. After the potential update, the slack becomes $s^+_{i,j} = s_{i,j} - \delta$ on $S \times T^c$, $s^+_{i,j} = s_{i,j} + \delta$ on $S^c \times T$, and remains $s^+_{i,j} = s_{i,j}$ on the other two blocks. The definition of $\delta$ makes every entry nonnegative. For $j \notin T$, all slacks from $S$ decrease by the same amount $\delta$, exactly as does $\ell_j$; hence the maintained-minimum identity remains valid. At least one maintained slack becomes zero, and its parent $p(j)$ identifies a corresponding zero-slack edge from $S$ to an unreached target. Thus dual feasibility is preserved and the next target selected by the algorithm is connected to the alternating tree. Moreover, every matched edge either has both endpoints in $S \times T$ or both endpoints outside the tree, so all matched edges remain tight.

Each inner iteration adds a new target to $T$. If that target is matched, its unique partner is added to $S$, and the slack update restores the maintained-minimum identity above. If it is unmatched, the pointers $p$ and $q$ trace an alternating path from $i_0$ to this target. Flipping the status of the edges on that path preserves the matching property, increases $|M|$ by one, and uses only equality edges. Since a target enters $T$ at most once, every phase reaches a free target after at most $n$ inner iterations. Starting from $M = \emptyset$, exactly $n$ phases therefore produce a perfect matching $M = \{(i, \sigma(i))\}_i$.

At termination, every matched edge is tight, so Proposition 1.5 directly proves that $\sigma$ is optimal and $(\mathrm{f}, \mathrm{g})$ is dual optimal. This is the finite assignment instance of the complementary-slackness mechanism proved in general in Proposition 4.3. Finally, initializing the slack and parent arrays costs $O(n)$ per phase. Each inner iteration scans the unreached targets to find $\delta$, updates the potentials and maintained slacks, and, when necessary, compares those slacks with one newly reached source row. These operations cost $O(n)$ per iteration. There are at most $n$ iterations per phase and $n$ phases, giving $O(n^3)$ arithmetic and comparison operations. The cost matrix uses $O(n^2)$ storage; the matching, reached sets, parents, and maintained slacks require only $O(n)$ additional storage. □

## 2 Monge Problem between Measures

### 2.1 Measures

**Histograms.**

**Definition 2.1** (Probability simplex)**.** The probability simplex of length $n$ is $\Sigma_n := \left\{\mathrm{a} \in \mathbb{R}_+^n \,;\, \sum_{i=1}^n \mathrm{a}_i = 1\right\}$. Its elements are also called probability vectors or histograms.

**Discrete measure, empirical measure.**

**Definition 2.2** (Discrete measure)**.** A discrete measure with weights a and locations $x_1, \ldots, x_n \in \mathcal{X}$ is

$$\alpha = \sum_{i=1}^{n} \mathrm{a}_i \delta_{x_i}, \tag{2.1}$$

where $\delta_x$ is the Dirac mass at position $x$. It is a probability measure if $\mathrm{a} \in \Sigma_n$, and a positive measure if all weights $\mathrm{a}_i$ are nonnegative.

**General measures.** A Dirac measure $\delta_x$ is then defined as $\delta_x(A) = 1$ if $x \in A$ and 0 otherwise, and this extends by linearity for discrete measures of the form (2.1) as
$\alpha(A) = \sum_{x_i \in A} \mathrm{a}_i$

**Polish metric spaces.**

**Definition 2.3** (Polish metric space)**.** A metric space $(\mathcal{X}, d)$ is Polish if it is complete and separable: every Cauchy sequence converges in $\mathcal{X}$, and $\mathcal{X}$ contains a countable dense subset. More generally, a topological space is called Polish if its topology can be induced by some complete separable metric.

$\mathcal{S} = C([0,1]; \mathbb{R}^d), \qquad d_\infty(\gamma, \eta) := \sup_{t \in [0,1]} \|\gamma(t) - \eta(t)\|$

**Definition 2.4** (Support of a measure)**.** Let $\alpha$ be a positive Borel measure on a metric space $\mathcal{X}$. Its support is $\mathrm{supp}(\alpha) := \{x \in \mathcal{X} \,;\, \alpha(B(x,r)) > 0 \text{ for every } r > 0\}$. If $\mathcal{X}$ is separable, this is the smallest closed set of full $\alpha$-mass.

**Radon measures.**

**Definition 2.5** (Radon and probability measures)**.** A locally finite positive Borel measure $\alpha$ on $\mathcal{X}$ is Radon if $\alpha(A) = \sup\{\alpha(K) \,;\, K \subset A \text{ compact}\}$ for every Borel set $A$. We write $\mathcal{M}_+(\mathcal{X})$ for finite positive Radon measures and $\mathcal{M}_+^1(\mathcal{X}) = \{\alpha \in \mathcal{M}_+(\mathcal{X}) \,;\, \alpha(\mathcal{X}) = 1\}$ for probability measures.

$\langle f, \alpha \rangle := \int_{\mathcal{X}} f(x) \mathrm{d}\alpha(x)$. $\int_{\mathcal{X}} f(x) \mathrm{d}\alpha(x) = \sum_{i=1}^n \mathrm{a}_i f(x_i)$.

**Relative densities.**

**Definition 2.6** (Relative density)**.** If $\alpha$ is absolutely continuous with respect to a positive reference measure $\lambda$, its relative density is the Radon–Nikodym derivative $\rho_\alpha := \frac{\mathrm{d}\alpha}{\mathrm{d}\lambda}, \qquad \mathrm{d}\alpha(x) = \rho_\alpha(x)\mathrm{d}\lambda(x)$. Equivalently, for every nonnegative measurable function $h$, and hence for every integrable $h$, $\int_{\mathcal{X}} h(x)\mathrm{d}\alpha(x) = \int_{\mathcal{X}} h(x)\rho_\alpha(x)\mathrm{d}\lambda(x)$.

**Total variation norm.**

**Definition 2.7** (Total variation)**.** For a finite signed Radon measure $\alpha$ on a compact space $\mathcal{X}$, $\|\alpha\|_{\mathrm{TV}} := \sup_{f \in \mathcal{C}(\mathcal{X})} \{\langle f, \alpha \rangle \,;\, \|f\|_\infty \leqslant 1\}$.

$|\alpha|(A) := \sup_{A = \cup_i B_i} \sum_i |\alpha(B_i)|$, where the supremum is over finite or countable measurable partitions of $A$. Thus, if $\alpha = \sum_i \mathrm{a}_i \delta_{x_i}$ with distinct atoms, $|\alpha| = \sum_i |\mathrm{a}_i| \delta_{x_i}$; if $\mathrm{d}\alpha(x) = \rho(x)\mathrm{d}\lambda(x)$, then $\mathrm{d}|\alpha|(x) = |\rho(x)|\mathrm{d}\lambda(x)$.

**Proposition 2.8** (Dual and measure definitions of total variation)**.** *For a finite signed Radon measure $\alpha$ on a compact space $\mathcal{X}$, one has $\|\alpha\|_{\mathrm{TV}} = |\alpha|(\mathcal{X})$.*

*Proof.* The inequality $\|\alpha\|_{\mathrm{TV}} \leqslant |\alpha|(\mathcal{X})$ follows from $\left|\int f \,\mathrm{d}\alpha\right| \leqslant \int |f| \,\mathrm{d}|\alpha| \leqslant \|f\|_\infty |\alpha|(\mathcal{X})$. For the reverse inequality, write the Jordan decomposition $\alpha = \alpha^+ - \alpha^-$, so that $|\alpha| = \alpha^+ + \alpha^-$. The measurable sign $s = \frac{\mathrm{d}\alpha}{\mathrm{d}|\alpha|}$ takes values in $\{-1, 1\}$ outside a $|\alpha|$-null set and satisfies $\mathrm{d}\alpha = s\,\mathrm{d}|\alpha|$. By regularity of Radon measures on compact spaces, $s$ can be approximated in $L^1(|\alpha|)$ by continuous functions $f_k$ with $\|f_k\|_\infty \leqslant 1$. Hence $\int f_k \,\mathrm{d}\alpha \to \int s \,\mathrm{d}\alpha = |\alpha|(\mathcal{X})$, which proves the equality. The final statement is the Riesz–Markov–Kakutani representation theorem with this norm. □

For two absolutely continuous measures $\mathrm{d}\alpha = \rho_\alpha \mathrm{d}\lambda$ and $\mathrm{d}\beta = \rho_\beta \mathrm{d}\lambda$, this gives the concrete formula
$\|\alpha - \beta\|_{\mathrm{TV}} = \int_{\mathcal{X}} |\rho_\alpha(x) - \rho_\beta(x)| \,\mathrm{d}\lambda(x)$. $\alpha = \sum_{i=1}^n \tilde{a}_i \delta_{x_i}, \qquad \beta = \sum_{j=1}^m \tilde{b}_j \delta_{y_j}$, $a_k := \sum_{i : x_i = z_k} \tilde{a}_i, \qquad b_k := \sum_{j : y_j = z_k} \tilde{b}_j$. Then $\alpha = \sum_{k=1}^r a_k \delta_{z_k}$ and $\beta = \sum_{k=1}^r b_k \delta_{z_k}$. This convention merges masses located at the same point before comparing the two measures.
$\|\alpha - \beta\|_{\mathrm{TV}} = \sum_{k=1}^r |a_k - b_k|$.

**Probabilistic interpretation.**

**Definition 2.9** (Random variable and law)**.** Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space and let $\mathcal{X}$ be Polish. An $\mathcal{X}$-valued random variable is a measurable map $X : (\Omega, \mathcal{F}) \to (\mathcal{X}, \mathcal{B}(\mathcal{X}))$. Its law is $\mathrm{Law}(X) \coloneqq X_\sharp \mathbb{P}, \qquad \mathrm{Law}(X)(A) = \mathbb{P}(X \in A)$ for every Borel set $A \subset \mathcal{X}$.

For every integrable function $f$, integration with respect to this law is expectation:
$\int_{\mathcal{X}} f(x)\,\mathrm{d}\,\mathrm{Law}(X)(x) = \mathbb{E}[f(X)]$.

## 2.2 Push-Forward

For a measurable map $T : \mathcal{X} \to \mathcal{Y}$, the push-forward operator $T_\sharp : \mathcal{M}(\mathcal{X}) \to \mathcal{M}(\mathcal{Y})$ records the distribution of the image points. For discrete measures (2.1), the push-forward operation consists simply in moving the positions of all the points in the support of the measure
$T_\sharp \alpha \coloneqq \sum_i \mathrm{a}_i \delta_{T(x_i)}$.

**Definition 2.10** (Push-forward)**.** For a measurable map $T : \mathcal{X} \to \mathcal{Y}$ and $\alpha \in \mathcal{M}(\mathcal{X})$, the push-forward measure $\beta = T_\sharp \alpha \in \mathcal{M}(\mathcal{Y})$ is defined, for every measurable set $B \subset \mathcal{Y}$, by

$$\beta(B) = \alpha\big(\{x \in \mathcal{X} \,;\, T(x) \in B\}\big). \tag{2.2}$$

Equivalently, for every bounded measurable function $h : \mathcal{Y} \to \mathbb{R}$,

$$\int_{\mathcal{Y}} h(y)\mathrm{d}\beta(y) = \int_{\mathcal{X}} h(T(x))\mathrm{d}\alpha(x). \tag{2.3}$$

Note that $T_\sharp$ preserves positivity and total mass, so that if $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ then $T_\sharp \alpha \in \mathcal{M}_+^1(\mathcal{Y})$.

**Proposition 2.11** (Push-forward formula for densities)**.** *Let $U, V \subset \mathbb{R}^d$ be open, let $\alpha$ and $\beta$ have Lebesgue densities $\rho_\alpha$ and $\rho_\beta$ on $U$ and $V$, and let $T : U \to V$ be a $C^1$ diffeomorphism such that $\beta = T_\sharp \alpha$. Then, for Lebesgue-almost every $x \in U$,*

$$\rho_\alpha(x) = |\det(T'(x))| \rho_\beta(T(x)). \tag{2.4}$$

*Equivalently, for Lebesgue-almost every $y \in V$, $\rho_\beta(y) = \rho_\alpha(T^{-1}y)\,\frac{1}{|\det(T'(T^{-1}y))|}$.*

*Proof.* Apply (2.3) to an arbitrary compactly supported continuous test function $h$ and use the change of variables $y = T(x)$, for which $\mathrm{d}y = |\det(T'(x))|\mathrm{d}x$.

$$\begin{aligned}\int_V h(y)\rho_\beta(y)\mathrm{d}y = \int_V h(y)\mathrm{d}\beta(y) = \int_U h(T(x))\mathrm{d}\alpha(x) = \int_U h(T(x))\rho_\alpha(x)\mathrm{d}x \\ = \int_V h(y)\rho_\alpha(T^{-1}y)\frac{\mathrm{d}y}{|\det(T'(T^{-1}y))|},\end{aligned}$$

Since the identity holds for every such $h$, uniqueness of Lebesgue densities gives the target-variable formula in the statement. Substituting $y = T(x)$ then yields (2.4), where $T'(x) \in \mathbb{R}^{d\times d}$ is the Jacobian matrix of $T$. □

## 2.3 Monge's Formulation

**Monge problem.**

**Definition 2.12** (Monge problem and Monge map)**.** For $\alpha \in \mathcal{M}_+^1(\mathcal{X})$, $\beta \in \mathcal{M}_+^1(\mathcal{Y})$ and measurable $c : \mathcal{X} \times \mathcal{Y} \to [0, +\infty]$, define

$$\mathcal{M}_c(\alpha, \beta) \coloneqq \inf_{T:\mathcal{X}\to\mathcal{Y} \text{ measurable}} \left\{ \int_{\mathcal{X}} c(x, T(x))\mathrm{d}\alpha(x) \,;\, T_\sharp \alpha = \beta \right\}. \tag{2.5}$$

A feasible $T$ is a Monge map; a minimizer is an optimal Monge map. The value is $+\infty$ if no feasible map exists.

$T = \mathcal{C}_\beta^{-1} \circ \mathcal{C}_\alpha$,

**Proposition 2.13** (Empirical Monge maps and matchings)**.** *Assume that the source atoms $x_1, \ldots, x_n$ are distinct and that $\alpha = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}, \qquad \beta = \frac{1}{n}\sum_{j=1}^n \delta_{y_j}$. If $T_\sharp \alpha = \beta$, then for each distinct target value $z$ in the support of $\beta$, exactly $n\beta(\{z\})$ source atoms are mapped to $z$. In particular, if the $y_j$ are distinct, then there exists a permutation $\sigma \in \mathrm{Perm}(n)$ such that $T(x_i) = y_{\sigma(i)}$ for all $i$. Conversely, every such assignment of source atoms to target atoms with the correct masses defines a feasible Monge map on the support of $\alpha$, and in the distinct-target case $\int_{\mathcal{X}} c(x, T(x))\mathrm{d}\alpha(x) = \frac{1}{n}\sum_{i=1}^n c(x_i, y_{\sigma(i)})$. The values of a feasible map away from $\{x_1, \ldots, x_n\}$ may be chosen arbitrarily, provided the resulting extension is measurable. If source locations are repeated, they should first be merged into atoms with larger masses; such atoms cannot be split by a Monge map.*

*Proof.* Since $T_\sharp \alpha = \beta$, all images $T(x_i)$ must belong to the support of $\beta$; otherwise the push-forward would give positive mass to a point outside that support. For any target atom $z$, $\beta(\{z\}) = \alpha(T^{-1}(\{z\})) = \frac{1}{n}\#\{i \,;\, T(x_i) = z\}$. This proves the counting statement. If all target atoms have mass $1/n$, each target receives exactly one source atom, which is a permutation. The converse and the cost identity follow by direct substitution. □

**Proposition 2.14** (Existence of transport maps from atomless sources)**.** *Let $\alpha$ and $\beta$ be Borel probability measures on Polish spaces, and assume that $\alpha$ is atomless. Then there exists a measurable map $T$ such that $T_\sharp \alpha = \beta$.*

*Proof.* A standard measure-isomorphism theorem identifies the atomless standard probability space $(\mathcal{X}, \alpha)$ with $([0,1], \mathrm{Leb})$, modulo null sets. It is therefore enough to construct a map from $[0,1]$ to the target law $\beta$. Since the target is Polish, it is a standard Borel space; choose a Borel isomorphism $i$ from a full-$\beta$ Borel subset of $\mathcal{Y}$ onto a Borel subset of $[0,1]$. The generalized quantile of $i_\sharp \beta$ pushes Lebesgue measure to $i_\sharp \beta$. Composing it with $i^{-1}$ and with the source isomorphism gives the desired measurable map, after arbitrary definitions on the discarded null sets. □

**Monge distance.**

**Definition 2.15** (Directed Monge distance)**.** Let $(\mathcal{X}, d)$ be a metric space and $1 \leqslant p < +\infty$. For probability measures with finite $p$th moments, define

$$\tilde{\mathcal{W}}_p(\alpha, \beta)^p := \inf_{T:\mathcal{X}\to\mathcal{X} \text{ measurable}} \left\{ \int_{\mathcal{X}} d(x, T(x))^p \mathrm{d}\alpha(x) \; ; \; T_\sharp \alpha = \beta \right\}. \tag{2.6}$$

The value is $+\infty$ when the constraint set is empty.

**Proposition 2.16** (Directed Monge distance)**.** *Let $\mathcal{X} = \mathcal{Y}$ be a Polish metric space and $1 \leqslant p < +\infty$. On probability measures with finite p-th moment, the quantity $\tilde{\mathcal{W}}_p$ is nonnegative, vanishes only on the diagonal, and satisfies the triangle inequality. It is therefore a directed extended distance: it need not be symmetric and may take the value $+\infty$.*

*Proof.* Nonnegativity is immediate. If $\tilde{\mathcal{W}}_p(\alpha, \beta) = 0$, choose feasible maps $T_k$ with $\int d(x, T_k(x))^p \mathrm{d}\alpha(x) \to 0$. For every bounded 1-Lipschitz function $h$,

$$\left| \int h \mathrm{d}\beta - \int h \mathrm{d}\alpha \right| = \left| \int h(T_k(x)) - h(x) \mathrm{d}\alpha(x) \right| \leqslant \left( \int d(x, T_k(x))^p \mathrm{d}\alpha(x) \right)^{1/p} \longrightarrow 0.$$

Since bounded Lipschitz functions separate probability measures on Polish metric spaces, $\alpha = \beta$. Conversely, the identity map shows that $\tilde{\mathcal{W}}_p(\alpha, \alpha) = 0$.

If either $\tilde{\mathcal{W}}_p(\alpha, \gamma)$ or $\tilde{\mathcal{W}}_p(\gamma, \beta)$ is infinite, the claim is automatic. If both are finite, feasible maps $S_\sharp \alpha = \gamma$ and $T_\sharp \gamma = \beta$ exist, so their composition is feasible from $\alpha$ to $\beta$; in particular, the left-hand side is finite. Fix $\varepsilon > 0$ and choose $\varepsilon$-minimizers satisfying $\mathcal{E}_\alpha(S)^{1/p} \leqslant \tilde{\mathcal{W}}_p(\alpha, \gamma) + \varepsilon, \qquad \mathcal{E}_\gamma(T)^{1/p} \leqslant \tilde{\mathcal{W}}_p(\gamma, \beta) + \varepsilon$. The composed map is feasible from $\alpha$ to $\beta$, and Minkowski's inequality gives

$$\begin{aligned}
\tilde{\mathcal{W}}_p(\alpha, \beta) &\leqslant \left( \int d(x, T(S(x)))^p \mathrm{d}\alpha(x) \right)^{1/p} \\
&\leqslant \left( \int d(x, S(x))^p \mathrm{d}\alpha(x) \right)^{1/p} + \left( \int d(S(x), T(S(x)))^p \mathrm{d}\alpha(x) \right)^{1/p} \\
&= \mathcal{E}_\alpha(S)^{1/p} + \mathcal{E}_\gamma(T)^{1/p} \leqslant \tilde{\mathcal{W}}_p(\alpha, \gamma) + \tilde{\mathcal{W}}_p(\gamma, \beta) + 2\varepsilon.
\end{aligned}$$

Letting $\varepsilon \to 0$ gives the result. □

## 2.4 Existence and Uniqueness of the Monge Map

**Brenier's theorem.**

**Theorem 2.17** (Brenier)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ have finite second moments, and assume that $\alpha$ is absolutely continuous with respect to Lebesgue measure. For the quadratic cost $c(x, y) = \|x - y\|^2$, there exists a convex function $\varphi : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ such that $T = \nabla\varphi, \qquad T_\sharp \alpha = \beta$, and $T$ is the unique optimal Monge map $\alpha$-almost everywhere. The optimal Kantorovich plan is $(\mathrm{Id}, T)_\sharp \alpha$.*

*Proof.* Choose optimal $c$-concave potentials $(f, g)$ for the cost $\|x-y\|^2$ and define $\varphi(x) := \frac{1}{2}\|x\|^2 - \frac{1}{2}f(x), \qquad \psi(y) := \frac{1}{2}\|y\|^2 - \frac{1}{2}g(y)$. The $c$-concavity of $f$ makes $\varphi$ convex. Dual feasibility is equivalent to $\varphi(x) + \psi(y) \geqslant \langle x, y \rangle$, and equality holds for almost every pair under any optimal plan. Replacing $\psi$ by the convex conjugate $\varphi^*$ therefore yields the Fenchel equality $\varphi(x) + \varphi^*(y) = \langle x, y \rangle$ on the optimal relation, hence $y \in \partial\varphi(x)$. Since $\alpha$ has a density and convex functions are differentiable Lebesgue-almost everywhere, $\partial\varphi(x) = \{\nabla\varphi(x)\}$ for $\alpha$-almost every $x$. Every optimal plan is therefore concentrated on the same graph, which proves existence and uniqueness of both the optimal plan and the Monge map. □

$\langle \nabla\varphi(x) - \nabla\varphi(x'), \, x - x' \rangle \geqslant 0.$

**Definition 2.18** (Measures not charging hypersurfaces)**.** A Borel measure $\alpha$ on $\mathbb{R}^d$ does not charge hypersurfaces if $\alpha(S) = 0$ for every countably $(d-1)$-rectifiable set $S$, i.e. every set that can be covered, up to an $\mathcal{H}^{d-1}$-null set, by countably many $C^1$ hypersurfaces.

**Radial measures.** Radial symmetry gives a useful higher-dimensional case where the Brenier map reduces to a one-dimensional monotone rearrangement. A measure $\alpha$ on $\mathbb{R}^d$ is radial if $Q_\sharp \alpha = \alpha$ for every orthogonal map $Q$.

**Proposition 2.19** (Optimal transport between radial measures)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ be radial probability measures with finite second moments, and assume that $\alpha$ is absolutely continuous with respect to Lebesgue measure. Define their radial distribution functions $F_\alpha(r) := \alpha(\{x : \|x\| \leqslant r\}), \qquad F_\beta(r) := \beta(\{y : \|y\| \leqslant r\}), \qquad r \geqslant 0$, and let $F_\beta^{-1}(u) := \inf\{r \geqslant 0 : F_\beta(r) \geqslant u\}$ be the generalized inverse. Set $\tau(r) := F_\beta^{-1}(F_\alpha(r))$. Then the quadratic Brenier map from $\alpha$ to $\beta$ is the radial map $T(0) = 0, \qquad T(x) = \frac{\tau(\|x\|)}{\|x\|} x$ for $x \neq 0$. Moreover, if $\alpha_R = (x \mapsto \|x\|)_\sharp \alpha$ is the law of the source radius, then $\mathcal{W}_2^2(\alpha, \beta) = \int_0^\infty |r - \tau(r)|^2 \, \mathrm{d}\alpha_R(r)$.*

*Proof.* Let $\alpha_R = (\|\cdot\|)_\sharp \alpha$ and $\beta_R = (\|\cdot\|)_\sharp \beta$ be the radial laws. Since $\alpha$ is absolutely continuous, $\alpha_R$ has no atoms. The map $\tau = F_\beta^{-1} \circ F_\alpha$ is therefore the monotone one-dimensional transport from $\alpha_R$ to $\beta_R$. If $X \sim \alpha$ and $R = \|X\|$, then $\tau(R)$ has law $\beta_R$. Moreover, by radiality, the conditional direction $X/\|X\|$ given $R = r$ is uniform on the sphere for $\alpha_R$-a.e. $r > 0$. Thus $T$ changes only the radius, from $R$ to $\tau(R)$, and keeps the uniform angular distribution. It follows that $T_\sharp \alpha$ is radial with radial law $\beta_R$, hence $T_\sharp \alpha = \beta$.

The function $\tau$ is nondecreasing and nonnegative. Define $\psi(r) := \int_0^r \tau(s) \mathrm{d}s$ on $\mathbb{R}_+$. Then $\psi$ is convex and nondecreasing, so $\varphi(x) := \psi(\|x\|)$ is convex on $\mathbb{R}^d$. Since $\alpha$ is absolutely continuous, it does not charge the origin nor the radii where the monotone function $\tau$ is discontinuous. Thus $\nabla\varphi(x) = \tau(\|x\|)x/\|x\| = T(x)$ for $\alpha$-a.e. $x$. The map $T$ is therefore a gradient of a convex potential and pushes $\alpha$ to $\beta$. Brenier's theorem identifies it as the unique quadratic optimal Monge map. The displayed cost formula follows by substituting $T(x)$ in $\int \|x - T(x)\|^2 \mathrm{d}\alpha(x)$. □

If $\alpha$ is not absolutely continuous, the same radial idea is still useful but has to be interpreted at the level of couplings of the radial variables: atoms on spheres may have to be split, so a Monge map need not exist.

**Polar factorization.** Brenier's theorem does more than solve one transport problem: it provides a canonical way to extract the "monotone part" of a nondegenerate map. Suppose one starts from a square-integrable deformation $u : \Omega \to \mathbb{R}^d$, for instance a velocity snapshot, an image deformation or a generative map.

**Proposition 2.20** (Polar factorization). *Let $\Omega \subset \mathbb{R}^d$ be a bounded domain endowed with normalized Lebesgue measure $\lambda$, and let $u \in L^2(\Omega; \mathbb{R}^d)$. Assume that the image law $\beta = u_\sharp\lambda$ is absolutely continuous with respect to Lebesgue measure. Then there exist a measure-preserving map $s : \Omega \to \Omega$ and a convex function $\varphi$ such that $u = \nabla\varphi \circ s \quad \lambda$-a.e. The Brenier factor $\nabla\varphi$ and the rearrangement $s$ are unique $\lambda$-almost everywhere.*

*Proof.* By Brenier's theorem there is a unique convex-gradient map $T = \nabla\varphi$ transporting $\lambda$ to $\beta$. Since $\beta$ is also absolutely continuous, the reverse Brenier map $S$ transporting $\beta$ to $\lambda$ exists, and the two maps are inverse almost everywhere: $T \circ S = \mathrm{Id}$ $\beta$-almost everywhere and $S \circ T = \mathrm{Id}$ $\lambda$-almost everywhere. Set $s := S \circ u$. Then $s_\sharp\lambda = S_\sharp(u_\sharp\lambda) = S_\sharp\beta = \lambda, \qquad T \circ s = T \circ S \circ u = u \quad \lambda$-a.e. Thus $s$ is measure preserving and gives the announced factorization. Uniqueness of $T$ follows from Brenier's theorem, and then $s = S \circ u$ is unique almost everywhere. □

The name is meant to echo the usual polar decomposition of matrices. This analogy becomes literal for linear maps under the Gaussian reference measure.

**Displacement interpolation.**

**Definition 2.21** (Monge and McCann displacement interpolation). If $T_\sharp\alpha = \beta$, the Monge interpolation generated by $T$ is the curve $T_t(x) := (1-t)x + tT(x), \qquad \alpha_t := (T_t)_\sharp\alpha, \qquad t \in [0,1]$. For the quadratic cost, when $T$ is the optimal Brenier map, this curve is called McCann's displacement interpolation.

**Proposition 2.22** (Directed Monge displacement geodesics). *Let $1 \leqslant p < +\infty$, let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ have finite $p$-th moments, and let $T$ be an optimal map for $\tilde{\mathcal{W}}_p(\alpha,\beta)$ in (2.6). Set $\alpha_t = (T_t)_\sharp\alpha$ with $T_t = (1-t)\mathrm{Id} + tT$. Assume that, for every $t < 1$, $T_t$ is one-to-one on a full $\alpha$-measure Borel set. Its inverse on the image of that set is then measurable and defined $\alpha_t$-almost everywhere. For $0 \leqslant s \leqslant t \leqslant 1$, $\tilde{\mathcal{W}}_p(\alpha_s,\alpha_t) = (t-s)\tilde{\mathcal{W}}_p(\alpha,\beta)$. Thus $t \mapsto \alpha_t$ is an oriented constant-speed geodesic for the directed Monge distance. In particular, for $p = 2$, this applies to the Brenier map under the hypotheses of Theorem 2.17.*

*Proof.* The case $s = t$ is trivial, so assume $s < t$. Since $s < 1$, the inverse $T_s^{-1}$ is defined $\alpha_s$-almost everywhere along the transported particles. Define $S_{s,t} := T_t \circ T_s^{-1}$ $\alpha_s$-a.e. Then $(S_{s,t})_\sharp\alpha_s = \alpha_t$, and, using the optimality of $T$, $\tilde{\mathcal{W}}_p(\alpha_s,\alpha_t)^p \leqslant \int \|S_{s,t}(z) - z\|^p \mathrm{d}\alpha_s(z) = \int \|T_t(x) - T_s(x)\|^p \mathrm{d}\alpha(x) = (t-s)^p \tilde{\mathcal{W}}_p(\alpha,\beta)^p$. The same particle construction gives $\tilde{\mathcal{W}}_p(\alpha,\alpha_s) \leqslant s\tilde{\mathcal{W}}_p(\alpha,\beta)$. If $t < 1$, the map $T \circ T_t^{-1}$ sends $\alpha_t$ to $\beta$ and gives $\tilde{\mathcal{W}}_p(\alpha_t,\beta) \leqslant (1-t)\tilde{\mathcal{W}}_p(\alpha,\beta)$; if $t = 1$, this latter distance is zero. Using the triangle inequality from Proposition 2.16,

$$\tilde{\mathcal{W}}_p(\alpha,\beta) \leqslant \tilde{\mathcal{W}}_p(\alpha,\alpha_s) + \tilde{\mathcal{W}}_p(\alpha_s,\alpha_t) + \tilde{\mathcal{W}}_p(\alpha_t,\beta) \leqslant s\tilde{\mathcal{W}}_p(\alpha,\beta) + \tilde{\mathcal{W}}_p(\alpha_s,\alpha_t) + (1-t)\tilde{\mathcal{W}}_p(\alpha,\beta).$$

Hence $\tilde{\mathcal{W}}_p(\alpha_s,\alpha_t) \geqslant (t-s)\tilde{\mathcal{W}}_p(\alpha,\beta)$, which proves equality. For a Brenier map $T = \nabla\varphi$, the map $T_t$ is the gradient of $x \mapsto (1-t)\frac{\|x\|^2}{2} + t\varphi(x)$, which is $(1-t)$-strongly convex for every $t < 1$. On the full-measure set where $\varphi$ is differentiable, this gives $\langle T_t(x) - T_t(y),\, x - y\rangle \geqslant (1-t)\|x-y\|^2$, so $T_t$ is injective there. □

**Regularity and the Monge–Ampère equation.**

**Proposition 2.23** (Caffarelli regularity). *Let $\Omega, \Lambda \subset \mathbb{R}^d$ be bounded uniformly convex domains with $C^2$ boundaries. Let $\alpha = \rho_\alpha(x)\mathrm{d}x$ and $\beta = \rho_\beta(y)\mathrm{d}y$ be supported on $\Omega$ and $\Lambda$, respectively, with $0 < m \leqslant \rho_\alpha, \rho_\beta \leqslant M < +\infty$. If $\rho_\alpha \in C^\alpha(\overline{\Omega})$ and $\rho_\beta \in C^\alpha(\overline{\Lambda})$ for some $\alpha \in (0,1)$, then the Brenier potential $\varphi$ is strictly convex in $\Omega$ and belongs to $C^{2,\alpha}_{\mathrm{loc}}(\Omega)$. Under the standard compatibility and higher boundary-regularity assumptions, the same regularity extends to the boundary.*

*Proof.* The potential solves the Monge–Ampère equation $\det(\nabla^2\varphi(x)) = \frac{\rho_\alpha(x)}{\rho_\beta(\nabla\varphi(x))}$ in the Alexandrov sense, with second boundary condition $\nabla\varphi(\Omega) = \Lambda$. The density bounds and convexity of the domains give strict convexity and localization of sections. Caffarelli's interior theory then yields $C^{2,\alpha}_{\mathrm{loc}}$ estimates for $\varphi$; the boundary statement follows from the boundary regularity theory under uniform convexity and compatibility assumptions. □

For sufficiently smooth positive densities and a classical Brenier map, the change-of-variables formula (2.4) gives the Monge–Ampère equation

$$\det(\nabla^2\varphi(x))\rho_\beta(\nabla\varphi(x)) = \rho_\alpha(x). \tag{2.7}$$

**Proposition 2.24** (Linearization of the Monge–Ampère equation). *Let $\rho_\varepsilon = \rho_0 + \varepsilon r + o(\varepsilon)$ be a smooth perturbation of a positive reference density $\rho_0$ on a smooth bounded domain $\Omega$, with $\int_\Omega r\mathrm{d}x = 0$. Assume that the expansions hold in a topology strong enough to differentiate the change-of-variables identity. If $T_\varepsilon(x) = x + \varepsilon\nabla u(x) + o(\varepsilon)$ transports $\rho_0\mathrm{d}x$ to $\rho_\varepsilon\mathrm{d}x$, then, to first order, $-\nabla\cdot(\rho_0\nabla u) = r$. In particular, when $\rho_0$ is constant, the linearized equation is $-\Delta u = r/\rho_0$. If the source and target occupy the same fixed domain and no mass crosses its boundary, the accompanying first-order boundary condition is $\rho_0\partial_n u = 0$ on $\partial\Omega$; together with $\int_\Omega u\mathrm{d}x = 0$, it fixes the additive constant.*

*Proof.* The change-of-variables equation for $T_\varepsilon$ is $\rho_0(x) = \rho_\varepsilon(T_\varepsilon(x))\det(\nabla T_\varepsilon(x))$. Expanding $\rho_\varepsilon(x + \varepsilon\nabla u) = \rho_0(x) + \varepsilon r(x) + \varepsilon\langle\nabla\rho_0,\, \nabla u\rangle + o(\varepsilon)$ and $\det(\mathrm{Id} + \varepsilon\nabla^2 u) = 1 + \varepsilon\Delta u + o(\varepsilon)$ gives

$$\rho_0 = \rho_0 + \varepsilon\big(r + \langle\nabla\rho_0,\, \nabla u\rangle + \rho_0\Delta u\big) + o(\varepsilon) = \rho_0 + \varepsilon\big(r + \nabla\cdot(\rho_0\nabla u)\big) + o(\varepsilon).$$

The first-order term must vanish. □

## 2.5 Beyond the Quadratic Euclidean Cost

**$\mathcal{W}_p$ costs..** The quadratic cost is special because it identifies the optimal map with the Euclidean gradient of an ordinary convex potential. For the Wasserstein cost, normalized as $c(x,y) = \|x-y\|^p/p$ with $p > 1$, the same Monge-map picture survives, but the potential is adapted to the cost.

$\nabla f(x) = \nabla_x c(x, T(x))$. $T(x) = x - \|\nabla f(x)\|^{q-2}\nabla f(x), \qquad \frac{1}{p} + \frac{1}{q} = 1.$

**Squared geodesic distance.** $T(x) = \exp_x(-\nabla\varphi(x))$,

$$T_t(x) = \exp_x\big(t\exp_x^{-1}(T(x))\big) = \exp_x(-t\nabla\varphi(x)), \qquad \alpha_t = (T_t)_\sharp\alpha, \qquad t \in [0,1].$$

**Poincaré disk.**
**Twist condition.**

**Definition 2.25** (Twist condition)**.** Let $X, Y \subset \mathbb{R}^d$ and let $c \in C^1(X \times Y)$. The cost $c$ satisfies the twist condition from $X$ to $Y$ if, for every fixed $x \in X$, the map $y \in Y \longmapsto \nabla_x c(x,y) \in \mathbb{R}^d$ is injective. The dual twist condition is the analogous injectivity of $x \mapsto \nabla_y c(x,y)$ for every fixed $y$.

**Proposition 2.26** (Twist prevents splitting)**.** *Assume that $c \in C^1(X \times Y)$ satisfies the twist condition from $X$ to $Y$. Suppose that optimal pairs for the cost $c$ admit a source potential $f$ with the following first-order certificate: $f$ is differentiable on a full $\alpha$-measure subset of $X$, and whenever $x$ belongs to this differentiability set and $y$ can be paired optimally with $x$, one has $\nabla f(x) = \nabla_x c(x,y)$. Then, for $\alpha$-almost every source point $x$, there is at most one target point $y$ that can be paired optimally with $x$.*

*Proof.* Let $(f,g)$ be optimal Kantorovich potentials and let $\pi$ be an optimal plan. Complementary slackness gives $f(x)+g(y) = c(x,y)$ for $\pi$-almost every $(x,y)$, while dual feasibility gives $f(z)+g(y) \leqslant c(z,y)$ for all $z$. Thus, for almost every contact pair $(x,y)$, the function $z \mapsto c(z,y)-g(y)$ touches $f$ from above at $x$. At a point where $f$ is differentiable, this gives $\nabla f(x) = \nabla_x c(x,y)$. If the cost is twisted, this equation determines at most one $y$. Hence no optimal plan can split the mass of such an $x$ between several target points. Since differentiability holds on a full $\alpha$-measure set, the plan is concentrated on the graph of a measurable map there. ☐

The quadratic cost satisfies twist since $\nabla_x\|x-y\|^2 = 2(x-y)$; the bilinear cost $c(x,y) = -\langle x,\, y\rangle$ satisfies twist since $\nabla_x c(x,y) = -y$; more generally $c(x,y) = h(x-y)$ is twisted when $h$ is smooth strictly convex and $\nabla h$ is injective. On a Riemannian manifold, the squared geodesic cost is twisted locally away from the cut locus.

**Ma–Trudinger–Wang curvature.**

**Definition 2.27** (Weak MTW condition)**.** Assume that $c \in C^4(X \times Y)$, that $X, Y \subset \mathbb{R}^d$, that $c$ is twisted, and that the mixed Hessian $\nabla^2_{xy}c(x,y)$ is invertible. For $(x,p)$ in the image of the change of variables $(x,y) \mapsto (x, -\nabla_x c(x,y))$, let $Y(x,p)$ be defined by $-\nabla_x c(x, Y(x,p)) = p$, and set $A_{i,j}(x,p) \coloneqq -\partial^2_{x_i x_j} c(x, Y(x,p))$. The cost satisfies the weak MTW condition if $\sum_{i,j,k,l} \partial^2_{p_k p_l} A_{i,j}(x,p)\, \xi_i \xi_j \eta_k \eta_l \geqslant 0 \qquad$ whenever $\langle \xi,\, \eta\rangle = 0$. A strong MTW condition asks for a positive lower bound, proportional to $\|\xi\|^2\|\eta\|^2$, on the same orthogonal directions.

**Proposition 2.28** (MTW condition and regularity)**.** *Assume that $c$ is smooth, twisted and non-degenerate, that the source and target domains satisfy the appropriate mutual $c$-convexity assumptions, and that the source and target densities are positive and bounded above and below. Under the weak MTW condition, the optimal map is locally Hölder continuous; with the corresponding global domain hypotheses, it is continuous up to the boundary. Under the strong MTW condition and higher smoothness of the data and domains, one obtains higher regularity estimates.*

*Proof.* This is the regularity theory of Ma–Trudinger–Wang, Trudinger–Wang and Loeper, stated in structural form rather than with its full boundary hypotheses. The $c$-convex potential solves a generated Jacobian equation. Twist and non-degeneracy turn its optimal relation into a single-valued map, while the weak MTW inequality controls the geometry of contact sets and yields interior Hölder estimates. Strong MTW provides a quantitative curvature lower bound; combined with smooth densities, domain geometry and elliptic estimates, it yields higher regularity. Conversely, Loeper proved that weak MTW is necessary for continuity for arbitrary smooth positive data. ☐

## 2.6 One-Dimensional Transport and Quantiles

**Cumulative and quantile functions.**

**Definition 2.29** (Cumulative and quantile functions)**.** For $\alpha \in \mathcal{M}^1_+(\mathbb{R})$, its cumulative distribution function is

$$\mathcal{C}_\alpha(x) \coloneqq \alpha((-\infty, x]). \tag{2.8}$$

Its generalized inverse, or quantile function, is

$$\mathcal{C}_\alpha^{-1}(r) \coloneqq \inf\left\{x \in \mathbb{R}\,;\, \mathcal{C}_\alpha(x) \geqslant r\right\}, \qquad r \in (0,1). \tag{2.9}$$

**Proposition 2.30** (Quantile push-forward)**.** *One has $(\mathcal{C}_\alpha^{-1})_\sharp \mathrm{Leb}_{[0,1]} = \alpha$. If $\alpha$ has no atoms, then $(\mathcal{C}_\alpha)_\sharp\alpha = \mathrm{Leb}_{[0,1]}$.*

*Proof.* For simplicity, assume first that $\alpha$ has a strictly positive density, so that $\mathcal{C}_\alpha$ is strictly increasing and continuous. Denote $\gamma \coloneqq (\mathcal{C}_\alpha^{-1})_\sharp \mathrm{Leb}_{[0,1]}$. It is enough to prove that $\mathcal{C}_\gamma = \mathcal{C}_\alpha$. For every $x$,

$$\mathcal{C}_\gamma(x) = \int_0^1 \mathbf{1}_{(-\infty,x]}(\mathcal{C}_\alpha^{-1}(z))\mathrm{d}z = \int_0^1 \mathbf{1}_{[0,\mathcal{C}_\alpha(x)]}(z)\mathrm{d}z = \mathcal{C}_\alpha(x),$$

where we used $\mathcal{C}_\alpha^{-1}(z) \leqslant x$ if and only if $z \leqslant \mathcal{C}_\alpha(x)$. General measures follow from the same argument with generalized inverses and right-continuity of the cumulative distribution function. If $\alpha$ has no atoms, the probability integral transform gives $(\mathcal{C}_\alpha)_\sharp\alpha = \mathrm{Leb}_{[0,1]}$. ☐

**1-D Monge solutions.**

**Theorem 2.31** (One-dimensional Monge solution)**.** *Let $\alpha, \beta \in \mathcal{M}^1_+(\mathbb{R})$, assume that $\alpha$ has no atoms, and let $h : \mathbb{R} \to \mathbb{R}$ be convex. Consider the cost $c(x,y) = h(x-y)$. Write $q_\alpha \coloneqq \mathcal{C}_\alpha^{-1}$ and $q_\beta \coloneqq \mathcal{C}_\beta^{-1}$, and assume that $h(q_\alpha - q_\beta) \in L^1(0,1)$. Then*

$$T = q_\beta \circ \mathcal{C}_\alpha = \mathcal{C}_\beta^{-1} \circ \mathcal{C}_\alpha \tag{2.10}$$

*satisfies $T_\sharp\alpha = \beta$ and minimizes $\min_{S_\sharp\alpha=\beta} \int h(x - S(x))\mathrm{d}\alpha(x)$. In particular, $h(t) = |t|^p$ gives the usual one-dimensional Monge map for every $1 \leqslant p < +\infty$ whenever the source has no atoms.*

*Proof.* By Proposition 2.30, atomlessness of $\alpha$ gives $(\mathcal{C}_\alpha)_\sharp\alpha = \mathrm{Leb}_{[0,1]}$, while $(q_\beta)_\sharp\mathrm{Leb}_{[0,1]} = \beta$. Hence $T_\sharp\alpha = \beta$.
Let $S$ be any admissible map and set $\pi_S = (\mathrm{Id}, S)_\sharp\alpha$. The one-dimensional uncrossing theorem for relaxed couplings, proved later as Theorem 3.28, shows that the quantile coupling $\pi^\star = (q_\alpha, q_\beta)_\sharp\mathrm{Leb}_{[0,1]}$ is optimal among all couplings. Since $\alpha$ has no atoms, $(\mathrm{Id}, T)_\sharp\alpha = \pi^\star$, and therefore $\int h(x - T(x))\mathrm{d}\alpha(x) = \int h(x-y)\mathrm{d}\pi^\star(x,y) \leqslant \int h(x-y)\mathrm{d}\pi_S(x,y) = \int h(x - S(x))\mathrm{d}\alpha(x)$. This proves optimality of $T$ among Monge maps. □

For discrete measures, one cannot directly apply the map formula when the source has atoms, but if the measures are uniform on the same number of Dirac masses, then it is exactly the sorting formula of Proposition 1.2.

**Proposition 2.32** (One-dimensional Wasserstein formulas)**.** *Let $1 \leqslant p < +\infty$ and let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R})$ have finite $p$-th moments. Then*

$$\mathcal{W}_p(\alpha,\beta)^p = \int_0^1 \left|\mathcal{C}_\alpha^{-1}(r) - \mathcal{C}_\beta^{-1}(r)\right|^p \mathrm{d}r = \|\mathcal{C}_\alpha^{-1} - \mathcal{C}_\beta^{-1}\|_{L^p([0,1])}^p. \tag{2.11}$$

*For $p = 1$, this is equivalently*

$$\mathcal{W}_1(\alpha,\beta) = \|\mathcal{C}_\alpha - \mathcal{C}_\beta\|_{L^1(\mathbb{R})} = \int_{\mathbb{R}} |\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x)|\,\mathrm{d}x = \int_{\mathbb{R}} \left|\int_{-\infty}^x \mathrm{d}(\alpha-\beta)\right| \mathrm{d}x. \tag{2.12}$$

*Proof.* The first formula follows from Theorem 3.28: the optimal coupling is obtained by taking the same quantile level $r$ for both measures. For $p = 1$, use the layer-cake identity. If $q_\alpha$ and $q_\beta$ are the two quantile functions, then $\int_0^1 |q_\alpha(r) - q_\beta(r)|\mathrm{d}r = \int_{\mathbb{R}} \lambda\big(\{r : q_\alpha(r) \leqslant x < q_\beta(r)\} \cup \{r : q_\beta(r) \leqslant x < q_\alpha(r)\}\big)\mathrm{d}x$. The measure of the displayed set is exactly $|\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(x)|$ for almost every $x$. □

**Proposition 2.33** (Bobkov–Ledoux cumulative formula)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1([a,b])$ and write $s_+ := \max(s, 0)$. Then*

$$\mathcal{W}_2^2(\alpha,\beta) = 2\int_a^b \int_x^b \Big[\big(\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(y)\big)_+ + \big(\mathcal{C}_\beta(x) - \mathcal{C}_\alpha(y)\big)_+\Big]\mathrm{d}y\,\mathrm{d}x. \tag{2.13}$$

*Proof.* For any $u, v \in [a,b]$, one has $|u-v|^2 = 2\int_a^b\int_x^b \Big[\mathbf{1}_{\{u \leqslant x \leqslant y < v\}} + \mathbf{1}_{\{v \leqslant x \leqslant y < u\}}\Big]\mathrm{d}y\,\mathrm{d}x$, because, when $u < v$, the integration domain is the triangle $u \leqslant x \leqslant y < v$, whose area is $(v-u)^2/2$, and the case $v < u$ is symmetric. Apply this identity with $u = q_\alpha(r)$ and $v = q_\beta(r)$, where $q_\alpha = \mathcal{C}_\alpha^{-1}$ and $q_\beta = \mathcal{C}_\beta^{-1}$, and integrate with respect to $r \in (0,1)$. Fubini's theorem applies since the integrand is bounded. For $x \leqslant y$, the generalized inverse identities give, up to endpoint sets of zero Lebesgue measure, $\lambda\{r\,;\,q_\alpha(r) \leqslant x \text{ and } q_\beta(r) > y\} = \big(\mathcal{C}_\alpha(x) - \mathcal{C}_\beta(y)\big)_+$, and the same argument with $\alpha$ and $\beta$ exchanged gives the second positive part. □

$\mathcal{C}_{\alpha_t}^{-1}(r) = (1-t)\mathcal{C}_\alpha^{-1}(r) + t\mathcal{C}_\beta^{-1}(r), \qquad r \in (0,1).$

**OT on trees.**

**Proposition 2.34** (Cumulative formula on a tree)**.** *Let $\mathsf{T} = (V, E, \ell)$ be a finite weighted tree, where each edge $e$ has length $\ell_e > 0$, and let $d_\mathsf{T}$ be its geodesic distance. Choose a root $o$. For an edge $e = (u,v)$ oriented away from $o$, let $V_e$ be the set of vertices in the connected component containing $v$ after removing $e$. If $\alpha, \beta \in \mathcal{P}(V)$, then*

$$\mathcal{W}_{1,d_\mathsf{T}}(\alpha,\beta) = \min_{\pi \in \mathcal{U}(\alpha,\beta)} \sum_{x,y \in V} d_\mathsf{T}(x,y)\pi_{xy} = \sum_{e \in E} \ell_e\,|\alpha(V_e) - \beta(V_e)|\,. \tag{2.14}$$

*The value is computable in $O(|V|)$ arithmetic operations once the signed masses $\alpha - \beta$ are stored on the vertices, by one postorder traversal of the rooted tree.*

*Proof.* Let $\sigma = \alpha - \beta$. Removing an edge $e$ splits the tree into $V_e$ and its complement. For any coupling $\pi$, the net amount of mass crossing $e$ from $V_e$ to $V \setminus V_e$ minus the amount crossing in the reverse direction is fixed and equals $\sigma(V_e)$. Hence the total amount of transported mass whose path uses $e$ is at least $|\sigma(V_e)|$. Since every unit of mass crossing $e$ pays the length $\ell_e$, summing over edges gives $\sum_{x,y} d_\mathsf{T}(x,y)\pi_{x,y} = \sum_{e\in E} \ell_e \sum_{x,y\colon\, e\in[x,y]} \pi_{x,y} \geqslant \sum_{e \in E} \ell_e|\sigma(V_e)|$, where $[x,y]$ is the unique path between $x$ and $y$.
Process the rooted tree from the leaves to the root. At a vertex $v \neq o$, compute the total signed mass in the subtree below $v$, namely $F_e = \sigma(V_e)$ for the edge $e$ joining $v$ to its parent. If $F_e > 0$, send $F_e$ units of surplus from this subtree to the parent side; if $F_e < 0$, import $-F_e$ units from the parent side into the subtree. After this operation the subtree has zero net imbalance and can be collapsed into its parent. Continuing upward balances all subtrees because $\sigma(V) = 0$. Decomposing these edge fluxes into source-to-target paths yields a coupling whose traffic across each edge is exactly $|F_e|$, and therefore whose cost is the right-hand side of (2.14). The same postorder pass computes all $F_e$, proving the complexity claim. □

**Triangular rearrangements.**

**Proposition 2.35** (Knothe–Rosenblatt triangular rearrangement)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$. Assume that the first marginal of $\alpha$ and, recursively, the one-dimensional conditional source laws used below are atomless almost everywhere. Fix measurable versions of all regular conditional distributions. Then there is a triangular map $T(x_1, \dots, x_d) = (T_1(x_1), T_2(x_1, x_2), \dots, T_d(x_1, \dots, x_d))$ such that $T_\sharp\alpha = \beta$ and, for each $k$, the function $x_k \mapsto T_k(x_1, \dots, x_k)$ is nondecreasing for $\alpha_{<k}$-almost every $x_{<k} = (x_1, \dots, x_{k-1})$.*

*Proof.* The construction is recursive. For $k = 1$, let $T_1$ be the monotone rearrangement between the first marginal of $\alpha$ and the first marginal of $\beta$. Suppose that $T_1, \dots, T_{k-1}$ have already been constructed. Write $x_{<k} = (x_1, \dots, x_{k-1})$ and $T_{<k} = (T_1, \dots, T_{k-1})$. By induction, $(T_{<k})_\sharp\alpha_{<k} = \beta_{<k}$, where $\alpha_{<k}$ and $\beta_{<k}$ are the first $(k-1)$-coordinate marginals. Let $\alpha^k_{x_{<k}}$ and $\beta^k_{y_{<k}}$ be regular conditional laws of the $k$-th coordinate given the previous coordinates. Define $T_k(x_{<k}, \cdot)$ as the one-dimensional monotone rearrangement from $\alpha^k_{x_{<k}}$ to $\beta^k_{T_{<k}(x_{<k})}$.
The map $T_k(x_{<k}, \cdot)$ is nondecreasing by the one-dimensional rearrangement theorem. The chain rule for disintegrations then shows that, after step $k$, the first $k$ coordinates of $T_\sharp\alpha$ have the same law as the first $k$ coordinates of $\beta$. At $k = d$ this gives $T_\sharp\alpha = \beta$. Target atoms are handled by generalized quantiles. If a source conditional law has atoms that must be split, the same recursive construction defines a triangular Markov kernel rather than a deterministic map. □

**Proposition 2.36** (Anisotropic Brenier maps converge to Knothe–Rosenblatt)**.** *Let $\alpha, \beta$ be compactly supported probability measures on $\mathbb{R}^d$ with densities bounded above and below on their rectangular supports. In particular, the source and target conditional laws entering the Knothe–Rosenblatt construction are atomless almost everywhere. For $\varepsilon > 0$, set $c_\varepsilon(x,y) := \sum_{k=1}^d \varepsilon^{k-1} |x_k - y_k|^2$. Let $T_\varepsilon$ be the Monge map from $\alpha$ to $\beta$ for the cost $c_\varepsilon$, and let $T_{\mathrm{KR}}$ be the triangular Knothe–Rosenblatt rearrangement with the coordinate order used above. Then $T_\varepsilon \longrightarrow T_{\mathrm{KR}}$ in $L^2(\alpha;\mathbb{R}^d)$ as $\varepsilon \to 0$.*

*Proof.* We summarize the scale-separated argument. Let $\pi_\varepsilon = (\mathrm{Id}, T_\varepsilon)_\sharp \alpha$. For a coupling $\gamma$, set $I_k(\gamma) := \int |x_k - y_k|^2 \, \mathrm{d}\gamma(x,y)$ and $F_\varepsilon(\gamma) := \sum_{k=1}^d \varepsilon^{k-1} I_k(\gamma)$. Since the supports are compact, every sequence $\varepsilon \downarrow 0$ has a subsequence for which $\pi_\varepsilon$ converges weakly to a coupling $\pi$ between $\alpha$ and $\beta$. The coordinate costs are bounded and continuous on the common compact support. Dividing $F_\varepsilon$ by its leading coefficient and passing to the limit shows that $\pi$ minimizes $I_1$. The first source marginal is atomless, so the $(x_1, y_1)$-marginal of $\pi$ is the unique monotone coupling; equivalently, $y_1 = T_1(x_1)$ under $\pi$.
Optimality against the Knothe coupling $\pi_{\mathrm{KR}}$ gives $F_\varepsilon(\pi_\varepsilon) \leqslant F_\varepsilon(\pi_{\mathrm{KR}})$. Since $I_1(\pi_\varepsilon)$ is bounded below by the optimal first-coordinate value $I_1(\pi_{\mathrm{KR}})$, cancellation and division by $\varepsilon$ yield $\sum_{k=2}^d \varepsilon^{k-2} I_k(\pi_\varepsilon) \leqslant \sum_{k=2}^d \varepsilon^{k-2} I_k(\pi_{\mathrm{KR}})$. Passing to the limit shows that $\pi$ minimizes $I_2$ among couplings with the already identified first-coordinate marginal. Disintegration with respect to $(x_1, y_1)$ turns this into the monotone transport between the conditional laws of $x_2$ and $y_2$, so $y_2 = T_2(x_1, x_2)$ under $\pi$.
The higher-coordinate induction repeats this comparison after disintegrating over the already identified coordinates. Its technical point is to control the residual suboptimality of the earlier coordinates before dividing by the next power of $\varepsilon$; the atomlessness of both source and target conditionals makes the preceding monotone couplings invertible almost everywhere. The scale-separated estimate proved in then gives successively $y_k = T_k(x_1, \dots, x_k)$ for every $k$.
Finally, let $X \sim \alpha$. The graph couplings are the laws of $(X, T_\varepsilon(X))$, and they converge weakly to the law of $(X, T_{\mathrm{KR}}(X))$. To turn this into convergence of maps, fix $\zeta > 0$. By Lusin's theorem, there is a compact set $K$ with $\alpha(K) > 1 - \zeta$ on which $T_{\mathrm{KR}}$ is continuous. On $K$, the set $\{(x,y) \,;\, x \in K, \ \|y - T_{\mathrm{KR}}(x)\| \geqslant \delta\}$ is closed and has zero mass under the limiting graph coupling. Portmanteau's theorem gives

$$\limsup_{\varepsilon \to 0} \alpha(\{x \,;\, x \in K, \ \|T_\varepsilon(x) - T_{\mathrm{KR}}(x)\| \geqslant \delta\}) = 0.$$

Adding the complement of $K$ and letting $\zeta \to 0$ proves convergence in $\alpha$-probability. Since all maps take values in a common compact set, their squared differences are uniformly integrable, and convergence therefore holds in $L^2(\alpha)$. ☐

## 2.7 Gaussian Measures and the Bures Metric

**One-dimensional Gaussians.** Let $\alpha = \mathcal{N}(m_\alpha, \sigma_\alpha^2)$ and $\beta = \mathcal{N}(m_\beta, \sigma_\beta^2)$ be two nondegenerate Gaussians on $\mathbb{R}$. Then $T(x) = \frac{\sigma_\beta}{\sigma_\alpha}(x - m_\alpha) + m_\beta$ satisfies $T_\sharp \alpha = \beta$. It is the derivative of the convex function
$\varphi(x) = \frac{\sigma_\beta}{2\sigma_\alpha}(x - m_\alpha)^2 + m_\beta x$,

$$\mathcal{W}_2(\alpha,\beta)^2 = \int_{\mathbb{R}} \left( \frac{\sigma_\beta}{\sigma_\alpha}(x - m_\alpha) + m_\beta - x \right)^2 \mathrm{d}\alpha(x) = (m_\alpha - m_\beta)^2 + (\sigma_\alpha - \sigma_\beta)^2.$$

The formula extends by continuity to Dirac masses, although the affine Monge map itself only pushes a Dirac source to another Dirac. Thus the OT geometry of one-dimensional Gaussians is the Euclidean geometry of the closed half-plane $(m, \sigma) \in \mathbb{R} \times \mathbb{R}_+$.
**Multivariate Gaussians.**

$$\alpha = \mathcal{N}(m_\alpha, \Sigma_\alpha), \qquad \beta = \mathcal{N}(m_\beta, \Sigma_\beta), \qquad T(x) = m_\beta + A(x - m_\alpha), \tag{2.15}$$

**Proposition 2.37** (Affine push-forward of Gaussians)**.** *One has $T_\sharp \alpha = \beta$ if and only if*

$$A \Sigma_\alpha A^\top = \Sigma_\beta. \tag{2.16}$$

*Proof.* An affine function maps a Gaussian to a Gaussian, so the law of $T(X)$ is determined by its mean and covariance. If $X \sim \alpha$ and $Y = T(X)$, then $\mathbb{E}(Y) = m_\beta + A\mathbb{E}(X - m_\alpha) = m_\beta$, and $\mathbb{E}((Y - m_\beta)(Y - m_\beta)^\top) = A\mathbb{E}((X - m_\alpha)(X - m_\alpha)^\top)A^\top = A\Sigma_\alpha A^\top$. Thus $A\Sigma_\alpha A^\top = \Sigma_\beta$ is necessary and sufficient for $Y$ to have the same mean and covariance as $\beta$; since both laws are Gaussian, this is equivalent to equality in distribution. ☐

**Definition 2.38** (Bures metric)**.** For positive semidefinite covariance matrices $\Sigma$ and $\Lambda$, the Bures metric is

$$\mathcal{B}(\Sigma, \Lambda)^2 := \operatorname{tr}\left( \Sigma + \Lambda - 2(\Sigma^{1/2} \Lambda \Sigma^{1/2})^{1/2} \right). \tag{2.17}$$

**Proposition 2.39** (Gaussian $\mathcal{W}_2$ formula and Bures covariance term)**.** *Assume that $\Sigma_\alpha$ and $\Sigma_\beta$ are positive definite. The covariance equation and its unique symmetric positive-definite solution are*

$$A\Sigma_\alpha A = \Sigma_\beta, \quad A \succ 0, \qquad \Longleftrightarrow \qquad A = \Sigma_\alpha^{-1/2} \left( \Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2} \right)^{1/2} \Sigma_\alpha^{-1/2}. \tag{2.18}$$

*The affine map $T(x) = m_\beta + A(x - m_\alpha)$ is the optimal quadratic-cost transport from $\mathcal{N}(m_\alpha, \Sigma_\alpha)$ to $\mathcal{N}(m_\beta, \Sigma_\beta)$, and*

$$\mathcal{W}_2(\alpha, \beta)^2 = \|m_\alpha - m_\beta\|^2 + \mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2, \tag{2.19}$$

*where $\mathcal{B}$ is the Bures metric of Definition 2.38.*

*Proof.* Multiplying $A\Sigma_\alpha A = \Sigma_\beta$ on the left and right by $\Sigma_\alpha^{1/2}$ gives $(\Sigma_\alpha^{1/2} A \Sigma_\alpha^{1/2})^2 = \Sigma_\alpha^{1/2} \Sigma_\beta \Sigma_\alpha^{1/2}$. Since $A$ is symmetric positive, $\Sigma_\alpha^{1/2} A \Sigma_\alpha^{1/2}$ is symmetric positive and is therefore the unique positive square root of the right-hand side. Conversely, the matrix in (2.18) is symmetric positive and satisfies the covariance equation.

By Proposition 2.37, this affine map pushes $\alpha$ to $\beta$. It is the gradient of a convex quadratic potential, so Brenier's theorem implies optimality. If $X \sim \alpha$, then

$$\begin{aligned}\mathbb{E}\|X - T(X)\|^2 &= \|m_\alpha - m_\beta\|^2 + \mathbb{E}\|(\mathrm{Id} - A)(X - m_\alpha)\|^2 \\ &= \|m_\alpha - m_\beta\|^2 + \mathrm{tr}((\mathrm{Id} - A)\Sigma_\alpha(\mathrm{Id} - A)^\top) \\ &= \|m_\alpha - m_\beta\|^2 + \mathrm{tr}(\Sigma_\alpha) + \mathrm{tr}(A\Sigma_\alpha A) - 2\,\mathrm{tr}(A\Sigma_\alpha) \\ &= \|m_\alpha - m_\beta\|^2 + \mathrm{tr}(\Sigma_\alpha) + \mathrm{tr}(\Sigma_\beta) - 2\,\mathrm{tr}((\Sigma_\alpha^{1/2}\Sigma_\beta\Sigma_\alpha^{1/2})^{1/2}),\end{aligned}$$

which is the desired expression. The formula for singular covariance matrices follows by adding $\eta\mathrm{Id}$ and letting $\eta \downarrow 0$. □

**Alternate formulation of the Bures metric.**

**Proposition 2.40** (Bures metric as a second-moment quotient)**.** *For $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, set its raw second-moment matrix to be $\Phi_2(\alpha) := \int_{\mathbb{R}^d} uu^\top \mathrm{d}\alpha(u) \in \mathbb{S}_+^d$. For $\Sigma, \Lambda \in \mathbb{S}_+^d$, $\min_{\Phi_2(\alpha)=\Sigma,\ \Phi_2(\beta)=\Lambda} \mathcal{W}_2(\alpha,\beta)^2 = \mathcal{B}(\Sigma,\Lambda)^2$. In other words, the Bures metric is the Wasserstein quotient distance induced on positive semidefinite matrices by identifying all probability measures with the same raw second-moment matrix.*

*Proof.* Assume first that $\Sigma$ and $\Lambda$ are positive definite. Fix two admissible laws $\alpha, \beta$ and an arbitrary coupling $\pi$ between them, and write $K := \int_{\mathbb{R}^d\times\mathbb{R}^d} uv^\top \mathrm{d}\pi(u,v)$. Then $\int \|u - v\|^2 \mathrm{d}\pi(u,v) = \mathrm{tr}(\Sigma) + \mathrm{tr}(\Lambda) - 2\,\mathrm{tr}(K)$. The block second-moment matrix

$$\int \begin{pmatrix} u \\ v \end{pmatrix}\begin{pmatrix} u \\ v \end{pmatrix}^\top \mathrm{d}\pi(u,v) = \begin{pmatrix} \Sigma & K \\ K^\top & \Lambda \end{pmatrix}$$

is positive semidefinite. The Schur complement gives $R := \Sigma^{-1/2}K\Lambda^{-1/2}$ with $RR^\top \preceq \mathrm{Id}$, so $K = \Sigma^{1/2}R\Lambda^{1/2}$ with $\|R\|_{\mathrm{op}} \leqslant 1$. By duality between the nuclear and operator norms,

$$\mathrm{tr}(K) \leqslant \sup_{\|R\|_{\mathrm{op}}\leqslant 1} \mathrm{tr}(\Sigma^{1/2}R\Lambda^{1/2}) = \|\Lambda^{1/2}\Sigma^{1/2}\|_* = \mathrm{tr}\left((\Sigma^{1/2}\Lambda\Sigma^{1/2})^{1/2}\right).$$

For singular matrices, fix $\eta > 0$. Adding $\eta\mathrm{Id}$ to the two diagonal blocks preserves block positivity, so the same Schur-complement argument gives $\mathrm{tr}(K) \leqslant \mathrm{tr}\left(((\Sigma+\eta\mathrm{Id})^{1/2}(\Lambda+\eta\mathrm{Id})(\Sigma+\eta\mathrm{Id})^{1/2})^{1/2}\right)$. Letting $\eta \downarrow 0$ and using continuity of the matrix square root gives the same trace bound with $\Sigma, \Lambda$. Hence every admissible coupling has cost at least $\mathcal{B}(\Sigma,\Lambda)^2$.
Conversely, choose the centered Gaussian laws $\mathcal{N}(0,\Sigma)$ and $\mathcal{N}(0,\Lambda)$. They satisfy the two second-moment constraints, and Proposition 2.39 gives $\mathcal{W}_2(\mathcal{N}(0,\Sigma),\mathcal{N}(0,\Lambda))^2 = \mathcal{B}(\Sigma,\Lambda)^2$, again by continuity in the singular case. This admissible pair attains the lower bound, proving the claim. □

**Proposition 2.41** (Bures metric as a Procrustes metric)**.** *Let $\Sigma, \Lambda \in \mathbb{S}_+^d$, let $d' \geqslant d$, and fix any $A_0, B_0 \in \mathbb{R}^{d\times d'}$ such that $A_0A_0^\top = \Sigma$ and $B_0B_0^\top = \Lambda$. Then*

$$\mathcal{B}(\Sigma,\Lambda)^2 = \min_{\substack{A,B\in\mathbb{R}^{d\times d'} \\ AA^\top=\Sigma,\ BB^\top=\Lambda}} \|A - B\|_{\mathrm{F}}^2 = \min_{R\in\mathrm{O}(d')} \|A_0 - B_0R\|_{\mathrm{F}}^2. \tag{2.20}$$

*For $d' = d$, one may in particular take $A_0 = \Sigma^{1/2}$ and $B_0 = \Lambda^{1/2}$; the second minimum is then over $\mathrm{O}(d)$.*

*Proof.* Two matrices in $\mathbb{R}^{d\times d'}$ have the same row-Gram matrix if and only if they differ by right multiplication by an orthogonal matrix. Indeed, equality of the Gram matrices makes the linear map sending each row of one matrix to the corresponding row of the other well defined and isometric on their row spans; this isometry extends to an element of $\mathrm{O}(d')$. Therefore every admissible pair has the form $A = A_0U$ and $B = B_0V$ for some $U, V \in \mathrm{O}(d')$. Right invariance of the Frobenius norm gives $\min_{U,V\in\mathrm{O}(d')} \|A_0U - B_0V\|_{\mathrm{F}}^2 = \min_{R\in\mathrm{O}(d')} \|A_0 - B_0R\|_{\mathrm{F}}^2$, where one may take $R = VU^\top$. Thus the two factor minima in (2.20) coincide and are independent of the chosen representatives $A_0, B_0$.
Expanding the last objective and applying the orthogonal Procrustes formula, which follows directly from a singular-value decomposition, yields $\min_{R\in\mathrm{O}(d')} \|A_0 - B_0R\|_{\mathrm{F}}^2 = \mathrm{tr}(\Sigma) + \mathrm{tr}(\Lambda) - 2\|A_0^\top B_0\|_*$. Moreover, $(A_0^\top B_0)^\top(A_0^\top B_0) = B_0^\top\Sigma B_0$, whose nonzero eigenvalues are those of $\Sigma^{1/2}B_0B_0^\top\Sigma^{1/2} = \Sigma^{1/2}\Lambda\Sigma^{1/2}$. $\|A_0^\top B_0\|_* = \mathrm{tr}((\Sigma^{1/2}\Lambda\Sigma^{1/2})^{1/2})$. Definition 2.38 now identifies the resulting value with $\mathcal{B}(\Sigma,\Lambda)^2$. □

Thus the map $A \mapsto AA^\top$ identifies $\mathbb{S}_+^d$ with the orbit space $\mathbb{R}^{d\times d'}/\mathrm{O}(d')$, and $\mathcal{B}$ is the quotient distance induced by the Frobenius metric. This is the matrix counterpart of the quotient transport geometries developed in Section 10.3.

**Proposition 2.42** (Metric and convexity properties of the Bures term)**.** *The function $\mathcal{B}$ is a distance on positive semidefinite covariance matrices. Moreover, $\mathcal{B}^2$ is jointly convex: for $t \in [0,1]$, $\mathcal{B}^2((1-t)\Sigma_0 + t\Sigma_1, (1-t)\Lambda_0 + t\Lambda_1) \leqslant (1-t)\mathcal{B}^2(\Sigma_0,\Lambda_0) + t\mathcal{B}^2(\Sigma_1,\Lambda_1)$.*

*Proof.* The free-factor formula in Proposition 2.41 is symmetric and nonnegative. Since its minima are attained, it vanishes if and only if two factors coincide, which is equivalent to $\Sigma = \Lambda$. For the triangle inequality, use the square-factor specialization and let $Q_1, Q_2$ be optimal orthogonal alignments for $(\Sigma,\Lambda)$ and $(\Lambda,\Gamma)$, respectively. Since $Q_2Q_1$ is admissible for $(\Sigma,\Gamma)$,

$$\mathcal{B}(\Sigma,\Gamma) \leqslant \|\Sigma^{1/2} - \Gamma^{1/2}Q_2Q_1\|_{\mathrm{F}} \leqslant \|\Sigma^{1/2} - \Lambda^{1/2}Q_1\|_{\mathrm{F}} + \|\Lambda^{1/2} - \Gamma^{1/2}Q_2\|_{\mathrm{F}} = \mathcal{B}(\Sigma,\Lambda) + \mathcal{B}(\Lambda,\Gamma).$$

For convexity, use the free-factor formula in Proposition 2.41. For $i \in \{0,1\}$, choose optimal square factors $(U_i, V_i)$ for $(\Sigma_i, \Lambda_i)$ and define the block factors $U_t = [\sqrt{1-t}\,U_0, \sqrt{t}\,U_1]$, $\quad V_t = [\sqrt{1-t}\,V_0, \sqrt{t}\,V_1]$. Set $\Sigma_t = (1-t)\Sigma_0 + t\Sigma_1$ and $\Lambda_t = (1-t)\Lambda_0 + t\Lambda_1$. Then $U_tU_t^\top = \Sigma_t$ and $V_tV_t^\top = \Lambda_t$. Applying the free-factor formula with $2d$ columns gives

$$\mathcal{B}^2(\Sigma_t,\Lambda_t) \leqslant \|U_t - V_t\|_{\mathrm{F}}^2 = (1-t)\|U_0 - V_0\|_{\mathrm{F}}^2 + t\|U_1 - V_1\|_{\mathrm{F}}^2 = (1-t)\mathcal{B}^2(\Sigma_0,\Lambda_0) + t\mathcal{B}^2(\Sigma_1,\Lambda_1),$$

which proves joint convexity. □

**Comparison with the Fisher–Rao metric for zero-mean Gaussians.**

$$\mathrm{KL}(\Sigma \mid \Sigma') := \mathrm{KL}\left(\mathcal{N}(0,\Sigma) \mid \mathcal{N}(0,\Sigma')\right) = \frac{1}{2}\left(\operatorname{tr}((\Sigma')^{-1}\Sigma) - d - \log\det((\Sigma')^{-1}\Sigma)\right).$$

If $D = D^\top$ and $\Sigma + \varepsilon D$ remains positive definite, then

$$\mathrm{KL}(\Sigma + \varepsilon D \mid \Sigma) = \frac{\varepsilon^2}{2}\langle Q_\Sigma(D), D\rangle_F + o(\varepsilon^2), \qquad Q_\Sigma(D) = \frac{1}{2}\Sigma^{-1}D\Sigma^{-1},$$

where $\langle A, B\rangle_F = \operatorname{tr}(A^\top B)$ is the Frobenius pairing. Thus the local quadratic form is $g_\Sigma(D,E) = \frac{1}{2}\operatorname{tr}(\Sigma^{-1}D\Sigma^{-1}E)$. $d_{\mathrm{FR}}(\Sigma,\Sigma')^2 = \frac{1}{2}\|\log(\Sigma^{-1/2}\Sigma'\Sigma^{-1/2})\|_F^2$, where log denotes the principal matrix logarithm, and the corresponding geodesic is

$$\Sigma_t^{\mathrm{FR}} = \Sigma^{1/2}\left(\Sigma^{-1/2}\Sigma'\Sigma^{-1/2}\right)^t\Sigma^{1/2}.$$

The contrast with Bures is especially visible near the boundary of the covariance cone. Fisher–Rao is different. $\Delta_z = t^2 - u^2 - v^2, \qquad \langle z, z'\rangle_{\mathrm{E}} = tt' + uu' + vv', \qquad \langle z, z'\rangle_{\mathrm{L}} = tt' - uu' - vv'.$

$$\mathcal{B}(\Sigma,\Sigma')^2 = \sqrt{2}(t+t') - 2\sqrt{\langle z, z'\rangle_{\mathrm{E}} + \sqrt{\Delta_z\Delta_{z'}}}, \tag{2.21}$$

$$d_{\mathrm{FR}}(\Sigma,\Sigma')^2 = \frac{1}{2}\left((\log\lambda_+)^2 + (\log\lambda_-)^2\right), \qquad \lambda_\pm = \frac{\langle z, z'\rangle_{\mathrm{L}} \pm \sqrt{\langle z, z'\rangle_{\mathrm{L}}^2 - \Delta_z\Delta_{z'}}}{\Delta_z}, \tag{2.22}$$

where the Fisher–Rao formula requires $\Delta_z, \Delta_{z'} > 0$ and $\lambda_\pm$ are the eigenvalues of $\Sigma^{-1}\Sigma'$.

# 3 Kantorovich Relaxation

## 3.1 Discrete Relaxation

**Mass splitting and couplings.** $\alpha = \sum_i \mathrm{a}_i\delta_{x_i}, \qquad \beta = \sum_j \mathrm{b}_j\delta_{y_j}.$

**Definition 3.1** (Discrete couplings and mass conservation)**.** Admissible couplings are only constrained to satisfy conservation of mass:

$$\mathrm{U}(\mathrm{a},\mathrm{b}) := \left\{\mathrm{P} \in \mathbb{R}_+^{n\times m}\ ;\ \mathrm{P}\mathbb{1}_m = \mathrm{a} \quad \text{and} \quad \mathrm{P}^\top\mathbb{1}_n = \mathrm{b}\right\}. \tag{3.1}$$

Equivalently, $\mathrm{P}\mathbb{1}_m = \left(\sum_j \mathrm{P}_{i,j}\right)_i \in \mathbb{R}^n, \qquad \mathrm{P}^\top\mathbb{1}_n = \left(\sum_i \mathrm{P}_{i,j}\right)_j \in \mathbb{R}^m.$

**Definition 3.2** (Discrete product coupling)**.** Given weights $\mathrm{a} \in \Sigma_n$ and $\mathrm{b} \in \Sigma_m$, the discrete product, or trivial, coupling is $(\mathrm{a}\otimes\mathrm{b})_{i,j} := \mathrm{a}_i\mathrm{b}_j$. It belongs to $\mathrm{U}(\mathrm{a},\mathrm{b})$ and corresponds to choosing the source and target labels independently.

**Proposition 3.3** (Discrete product optimality is degenerate)**.** *Assume that the zero-mass rows and columns have been removed, so that* $\mathrm{a}_i > 0$ *and* $\mathrm{b}_j > 0$*, and let* $\mathrm{C}$ *be a finite cost matrix. The product plan* $\mathrm{a}\otimes\mathrm{b}$ *minimizes* $\mathrm{P} \mapsto \langle \mathrm{C}, \mathrm{P}\rangle$ *over* $\mathrm{U}(\mathrm{a},\mathrm{b})$ *if and only if every coupling* $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$ *minimizes it.*

*Proof.* The reverse implication is immediate. Assume conversely that $\mathrm{a}\otimes\mathrm{b}$ is optimal and let $\mathrm{Q} \in \mathrm{U}(\mathrm{a},\mathrm{b})$ be arbitrary. Since all entries of $\mathrm{a}\otimes\mathrm{b}$ are positive, there exists $t > 0$ small enough that $\mathrm{R} := (1+t)(\mathrm{a}\otimes\mathrm{b}) - t\mathrm{Q}$ has nonnegative entries. Its row and column sums are $\mathrm{R}\mathbb{1}_m = (1+t)\mathrm{a} - t\mathrm{a} = \mathrm{a}$ and $\mathrm{R}^\top\mathbb{1}_n = (1+t)\mathrm{b} - t\mathrm{b} = \mathrm{b}$, so $\mathrm{R} \in \mathrm{U}(\mathrm{a},\mathrm{b})$. Moreover, $\mathrm{a}\otimes\mathrm{b} = \frac{1}{1+t}\mathrm{R} + \frac{t}{1+t}\mathrm{Q}$. By optimality of $\mathrm{a}\otimes\mathrm{b}$, both $\langle \mathrm{C}, \mathrm{R}\rangle$ and $\langle \mathrm{C}, \mathrm{Q}\rangle$ are at least $\langle \mathrm{C}, \mathrm{a}\otimes\mathrm{b}\rangle$. Taking the scalar product of the convex-combination identity with $\mathrm{C}$ gives $\langle \mathrm{C}, \mathrm{a}\otimes\mathrm{b}\rangle = \frac{1}{1+t}\langle \mathrm{C}, \mathrm{R}\rangle + \frac{t}{1+t}\langle \mathrm{C}, \mathrm{Q}\rangle$. A convex average of two numbers not smaller than $\langle \mathrm{C}, \mathrm{a}\otimes\mathrm{b}\rangle$ can equal $\langle \mathrm{C}, \mathrm{a}\otimes\mathrm{b}\rangle$ only if both numbers are equal to it. Hence $\langle \mathrm{C}, \mathrm{Q}\rangle = \langle \mathrm{C}, \mathrm{a}\otimes\mathrm{b}\rangle$, and $\mathrm{Q}$ is optimal. Since $\mathrm{Q}$ was arbitrary, all couplings are optimal. ◻

Thus the product plan is mainly a feasibility witness. Except in the degenerate situation where the linear cost is constant on the whole transportation polytope, it is not expected to solve optimal transport.

**Linear-programming structure.**

**Definition 3.4** (Discrete Kantorovich problem)**.** For $\mathrm{a} \in \Sigma_n$, $\mathrm{b} \in \Sigma_m$, and $\mathrm{C} \in \mathbb{R}^{n\times m}$, define

$$\mathrm{L}_{\mathrm{C}}(\mathrm{a},\mathrm{b}) := \min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})}\langle \mathrm{C}, \mathrm{P}\rangle = \min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})}\sum_{i,j}\mathrm{C}_{i,j}\mathrm{P}_{i,j}. \tag{3.2}$$

A minimizer $\mathrm{P}$ is an optimal coupling, or optimal transport plan.

**Proposition 3.5** (Convexity in the marginals and concavity in the cost)**.** *For every fixed cost matrix* $\mathrm{C} \in \mathbb{R}^{n\times m}$*, the map* $(\mathrm{a},\mathrm{b}) \longmapsto \mathrm{L}_{\mathrm{C}}(\mathrm{a},\mathrm{b})$ *is jointly convex on* $\Sigma_n\times\Sigma_m$*. For every fixed* $(\mathrm{a},\mathrm{b}) \in \Sigma_n\times\Sigma_m$*, the map* $\mathrm{C} \longmapsto \mathrm{L}_{\mathrm{C}}(\mathrm{a},\mathrm{b})$ *is concave on* $\mathbb{R}^{n\times m}$*.*

*Proof.* For $r \in \{0,1\}$, let $\mathrm{P}_r$ be optimal between $\mathrm{a}_r$ and $\mathrm{b}_r$. For $t \in [0,1]$, the matrix $\mathrm{P}_t = (1-t)\mathrm{P}_0 + t\mathrm{P}_1$ belongs to $\mathrm{U}((1-t)\mathrm{a}_0 + t\mathrm{a}_1, (1-t)\mathrm{b}_0 + t\mathrm{b}_1)$. Therefore $\mathrm{L}_{\mathrm{C}}\left((1-t)\mathrm{a}_0 + t\mathrm{a}_1, (1-t)\mathrm{b}_0 + t\mathrm{b}_1\right) \leqslant \langle \mathrm{C}, \mathrm{P}_t\rangle = (1-t)\mathrm{L}_{\mathrm{C}}(\mathrm{a}_0,\mathrm{b}_0) + t\mathrm{L}_{\mathrm{C}}(\mathrm{a}_1,\mathrm{b}_1)$. For fixed marginals, every $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$ satisfies $\langle (1-t)\mathrm{C}_0 + t\mathrm{C}_1, \mathrm{P}\rangle \geqslant (1-t)\mathrm{L}_{\mathrm{C}_0}(\mathrm{a},\mathrm{b}) + t\mathrm{L}_{\mathrm{C}_1}(\mathrm{a},\mathrm{b})$. Minimizing the left-hand side over $\mathrm{P}$ proves concavity in $\mathrm{C}$. ◻

**Proposition 3.6** (Rank-controlled sparse minimizers)**.** *Let* $\mathcal{A} : \mathbb{R}^{n\times m} \to \mathbb{R}^q$ *be linear, let* $\mathbf{d} \in \mathbb{R}^q$ *and* $\mathrm{C} \in \mathbb{R}^{n\times m}$*, and consider*

$$\inf_{\mathrm{P}\geqslant 0,\ \mathcal{A}(\mathrm{P})\leqslant\mathbf{d}}\langle \mathrm{C}, \mathrm{P}\rangle, \tag{3.3}$$

*where the inequality is componentwise. If the feasible set is nonempty and the objective is bounded below on it, then* (3.3) *admits a minimizer* $\mathrm{P}^\star$ *satisfying*

$$\#\left\{(i,j)\ ;\ \mathrm{P}^\star_{i,j} > 0\right\} \leqslant \operatorname{rank}(\mathcal{A}). \tag{3.4}$$

*Here* $\operatorname{rank}(\mathcal{A}) = \dim(\operatorname{Im}\mathcal{A})$*. The same conclusion holds when* $\mathcal{A}(\mathrm{P}) \leqslant \mathbf{d}$ *is replaced by* $\mathcal{A}(\mathrm{P}) = \mathbf{d}$*.*

*Proof.* The feasible region is a nonempty polyhedron, and the objective is bounded below on it. The attainment theorem for linear programs therefore provides a minimizer. Among all minimizers, choose $\mathrm{P}^\star$ with the smallest number of positive entries, and denote its support by $S$.

Suppose that $\#S > \operatorname{rank}(\mathcal{A})$. The restriction of $\mathcal{A}$ to the $\#S$-dimensional space of matrices supported on $S$ then has a nontrivial kernel. Hence there is a nonzero matrix $H$, supported on $S$, such that $\mathcal{A}(H) = 0$. Since $\mathrm{P}^\star$ is strictly positive on $S$, both $\mathrm{P}^\star + tH$ and $\mathrm{P}^\star - tH$ remain nonnegative for all sufficiently small $t > 0$. Moreover, $\mathcal{A}(\mathrm{P}^\star \pm tH) = \mathcal{A}(\mathrm{P}^\star)$, so both perturbations satisfy exactly the same linear inequalities, or equalities, as $\mathrm{P}^\star$. Optimality forces $\langle \mathrm{C},\, H\rangle = 0$; otherwise one of the two signs would strictly decrease the objective.

Choose $\sigma \in \{-1, 1\}$ such that $\sigma H$ has a negative entry, and set $t_\star := \min_{(i,j)\colon \sigma H_{i,j}<0} \mathrm{P}^\star_{i,j}/(-\sigma H_{i,j}) > 0$. Then $\mathrm{P}^\star + t_\star \sigma H$ is nonnegative, feasible, and has the same cost as $\mathrm{P}^\star$. At least one of its positive entries has become zero, while no new entry has appeared outside $S$. This contradicts the minimality of $S$ and proves (3.4). ☐

**Proposition 3.7** (Sparse optimal plans)**.** *Assume $\mathrm{a}_i > 0$, $\mathrm{b}_j > 0$ and $\sum_i \mathrm{a}_i = \sum_j \mathrm{b}_j = 1$. The linear program (3.2) admits an optimal coupling with at most $n + m - 1$ nonzero entries.*

*Proof.* The transportation polytope is nonempty and compact, so the minimum is finite. Apply the equality case of Proposition 3.6 to the marginal operator $\mathcal{A}_{\mathrm{marg}}(\mathrm{P}) := (\mathrm{P}\mathbb{1}_m, \mathrm{P}^\top \mathbb{1}_n) \in \mathbb{R}^{n+m}$. Its rank is $n + m - 1$. Indeed, every matrix satisfies $\sum_i (\mathrm{P}\mathbb{1}_m)_i = \sum_j (\mathrm{P}^\top \mathbb{1}_n)_j$, so there is one linear dependence among the $n + m$ marginal coordinates. It is the only one: if $u \in \mathbb{R}^n$ and $v \in \mathbb{R}^m$ satisfy $\langle u,\, \mathrm{P}\mathbb{1}_m\rangle + \langle v,\, \mathrm{P}^\top \mathbb{1}_n\rangle = 0$ for every $\mathrm{P}$, then testing this identity on each elementary matrix gives $u_i + v_j = 0$ for every $(i, j)$. Thus all $u_i$ equal one scalar and all $v_j$ equal its opposite, so the annihilator of $\operatorname{Im}(\mathcal{A}_{\mathrm{marg}})$ is one-dimensional. It follows that $\operatorname{rank}(\mathcal{A}_{\mathrm{marg}}) = n + m - 1$, and Proposition 3.6 supplies the claimed optimal coupling. ☐

**Proposition 3.8** (North-west corner feasible plan)**.** *Let $\mathrm{a} \in \mathbb{R}^n_+$ and $\mathrm{b} \in \mathbb{R}^m_+$ have the same positive total mass. Starting from $(i, j) = (1, 1)$ with residual masses $r_i = \mathrm{a}_i$ and $s_j = \mathrm{b}_j$, skip zero residuals, set $\mathrm{P}_{i,j} = \min(r_i, s_j)$, subtract this value from both residuals, and advance every index whose residual has become zero. Repeat until all residual masses are exhausted.*

*Proof.* All assignments are nonnegative. At each step, the mass placed in entry $(i, j)$ is subtracted from exactly one current row residual and one current column residual, so no row or column can receive more mass than prescribed. Conversely, an index is advanced only when its residual has been fully filled. When the algorithm stops, the total assigned mass is $\sum_i \mathrm{a}_i = \sum_j \mathrm{b}_j$, hence all row and column sums are exactly $\mathrm{a}$ and $\mathrm{b}$.

Each positive assignment exhausts at least one current row or one current column. Before the final assignment, at most $n-1$ row advances and $m - 1$ column advances can occur without terminating the construction. Hence the number of positive entries is at most $(n - 1) + (m - 1) + 1 = n + m - 1$. For acyclicity, view the positive support as a bipartite graph. Once a row or column index is advanced, it never appears again, so each new positive edge either starts a new component or attaches at least one new vertex to the component currently being swept. No edge is ever added between two old vertices of the same component, so no cycle can be created. ☐

**One-dimensional cases.**

**Proposition 3.9** (One-dimensional weighted sweep)**.** *Let $x_1 \leqslant \cdots \leqslant x_n$ and $y_1 \leqslant \cdots \leqslant y_m$ be points on the line, and let $c(x, y) = h(x - y)$ with $h$ convex. The north-west corner plan between the sorted weighted atoms is optimal for (3.2).*

*Proof.* The north-west plan is monotone: if $i < i'$ and $j > j'$, it cannot put positive mass on both $(i, j)$ and $(i', j')$, because the sweep exhausts rows and columns in increasing order. To prove optimality, start from an arbitrary feasible plan $\mathrm{P}$. If $\mathrm{P}_{1,1} < \min(\mathrm{a}_1, \mathrm{b}_1)$, then row 1 sends positive mass to some column $j > 1$, while column 1 receives positive mass from some row $i > 1$. Moving $\eta = \min(\mathrm{P}_{1,j}, \mathrm{P}_{i,1})$ from $(1, j)$ and $(i, 1)$ to $(1, 1)$ and $(i, j)$ preserves both marginals. Convexity of $h$ gives the Monge inequality $h(x_1 - y_j) + h(x_i - y_1) \geqslant h(x_1 - y_1) + h(x_i - y_j)$, so this exchange does not increase the cost. Each exchange either removes the chosen entry in row 1 or removes the chosen entry in column 1. Repeating finitely many times forces $\mathrm{P}_{1,1} = \min(\mathrm{a}_1, \mathrm{b}_1)$, exactly the first north-west assignment. It exhausts row 1, column 1, or both. Removing every exhausted index reduces the problem to a smaller ordered transportation problem. Induction on $n + m$ produces the complete north-west plan without increasing cost. Sorting costs $O(n \log n + m \log m)$ and the sweep uses at most $n + m - 1$ positive assignments. ☐

**Permutation matrices as couplings.**

**Definition 3.10** (Permutation matrices)**.** For a permutation $\sigma \in \operatorname{Perm}(n)$, its permutation matrix $\mathrm{P}_\sigma$ is

$$(\mathrm{P}_\sigma)_{i,j} = \begin{cases} 1 & \text{if} \quad j = \sigma(i) \\ 0 & \text{otherwise.} \end{cases}$$

The set of all permutation matrices is $\mathcal{P}_n^{\mathrm{perm}} := \{\mathrm{P}_\sigma \ ; \ \sigma \in \operatorname{Perm}(n)\}$.

$\langle \mathrm{C},\, \mathrm{P}_\sigma/n\rangle = \frac{1}{n}\sum_{i=1}^n \mathrm{C}_{i,\sigma(i)}$.

**Definition 3.11** (Birkhoff polytope)**.** The Birkhoff polytope is the convex set of bistochastic matrices $\mathcal{B}_n := \left\{\mathrm{P} \in \mathbb{R}_+^{n\times n} \ ; \ \mathrm{P}\mathbb{1}_n = \mathbb{1}_n \quad \text{and} \quad \mathrm{P}^\top \mathbb{1}_n = \mathbb{1}_n\right\}$.

Then $\mathrm{U}(\mathbb{1}_n/n, \mathbb{1}_n/n) = \mathcal{B}_n/n$, and permutation couplings are included in this convex relaxation. More precisely, $\mathcal{P}_n^{\mathrm{perm}} = \mathcal{B}_n \cap \{0, 1\}^{n\times n} \subset \mathcal{B}_n$, $\min_{\mathrm{P}\in\mathcal{B}_n} \langle \mathrm{C},\, \mathrm{P}\rangle \leqslant \min_{\mathrm{P}\in\mathcal{P}_n^{\mathrm{perm}}} \langle \mathrm{C},\, \mathrm{P}\rangle$.

**Definition 3.12** (Extreme points)**.** For a convex set $\mathcal{C}$ in a finite-dimensional vector space, $\operatorname{Extr}(\mathcal{C}) := \{x \in \mathcal{C} \ ; \ x = (y + z)/2,\ y, z \in \mathcal{C} \Rightarrow y = z = x\}$.

**Proposition 3.13** (Existence of extreme points)**.** *If $\mathcal{C}$ is a non-empty compact convex subset of a finite-dimensional vector space, then* $\operatorname{Extr}(\mathcal{C})$ *is non-empty.*

*Proof.* Among all non-empty faces of $\mathcal{C}$, choose one of minimal affine dimension. If this face contained two distinct points, maximizing a linear functional that is not constant on the face would produce a non-empty proper exposed subface, contradicting minimality. Hence the minimal face is a singleton, and its point is extreme. ☐

**Proposition 3.14** (Linear programs have extreme minimizers)**.** *Let $\mathcal{C}$ be a non-empty compact convex set. For every linear form $\ell$, $\operatorname{Extr}(\mathcal{C}) \cap \operatorname{argmin}_{x\in\mathcal{C}} \ell(x) \neq \emptyset$.*

*Proof.* The set $S = \operatorname{argmin}_{x\in\mathcal{C}} \ell(x)$ is non-empty, compact and convex. By Proposition 3.13, it has an extreme point $x$. If $x = (y+z)/2$ with $y, z \in \mathcal{C}$, then by linearity and optimality of $x$, both $y$ and $z$ also minimize $\ell$ on $\mathcal{C}$, hence $y, z \in S$. Since $x$ is extreme in $S$, $y = z = x$. Thus $x$ is extreme in $\mathcal{C}$. □

**Theorem 3.15** (Birkhoff–von Neumann)**.** *The extreme points of $\mathcal{B}_n$ are exactly the permutation matrices.*

*Proof.* We first prove that permutation matrices are extreme. Let $\mathrm{P}_\sigma \in \mathcal{P}_n^{\text{perm}}$ and assume that $\mathrm{P}_\sigma = (\mathrm{Q}+\mathrm{R})/2$ with $\mathrm{Q}, \mathrm{R} \in \mathcal{B}_n$. Every bistochastic matrix has entries in $[0, 1]$. Since the only extreme points of $[0, 1]$ are 0 and 1, each entry of $\mathrm{P}_\sigma$ fixes the corresponding entries of Q and R: if $(\mathrm{P}_\sigma)_{i,j} = 0$, then $\mathrm{Q}_{i,j} = \mathrm{R}_{i,j} = 0$, while if $(\mathrm{P}_\sigma)_{i,j} = 1$, then $\mathrm{Q}_{i,j} = \mathrm{R}_{i,j} = 1$. Hence $\mathrm{Q} = \mathrm{R} = \mathrm{P}_\sigma$, so $\mathrm{P}_\sigma$ is extreme.
Pick $\mathrm{P} \in \mathcal{B}_n \setminus \mathcal{P}_n^{\text{perm}}$. Since an integral bistochastic matrix is necessarily a permutation matrix, P has at least one fractional entry. We shall split $\mathrm{P} = \frac{\mathrm{Q}+\mathrm{R}}{2}$ with $\mathrm{Q}, \mathrm{R} \in \mathcal{B}_n$ and $\mathrm{Q} \neq \mathrm{R}$, proving that P is not extreme.
Associate with P the bipartite graph whose left vertices are the rows, whose right vertices are the columns, and whose edges are the fractional entries $0 < \mathrm{P}_{i,j} < 1$. An entry equal to 1 uses the whole mass of its row and column, so it is isolated in the positive support and does not appear in this fractional graph. If a left vertex $i$ is incident to a fractional edge $(i, j_1)$, then it must be incident to at least one other fractional edge. Indeed, the row sum is one; after the contribution $\mathrm{P}_{i,j_1} \in (0, 1)$, a positive amount $1 - \mathrm{P}_{i,j_1}$ remains in the same row, and it cannot be carried by an entry equal to 1. The same argument applies to right vertices, using the column sums. Thus every non-isolated vertex of the fractional graph has degree at least two.
Starting from any fractional edge, one may therefore walk through adjacent fractional edges without immediately backtracking and without getting stuck. Since the graph is finite, some vertex is eventually visited twice; the portion of the walk between the two visits contains a cycle. Choose a shortest such cycle and write it in alternating form $(i_1, j_1, i_2, j_2, \ldots, i_p, j_p)$, with $i_{p+1} = i_1$, where both $(i_s, j_s)$ and $(i_{s+1}, j_s)$ are fractional for every $s$. The minimality of the cycle implies that the vertices $i_s$ are all distinct and that the vertices $j_s$ are all distinct. In particular, $0 < \mathrm{P}_{i_s,j_s} < 1$ and $0 < \mathrm{P}_{i_{s+1},j_s} < 1$. Define $\varepsilon \coloneqq \min_{1\leqslant s\leqslant p}\{\mathrm{P}_{i_s,j_s}, \mathrm{P}_{i_{s+1},j_s}, 1 - \mathrm{P}_{i_s,j_s}, 1 - \mathrm{P}_{i_{s+1},j_s}\}$. All these numbers are positive, so $\varepsilon > 0$. Split the cycle edges into the two alternating families $A \coloneqq \{(i_s, j_s)\}_{s=1}^p$, $\quad B \coloneqq \{(i_{s+1}, j_s)\}_{s=1}^p$.

$$\mathrm{Q}_{i,j} \coloneqq \begin{cases} \mathrm{P}_{i,j}, & (i,j) \notin A \cup B, \\ \mathrm{P}_{i,j} + \varepsilon/2, & (i,j) \in A, \\ \mathrm{P}_{i,j} - \varepsilon/2, & (i,j) \in B, \end{cases} \qquad \mathrm{R}_{i,j} \coloneqq \begin{cases} \mathrm{P}_{i,j}, & (i,j) \notin A \cup B, \\ \mathrm{P}_{i,j} - \varepsilon/2, & (i,j) \in A, \\ \mathrm{P}_{i,j} + \varepsilon/2, & (i,j) \in B. \end{cases}$$

By the definition of $\varepsilon$, all modified entries stay in $[0, 1]$, so Q and R are nonnegative. Each row vertex $i_s$ of the cycle is incident to exactly one edge of $A$ and one edge of $B$; the $+\varepsilon/2$ and $-\varepsilon/2$ perturbations therefore cancel in that row. The same cancellation holds in each column vertex $j_s$, and all other rows and columns are unchanged. Finally, $\mathrm{Q} \neq \mathrm{R}$ because $\varepsilon > 0$ and the cycle is non-empty, while by construction $\mathrm{P} = (\mathrm{Q} + \mathrm{R})/2$. Thus P is not extreme. □

**Corollary 3.16** (Kantorovich for matching)**.** *If $m = n$ and $\mathrm{a} = \mathrm{b} = \mathbb{1}_n/n$, then the discrete Kantorovich problem (3.2) admits an optimal solution of the form $\mathrm{P}_\sigma/n$. The associated permutation $\sigma$ solves the assignment problem of Section 1.1.*

*Proof.* The feasible set is $\mathcal{B}_n/n$. By Proposition 3.14, the linear objective has an optimal extreme point. Since scaling preserves extreme points and Theorem 3.15 identifies the extreme points of $\mathcal{B}_n$, this optimizer is $\mathrm{P}_\sigma/n$ for some permutation $\sigma$. Its cost is exactly $n^{-1}\sum_i \mathrm{C}_{i,\sigma(i)}$, so $\sigma$ is an optimal assignment. □

**Constructive Birkhoff–von Neumann decomposition.**

**Corollary 3.17** (Birkhoff–von Neumann decomposition)**.** *Every $\mathrm{P} \in \mathcal{B}_n$ admits a representation $\mathrm{P} = \sum_{r=1}^L \lambda_r \mathrm{P}_{\sigma_r}$, $\quad \lambda_r > 0$, $\quad \sum_{r=1}^L \lambda_r = 1$, $\quad L \leqslant (n-1)^2 + 1$, where each $\mathrm{P}_{\sigma_r}$ is a permutation matrix.*

*Proof.* Let $\mathcal{K}$ be the convex hull of the permutation matrices. Clearly $\mathcal{K} \subset \mathcal{B}_n$. Conversely, if some $\mathrm{P} \in \mathcal{B}_n$ did not belong to the compact convex set $\mathcal{K}$, strict separation would provide a linear form $\ell$ such that $\ell(\mathrm{P}) < \min_{\mathrm{Q}\in\mathcal{K}} \ell(\mathrm{Q})$. Proposition 3.14 and Theorem 3.15 give $\min_{\mathrm{Q}\in\mathcal{B}_n} \ell(\mathrm{Q}) = \min_{\sigma\in\mathrm{Perm}(n)} \ell(\mathrm{P}_\sigma) = \min_{\mathrm{Q}\in\mathcal{K}} \ell(\mathrm{Q})$, contradicting $\mathrm{P} \in \mathcal{B}_n$. Hence $\mathcal{B}_n = \mathcal{K}$.
The $2n$ row- and column-sum equations have rank $2n - 1$, and the strictly positive matrix $\mathbb{1}_n\mathbb{1}_n{}^\top/n$ lies in their affine space. Therefore $\mathcal{B}_n$ has affine dimension $n^2 - (2n-1) = (n-1)^2$. Carathéodory's theorem in this affine hull now reduces any convex decomposition to at most $(n-1)^2 + 1$ permutation matrices; zero coefficients can be discarded. □

$s|I| = \sum_{i\in I}\sum_{j\in N(I)} \mathrm{R}_{i,j} \leqslant \sum_{j\in N(I)}\sum_i \mathrm{R}_{i,j} = s|N(I)|$. Thus $|N(I)| \geqslant |I|$, so $G_{\mathrm{R}}$ contains a perfect matching.
$\widehat{\mathrm{P}}' \coloneqq \frac{\mathrm{R}-\lambda\mathrm{P}_\sigma}{s-\lambda}$ If the current support graph has $E$ edges, the elementary matching subroutine performs at most $n$ breadth-first searches and costs $O(nE)$. Thus a dense implementation costs $O(n^3)$ per extracted permutation and, using the term bound above, at most $O(n^5)$ arithmetic and graph operations overall, with $O(n^2)$ memory.

**Rational weights.**

**Proposition 3.18** (Rational weights as duplicated uniform matching)**.** *Let $\alpha = \sum_{i=1}^n \frac{k_i}{N}\delta_{x_i}$, $\quad \beta = \sum_{j=1}^m \frac{\ell_j}{N}\delta_{y_j}$, $\quad \sum_i k_i = \sum_j \ell_j = N$, with positive integers $k_i, \ell_j$. Replace each $x_i$ by $k_i$ identical copies and each $y_j$ by $\ell_j$ identical copies, producing two uniform $N$-point clouds. Then the duplicated assignment problem and the discrete Kantorovich problem between $\alpha$ and $\beta$ have the same optimal value. Moreover, there exists an optimal coupling of the form $\mathrm{P}_{i,j} = \frac{n_{i,j}}{N}$, $\quad n_{i,j} \in \mathbb{N}$, $\quad \sum_j n_{i,j} = k_i$, $\quad \sum_i n_{i,j} = \ell_j$, and couplings whose scaled entries $N\mathrm{P}_{i,j}$ are integral are exactly the collapsed assignments between duplicated clouds. Fractional optimal couplings may nevertheless coexist when the optimum is degenerate.*

*Proof.* Any assignment between the duplicated source and target clouds defines integers $n_{i,j}$ counting how many copied particles of type $x_i$ are matched to copied particles of type $y_j$. These counts satisfy $\sum_j n_{i,j} = k_i$ and $\sum_i n_{i,j} = \ell_j$, and the associated coupling $\mathrm{P}_{i,j} = n_{i,j}/N$ has marginals $k_i/N$ and $\ell_j/N$. The assignment cost is $\frac{1}{N}\sum_{i,j} n_{i,j} c(x_i, y_j) = \sum_{i,j} \mathrm{P}_{i,j} c(x_i, y_j)$. Conversely, any nonnegative integer count matrix with those row and column sums can be realized by allocating the $k_i$ copies of each $x_i$ among the target copies according to $(n_{i,j})_j$. Scale a feasible coupling by $N$ and write $\mathrm{Q} = N\mathrm{P}$. The constraints on Q have integer right-hand sides $(k_1, \ldots, k_n, \ell_1, \ldots, \ell_m)$. After multiplying the target rows by $-1$, their coefficient matrix is the oriented node-edge incidence matrix of a bipartite graph and is therefore totally unimodular. Every vertex of this transportation polytope is integral. Since a linear objective attains its minimum at a vertex, an integral optimal Q exists, proving equality of the two optimal values. The argument establishes existence, not integrality of every optimizer; convex combinations of distinct integral optima give fractional optima. □

**Applications of discrete transport.**

### 3.2 Linear-Programming Algorithms

**Transportation simplex and network simplex.**

**Interior-point methods.**

$$\mathrm{P}_\varepsilon := \underset{\substack{\mathrm{P}\mathbb{1}_m=\mathrm{a},\ \mathrm{P}^\top\mathbb{1}_n=\mathrm{b}\\ \mathrm{P}_{i,j}>0}}{\mathrm{argmin}} \quad \langle \mathrm{C},\, \mathrm{P}\rangle - \varepsilon \sum_{i,j} \log \mathrm{P}_{i,j}, \tag{3.5}$$

where $\varepsilon > 0$ is decreased along the algorithm. The barrier is singular at the boundary, so each iterate stays strictly inside the transportation polytope; as $\varepsilon \downarrow 0$, the central path approaches the set of LP minimizers.

### 3.3 Relaxation for Arbitrary Measures

**Continuous couplings.**

**Definition 3.19** (Marginals of a joint measure)**.** Let $\pi \in \mathcal{M}_+^1(\mathcal{X}\times\mathcal{Y})$ and let $P_\mathcal{X}(x,y) = x$, $P_\mathcal{Y}(x,y) = y$ be the coordinate projections. The marginals of $\pi$ are $\pi_1 := (P_\mathcal{X})_\sharp\pi \in \mathcal{M}_+^1(\mathcal{X})$, $\quad \pi_2 := (P_\mathcal{Y})_\sharp\pi \in \mathcal{M}_+^1(\mathcal{Y})$. Equivalently, for all bounded continuous test functions $f$ on $\mathcal{X}$ and $g$ on $\mathcal{Y}$, $\int_{\mathcal{X}\times\mathcal{Y}} f(x)\mathrm{d}\pi(x,y) = \int_\mathcal{X} f\mathrm{d}\pi_1$, $\quad \int_{\mathcal{X}\times\mathcal{Y}} g(y)\mathrm{d}\pi(x,y) = \int_\mathcal{Y} g\mathrm{d}\pi_2$.

$\int_\mathcal{Y} \mathrm{d}\pi(x,y) = \mathrm{d}\alpha(x)$, $\quad \int_\mathcal{X} \mathrm{d}\pi(x,y) = \mathrm{d}\beta(y)$, which is made rigorous by Definition 3.19, or equivalently by the identities $\pi(A\times\mathcal{Y}) = \alpha(A)$ and $\pi(\mathcal{X}\times B) = \beta(B)$ for measurable sets $A\subset\mathcal{X}$ and $B\subset\mathcal{Y}$.

**Definition 3.20** (Couplings)**.** Given $\alpha\in\mathcal{M}_+^1(\mathcal{X})$ and $\beta\in\mathcal{M}_+^1(\mathcal{Y})$, the set of couplings between $\alpha$ and $\beta$ is

$$\mathcal{U}(\alpha,\beta) := \left\{\pi\in\mathcal{M}_+^1(\mathcal{X}\times\mathcal{Y})\ ;\ \pi_1=\alpha \quad\text{and}\quad \pi_2=\beta\right\}. \tag{3.6}$$

This is the continuous analogue of the transportation polytope (3.1).

**Definition 3.21** (Tensor product and trivial coupling)**.** Given $\alpha\in\mathcal{M}_+^1(\mathcal{X})$ and $\beta\in\mathcal{M}_+^1(\mathcal{Y})$, the tensor product coupling $\alpha\otimes\beta$ is the probability measure on $\mathcal{X}\times\mathcal{Y}$ defined by

$$\int_{\mathcal{X}\times\mathcal{Y}} h(x,y)\mathrm{d}(\alpha\otimes\beta)(x,y) = \int_\mathcal{X}\left(\int_\mathcal{Y} h(x,y)\mathrm{d}\beta(y)\right)\mathrm{d}\alpha(x)$$

for every bounded measurable $h$. It is also called the trivial coupling because it makes the two coordinates independent.

$$\int_{\mathcal{X}\times\mathcal{Y}} f(x)\mathrm{d}(\alpha\otimes\beta)(x,y) = \left(\int_\mathcal{X} f(x)\mathrm{d}\alpha(x)\right)\left(\int_\mathcal{Y}\mathrm{d}\beta(y)\right) = \int f\mathrm{d}\alpha,$$

**Proposition 3.22** (Product optimality is degenerate)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c\in\mathcal{C}(\mathcal{X}\times\mathcal{Y})$.* $\alpha\otimes\beta \in \underset{\pi\in\mathcal{U}(\alpha,\beta)}{\mathrm{argmin}} \int c\,\mathrm{d}\pi$, $\quad$ *every coupling in $\mathcal{U}(\alpha,\beta)$ is optimal. They are also equivalent to the additive decomposition of the cost on the product support,* $c(x,y) = u(x)+v(y)$, $\quad u\in\mathcal{C}(\mathrm{supp}(\alpha))$, $\quad v\in\mathcal{C}(\mathrm{supp}(\beta))$.

*Proof.* If every coupling is optimal, then $\alpha\otimes\beta$ is optimal. Conversely, assume that $\alpha\otimes\beta$ is optimal. We first show that, for every $x_0,x_1\in\mathrm{supp}(\alpha)$ and $y_0,y_1\in\mathrm{supp}(\beta)$, $c(x_0,y_0)+c(x_1,y_1) = c(x_0,y_1)+c(x_1,y_0)$. Indeed, if this equality failed, after exchanging $y_0$ and $y_1$ if necessary one would have a strict inequality $c(x_0,y_0)+c(x_1,y_1) > c(x_0,y_1)+c(x_1,y_0)$. Failure of the equality forces $x_0\neq x_1$ and $y_0\neq y_1$. By continuity, the strict inequality persists with a uniform margin on small neighborhoods $U_0,U_1$ of $x_0,x_1$ and $V_0,V_1$ of $y_0,y_1$, chosen disjoint within each pair. Since the four points lie in the supports, these neighborhoods have positive marginal mass. Denote by $\alpha_i$ and $\beta_i$ the normalized restrictions of $\alpha$ to $U_i$ and of $\beta$ to $V_i$, and choose $0<\lambda\leqslant\min\{\alpha(U_0)\beta(V_0),\alpha(U_1)\beta(V_1)\}$. The exchanged measure $\tilde\pi = \alpha\otimes\beta - \lambda\,\alpha_0\otimes\beta_0 - \lambda\,\alpha_1\otimes\beta_1 + \lambda\,\alpha_0\otimes\beta_1 + \lambda\,\alpha_1\otimes\beta_0$ is nonnegative and has the same two marginals as $\alpha\otimes\beta$. The uniform strict inequality on the neighborhoods implies that $\int c\,\mathrm{d}\tilde\pi < \int c\,\mathrm{d}(\alpha\otimes\beta)$, contradicting optimality.
Fixing any $x_\star\in\mathrm{supp}(\alpha)$ and $y_\star\in\mathrm{supp}(\beta)$, the equality of cross differences gives, for all $(x,y)\in\mathrm{supp}(\alpha)\times\mathrm{supp}(\beta)$, $c(x,y) = c(x,y_\star)+c(x_\star,y)-c(x_\star,y_\star)$. Thus $c=u+v$ on the product support, with $u(x) = c(x,y_\star)-c(x_\star,y_\star)$ and $v(y)=c(x_\star,y)$. These functions are continuous on the corresponding supports. Every coupling is concentrated on this product support, so for any $\pi\in\mathcal{U}(\alpha,\beta)$, $\int c\,\mathrm{d}\pi = \int u\,\mathrm{d}\alpha + \int v\,\mathrm{d}\beta$, which depends only on the marginals. Hence all couplings are optimal. □

The tensor product is therefore a trivial feasible coupling, not a typical optimizer. Product optimality means that the cost cannot distinguish between dependences once the marginals are fixed.
If there exists a map $T:\mathcal{X}\to\mathcal{Y}$ such that $T_\sharp\alpha=\beta$, then the Monge map induces the graph coupling $\pi = (\mathrm{Id},T)_\sharp\alpha\in\mathcal{U}(\alpha,\beta)$, characterized by
$\int_{\mathcal{X}\times\mathcal{Y}} h(x,y)\mathrm{d}\pi(x,y) = \int_\mathcal{X} h(x,T(x))\mathrm{d}\alpha(x)$. Applying this identity to $h(x,y)=f(x)$ or $h(x,y)=g(y)$ gives respectively $\pi_1=\alpha$ and $\pi_2=\beta$. Thus graph couplings are precisely the Kantorovich representation of deterministic Monge maps.

**Continuous Kantorovich problem.**

**Definition 3.23** (Continuous Kantorovich problem)**.** For $\alpha\in\mathcal{M}_+^1(\mathcal{X})$, $\beta\in\mathcal{M}_+^1(\mathcal{Y})$, and a nonnegative Borel cost $c:\mathcal{X}\times\mathcal{Y}\to[0,+\infty]$, define

$$\mathcal{L}_c(\alpha,\beta) := \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y). \tag{3.7}$$

A minimizer is an optimal coupling, or optimal transport plan. The value is understood in $[0,+\infty]$.

**Proposition 3.24** (Convexity in the marginals and concavity in the cost)**.** *Fix a nonnegative Borel cost $c$. The extended-valued map $(\alpha,\beta)\longmapsto\mathcal{L}_c(\alpha,\beta)$ is jointly convex on pairs of probability measures. Conversely, for fixed marginals $(\alpha,\beta)$, the map $c\mapsto\mathcal{L}_c(\alpha,\beta)$ is concave on the cone of nonnegative Borel costs: for any such $c_0,c_1$ and $t\in[0,1]$, $\mathcal{L}_{(1-t)c_0+tc_1}(\alpha,\beta)\geqslant(1-t)\mathcal{L}_{c_0}(\alpha,\beta)+t\mathcal{L}_{c_1}(\alpha,\beta)$. The inequality is understood in $[0,+\infty]$; in particular, the map is concave on every convex family of costs on which it is finite.*

*Proof.* For joint convexity, let $(\alpha_0,\beta_0)$ and $(\alpha_1,\beta_1)$ be two pairs of probability measures. The claim is immediate at $t\in\{0,1\}$ or if one of the two values on the right-hand side is infinite. Otherwise, for $\eta>0$, choose $\pi_i\in\mathcal{U}(\alpha_i,\beta_i)$ such that $\int c\mathrm{d}\pi_i\leqslant\mathcal{L}_c(\alpha_i,\beta_i)+\eta$ for $i\in\{0,1\}$. Then $(1-t)\pi_0+t\pi_1$ couples $(1-t)\alpha_0+t\alpha_1$ and $(1-t)\beta_0+t\beta_1$, and hence $\mathcal{L}_c\big((1-t)\alpha_0+t\alpha_1,(1-t)\beta_0+t\beta_1\big)\leqslant(1-t)\mathcal{L}_c(\alpha_0,\beta_0)+t\mathcal{L}_c(\alpha_1,\beta_1)+\eta$. Letting $\eta\to0$ proves joint convexity.
For concavity, the endpoint cases are immediate, so assume $t\in(0,1)$ and fix $\pi\in\mathcal{U}(\alpha,\beta)$. Linearity of integration for nonnegative functions gives $\int\big((1-t)c_0+tc_1\big)\mathrm{d}\pi\geqslant(1-t)\mathcal{L}_{c_0}(\alpha,\beta)+t\mathcal{L}_{c_1}(\alpha,\beta)$. Taking the infimum over $\pi$ proves the stated inequality. ☐

**Proposition 3.25** (Existence for lower-semicontinuous costs)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are Polish spaces and that $c:\mathcal{X}\times\mathcal{Y}\to[0,+\infty]$ is lower semicontinuous. Then the Kantorovich problem (3.7) admits at least one minimizer. In particular, this applies to every continuous nonnegative cost on compact metric spaces.*

*Proof.* The constraint set is non-empty because it contains the product coupling $\alpha\otimes\beta$. It is uniformly tight: given $\varepsilon>0$, choose compact sets $K_{\mathcal{X}}\subset\mathcal{X}$ and $K_{\mathcal{Y}}\subset\mathcal{Y}$ such that $\alpha(K_{\mathcal{X}})\geqslant1-\varepsilon/2$ and $\beta(K_{\mathcal{Y}})\geqslant1-\varepsilon/2$. Every $\pi\in\mathcal{U}(\alpha,\beta)$ then satisfies $\pi(K_{\mathcal{X}}\times K_{\mathcal{Y}})\geqslant1-\varepsilon$. By Prokhorov's theorem, the set of couplings is relatively compact for weak convergence; the weak topology is reviewed in Section **??**. It is also weakly closed, because the coordinate projections are continuous and therefore preserve the marginal identities in the limit. Hence $\mathcal{U}(\alpha,\beta)$ is weakly compact. Finally, Portmanteau's theorem makes $\pi\mapsto\int c\mathrm{d}\pi$ weakly lower semicontinuous. The direct method gives a minimizer. ☐

**Axiomatic characterization of the relaxation.**
**Theorem 3.26** (Kantorovich cost as the maximal convex relaxation)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c:\mathcal{X}\times\mathcal{Y}\to[0,+\infty)$ is continuous. Then $\mathcal{L}_c$ is the pointwise largest proper functional $\mathsf{F}:\mathcal{M}_+^1(\mathcal{X})\times\mathcal{M}_+^1(\mathcal{Y})\longrightarrow(-\infty,+\infty]$ that is jointly convex, weakly lower semicontinuous, and satisfies $\mathsf{F}(\delta_x,\delta_y)\leqslant c(x,y)\qquad$ for every $(x,y)\in\mathcal{X}\times\mathcal{Y}$. More precisely, every such $\mathsf{F}$ satisfies $\mathsf{F}(\alpha,\beta)\leqslant\mathcal{L}_c(\alpha,\beta)$ for all $(\alpha,\beta)$, while $\mathcal{L}_c$ itself has these properties and obeys $\mathcal{L}_c(\delta_x,\delta_y)=c(x,y)$.*

*Proof.* Joint convexity follows from Proposition 3.24. For weak lower semicontinuity, consider converging marginals and optimal plans along a subsequence realizing the lower limit. Compactness gives a further weakly convergent subsequence; its limit has the limiting marginals, and lower semicontinuity of the integral gives the claim, exactly as in Proposition 3.25. The only coupling of $(\delta_x,\delta_y)$ is $\delta_{(x,y)}$, which proves the identity on Dirac pairs.
Fix $\pi\in\mathcal{U}(\alpha,\beta)$. Its marginal constraints are exactly the barycentric identity $(\alpha,\beta)=\int(\delta_x,\delta_y)\mathrm{d}\pi$. Jensen's inequality for lower-semicontinuous convex functionals therefore gives $\mathsf{F}(\alpha,\beta)\leqslant\int\mathsf{F}(\delta_x,\delta_y)\mathrm{d}\pi(x,y)\leqslant\int c(x,y)\mathrm{d}\pi(x,y)$. Taking the infimum over $\pi\in\mathcal{U}(\alpha,\beta)$ proves maximality. ☐

**Monge–Kantorovich equivalence.**
**Corollary 3.27** (Monge–Kantorovich equivalence under Brenier)**.** *Let $\alpha,\beta\in\mathcal{P}_2(\mathbb{R}^d)$, assume that $\alpha$ is absolutely continuous with respect to Lebesgue measure, and let $c(x,y)=\|x-y\|^2$. If $T$ is the Brenier map solving Monge's problem, then $\pi=(\mathrm{Id},T)_\sharp\alpha$ is the unique optimal coupling solving the Kantorovich problem. In particular, Monge and Kantorovich costs are the same.*

*Proof.* The proof of Brenier's theorem shows that the support of any optimal Kantorovich plan is contained in the subdifferential $\partial\varphi$ of a convex function $\varphi$. When $\alpha$ has a density, $\varphi$ is differentiable $\alpha$-almost everywhere, so $\partial\varphi(x)=\{\nabla\varphi(x)\}$ for $\alpha$-almost every $x$. Thus every optimal coupling is concentrated on the graph of $T=\nabla\varphi$ and must equal $(\mathrm{Id},T)_\sharp\alpha$. The graph coupling is feasible and optimal, and the two formulations have the same value. ☐

**Kantorovich solution in 1-D.**
**Theorem 3.28** (One-dimensional Kantorovich solution)**.** *Let $\alpha,\beta\in\mathcal{M}_+^1(\mathbb{R})$ and let $h:\mathbb{R}\to[0,+\infty)$ be convex. Consider the cost $c(x,y)=h(x-y)$. Write $q_\alpha\coloneqq\mathcal{C}_\alpha^{-1}$ and $q_\beta\coloneqq\mathcal{C}_\beta^{-1}$, and assume that $h(q_\alpha-q_\beta)\in L^1(0,1)$. Then the quantile coupling $\pi^\star=(q_\alpha,q_\beta)_\sharp\mathrm{Leb}_{[0,1]}=(\mathcal{C}_\alpha^{-1},\mathcal{C}_\beta^{-1})_\sharp\mathrm{Leb}_{[0,1]}$ solves the Kantorovich problem (3.7) for $c(x,y)=h(x-y)$. In particular, $h(t)=|t|^p$ gives the usual one-dimensional optimal coupling for every $p\geqslant1$.*

*Proof.* The push-forward statement $\pi^\star\in\mathcal{U}(\alpha,\beta)$ follows from Proposition 2.30.
The key point is the one-dimensional uncrossing inequality. If $x<x'$ and $y>y'$, set $a=x-y$, $\delta=x'-x>0$ and $\eta=y-y'>0$. Convexity of $h$ implies that increments are monotone, hence $h(a+\eta)-h(a)\leqslant h(a+\delta+\eta)-h(a+\delta)$, which is exactly $h(x-y)+h(x'-y')\geqslant h(x-y')+h(x'-y)$. Thus removing a crossing never increases the cost. For a finite transport matrix on two ordered grids, if $i<i'$ and $j>j'$ carry crossed masses $\mathrm{P}_{i,j}$ and $\mathrm{P}_{i',j'}$, one may move $\theta=\min(\mathrm{P}_{i,j},\mathrm{P}_{i',j'})$ units of mass from the crossed entries $(i,j)$, $(i',j')$ to the uncrossed entries $(i,j')$, $(i',j)$. The row and column sums are unchanged, and the preceding inequality shows that the cost does not increase. Repeating this elementary uncrossing step yields an ordered plan; on an ordered uniform quantile grid, this is the diagonal plan.
For general measures, apply the same argument in quantile coordinates. Let $\pi\in\mathcal{U}(\alpha,\beta)$. Let $U_\alpha$ and $U_\beta$ be uniform random variables on $(0,1)$, and denote by $K_\alpha(x,\mathrm{d}r)$ and $K_\beta(y,\mathrm{d}s)$ regular conditional laws of $U_\alpha$ given $q_\alpha(U_\alpha)=x$ and of $U_\beta$ given $q_\beta(U_\beta)=y$. Given $(X,Y)\sim\pi$, draw conditionally $R\sim K_\alpha(X,\cdot)$ and $S\sim K_\beta(Y,\cdot)$. The law $\gamma$ of $(R,S)$ has uniform marginals and satisfies $\pi=(q_\alpha,q_\beta)_\sharp\gamma$.
Let $\kappa_M(r)=\max(-M,\min(r,M))$ and set $q_{\alpha,M}=\kappa_M\circ q_\alpha$ and $q_{\beta,M}=\kappa_M\circ q_\beta$. These functions are bounded and nondecreasing. Approximate them almost everywhere by nondecreasing step functions $q_{\alpha,M}^N$ and $q_{\beta,M}^N$ that are constant on the uniform intervals $I_k=((k-1)/N,k/N]$. The matrix $G_{k,\ell}^N=\gamma(I_k\times I_\ell)$ is a coupling of the two uniform histograms. Proposition 3.9, applied to the ordered step values, therefore gives $\int_0^1h(q_{\alpha,M}^N(r)-q_{\beta,M}^N(r))\mathrm{d}r\leqslant\int_{(0,1)^2}h(q_{\alpha,M}^N(r)-q_{\beta,M}^N(s))\mathrm{d}\gamma(r,s)$. For fixed $M$, the arguments of $h$ remain in the compact interval $[-2M,2M]$. Dominated convergence as $N\to\infty$ yields the same inequality for $q_{\alpha,M}$ and $q_{\beta,M}$.
Finally, $\kappa_M$ is nondecreasing and 1-Lipschitz. Hence, for every $x,y\in\mathbb{R}$, there exists $t_M(x,y)\in[0,1]$ such that $\kappa_M(x)-\kappa_M(y)=t_M(x,y)(x-y)$. Convexity and nonnegativity imply $h(t_M(x,y)(x-y))\leqslant(1-t_M(x,y))h(0)+t_M(x,y)h(x-y)\leqslant h(0)+h(x-y)$. The assumed integrability controls the diagonal term. If the cost of $\pi$ is finite, the same bound controls the right-hand side; if

it is infinite, the desired comparison is immediate. Dominated convergence as $M \to \infty$ therefore gives $\int_0^1 h(q_\alpha(r) - q_\beta(r))\mathrm{d}r \leqslant \int h(x-y)\mathrm{d}\pi(x,y)$ for every $\pi \in \mathcal{U}(\alpha,\beta)$. □

### 3.4 Cone of Convex Functions and Monge Gap

**Definition and optimality certificate.**

**Definition 3.29** (Monge gap)**.** Let $\alpha \in \mathcal{M}_+^1(\mathcal{X})$, let $T : \mathcal{X} \to \mathcal{Y}$ be measurable, and assume that the following costs are finite. The $c$-Monge gap of $T$ relative to $\alpha$ is

$$\mathcal{M}_\alpha^c(T) := \int c(x,T(x))\mathrm{d}\alpha(x) - \mathcal{L}_c(\alpha, T_\sharp\alpha). \tag{3.8}$$

**Proposition 3.30** (Elementary properties of the Monge gap)**.** *Under the standing Polish-space assumptions, suppose that the relevant integrals are finite. Then $\mathcal{M}_\alpha^c(T) \geqslant 0$, with equality if and only if $(\mathrm{Id}, T)_\sharp\alpha$ is an optimal coupling between $\alpha$ and $T_\sharp\alpha$. Moreover,*

$$\mathcal{M}_\alpha^c(T) = \sup_{\eta\in\mathcal{U}(\alpha,\alpha)} \int \big[c(x,T(x)) - c(x,T(z))\big]\mathrm{d}\eta(x,z). \tag{3.9}$$

*Finally, the gap is unchanged when $c(x,y)$ is replaced by $c(x,y)+r(x)+s(y)$, provided the modified cost remains admissible and all marginal terms are integrable.*

*Proof.* The graph plan $(\mathrm{Id},T)_\sharp\alpha$ is feasible for $\mathcal{L}_c(\alpha,T_\sharp\alpha)$, which proves nonnegativity and the equality criterion. Every $\eta \in \mathcal{U}(\alpha,\alpha)$ induces the coupling $(x,z) \mapsto (x,T(z))$. Conversely, disintegrating $\alpha$ along $T$ lifts every coupling in $\mathcal{U}(\alpha,T_\sharp\alpha)$ to such a self-coupling. Subtracting the resulting infimum from the graph cost gives (3.9). Adding $r(x)+s(y)$ shifts both terms in (3.8) by $\int r\mathrm{d}\alpha + \int s\mathrm{d}(T_\sharp\alpha)$. □

**Convex gaps and convex potentials.**

**Proposition 3.31** (Convex Monge gaps for affine-difference costs)**.** *Let $\mathcal{Y} \subset \mathbb{R}^m$ be convex and let the nonnegative cost $c$ have the form*

$$c(x,y) = a(x) + b(y) - \langle A(x),\, y\rangle, \tag{3.10}$$

*where $A, T \in L^2(\alpha;\mathbb{R}^m)$ and all displayed terms are finite. Then*

$$\mathcal{M}_\alpha^c(T) = \sup_{\eta\in\mathcal{U}(\alpha,\alpha)} \int \langle A(x),\, T(z)-T(x)\rangle\mathrm{d}\eta(x,z). \tag{3.11}$$

*Moreover, $\mathcal{M}_\alpha^c(T) = 0$ if and only if there is a proper lower-semicontinuous convex function $\varphi$ such that*

$$T(x) \in \partial\varphi(A(x)) \qquad \text{for } \alpha\text{-almost every } x. \tag{3.12}$$

*Proof.* The marginal terms $a$ and $b$ cancel in (3.9), giving (3.11); this is a supremum of linear functionals of $T$. Next, disintegrating $\alpha$ along $A$ lifts every coupling between $A_\sharp\alpha$ and $T_\sharp\alpha$. Hence zero gap is equivalent to optimality of $(A,T)_\sharp\alpha$ for the bilinear cost $(u,y) \mapsto -\langle u,\, y\rangle$. Up to marginal terms, this is the quadratic cost $\frac{1}{2}\|u-y\|^2$. By Rockafellar's characterization, recalled in Section 3.5, its optimal plans are exactly those concentrated on the subdifferential of a proper lower-semicontinuous convex function. □

$B_\Phi(y|x) = \Phi(y) - \Phi(x) - \langle\nabla\Phi(x),\, y-x\rangle$, where $\Phi$ is differentiable and convex: here $a(x) = \langle\nabla\Phi(x),\, x\rangle - \Phi(x)$, $b(y) = \Phi(y)$ and $A(x) = \nabla\Phi(x)$. The argument order is essential: on a full-dimensional domain, the reversed divergence $B_\Phi(x|y)$ is generally not of the form (3.10) unless $\Phi$ is quadratic.

**One-dimensional order and isotonicity.**

**Definition 3.32** (Increasing rearrangement relative to a source)**.** Let $\alpha \in \mathcal{M}_+^1(\mathbb{R})$ be atomless and let $T : \mathbb{R} \to \mathbb{R}$ be measurable. The increasing rearrangement of $T$ relative to $\alpha$ is

$$T^{\uparrow,\alpha} := \mathcal{C}_{T_\sharp\alpha}^{-1} \circ \mathcal{C}_\alpha. \tag{3.13}$$

It is the unique $\alpha$-almost-everywhere nondecreasing map satisfying $(T^{\uparrow,\alpha})_\sharp\alpha = T_\sharp\alpha$.

**Proposition 3.33** (One-dimensional Monge-gap formulas)**.** *Let $\alpha \in \mathcal{M}_+^1(\mathbb{R})$ be atomless, let $c(x,y) = h(x-y)$ with $h$ convex, and assume the relevant terms are integrable. Then*

$$\mathcal{M}_\alpha^c(T) = \int \big[h(x-T(x)) - h(x-T^{\uparrow,\alpha}(x))\big]\mathrm{d}\alpha(x). \tag{3.14}$$

*If $h$ is strictly convex, this gap vanishes if and only if $T$ has a nondecreasing representative $\alpha$-almost everywhere. For the equal-weight atomic measure $\alpha_n = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$ on the uniform grid $x_i = x_1 + (i-1)\Delta$, let $t_i = T(x_i)$ and let $t_{(1)} \leqslant \cdots \leqslant t_{(n)}$ be their order statistics. For $c(x,y) = |x-y|^2$,*

$$\mathcal{M}_{\alpha_n}^{|\cdot|^2}(T) = \frac{1}{n}\sum_{i=1}^n \big[(x_i-t_i)^2 - (x_i - t_{(i)})^2\big] = \frac{2\Delta}{n}\sum_{1\leqslant i<j\leqslant n}(t_i-t_j)_+. \tag{3.15}$$

*Proof.* The one-dimensional rearrangement result of Theorem 2.31, together with its coupling extension in Theorem 3.28, shows that $T^{\uparrow,\alpha}$ is optimal between $\alpha$ and $T_\sharp\alpha$. If $h$ is strictly convex, atomlessness makes this optimal map unique $\alpha$-almost everywhere, so zero gap is equivalent to $T = T^{\uparrow,\alpha}$ $\alpha$-almost everywhere. For the empirical formula, sorting $(t_i)_i$ gives the increasing optimal assignment and hence the first equality. An adjacent exchange of an inversion $u > v$ decreases the average squared cost by $2\Delta(u-v)/n$; bubble sort exchanges every inverted pair once, which gives the second equality. □

**Monge-gap-regularized kernel regression.** The hinge representation above turns transport optimality into a tractable soft monotonicity prior. Let $x_i = x_1 + (i-1)\Delta$ and let $y_i \in \mathbb{R}$ be noisy observations.

$$\min_{T\in\mathcal{H}_k} \left\{ \frac{1}{n}\sum_{i=1}^{n} |T(x_i)-y_i|^2 + \lambda\|T\|^2_{\mathcal{H}_k} + \gamma\mathcal{M}^{|\cdot|^2}_{\alpha_n}(T) \right\}, \qquad \lambda > 0, \quad \gamma \geqslant 0, \tag{3.16}$$

where $\alpha_n = n^{-1}\sum_i \delta_{x_i}$. The usual orthogonal-projection argument shows that a minimizer has the representer form $T_{\mathrm{q}}(\cdot) = \sum_{j=1}^{n} \mathrm{q}_j k(x_j, \cdot)$.

$$\min_{\mathrm{q}\in\mathbb{R}^n} \left\{ \frac{1}{n}\|\mathrm{Kq}-\mathrm{y}\|^2 + \lambda\langle \mathrm{q}, \mathrm{Kq}\rangle + \frac{2\gamma\Delta}{n}\sum_{1\leqslant i<j\leqslant n} \left[(\mathrm{Kq})_i - (\mathrm{Kq})_j\right]_+ \right\}. \tag{3.17}$$

This is a finite-dimensional convex program: the first two terms are convex quadratics because $\mathrm{K} \succeq 0$, and the last term is a sum of hinges of affine functionals. Thus every point along the regularization path can be computed globally, without alternating between regression and transport.

### 3.5 $c$-Cyclical Monotonicity

**Support and $c$-cyclical monotonicity.** The support of a coupling is the topological support introduced in Definition 2.4, now applied to a Radon measure on $\mathcal{X}\times\mathcal{Y}$. Thus $(x, y) \in \mathrm{supp}(\pi)$ exactly when every open neighborhood of $(x, y)$ has positive $\pi$-mass.

**Definition 3.34** ($c$-cyclical monotonicity)**.** A set $\Gamma \subset \mathcal{X}\times\mathcal{Y}$ is $c$-cyclically monotone if, for every $k \geqslant 2$, every finite family $(x_i, y_i)_{i=1}^k \subset \Gamma$ and every permutation $\sigma$ of $\{1, \dots, k\}$, $\sum_{i=1}^k c(x_i, y_i) \leqslant \sum_{i=1}^k c(x_i, y_{\sigma(i)})$.

$\sum_{i=1}^k c(x_i, y_i) \leqslant \sum_{i=1}^k c(x_i, y_{i+1}), \qquad y_{k+1} = y_1.$

**Optimal matching to optimal transport.** Let the marginals be uniform on $n$ points, $\alpha = \frac{1}{n}\sum_{i=1}^n \delta_{x_i}$ and $\beta = \frac{1}{n}\sum_{i=1}^n \delta_{y_i}$. By Corollary 3.16, there exists an optimal plan induced by a permutation. Its support $\Gamma = \{(x_i, y_{\sigma(i)})\}_i$ is $c$-cyclically monotone: otherwise exchanging the finitely many targets along a violating cycle would lower the matching cost.

**Theorem 3.35** (Optimal plans are $c$-cyclically monotone)**.** *Assume that $c : \mathcal{X}\times\mathcal{Y} \to [0, +\infty)$ is continuous and that the Kantorovich value is finite. For any optimal plan $\pi$ solving* (3.7)*,* $\mathrm{supp}(\pi)$ *is $c$-cyclically monotone.*

*Proof.* Suppose that $\mathrm{supp}(\pi)$ is not $c$-cyclically monotone. Then there exist points $(x_i, y_i)_{i=1}^k$ in the support and a permutation $\sigma$ such that $\sum_i c(x_i, y_i) > \sum_i c(x_i, y_{\sigma(i)})$. By continuity of $c$, after shrinking neighborhoods $U_i \ni x_i$ and $V_i \ni y_i$, the same strict inequality holds uniformly for every choice of points in these neighborhoods: $\sum_i c(u_i, v_i) > \sum_i c(u_i, \tilde{v}_{\sigma(i)}) \quad (u_i \in U_i,\ v_i \in V_i,\ \tilde{v}_{\sigma(i)} \in V_{\sigma(i)})$. Choose the sets so that $m_i := \pi(U_i \times V_i) > 0$. Taking $0 < \lambda \leqslant \left(\sum_{i=1}^k \frac{1}{m_i}\right)^{-1}$ ensures that the scaled restrictions $\pi_i = \lambda\frac{\pi|_{U_i\times V_i}}{m_i}$ have common mass $\lambda$ and satisfy $\sum_i \pi_i \leqslant \pi$, even if some rectangles overlap. Let $\alpha_i = (P_\mathcal{X})_\sharp\pi_i$ and $\beta_i = (P_\mathcal{Y})_\sharp\pi_i$, and define $\tilde{\pi} = \pi - \sum_i \pi_i + \sum_i(\alpha_i \otimes \beta_{\sigma(i)})/\lambda$. The removed and reinserted first marginals are both $\sum_i \alpha_i$, and the removed and reinserted second marginals are both $\sum_i \beta_i$ because $\sigma$ is a permutation. Hence $\tilde{\pi} \in \mathcal{U}(\alpha, \beta)$. Integrating the uniform strict inequality against the product probability $\otimes_i(\pi_i/\lambda)$ shows that the reinserted crossed terms have strictly smaller cost than the removed diagonal terms. This contradicts the optimality of $\pi$. ☐

**Monotonicity.** Assume the optimal plan is induced by a measurable map $T : \mathcal{X} \to \mathcal{Y}$, meaning that $\pi = (\mathrm{Id}, T)_\sharp\alpha$. There is a set $G \subset \mathcal{X}$ of full $\alpha$-measure such that $(x, T(x)) \in \mathrm{supp}(\pi)$ for every $x \in G$. For any $x_1, \dots, x_k \in G$, cyclical monotonicity reads $\sum_{i=1}^k c(x_i, T(x_i)) \leqslant \sum_{i=1}^k c(x_i, T(x_{i+1})), \qquad x_{k+1} = x_1.$

**One dimension.** $|x - T(x)|^p + |y - T(y)|^p \leqslant |x - T(y)|^p + |y - T(x)|^p$, For $p > 1$, strict convexity implies that a reversed pair $x < y$ and $T(x) > T(y)$ violates this inequality. Hence every optimal map is nondecreasing on the full-measure transport set, recovering the classical monotone rearrangement.

## 4 Dual Problem

### 4.1 Discrete dual

The Kantorovich problem (3.2) is a linear program so that one can equivalently compute its value by solving a dual linear program.

**Definition 4.1** (Admissible potentials)**.** For a discrete problem with marginal sizes $n, m$ and cost matrix $\mathrm{C} \in \mathbb{R}^{n\times m}$, a pair $(\mathrm{f}, \mathrm{g}) \in \mathbb{R}^n \times \mathbb{R}^m$ is admissible if it lies below the cost:

$$\mathrm{R}(\mathrm{C}) := \{(\mathrm{f}, \mathrm{g}) \in \mathbb{R}^n \times \mathbb{R}^m \,;\, \forall\,(i, j) \in [\![n]\!] \times [\![m]\!], \mathrm{f}_i + \mathrm{g}_j \leqslant \mathrm{C}_{i,j}\}\,. \tag{4.1}$$

Equivalently, $\mathrm{f} \oplus \mathrm{g} \leqslant \mathrm{C}$ entrywise. In particular, the feasible set depends on the cost matrix, not on the marginal weights.

**Proposition 4.2** (Discrete Kantorovich duality)**.** *One has*

$$\mathrm{L}_\mathrm{C}(\mathrm{a}, \mathrm{b}) = \max_{(\mathrm{f},\mathrm{g})\in\mathrm{R}(\mathrm{C})} \langle \mathrm{f}, \mathrm{a}\rangle + \langle \mathrm{g}, \mathrm{b}\rangle. \tag{4.2}$$

*Proof.* For completeness, introduce multipliers $(\mathrm{f}, \mathrm{g})$ for the two marginal constraints. The Lagrangian is

$$\mathcal{L}(\mathrm{P}, \mathrm{f}, \mathrm{g}) = \langle \mathrm{C}, \mathrm{P}\rangle + \langle \mathrm{a} - \mathrm{P}\mathbb{1}_m, \mathrm{f}\rangle + \langle \mathrm{b} - \mathrm{P}^\top\mathbb{1}_n, \mathrm{g}\rangle. \tag{4.3}$$

For fixed potentials, the dual function is $\inf_{\mathrm{P}\geqslant 0}\mathcal{L}(\mathrm{P}, \mathrm{f}, \mathrm{g}) = \langle \mathrm{a}, \mathrm{f}\rangle + \langle \mathrm{b}, \mathrm{g}\rangle + \inf_{\mathrm{P}\geqslant 0}\langle \mathrm{C} - (\mathrm{f}\oplus\mathrm{g}), \mathrm{P}\rangle$. The last infimum is

$$\inf_{\mathrm{P}\geqslant 0}\langle \mathrm{Q}, \mathrm{P}\rangle = \begin{cases} 0 & \text{if } \mathrm{Q} \geqslant 0, \\ -\infty & \text{otherwise,} \end{cases}$$

and is therefore finite exactly when $(\mathrm{f}, \mathrm{g}) \in \mathrm{R}(\mathrm{C})$. The primal polytope is nonempty because it contains the product coupling $\mathrm{a}\otimes\mathrm{b}$, and it is compact; hence the primal value is finite and attained. Finite-dimensional linear-programming strong duality then identifies this value with the maximum of the dual function and also gives dual attainment. ☐

**Proposition 4.3** (Discrete complementary slackness)**.** *For every* $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$ *and* $(\mathrm{f},\mathrm{g}) \in \mathrm{R}(\mathrm{C})$, $\langle \mathrm{C}, \mathrm{P}\rangle - \langle \mathrm{f}, \mathrm{a}\rangle - \langle \mathrm{g}, \mathrm{b}\rangle = \sum_{i,j} \mathrm{P}_{i,j}\big(\mathrm{C}_{i,j} - \mathrm{f}_i - \mathrm{g}_j\big) \geqslant 0$. *The plan and potentials are respectively primal and dual optimal if and only if*

$$\mathrm{Supp}(\mathrm{P}) \subset \{(i,j) \in [\![n]\!] \times [\![m]\!] \;;\; \mathrm{f}_i + \mathrm{g}_j = \mathrm{C}_{i,j}\}\,. \tag{4.4}$$

*Proof.* The marginal constraints give $\langle \mathrm{f}, \mathrm{a}\rangle + \langle \mathrm{g}, \mathrm{b}\rangle = \sum_{i,j} \mathrm{P}_{i,j}(\mathrm{f}_i + \mathrm{g}_j)$, which proves the identity and nonnegativity. If both feasible objects are optimal, their values agree by Proposition 4.2; every nonnegative summand with $\mathrm{P}_{i,j} > 0$ must therefore vanish. Conversely, the support condition makes the gap zero, so weak duality forces both objects to be optimal. □

## 4.2 General formulation

**Continuous duality.**

**Definition 4.4** (Continuous admissible potentials)**.** For $c \in \mathcal{C}(\mathcal{X} \times \mathcal{Y})$, define

$$\mathcal{R}(c) \coloneqq \{(f,g) \in \mathcal{C}(\mathcal{X}) \times \mathcal{C}(\mathcal{Y}) \;;\; f(x) + g(y) \leqslant c(x,y) \text{ for all } (x,y)\}\,. \tag{4.5}$$

Its elements are admissible continuous potentials.

**Proposition 4.5** (Kantorovich duality)**.** *Assume that* $\mathcal{X}$ *and* $\mathcal{Y}$ *are compact metric spaces and that* $c \in \mathcal{C}(\mathcal{X} \times \mathcal{Y})$. *Then*

$$\mathcal{L}_c(\alpha,\beta) = \max_{(f,g)\in\mathcal{R}(c)} \int_{\mathcal{X}} f(x)\mathrm{d}\alpha(x) + \int_{\mathcal{Y}} g(y)\mathrm{d}\beta(y), \tag{4.6}$$

*Here,* $(f,g)$ *is a pair of continuous functions, often called "Kantorovich potentials". The same formula extends under the usual lower-semicontinuity and integrability assumptions, replacing maxima by suprema when dual optimizers need not exist.*

*Proof.* Weak duality is immediate: if $f(x)+g(y) \leqslant c(x,y)$ and $\pi \in \mathcal{U}(\alpha,\beta)$, then $\int f\mathrm{d}\alpha + \int g\mathrm{d}\beta = \int (f(x)+g(y))\mathrm{d}\pi(x,y) \leqslant \int c\mathrm{d}\pi$. Taking the supremum over admissible potentials and the infimum over couplings gives the first inequality.
For the reverse inequality, write $V = \mathcal{L}_c(\alpha,\beta)$ and consider the convex cone $\mathcal{K} \coloneqq \big\{(\pi_1, \pi_2, \int c\mathrm{d}\pi + r) : \pi \in \mathcal{M}_+(\mathcal{X}\times\mathcal{Y}),\ r \geqslant 0\big\}$ in $\mathcal{M}(\mathcal{X}) \times \mathcal{M}(\mathcal{Y}) \times \mathbb{R}$, equipped with the weak topologies on the measure factors. This cone is closed. Indeed, convergence of the first marginal implies a uniform bound on the masses of the underlying nonnegative measures. Compactness of $\mathcal{X}\times\mathcal{Y}$ then gives a convergent subnet $\pi_k \rightharpoonup \pi$; continuity of marginalization and of $\pi \mapsto \int c\mathrm{d}\pi$ identifies the limit with an element of $\mathcal{K}$, while the limiting epigraph remainder remains nonnegative.
For every $t < V$, the point $z_t = (\alpha, \beta, t)$ does not belong to $\mathcal{K}$. Strong separation of this point from the closed convex cone gives a continuous linear functional. Because $\mathcal{K}$ is a cone containing the origin, the functional can be oriented so that, for some $F \in \mathcal{C}(\mathcal{X})$, $G \in \mathcal{C}(\mathcal{Y})$ and $\lambda \in \mathbb{R}$, $\int F\mathrm{d}\alpha + \int G\mathrm{d}\beta + \lambda t < 0 \leqslant \int F\mathrm{d}\eta + \int G\mathrm{d}\zeta + \lambda s \quad$ for all $(\eta,\zeta,s) \in \mathcal{K}$. Testing $(0,0,r) \in \mathcal{K}$ gives $\lambda \geqslant 0$, while testing the element generated by $\pi = \delta_{(x,y)}$ and $r = 0$ gives $F(x) + G(y) + \lambda c(x,y) \geqslant 0$. The multiplier cannot vanish: if $\lambda = 0$, integrating this last inequality against $\alpha \otimes \beta$ would contradict the strict inequality at $z_t$. Thus $\lambda > 0$, and $f \coloneqq -F/\lambda, \quad g \coloneqq -G/\lambda$ satisfy $f \oplus g \leqslant c$ and $\int f\mathrm{d}\alpha + \int g\mathrm{d}\beta > t$. Letting $t \uparrow V$ proves the reverse inequality.
Take a maximizing sequence $(f_k, g_k)$. Successively replace it by $\widetilde{g}_k(y) = \min_x\{c(x,y) - f_k(x)\}, \quad \widetilde{f}_k(x) = \min_y\{c(x,y) - \widetilde{g}_k(y)\}$. Each replacement preserves feasibility and cannot decrease the objective. Fixing $x_0 \in \mathcal{X}$, use the gauge freedom $(\widetilde{f}_k, \widetilde{g}_k) \mapsto (\widetilde{f}_k - a_k, \widetilde{g}_k + a_k)$ to impose $\widetilde{f}_k(x_0) = 0$. Uniform continuity of $c$ shows that $\widetilde{g}_k$ inherits a common modulus in the second variable and $\widetilde{f}_k$ a common modulus in the first. The gauge gives a uniform bound on $\widetilde{f}_k$, and the first envelope then gives one on $\widetilde{g}_k$. Arzelà–Ascoli yields uniformly convergent subsequences. Their limit remains admissible and attains the dual supremum. □

**Complementary slackness.**

**Proposition 4.6** (Continuous complementary slackness)**.** *For* $\pi \in \mathcal{U}(\alpha,\beta)$ *and* $(f,g) \in \mathcal{R}(c)$, $\int c\mathrm{d}\pi - \int f\mathrm{d}\alpha - \int g\mathrm{d}\beta = \int \big(c(x,y) - f(x) - g(y)\big)\mathrm{d}\pi(x,y) \geqslant 0$. *The coupling and potentials are respectively primal and dual optimal if and only if this gap vanishes. In that case,*

$$\mathrm{Supp}(\pi) \subset \{(x,y) \in \mathcal{X}\times\mathcal{Y} \;;\; f(x) + g(y) = c(x,y)\}\,. \tag{4.7}$$

*Proof.* The marginal identities give the equality, and admissibility gives nonnegativity. A zero gap is equivalent to equality of the feasible primal and dual values, hence to joint optimality by Proposition 4.5. Finally, the slack is continuous and nonnegative. If it were positive at a point of $\mathrm{Supp}(\pi)$, it would be bounded below by a positive constant on a neighborhood of positive $\pi$-mass, contradicting its zero integral. □

$s(x,y) \coloneqq c(x,y) - f(x) - g(y)$. Dual feasibility is the statement $s \geqslant 0$, while complementary slackness says that an optimal coupling can only live where this slack vanishes.

## 4.3 $c$-transforms

**Best-response potentials and the $c$-transform.** $\sup_{g\in\mathcal{C}(\mathcal{Y})} \big\{\int g\mathrm{d}\beta \;;\; \forall\,(x,y), g(y) \leqslant c(x,y) - f(x)\big\}\,.$

**Definition 4.7** ($c$-transform)**.** For a real-valued function $f : \mathcal{X} \to \mathbb{R}$, its $c$-transform is the extended-real-valued function

$$\forall\, y \in \mathcal{Y}, \quad f^c(y) \coloneqq \inf_{x\in\mathcal{X}} c(x,y) - f(x). \tag{4.8}$$

For a real-valued function $g : \mathcal{Y} \to \mathbb{R}$, the $\bar{c}$-transform associated with $\bar{c}(y,x) = c(x,y)$ is $\forall\, x \in \mathcal{X}, \quad g^{\bar{c}}(x) \coloneqq \inf_{y\in\mathcal{Y}} c(x,y) - g(y)$.
When $\mathcal{X}, \mathcal{Y}$ are compact, $c$ is continuous and the input potential is continuous, these infima are finite minima and the transformed potential is continuous.

**Proposition 4.8** ($c$-transforms solve the semi-relaxed problems)**.** *Assume the compactness and continuity hypotheses of Proposition 4.5. For fixed* $f \in \mathcal{C}(\mathcal{X})$*, the maximizers of the dual objective over all* $g \in \mathcal{C}(\mathcal{Y})$ *such that* $f \oplus g \leqslant c$ *are exactly the functions satisfying* $g = f^c$ $\beta$*-almost everywhere. Symmetrically, for fixed* $g$*, the maximizers over* $f$ *are exactly the functions satisfying* $f = g^{\bar{c}}$ $\alpha$*-almost everywhere.*

*Proof.* The constraint $f(x)+g(y) \leqslant c(x,y)$ for every $x$ is equivalent to $g(y) \leqslant \inf_x\{c(x,y)-f(x)\} = f^c(y)$. Thus $f^c$ is feasible and pointwise dominates every feasible $g$. Integrating against $\beta$ proves optimality, and equality holds exactly when $g = f^c$ $\beta$-almost everywhere. The other assertion follows by exchanging the two marginals. ☐

**Proposition 4.9** (Modulus stability of $c$-transforms)**.** *Suppose that $\omega$ is a modulus such that $|c(x,y) - c(x,y')| \leqslant \omega\big(d_{\mathcal{Y}}(y,y')\big)$ for all $x,y,y'$. Then every finite $f^c$ has modulus $\omega$. Symmetrically, $g^{\bar{c}}$ inherits any uniform modulus of $c$ in its first variable. In particular, a uniform L-Lipschitz bound on the cost yields an L-Lipschitz bound on the corresponding transform.*

*Proof.* For each $x$, set $F_x(y) = c(x,y) - f(x)$ and $F(y) = f^c(y) = \inf_x F_x(y)$. Since all the functions $F_x$ share the modulus $\omega$,

$$|F(y) - F(y')| = \left|\inf_x F_x(y) - \inf_x F_x(y')\right| \leqslant \sup_x |F_x(y) - F_x(y')| \leqslant \omega\big(d_{\mathcal{Y}}(y,y')\big).$$

The proof in the first variable is identical. ☐

**Discrete $c$-transform.** If $\alpha = \sum_{i=1}^n a_i\delta_{x_i}$ and $\beta = \sum_{j=1}^m b_j\delta_{y_j}$, the same definition applies to the finite spaces $\mathcal{X} = \{x_i\}_i$ and $\mathcal{Y} = \{y_j\}_j$. The dual potentials are then vectors $(\mathrm{f},\mathrm{g}) \in \mathbb{R}^n \times \mathbb{R}^m$, as in Section 4.1, and the cost is the matrix $\mathrm{C} \in \mathbb{R}^{n\times m}$ with entries
$\mathrm{C}_{i,j} = c(x_i,y_j)$. With the present sign convention $f \oplus g \leqslant c$, the discrete transform of f is the vector $\mathrm{f}^{\mathrm{C}} \in \mathbb{R}^m$ defined by
$(\mathrm{f}^{\mathrm{C}})_j = \min_{1\leqslant i\leqslant n} \mathrm{C}_{i,j} - \mathrm{f}_i$. This is a minimum because feasibility imposes the upper bound $\mathrm{g}_j \leqslant \mathrm{C}_{i,j} - \mathrm{f}_i$ for every $i$. Thus the largest feasible best response is the minimum of these upper bounds; formulas with a maximum correspond to a different sign convention, not to the dual constraint (4.2).
$\mathrm{g}^{\bar{\mathrm{C}}} := \mathrm{g}^{\mathrm{C}^\top} \in \mathbb{R}^n$, $\qquad (\mathrm{g}^{\bar{\mathrm{C}}})_i = \min_{1\leqslant j\leqslant m} \mathrm{C}_{i,j} - \mathrm{g}_j$. Thus the continuous best-response operation reduces exactly to taking column minima for $\mathrm{f}^{\mathrm{C}}$, or row minima for $\mathrm{g}^{\bar{\mathrm{C}}}$, in shifted cost matrices.
**Euclidean case.** The Euclidean quadratic cost is the model case where $c$-transforms become ordinary convex conjugates after removing the quadratic terms. This is the algebraic bridge between Kantorovich duality and Brenier maps.
$\int c(x,y)\mathrm{d}\pi(x,y) = \frac{1}{2}\int \|x\|^2\mathrm{d}\alpha(x) + \frac{1}{2}\int \|y\|^2\mathrm{d}\beta(y) - \int\langle x,\, y\rangle\mathrm{d}\pi(x,y)$.

**Definition 4.10** (Legendre–Fenchel transform and biconjugate)**.** For $h : \mathbb{R}^d \to \mathbb{R}\cup\{+\infty\}$, define $h^*(y) := \sup_{x\in\mathbb{R}^d}\{\langle x,\, y\rangle - h(x)\}$, $\qquad h^{**} := (h^*)^*$.

$f^c(y) = \frac{1}{2}\|y\|^2 - \varphi^*(y)$, $\qquad f^{c\bar{c}}(x) = \frac{1}{2}\|x\|^2 - \varphi^{**}(x)$. On the full Euclidean space, the $c$-closed functions are therefore those for which $x \mapsto \frac{1}{2}\|x\|^2 - f(x)$ is lower-semicontinuous and convex; equivalently, $f$ is 1-semiconcave. The double transform is the smallest $\bar{c}$-concave majorant of $f$.
**Why hard alternating optimization stops.** $(f,g) \longmapsto (f,f^c) \longmapsto (f^{c\bar{c}},f^c)$
**Proposition 4.11** (Algebra of $c$-transforms)**.** *Writing $f^{c\bar{c}} := (f^c)^{\bar{c}}$, the following pointwise identities hold whenever the transforms are well defined:*

$$\text{(i) } f \leqslant f' \Rightarrow f^c \geqslant f'^c, \qquad \text{(ii) } f^{c\bar{c}} \geqslant f, \qquad \text{(iii) } g^{\bar{c}c} \geqslant g, \qquad \text{(iv) } f^{c\bar{c}c} = f^c.$$

*Proof.* The order reversal in (i) follows directly from the minus sign in the definition. For every $x$ and $y$, $f^c(y) = \inf_{x'}\{c(x',y) - f(x')\} \leqslant c(x,y) - f(x)$, so $c(x,y) - f^c(y) \geqslant f(x)$. Taking the infimum over $y$ proves (ii), and exchanging the two spaces proves (iii). Finally, (ii) and the order reversal imply $f^c \geqslant f^{c\bar{c}c}$, whereas (iii) applied to $g = f^c$ gives $f^{c\bar{c}c} \geqslant f^c$. This proves (iv). ☐

$$(f^{c\bar{c}},f^c) \longmapsto (f^{c\bar{c}},f^{c\bar{c}c}) = (f^{c\bar{c}},f^c). \tag{4.9}$$

Thus exact block maximization reaches a coordinatewise fixed point after one full cycle. The resulting pair is dual feasible, but it need not maximize the joint dual objective because its value still depends on the arbitrary initialization $f$.
For the bilinear cost $c(x,y) = -xy$ on a compact interval, $f^{c\bar{c}}$ is the smallest concave majorant representable with slopes in the opposite interval. A hard transform removes non-concave oscillations in one step rather than producing a gradual ascent.
**$c$-concave functions.** The ranges of the two $c$-transform operators define the natural classes of dual potentials.

**Definition 4.12** ($c$-concave functions)**.** A function on $\mathcal{Y}$ is $c$-concave if it equals $f^c$ for some function $f$ on $\mathcal{X}$. A function on $\mathcal{X}$ is $\bar{c}$-concave if it equals $g^{\bar{c}}$ for some function $g$ on $\mathcal{Y}$.

**Definition 4.13** (Finite-valued $c$-concave class)**.** The finite-valued $c$-concave class on $\mathcal{Y}$ is $\mathrm{Conc}_c(\mathcal{Y}) := \{u : \mathcal{Y} \to \mathbb{R}\,;\ u = f^c \text{ for some } f : \mathcal{X} \to \mathbb{R}\cup\{-\infty\}\}$.

**Proposition 4.14** (Stability under tropical barycenters)**.** *If $u_r = f_r^c \in \mathrm{Conc}_c(\mathcal{Y})$ and $a_r \in \mathbb{R}$ for $1 \leqslant r \leqslant R$, then*

$$\min_{1\leqslant r\leqslant R}\{u_r + a_r\} = \Big(\max_{1\leqslant r\leqslant R}\{f_r - a_r\}\Big)^c \in \mathrm{Conc}_c(\mathcal{Y}).$$

*Proof.* Set $f = \max_r(f_r - a_r)$. For every $y \in \mathcal{Y}$, $f^c(y) = \inf_x \min_r\{c(x,y) - f_r(x) + a_r\} = \min_r \inf_x\{c(x,y) - f_r(x) + a_r\} = \min_r\{u_r(y) + a_r\}$. The middle equality holds because the minimum ranges over finitely many indices: both sides equal the infimum over $(x,r) \in \mathcal{X} \times \{1,\dots,R\}$. ☐

Thus $\mathrm{Conc}_c(\mathcal{Y})$ is a min-plus, or tropical, convex cone: tropical addition is pointwise minimum, while tropical scalar multiplication adds a constant. This universal closure does not imply stability under ordinary averages.
$u \in \mathrm{Conc}_{c_\tau}(\mathbb{R}^d) \quad\Longleftrightarrow\quad y \mapsto u(y) - \frac{\|y\|^2}{2\tau}$ is concave, $\mathrm{Conc}_d(\mathcal{X}) = \{u : \mathcal{X} \to \mathbb{R}\,;\ \mathrm{Lip}(u) \leqslant 1\}$,

# 5 Semi-discrete and $\mathcal{W}_1$

## 5.1 Semi-dual

**General measure semi-dual.**

**Definition 5.1** (Full dual and semi-dual functionals)**.** Under the compactness and continuity assumptions of Chapter 4, define

$$\mathcal{D}_0(f,g) \coloneqq \begin{cases} \int_{\mathcal{X}} f \mathrm{d}\alpha + \int_{\mathcal{Y}} g \mathrm{d}\beta, & (f,g) \in \mathcal{R}(c), \\ -\infty, & \text{otherwise.} \end{cases} \tag{5.1}$$

$$\mathcal{E}_0(g) \coloneqq \mathcal{D}_0(g^{\bar{c}}, g) = \int_{\mathcal{X}} g^{\bar{c}} \mathrm{d}\alpha + \int_{\mathcal{Y}} g \mathrm{d}\beta. \tag{5.2}$$

Thus the dual problem (4.6) is $\mathcal{L}_c(\alpha,\beta) = \max_{f,g} \mathcal{D}_0(f,g)$. For fixed $g$, feasibility is equivalent to $f \leqslant g^{\bar{c}}$.
$\mathcal{L}_c(\alpha,\beta) = \sup_{g\in\mathcal{C}(\mathcal{Y})} \mathcal{E}_0(g)$, $\qquad \mathcal{E}_0(g) = \sup_{f\in\mathcal{C}(\mathcal{X})} \mathcal{D}_0(f,g)$.
**Discrete semi-dual.** When both measures are discrete, the same elimination produces a finite-dimensional concave objective whose active affine pieces are selected by the discrete $c$-transform.
Let $\alpha = \sum_{i=1}^n \mathrm{a}_i \delta_{x_i}$, $\beta = \sum_{j=1}^m \mathrm{b}_j \delta_{y_j}$, and $\mathrm{C}_{i,j} = c(x_i, y_j)$, where $\mathrm{a} \in \mathbb{R}_+^n$ and $\mathrm{b} \in \mathbb{R}_+^m$ have the same total mass $M$. We use the same notation for functions and vectors: in the discrete setting,

$$\mathcal{D}_0(\mathrm{f},\mathrm{g}) \coloneqq \begin{cases} \langle \mathrm{f},\, \mathrm{a}\rangle + \langle \mathrm{g},\, \mathrm{b}\rangle, & \mathrm{f}\oplus\mathrm{g} \leqslant \mathrm{C}, \\ -\infty, & \text{otherwise.} \end{cases}$$

$$\mathrm{L}_{\mathrm{C}}(\mathrm{a},\mathrm{b}) = \max_{\mathrm{g}\in\mathbb{R}^m} \mathcal{E}_0(\mathrm{g}), \qquad \mathcal{E}_0(\mathrm{g}) \coloneqq \mathcal{D}_0(\mathrm{g}^{\bar{\mathrm{C}}}, \mathrm{g}) = \sum_{i=1}^n \mathrm{a}_i (\mathrm{g}^{\bar{\mathrm{C}}})_i + \sum_{j=1}^m \mathrm{b}_j \mathrm{g}_j, \tag{5.3}$$

$(\mathrm{g}^{\bar{\mathrm{C}}})_i = \min_{1\leqslant j\leqslant m} (\mathrm{C}_{i,j} - \mathrm{g}_j)$. $\sigma_{\mathrm{g}}(i) \in \operatorname{argmin}_j (\mathrm{C}_{i,j} - \mathrm{g}_j)$,
**Proposition 5.2** (Supergradients of the discrete semi-dual)**.** *For any selection $\sigma_{\mathrm{g}}(i) \in \operatorname{argmin}_j(\mathrm{C}_{i,j} - \mathrm{g}_j)$, define $\widehat{\mathrm{b}}_j(\mathrm{g}) \coloneqq \sum_{i:\, \sigma_{\mathrm{g}}(i)=j} \mathrm{a}_i$. Then $\mathrm{b} - \widehat{\mathrm{b}}(\mathrm{g})$ is a supergradient of $\mathcal{E}_0$ at $\mathrm{g}$. If every row has a unique minimizer, $\mathcal{E}_0$ is differentiable at $\mathrm{g}$ and $\nabla\mathcal{E}_0(\mathrm{g}) = \mathrm{b} - \widehat{\mathrm{b}}(\mathrm{g})$.*

*Proof.* Let $\mathrm{h} \in \mathbb{R}^m$. Since $\sigma_{\mathrm{g}}(i)$ is active at $\mathrm{g}$,

$$\min_j (\mathrm{C}_{i,j} - \mathrm{h}_j) \leqslant \mathrm{C}_{i,\sigma_{\mathrm{g}}(i)} - \mathrm{h}_{\sigma_{\mathrm{g}}(i)} = \min_j (\mathrm{C}_{i,j} - \mathrm{g}_j) - \big(\mathrm{h}_{\sigma_{\mathrm{g}}(i)} - \mathrm{g}_{\sigma_{\mathrm{g}}(i)}\big).$$

Multiplying by $\mathrm{a}_i$, summing over $i$, and adding $\langle \mathrm{b},\, \mathrm{h}-\mathrm{g}\rangle$ gives $\mathcal{E}_0(\mathrm{h}) \leqslant \mathcal{E}_0(\mathrm{g}) \langle \mathrm{b} - \widehat{\mathrm{b}}(\mathrm{g}),\, \mathrm{h}-\mathrm{g}\rangle$, which is the supergradient inequality. Uniqueness of all active indices makes the same affine branch active in a neighborhood of g, yielding the gradient formula. ☐

## 5.2 Auction Algorithm

The auction algorithm is derived from coordinate maximization of the semi-dual of the linear assignment problem. Its practical form uses bidder-specific dual-weight updates that cross the selected row's next indifference threshold by $\varepsilon$; this controlled relaxation prevents jamming at nonsmooth ties.

**Coordinate ascent and discrete Laguerre cells.**

$$\mathcal{E}_0(\mathrm{g}) = \frac{1}{n} \sum_{i=1}^n (\mathrm{g}^{\bar{\mathrm{C}}})_i + \frac{1}{n} \sum_{j=1}^n \mathrm{g}_j, \qquad (\mathrm{g}^{\bar{\mathrm{C}}})_i = \min_{1\leqslant j\leqslant n} (\mathrm{C}_{i,j} - \mathrm{g}_j). \tag{5.4}$$

Thus $(\mathrm{g}^{\bar{\mathrm{C}}}, \mathrm{g})$ is dual feasible. This is precisely the discrete $\bar{C}$-transform described in Section 4.3, with the same sign convention as in the general and semi-discrete formulations.

$$\mathbb{L}_j^{\mathrm{D}}(\mathrm{g}) \coloneqq \{ i \in [\![n]\!] \,;\, \mathrm{C}_{i,j} - \mathrm{g}_j \leqslant \mathrm{C}_{i,k} - \mathrm{g}_k \text{ for every } k \}\,. \tag{5.5}$$

$$\frac{\partial}{\partial \mathrm{g}_j} \mathcal{E}_0(\mathrm{g}) = \frac{1 - \big|\mathbb{L}_j^{\mathrm{D}}(\mathrm{g})\big|}{n}. \tag{5.6}$$

Hence an overfull cell calls for a decrease of its dual weight. More generally, Proposition 1.5 shows that a perfect matching in the contact graph $i \in \mathbb{L}_j^{\mathrm{D}}(\mathrm{g})$ certifies optimality of g in (5.4); conversely, assignment duality and complementary slackness produce such a matching at every maximizer.
For $i \in \mathbb{L}_j^{\mathrm{D}}(\mathrm{g})$, define the bid needed to make row $i$ indifferent between $j$ and its best alternative,

$$\mathrm{bid}_j(\mathrm{g}, i) \coloneqq \min_{k\neq j} (\mathrm{C}_{i,k} - \mathrm{g}_k) - (\mathrm{C}_{i,j} - \mathrm{g}_j). \tag{5.7}$$

**Bids and relaxed contacts.** Exact coordinate maximization can stall at a nonsmooth tie, so auction deliberately moves a bidder $\varepsilon$ beyond its next indifference point. Suppose row $i$ is currently unassigned.

$$j_0 \in \operatorname{argmin}_j (\mathrm{C}_{i,j} - \mathrm{g}_j), \qquad j_1 \in \operatorname{argmin}_{j\neq j_0} (\mathrm{C}_{i,j} - \mathrm{g}_j), \tag{5.8}$$

$$\Delta_i \coloneqq \mathrm{bid}_{j_0}(\mathrm{g}, i) + \varepsilon = (\mathrm{C}_{i,j_1} - \mathrm{g}_{j_1}) - (\mathrm{C}_{i,j_0} - \mathrm{g}_{j_0}) + \varepsilon, \qquad \mathrm{g}_{j_0} \leftarrow \mathrm{g}_{j_0} - \Delta_i. \tag{5.9}$$

**Definition 5.3** ($\varepsilon$-complementary slackness)**.** A partial permutation matrix $\mathrm{P} \in \{0,1\}^{n\times n}$ has at most one nonzero entry in each row and column. Such a matrix and target potential g satisfy $\varepsilon$-complementary slackness if

$$\mathrm{P}_{i,j} = 1 \implies \mathrm{C}_{i,j} - \mathrm{g}_j \leqslant (\mathrm{g}^{\bar{\mathrm{C}}})_i + \varepsilon. \tag{5.10}$$

$\mathrm{r}_{i,j}(\mathrm{g}) \coloneqq \mathrm{C}_{i,j} - \mathrm{g}_j - (\mathrm{g}^{\bar{\mathrm{C}}})_i \geqslant 0.$

**Fixed tolerance.**
**Proposition 5.4** (Fixed-$\varepsilon$ auction convergence and complexity). *Let $R_{\mathrm{C}} \coloneqq \max_{i,j} \mathrm{C}_{i,j} - \min_{i,j} \mathrm{C}_{i,j}$. Started from $\mathrm{g} = 0$, Algorithm ?? terminates after at most $n\big(\lfloor R_{\mathrm{C}}/\varepsilon \rfloor + 1\big)$ bids. It returns a permutation matrix $\mathrm{P}$ satisfying $\varepsilon$-complementary slackness and*

$$0 \leqslant \frac{1}{n}\langle \mathrm{C}, \mathrm{P}\rangle - \min_{\mathrm{P}' \in \mathcal{P}_n^{\mathrm{perm}}} \frac{1}{n}\langle \mathrm{C}, \mathrm{P}'\rangle \leqslant \varepsilon. \tag{5.11}$$

*With dense scans for the two smallest reduced costs, the algorithm requires $O\big(n^2(1+R_{\mathrm{C}}/\varepsilon)\big)$ arithmetic and comparison operations and $O(n^2)$ storage, including $\mathrm{C}$. If $\mathrm{C}$ has integer entries and $\varepsilon < 1/n$, the returned assignment is exactly optimal.*

*Proof.* At every bid, (5.9) decreases one target potential by at least $\varepsilon$. Once a target is assigned, it remains assigned: a later bid can change its owner but never leaves it empty. While the algorithm is incomplete, there is therefore an unassigned target whose potential is still zero. If the selected target $j_0$ is already assigned, comparison with such a zero-potential target gives $\mathrm{g}_{j_0} \geqslant -R_{\mathrm{C}}$; if it is unassigned, its potential is itself zero. Thus a target can be selected at most $\lfloor R_{\mathrm{C}}/\varepsilon \rfloor + 1$ times. Summing over the $n$ targets proves finite termination and the bid bound. At termination, every row and column has exactly one nonzero entry, so $\mathrm{P}$ is a permutation matrix.
The $\varepsilon$-complementary-slackness invariant was established above. Summing it over the returned permutation gives the first inequality below. The second is the dual lower bound of Proposition 1.5, applied to the feasible pair $(\mathrm{g}^{\bar{\mathrm{C}}}, \mathrm{g})$:

$$\frac{1}{n}\langle \mathrm{C}, \mathrm{P}\rangle \leqslant \frac{1}{n}\sum_i (\mathrm{g}^{\bar{\mathrm{C}}})_i + \frac{1}{n}\sum_j \mathrm{g}_j + \varepsilon \leqslant \min_{\mathrm{P}' \in \mathcal{P}_n^{\mathrm{perm}}} \frac{1}{n}\langle \mathrm{C}, \mathrm{P}'\rangle + \varepsilon,$$

which proves (5.11). Each bid scans one row of $\mathrm{C}$ and otherwise uses constant work if the current owner of every target is stored, proving the complexity bound. For integer costs, the unnormalized assignment gap is an integer bounded by $n\varepsilon < 1$, hence it vanishes. $\square$

**$\varepsilon$-scaling.**
**Proposition 5.5** (Complexity of $\varepsilon$-scaling). *For a final tolerance $\eta > 0$, Algorithm ?? returns a permutation matrix satisfying $\eta$-complementary slackness and (5.11) with $\varepsilon$ replaced by $\eta$. If $R_{\mathrm{C}} > 0$, it uses at most $1 + \lceil \log_2^+(R_{\mathrm{C}}/\eta)\rceil$ auction phases, where $\log_2^+(s) \coloneqq \max\{0, \log_2 s\}$. Its dense worst-case complexity is*

$$O\left(n^3\left(1 + \log_+ \frac{R_{\mathrm{C}}}{\eta}\right)\right), \qquad \log_+(s) \coloneqq \max\{0, \log s\}.$$

*When $R_{\mathrm{C}} = 0$, the algorithm terminates immediately. If $\mathrm{C}$ has integer entries and $\eta < 1/n$, the returned assignment is exactly optimal.*

*Proof.* Assume $R_{\mathrm{C}} > 0$. The first tolerance is $\varepsilon_0 = \max\{R_{\mathrm{C}}, \eta\}$. If $\varepsilon_0 = \eta$, Proposition 5.4 directly gives the claim with one phase. Otherwise, the first phase starts from the zero potential and costs $O(n^2)$ operations because $R_{\mathrm{C}}/\varepsilon_0 = 1$.
Consider a later phase with tolerance $\varepsilon$, initialized from a target potential produced at tolerance $\lambda$. Expressed in the present dual-potential convention, the warm-start estimate of shows that the new phase decreases each component of $\mathrm{g}$ by at most $n(\lambda+\varepsilon)$. Since every bid decreases one component by at least $\varepsilon$, the phase performs at most $n^2\left(1 + \frac{\lambda}{\varepsilon}\right)$ bids. Consecutive tolerances in Algorithm ?? satisfy $\lambda/\varepsilon \leqslant 2$, so a warm-started phase uses $O(n^2)$ bids and $O(n^3)$ dense operations.
Halving from $R_{\mathrm{C}}$ to $\eta$ gives the announced phase count. The final phase enforces $\eta$-complementary slackness, and Proposition 5.4 gives the $\eta$-optimality certificate. Its integer-cost conclusion also applies with $\varepsilon = \eta$. $\square$

### 5.3 Semi-discrete

**Discrete target and Laguerre cells.** One can adapt Definition 4.7 of the $\bar{c}$-transform to this setting by restricting the minimization to the finite support $\{y_j\}_{j=1}^m$ of $\beta$. Equivalently, one applies the same definition with the discrete target space $\mathcal{Y} = \{y_j\}_{j=1}^m$ and identifies a vector $\mathrm{g} \in \mathbb{R}^m$ with the function $g : \mathcal{Y} \to \mathbb{R}$ defined by $g(y_j) = \mathrm{g}_j$:

$$\forall\, \mathrm{g} \in \mathbb{R}^m,\ \forall\, x \in \mathcal{X}, \quad \mathrm{g}^{\bar{c}}(x) \coloneqq \min_{j \in [\![m]\!]} c(x, y_j) - \mathrm{g}_j. \tag{5.12}$$

$$\mathcal{L}_c(\alpha, \beta) = \max_{\mathrm{g} \in \mathbb{R}^m} \mathcal{E}_0(\mathrm{g}) \coloneqq \mathcal{D}_0(\mathrm{g}^{\bar{c}}, \mathrm{g}) = \int_{\mathcal{X}} \mathrm{g}^{\bar{c}}(x)\mathrm{d}\alpha(x) + \sum_j \mathrm{g}_j \mathrm{b}_j. \tag{5.13}$$

The objective satisfies $\mathcal{E}_0(\mathrm{g} + s\mathbb{1}) = \mathcal{E}_0(\mathrm{g})$ for every $s \in \mathbb{R}$. One may therefore impose the gauge $\sum_j \mathrm{g}_j = 0$ without changing the problem.

**Definition 5.6** (Laguerre cells and power diagrams). For sites $(y_j)_{j=1}^m$ and weights $\mathrm{g} \in \mathbb{R}^m$, the Laguerre cell associated with $y_j$ is

$$\mathbb{L}_j(\mathrm{g}) \coloneqq \{x \in \mathcal{X}\ ;\ \forall\, j' \neq j, c(x, y_j) - \mathrm{g}_j \leqslant c(x, y_{j'}) - \mathrm{g}_{j'}\}. \tag{5.14}$$

The cells cover $\mathcal{X}$; after arbitrary tie-breaking on common boundaries, they induce a disjoint partition. When $c(x,y) = \|x-y\|^2$, this Laguerre decomposition is also called a power diagram. If $\mathrm{g}$ is constant, it reduces to the ordinary Voronoi diagram.

**One-dimensional semi-discrete OT.** In one dimension, the Laguerre geometry of Definition 5.6 becomes explicit. Consider the quadratic cost $c(x,y) = |x-y|^2$, assume that $\alpha$ has a density and that $y_1 < \cdots < y_m$, and set $B_0 = 0$ and $B_j = \sum_{k=1}^j \mathrm{b}_k$.

$$\forall\, j \in [\![m]\!], \qquad \alpha\left(\mathbb{L}_j(\mathrm{g}^\star) \bigtriangleup I_j\right) = 0, \qquad I_j \coloneqq (q_\alpha(B_{j-1}), q_\alpha(B_j)], \tag{5.15}$$

with the harmless endpoint conventions $q_\alpha(0) = -\infty$ and $q_\alpha(1) = +\infty$. Thus the optimal semi-discrete map is $T^\star(x) = y_j$ on $I_j$; this is the Laguerre-cell realization of the monotone rearrangement in Theorem 2.31.

**Proposition 5.7** (Closed-form one-dimensional semi-discrete transport). *Let $\alpha = \rho_\alpha\, \mathrm{d}x \in \mathcal{P}_2(\mathbb{R})$, write $q_\alpha = \mathcal{C}_\alpha^{-1}$, and define its integrated quantile by $M_\alpha(r) := \int_0^r q_\alpha(s)\mathrm{d}s$. For $\beta = m^{-1}\sum_{j=1}^m \delta_{y_j}$ with $y_1 < \cdots < y_m$, one has*

$$\mathcal{W}_2(\alpha,\beta)^2 = \int_{\mathbb{R}} x^2 \mathrm{d}\alpha(x) + \frac{1}{m}\sum_{j=1}^m y_j^2 - 2\sum_{j=1}^m y_j \left[ M_\alpha\left(\frac{j}{m}\right) - M_\alpha\left(\frac{j-1}{m}\right)\right]. \tag{5.16}$$

*Proof.* Proposition 2.32 gives $\mathcal{W}_2(\alpha,\beta)^2 = \int_0^1 |q_\alpha - q_\beta|^2$. The target quantile satisfies $q_\beta(r) = y_j$ for almost every $r \in ((j-1)/m, j/m]$. Expanding the square, using $\int_0^1 q_\alpha(r)^2 \mathrm{d}r = \int_{\mathbb{R}} x^2 \mathrm{d}\alpha(x)$, and integrating $q_\alpha$ over each interval gives the stated formula. □

**Mass balance.**

$$\mathcal{E}_0(\mathrm{g}) = \sum_{j=1}^m \int_{\mathbb{L}_j(\mathrm{g})} (c(x,y_j) - \mathrm{g}_j)\, \mathrm{d}\alpha(x) + \langle \mathrm{g}, \mathrm{b}\rangle. \tag{5.17}$$

**Proposition 5.8** (Gradient of the semi-discrete dual). *If the minimizing index in (5.12) is unique for $\alpha$-almost every $x$, then $\mathcal{E}_0$ is differentiable at* $\mathrm{g}$ *and* $\forall\, j \in [\![m]\!]$, $\quad \nabla \mathcal{E}_0(\mathrm{g})_j = \mathrm{b}_j - \int_{\mathbb{L}_j(\mathrm{g})} \mathrm{d}\alpha$.

*Proof.* For $\alpha$-almost every $x$, the minimizing index in $\min_j c(x,y_j) - \mathrm{g}_j$ is unique. If this index is $j(x)$, then the directional derivative in a direction $h \in \mathbb{R}^m$ is

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} \min_j \big(c(x,y_j) - (\mathrm{g}_j + t h_j)\big) = -h_{j(x)}.$$

The corresponding difference quotients are bounded by $\|h\|_\infty$. Dominated convergence therefore gives $\mathrm{d}\mathcal{E}_0(\mathrm{g})[h] = -\sum_j h_j \int_{\mathbb{L}_j(\mathrm{g})} \mathrm{d}\alpha + \sum_j h_j \mathrm{b}_j$, which is the announced gradient formula. □

**Stochastic optimization.**

$$\mathcal{E}_0(\mathrm{g}) = \int_{\mathcal{X}} E(\mathrm{g},x)\mathrm{d}\alpha(x) = \mathbb{E}_X(E(\mathrm{g},X)), \qquad E(\mathrm{g},x) := \mathrm{g}^{\bar{c}}(x) + \langle \mathrm{g}, \mathrm{b}\rangle. \tag{5.18}$$

$$\nabla_{\mathrm{g}} E(\mathrm{g},x) = \big(\mathrm{b}_j - \mathbb{1}_{\mathbb{L}_j(\mathrm{g})}(x)\big)_{j=1}^m,$$
$$\mathrm{g}^{(\ell+1)} := \mathrm{g}^{(\ell)} + \tau^{(\ell)} \nabla_{\mathrm{g}} E(\mathrm{g}^{(\ell)}, x^{(\ell)}). \tag{5.19}$$

$j^{(\ell)} \in \mathrm{argmin}_j \big(c(x^{(\ell)}, y_j) - \mathrm{g}_j^{(\ell)}\big)$, $\mathrm{g}_j^{(\ell+1)} = \mathrm{g}_j^{(\ell)} + \tau^{(\ell)}\big(\mathrm{b}_j - \mathbb{1}_{\{j=j^{(\ell)}\}}\big)$.

$$\sum_{\ell=0}^{+\infty} \tau^{(\ell)} = +\infty, \qquad \sum_{\ell=0}^{+\infty} (\tau^{(\ell)})^2 < +\infty, \tag{5.20}$$

**Proposition 5.9** (Averaged stochastic semi-dual rate). *Let* $\mathrm{g}^\star$ *maximize $\mathcal{E}_0$, set $R = \|\mathrm{g}^{(0)} - \mathrm{g}^\star\|_2$, and suppose the stochastic supergradients are conditionally unbiased and satisfy $\|\nabla_{\mathrm{g}} E(\mathrm{g},x)\|_2 \leqslant G$. For a horizon $L \geqslant 1$, use the constant step $\tau = R/(G\sqrt{L})$ and the average $\bar{\mathrm{g}}_L = L^{-1}\sum_{\ell=0}^{L-1} \mathrm{g}^{(\ell)}$. If $R = 0$, the conclusion is immediate; otherwise, $\mathcal{E}_0(\mathrm{g}^\star) - \mathbb{E}\big[\mathcal{E}_0(\bar{\mathrm{g}}_L)\big] \leqslant \frac{RG}{\sqrt{L}}$.*

*Proof.* Conditionally on $\mathrm{g}^{(\ell)}$, the stochastic supergradient is unbiased. Concavity and the squared-distance recursion give

$$2\tau\, \mathbb{E}\Big[\mathcal{E}_0(\mathrm{g}^\star) - \mathcal{E}_0(\mathrm{g}^{(\ell)})\Big] \leqslant \mathbb{E}\|\mathrm{g}^{(\ell)} - \mathrm{g}^\star\|_2^2 - \mathbb{E}\|\mathrm{g}^{(\ell+1)} - \mathrm{g}^\star\|_2^2 + \tau^2 G^2.$$

Summing from $0$ to $L-1$, discarding the final nonnegative squared distance, and using concavity at $\bar{\mathrm{g}}_L$ yields

$$\mathcal{E}_0(\mathrm{g}^\star) - \mathbb{E}[\mathcal{E}_0(\bar{\mathrm{g}}_L)] \leqslant \frac{R^2}{2\tau L} + \frac{\tau G^2}{2}.$$

The stated choice of $\tau$ gives the result. □

### 5.4 Optimal Quantization

**Free masses and prescribed weights.**

**Definition 5.10** (Optimal quantization problem). For $\alpha \in \mathcal{P}_p(\mathbb{R}^d)$ and $m \in \mathbb{N}\setminus\{0\}$, define

$$\mathcal{Q}_m(\alpha) := \min_{Y=(y_j)_{j=1}^m,\ \mathrm{b}\in\Sigma_m} \mathcal{W}_p\Big(\alpha, \sum_{j=1}^m \mathrm{b}_j \delta_{y_j}\Big). \tag{5.21}$$

Zero weights and coincident codepoints are allowed, so the approximating measure has at most $m$ atoms.

**Proposition 5.11** (Quantization rate and curse of dimensionality). *Let $\alpha$ be supported in a bounded subset $\Omega \subset \mathbb{R}^d$ and assume that $\alpha = \rho\, \mathrm{d}x$ with $\rho \leqslant \rho_+ < +\infty$. Then, for fixed $p \geqslant 1$, there exist constants $0 < c \leqslant C < +\infty$ such that $c\, m^{-1/d} \leqslant \mathcal{Q}_m(\alpha) \leqslant C\, m^{-1/d}$.*

*Proof.* Enclose $\Omega$ in a cube. With $k = \lfloor m^{1/d}\rfloor$, subdivide this cube into $k^d \leqslant m$ congruent subcubes and place one codepoint in each subcube meeting $\Omega$. Every source point then lies within distance $C/k \leqslant C' m^{-1/d}$ of a codepoint. The nearest-codepoint coupling proves the upper bound; unused codepoints may be assigned zero mass.
For the lower bound, fix any set $Y$ of at most $m$ codepoints and write $d_Y(x) = \min_j \|x - y_j\|$. If $\omega_d$ is the volume of the Euclidean unit ball, then $\alpha(\{d_Y \leqslant t\}) \leqslant \rho_+ m \omega_d t^d$. Set $t_0 = (2\rho_+ m\omega_d)^{-1/d}$. Then $\alpha(\{d_Y > t\}) \geqslant 1/2$ for $0 < t < t_0$, and the layer-cake formula gives $\int d_Y(x)^p \mathrm{d}\alpha(x) = \int_0^{+\infty} p t^{p-1} \alpha(\{d_Y > t\})\mathrm{d}t \geqslant \frac{t_0^p}{2} \simeq c m^{-p/d}$. Taking the $p$-th root and minimizing over $Y$ proves the lower bound. □

**Lloyd algorithm.**
**Proposition 5.12** (Free masses give Voronoi cells). *For the cost $c(x,y) = d(x,y)^p$, fix distinct codepoints $Y = (y_j)_{j=1}^m$. Duplicate codepoints can be merged beforehand. Minimizing over the weights $\mathrm{b} \in \Sigma_m$ gives $\min_{\mathrm{b}\in\Sigma_m} \mathcal{W}_p^p(\alpha, \sum_j \mathrm{b}_j\delta_{y_j}) = \int_{\mathcal{X}} \min_{1\leqslant j\leqslant m} c(x,y_j)\mathrm{d}\alpha(x)$. An optimal coupling is induced by sending each $x$ to a nearest codepoint. The corresponding cells are the Voronoi cells $\mathbb{V}_j(Y) := \{x\ ;\ \forall j',\ c(x,y_j) \leqslant c(x,y_{j'})\}$, up to arbitrary tie-breaking on common boundaries.*

*Proof.* For any coupling between $\alpha$ and a measure supported on $Y$, the conditional destination of a point $x$ belongs to $Y$, hence its conditional cost is at least $\min_j c(x,y_j)$. Integrating gives the lower bound. Conversely, choose a measurable nearest-codepoint map $T_Y(x) \in \operatorname{argmin}_j c(x,y_j)$, breaking ties measurably, and set $\mathrm{b}_j = \alpha(T_Y^{-1}(y_j))$. Then $(T_Y)_\sharp\alpha = \sum_j \mathrm{b}_j\delta_{y_j}$ and the induced transport reaches the displayed lower bound. □

$\mathcal{Q}_m(\alpha)^p = \min_Y \mathcal{F}(Y), \qquad \mathcal{F}(Y) := \int_{\mathcal{X}} \min_{1\leqslant j\leqslant m} c(x,y_j)\mathrm{d}\alpha(x)$. $y_j \in \underset{y}{\operatorname{argmin}} \int_{\mathbb{V}_j(Y)} c(x,y)\mathrm{d}\alpha(x)$. $y_j = \frac{\int_{\mathbb{V}_j(Y)} x\mathrm{d}\alpha(x)}{\int_{\mathbb{V}_j(Y)} \mathrm{d}\alpha}$.

**Continuous Lloyd flow.** $a_j(Y) = \alpha(\mathbb{V}_j(Y)), \qquad b_j(Y) = \frac{1}{a_j(Y)}\int_{\mathbb{V}_j(Y)} x\,\mathrm{d}\alpha(x)$ $y_j^{(\ell+1)} = y_j^{(\ell)} + \tau\big(b_j(Y^{(\ell)}) - y_j^{(\ell)}\big), \qquad 0<\tau\leqslant 1$, $\dot{y}_j(t) = b_j(Y_t) - y_j(t)$. $\nabla_{y_j}\mathcal{F}(Y) = 2\int_{\mathbb{V}_j(Y)}(y_j - x)\mathrm{d}\alpha(x) = 2a_j(Y)(y_j - b_j(Y))$, $\dot{y}_j(t) = -\frac{1}{2a_j(Y_t)}\nabla_{y_j}\mathcal{F}(Y_t)$. $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{F}(Y_t) = -2\sum_j a_j(Y_t)\|b_j(Y_t) - y_j(t)\|^2 \leqslant 0$. If $\eta_t = \sum_j w_j\delta_{y_j(t)}$ carries fixed positive weights, independent of the Voronoi masses $a_j(Y_t)$, Proposition 13.2 turns this labelled particle ODE into the weak continuity equation
$\partial_t\eta_t + \operatorname{div}(v_t\eta_t) = 0, \qquad v_t(y_j(t)) = b_j(Y_t) - y_j(t)$, in the sense of the measure evolutions of Section 13. The weights $w_j$ in this transport equation are auxiliary weights for the moving labelled particles; they are not the Voronoi masses used to define the quantization energy.
$\nu_{Y_t} = \sum_j a_j(Y_t)\delta_{y_j(t)}$, $\partial_t\nu_{Y_t} + \operatorname{div}(v_t\nu_{Y_t}) = \sum_j \dot{a}_j(Y_t)\delta_{y_j(t)}, \qquad v_t(y_j(t)) = \dot{y}_j(t)$.
**Mean-field limit and ultrafast diffusion.** $\mathcal{G}_\rho(\sigma) := \int_\Omega \rho(x)\sigma(x)^{-r}\mathrm{d}x, \qquad \int_\Omega \sigma\,\mathrm{d}x = 1$,

$$\partial_t\sigma = -r\,\operatorname{div}\Big(\sigma\nabla\Big(\frac{\rho}{\sigma^{r+1}}\Big)\Big),$$

**Quantization with fixed equal weights.** $\nu_Y := \frac{1}{m}\sum_{j=1}^m \delta_{y_j}, \qquad \mathcal{F}_{\mathrm{eq}}(Y) := \frac{1}{2}\mathcal{W}_2^2(\alpha,\nu_Y)$, $\bar{x}_j(Y) := m\int_{C_j(Y)} x\mathrm{d}\alpha(x)$. $\nabla_{y_j}\mathcal{F}_{\mathrm{eq}}(Y) = \frac{1}{m}\big(y_j - \bar{x}_j(Y)\big)$. As long as the optimal matching between two nearby labelled configurations preserves labels, the $\mathcal{W}_2$ metric on equal-weight empirical measures induces the local particle metric $g_Y(U,V) = m^{-1}\sum_j\langle u_j, v_j\rangle$. Hence the associated $\mathcal{W}_2$ gradient flow is the coupled system
$\dot{y}_j(t) = \bar{x}_j(Y_t) - y_j(t), \qquad j = 1,\dots,m$. Equivalently, $\nu_{Y_t}$ satisfies a continuity equation with velocity $v_t(y_j(t)) = \bar{x}_j(Y_t) - y_j(t)$. This is the so-called finite-particle $\mathcal{W}_2$ gradient-flow viewpoint developed more systematically in Chapter 14; the fixed-weight cells are Laguerre cells rather than the free-mass Voronoi cells used by Lloyd's method.
**Equal-weight quantization on the line.**
**Proposition 5.13** (One-dimensional equal-weight quantization). *Let $\alpha \in \mathcal{M}_+^1(\mathbb{R})$ have finite second moment and quantile function $Q = \mathcal{C}_\alpha^{-1}$. For $\mathcal{Q}_m^{\mathrm{eq}}(\alpha)^2 := \min_{y_1\leqslant\cdots\leqslant y_m}\mathcal{W}_2^2(\alpha, \frac{1}{m}\sum_{i=1}^m\delta_{y_i})$, set $I_i = ((i-1)/m, i/m]$. Then the sorted minimizer is unique and its ith atom is $y_i^\star = m\int_{I_i} Q(u)\mathrm{d}u$, and $\mathcal{Q}_m^{\mathrm{eq}}(\alpha)^2 = \sum_{i=1}^m\int_{I_i}|Q(u) - y_i^\star|^2\,\mathrm{d}u$. If $Q\in C^1([0,1])$, then $m^2\mathcal{Q}_m^{\mathrm{eq}}(\alpha)^2 \longrightarrow \frac{1}{12}\int_0^1|Q'(u)|^2\,\mathrm{d}u$.*

*Proof.* After sorting the atoms, the quantile formula for $\mathcal{W}_2$ gives $\mathcal{W}_2^2(\alpha, \frac{1}{m}\sum_{i=1}^m\delta_{y_i}) = \sum_{i=1}^m\int_{I_i}|Q(u) - y_i|^2\,\mathrm{d}u$. The minimization therefore decouples over the intervals $I_i$, and the best constant approximation of $Q$ on $I_i$ in $L^2$ is its average on $I_i$. Since $Q$ is nondecreasing, these interval averages are nondecreasing, so they satisfy the sorting constraint. Strict convexity in each scalar $y_i$ proves uniqueness and gives the displayed energy.
Denote by $\bar{Q}_{I_i} = m\int_{I_i}Q(u)\mathrm{d}u$ the interval average. If $Q$ is $C^1$, set $h = 1/m$ and write $I_i = (a_i, a_i+h]$. Uniform Taylor expansion gives, for $v\in[0,1]$, $Q(a_i+hv) = Q(a_i) + hvQ'(a_i) + h\,r_i(v), \qquad \max_i\sup_{v\in[0,1]}|r_i(v)|\to 0$. Subtracting the average over $v\in[0,1]$ and integrating gives $\int_{I_i}\big|Q(u) - \bar{Q}_{I_i}\big|^2\,\mathrm{d}u = \frac{h^3}{12}|Q'(a_i)|^2 + o(h^3)$, uniformly in $i$. Summing over $i$ yields a Riemann sum for $\frac{1}{12}\int_0^1|Q'(u)|^2\,\mathrm{d}u$. □

Thus the common deterministic rule $m^{-1}\sum_i\delta_{Q((i-1/2)/m)}$ should be read as a midpoint approximation of the optimal bin-average formula. It has the same leading squared error under the same smoothness assumptions.
$\int_{I_i}|Q(u) - z|^2\mathrm{d}u = \int_{I_i}|Q(u) - y_i^\star|^2\mathrm{d}u + \frac{1}{m}|z - y_i^\star|^2$,
**Proposition 5.14** (Quantile-process asymptotics for random placement). *Let $\alpha\in\mathcal{M}_+^1(\mathbb{R})$ have quantile $Q\in C^1([0,1])$, and let $\widehat{\alpha}_m = m^{-1}\sum_{i=1}^m\delta_{X_i}$ with $X_i$ i.i.d. with law $\alpha$. If $B$ denotes the standard Brownian bridge on $[0,1]$, then $m\,\mathcal{W}_2^2(\alpha,\widehat{\alpha}_m) \xrightarrow{\text{law}} \int_0^1 B(u)^2\,|Q'(u)|^2\,\mathrm{d}u$, and, in expectation,*

$$m\,\mathbb{E}\big[\mathcal{W}_2^2(\alpha,\widehat{\alpha}_m)\big] \longrightarrow \int_0^1 u(1-u)\,|Q'(u)|^2\,\mathrm{d}u.$$

*Proof.* Let $\widehat{Q}_m$ be the empirical quantile function. The one-dimensional formula gives $\mathcal{W}_2^2(\alpha,\widehat{\alpha}_m) = \int_0^1\Big|\widehat{Q}_m(u) - Q(u)\Big|^2\,\mathrm{d}u$. Write $X_i = Q(U_i)$ with $U_i$ i.i.d. uniform on $[0,1]$, and let $\widehat{U}_m^{-1}$ be the empirical quantile function of the $U_i$. Then $\widehat{Q}_m = Q\circ\widehat{U}_m^{-1}$. The classical uniform quantile-process theorem gives $\sqrt{m}\,(\widehat{U}_m^{-1} - \mathrm{Id}) \xrightarrow{\text{law}} -B$ in $L^2(0,1)$. Because $Q'$ is uniformly continuous and $\widehat{U}_m^{-1}\to\mathrm{Id}$ uniformly in probability, the functional delta method gives $\sqrt{m}\,(\widehat{Q}_m - Q) \xrightarrow{\text{law}} -B\,Q'$ in $L^2(0,1)$. Since $B$ and $-B$ have the same law, the sign disappears in the squared $L^2$ norm. The continuous mapping theorem gives the distributional convergence. Standard fourth-moment bounds for the uniform quantile process, together with the boundedness of $Q'$, give uniform integrability of the squared $L^2$ norms and hence convergence of expectations. Since $\mathbb{E}[B(u)^2] = u(1-u)$, Fubini's theorem gives the displayed expectation limit. □

### 5.5 Wasserstein-1 norm

**$c$-transform for $\mathcal{W}_1$.** Assume that $d$ is a distance on $\mathcal{X} = \mathcal{Y}$ and take the ground cost $c(x, y) = d(x, y)$. We denote the Lipschitz constant of $f \in \mathcal{C}(\mathcal{X})$ by

**Definition 5.15** (Lipschitz constant)**.** For a function $f : \mathcal{X} \to \mathbb{R}$ on a metric space $(\mathcal{X}, d)$, its Lipschitz constant is

$$\mathrm{Lip}(f) := \sup \left\{ \frac{|f(x) - f(y)|}{d(x, y)} \; ; \; x \neq y \right\}. \tag{5.22}$$

The function is 1-Lipschitz when $\mathrm{Lip}(f) \leqslant 1$.

**Proposition 5.16** ($c$-transforms and 1-Lipschitz functions)**.** *Suppose $\mathcal{X} = \mathcal{Y}$ and $c(x, y) = d(x, y)$. Then there exists $g$ such that $f = g^c$ if and only if $\mathrm{Lip}(f) \leqslant 1$. Furthermore, if $\mathrm{Lip}(f) \leqslant 1$, then $f^c = -f$.*

*Proof.* First suppose $f = g^c$ for some $g$. For $x, y \in \mathcal{X}$,

$$|f(x) - f(y)| = \left| \inf_z [d(x, z) - g(z)] - \inf_z [d(y, z) - g(z)] \right| \leqslant \sup_z |d(x, z) - d(y, z)| \leqslant d(x, y),$$

where the last inequality is the reverse triangle inequality. Thus $\mathrm{Lip}(f) \leqslant 1$.
If $\mathrm{Lip}(f) \leqslant 1$, then $f(x) \leqslant f(y) + d(x, y)$, so $d(x, y) - f(x) \geqslant -f(y)$ for all $x$ and hence $f^c(y) \geqslant -f(y)$. Taking $x = y$ gives $f^c(y) \leqslant -f(y)$. Therefore $f^c = -f$. Applying the same property to $-f$ gives $(-f)^c = f$, so every 1-Lipschitz function is $c$-concave. $\square$

$$\mathcal{W}_1(\alpha, \beta) = \max_f \left\{ \int_{\mathcal{X}} f \mathrm{d}(\alpha - \beta) \; ; \; \mathrm{Lip}(f) \leqslant 1 \right\} =: \|\alpha - \beta\|_{W_1}. \tag{5.23}$$

For a discrete signed measure $\alpha - \beta = \sum_{k=1}^N r_k \delta_{z_k}$ on distinct sites, with $\sum_k r_k = 0$, (5.23) becomes the finite-dimensional linear program

$$\mathcal{W}_1(\alpha, \beta) = \max_{(f_k)_k} \left\{ \sum_k f_k r_k \; ; \; \forall\, k, \ell, \; |f_k - f_\ell| \leqslant d(z_k, z_\ell) \right\}. \tag{5.24}$$

This linear program can be solved by generic interior-point or first-order methods. If $N$ support points are involved, however, it still contains $O(N^2)$ Lipschitz constraints, mirroring the $O(nm)$ coupling variables of the original discrete Kantorovich LP; the dual formulation alone does not remove the all-pairs structure.
$\mathcal{W}_1(\alpha, \beta) = \max_{(f_k)_k} \left\{ \sum_k f_k r_k \; ; \; \forall\, k, \; |f_{k+1} - f_k| \leqslant z_{k+1} - z_k \right\}.$

**$\mathcal{W}_1$ on Euclidean spaces.** Let $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ have finite first moments and set $\xi = \alpha - \beta$. Rademacher's theorem identifies 1-Lipschitz functions with locally Sobolev functions whose Euclidean gradient has norm at most one almost everywhere. Hence

$$\mathcal{W}_1(\alpha, \beta) = \sup_{f \in W^{1,\infty}_{\mathrm{loc}}(\mathbb{R}^d)} \left\{ \int_{\mathbb{R}^d} f \mathrm{d}\xi \; ; \; \|\nabla f\|_{L^\infty(\mathbb{R}^d)} \leqslant 1 \right\}. \tag{5.25}$$

**Definition 5.17** (Total variation norm of a vector-valued measure)**.** Let $\mathcal{M}(\mathbb{R}^d; \mathbb{R}^d)$ denote the finite $\mathbb{R}^d$-valued Radon measures. For $s \in \mathcal{M}(\mathbb{R}^d; \mathbb{R}^d)$, define $\|s\|_{\mathrm{TV}} := \sup_{\zeta \in C_c(\mathbb{R}^d; \mathbb{R}^d)} \left\{ \int_{\mathbb{R}^d} \langle \zeta, \mathrm{d}s \rangle \; ; \; \sup_{x \in \mathbb{R}^d} \|\zeta(x)\|_2 \leqslant 1 \right\}.$

For $s \in \mathcal{M}(\mathbb{R}^d; \mathbb{R}^d)$, its divergence is defined distributionally by
$\langle \operatorname{div} s, \varphi \rangle := -\int_{\mathbb{R}^d} \langle \nabla \varphi, \mathrm{d}s \rangle, \qquad \varphi \in C_c^1(\mathbb{R}^d).$

**Definition 5.18** (Beckmann problem)**.** For $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$, define $\mathrm{Beck}(\alpha, \beta) := \inf_{s \in \mathcal{M}(\mathbb{R}^d; \mathbb{R}^d)} \{ \|s\|_{\mathrm{TV}} \; ; \; \operatorname{div} s = \alpha - \beta \}.$

**Proposition 5.19** (Euclidean Beckmann formula)**.** *For $\alpha, \beta \in \mathcal{M}_+^1(\mathbb{R}^d)$ with finite first moments,*

$$\mathcal{W}_1(\alpha, \beta) = \mathrm{Beck}(\alpha, \beta). \tag{5.26}$$

*Proof.* Let $s$ be feasible. For a smooth 1-Lipschitz test function, $\int f \mathrm{d}(\alpha - \beta) = -\int \langle \nabla f, \mathrm{d}s \rangle \leqslant \|s\|_{\mathrm{TV}}$. Truncation followed by mollification approximates normalized Lipschitz functions by smooth test functions without increasing the gradient bound asymptotically. Equation (5.25) therefore gives $\mathcal{W}_1(\alpha, \beta) \leqslant \|s\|_{\mathrm{TV}}$.
Conversely, let $\pi$ be an optimal coupling and define a vector measure by its action on $\zeta \in C_c(\mathbb{R}^d; \mathbb{R}^d)$: $\int \langle \zeta(z), \mathrm{d}s(z) \rangle := \int_{\mathbb{R}^d \times \mathbb{R}^d} \int_0^1 \langle \zeta((1-t)x + ty), y - x \rangle \mathrm{d}t \mathrm{d}\pi(x, y)$. For $\varphi \in C_c^1(\mathbb{R}^d)$, the fundamental theorem of calculus gives $\langle \operatorname{div} s, \varphi \rangle = -\int \int_0^1 \langle \nabla \varphi((1-t)x + ty), y - x \rangle \mathrm{d}t \mathrm{d}\pi(x, y) = \int \varphi \mathrm{d}(\alpha - \beta)$. Moreover, $\|s\|_{\mathrm{TV}} \leqslant \int \|x - y\|_2 \mathrm{d}\pi(x, y) = \mathcal{W}_1(\alpha, \beta)$. Together with the first inequality, this proves equality and attainment. $\square$

**Graph distances and Beckmann flows.**

**Definition 5.20** (Graph geodesic distance)**.** Let $G = (V, E)$ be a connected finite graph with positive edge lengths $(\ell_e)_{e \in E}$. The graph geodesic distance between two vertices is $d_G(i, j) = \min_{\gamma : i \rightsquigarrow j} \sum_{e \in \gamma} \ell_e$. The minimum is over all paths $\gamma$ joining $i$ to $j$.

**Proposition 5.21** ($\mathcal{W}_1$ and Beckmann flow on a graph)**.** *Let $G = (V, E)$ be a connected finite graph with positive edge lengths $(\ell_e)_{e \in E}$ and graph geodesic distance $d_G$. For two probability vectors* $\mathrm{a}, \mathrm{b}$ *on $V$, set $r = \mathrm{a} - \mathrm{b}$ and orient each edge $e = (i, j)$. If*

$(\nabla_G f)_e = f_j - f_i, \qquad \mathrm{div}_G = -\nabla_G^*$ *are the finite-difference gradient and its negative adjoint, then a positive flow on the oriented edge $i \to j$ has positive divergence at $i$ and negative divergence at $j$. With this convention,*

$$\mathcal{W}_{1,G}(\mathrm{a}, \mathrm{b}) = \max_f \left\{ \sum_{i \in V} f_i r_i \; ; \; |f_i - f_j| \leqslant \ell_e \quad \forall e = (i,j) \right\} = \min_m \left\{ \sum_{e \in E} \ell_e |m_e| \; ; \; \mathrm{div}_G\, m = r \right\}.$$

*The vector $m_e$ is an oriented edge flow, and the constraint $\mathrm{div}_G\, m = r$ is conservation of mass at each vertex.*

*Proof.* The edge constraints imply, by summing along every path and then minimizing over paths, that $|f_i - f_j| \leqslant d_G(i,j)$ for all vertices. Conversely, any 1-Lipschitz function for $d_G$ satisfies $|f_i - f_j| \leqslant d_G(i,j) \leqslant \ell_e$ on every edge. The first equality is therefore the Kantorovich–Rubinstein formula on the metric space $(V, d_G)$.
For the second equality, write the graph Beckmann problem and dualize its equality constraint with a potential $f$: $\inf_m \sum_e \ell_e |m_e| + \sup_f \sum_i f_i (r_i - (\mathrm{div}_G\, m)_i)$. Using $\mathrm{div}_G = -\nabla_G^*$, the coupling term is $\sum_e m_e (\nabla_G f)_e$. The minimization over each scalar flow $m_e$ is finite exactly when $|(\nabla_G f)_e| \leqslant \ell_e$, and is then equal to zero. The dual problem is therefore precisely the graph Lipschitz dual above. Strong duality holds because this is a finite-dimensional linear program with a nonempty feasible set: connectedness and $\sum_i r_i = 0$ allow the signed surplus to be routed along paths. This proves the graph Beckmann formula. □

# 6 Divergences and Dual Norms

## 6.1 Dual norms (Integral Probability Metrics)

**Integral probability metrics.**

**Definition 6.1** (Dual seminorm and integral probability metric)**.** Let $B$ be a nonempty symmetric convex class of measurable functions that are integrable against the measures under consideration. For a signed measure $\xi$, define the extended dual seminorm

$$\|\xi\|_B \coloneqq \sup_f \left\{ \int_{\mathcal{X}} f(x) \mathrm{d}\xi(x) \; ; \; f \in B \right\}. \tag{6.1}$$

It is a norm when it is finite and $B$ separates signed measures. When applied to $\alpha - \beta$ for probability measures, it is called an integral probability metric (IPM); it is a genuine metric precisely when $B$ separates probability measures.

**Proposition 6.2** (Metrization by dual norms)**.** *Assume that $\mathcal{X}$ is compact, that $B = -B$, and that the measures considered are probability measures.*
*1. If every function in $\mathcal{C}(\mathcal{X})$ can be uniformly approximated by elements of* $\mathrm{Span}(B)$*, then $\|\alpha_n - \alpha\|_B \to 0$ implies $\alpha_n \rightharpoonup \alpha$.*
*2. If $B \subset \mathcal{C}(\mathcal{X})$ is compact for $\|\cdot\|_\infty$, then $\alpha_n \rightharpoonup \alpha$ implies $\|\alpha_n - \alpha\|_B \to 0$.*

*Proof.* For the first implication, $\|\alpha_n - \alpha\|_B \to 0$ and the symmetry of $B$ imply $|\langle f, \alpha_n - \alpha \rangle| \leqslant \|\alpha_n - \alpha\|_B$ for $f \in B$. If $h = \sum_{j=1}^J c_j f_j \in \mathrm{Span}(B)$, then $|\langle h, \alpha_n - \alpha \rangle| \leqslant (\sum_{j=1}^J |c_j|) \|\alpha_n - \alpha\|_B$, so integrals converge for every $h \in \mathrm{Span}(B)$. Let $u \in \mathcal{C}(\mathcal{X})$ and choose such an $h$ with $\|u - h\|_\infty \leqslant \eta$. Since $\alpha_n$ and $\alpha$ are probabilities, $|\langle u, \alpha_n - \alpha \rangle| \leqslant |\langle h, \alpha_n - \alpha \rangle| + 2\eta$. Taking the limsup as $n \to \infty$ and then letting $\eta \to 0$ gives $\langle u, \alpha_n \rangle \to \langle u, \alpha \rangle$ for all $u \in \mathcal{C}(\mathcal{X})$, which is weak convergence.
For the second implication, assume $\alpha_n \rightharpoonup \alpha$ and choose a subsequence $(\alpha_{n_k})_k$ such that $\|\alpha_{n_k} - \alpha\|_B \to \limsup_n \|\alpha_n - \alpha\|_B$. Since $B$ is compact and the map $f \mapsto \langle f, \alpha_{n_k} - \alpha \rangle$ is continuous on $B$, the supremum is attained by some $f_{n_k} \in B$. Extracting a further subsequence if needed, $f_{n_k} \to f$ uniformly for some $f \in B$. Then $\langle f_{n_k}, \alpha_{n_k} - \alpha \rangle = \langle f, \alpha_{n_k} - \alpha \rangle + \langle f_{n_k} - f, \alpha_{n_k} \rangle - \langle f_{n_k} - f, \alpha \rangle$. The first term tends to zero by weak convergence and the last two by uniform convergence. Hence every limsup subsequence has limit zero, proving $\|\alpha_n - \alpha\|_B \to 0$. □

**Corollary 6.3** (Wasserstein metrizes weak convergence)**.** *On a compact metric space, $\mathcal{W}_p$ metrizes weak convergence on probability measures for every $p \geqslant 1$.*

*Proof.* For $p = 1$, take $B = \{f \; ; \; \mathrm{Lip}(f) \leqslant 1\}$. The span of $B$ contains all Lipschitz functions, and Lipschitz functions are dense in $\mathcal{C}(\mathcal{X})$ on compact metric spaces.
Conversely, constants do not change the pairing with $\alpha_n - \alpha$. Fix $x_0 \in \mathcal{X}$ and normalize potentials by $f(x_0) = 0$. The normalized unit Lipschitz ball is uniformly bounded by $\mathrm{diam}(\mathcal{X})$ and equicontinuous, hence compact in $\|\cdot\|_\infty$ by Arzelà–Ascoli. Proposition 6.2 gives $\mathcal{W}_1(\alpha_n, \alpha) \to 0$. Proposition **??** shows that all $\mathcal{W}_p$ distances induce the same topology on a compact space, so the result follows for every $p \geqslant 1$. □

## 6.2 Dual RKHS Norms and Maximum Mean Discrepancies

**Definition 6.4** (Positive and conditionally definite kernels)**.** A symmetric function $k : \mathcal{X} \times \mathcal{X} \to \mathbb{R}$ is positive definite if for every $n \geqslant 1$, every $x_1, \ldots, x_n \in \mathcal{X}$ and every $r \in \mathbb{R}^n$,

$$\sum_{i,j=1}^n r_i r_j k(x_i, x_j) \geqslant 0. \tag{6.2}$$

It is conditionally positive definite of order one if the same inequality is required only for zero-sum vectors, $\langle r, \mathbb{1}_n \rangle = 0$. It is conditionally negative definite of order one if the reverse inequality holds for every such vector, or equivalently if $-k$ is conditionally positive definite.

The conditional notions are the right ones for probability distances, because one applies quadratic forms to signed measures $\xi = \alpha - \beta$ of total mass zero. Adding a term of the form $a(x) + a(y)$ to a kernel does not change $\iint k(x,y) \mathrm{d}\xi(x) \mathrm{d}\xi(y)$ on such measures.

**Definition 6.5** (Kernel seminorm and MMD)**.** Let $k$ be positive definite. More generally, let $k$ be conditionally positive definite and restrict attention to signed measures of total mass zero. For a signed measure $\xi$ with finite kernel energy, the associated seminorm is

$$\|\xi\|_k^2 \coloneqq \iint_{\mathcal{X} \times \mathcal{X}} k(x,y) \mathrm{d}\xi(x) \mathrm{d}\xi(y). \tag{6.3}$$

For two probability measures, the maximum mean discrepancy associated with $k$ is $\mathrm{MMD}_k(\alpha,\beta) := \|\alpha-\beta\|_k$. It is a genuine distance when the kernel is characteristic, meaning that $\|\alpha-\beta\|_k = 0$ implies $\alpha=\beta$.

$\widetilde{k}(x,y) = k(x,y) - k(x,x_0) - k(x_0,y) + k(x_0,x_0)$

**Proposition 6.6** (Kernel seminorm as an RKHS dual norm)**.** *Let $\mathcal{H}$ be the RKHS with reproducing kernel $k$, and assume that the kernel mean embedding $m_\xi := \int k(x,\cdot)\mathrm{d}\xi(x)$ is well-defined for the signed measure $\xi$. Then $\|\xi\|_k = \sup_{\|h\|_{\mathcal{H}}\leqslant 1}\int h(x)\mathrm{d}\xi(x)$, so $\|\cdot\|_k$ is the dual seminorm in the sense of (6.1) associated with the RKHS unit ball.*

*Proof.* By the reproducing property, $\int h(x)\mathrm{d}\xi(x) = \langle h, \int k(x,\cdot)\mathrm{d}\xi(x)\rangle_{\mathcal{H}} = \langle h, m_\xi\rangle_{\mathcal{H}}$. Cauchy–Schwarz gives $\sup_{\|h\|_{\mathcal{H}}\leqslant 1}\int h\mathrm{d}\xi = \|m_\xi\|_{\mathcal{H}}$. Finally, $\|m_\xi\|_{\mathcal{H}}^2 = \iint k(x,y)\mathrm{d}\xi(x)\mathrm{d}\xi(y)$, which is exactly (6.3). □

**Proposition 6.7** (Universal kernels metrize weak convergence)**.** *Assume that $\mathcal{X}$ is compact and that the RKHS generated by the continuous kernel $k$ is dense in $\mathcal{C}(\mathcal{X})$ for the uniform norm. Then*

$$\mathrm{MMD}_k(\alpha_n,\alpha)\to 0 \quad\Longleftrightarrow\quad \alpha_n \rightharpoonup \alpha$$

*for probability measures on $\mathcal{X}$.*

*Proof.* If $\mathrm{MMD}_k(\alpha_n,\alpha)\to 0$, then for every $g\in\mathcal{H}$, $\left|\int g\mathrm{d}(\alpha_n-\alpha)\right| \leqslant \|g\|_{\mathcal{H}}\,\mathrm{MMD}_k(\alpha_n,\alpha)\to 0$. For any $h\in\mathcal{C}(\mathcal{X})$ and any $\eta>0$, choose $g\in\mathcal{H}$ with $\|h-g\|_\infty\leqslant\eta$. Since $\alpha_n$ and $\alpha$ are probabilities,

$$\left|\int h\,\mathrm{d}(\alpha_n-\alpha)\right| \leqslant 2\eta + \left|\int g\,\mathrm{d}(\alpha_n-\alpha)\right|,$$

and the last term tends to zero. This proves weak convergence. Conversely, if $\alpha_n\rightharpoonup\alpha$, then $\alpha_n\otimes\alpha_n$, $\alpha_n\otimes\alpha$ and $\alpha\otimes\alpha$ converge weakly on the compact product space. Applying this to the continuous bounded function $k$ in the identity $\mathrm{MMD}_k(\alpha_n,\alpha)^2 = \iint k\,\mathrm{d}\alpha_n\mathrm{d}\alpha_n - 2\iint k\,\mathrm{d}\alpha_n\mathrm{d}\alpha + \iint k\,\mathrm{d}\alpha\mathrm{d}\alpha$ gives convergence to zero. □

**Definition 6.8** (Universal kernel)**.** A continuous positive-definite kernel $k$ on a compact space $\mathcal{X}$ is universal if finite sums of sections $\sum_{i=1}^n a_i k(x_i,\cdot)$ are dense in $\mathcal{C}(\mathcal{X})$ for the uniform norm.

In the special case where $\alpha$ is a discrete measure, one thus has the simple expression

$$\|\alpha\|_k^2 = \sum_{i=1}^n\sum_{i'=1}^n \mathrm{a}_i\mathrm{a}_{i'}\mathrm{k}_{i,i'} = \langle \mathrm{ka},\,\mathrm{a}\rangle \quad\text{where}\quad \mathrm{k}_{i,i'} := k(x_i,x_{i'}).$$

$$\|\alpha-\beta\|_k^2 = \sum_{i,i'}\mathrm{a}_i\mathrm{a}_{i'}k(x_i,x_{i'}) + \sum_{j,j'}\mathrm{b}_j\mathrm{b}_{j'}k(y_j,y_{j'}) - 2\sum_{i,j}\mathrm{a}_i\mathrm{b}_j k(x_i,y_j). \tag{6.4}$$

### 6.3 $\varphi$-divergences

**Definition by density ratios.**

**Definition 6.9** (Entropy function)**.** A function $\varphi:\mathbb{R}\to\mathbb{R}\cup\{+\infty\}$ is an entropy function if it is proper, lower semicontinuous and convex, with $\operatorname{dom}\varphi\subset[0,+\infty)$ and $\operatorname{dom}\varphi\cap(0,+\infty)\neq\emptyset$. Its recession slope is

$$\varphi'_\infty = \lim_{s\to+\infty}\frac{\varphi(s)}{s}\in\mathbb{R}\cup\{+\infty\}.$$

If $\varphi'_\infty=\infty$, then $\varphi$ grows faster than any linear function and $\varphi$ is said to be *superlinear*.

**Definition 6.10** ($\varphi$-Divergences)**.** Let $\varphi$ be an entropy function and let $\alpha,\beta\in\mathcal{M}_+(\mathcal{X})$. Write the Lebesgue decomposition of $\alpha$ with respect to $\beta$ as $\alpha = (\mathrm{d}\alpha/\mathrm{d}\beta)\beta + \alpha^\perp$, where $\alpha^\perp\perp\beta$. The associated divergence is

$$\mathcal{D}_\varphi(\alpha|\beta) := \int_{\mathcal{X}}\varphi\left(\frac{\mathrm{d}\alpha}{\mathrm{d}\beta}\right)\mathrm{d}\beta + \varphi'_\infty\alpha^\perp(\mathcal{X}), \tag{6.5}$$

with the convention $0\cdot(+\infty)=0$. It is extended by $+\infty$ whenever either argument is not a nonnegative measure.

$$\alpha = \sum_i \mathrm{a}_i\delta_{x_i} \quad\text{and}\quad \beta = \sum_i \mathrm{b}_i\delta_{x_i} \tag{6.6}$$

$$\mathrm{D}_\varphi(\mathrm{a}|\mathrm{b}) = \sum_{i\in\mathrm{Supp}(\mathrm{b})}\varphi\left(\frac{\mathrm{a}_i}{\mathrm{b}_i}\right)\mathrm{b}_i + \varphi'_\infty\sum_{i\notin\mathrm{Supp}(\mathrm{b})}\mathrm{a}_i, \tag{6.7}$$

where $\mathrm{Supp}(\mathrm{b}) := \{i\in[\![n]\!]\;;\;\mathrm{b}_i\neq 0\}$.

**Proposition 6.11** (Basic properties of $\varphi$-divergences)**.** *If $\varphi$ is an entropy function, then $\mathcal{D}_\varphi$ is jointly 1-homogeneous, convex and weak-* lower semicontinuous in $(\alpha,\beta)$.*

*Proof.* Introduce the lower-semicontinuous perspective

$$\forall\,(u,v)\in(\mathbb{R}_+)^2,\quad \psi(u,v) = \begin{cases} v\varphi(u/v) & \text{if}\quad v>0\\ u\varphi'_\infty & \text{if}\quad v=0.\end{cases}$$

The joint 1-homogeneity follows directly from this formula. To see convexity, first take $v_0,v_1>0$ and $\tau\in[0,1]$. With $\bar v = (1-\tau)v_0+\tau v_1$, $\theta_0 = (1-\tau)v_0/\bar v$, and $\theta_1 = \tau v_1/\bar v$, one has $\theta_0+\theta_1=1$ and $((1-\tau)u_0+\tau u_1)/\bar v = \theta_0 u_0/v_0 + \theta_1 u_1/v_1$. Jensen's inequality for $\varphi$, followed by multiplication by $\bar v$, proves convexity of $\psi$ when both second arguments are positive. Its recession value at $v = 0$ gives the lower-semicontinuous convex extension to the closed quadrant. In the discrete case, $\mathrm{D}_\varphi(\mathrm{a}|\mathrm{b}) = \sum_i \psi(\mathrm{a}_i, \mathrm{b}_i)$, so homogeneity and convexity follow by summation. For general finite measures, representing both arguments with respect to a common dominating measure yields the integral of $\psi$; the standard lower-semicontinuity theorem for convex integral functionals gives weak-* lower semicontinuity, with the recession value accounting for singular mass. □

**Proposition 6.12** (Non-negativity of $\varphi$-divergences)**.** *Assume that $\varphi$ is normalized by $\varphi(1) = 0$. For probability distributions $(\alpha, \beta) \in \mathcal{M}_+^1(\mathcal{X})$, one has $\mathcal{D}_\varphi(\alpha|\beta) \geqslant 0$. If $\varphi$ is strictly convex, then one has $\mathcal{D}_\varphi(\alpha|\beta) = 0$ if and only if $\alpha = \beta$.*

*Proof.* Let $m = \alpha + \beta$ and write $a = \mathrm{d}\alpha/\mathrm{d}m$ and $b = \mathrm{d}\beta/\mathrm{d}m$. Then $a + b = 1$ $m$-almost everywhere and, using the perspective $\psi$ above, $\mathcal{D}_\varphi(\alpha|\beta) = \int \psi(a, b)\mathrm{d}m$. Since $m(\mathcal{X}) = 2$, the measure $m/2$ is a probability. Jensen's inequality, the 1-homogeneity of $\psi$, and $\varphi(1) = 0$ give $\frac{1}{2}\mathcal{D}_\varphi(\alpha|\beta) \geqslant \psi\left(\frac{1}{2}\int a\mathrm{d}m, \frac{1}{2}\int b\mathrm{d}m\right) = \psi(1/2, 1/2) = 0$. If $\varphi$ is strictly convex, the map $s \mapsto \psi(s, 1-s)$ is strictly convex on its effective domain: distinct points on the line $u + v = 1$ cannot lie on the same ray, which is the only degeneracy of a strict perspective. Equality in Jensen therefore forces $a = 1/2$, hence $a = b$ and $\alpha = \beta$. For arbitrary finite nonnegative measures, $\varphi \geqslant 0$ also implies $\varphi'_\infty \geqslant 0$, so every term in (6.5) is nonnegative. □

**Classical examples and topology.**

**Theorem 6.13** (Pinsker inequality)**.** *Let $\alpha, \beta \in \mathcal{M}_+^1(\mathcal{X})$ be probability measures on a measurable space. With the full-variation convention (??), $\mathrm{TV}(\alpha|\beta)^2 = \|\alpha - \beta\|_{\mathrm{TV}}^2 \leqslant 2\,\mathrm{KL}(\alpha|\beta)$. In particular, on a finite space, for $\mathrm{a}, \mathrm{b} \in \Sigma_n$, $\|\mathrm{a} - \mathrm{b}\|_{\ell^1}^2 \leqslant 2\mathrm{KL}(\mathrm{a}|\mathrm{b})$.*

*Proof.* The claim is immediate if $\mathrm{KL}(\alpha|\beta) = +\infty$, so assume $\alpha \ll \beta$. Let $A$ be a positive Hahn set for the signed measure $\alpha - \beta$, and set $a = \alpha(A)$ and $b = \beta(A)$. Since $(\alpha - \beta)(\mathcal{X}) = 0$, $a - b = \frac{1}{2}\|\alpha - \beta\|_{\mathrm{TV}}$. Applying Jensen's inequality separately on $A$ and $A^c$, or equivalently coarse-graining onto the partition $(A, A^c)$, gives $\mathrm{KL}(\alpha|\beta) \geqslant a \log \frac{a}{b} + (1-a) \log \frac{1-a}{1-b}$. For fixed $b \in (0, 1)$, subtract $2(a - b)^2$ from the binary relative entropy on the right. The resulting function of $a$ has value and first derivative zero at $a = b$, while its second derivative is $\frac{1}{a} + \frac{1}{1-a} - 4 \geqslant 0$. Hence the binary relative entropy is at least $2(a - b)^2 = \frac{1}{2}\|\alpha - \beta\|_{\mathrm{TV}}^2$. The boundary cases follow by lower semicontinuity. For a finite space, the identity $\|\alpha - \beta\|_{\mathrm{TV}} = \|\mathrm{a} - \mathrm{b}\|_{\ell^1}$ gives the second formula. □

**Main families of $\varphi$-divergences.** $\varphi_\gamma(s) = \frac{s^\gamma - \gamma s + \gamma - 1}{\gamma(\gamma-1)}$ $\quad (\gamma \neq 0, 1)$ $\mathfrak{h}(\alpha, \beta) := \mathcal{D}_{\varphi_H}(\alpha|\beta)^{1/2} = \|\sqrt{\rho_\alpha} - \sqrt{\rho_\beta}\|_{L^2(\lambda)}$.

$$\mathrm{JS}(\alpha, \beta)^2 = \frac{1}{2}\,\mathrm{KL}\left(\alpha \middle| \frac{\alpha + \beta}{2}\right) + \frac{1}{2}\,\mathrm{KL}\left(\beta \middle| \frac{\alpha + \beta}{2}\right),$$

and satisfies $0 \leqslant \mathrm{JS}(\alpha, \beta)^2 \leqslant \log 2$. Its exact entropy generator is

$$\varphi_{\mathrm{JS}}(s) = \frac{1}{2}\left[s \log s - (s+1) \log\left(\frac{s+1}{2}\right)\right], \qquad \mathcal{D}_{\varphi_{\mathrm{JS}}}(\alpha|\beta) = \mathrm{JS}(\alpha, \beta)^2.$$

**Variational dual formula.**

**Proposition 6.14** (Variational representation of a $\varphi$-divergence)**.** *Assume that $\mathcal{X}$ is compact and let $\alpha, \beta \in \mathcal{M}_+(\mathcal{X})$. Define the Legendre transform of the entropy, already extended by $+\infty$ on negative arguments, by $\varphi^*(s) := \sup_{r \geqslant 0}\{sr - \varphi(r)\}$.*

*Then*
$$\mathcal{D}_\varphi(\alpha|\beta) = \sup_{f \in \mathcal{C}(\mathcal{X})} \left\{\int_{\mathcal{X}} f \mathrm{d}\alpha - \int_{\mathcal{X}} \varphi^*(f)\mathrm{d}\beta\right\}. \tag{6.8}$$

*Equivalently, the convex conjugate of $\mathcal{D}_\varphi(\cdot|\beta)$ on $\mathcal{M}(\mathcal{X})$ is*

$$\forall\, f \in \mathcal{C}(\mathcal{X}), \quad \mathcal{D}_\varphi^*(f|\beta) = \int_{\mathcal{X}} \varphi^*(f(x))\mathrm{d}\beta(x). \tag{6.9}$$

*Proof.* First suppose that $\varphi'_\infty = +\infty$. Finite divergence then forces $\alpha = \rho\beta$ for some $\rho \geqslant 0$, and pointwise convex duality gives

$$\begin{aligned}\mathcal{D}_\varphi^*(f|\beta) &= \sup_{\rho \geqslant 0} \int_{\mathcal{X}} \big(f(x)\rho(x) - \varphi(\rho(x))\big)\mathrm{d}\beta(x) \\ &= \int_{\mathcal{X}} \sup_{r \geqslant 0}\big(f(x)r - \varphi(r)\big)\mathrm{d}\beta(x) = \int_{\mathcal{X}} \varphi^*(f(x))\mathrm{d}\beta(x).\end{aligned}$$

For finite $\varphi'_\infty$, the effective domain of $\varphi^*$ is bounded above by $\varphi'_\infty$. Hence a singular mass contributes at most $\varphi'_\infty \alpha^\perp(\mathcal{X})$ to the pairing, exactly matching the recession term in (6.5); approximation by continuous test functions recovers this value. The same conjugate formula therefore holds in general. Proposition 6.11 gives convexity and weak-* lower semicontinuity, so Fenchel–Moreau yields (6.8). □

### 6.4 GANs via Duality

**Divergence-based adversarial losses.**

$$\min_\theta \mathcal{D}_\varphi(\alpha_\theta|\beta) = \min_\theta \sup_f \int_{\mathcal{X}} f(x)\mathrm{d}\alpha_\theta(x) - \mathcal{D}_\varphi^*(f|\beta) = \min_\theta \sup_f \int_{\mathcal{X}} f(g_\theta(z))\mathrm{d}\zeta(z) - \frac{1}{m}\sum_{j=1}^m \varphi^*(f(y_j)).$$

$\min_\theta \max_\xi \int_{\mathcal{Z}} f_\xi(g_\theta(z))\mathrm{d}\zeta(z) - \frac{1}{m}\sum_{j=1}^m \varphi^*(f_\xi(y_j))$. For fixed $\theta$, this restriction gives a lower bound on the exact divergence unless the discriminator class is rich enough to attain the supremum. This distinction is essential for empirical data: if $\beta$ is discrete and $\alpha_\theta$ is non-atomic, a superlinear divergence is $+\infty$, whereas its neural restricted objective can remain finite.
$\widehat{\varphi}_{\mathrm{JS}}(s) = s \log s - (s+1) \log \frac{s+1}{2}$, $\qquad \widehat{\varphi}_{\mathrm{JS}}^*(u) = -\log(2 - e^u)$, $\quad u < \log 2$. Thus $\mathcal{D}_{\widehat{\varphi}_{\mathrm{JS}}} = 2\,\mathrm{JS}^2$, consistently with the normalization above. In practice the min–max problem is solved by alternating stochastic gradient descent/ascent on $(\theta, \xi)$.

**Dual norms and integral probability metrics.**

$$\min_{\theta} \|\alpha_\theta - \beta\|_B = \min_{\theta} \sup_{f\in B} \int_{\mathcal{X}} f(x)\mathrm{d}(\alpha_\theta - \beta)(x) = \min_{\theta} \sup_{f\in B} \int_{\mathcal{Z}} f(g_\theta(z))\mathrm{d}\zeta - \frac{1}{m}\sum_{j=1}^{m} f(y_j).$$

# 7 Entropic Regularization: Sinkhorn Algorithm

## 7.1 Entropic Regularization for Discrete Measures

**Entropy penalty.**

**Definition 7.1** (Discrete Shannon–Boltzmann entropy)**.** For a nonnegative matrix $\mathrm{P}$, its Shannon–Boltzmann entropy is $\mathrm{H}(\mathrm{P}) \coloneqq -\sum_{i,j} \mathrm{P}_{i,j}\log(\mathrm{P}_{i,j})$, with the convention $0\log(0) = 0$.

**Definition 7.2** (Discrete entropic optimal transport)**.** For $\varepsilon > 0$, define

$$\mathrm{L}^{\varepsilon}_{\mathrm{C}}(\mathrm{a},\mathrm{b}) \coloneqq \min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle \mathrm{P},\, \mathrm{C}\rangle - \varepsilon\mathrm{H}(\mathrm{P}). \tag{7.1}$$

**Proposition 7.3** (Existence and uniqueness of entropic OT)**.** *Assume that* $\mathrm{a},\mathrm{b}$ *are probability histograms and that* $\mathrm{C}$ *is finite. For every* $\varepsilon > 0$*, problem* (7.1) *admits a unique minimizer. If all entries of* $\mathrm{a}$ *and* $\mathrm{b}$ *are positive, then this minimizer is positive on every entry.*

*Proof.* The transport polytope $\mathrm{U}(\mathrm{a},\mathrm{b})$ is non-empty and compact, and the objective is continuous on it with the convention $0\log 0 = 0$, so a minimizer exists. The scalar function $r \mapsto r\log r$ is strictly convex on $[0,+\infty)$. On the relative interior, the Hessian $-\partial^2\mathrm{H}(\mathrm{P}) = \mathrm{diag}(1/\mathrm{P}_{i,j})$ is positive definite. Hence $-\mathrm{H}$ is strictly convex on every nontrivial feasible segment, and $\langle \mathrm{P},\, \mathrm{C}\rangle - \varepsilon\mathrm{H}(\mathrm{P})$ has at most one minimizer.
If $\mathrm{a}_i, \mathrm{b}_j > 0$ and a minimizer had $\mathrm{P}_{i,j} = 0$, then for small $t > 0$ the perturbation $\mathrm{P}_t = (1-t)\mathrm{P} + t\,\mathrm{a}\otimes\mathrm{b}$ remains feasible. The directional derivative of the entropic part at $t = 0$ is $-\infty$ because the derivative of $r\log r$ at $0$ is $-\infty$ along a positive direction. Thus the objective decreases for small $t$, contradicting optimality. Therefore the minimizer is strictly positive. □

**Smoothing effect.**

## 7.2 Sinkhorn's Algorithm

**Proposition 7.4** (Scaling form of entropic OT)**.** $\mathrm{P}$ *is the unique solution to* (7.1) *if and only if there exists* $(\mathrm{u},\mathrm{v}) \in \mathbb{R}^n_+ \times \mathbb{R}^m_+$ *such that*

$$\forall\,(i,j) \in [\![n]\!] \times [\![m]\!], \quad \mathrm{P}_{i,j} = \mathrm{u}_i\mathrm{K}_{i,j}\mathrm{v}_j \quad \textit{where} \quad \mathrm{K}_{i,j} \coloneqq e^{-\frac{\mathrm{C}_{i,j}}{\varepsilon}}, \tag{7.2}$$

*and* $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$.

*Proof.* Without loss of generality, we assume $\mathrm{a}_i, \mathrm{b}_j > 0$ (otherwise, rows or columns with zero mass are fixed to zero and can be removed). By Proposition 7.3, the minimizer is strictly positive on the remaining support.
We can thus ignore the positivity constraint when introducing two dual variables $\mathrm{f} \in \mathbb{R}^n, \mathrm{g} \in \mathbb{R}^m$ for each marginal constraint so that the Lagrangian of (7.1) reads

$$\mathcal{E}(\mathrm{P},\mathrm{f},\mathrm{g}) = \langle \mathrm{P},\, \mathrm{C}\rangle + \varepsilon\sum_{i,j} \mathrm{P}_{i,j}\log(\mathrm{P}_{i,j}) + \langle \mathrm{f},\, \mathrm{a} - \mathrm{P}\mathbb{1}_m\rangle + \langle \mathrm{g},\, \mathrm{b} - \mathrm{P}^\top\mathbb{1}_n\rangle.$$

The first-order conditions are
$\frac{\partial\mathcal{E}(\mathrm{P},\mathrm{f},\mathrm{g})}{\partial\mathrm{P}_{i,j}} = \mathrm{C}_{i,j} + \varepsilon(\log(\mathrm{P}_{i,j}) + 1) - \mathrm{f}_i - \mathrm{g}_j = 0$. Thus $\mathrm{P}_{i,j} = e^{(\mathrm{f}_i + \mathrm{g}_j - \mathrm{C}_{i,j})/\varepsilon - 1}$, which has the stated form with the positive scalings $\mathrm{u}_i \coloneqq e^{\mathrm{f}_i/\varepsilon - 1}$ and $\mathrm{v}_j \coloneqq e^{\mathrm{g}_j/\varepsilon}$. □

$$\mathrm{diag}(\mathrm{u})\mathrm{K}\,\mathrm{diag}(\mathrm{v})\mathbb{1}_m = \mathrm{a}, \quad \text{and} \quad \mathrm{diag}(\mathrm{v})\mathrm{K}^\top\,\mathrm{diag}(\mathrm{u})\mathbb{1}_n = \mathrm{b}, \tag{7.3}$$

$$\mathrm{u}\odot(\mathrm{K}\mathrm{v}) = \mathrm{a} \quad \text{and} \quad \mathrm{v}\odot(\mathrm{K}^\top\mathrm{u}) = \mathrm{b} \tag{7.4}$$

where $\odot$ denotes entrywise multiplication.
When $n = m$ and $\mathrm{a},\mathrm{b}$ are uniform, this is diagonal scaling toward bistochasticity. Proposition 7.3 guarantees a unique scaled matrix $\mathrm{P}$, although the two scaling vectors retain one multiplicative gauge degree of freedom. If some entries of $\mathrm{K}$ vanish, equivalently if $\mathrm{C}$ can take the value $+\infty$, existence and uniqueness require additional support conditions; here we keep $\mathrm{K} > 0$.

$$\mathrm{u}^{(\ell+1)} \coloneqq \frac{\mathrm{a}}{\mathrm{K}\mathrm{v}^{(\ell)}} \quad \text{and} \quad \mathrm{v}^{(\ell+1)} \coloneqq \frac{\mathrm{b}}{\mathrm{K}^\top\mathrm{u}^{(\ell+1)}}, \tag{7.5}$$

## 7.3 Reformulation Using Relative Entropy

**Relative entropy.**

**Definition 7.5** (Discrete relative entropy)**.** For nonnegative matrices $\mathrm{P},\mathrm{Q}$ of the same size, the generalized relative entropy, or Kullback–Leibler divergence, is

$$\mathrm{KL}(\mathrm{P}|\mathrm{Q}) \coloneqq \sum_{i,j} \mathrm{P}_{i,j}\log\left(\frac{\mathrm{P}_{i,j}}{\mathrm{Q}_{i,j}}\right) - \mathrm{P}_{i,j} + \mathrm{Q}_{i,j}. \tag{7.6}$$

The convention is $0\log(0) = 0$, and $\mathrm{KL}(\mathrm{P}|\mathrm{Q}) = +\infty$ if there exists $(i,j)$ such that $\mathrm{Q}_{i,j} = 0$ but $\mathrm{P}_{i,j} \neq 0$.

For the specific case of comparing probability distributions, where $\mathrm{P}$ and $\mathrm{Q}$ have the same total mass, this further simplifies to $\mathrm{KL}(\mathrm{P}|\mathrm{Q}) = \sum_{i,j}\mathrm{P}_{i,j}\log\left(\frac{\mathrm{P}_{i,j}}{\mathrm{Q}_{i,j}}\right)$. For the reference matrix $\mathrm{Q} = \mathbb{1}_{n\times m}$, one has
$-\mathrm{KL}(\mathrm{P}|\mathbb{1}_{n\times m}) = \mathrm{H}(\mathrm{P}) + \sum_{i,j}\mathrm{P}_{i,j} - n\,m$. On fixed-mass couplings the last two terms are constant, so KL regularization with reference $\mathbb{1}_{n\times m}$ is equivalent to subtracting Shannon–Boltzmann entropy. KL is a particular instance of both a $\varphi$-divergence (as defined in Section 6.3) and a Bregman divergence; up to standard affine rescalings, it is the canonical overlap between these two families.

**Proposition 7.6** (Non-negativity and definiteness of relative entropy)**.** *For all* $\mathrm{P}, \mathrm{Q} \in \mathbb{R}_+^{n\times m}$*, one has* $\mathrm{KL}(\mathrm{P}|\mathrm{Q}) \geqslant 0$*, with equality if and only if* $\mathrm{P} = \mathrm{Q}$*.*

*Proof.* Write $\varphi(s) = s\log s - s + 1$. Convexity gives $\varphi(s) \geqslant \varphi(1) + \varphi'(1)(s-1) = 0$, and strict convexity gives equality only at $s = 1$. Hence $\mathrm{KL}(\mathrm{P}|\mathrm{Q}) = \sum_{(i,j):\, \mathrm{Q}_{i,j}>0} \mathrm{Q}_{i,j}\varphi(\mathrm{P}_{i,j}/\mathrm{Q}_{i,j}) \geqslant 0$, whenever $\mathrm{P} \ll \mathrm{Q}$ entrywise; otherwise the divergence is $+\infty$. Equality forces $\mathrm{P}_{i,j} = \mathrm{Q}_{i,j}$ wherever $\mathrm{Q}_{i,j} > 0$, while entries with $\mathrm{Q}_{i,j} = 0$ vanish in both matrices. Thus equality is equivalent to $\mathrm{P} = \mathrm{Q}$. □

When P and Q are probability matrices and $\mathrm{P} \ll \mathrm{Q}$, the same divergence can also be written
$\mathrm{KL}(\mathrm{P}|\mathrm{Q}) = \sum_{i,j} \psi(\mathrm{P}_{i,j}/\mathrm{Q}_{i,j})\mathrm{Q}_{i,j}, \qquad \psi(s) = s\log s.$

$$\sum_{i,j} \psi(\mathrm{P}_{i,j}/\mathrm{Q}_{i,j})\mathrm{Q}_{i,j} \geqslant \psi\left(\sum_{i,j} \mathrm{P}_{i,j}\right) = \psi(1) = 0.$$

**KL reformulation of regularized OT.**

$$\min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle \mathrm{P},\, \mathrm{C}\rangle + \varepsilon\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}). \tag{7.7}$$

For every $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$, the two objectives satisfy
$\langle \mathrm{P},\, \mathrm{C}\rangle + \varepsilon\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) = \langle \mathrm{P},\, \mathrm{C}\rangle - \varepsilon\mathrm{H}(\mathrm{P}) + \varepsilon\big(\mathrm{H}(\mathrm{a}) + \mathrm{H}(\mathrm{b})\big)$. Thus (7.7) has exactly the same minimizer as the original entropic problem (7.1), while its optimal value is $\mathrm{L}_\mathrm{C}^\varepsilon(\mathrm{a},\mathrm{b}) + \varepsilon\big(\mathrm{H}(\mathrm{a}) + \mathrm{H}(\mathrm{b})\big)$. This normalization will matter again for unbalanced OT, where changing the reference measure is no longer merely a harmless notational choice.

**Proposition 7.7** (Reference measure shift for KL)**.** *After removing zero-mass rows and columns, assume that* $\mathrm{a}, \mathrm{a}' \in \Sigma_n$ *and* $\mathrm{b}, \mathrm{b}' \in \Sigma_m$ *have positive entries. For every* $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$*, one has* $\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) = \mathrm{KL}(\mathrm{P}|\mathrm{a}'\otimes\mathrm{b}') - \mathrm{KL}(\mathrm{a}|\mathrm{a}') - \mathrm{KL}(\mathrm{b}|\mathrm{b}')$*. In particular,* (7.7) *and* (7.1) *have the same unique minimizer.*

*Proof.* Expanding the logarithm and using the marginal constraints gives

$$\begin{aligned}\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) &= \mathrm{KL}(\mathrm{P}|\mathrm{a}'\otimes\mathrm{b}') + \sum_i \mathrm{a}_i \log\frac{\mathrm{a}_i'}{\mathrm{a}_i} + \sum_j \mathrm{b}_j\log\frac{\mathrm{b}_j'}{\mathrm{b}_j} \\ &= \mathrm{KL}(\mathrm{P}|\mathrm{a}'\otimes\mathrm{b}') - \mathrm{KL}(\mathrm{a}|\mathrm{a}') - \mathrm{KL}(\mathrm{b}|\mathrm{b}').\end{aligned}$$

□

The tensor-product reference is nevertheless useful when supports vary, because it makes explicit which entries are allowed to vanish. It is also the normalization that passes cleanly to the continuous formulation below, where the reference measure is $\alpha\otimes\beta$ rather than an ambient Lebesgue measure.

**Proposition 7.8** (Convergence with $\varepsilon$)**.** *Assume, after removing zero-mass rows and columns, that* a *and* b *are positive and that* C *is finite. The unique solution* $\mathrm{P}_\varepsilon$ *of* (7.1) *converges to the optimal solution with maximal entropy within the set of all optimal solutions of the Kantorovich problem, namely*

$$\mathrm{P}_\varepsilon \overset{\varepsilon\to 0}{\longrightarrow} \underset{\mathrm{P}}{\operatorname{argmin}}\ \{-\mathrm{H}(\mathrm{P})\ ;\ \mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b}), \langle \mathrm{P},\, \mathrm{C}\rangle = \mathrm{L}_\mathrm{C}(\mathrm{a},\mathrm{b})\} \tag{7.8}$$

*so that in particular*

$$\mathrm{L}_\mathrm{C}^\varepsilon(\mathrm{a},\mathrm{b}) \overset{\varepsilon\to 0}{\longrightarrow} \mathrm{L}_\mathrm{C}(\mathrm{a},\mathrm{b}).$$

*Moreover,*

$$\mathrm{P}_\varepsilon \overset{\varepsilon\to\infty}{\longrightarrow} \mathrm{a}\otimes\mathrm{b}. \tag{7.9}$$

*Proof.* **Case $\varepsilon \to 0$.** We denote $\mathrm{P}_\ell$ the solution of (7.1) for $\varepsilon = \varepsilon_\ell$.
Since $\mathrm{U}(\mathrm{a},\mathrm{b})$ is bounded, we can extract a sequence (that we do not relabel for the sake of simplicity) such that $\mathrm{P}_\ell \to \mathrm{P}^\star$. Since $\mathrm{U}(\mathrm{a},\mathrm{b})$ is closed, $\mathrm{P}^\star \in \mathrm{U}(\mathrm{a},\mathrm{b})$. Using the equivalent KL-normalized formulation (7.7), optimality of P and $\mathrm{P}_\ell$ for their respective optimization problems (for $\varepsilon = 0$ and $\varepsilon = \varepsilon_\ell$) gives

$$0 \leqslant \langle \mathrm{C},\, \mathrm{P}_\ell\rangle - \langle \mathrm{C},\, \mathrm{P}\rangle \leqslant \varepsilon_\ell(\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) - \mathrm{KL}(\mathrm{P}_\ell|\mathrm{a}\otimes\mathrm{b})). \tag{7.10}$$

Since KL is continuous, taking the limit $\ell \to +\infty$ in this expression shows that $\langle \mathrm{C},\, \mathrm{P}^\star\rangle = \langle \mathrm{C},\, \mathrm{P}\rangle$ so that $\mathrm{P}^\star$ is a feasible point of (7.8). Furthermore, dividing by $\varepsilon_\ell$ in (7.10) and taking the limit shows that $\mathrm{KL}(\mathrm{P}^\star|\mathrm{a}\otimes\mathrm{b}) \leqslant \mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b})$, which shows that $\mathrm{P}^\star$ is a solution of (7.8). Since the solution $\mathrm{P}_0^\star$ to this program is unique by strict convexity of $\mathrm{KL}(\cdot|\mathrm{a}\otimes\mathrm{b})$ on the optimal face, one has $\mathrm{P}^\star = \mathrm{P}_0^\star$, and the whole sequence is converging.
**Case $\varepsilon \to +\infty$.** Subtracting $\min_{i,j} \mathrm{C}_{i,j}$ from the cost changes every feasible objective by the same constant, so it does not change the minimizer. We can therefore assume $\mathrm{C} \geqslant 0$. Evaluating the energy at $\mathrm{a}\otimes\mathrm{b}$ (which belongs to the constraint set $\mathrm{U}(\mathrm{a},\mathrm{b})$), one has $\langle \mathrm{C},\, \mathrm{P}_\varepsilon\rangle + \varepsilon\mathrm{KL}(\mathrm{P}_\varepsilon|\mathrm{a}\otimes\mathrm{b}) \leqslant \langle \mathrm{C},\, \mathrm{a}\otimes\mathrm{b}\rangle + \varepsilon\times 0$ and since $\langle \mathrm{C},\, \mathrm{P}_\varepsilon\rangle \geqslant 0$, this leads to $\mathrm{KL}(\mathrm{P}_\varepsilon|\mathrm{a}\otimes\mathrm{b}) \leqslant \varepsilon^{-1}\langle \mathrm{C},\, \mathrm{a}\otimes\mathrm{b}\rangle \leqslant \frac{\|\mathrm{C}\|_\infty}{\varepsilon}$ so that $\mathrm{KL}(\mathrm{P}_\varepsilon|\mathrm{a}\otimes\mathrm{b}) \to 0$ and thus $\mathrm{P}_\varepsilon \to \mathrm{a}\otimes\mathrm{b}$ since KL is a valid divergence. □

### 7.4 General Formulation

**Measure formulation.**

**Definition 7.9** (Relative entropy of measures)**.** For finite nonnegative measures $\pi$ and $\xi$ on $\mathcal{X}\times\mathcal{Y}$, the relative entropy is

$$\mathrm{KL}(\pi|\xi) := \int_{\mathcal{X}\times\mathcal{Y}} \log\left(\frac{\mathrm{d}\pi}{\mathrm{d}\xi}\right)\mathrm{d}\pi + \xi(\mathcal{X}\times\mathcal{Y}) - \pi(\mathcal{X}\times\mathcal{Y}). \tag{7.11}$$

By convention, $\mathrm{KL}(\pi|\xi) = +\infty$ if $\pi$ is not absolutely continuous with respect to $\xi$.

**Definition 7.10** (Continuous entropic optimal transport)**.** For $\varepsilon > 0$, define

$$\mathcal{L}_c^\varepsilon(\alpha,\beta) := \min_{\pi\in\mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y) + \varepsilon\,\mathrm{KL}(\pi|\alpha\otimes\beta). \tag{7.12}$$

For fixed balanced marginals, the specific product reference in the entropy only matters up to additive constants, exactly as in Proposition 7.7, provided the alternative reference marginals are mutually absolutely continuous with $\alpha$ and $\beta$. Its support and absolute-continuity structure are nevertheless essential: they determine which couplings have finite entropy.

**Probabilistic interpretation.**

**Definition 7.11** (Mutual information)**.** If $(X,Y)\sim\pi$ have marginals $X\sim\alpha$ and $Y\sim\beta$, the mutual information of the pair is $\mathcal{I}(X,Y) := \mathrm{KL}(\pi|\alpha\otimes\beta)$. It is nonnegative and vanishes if and only if $X$ and $Y$ are independent.

$\inf_{X\sim\alpha,\,Y\sim\beta} \mathbb{E}\big(c(X,Y)\big) + \varepsilon\mathcal{I}(X,Y)$.

**Sinkhorn for general measures.**

$$\begin{aligned} k_\varepsilon(x,y) &:= \exp\left(-\frac{c(x,y)}{\varepsilon}\right), \\ (\mathcal{K}_\varepsilon v)(x) &:= \int_{\mathcal{Y}} k_\varepsilon(x,y)v(y)\mathrm{d}\beta(y), \quad (\mathcal{K}_\varepsilon^* u)(y) := \int_{\mathcal{X}} k_\varepsilon(x,y)u(x)\mathrm{d}\alpha(x). \end{aligned} \tag{7.13}$$

$$\frac{\mathrm{d}\pi_\varepsilon}{\mathrm{d}(\alpha\otimes\beta)}(x,y) = u_\varepsilon(x)k_\varepsilon(x,y)v_\varepsilon(y), \tag{7.14}$$

$u_\varepsilon\,\mathcal{K}_\varepsilon v_\varepsilon = 1 \quad \alpha\text{-a.e.}, \qquad v_\varepsilon\,\mathcal{K}_\varepsilon^* u_\varepsilon = 1 \quad \beta\text{-a.e.}$

$$u^{(\ell+1)} = \frac{1}{\mathcal{K}_\varepsilon v^{(\ell)}}, \qquad v^{(\ell+1)} = \frac{1}{\mathcal{K}_\varepsilon^* u^{(\ell+1)}}. \tag{7.15}$$

$$u^{(\ell+1)}(x) = \frac{1}{\displaystyle\int_{\mathcal{Y}} k_\varepsilon(x,y)v^{(\ell)}(y)\mathrm{d}\beta(y)}, \qquad v^{(\ell+1)}(y) = \frac{1}{\displaystyle\int_{\mathcal{X}} k_\varepsilon(x,y)u^{(\ell+1)}(x)\mathrm{d}\alpha(x)}.$$

Given $v^{(\ell)}$, the first update produces an intermediate coupling with $\mathcal{X}$-marginal $\alpha$; the second produces one with $\mathcal{Y}$-marginal $\beta$, while generally perturbing the first marginal again. As in the discrete case, $(u,v)$ is defined only up to the gauge $(u,v)\mapsto(\lambda u, v/\lambda)$.

**Proposition 7.12** (Closed-form Gibbs coupling)**.** *Let $\beta_0$ be a sigma-finite reference measure on $\mathcal{Y}$ and suppose that $Z_\varepsilon(x) := \int_{\mathcal{Y}} k_\varepsilon(x,y)\mathrm{d}\beta_0(y) \in (0,+\infty)$ for $\alpha$-a.e. $x$. Define the normalized Gibbs transition and its output density by*

$$p_\varepsilon(x,y) := \frac{k_\varepsilon(x,y)}{Z_\varepsilon(x)}, \qquad q_\varepsilon(y) := \int_{\mathcal{X}} p_\varepsilon(x,y)\mathrm{d}\alpha(x), \qquad \mathrm{d}\beta_\varepsilon = q_\varepsilon\,\mathrm{d}\beta_0. \tag{7.16}$$

*Assume that $\log Z_\varepsilon \in L^1(\alpha)$, $\log q_\varepsilon \in L^1(\beta_\varepsilon)$, and the plan below has finite entropic objective. Then the unique solution of (7.12) between $\alpha$ and $\beta_\varepsilon$ is*

$$\mathrm{d}\pi_\varepsilon(x,y) = p_\varepsilon(x,y)\mathrm{d}\alpha(x)\mathrm{d}\beta_0(y). \tag{7.17}$$

*Relative to $\alpha\otimes\beta_\varepsilon$, its Sinkhorn scalings are explicitly*

$$\frac{\mathrm{d}\pi_\varepsilon}{\mathrm{d}(\alpha\otimes\beta_\varepsilon)}(x,y) = \frac{k_\varepsilon(x,y)}{Z_\varepsilon(x)q_\varepsilon(y)} = u_\varepsilon(x)k_\varepsilon(x,y)v_\varepsilon(y), \qquad u_\varepsilon = \frac{1}{Z_\varepsilon}, \quad v_\varepsilon = \frac{1}{q_\varepsilon}.$$

*Proof.* By construction, $p_\varepsilon(x,\cdot)\mathrm{d}\beta_0$ is a probability measure. Fubini's theorem therefore shows that (7.17) has first marginal $\alpha$ and second marginal $\beta_\varepsilon$.

Let $\gamma\in\mathcal{U}(\alpha,\beta_\varepsilon)$ have finite objective. Since $-\varepsilon\log k_\varepsilon = c$, the displayed density of $\pi_\varepsilon$ gives

$$\varepsilon\,\mathrm{KL}(\gamma|\pi_\varepsilon) = \int c\,\mathrm{d}\gamma + \varepsilon\,\mathrm{KL}(\gamma|\alpha\otimes\beta_\varepsilon) + \varepsilon\int \log Z_\varepsilon\,\mathrm{d}\alpha + \varepsilon\int \log q_\varepsilon\,\mathrm{d}\beta_\varepsilon.$$

The last two terms depend only on the prescribed marginals. Thus minimizing the entropic objective is equivalent to minimizing $\mathrm{KL}(\gamma|\pi_\varepsilon)$, whose unique minimizer is $\gamma = \pi_\varepsilon$. □

If $k_\varepsilon(x,\cdot)$ is already normalized with respect to $\beta_0$, then $Z_\varepsilon = 1$ and (7.16) reduces to the particularly simple formula $\frac{\mathrm{d}\beta_\varepsilon}{\mathrm{d}\beta_0}(y) = \int_{\mathcal{X}} k_\varepsilon(x,y)\mathrm{d}\alpha(x)$.

$$p_\varepsilon(x,y) = (\pi\varepsilon)^{-d/2}e^{-\|x-y\|^2/\varepsilon}, \qquad \beta_\varepsilon = \alpha * \mathcal{N}\left(0,\frac{\varepsilon}{2}\mathrm{Id}\right).$$

**Convergence with $\varepsilon$.**

**Proposition 7.13** (Convergence with $\varepsilon$ for measures)**.** *Let $\mathcal{X},\mathcal{Y}$ be compact metric spaces, let $c\in\mathcal{C}(\mathcal{X}\times\mathcal{Y})$, and let $\alpha\in\mathcal{P}(\mathcal{X})$, $\beta\in\mathcal{P}(\mathcal{Y})$. Assume that finite-entropy couplings are dense in $\mathcal{U}(\alpha,\beta)$ for weak convergence with convergence of the cost integral. For $\varepsilon>0$, let $\pi_\varepsilon$ be a minimizer of (7.12). Then $\mathcal{L}_c^\varepsilon(\alpha,\beta)\longrightarrow\mathcal{L}_c(\alpha,\beta)$ as $\varepsilon\downarrow 0$, and every weak cluster point of $(\pi_\varepsilon)_\varepsilon$ is an optimal plan for the unregularized Kantorovich problem. If the latter optimal plan is unique, then $\pi_\varepsilon$ converges weakly to it. In particular, for $\mathcal{X}=\mathcal{Y}\subset\mathbb{R}^d$, $c(x,y)=\|x-y\|^2$ and $\alpha$ absolutely continuous, $\pi_\varepsilon\rightharpoonup(\mathrm{Id},T)_\sharp\alpha$, where $T$ is the Brenier map. As $\varepsilon\to+\infty$,*

$$\pi_\varepsilon \longrightarrow \alpha\otimes\beta \quad \textit{in total variation}, \qquad \mathcal{L}_c^\varepsilon(\alpha,\beta)\longrightarrow\int_{\mathcal{X}\times\mathcal{Y}} c\,\mathrm{d}\alpha\mathrm{d}\beta.$$

*Proof.* Existence follows from compactness of $\mathcal{U}(\alpha,\beta)$ and lower semicontinuity of the relative entropy. For the limit $\varepsilon \downarrow 0$, take any sequence $\varepsilon_k \downarrow 0$ and a weak cluster point $\pi^\star$ of $\pi_{\varepsilon_k}$. Since $\mathrm{KL}(\pi|\alpha\otimes\beta) \geqslant 0$ on probability couplings, $\liminf_k \mathcal{L}_c^{\varepsilon_k}(\alpha,\beta) \geqslant \liminf_k \int c\,\mathrm{d}\pi_{\varepsilon_k} = \int c\,\mathrm{d}\pi^\star \geqslant \mathcal{L}_c(\alpha,\beta)$. Conversely, by the density assumption, for every $\delta > 0$ there exists a coupling $\pi^\delta$ with finite relative entropy and $\int c\,\mathrm{d}\pi^\delta \leqslant \mathcal{L}_c(\alpha,\beta)+\delta$. Testing (7.12) with $\pi^\delta$ gives $\limsup_k \mathcal{L}_c^{\varepsilon_k}(\alpha,\beta) \leqslant \int c\,\mathrm{d}\pi^\delta$. Letting $\delta \downarrow 0$ proves convergence of the values and forces $\int c\,\mathrm{d}\pi^\star = \mathcal{L}_c(\alpha,\beta)$. Uniqueness of the unregularized optimizer then identifies all cluster points. Brenier's theorem gives the stated quadratic consequence when $\alpha$ is absolutely continuous.

For the limit $\varepsilon \to +\infty$, subtract a constant from $c$ and assume $c \geqslant 0$. Since $\alpha\otimes\beta$ is admissible and has zero relative entropy with respect to itself, $\int c\,\mathrm{d}\pi_\varepsilon + \varepsilon\,\mathrm{KL}(\pi_\varepsilon|\alpha\otimes\beta) \leqslant \int c\,\mathrm{d}\alpha\mathrm{d}\beta$. Hence $\mathrm{KL}(\pi_\varepsilon|\alpha\otimes\beta) \leqslant \varepsilon^{-1}\int c\,\mathrm{d}\alpha\mathrm{d}\beta$. Pinsker's inequality from Theorem 6.13 gives total-variation convergence to $\alpha\otimes\beta$, and the convergence of the values follows by continuity of $c$. ☐

**Large-temperature expansion.** When $\varepsilon$ is large, the entropy dominates and the optimal plan is a small perturbation of the product coupling. The useful quantity is the component of the cost that cannot be absorbed into row and column potentials.

**Proposition 7.14** (Large-temperature expansion)**.** *Let $r = \alpha\otimes\beta$ and assume $c \in L^\infty(r)$. Define the product averages $\bar c_{\mathcal{X}}(x) := \int_{\mathcal{Y}} c(x,y)\,\mathrm{d}\beta(y)$, $\bar c_{\mathcal{Y}}(y) := \int_{\mathcal{X}} c(x,y)\,\mathrm{d}\alpha(x)$, $\bar c := \int_{\mathcal{X}\times\mathcal{Y}} c\,\mathrm{d}r$, and the centered interaction $c_0(x,y) := c(x,y) - \bar c_{\mathcal{X}}(x) - \bar c_{\mathcal{Y}}(y) + \bar c$. Assume that the large-temperature branch $p_\varepsilon = \mathrm{d}\pi_\varepsilon/\mathrm{d}r$, $f_\varepsilon, g_\varepsilon$ admits an expansion to third order in $\varepsilon^{-1}$ near $0$ in $L^\infty(r)$, with gauge $\int g_\varepsilon\,\mathrm{d}\beta = 0$. Then $p_\varepsilon(x,y) = 1 - \frac{c_0(x,y)}{\varepsilon} + O(\varepsilon^{-2})$ in $L^2(r)$, and*

$$\mathcal{L}_c^\varepsilon(\alpha,\beta) = \bar c - \frac{1}{2\varepsilon}\int_{\mathcal{X}\times\mathcal{Y}} c_0(x,y)^2\,\mathrm{d}r(x,y) + \frac{1}{6\varepsilon^2}\int_{\mathcal{X}\times\mathcal{Y}} c_0(x,y)^3\,\mathrm{d}r(x,y) + O(\varepsilon^{-3}).$$

*Moreover, with $A(x) := \int_{\mathcal{Y}} c_0(x,y)^2\,\mathrm{d}\beta(y)$, $B(y) := \int_{\mathcal{X}} c_0(x,y)^2\,\mathrm{d}\alpha(x)$, $\sigma^2 := \int_{\mathcal{X}\times\mathcal{Y}} c_0^2\,\mathrm{d}r$, one has $f_\varepsilon(x) = \bar c_{\mathcal{X}}(x) - \frac{A(x)}{2\varepsilon} + O(\varepsilon^{-2})$, and $g_\varepsilon(y) = \bar c_{\mathcal{Y}}(y) - \bar c + \frac{\sigma^2 - B(y)}{2\varepsilon} + O(\varepsilon^{-2})$ in the same gauge.*

*Proof.* The optimality equation can be written $p_\varepsilon(x,y) = \exp\left(\frac{f_\varepsilon(x)+g_\varepsilon(y)-c(x,y)}{\varepsilon}\right)$, with the constraints $\int p_\varepsilon(x,y)\,\mathrm{d}\beta(y) = 1$ and $\int p_\varepsilon(x,y)\,\mathrm{d}\alpha(x) = 1$. Write $p_\varepsilon = 1 + \varepsilon^{-1}q + O(\varepsilon^{-2})$. The linearized marginal constraints impose $\int q(x,y)\,\mathrm{d}\beta(y) = 0$, $\int q(x,y)\,\mathrm{d}\alpha(x) = 0$. Expanding the objective gives

$$\int c\,p_\varepsilon\,\mathrm{d}r + \varepsilon\int p_\varepsilon\log(p_\varepsilon)\,\mathrm{d}r = \bar c + \frac{1}{\varepsilon}\left(\int c\,q\,\mathrm{d}r + \frac{1}{2}\int q^2\,\mathrm{d}r\right) + O(\varepsilon^{-2}).$$

Thus $q$ is the minimizer of the quadratic expression over zero-row and zero-column perturbations. Orthogonal projection in $L^2(r)$ gives $q = -c_0$, because $c_0$ has zero conditional means. This proves the first-order density and value expansions.

To identify the next displayed terms, write $p_\varepsilon = 1+\varepsilon^{-1}q+\varepsilon^{-2}s+O(\varepsilon^{-3})$ and $f_\varepsilon = f_0+\varepsilon^{-1}f_1+O(\varepsilon^{-2})$, $g_\varepsilon = g_0+\varepsilon^{-1}g_1+O(\varepsilon^{-2})$, where $f_0 = \bar c_{\mathcal{X}}$ and $g_0 = \bar c_{\mathcal{Y}} - \bar c$. Since $p_\varepsilon = \exp\left(-\frac{c_0}{\varepsilon} + \frac{f_1(x)+g_1(y)}{\varepsilon^2} + O(\varepsilon^{-3})\right)$, one has $s = f_1 + g_1 + c_0^2/2$. The second-order marginal constraints and the gauge $\int g_1\,\mathrm{d}\beta = 0$ give $f_1 = -A/2$ and $g_1 = (\sigma^2 - B)/2$. Finally, $p_\varepsilon\log p_\varepsilon = \varepsilon^{-1}q + \varepsilon^{-2}\left(s + \frac{1}{2}q^2\right) + \varepsilon^{-3}\left(qs - \frac{1}{6}q^3\right) + O(\varepsilon^{-4})$, and the zero-conditional-mean property of $c_0$ cancels all additive terms in $s$. The order-$\varepsilon^{-2}$ contribution to the value is therefore $\frac{1}{2}\int c_0^3\,\mathrm{d}r - \frac{1}{3}\int c_0^3\,\mathrm{d}r = \frac{1}{6}\int c_0^3\,\mathrm{d}r$. ☐

**Small-temperature expansion for smooth densities.**

**Proposition 7.15** (Small-temperature quadratic expansion)**.** *Let $\alpha = \rho_\alpha\,\mathrm{d}x$ and $\beta = \rho_\beta\,\mathrm{d}x$ be probability measures on $\mathbb{R}^d$ with bounded compactly supported densities. Let $\alpha_t = \rho_t\,\mathrm{d}x$ be their quadratic displacement interpolation and assume $\mathcal{I}_{\mathrm{geo}}(\alpha,\beta) := \int_0^1\int_{\mathbb{R}^d}\|\nabla\log\rho_t(x)\|^2\rho_t(x)\,\mathrm{d}x\,\mathrm{d}t < +\infty$. For $c(x,y) = \|x-y\|^2$, define*

$$\mathrm{H}(\alpha) := \int_{\mathbb{R}^d}\rho_\alpha\log\rho_\alpha\,\mathrm{d}x, \qquad \mathrm{H}(\beta) := \int_{\mathbb{R}^d}\rho_\beta\log\rho_\beta\,\mathrm{d}x.$$

*Then, as $\varepsilon \downarrow 0$,*

$$\mathcal{L}_{\|\cdot\|^2}^\varepsilon(\alpha,\beta) = \mathcal{W}_2^2(\alpha,\beta) - \frac{d\varepsilon}{2}\log(\pi\varepsilon) - \frac{\varepsilon}{2}\left(\mathrm{H}(\alpha)+\mathrm{H}(\beta)\right) + \frac{\varepsilon^2}{16}\mathcal{I}_{\mathrm{geo}}(\alpha,\beta) + o(\varepsilon^2).$$

*If, in addition, the endpoint densities are smooth and positive on their supports and the quadratic optimal map is a smooth non-degenerate diffeomorphism, let $f_\varepsilon, g_\varepsilon$ be optimal Sinkhorn potentials and $f_0, g_0$ Kantorovich potentials, normalized by $\int g_\varepsilon\,\mathrm{d}\beta = \int g_0\,\mathrm{d}\beta = 0$. Then $f_\varepsilon \to f_0$ and $g_\varepsilon \to g_0$ locally uniformly on the interiors of the supports. Moreover, the scalar part of the first potential is fixed by*

$$\int f_\varepsilon\,\mathrm{d}\alpha = \mathcal{W}_2^2(\alpha,\beta) - \frac{d\varepsilon}{2}\log(\pi\varepsilon) - \frac{\varepsilon}{2}\big(\mathrm{H}(\alpha)+\mathrm{H}(\beta)\big) + \frac{\varepsilon^2}{16}\mathcal{I}_{\mathrm{geo}}(\alpha,\beta) + o(\varepsilon^2).$$

*The spatial order-$\varepsilon$ corrections follow from the Laplace prefactors in the two soft $c$-transform equations and depend on the endpoint densities and the Hessian of the quadratic contact phase.*

*Proof.* For the Brownian reference with heat kernel variance $T$, the dynamic Schrödinger formulation gives

$$T\mathcal{C}_T(\alpha,\beta) = \frac{1}{2}\mathcal{W}_2^2(\alpha,\beta) + \frac{T}{2}\left(\mathrm{H}(\alpha)+\mathrm{H}(\beta)\right) + \frac{T^2}{8}\mathcal{I}_{\mathrm{geo}}(\alpha,\beta) + o(T^2),$$

under the stated regularity assumptions. Here $\mathcal{C}_T$ uses entropy relative to the sigma-finite Brownian endpoint measure:

$$\mathcal{C}_T(\alpha,\beta) = \inf_{\pi\in\mathcal{U}(\alpha,\beta)}\mathcal{H}(\pi|p_T(x,y)\,\mathrm{d}x\mathrm{d}y), \qquad p_T(x,y) = (2\pi T)^{-d/2}e^{-\|x-y\|^2/(2T)},$$

where $\mathcal{H}(\pi|\xi) := \int\log(\mathrm{d}\pi/\mathrm{d}\xi)\mathrm{d}\pi$ for sigma-finite $\xi$. When $\xi$ is a probability measure, this agrees with Definition 7.9; unlike the generalized KL of that definition, it remains meaningful when the Brownian reference has infinite total mass. Expanding the heat kernel, multiplying by $2T$, and setting $\varepsilon = 2T$ gives $2T\mathcal{C}_T(\alpha,\beta) = \mathcal{L}^{\varepsilon}_{\|\cdot\|^2}(\alpha,\beta) + 2T\,(\mathrm{H}(\alpha)+\mathrm{H}(\beta)) + dT\log(2\pi T)$. Equivalently, $\mathcal{L}^{\varepsilon}_{\|\cdot\|^2}(\alpha,\beta) = 2T\mathcal{C}_T(\alpha,\beta) - 2T\,(\mathrm{H}(\alpha)+\mathrm{H}(\beta)) - dT\log(2\pi T)$, which is exactly the displayed expansion because $2T=\varepsilon$. At a dual optimum the exponential penalty integrates to zero, so the gauge $\int g_\varepsilon \mathrm{d}\beta = 0$ gives $\int f_\varepsilon \mathrm{d}\alpha = \mathcal{L}^{\varepsilon}_{\|\cdot\|^2}(\alpha,\beta)$. This proves the displayed scalar expansion. Local convergence and the spatial corrections follow from Laplace's method applied to the soft $c$-transform equations; changing the additive gauge redistributes scalar terms between the two potentials. ☐

### 7.5 Dual of Sinkhorn

**Discrete dual.**

**Proposition 7.16** (Dual of entropic OT)**.** *The optimal value of* (7.7) *is*

$$\min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle \mathrm{P},\, \mathrm{C}\rangle + \varepsilon \mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) = \max_{\mathrm{f}\in\mathbb{R}^n,\mathrm{g}\in\mathbb{R}^m} \langle \mathrm{f},\, \mathrm{a}\rangle + \langle \mathrm{g},\, \mathrm{b}\rangle - \varepsilon\sum_{i,j}\exp\left(\frac{\mathrm{f}_i+\mathrm{g}_j-\mathrm{C}_{i,j}}{\varepsilon}\right)\mathrm{a}_i\mathrm{b}_j + \varepsilon. \tag{7.18}$$

*The optimal* $(\mathrm{f},\mathrm{g})$ *are linked to scalings* $(\mathrm{u},\mathrm{v})$ *appearing in* (7.2) *through*

$$\mathrm{u}_i = \mathrm{a}_i e^{\mathrm{f}_i/\varepsilon} \quad \textit{and} \quad \mathrm{v}_j = \mathrm{b}_j e^{\mathrm{g}_j/\varepsilon}. \tag{7.19}$$

*Proof.* We introduce Lagrange multipliers and consider $\min\limits_{\mathrm{P}\geqslant 0}\max\limits_{\mathrm{f},\mathrm{g}} \langle \mathrm{C},\, \mathrm{P}\rangle + \varepsilon\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) + \langle \mathrm{a}-\mathrm{P}\mathbb{1},\, \mathrm{f}\rangle + \langle \mathrm{b}-\mathrm{P}^\top\mathbb{1},\, \mathrm{g}\rangle$. Finite-dimensional convex duality allows us to exchange the minimum over P with the maximum over $(\mathrm{f},\mathrm{g})$, giving

$$\max_{\mathrm{f},\mathrm{g}} \langle \mathrm{f},\, \mathrm{a}\rangle + \langle \mathrm{g},\, \mathrm{b}\rangle + \varepsilon\min_{\mathrm{P}\geqslant 0}\left(\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) - \langle\frac{\mathrm{f}\oplus\mathrm{g}-\mathrm{C}}{\varepsilon},\, \mathrm{P}\rangle\right) = \langle \mathrm{f},\, \mathrm{a}\rangle + \langle \mathrm{g},\, \mathrm{b}\rangle - \varepsilon\mathrm{KL}^*\left(\frac{\mathrm{f}\oplus\mathrm{g}-\mathrm{C}}{\varepsilon}|\mathrm{a}\otimes\mathrm{b}\right).$$

One concludes by using (6.9) for $\varphi(r) = r\log(r) - r + 1$ $\mathrm{KL}^*(H|\mathrm{a}\otimes\mathrm{b}) = \sum_{i,j}\varphi^*(H_{i,j})\mathrm{a}_i\mathrm{b}_j$. Indeed, the scalar maximization $\varphi^*(s) = \sup_{r\geqslant 0}\{rs - r\log r + r - 1\}$ has first-order condition $s - \log r = 0$, hence $r = e^s$ and $\varphi^*(s) = e^s - 1$. ☐

**Discrete soft $c$-transforms.**

$$\mathrm{a}_i - e^{\frac{\mathrm{f}_i}{\varepsilon}}\mathrm{a}_i\sum_j\exp\left(\frac{\mathrm{g}_j-\mathrm{C}_{i,j}}{\varepsilon}\right)\mathrm{b}_j = 0$$

$$\mathrm{f}_i = -\varepsilon\log\sum_j\exp\left(\frac{\mathrm{g}_j-\mathrm{C}_{i,j}}{\varepsilon}\right)\mathrm{b}_j.$$

**Definition 7.17** (Soft-min and discrete soft $c$-transform)**.** For $h\in\mathbb{R}^m$ and weights $\mathrm{b}\in\Sigma_m$, the weighted soft-min at temperature $\varepsilon>0$ is $\min^\varepsilon_{\mathrm{b}}(h) := -\varepsilon\log\sum_j e^{-h_j/\varepsilon}\mathrm{b}_j$. It converges to $\min_{j:\,\mathrm{b}_j>0} h_j$ as $\varepsilon\to 0$; for positive weights this is $\min_j h_j$. Given a cost matrix C, the discrete soft $c$-transforms are

$$\mathrm{f}_i = \min^\varepsilon_{\mathrm{b}}(\mathrm{C}_{i,\cdot}-\mathrm{g}), \qquad \mathrm{g}_j = \min^\varepsilon_{\mathrm{a}}(\mathrm{C}_{\cdot,j}-\mathrm{f}). \tag{7.20}$$

$\min^\varepsilon_{\mathrm{b}}(h-\mathrm{cst}) = \min^\varepsilon_{\mathrm{b}}(h) - \mathrm{cst}$

**Continuous dual and soft-transforms.**

**Proposition 7.18** (Continuous entropic duality)**.** *Let $\mathcal{X}$ and $\mathcal{Y}$ be compact metric spaces, let $\alpha\in\mathcal{P}(\mathcal{X})$ and $\beta\in\mathcal{P}(\mathcal{Y})$, and let $c\in\mathcal{C}(\mathcal{X}\times\mathcal{Y})$. For every $\varepsilon>0$, the KL-regularized problem* (7.12) *satisfies*

$$\mathcal{L}^\varepsilon_c(\alpha,\beta) = \sup_{f\in\mathcal{C}(\mathcal{X}),\, g\in\mathcal{C}(\mathcal{Y})} \mathcal{D}_\varepsilon(f,g), \tag{7.21}$$

*where the concave dual objective is*

$$\mathcal{D}_\varepsilon(f,g) := \int_{\mathcal{X}} f(x)\mathrm{d}\alpha(x) + \int_{\mathcal{Y}} g(y)\mathrm{d}\beta(y) - \varepsilon\int_{\mathcal{X}\times\mathcal{Y}}\left(e^{\frac{f(x)+g(y)-c(x,y)}{\varepsilon}} - 1\right)\mathrm{d}\alpha(x)\mathrm{d}\beta(y). \tag{7.22}$$

*If $\pi^\star$ and $(f^\star,g^\star)$ are primal and dual optimizers, then*

$$\frac{\mathrm{d}\pi^\star}{\mathrm{d}(\alpha\otimes\beta)}(x,y) = \exp\left(\frac{f^\star(x)+g^\star(y)-c(x,y)}{\varepsilon}\right) \qquad (\alpha\otimes\beta)\text{-a.e.} \tag{7.23}$$

*Proof.* Write $\xi = \alpha\otimes\beta$ and $\varphi(r) = r\log r - r + 1$ for $r\geqslant 0$. Since the primal objective is infinite unless $\pi\ll\xi$, put $r = \mathrm{d}\pi/\mathrm{d}\xi$. The marginal constraints are equivalent to $\int_{\mathcal{Y}} r(x,y)\mathrm{d}\beta(y) = 1\quad \alpha\text{-a.e.},\qquad \int_{\mathcal{X}} r(x,y)\mathrm{d}\alpha(x) = 1\quad \beta\text{-a.e.}$ Introduce potentials $f$ and $g$ for these two constraints. The Lagrangian is

$$\int_{\mathcal{X}} f\mathrm{d}\alpha + \int_{\mathcal{Y}} g\mathrm{d}\beta + \int_{\mathcal{X}\times\mathcal{Y}}\left[\varepsilon\varphi(r(x,y)) + \big(c(x,y)-f(x)-g(y)\big)r(x,y)\right]\mathrm{d}\xi(x,y).$$

The scalar conjugate of $\varphi$ is $\varphi^*(s) = e^s - 1$. Therefore pointwise minimization over $r\geqslant 0$ gives

$$\inf_{r\geqslant 0}\{\varepsilon\varphi(r) + (c-f-g)r\} = -\varepsilon\varphi^*\left(\frac{f+g-c}{\varepsilon}\right) = -\varepsilon\left(e^{(f+g-c)/\varepsilon} - 1\right).$$

Integrating this identity yields $\mathcal{D}_\varepsilon(f,g)$ and proves weak duality. Strong duality follows from Fenchel–Rockafellar entropy duality in the exponential Orlicz pair: the feasible density $r\equiv 1$, corresponding to the product coupling $\xi$, is strictly positive and supplies the qualification point, while the boundedness of $c$ makes its objective finite. Because $c$ is continuous on the compact product, the dual supremum can be restricted to continuous potentials; equivalently, alternating soft $c$-transforms regularize bounded potentials into continuous ones without decreasing the dual value. This proves (7.21).

Finally, equality in the pointwise Fenchel inequality forces $r^\star(x,y)$ to be the unique minimizer of the scalar problem above. Its first-order condition is $\varepsilon\log r^\star(x,y) + c(x,y) - f^\star(x) - g^\star(y) = 0$. Exponentiation gives (7.23). ☐

**Definition 7.19** (Continuous soft $c$-transforms)**.** For $f \in \mathcal{C}(\mathcal{X})$ and $g \in \mathcal{C}(\mathcal{Y})$, define

$$f^{c,\varepsilon}(y) \coloneqq -\varepsilon \log\left(\int_{\mathcal{X}} e^{\frac{f(x)-c(x,y)}{\varepsilon}} \mathrm{d}\alpha(x)\right), \qquad y \in \mathcal{Y}, \tag{7.24}$$

$$g^{\bar{c},\varepsilon}(x) \coloneqq -\varepsilon \log\left(\int_{\mathcal{Y}} e^{\frac{g(y)-c(x,y)}{\varepsilon}} \mathrm{d}\beta(y)\right), \qquad x \in \mathcal{X}. \tag{7.25}$$

**Proposition 7.20** (Existence and uniqueness of entropic dual potentials)**.** *Assume that $\mathcal{X} = \mathrm{supp}(\alpha)$ and $\mathcal{Y} = \mathrm{supp}(\beta)$ are compact and that $c$ is continuous. The dual problem (7.21) has solutions, and its set of solutions is $(f^\star + \lambda, g^\star - \lambda)$, $\qquad \lambda \in \mathbb{R}$.*

*Proof.* Normalize potentials by imposing $\int f \mathrm{d}\alpha = 0$. Replacing any pair $(f, g)$ by the corresponding soft $c$-transforms does not decrease the dual objective, because each soft transform is the exact maximizer in one block variable. For transformed potentials, the oscillations are bounded by the oscillation of the cost: $\|f\|_V + \|g\|_V \leqslant 2(\sup c - \inf c)$. Moreover the modulus of continuity of the soft transforms is controlled by that of $c$; for instance $|g^{\bar{c},\varepsilon}(x) - g^{\bar{c},\varepsilon}(x')| \leqslant \sup_y |c(x,y) - c(x',y)|$. After normalization, maximizing sequences are therefore uniformly bounded and equicontinuous. Arzelà–Ascoli gives a uniformly converging subsequence, and continuity of (7.22) gives a maximizer.
For uniqueness, use strict convexity of $H \mapsto \int e^{H/\varepsilon} \mathrm{d}(\alpha \otimes \beta)$ on the image of $(f, g) \mapsto H = f \oplus g - c$, modulo constants. If two optimal pairs exist, their midpoint is also optimal; strict convexity forces the two functions $f \oplus g$ to agree $\alpha \otimes \beta$-almost everywhere. Full support and continuity then imply equality everywhere on $\mathcal{X} \times \mathcal{Y}$, so $f - f'$ is constant and $g - g'$ is the opposite constant. Without replacing the ambient spaces by the marginal supports, uniqueness has this meaning only on those supports; values outside them do not enter the dual objective. □

**Dual Sinkhorn for general measures.**

$$\frac{\mathrm{d}}{\mathrm{d}s}\mathcal{D}_\varepsilon(f + sh, g)\bigg|_{s=0} = \int_{\mathcal{X}} h(x) \left[1 - e^{f(x)/\varepsilon} \int_{\mathcal{Y}} e^{(g(y)-c(x,y))/\varepsilon} \mathrm{d}\beta(y)\right] \mathrm{d}\alpha(x).$$

$$f^{(\ell+1)} = (g^{(\ell)})^{\bar{c},\varepsilon}, \qquad g^{(\ell+1)} = (f^{(\ell+1)})^{c,\varepsilon}. \tag{7.26}$$

$u^{(\ell)} = e^{f^{(\ell)}/\varepsilon}$, $\qquad v^{(\ell)} = e^{g^{(\ell)}/\varepsilon}$ $u^{(\ell+1)} = \frac{1}{\mathcal{K}_\varepsilon v^{(\ell)}}$, $\qquad v^{(\ell+1)} = \frac{1}{\mathcal{K}_\varepsilon^* u^{(\ell+1)}}$. Thus the coupling density reconstructed from the current potentials is $u^{(\ell)}(x) k_\varepsilon(x,y) v^{(\ell)}(y)$ with respect to $\alpha \otimes \beta$, exactly as in the scaling law (7.14). At a fixed point, its two marginals are $\alpha$ and $\beta$; equivalently, the potentials jointly maximize the continuous dual.
**Neural dual solvers.** $\Phi(x) + \Psi(y) \geqslant \langle x, y \rangle$. For the quadratic cost this is the same statement after subtracting the quadratic terms, using the convexity properties of soft transforms. One can therefore maximize the continuous dual (7.22) over parameterized convex potentials, estimating the integrals by stochastic samples.

## 7.6 Path-Space Schrödinger Problem

**Unregularized path-space transport.**

**Definition 7.21** (Unregularized path-space transport)**.** Let $\Omega = C([0,1]; \mathcal{X})$, let $e_t(\omega) = \omega_t$, and let $\mathcal{A} : \Omega \to [0, +\infty]$ be a path action. Define

$$\mathrm{PT}_{\mathcal{A}}(\alpha, \beta) \coloneqq \inf_{M \in \mathcal{P}(\Omega)} \left\{ \int_\Omega \mathcal{A}(\omega) \, \mathrm{d}M(\omega) \; ; \; (e_0)_\sharp M = \alpha, \; (e_1)_\sharp M = \beta \right\}. \tag{7.27}$$

$$\mathcal{A}(\omega) = \begin{cases} \displaystyle\int_0^1 \|\dot{\omega}_t\|^2 \, \mathrm{d}t, & \text{if } \omega \text{ is absolutely continuous,} \\ +\infty, & \text{otherwise.} \end{cases}$$

$$c_{\mathcal{A}}(x, y) \coloneqq \inf_{\omega \in \Omega} \{\mathcal{A}(\omega) \; ; \; e_0(\omega) = x, \; e_1(\omega) = y\}. \tag{7.28}$$

For the quadratic action above, the minimizing path is the straight segment $\omega_t = (1-t)x + ty$, and $c_{\mathcal{A}}(x,y) = \|x - y\|^2$.
**Proposition 7.22** (Endpoint reduction of path-space transport)**.** *Assume that minimizing paths in (7.28) can be selected measurably, or more generally that the infimum can be approximated by measurable selections. Then (7.27) has the same value as the Kantorovich problem $\inf_{\pi \in \mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{X}} c_{\mathcal{A}}(x,y) \, \mathrm{d}\pi(x,y)$. Moreover, if $\pi^\star$ is an optimal endpoint coupling and $\omega^{x,y}$ is an optimal path from $x$ to $y$, then $M^\star = \int_{\mathcal{X}\times\mathcal{X}} \delta_{\omega^{x,y}} \, \mathrm{d}\pi^\star(x,y)$ is an optimal path law.*

*Proof.* Let $M$ be any feasible path law and set $\pi = (e_0, e_1)_\sharp M$. Then $\pi \in \mathcal{U}(\alpha, \beta)$ and, by the definition of $c_{\mathcal{A}}$, $\int_\Omega \mathcal{A}(\omega) \, \mathrm{d}M(\omega) \geqslant \int_{\mathcal{X}\times\mathcal{X}} c_{\mathcal{A}}(x,y) \, \mathrm{d}\pi(x,y)$. This proves that the path-space value is at least the Kantorovich value. Conversely, given a coupling $\pi$ and a measurable selection $(x,y) \mapsto \omega^{x,y}$ with action arbitrarily close to $c_{\mathcal{A}}(x,y)$, the mixture of Dirac path laws $M = \int \delta_{\omega^{x,y}} \, \mathrm{d}\pi(x,y)$ has endpoints $\alpha$ and $\beta$ and action equal, up to the selected approximation error, to $\int c_{\mathcal{A}} \, \mathrm{d}\pi$. Optimizing over $\pi$ and letting the approximation error vanish proves equality. If exact minimizing paths are selected for an optimal $\pi^\star$, the displayed $M^\star$ is optimal. □

**Entropic path-space problem.** Let $\mathcal{R}^\varepsilon \in \mathcal{P}(\Omega)$ be a reference path law, for instance a Brownian or Langevin dynamics at noise level $\varepsilon$. The corresponding entropy projection defines the dynamic Schrödinger problem.

**Definition 7.23** (Schrödinger bridge)**.**

$$\mathrm{SB}_\varepsilon(\alpha, \beta) \coloneqq \inf_{M \in \mathcal{P}(\Omega)} \{\varepsilon \, \mathrm{KL}(M | \mathcal{R}^\varepsilon) \; ; \; (e_0)_\sharp M = \alpha, \; (e_1)_\sharp M = \beta\}. \tag{7.29}$$

**Stochastic-control interpretation.** For Brownian references, the entropy projection has a direct control meaning. To fix constants, suppose that $\mathcal{R}^\varepsilon$ is the law of
$\mathrm{d}X_t = \sqrt{\varepsilon}\,\mathrm{d}B_t, \qquad X_0 \sim \alpha$, $\mathrm{d}X_t = u_t\,\mathrm{d}t + \sqrt{\varepsilon}\,\mathrm{d}B_t, \qquad X_0 \sim \alpha.$

$$\mathrm{KL}(M|\mathcal{R}^\varepsilon) = \frac{1}{2\varepsilon}\mathbb{E}_M \int_0^1 \|u_t\|^2\,\mathrm{d}t, \qquad \varepsilon\,\mathrm{KL}(M|\mathcal{R}^\varepsilon) = \frac{1}{2}\mathbb{E}_M \int_0^1 \|u_t\|^2\,\mathrm{d}t.$$

If the reference initial law is not $\alpha$, an additional term $\varepsilon\,\mathrm{KL}(\alpha|\mathcal{R}_0^\varepsilon)$ appears, but it is fixed by the endpoint constraint. Hence, up to this fixed endpoint cost, the Schrödinger bridge can be read as
$\inf_u \left\{\frac{1}{2}\mathbb{E}\int_0^1 \|u_t\|^2\,\mathrm{d}t \;;\; \mathrm{d}X_t = u_t\,\mathrm{d}t + \sqrt{\varepsilon}\,\mathrm{d}B_t, \quad X_0 \sim \alpha,\; X_1 \sim \beta\right\}$, $u_t^\star(x) = \varepsilon\nabla\log h_t(x)$.

**Viscous Benamou–Brenier formulations.** $\partial_t\rho_t + \mathrm{div}(\rho_t v_t) = \frac{\sigma}{2}\Delta\rho_t$, $\int_0^1\int \frac{1}{2}\|v_t(x)\|^2\rho_t(x)\mathrm{d}x\mathrm{d}t$. $u_t = v_t - \frac{\sigma}{2}\nabla\log\rho_t, \qquad \partial_t\rho_t + \mathrm{div}(\rho_t u_t) = 0$.

$$\int_0^1\int \frac{1}{2}\|v_t\|^2\rho_t\,\mathrm{d}x\,\mathrm{d}t = \int_0^1\int \left(\frac{1}{2}\|u_t\|^2 + \frac{\sigma^2}{8}\|\nabla\log\rho_t\|^2\right)\rho_t\,\mathrm{d}x\,\mathrm{d}t + \frac{\sigma}{2}\left[\int \rho_1\log\rho_1\,\mathrm{d}x - \int \rho_0\log\rho_0\,\mathrm{d}x\right].$$

$$\int_0^1\int \left(\frac{1}{2}\|u_t(x)\|^2 + \frac{\sigma^2}{8}\|\nabla\log\rho_t(x)\|^2\right)\rho_t(x)\mathrm{d}x\mathrm{d}t.$$

Thus the Schrödinger bridge is a least-action interpolation with both transport kinetic energy and a Fisher-information penalty. If one instead writes the viscous equation with diffusion coefficient $\sigma\Delta\rho_t$, the same formula is obtained after replacing $\sigma$ above by $2\sigma$, so the Fisher coefficient becomes $\sigma^2/2$.

$$\mathcal{R}^\varepsilon(\mathrm{d}\omega) = \int \mathcal{R}^{\varepsilon,x,y}(\mathrm{d}\omega)\,\mathcal{R}^\varepsilon_{0,1}(\mathrm{d}x,\mathrm{d}y), \qquad \mathcal{R}^\varepsilon_{0,1} := (e_0,e_1)_\sharp\mathcal{R}^\varepsilon,$$

where $\mathcal{R}^{\varepsilon,x,y}$ is the reference bridge conditioned on endpoints $(x,y)$. Similarly, for any feasible $M$, write
$M(\mathrm{d}\omega) = \int M^{x,y}(\mathrm{d}\omega)\,\pi(\mathrm{d}x,\mathrm{d}y), \qquad \pi := (e_0,e_1)_\sharp M.$

**Proposition 7.24** (Endpoint reduction of the Schrödinger problem)**.** *Assume that the regular conditional laws above exist and that the relative-entropy chain rule applies, with value $+\infty$ when absolute continuity fails. Then*

$$\mathrm{SB}_\varepsilon(\alpha,\beta) = \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \varepsilon\,\mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{0,1}). \tag{7.30}$$

*For a fixed endpoint coupling $\pi$ with finite $\mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{0,1})$, the minimizing path law is the mixture of reference bridges*

$$M^\pi = \int \mathcal{R}^{\varepsilon,x,y}\,\mathrm{d}\pi(x,y). \tag{7.31}$$

*Proof.* If $M$ has finite relative entropy with respect to $\mathcal{R}^\varepsilon$, then $\pi = (e_0,e_1)_\sharp M$ is necessarily absolutely continuous with respect to $\mathcal{R}^\varepsilon_{0,1}$. The chain rule for relative entropy gives

$$\mathrm{KL}(M|\mathcal{R}^\varepsilon) = \mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{0,1}) + \int_{\mathcal{X}\times\mathcal{X}} \mathrm{KL}(M^{x,y}|\mathcal{R}^{\varepsilon,x,y})\,\mathrm{d}\pi(x,y). \tag{7.32}$$

The second term is nonnegative and vanishes exactly when $M^{x,y} = \mathcal{R}^{\varepsilon,x,y}$ for $\pi$-almost every $(x,y)$. Thus, once an endpoint coupling $\pi$ with finite $\mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{0,1})$ is fixed, the best path law is (7.31), and the remaining minimization is precisely (7.30). If no such endpoint coupling exists, both sides are $+\infty$. ☐

**Brownian bridges and Sinkhorn couplings.**

$$p_\varepsilon(x,y) \propto \exp\left(-\frac{\|x-y\|^2}{\varepsilon}\right).$$

$$\mathcal{R}^\varepsilon_{0,1}(\mathrm{d}x,\mathrm{d}y) \propto \exp\left(-\frac{c(x,y)}{\varepsilon}\right)\alpha(\mathrm{d}x)\beta(\mathrm{d}y).$$

This includes the usual heat-kernel reference after rewriting it with respect to $\alpha\otimes\beta$, whenever the one-time endpoint densities are fixed and mutually absolutely continuous. The additional one-body density factors only add constants under the marginal constraints.
$\varepsilon\,\mathrm{KL}(\pi|\mathcal{R}^\varepsilon_{0,1}) = \int c(x,y)\,\mathrm{d}\pi(x,y) + \varepsilon\,\mathrm{KL}(\pi|\alpha\otimes\beta) + \text{constant}$, where the constant does not depend on $\pi$. Hence (7.30) is exactly the continuous Sinkhorn problem (7.12), up to an additive constant in the value.
$M^\star_\varepsilon = \int \mathcal{R}^{\varepsilon,x,y}\,\mathrm{d}\pi^\star_\varepsilon(x,y)$.

## 7.7 Marginal-Dependent Problems

Let $\mathcal{F}$ and $\mathcal{G}$ be proper convex lower semicontinuous functionals on finite nonnegative measures on $\mathcal{X}$ and $\mathcal{Y}$. The unregularized marginal-dependent transport problem is

$$\inf_{\pi\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})} \int c(x,y)\mathrm{d}\pi(x,y) + \mathcal{F}(\pi_1) + \mathcal{G}(\pi_2), \tag{7.33}$$

where $\pi_1 = (\mathrm{p}_\mathcal{X})_\sharp\pi$ and $\pi_2 = (\mathrm{p}_\mathcal{Y})_\sharp\pi$ are the two marginals of $\pi$. Entropic regularization turns this problem into the scaling-friendly form below.

**Definition 7.25** (Marginal-dependent entropic transport)**.** For reference measures $\alpha, \beta$, proper convex marginal functionals $\mathcal{F}, \mathcal{G}$, and $\varepsilon > 0$, define

$$\inf_{\pi \in \mathcal{M}_+(\mathcal{X}\times\mathcal{Y})} \int c(x,y)\mathrm{d}\pi(x,y) + \mathcal{F}(\pi_1) + \mathcal{G}(\pi_2) + \varepsilon\,\mathrm{KL}(\pi|\alpha\otimes\beta). \tag{7.34}$$

For discrete reference weights a, b and convex functions $\mathsf{F} : \mathbb{R}^n_+ \to \mathbb{R}\cup\{+\infty\}$, $\mathsf{G} : \mathbb{R}^m_+ \to \mathbb{R}\cup\{+\infty\}$, its matrix form is

$$\inf_{\mathrm{P}\in\mathbb{R}^{n\times m}_+} \langle \mathrm{C},\, \mathrm{P}\rangle + \mathsf{F}(\mathrm{P}\mathbb{1}_m) + \mathsf{G}(\mathrm{P}^\top\mathbb{1}_n) + \varepsilon\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}). \tag{7.35}$$

In this subsection, when the total mass of $\pi$ is not fixed, the KL term is understood in the generalized sense associated with $\varphi(s) = s\log s - s + 1$. Thus, if $\lambda = \alpha\otimes\beta$ and $\pi = \rho\lambda + \pi^\perp$ is the Lebesgue decomposition of $\pi$ with respect to $\lambda$, then $\mathrm{KL}(\pi|\lambda) = \int\big(\rho\log\rho - \rho + 1\big)\mathrm{d}\lambda\ \inf_{\mathrm{P}\geqslant 0}\mathsf{F}(\mathrm{P}\mathbb{1}_m) + \mathsf{G}(\mathrm{P}^\top\mathbb{1}_n) + \varepsilon\mathrm{KL}(\mathrm{P}|\mathrm{K})$.

**Proposition 7.26** (Dual and scaling for marginal penalties)**.** *Assume $a_i, b_j > 0$, $\varepsilon > 0$, and a Fenchel qualification condition, for instance the existence of a matrix $\mathrm{P} > 0$ such that $\mathrm{P}\mathbb{1}_m$ and $\mathrm{P}^\top\mathbb{1}_n$ belong to the relative interiors of* dom(F) *and* dom(G)*. The Fenchel dual of* (7.35) *is*

$$\sup_{\mathrm{f}\in\mathbb{R}^n,\mathrm{g}\in\mathbb{R}^m} -\mathsf{F}^*(-\mathrm{f}) - \mathsf{G}^*(-\mathrm{g}) - \varepsilon\sum_{i,j} a_i b_j\left[\exp\left(\frac{\mathrm{f}_i + \mathrm{g}_j - \mathrm{C}_{i,j}}{\varepsilon}\right) - 1\right]. \tag{7.36}$$

*Equivalently, up to the additive constant $\varepsilon\sum_{i,j} a_i b_j$, the last term is*

$$-\varepsilon\sum_{i,j}\mathrm{K}_{i,j}\exp\left(\frac{\mathrm{f}_i + \mathrm{g}_j}{\varepsilon}\right).$$

*If $\mathrm{u} = e^{\mathrm{f}/\varepsilon}$ and $\mathrm{v} = e^{\mathrm{g}/\varepsilon}$, exact block ascent in the two dual variables is the generalized Sinkhorn cycle*

$$\begin{aligned} &r \leftarrow \mathrm{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(\mathrm{K}\mathrm{v}), && \mathrm{u} \leftarrow r \oslash (\mathrm{K}\mathrm{v}),\\ &s \leftarrow \mathrm{prox}^{\mathrm{KL}}_{\mathsf{G}/\varepsilon}(\mathrm{K}^\top\mathrm{u}), && \mathrm{v} \leftarrow s \oslash (\mathrm{K}^\top\mathrm{u}),\\ &\mathrm{P} = \mathrm{diag}(\mathrm{u})\mathrm{K}\,\mathrm{diag}(\mathrm{v}), \end{aligned} \tag{7.37}$$

*where divisions are entrywise and*

$$\mathrm{prox}^{\mathrm{KL}}_{\mathsf{h}}(z) \coloneqq \operatorname*{argmin}_{r\in\mathbb{R}^d_+} \mathsf{h}(r) + \mathrm{KL}(r|z). \tag{7.38}$$

*In particular, $\mathrm{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z)$ is the minimizer of $\mathsf{F}(r) + \varepsilon\mathrm{KL}(r|z)$.*

*Proof.* Introduce independent variables $r$ and $s$ for the two marginals and use dual variables f, g for the constraints $r = \mathrm{P}\mathbb{1}_m$ and $s = \mathrm{P}^\top\mathbb{1}_n$. The Lagrangian contains $\mathsf{F}(r) + \langle \mathrm{f},\, r\rangle + \mathsf{G}(s) + \langle \mathrm{g},\, s\rangle + \langle \mathrm{C},\, \mathrm{P}\rangle + \varepsilon\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) - \langle \mathrm{f}\oplus\mathrm{g},\, \mathrm{P}\rangle$. Minimizing over $r$ and $s$ gives $-\mathsf{F}^*(-\mathrm{f})$ and $-\mathsf{G}^*(-\mathrm{g})$. For each scalar entry, the convex conjugate of $p \mapsto \mathrm{C}_{i,j}p + \varepsilon(p\log(p/(a_ib_j)) - p + a_ib_j)$ is $q \mapsto \varepsilon a_ib_j(e^{(q-\mathrm{C}_{i,j})/\varepsilon} - 1)$. Minimizing over P with $q = \mathrm{f}_i + \mathrm{g}_j$ gives (7.36).
For the scaling form, fix $\mathrm{v} > 0$, set $\widetilde{\mathrm{K}} = \mathrm{K}\,\mathrm{diag}(\mathrm{v})$ and $z = \widetilde{\mathrm{K}}\mathbb{1}_m = \mathrm{K}\mathrm{v}$. Updating the row scaling means looking for $\mathrm{P} = \mathrm{diag}(\mathrm{u})\widetilde{\mathrm{K}}$, so its row marginal is $r = \mathrm{P}\mathbb{1}_m = \mathrm{u}\odot z$. The row-wise chain rule gives $\mathrm{KL}(\mathrm{diag}(\mathrm{u})\widetilde{\mathrm{K}}|\widetilde{\mathrm{K}}) = \sum_i\big(r_i\log(r_i/z_i) - r_i + z_i\big) = \mathrm{KL}(r|z)$. Thus exact optimization of this block is equivalent to minimizing $\mathsf{F}(r) + \varepsilon\mathrm{KL}(r|z)$ over $r \geqslant 0$, hence $r = \mathrm{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z)$ and $\mathrm{u} = r\oslash z$. The column update is identical with $w = \mathrm{K}^\top\mathrm{u}$. □

When $\mathsf{F} = \iota_{\{\mathrm{a}\}}$ and $\mathsf{G} = \iota_{\{\mathrm{b}\}}$, the first two dual terms are $\langle \mathrm{f},\, \mathrm{a}\rangle + \langle \mathrm{g},\, \mathrm{b}\rangle$, and one recovers the usual entropic dual (7.22). The classical Sinkhorn update is the special case in which the KL proximal maps return the prescribed marginals a and b.

- **Hard marginal constraint.** If $\mathsf{F} = \iota_{\{\mathrm{a}\}}$, then $\mathrm{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z) = \mathrm{a}$.
- **KL marginal relaxation.** If $\mathsf{F}(r) = \tau\mathrm{KL}(r|\mathrm{a})$ with $\tau > 0$, then, coordinatewise, $\mathrm{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z) = z^{\varepsilon/(\tau+\varepsilon)}\odot\mathrm{a}^{\tau/(\tau+\varepsilon)}$, and the scaling update becomes $\mathrm{u} \leftarrow (\mathrm{a}\oslash z)^{\tau/(\tau+\varepsilon)}$, the damped scaling of unbalanced Sinkhorn.
- **Pointwise bounds.** If $\mathsf{F} = \iota_{\{\ell\leqslant r\leqslant u\}}$, then the proximal map is the coordinatewise clipping $\mathrm{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z)_i = \min\{u_i, \max\{\ell_i, z_i\}\}$.
- **Total-variation marginal relaxation.** If $\mathsf{F}(r) = \tau\|r - \mathrm{a}\|_1$ and $\lambda = \tau/\varepsilon$, then the proximal map is coordinatewise

$$\mathrm{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z)_i = \begin{cases} z_ie^{\lambda}, & z_i < a_ie^{-\lambda},\\ a_i, & a_ie^{-\lambda} \leqslant z_i \leqslant a_ie^{\lambda},\\ z_ie^{-\lambda}, & z_i > a_ie^{\lambda}. \end{cases}$$

- **Fixed total mass.** If $\mathsf{F} = \iota_{\{\langle r,\,\mathbb{1}\rangle = m\}}$, then $\mathrm{prox}^{\mathrm{KL}}_{\mathsf{F}/\varepsilon}(z) = mz/\langle z,\, \mathbb{1}\rangle$.

## 7.8 Heat Kernels and Hopf–Cole Transforms

**Geodesics in heat.**

$$h_t(x,y) = (4\pi t)^{-d/2}\exp\left(-\frac{\|x-y\|^2}{4t}\right).$$

For the quadratic cost $c(x,y) = \|x-y\|^2$, the Gibbs kernel used by Sinkhorn satisfies

$$\mathrm{K}_\varepsilon(x,y) \coloneqq e^{-\|x-y\|^2/\varepsilon} = (\pi\varepsilon)^{d/2}h_{\varepsilon/4}(x,y). \tag{7.39}$$

$\mathsf{H}_\varepsilon \coloneqq e^{-(\varepsilon/4)L}$,

$$\mathrm{u}^{(\ell+1)} = \mathrm{a}\oslash\big(\mathsf{H}_\varepsilon\mathrm{v}^{(\ell)}\big), \qquad \mathrm{v}^{(\ell+1)} = \mathrm{b}\oslash\big(\mathsf{H}_\varepsilon^\top\mathrm{u}^{(\ell+1)}\big), \tag{7.40}$$

where $\mathsf{H}_\varepsilon$ denotes the discretized integral operator, including quadrature weights, and $\mathsf{H}_\varepsilon^\top$ its discrete adjoint. In a symmetric mass-normalized discretization, the two operators coincide.
$c_\varepsilon(x,y) := -\varepsilon \log h_{\varepsilon/4}(x,y)$. $c_\varepsilon(x,y) \longrightarrow d_M(x,y)^2 \qquad (\varepsilon \downarrow 0)$,

$$\mathsf{H}_\varepsilon = \lim_{q\to\infty} \left(\mathrm{Id} + \frac{\varepsilon}{4q} L\right)^{-q}.$$

$A_{\varepsilon,q} := \mathrm{Id} + \frac{\varepsilon}{4q} L_h$.
**Soft Hopf–Lax and Hopf–Cole.** $c(x,y) = \frac{1}{2}\|x-y\|^2$,

$$(-h)^{\bar{c}}(x) = \inf_y \left\{ h(y) + \frac{1}{2}\|x-y\|^2 \right\},$$

where the sign only reflects the convention of Definition 4.7. In the present entropic setting, the parameter of interest is instead the temperature $\varepsilon$.

$$(-h)^{\bar{c},\varepsilon}(x) = -\varepsilon \log \int \exp\left(-\frac{h(y) + \|x-y\|^2/2}{\varepsilon}\right) \mathrm{d}y. \tag{7.41}$$

$$G_\varepsilon(z) := (2\pi\varepsilon)^{-d/2} \exp\left(-\frac{\|z\|^2}{2\varepsilon}\right)$$

$$(-h)^{\bar{c},\varepsilon}(x) = -\varepsilon \log \left(G_\varepsilon * e^{-h/\varepsilon}\right)(x) - \frac{\varepsilon d}{2} \log(2\pi\varepsilon). \tag{7.42}$$

Thus a soft quadratic $c$-transform is a Gaussian convolution followed by a logarithm, up to an explicit additive constant independent of $x$. This is the bridge between soft-minimum operators, heat kernels and entropic transport potentials.
**Proposition 7.27** (Soft quadratic $c$-transform and Legendre approximation)**.** *Let $f : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be such that the integrals below are finite, and introduce the quadratic shift $\mathsf{S}f(y) := f(y) - \frac{1}{2}\|y\|^2$. For $\varepsilon > 0$, define the soft conjugate by applying the soft $\bar{c}$-transform to the shifted function:*

$$f^{*,\varepsilon}(p) := \frac{1}{2}\|p\|^2 - \big(-\mathsf{S}f\big)^{\bar{c},\varepsilon}(p). \tag{7.43}$$

*Then*

$$f^{*,\varepsilon}(p) = \varepsilon \log \int_{\mathbb{R}^d} \exp\left(\frac{\langle p,\, y\rangle - f(y)}{\varepsilon}\right) \mathrm{d}y, \tag{7.44}$$

*and equivalently*

$$f^{*,\varepsilon}(p) = \frac{1}{2}\|p\|^2 + \varepsilon \log \left(G_\varepsilon * e^{-\mathsf{S}f/\varepsilon}\right)(p) + \frac{\varepsilon d}{2} \log(2\pi\varepsilon). \tag{7.45}$$

*If, for instance, $f$ is proper, lower semicontinuous and superlinear, then $f^{*,\varepsilon}(p) \to f^*(p)$ for every $p$ as $\varepsilon \to 0$.*

*Proof.* The hard identity follows by completing the square:

$$\inf_y \left\{ \mathsf{S}f(y) + \frac{1}{2}\|p-y\|^2 \right\} = \frac{1}{2}\|p\|^2 - \sup_y\{\langle p,\, y\rangle - f(y)\} = \frac{1}{2}\|p\|^2 - f^*(p).$$

This explains the shift: it turns the Legendre–Fenchel transform into a quadratic $\bar{c}$-transform. Replacing the hard transform by its soft version gives (7.43). Expanding the square in (7.41) with the function $\mathsf{S}f$ gives

$$\exp\left(-\frac{\mathsf{S}f(y) + \|p-y\|^2/2}{\varepsilon}\right) = \exp\left(-\frac{\|p\|^2}{2\varepsilon}\right) \exp\left(\frac{\langle p,\, y\rangle - f(y)}{\varepsilon}\right),$$

which proves (7.44). Formula (7.45) is the Gaussian-convolution identity (7.42) applied to $\mathsf{S}f$. Under the stated superlinearity assumption, the functions $y \mapsto \langle p,\, y\rangle - f(y)$ have compact upper level sets, so Laplace's principle applied to (7.44) gives the convergence to $f^*(p)$. □

$\partial_s \varphi_s + \frac{1}{2}\|\nabla\varphi_s\|^2 = \frac{\varepsilon}{2}\Delta\varphi_s$, $\partial_s v_s + (v_s \cdot \nabla) v_s = \frac{\varepsilon}{2}\Delta v_s$.

### 7.9 Other Convex Regularizers

**$\varphi$-divergence regularization.** Let $\varphi$ be an entropy function in the sense of Definition 6.9, and recall the $\varphi$-divergence $\mathcal{D}_\varphi$ from Definition 6.10. For $\varepsilon > 0$, define the $\varphi$-regularized transport value.

**Definition 7.28** ($\varphi$-regularized optimal transport)**.**

$$\mathcal{L}^\varepsilon_{c,\varphi}(\alpha,\beta) := \min_{\pi\in\mathcal{U}(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y) + \varepsilon \mathcal{D}_\varphi(\pi|\alpha\otimes\beta). \tag{7.46}$$

**Proposition 7.29** (Dual and density law for $\varphi$-regularized OT)**.** *Assume that the Fenchel–Rockafellar qualification condition holds; for example, let the spaces be compact, let $c$ be continuous, and assume that the product coupling belongs to the continuity domain of the divergence term. Then*

$$\mathcal{L}^\varepsilon_{c,\varphi}(\alpha,\beta) = \sup_{f\in\mathcal{C}(\mathcal{X}),\, g\in\mathcal{C}(\mathcal{Y})} \int f \mathrm{d}\alpha + \int g \mathrm{d}\beta - \varepsilon \int \varphi^*\left(\frac{f(x) + g(y) - c(x,y)}{\varepsilon}\right) \mathrm{d}\alpha(x)\mathrm{d}\beta(y). \tag{7.47}$$

*If an optimal plan is absolutely continuous, with density $r^\star = \frac{\mathrm{d}\pi^\star}{\mathrm{d}(\alpha\otimes\beta)}$, and $(f^\star, g^\star)$ are optimal potentials, then*

$$\frac{f^\star(x) + g^\star(y) - c(x,y)}{\varepsilon} \in \partial\varphi(r^\star(x,y)) \qquad \alpha\otimes\beta\text{-a.e.} \tag{7.48}$$

*In the smooth interior this reads*

$$r^\star(x,y) = (\varphi')^{-1}\left(\frac{f^\star(x) + g^\star(y) - c(x,y)}{\varepsilon}\right).$$

*Proof.* Introduce dual variables $(f, g)$ for the two marginal constraints. For fixed $(f, g)$, the minimization over $\pi$ gives

$$\int f\mathrm{d}\alpha + \int g\mathrm{d}\beta + \inf_{\pi\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})}\left\{\int \big(c - (f\oplus g)\big)\mathrm{d}\pi + \varepsilon\mathcal{D}_\varphi(\pi|\alpha\otimes\beta)\right\}.$$

Using the Legendre formula (6.9) for the convex functional $\mathcal{D}_\varphi(\cdot|\alpha\otimes\beta)$, the infimum equals

$$-\varepsilon\int \varphi^*\left(\frac{f(x) + g(y) - c(x,y)}{\varepsilon}\right)\mathrm{d}\alpha(x)\mathrm{d}\beta(y),$$

which gives (7.47). Equality in the Fenchel inequality is equivalent to the subgradient inclusion $\frac{f^\star\oplus g^\star - c}{\varepsilon} \in \partial\varphi(r^\star)$, and inversion of $\varphi'$ gives the density law when $\varphi$ is differentiable and the optimizer is in the interior of its domain. $\square$

For the KL entropy $\varphi(r) = r\log r - r + 1$, one has $\varphi^*(s) = e^s - 1$, and (7.47) reduces exactly to the continuous Sinkhorn dual (7.22). Other choices replace the exponential density law by another scalar transfer function:

$$\begin{aligned}
\varphi(r) &= r\log r - r + 1 &&\Rightarrow\quad r^\star = e^s, \\
\varphi(r) &= r - \log r - 1 &&\Rightarrow\quad r^\star = (1-s)^{-1} \quad (s<1), \qquad s := \frac{f^\star\oplus g^\star - c}{\varepsilon}. \\
\varphi(r) &= \tfrac{1}{2}(r-1)^2 &&\Rightarrow\quad r^\star = (1+s)_+,
\end{aligned}$$

**Bregman-divergence regularization.**

**Definition 7.30** (Measure Bregman divergence)**.** If $\Phi$ is a differentiable convex functional on a convex class of nonnegative measures and $\xi$ is a reference measure in its domain, the measure Bregman divergence generated by $\Phi$ is

$$B_\Phi(\pi|\xi) := \Phi(\pi) - \Phi(\xi) - \int \delta\Phi(\xi)\mathrm{d}(\pi - \xi), \tag{7.49}$$

where $\delta\Phi(\xi)$ is the first variation in the sense of Definition 14.2, and the formula is understood whenever the right-hand side is well-defined.

**Definition 7.31** (Bregman-regularized optimal transport)**.** Fix $\xi = \alpha\otimes\beta$. For $\varepsilon > 0$, define

$$\mathcal{L}^\varepsilon_{c,\Phi}(\alpha,\beta) := \inf_{\pi\in\mathcal{U}(\alpha,\beta)}\left\{\int c\,\mathrm{d}\pi + \varepsilon B_\Phi(\pi|\xi)\right\}.$$

**Proposition 7.32** (Dual and density law for Bregman-regularized OT)**.** *Fix the marginals $\alpha, \beta$ and set $\xi := \alpha\otimes\beta$. Let $\Phi$ be a convex Gâteaux-differentiable functional on nonnegative measures on $\mathcal{X}\times\mathcal{Y}$. Its convex conjugate is, for a continuous test function $u$,*

$$\Phi^*(u) := \sup_{\pi\geqslant 0}\left\{\int u\,\mathrm{d}\pi - \Phi(\pi)\right\}.$$

*Assume that Fenchel duality is exact for this constrained problem, as happens in finite-dimensional discretizations and, more generally, under standard compactness and lower-semicontinuity hypotheses. Then*

$$\mathcal{L}^\varepsilon_{c,\Phi}(\alpha,\beta) = \sup_{f,g}\int f\mathrm{d}\alpha + \int g\mathrm{d}\beta - \varepsilon\left[\Phi^*\left(\delta\Phi(\xi) + \frac{f\oplus g - c}{\varepsilon}\right) - \Phi^*(\delta\Phi(\xi))\right]. \tag{7.50}$$

*If $(f^\star, g^\star)$ and $\pi^\star$ are optimal and the solution is interior, then*

$$\delta\Phi(\pi^\star) = \delta\Phi(\xi) + \frac{f^\star\oplus g^\star - c}{\varepsilon}. \tag{7.51}$$

*Proof.* Using (7.49), the Bregman-regularized objective can be written, up to a constant independent of $\pi$, as $\varepsilon\Phi(\pi) + \int\big(c - \varepsilon\delta\Phi(\xi)\big)\mathrm{d}\pi - \varepsilon\Phi(\xi) + \varepsilon\int\delta\Phi(\xi)\mathrm{d}\xi$. Introduce dual potentials $(f, g)$ for the two marginal constraints. The inner minimization over nonnegative measures $\pi$ gives

$$\inf_{\pi\geqslant 0}\left\{\varepsilon\Phi(\pi) + \int\big(c - (f\oplus g) - \varepsilon\delta\Phi(\xi)\big)\mathrm{d}\pi\right\} = -\varepsilon\Phi^*\left(\delta\Phi(\xi) + \frac{f\oplus g - c}{\varepsilon}\right).$$

Fenchel equality at $\xi$ gives $-\Phi(\xi) + \int\delta\Phi(\xi)\mathrm{d}\xi = \Phi^*(\delta\Phi(\xi))$, which yields (7.50). Equality in Fenchel's inequality gives the primal–dual relation (7.51). $\square$

When $\Phi$ is separable with respect to a fixed dominating measure $\xi_0$, say $\Phi(\pi) = \int h(\mathrm{d}\pi/\mathrm{d}\xi_0)\mathrm{d}\xi_0$ with $\xi \ll \xi_0$, the Bregman optimality condition becomes

$$h'\left(\frac{\mathrm{d}\pi^\star}{\mathrm{d}\xi_0}\right) = h'\left(\frac{\mathrm{d}\xi}{\mathrm{d}\xi_0}\right) + \frac{f^\star \oplus g^\star - c}{\varepsilon}.$$

This is an additive update in entropy coordinates. The density-ratio formulation instead uses the scalar law associated with $\varphi$ relative to $\alpha \otimes \beta$.

**Proposition 7.33** (KL is the common Bregman and $\varphi$ case)**.** *Let $\omega$ be a finite reference measure on $\mathcal{X}$ with a measurable set $A$ satisfying $0 < \omega(A) < \omega(\mathcal{X})$. Work on probability measures $\alpha = p\omega$ and $\beta = q\omega$ whose densities are bounded above and below away from 0. Assume that $\Phi$ is twice Gâteaux differentiable along bounded zero-mass density perturbations, and that $\varphi \in C^2(0, +\infty)$ is convex with $\varphi(1) = 0$. If $B_\Phi(\alpha|\beta) = \mathcal{D}_\varphi(\alpha|\beta)$ for all such $\alpha, \beta$, then there exist $\kappa \geqslant 0$ and $a \in \mathbb{R}$ such that $\varphi(t) = \kappa\, t \log t + a(t-1)$. Hence the common divergence is $\kappa\,\mathrm{KL}(\alpha|\beta)$, and $\Phi(p\omega)$ differs from $\kappa \int p \log p\, \mathrm{d}\omega$ by an affine functional on the positive probability simplex.*

*Proof.* Fix $\beta = q\omega$ and perturb it by $\alpha_t = (q + th)\omega$, where $h$ is bounded, $\int h \mathrm{d}\omega = 0$, and $t$ is small enough that $q + th > 0$. Differentiating twice at $t = 0$ gives $D^2\Phi(q)[h, h] = \varphi''(1) \int \frac{h^2}{q} \mathrm{d}\omega$. Indeed the left-hand side is the second variation of the Bregman error, while

$$\mathcal{D}_\varphi(\alpha_t|\beta) = \int q\, \varphi\left(1 + th/q\right) \mathrm{d}\omega$$

has second derivative $\varphi''(1) \int h^2/q\, \mathrm{d}\omega$. Setting $\kappa = \varphi''(1) \geqslant 0$, the functional $\Psi(p\omega) = \Phi(p\omega) - \kappa \int p \log p \mathrm{d}\omega$ has zero second variation along every zero-mass line segment in the positive simplex. It is therefore affine there, and $B_\Psi = 0$. Thus $B_\Phi = \kappa\,\mathrm{KL}$. Since $\mathcal{D}_\varphi = \kappa\,\mathrm{KL}$, the function $r(t) = \varphi(t) - \kappa t \log t$ generates the zero $\varphi$-divergence. Testing on densities for which the ratio $p/q$ takes two values $x < 1 < y$, with weights chosen so that the mean ratio is 1, gives $(y-1)r(x) + (1-x)r(y) = 0$. Hence $r(t)/(t-1)$ is constant on $(0, +\infty) \setminus \{1\}$, so $r(t) = a(t-1)$. This proves the claim. □

Thus, except for KL, the two generalizations lead to different duals and different algorithms. Bregman regularization by $B_\Phi(\pi|\xi)$ keeps the projection geometry of Section 8.1: linear costs tilt the reference in dual coordinates and alternating marginal updates are Bregman projections.

**Generalized soft $c$-transforms and alternate dual maximization method.**

$$\begin{aligned} g^{\bar{c},\varepsilon,\varphi}(x) &\in \underset{u\in\mathbb{R}}{\operatorname{argmin}} \left\{\varepsilon \int_{\mathcal{Y}} \varphi^*\left(\frac{u + g(y) - c(x,y)}{\varepsilon}\right) \mathrm{d}\beta(y) - u\right\}, \\ f^{c,\varepsilon,\varphi}(y) &\in \underset{v\in\mathbb{R}}{\operatorname{argmin}} \left\{\varepsilon \int_{\mathcal{X}} \varphi^*\left(\frac{f(x) + v - c(x,y)}{\varepsilon}\right) \mathrm{d}\alpha(x) - v\right\}. \end{aligned} \tag{7.52}$$

When $\varphi^*$ is differentiable, the first scalar minimizer is characterized by

$$\int_{\mathcal{Y}} (\varphi^*)'\left(\frac{u + g(y) - c(x,y)}{\varepsilon}\right) \mathrm{d}\beta(y) = 1,$$

The Bregman dual (7.50) has analogous block transforms, but the global conjugate $\Phi^*$ prevents a pointwise definition in general. With $\xi = \alpha \otimes \beta$, define

$$\begin{aligned} g^{\bar{c},\varepsilon,\Phi} &\in \underset{u\in\mathcal{C}(\mathcal{X})}{\operatorname{argmin}} \left\{\varepsilon\Phi^*\left(\delta\Phi(\xi) + \frac{u \oplus g - c}{\varepsilon}\right) - \int u \mathrm{d}\alpha\right\}, \\ f^{c,\varepsilon,\Phi} &\in \underset{v\in\mathcal{C}(\mathcal{Y})}{\operatorname{argmin}} \left\{\varepsilon\Phi^*\left(\delta\Phi(\xi) + \frac{f \oplus v - c}{\varepsilon}\right) - \int v \mathrm{d}\beta\right\}. \end{aligned} \tag{7.53}$$

$$\begin{aligned} f^{(\ell+1)} &= \left(g^{(\ell)}\right)^{\bar{c},\varepsilon,\varphi}, \quad g^{(\ell+1)} = \left(f^{(\ell+1)}\right)^{c,\varepsilon,\varphi}, \\ f^{(\ell+1)} &= \left(g^{(\ell)}\right)^{\bar{c},\varepsilon,\Phi}, \quad g^{(\ell+1)} = \left(f^{(\ell+1)}\right)^{c,\varepsilon,\Phi}, \end{aligned} \tag{7.54}$$

$$\mathrm{P}^\star_{i,j} = a_i b_j \left(1 + \frac{f^\star_i + g^\star_j - \mathrm{C}_{i,j}}{\varepsilon}\right)_+, \qquad \mathrm{P}^\star_{i,j} = \left(a_i b_j + \frac{f^\star_i + g^\star_j - \mathrm{C}_{i,j}}{\varepsilon}\right)_+. \tag{7.55}$$

$$\sum_j b_j \left(1 + \frac{u + g_j - \mathrm{C}_{i,j}}{\varepsilon}\right)_+ = 1.$$

This left-hand side is continuous, nondecreasing, and piecewise affine. Sorting its breakpoints computes the update in $O(m \log m)$ operations for a row with $m$ entries; equivalently, it is a weighted simplex projection.

## 7.10 Sinkhorn Divergences

**Entropic bias.** $\beta_\varepsilon \in \underset{\beta}{\operatorname{argmin}}\, \mathcal{L}^\varepsilon_c(\alpha, \beta)$

**Proposition 7.34** (Large-temperature entropic bias)**.** *Let $\mathcal{X}, \mathcal{Y}$ be compact and assume that $c$ is continuous. Then $\mathcal{L}^\varepsilon_c(\alpha, \beta) \to \iint c(x, y)\mathrm{d}\alpha(x)\mathrm{d}\beta(y)$ as $\varepsilon \to +\infty$.*

*Proof.* Let $(f_\varepsilon, g_\varepsilon)$ be optimal dual potentials, normalized by $\int g_\varepsilon \mathrm{d}\beta = 0$. The soft $c$-transform equation gives

$$f_\varepsilon(x) = -\varepsilon \log \int \exp\left(\frac{g_\varepsilon(y) - c(x,y)}{\varepsilon}\right) \mathrm{d}\beta(y).$$

For bounded $c$, the oscillations of normalized entropic potentials are bounded uniformly in $\varepsilon$ by the oscillation of $c$. Hence the log-sum-exp expansion is uniform: $f_\varepsilon(x) = -\int (g_\varepsilon(y) - c(x, y))\mathrm{d}\beta(y) + O(\varepsilon^{-1}) = \int c(x, y)\mathrm{d}\beta(y) + O(\varepsilon^{-1})$. At optimality the exponential penalty in the dual integrates to zero, so $\mathcal{L}^\varepsilon_c(\alpha, \beta) = \int f_\varepsilon \mathrm{d}\alpha + \int g_\varepsilon \mathrm{d}\beta = \iint c(x, y)\mathrm{d}\alpha(x)\mathrm{d}\beta(y) + O(\varepsilon^{-1})$. This proves the limit. □

Thus, at large $\varepsilon$, the raw entropic cost approaches the bilinear cross-energy $(\alpha, \beta) \mapsto \iint c\,\mathrm{d}\alpha\mathrm{d}\beta$, not a distance.
**Sinkhorn divergences.**

**Definition 7.35** (Sinkhorn divergence)**.** Let $c$ be a symmetric cost on a common state space. For $\varepsilon > 0$, the debiased Sinkhorn divergence associated with the entropic OT value $\mathcal{L}_c^\varepsilon$ is

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) := \mathcal{L}_c^\varepsilon(\alpha, \beta) - \frac{1}{2}\mathcal{L}_c^\varepsilon(\alpha, \alpha) - \frac{1}{2}\mathcal{L}_c^\varepsilon(\beta, \beta). \tag{7.56}$$

**Lemma 7.36** (Entropic dual cost at optimum)**.** *Let $(f_{\alpha,\beta}, g_{\alpha,\beta})$ be optimal dual potentials, normalized arbitrarily. Then*

$$\mathcal{L}_c^\varepsilon(\alpha, \beta) = \langle f_{\alpha,\beta},\, \alpha\rangle + \langle g_{\alpha,\beta},\, \beta\rangle. \tag{7.57}$$

*Proof.* We first notice that at optimality, the relation $f_{\alpha,\beta} = -\varepsilon \log \int_{\mathcal{Y}} e^{\frac{g_{\alpha,\beta}(y) - c(x,y)}{\varepsilon}} \mathrm{d}\beta(y)$ after taking the exponential, equivalently reads

$$1 = \int_{\mathcal{Y}} e^{\frac{f_{\alpha,\beta}(x) + g_{\alpha,\beta}(y) - c(x,y)}{\varepsilon}} \mathrm{d}\beta(y) \quad\Longrightarrow\quad \int_{\mathcal{X}\times\mathcal{Y}} \left(e^{\frac{f_{\alpha,\beta}\oplus g_{\alpha,\beta} - c}{\varepsilon}} - 1\right)\mathrm{d}(\alpha\otimes\beta) = 0.$$

Substituting this identity in (7.21) gives the result. ◻

**Proposition 7.37** (Asymptotics of Sinkhorn divergences)**.** *Let $\mathcal{X}$ be a compact metric space, let $\alpha, \beta \in \mathcal{P}(\mathcal{X})$, and assume that $c : \mathcal{X}\times\mathcal{X} \to \mathbb{R}_+$ is continuous and symmetric, with $c(x,x) = 0$. Then $\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta) \to \mathcal{L}_c(\alpha,\beta)$ when $\varepsilon \to 0$ and*

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) \to \frac{1}{2}\int -c\mathrm{d}(\alpha-\beta)\otimes\mathrm{d}(\alpha-\beta) \quad \textit{when} \quad \varepsilon \to +\infty.$$

*Proof.* The discrete convergence result above gives the finite-dimensional analogue; we now pass to continuous measures. **Case $\varepsilon \to 0$.** The first limit is the $\Gamma$-convergence of entropic OT to the Kantorovich problem. Non-negativity of the relative entropy and continuity of $c$ give the liminf inequality. For the matching limsup, one approximates an optimal coupling by finite-entropy couplings and chooses the approximation scale so that its entropy multiplied by $\varepsilon$ tends to zero. Since $c(x,x) = 0$, the two self-costs in the debiased expression converge to zero, while the cross term converges to $\mathcal{L}_c(\alpha,\beta)$.
**Case $\varepsilon \to +\infty$.** We denote by $(f_\varepsilon, g_\varepsilon)$ optimal dual potentials. After normalizing them and using boundedness of $c$, their oscillations stay uniformly bounded, so the following expansion is uniform. The optimality condition on $f_\varepsilon$ (equivalently the Sinkhorn fixed point on $f_\varepsilon$) reads

$$\begin{aligned} f_\varepsilon &= -\varepsilon\log\int \exp\left(\frac{g_\varepsilon(y) - c(\cdot,y)}{\varepsilon}\right)\mathrm{d}\beta(y) = -\varepsilon\log\int\left(1 + \frac{g_\varepsilon(y) - c(\cdot,y)}{\varepsilon} + o(1/\varepsilon)\right)\mathrm{d}\beta(y) \\ &= -\varepsilon\log\left(1 + \frac{1}{\varepsilon}\int (g_\varepsilon(y) - c(\cdot,y))\mathrm{d}\beta(y) + o(1/\varepsilon)\right) = -\int g_\varepsilon\mathrm{d}\beta + \int c(\cdot,y)\mathrm{d}\beta(y) + o(1). \end{aligned}$$

Plugging this relation in the dual expression (7.57) $\mathcal{L}_c^\varepsilon(\alpha,\beta) = \int f_\varepsilon\mathrm{d}\alpha + \int g_\varepsilon\mathrm{d}\beta = \iint c(x,y)\mathrm{d}\alpha(x)\mathrm{d}\beta(y) + o(1)$. Applying this expansion to $(\alpha,\beta)$, $(\alpha,\alpha)$ and $(\beta,\beta)$ gives

$$\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta) \to \int c\,\mathrm{d}\alpha\otimes\mathrm{d}\beta - \frac{1}{2}\int c\,\mathrm{d}\alpha\otimes\mathrm{d}\alpha - \frac{1}{2}\int c\,\mathrm{d}\beta\otimes\mathrm{d}\beta = -\frac{1}{2}\int c\,\mathrm{d}(\alpha-\beta)\otimes\mathrm{d}(\alpha-\beta).$$

◻

**Proposition 7.38** (Positivity of Sinkhorn divergences)**.** *Assume that the entropic dual admits optimal potentials and that the symmetric kernel $k_\varepsilon(x,y) = e^{-c(x,y)/\varepsilon}$ is positive semidefinite in the sense of Definition 6.4. Then $\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta) \geqslant 0$. If, in addition, the common state space is compact, $c$ is continuous, and $k_\varepsilon$ is universal in the sense of Definition 6.8, then $\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta) = 0 \iff \alpha = \beta$, and $\bar{\mathcal{L}}_c^\varepsilon(\alpha_n,\alpha) \to 0$ is equivalent to $\alpha_n \rightharpoonup \alpha$.*

*Proof.* In the following, we denote by $(f_{\alpha,\beta}, g_{\alpha,\beta})$ optimal dual potentials for the dual Schrödinger problem between $\alpha$ and $\beta$. We denote by $f_{\alpha,\alpha} = g_{\alpha,\alpha}$ (one can assume they are equal by symmetry) the solution for the problem between $\alpha$ and itself.
Using the suboptimal function $(f_{\alpha,\alpha}, g_{\beta,\beta})$ in the dual maximization problem, and using relation (7.57) for the simplified expression of the dual cost, one obtains

$$\mathcal{L}_c^\varepsilon(\alpha,\beta) \geqslant \langle f_{\alpha,\alpha},\,\alpha\rangle + \langle g_{\beta,\beta},\,\beta\rangle - \varepsilon\langle e^{\frac{f_{\alpha,\alpha}\oplus g_{\beta,\beta} - c}{\varepsilon}} - 1,\, \alpha\otimes\beta\rangle$$

Moreover $\langle f_{\alpha,\alpha},\,\alpha\rangle = \frac{1}{2}\mathcal{L}_c^\varepsilon(\alpha,\alpha)$, and similarly for $\beta$, so the previous inequality equivalently reads

$$\frac{1}{\varepsilon}\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta) \geqslant 1 - \langle e^{\frac{f_{\alpha,\alpha}\oplus g_{\beta,\beta} - c}{\varepsilon}},\, \alpha\otimes\beta\rangle = 1 - \langle\tilde{\alpha},\,\tilde{\beta}\rangle_{k_\varepsilon}$$

where $\tilde{\alpha} = e^{f_{\alpha,\alpha}/\varepsilon}\alpha$, $\tilde{\beta} = e^{f_{\beta,\beta}/\varepsilon}\beta$ and we introduced the positive-semidefinite kernel pairing $\langle\tilde{\alpha},\,\tilde{\beta}\rangle_{k_\varepsilon} := \int k_\varepsilon(x,y)\mathrm{d}\tilde{\alpha}(x)\mathrm{d}\tilde{\beta}(y)$. The self Sinkhorn fixed point equation, once exponentiated, reads pointwise $e^{f_{\alpha,\alpha}(x)/\varepsilon}\int k_\varepsilon(x,y)\mathrm{d}\tilde{\alpha}(y) = 1 \quad$ for $\alpha$-a.e. $x$, and hence

$$\|\tilde{\alpha}\|_{k_\varepsilon}^2 = \langle\tilde{\alpha},\,\tilde{\alpha}\rangle_{k_\varepsilon} = \int e^{f_{\alpha,\alpha}(x)/\varepsilon}\left(\int k_\varepsilon(x,y)\mathrm{d}\tilde{\alpha}(y)\right)\mathrm{d}\alpha(x) = 1$$

and similarly $\|\tilde{\beta}\|_{k_\varepsilon}^2 = 1$. Therefore, by Cauchy–Schwarz, one has $1 - \langle\tilde{\alpha},\,\tilde{\beta}\rangle_{k_\varepsilon} \geqslant 0$.
Suppose now that $k_\varepsilon$ is universal and $\bar{\mathcal{L}}_c^\varepsilon(\alpha,\beta) = 0$. Both inequalities above are then equalities. Equality in Cauchy–Schwarz, together with the unit norms, implies that $\tilde{\alpha}$ and $\tilde{\beta}$ have the same kernel mean embedding; universality gives $\tilde{\alpha} = \tilde{\beta} := \tilde{\mu}$. The two self-consistency equations yield $\mathrm{d}\alpha = (k_\varepsilon\tilde{\mu})\,\mathrm{d}\tilde{\mu} = \mathrm{d}\beta, \quad (k_\varepsilon\tilde{\mu})(x) := \int k_\varepsilon(x,y)\mathrm{d}\tilde{\mu}(y)$, so $\alpha = \beta$. The converse follows directly from the definition. Finally, on a compact space, continuity of the entropic value under weak convergence and compactness of $\mathcal{P}(\mathcal{X})$ turn this definiteness into the stated equivalence of convergences; see. ◻

### 7.11 Complex $\varepsilon$

**Measure fixed point.** Let $\alpha \in \mathcal{P}(\mathcal{X})$ and $\beta \in \mathcal{P}(\mathcal{Y})$. For $\varepsilon \in \mathbb{C} \setminus \{0\}$, reuse the Gibbs kernel and integral operators of (7.13) whenever the defining integrals exist.

**Discrete histograms.** For discrete histograms, evaluate the Gibbs kernel entrywise at $(x_i, y_j)$. The scaling factorization (7.2) and Sinkhorn updates (7.5) are then used verbatim over $\mathbb{C}$, with the same marginal constraints and multiplicative gauge, whenever $\varepsilon \neq 0$ and all divisions are defined.

**Theorem 7.39** (Local holomorphic continuation of Sinkhorn scalings)**.** *Fix positive histograms $\mathrm{a}^0 \in \Sigma_n$, $\mathrm{b}^0 \in \Sigma_m$, a finite real cost matrix $\mathrm{C}^0$, and a real temperature $\varepsilon_0 > 0$. Choose positive scalings $(\mathrm{u}^0, \mathrm{v}^0)$ of the corresponding Sinkhorn coupling. Then there are complex neighborhoods of $(\varepsilon_0, \mathrm{a}^0, \mathrm{b}^0, \mathrm{C}^0)$, inside the affine constraint $\sum_i \mathrm{a}_i = \sum_j \mathrm{b}_j$, and a unique holomorphic map $(\varepsilon, \mathrm{a}, \mathrm{b}, \mathrm{C}) \mapsto (\mathrm{u}_\varepsilon, \mathrm{v}_\varepsilon) \in \mathbb{C}^n \times \mathbb{C}^m$ satisfying the linear gauge $\sum_i \mathrm{a}_i^0 u_{\varepsilon,i} = \sum_i \mathrm{a}_i^0 u_i^0$, such that $\mathrm{P}_\varepsilon = \mathrm{diag}(\mathrm{u}_\varepsilon)\mathrm{K}_\varepsilon(\mathrm{C})\,\mathrm{diag}(\mathrm{v}_\varepsilon)$ has marginals $(\mathrm{a}, \mathrm{b})$. In particular, the gauge-fixed scalings and $\mathrm{P}_\varepsilon$ are holomorphic in all four arguments near the base point.*

*Proof.* Apply the holomorphic implicit-function theorem to the $n$ row residuals, the first $m-1$ column residuals, and the gauge residual $G(\mathrm{u}) = \sum_i \mathrm{a}_i^0(u_i - u_i^0)$. There are $n+m$ equations for the $n+m$ scaling coordinates.

Let $(\delta\mathrm{u}, \delta\mathrm{v})$ belong to its kernel and write $r_i := \frac{\delta u_i}{u_i^0}$, $\quad s_j := \frac{\delta v_j}{v_j^0}$. Since $\mathrm{P}^0 = \mathrm{diag}(\mathrm{u}^0)\mathrm{K}_{\varepsilon_0}(\mathrm{C}^0)\,\mathrm{diag}(\mathrm{v}^0)$, the induced perturbation is $\delta\mathrm{P}_{i,j} = \mathrm{P}_{i,j}^0(r_i + s_j)$. The linearized row sums vanish. The first $m-1$ linearized column sums vanish by definition of the residuals, and the last one vanishes because the total row sum equals the total column sum. Therefore $0 = \sum_i \overline{r_i} \sum_j \delta\mathrm{P}_{i,j} + \sum_j \overline{s_j} \sum_i \delta\mathrm{P}_{i,j} = \sum_{i,j} \mathrm{P}_{i,j}^0 \left|r_i + s_j\right|^2$. Strict positivity of $\mathrm{P}^0$ gives $r_i = \theta$ and $s_j = -\theta$ for some $\theta \in \mathbb{C}$. The linearized gauge is $0 = \sum_i \mathrm{a}_i^0 \delta u_i = \theta \sum_i \mathrm{a}_i^0 u_i^0$; its last factor is positive, hence $\theta = 0$. The derivative is injective and therefore invertible. The implicit-function theorem now gives the asserted holomorphic scalings, and their matrix product gives the holomorphic coupling. $\square$

## 8 Entropic Regularization: Convergence

The chapter revisits Sinkhorn convergence through several complementary lenses. Bregman projections explain the alternating-projection geometry, while Fortet's order argument gives qualitative fixed-point convergence.

### 8.1 Sinkhorn Convergence: Bregman Point of View

**Alternating KL projections.**

**Definition 8.1** (Bregman divergence)**.** Let $\Omega \subset \mathbb{R}^{n\times m}$ be convex with nonempty interior, and let $\Phi : \mathbb{R}^{n\times m} \to \mathbb{R} \cup \{+\infty\}$ be a proper lower-semicontinuous Legendre function with $\mathrm{dom}(\Phi) = \Omega$. Thus $\Phi = +\infty$ outside $\Omega$, while $\Phi$ is differentiable and strictly convex on $\mathrm{int}(\Omega)$. For $\mathrm{P}, \mathrm{Q} \in \mathrm{int}(\Omega)$, the Bregman divergence generated by $\Phi$ is $B_\Phi(\mathrm{P}|\mathrm{Q}) := \Phi(\mathrm{P}) - \Phi(\mathrm{Q}) - \langle \nabla\Phi(\mathrm{Q}), \mathrm{P} - \mathrm{Q}\rangle$.

**Definition 8.2** (Bregman projection)**.** For a nonempty convex set $\mathcal{C}$, define $\mathrm{Proj}_{\mathcal{C}}^{B_\Phi}(\mathrm{Q}) := \mathrm{argmin}_{\mathrm{P}\in\mathcal{C}} B_\Phi(\mathrm{P}|\mathrm{Q})$, which is set-valued if the minimizer is not unique.

**Linear tilts and Gibbs references.**

**Proposition 8.3** (Linear tilts of Bregman penalties)**.** *Let $\Phi$ be differentiable and strictly convex, and let $B_\Phi$ be its Bregman divergence. Fix a reference point $\mathrm{Q}$ in the interior of the domain. Assume that there exists $\mathrm{Q}^{\mathrm{C}}$ such that $\nabla\Phi(\mathrm{Q}^{\mathrm{C}}) = \nabla\Phi(\mathrm{Q}) - \mathrm{C}/\varepsilon$. Then, for all $\mathrm{P}$ in the domain, $\langle \mathrm{P}, \mathrm{C}\rangle + \varepsilon B_\Phi(\mathrm{P}|\mathrm{Q}) = \varepsilon B_\Phi(\mathrm{P}|\mathrm{Q}^{\mathrm{C}}) + \mathrm{cst}$, where the constant does not depend on $\mathrm{P}$.*

*Proof.* Subtract the two Bregman divergences: $B_\Phi(\mathrm{P}|\mathrm{Q}^{\mathrm{C}}) - B_\Phi(\mathrm{P}|\mathrm{Q}) = \langle \nabla\Phi(\mathrm{Q}) - \nabla\Phi(\mathrm{Q}^{\mathrm{C}}), \mathrm{P}\rangle + \mathrm{cst}$. Using $\nabla\Phi(\mathrm{Q}) - \nabla\Phi(\mathrm{Q}^{\mathrm{C}}) = \mathrm{C}/\varepsilon$ and multiplying by $\varepsilon$ gives the claim. $\square$

For the negative entropy $\Phi(\mathrm{P}) = \sum_{i,j} \mathrm{P}_{i,j} \log \mathrm{P}_{i,j}$, one has $B_\Phi = \mathrm{KL}$. Taking $\mathrm{Q} = \mathrm{a} \otimes \mathrm{b}$ gives the tilted reference $\mathrm{K}_{\mathrm{a,b}}^\varepsilon := (\mathrm{a} \otimes \mathrm{b}) \odot e^{-\mathrm{C}/\varepsilon}$. $\langle \mathrm{P}, \mathrm{C}\rangle + \varepsilon \mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) = \varepsilon \mathrm{KL}(\mathrm{P}|\mathrm{K}_{\mathrm{a,b}}^\varepsilon) + \mathrm{cst}$. Thus the unique solution $\mathrm{P}_\varepsilon$ of (7.1) is the KL projection of the tilted Gibbs reference onto $\mathrm{U}(\mathrm{a}, \mathrm{b})$:

$$\mathrm{P}_\varepsilon = \mathrm{Proj}_{\mathrm{U(a,b)}}^{\mathrm{KL}}(\mathrm{K}_{\mathrm{a,b}}^\varepsilon) := \underset{\mathrm{P}\in\mathrm{U(a,b)}}{\mathrm{argmin}}\ \mathrm{KL}(\mathrm{P}|\mathrm{K}_{\mathrm{a,b}}^\varepsilon). \tag{8.1}$$

**Cyclic projection convergence.** Given two closed convex constraint sets $\mathcal{C}_1$ and $\mathcal{C}_2$, the cyclic method projects first onto $\mathcal{C}_1$, then onto $\mathcal{C}_2$, and repeats. Full iterates satisfy the second constraint and half-iterates satisfy the first; both constraints are attained together only in the limit.

**Proposition 8.4** (Convergence of cyclic Bregman projections)**.** *Let $\Phi$ be a Legendre generator on a finite-dimensional convex domain, and let $\mathcal{C}_1, \mathcal{C}_2$ be closed convex sets whose intersection meets the interior of the domain. Generate $(\mathrm{P}^{(\ell-1/2)}, \mathrm{P}^{(\ell)})$ by Algorithm ?? from an interior point $\mathrm{P}^{(0)}$. Assume that the projections are well-defined and that the full half-step sequence remains in a compact subset of the interior of the domain. Then there exists $\bar{\mathrm{P}} \in \mathcal{C}_1 \cap \mathcal{C}_2$ such that $\mathrm{P}^{(\ell)} \to \bar{\mathrm{P}}$ and $\mathrm{P}^{(\ell-1/2)} \to \bar{\mathrm{P}}$. If, in addition, $\mathcal{C}_1$ and $\mathcal{C}_2$ are affine, then $\bar{\mathrm{P}} = \mathrm{Proj}_{\mathcal{C}_1\cap\mathcal{C}_2}^{B_\Phi}(\mathrm{P}^{(0)})$.*

*Proof.* For three interior points, the Bregman three-point formula reads $B_\Phi(\mathrm{Q}|\mathrm{P}) = B_\Phi(\mathrm{Q}|\mathrm{P}^+) + B_\Phi(\mathrm{P}^+|\mathrm{P}) + \langle \nabla\Phi(\mathrm{P}^+) - \nabla\Phi(\mathrm{P}), \mathrm{Q} - \mathrm{P}^+\rangle$. If $\mathrm{P}^+ = \mathrm{Proj}_{\mathcal{C}}^{B_\Phi}(\mathrm{P})$ and $\mathcal{C}$ is closed and convex, first-order optimality gives $\langle \nabla\Phi(\mathrm{P}^+) - \nabla\Phi(\mathrm{P}), \mathrm{Q} - \mathrm{P}^+\rangle \geqslant 0 \quad \forall \mathrm{Q} \in \mathcal{C}$. $B_\Phi(\mathrm{Q}|\mathrm{P}) \geqslant B_\Phi(\mathrm{Q}|\mathrm{P}^+) + B_\Phi(\mathrm{P}^+|\mathrm{P}) \qquad \forall \mathrm{Q} \in \mathcal{C}$.

Let $(Z^{(r)})_r$ be the half-step sequence, so that $Z^{(2\ell)} = \mathrm{P}^{(\ell)}$ and $Z^{(2\ell+1)} = \mathrm{P}^{(\ell+1/2)}$. Fix $\mathrm{Q} \in \mathcal{C}_1 \cap \mathcal{C}_2$. Applying the inequality at each half-step gives $B_\Phi(\mathrm{Q}|Z^{(r)}) - B_\Phi(\mathrm{Q}|Z^{(r+1)}) \geqslant B_\Phi(Z^{(r+1)}|Z^{(r)}) \geqslant 0$. Thus $B_\Phi(\mathrm{Q}|Z^{(r)})$ decreases and $\sum_r B_\Phi(Z^{(r+1)}|Z^{(r)}) < +\infty$. The projection drops tend to zero; compactness and strict convexity imply $\|Z^{(r+1)} - Z^{(r)}\| \to 0$. Every cluster point of either subsequence consequently belongs to both closed sets. Let $\bar{\mathrm{P}} \in \mathcal{C}_1 \cap \mathcal{C}_2$ be one such cluster point. The sequence $B_\Phi(\bar{\mathrm{P}}|Z^{(r)})$ is nonincreasing and tends to zero along a convergent subsequence, hence tends to zero altogether. Compactness and strict convexity then give $Z^{(r)} \to \bar{\mathrm{P}}$, proving the first claim.

Suppose now that $\mathcal{C}_1$ and $\mathcal{C}_2$ are affine. The first-order inequality above is then an equality because both signs of every feasible direction are admissible. Moreover, each dual displacement $\nabla\Phi(Z^{(r+1)}) - \nabla\Phi(Z^{(r)})$ belongs to the normal space of the affine set used at that half-step. Telescoping and passing to the limit gives $\nabla\Phi(\bar{\mathrm{P}}) - \nabla\Phi(\mathrm{P}^{(0)}) \in N_{\mathcal{C}_1} + N_{\mathcal{C}_2} = N_{\mathcal{C}_1\cap\mathcal{C}_2}$, where the last equality uses affinity and the nonempty intersection. This is precisely the first-order optimality condition for minimizing $\mathrm{Q} \mapsto B_\Phi(\mathrm{Q}|\mathrm{P}^{(0)})$ over $\mathcal{C}_1 \cap \mathcal{C}_2$, proving the displayed projection identity. $\square$

**Row and column scalings.**

$$\mathcal{C}^1_{\mathrm{a}} := \left\{\mathrm{P} \in \mathbb{R}^{n\times m} \; ; \; \mathrm{P}\mathbb{1}_m = \mathrm{a}\right\} \quad \text{and} \quad \mathcal{C}^2_{\mathrm{b}} := \left\{\mathrm{P} \in \mathbb{R}^{n\times m} \; ; \; \mathrm{P}^\top\mathbb{1}_n = \mathrm{b}\right\}.$$

$\mathrm{U}(\mathrm{a},\mathrm{b}) = \mathcal{C}^1_{\mathrm{a}}\cap\mathcal{C}^2_{\mathrm{b}}\cap\mathbb{R}^{n\times m}_+$. For negative entropy, however, the effective domain of the Legendre generator is $\Omega = \mathbb{R}^{n\times m}_+$ and $\Phi = +\infty$ outside $\Omega$. Thus the two half-steps of Algorithm **??** specialize to

$$\mathrm{P}^{(\ell+1)} := \mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}^1_{\mathrm{a}}}(\mathrm{P}^{(\ell)}) \quad \text{and} \quad \mathrm{P}^{(\ell+2)} := \mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}^2_{\mathrm{b}}}(\mathrm{P}^{(\ell+1)}). \tag{8.2}$$

**Proposition 8.5** (KL projections are scalings)**.** *One has* $\mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}^1_{\mathrm{a}}}(\mathrm{P}) = \mathrm{diag}\left(\frac{\mathrm{a}}{\mathrm{P}\mathbb{1}_m}\right)\mathrm{P}$ *and* $\mathrm{Proj}^{\mathrm{KL}}_{\mathcal{C}^2_{\mathrm{b}}}(\mathrm{P}) = \mathrm{P}\,\mathrm{diag}\left(\frac{\mathrm{b}}{\mathrm{P}^\top\mathbb{1}_n}\right)$.

*Proof.* Consider the problem along each row or column vector to impose a fixed sum $s \in \mathbb{R}_+$ $\min_p \{\mathrm{KL}(p|q) \; ; \; \langle p, \mathbb{1}\rangle = s\}$. The Lagrange multiplier equation for this problem reads $\log(p/q) + \lambda\mathbb{1} = 0 \implies p = uq$ where $u = e^{-\lambda} > 0$. The constraint $\langle p, \mathbb{1}\rangle = s$ is equivalent to $\langle uq, \mathbb{1}\rangle = s$, i.e. $u = s/\sum_i q_i$, which gives the desired scaling formula $p = sq/\sum_i q_i$. ☐

$$\mathrm{P}^{(2\ell)} := \mathrm{diag}(\mathrm{u}^{(\ell)})\mathrm{K}\,\mathrm{diag}(\mathrm{v}^{(\ell)}),$$

$$\begin{aligned} &\mathrm{P}^{(2\ell+1)} := \mathrm{diag}(\mathrm{u}^{(\ell+1)})\mathrm{K}\,\mathrm{diag}(\mathrm{v}^{(\ell)}) \\ \text{and} \quad &\mathrm{P}^{(2\ell+2)} := \mathrm{diag}(\mathrm{u}^{(\ell+1)})\mathrm{K}\,\mathrm{diag}(\mathrm{v}^{(\ell+1)}) \end{aligned}$$

**Other divergences.**

$$\mathcal{C}^1_{\mathrm{a},+} := \mathcal{C}^1_{\mathrm{a}} \cap \mathbb{R}^{n\times m}_+ \quad \text{and} \quad \mathcal{C}^2_{\mathrm{b},+} := \mathcal{C}^2_{\mathrm{b}} \cap \mathbb{R}^{n\times m}_+, \tag{8.3}$$

$\left[\mathrm{Proj}^{B_\Phi}_{\mathcal{C}^1_{\mathrm{a},+}}(\mathrm{Q})\right]_{i,j} = (\mathrm{Q}_{i,j} - \tau_i)_+$, $\quad \sum_j(\mathrm{Q}_{i,j} - \tau_i)_+ = a_i$, For the quadratic generator and the product reference $\xi = \mathrm{a}\otimes\mathrm{b}$, the Bregman penalty is

$B_\Phi(\mathrm{P}|\xi) = \frac{1}{2}\sum_{i,j}(\mathrm{P}_{i,j} - a_ib_j)^2$. $\mathcal{D}_\varphi(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}) = \frac{1}{2}\sum_{i,j}\frac{(\mathrm{P}_{i,j}-a_ib_j)^2}{a_ib_j}$.

## 8.2 Sinkhorn Convergence: Monotone Point of View

**Variation seminorm and topical maps.**

**Definition 8.6** (Variation seminorm)**.** For a bounded real-valued function $h$, its variation seminorm is $\|h\|_V := \sup h - \inf h$. It vanishes exactly on constant functions and therefore induces a norm after quotienting by additive constants.

**Definition 8.7** (Topical map)**.** Let $E$ be an ordered vector space of functions containing the constants. A map $\mathcal{T} : E \to E$ is *topical* if $f \leqslant g \Longrightarrow \mathcal{T}(f) \leqslant \mathcal{T}(g)$, $\qquad \mathcal{T}(f+s) = \mathcal{T}(f) + s \quad (s \in \mathbb{R})$.

**Proposition 8.8** (Topical maps are variation-nonexpansive)**.** *Let $E$ be a vector space of bounded real-valued functions, ordered pointwise. Every topical map $\mathcal{T} : E \to E$ satisfies $\|\mathcal{T}(f) - \mathcal{T}(g)\|_V \leqslant \|f - g\|_V$. The same conclusion holds if $\mathcal{T}$ is order reversing and additively anti-homogeneous, meaning $\mathcal{T}(f+s) = \mathcal{T}(f) - s$.*

*Proof.* Set $a = \inf(f-g)$ and $b = \sup(f-g)$, so that $g + a \leqslant f \leqslant g + b$. If $\mathcal{T}$ is topical, then $\mathcal{T}(g) + a \leqslant \mathcal{T}(f) \leqslant \mathcal{T}(g) + b$. Hence the oscillation of $\mathcal{T}(f) - \mathcal{T}(g)$ is at most $b - a = \|f - g\|_V$. In the order-reversing, additively anti-homogeneous case, the corresponding bounds are $\mathcal{T}(g) - b \leqslant \mathcal{T}(f) \leqslant \mathcal{T}(g) - a$ and give the same conclusion. ☐

**Generalized Sinkhorn maps.** Consider the $\varphi$-divergence regularized transport problem of Section 7.9. Its one-sided block updates, expressed using the Legendre transform $\varphi^*$ defined in (6.9), are the generalized soft $c$-transforms of (7.52); their alternating dual-maximization iteration is (7.54).

$$\mathcal{A}_\varphi(f) := \left(f^{c,\varepsilon,\varphi}\right)^{\bar{c},\varepsilon,\varphi}. \tag{8.4}$$

**Proposition 8.9** (Generalized Sinkhorn maps are topical)**.** *Assume that the scalar minimizer sets in (7.52) are nonempty compact intervals, and select either their smallest elements everywhere or their largest elements everywhere. Each selected one-sided transform is order reversing and additively anti-homogeneous:*

$$g \leqslant g' \Longrightarrow g^{\bar{c},\varepsilon,\varphi} \geqslant g'^{\bar{c},\varepsilon,\varphi}, \qquad (g+s)^{\bar{c},\varepsilon,\varphi} = g^{\bar{c},\varepsilon,\varphi} - s,$$

*and likewise for $f \mapsto f^{c,\varepsilon,\varphi}$. $\|\mathcal{A}_\varphi(f) - \mathcal{A}_\varphi(h)\|_V \leqslant \|f - h\|_V$. No selection convention is needed when the scalar minimizers are unique.*

*Proof.* Since $\varphi$ is extended by $+\infty$ on $(-\infty, 0)$, its Legendre transform $\varphi^*$ from (6.9) is convex and nondecreasing. For fixed $x$, let $H^x_g(u)$ denote the scalar objective in the first line of (7.52). If $g \leqslant g'$, then $u \mapsto H^x_{g'}(u) - H^x_g(u)$ is nondecreasing: this follows by integrating the fact that $s \mapsto \varphi^*(s+\delta) - \varphi^*(s)$ is nondecreasing for every $\delta \geqslant 0$.
Let $I = \mathrm{argmin}\, H^x_g$ and $I' = \mathrm{argmin}\, H^x_{g'}$. If $a = \min I$ and $a' = \min I'$ satisfied $a' > a$, the optimality inequalities $H^x_g(a) \leqslant H^x_g(a')$ and $H^x_{g'}(a') \leqslant H^x_{g'}(a)$ would contradict the preceding monotonicity; in the equality case, $a$ would also belong to $I'$, contradicting the definition of $a'$. Hence $\min I' \leqslant \min I$. The same argument with the maximal minimizers gives $\max I' \leqslant \max I$. Finally, $H^x_{g+s}(u) = H^x_g(u+s) + s$, so either extremal selection is additively anti-homogeneous. The second one-sided transform is treated symmetrically. Their composition is topical, and Proposition 8.8 gives the final estimate. ☐

**Monotone convergence for generalized Sinkhorn.**

**Proposition 8.10** (Monotone convergence of generalized Sinkhorn cycles)**.** *Let $\mathcal{X}$ and $\mathcal{Y}$ be compact metric spaces and $c \in \mathcal{C}(\mathcal{X}\times\mathcal{Y})$. Suppose that the assumptions of Proposition 8.9 hold, that the selected transforms are finite valued, and that $\mathcal{A}_\varphi$ has a fixed point $f^\star \in \mathcal{C}(\mathcal{X})$. If $f^{(0)} \in \mathcal{C}(\mathcal{X})$ is a subsolution, $f^{(0)} \leqslant \mathcal{A}_\varphi(f^{(0)})$, then the iterates $f^{(\ell+1)} = \mathcal{A}_\varphi(f^{(\ell)})$ converge uniformly to a fixed point of $\mathcal{A}_\varphi$. The analogous conclusion holds for a supersolution, with a decreasing sequence. If the fixed point is unique modulo additive constants, the corresponding potential classes converge to this unique class.*

*Proof.* Because $\mathcal{A}_\varphi$ is additively homogeneous, shifting a subsolution preserves the subsolution inequality. After such a shift, compactness of $\mathcal{X}$ allows us to assume $f^{(0)} \leqslant f^\star$. Topicality then gives $f^{(0)} \leqslant f^{(1)} \leqslant \cdots \leqslant f^\star$, so the orbit converges pointwise. The one-sided transforms inherit the modulus of continuity of the cost. Indeed, if $\delta(x,x') := \sup_{y\in\mathcal{Y}} |c(x,y)-c(x',y)|$, order reversal and additive anti-homogeneity give

$$\left|g^{\bar{c},\varepsilon,\varphi}(x) - g^{\bar{c},\varepsilon,\varphi}(x')\right| \leqslant \delta(x,x').$$

Thus $(f^{(\ell)})_{\ell\geqslant 1}$ is uniformly bounded and equicontinuous. Arzelà–Ascoli makes the orbit relatively compact in the uniform norm, and its monotonicity forces every uniformly convergent subsequence to have the same pointwise limit. Hence the full sequence converges uniformly.

Moreover, every selected one-sided transform is nonexpansive for the uniform norm. If $\|g-h\|_\infty \leqslant r$, then $g-r \leqslant h \leqslant g+r$; order reversal and additive anti-homogeneity imply $\|g^{\bar{c},\varepsilon,\varphi} - h^{\bar{c},\varepsilon,\varphi}\|_\infty \leqslant r$, and similarly for the other block. Therefore $\mathcal{A}_\varphi$ is uniformly continuous, and the limit is a fixed point. The supersolution argument reverses all inequalities. Uniqueness modulo constants gives the last claim. □

## 8.3 Sinkhorn Convergence: Sublinear Robust Rate

**Proposition 8.11** (Robust $O(1/\ell)$ dual rate for discrete Sinkhorn)**.** *Let $\mathrm{a} \in \Sigma_n$ and $\mathrm{b} \in \Sigma_m$ be positive histograms, and let $\mathrm{C} \in \mathbb{R}_+^{n\times m}$ be finite. Initialize $\mathrm{g}^{(0)} = 0$, perform complete Sinkhorn dual cycles, and denote the resulting potentials by $(\mathrm{f}^{(\ell)}, \mathrm{g}^{(\ell)})$ for $\ell \geqslant 1$. Let $\Delta^{(\ell)} := \mathcal{D}_\varepsilon(\mathrm{f}^\star,\mathrm{g}^\star) - \mathcal{D}_\varepsilon(\mathrm{f}^{(\ell)},\mathrm{g}^{(\ell)})$ be the dual suboptimality gap for the entropic dual objective (7.22). Then $0 \leqslant \Delta^{(\ell)} \leqslant \frac{2\|\mathrm{C}\|_\infty^2}{\varepsilon\ell}, \qquad \ell \geqslant 1.$*

*Proof.* Write $M := \|\mathrm{C}\|_\infty$; the case $M=0$ is immediate because one complete cycle produces $\mathrm{a}\otimes\mathrm{b}$. We first prove the oscillation estimate used below. If f is the soft transform of g, as in (7.20), set

$$Z_i := \sum_j b_j \exp\left(\frac{g_j - C_{i,j}}{\varepsilon}\right), \qquad f_i = -\varepsilon \log Z_i.$$

Since $0 \leqslant C_{i,j} \leqslant M$, one has $e^{-M/\varepsilon} Z_{i'} \leqslant Z_i \leqslant e^{M/\varepsilon} Z_{i'}$ for every $i, i'$. Therefore $|f_i - f_{i'}| \leqslant M$ and $\|\mathrm{f}\|_V \leqslant M$. The same argument applies to the other soft transform. Every Sinkhorn potential is produced by such a transform, and the block optimality equations imply the same property for $(\mathrm{f}^\star,\mathrm{g}^\star)$. $\|\mathrm{f}^{(\ell)}\|_V,\ \|\mathrm{g}^{(\ell)}\|_V,\ \|\mathrm{f}^\star\|_V,\ \|\mathrm{g}^\star\|_V \leqslant M$.

Associate with any potentials the nonnegative matrix

$$\mathrm{P}(\mathrm{f},\mathrm{g}) := (\mathrm{a}\otimes\mathrm{b}) \odot \exp\left(\frac{\mathrm{f}\oplus\mathrm{g} - \mathrm{C}}{\varepsilon}\right).$$

At the end of cycle $\ell$, the matrix $\mathrm{P}^{(\ell)} := \mathrm{P}(\mathrm{f}^{(\ell)},\mathrm{g}^{(\ell)})$ has column marginal b and hence total mass one. Denote its row marginal by $r^{(\ell)} := \mathrm{P}^{(\ell)}\mathbb{1}_m$ and set $\delta_\ell := \|\mathrm{a} - r^{(\ell)}\|_1$. The dual gradient at this point is $(\mathrm{a}-r^{(\ell)}, 0)$. Concavity of $\mathcal{D}_\varepsilon$ therefore gives $\Delta^{(\ell)} \leqslant \langle \mathrm{f}^\star - \mathrm{f}^{(\ell)},\, \mathrm{a} - r^{(\ell)}\rangle$. For every vector $h$ and every zero-sum vector $z$, subtracting $(\max h + \min h)/2$ from $h$ proves $|\langle h,\, z\rangle| \leqslant \frac{1}{2}\|h\|_V\|z\|_1$. The oscillation estimate and the triangle inequality for $\|\cdot\|_V$ thus yield

$$\Delta^{(\ell)} \leqslant M\delta_\ell. \tag{8.5}$$

Its row update satisfies

$$f_i^{(\ell+1)} - f_i^{(\ell)} = \varepsilon \log\left(\frac{a_i}{r_i^{(\ell)}}\right).$$

The matrices before and after this row update both have total mass one. Substitution in (7.22) therefore gives the exact gain $\varepsilon\mathrm{KL}(\mathrm{a}|r^{(\ell)})$. The subsequent column update can only increase the dual objective, so Pinsker's inequality, Theorem 6.13, and (8.5) give

$$\Delta^{(\ell)} - \Delta^{(\ell+1)} \geqslant \varepsilon\mathrm{KL}(\mathrm{a}|r^{(\ell)}) \geqslant \frac{\varepsilon}{2}\delta_\ell^2 \geqslant \frac{\varepsilon}{2M^2}(\Delta^{(\ell)})^2.$$

Set $x = M/\varepsilon$, and let $\widehat{\mathrm{P}}^{(1)}$ be the matrix after the first row update from $\mathrm{g}^{(0)} = 0$. Directly from the update, $e^{-x} \leqslant \frac{\widehat{P}_{i,j}^{(1)}}{a_i b_j} \leqslant e^x$. Its column marginal $\widehat{s}$ consequently satisfies $e^{-x} \leqslant \widehat{s}_j/b_j \leqslant e^x$. The first column update gives $P_{i,j}^{(1)} = \widehat{P}_{i,j}^{(1)} b_j/\widehat{s}_j$, and hence $\frac{r_i^{(1)}}{a_i} = \sum_j \frac{\widehat{P}_{i,j}^{(1)}}{a_i}\frac{b_j}{\widehat{s}_j} \in [e^{-x}, e^x]$, because $(\widehat{P}_{i,j}^{(1)}/a_i)_j$ is a probability vector. Thus $\delta_1 \leqslant \min\{2, e^x - 1\} \leqslant 2\min\{1, x\}$, where the last inequality uses $e^x - 1 \leqslant 2x$ for $0 \leqslant x \leqslant 1$. Equation (8.5) gives $\Delta^{(1)} \leqslant 2M^2/\varepsilon$.

Finally, whenever $\Delta^{(\ell+1)} > 0$,

$$\frac{1}{\Delta^{(\ell+1)}} - \frac{1}{\Delta^{(\ell)}} = \frac{\Delta^{(\ell)} - \Delta^{(\ell+1)}}{\Delta^{(\ell)}\Delta^{(\ell+1)}} \geqslant \frac{\varepsilon}{2M^2}.$$

Together with the bound on $\Delta^{(1)}$, telescoping proves $\Delta^{(\ell)} \leqslant 2M^2/(\varepsilon\ell)$. If a gap vanishes, all subsequent gaps vanish and the conclusion is immediate. □

**Corollary 8.12** (Approximating unregularized OT by regularized dual costs)**.** *Under the assumptions of Proposition 8.11, suppose $nm > 1$ and define the unregularized cost $L_0 := \mathrm{L}_\mathrm{C}(\mathrm{a},\mathrm{b})$. For any $\delta > 0$, choose $\varepsilon = \delta/\log(nm)$. Then $\ell \geqslant \frac{2\|\mathrm{C}\|_\infty^2 \log(nm)}{\delta^2} \implies \left|L_0 - \mathcal{D}_\varepsilon(\mathrm{f}^{(\ell)},\mathrm{g}^{(\ell)})\right| \leqslant \delta.$*

*Proof.* Let $L_\varepsilon := \max_{\mathrm{f},\mathrm{g}} \mathcal{D}_\varepsilon(\mathrm{f},\mathrm{g})$ and let $\mathrm{P}^0$ solve the unregularized problem. The KL-normalized primal formulation (7.7) gives $0 \leqslant L_\varepsilon - L_0 \leqslant \varepsilon \mathrm{KL}(\mathrm{P}^0|\mathrm{a}\otimes\mathrm{b}) \leqslant \varepsilon \log(nm)$. The last inequality follows from $\mathrm{KL}(\mathrm{P}^0|\mathrm{a}\otimes\mathrm{b}) = \mathrm{H}(\mathrm{a}) + \mathrm{H}(\mathrm{b}) - \mathrm{H}(\mathrm{P}^0) \leqslant \log(nm)$. Moreover, Proposition 8.11 gives $0 \leqslant L_\varepsilon - \mathcal{D}_\varepsilon(\mathrm{f}^{(\ell)},\mathrm{g}^{(\ell)}) \leqslant 2\|\mathrm{C}\|_\infty^2/(\varepsilon\ell)$. Since the difference between the computed dual value and $L_0$ is the difference of these two nonnegative errors, its absolute value is bounded by their maximum. The prescribed $\varepsilon$ and $\ell$ make both errors at most $\delta$. □

For dense $n\times n$ problems, one Sinkhorn cycle costs $O(n^2)$ operations. Corollary 8.12 therefore yields the arithmetic complexity

$$O\left(\frac{n^2\|\mathrm{C}\|_\infty^2 \log n}{\delta^2}\right)$$

### 8.4 Sinkhorn Convergence: Linear Hilbert Metric Rate

**Projective contraction.**

**Definition 8.13** (Hilbert metric)**.** On $\mathbb{R}^n_{+,*}$, Hilbert's projective metric is

$$\forall\,(\mathrm{u},\mathrm{u}') \in (\mathbb{R}^n_{+,*})^2, \quad d_{\mathcal{H}}(\mathrm{u},\mathrm{u}') := \|\log(\mathrm{u}) - \log(\mathrm{u}')\|_V. \tag{8.6}$$

Here, $\|\cdot\|_V$ is the variation seminorm of Definition 8.6, specialized to functions on the finite space $\{1,\dots,n\}$: $\|z\|_V = \max_i z_i - \min_i z_i$.

**Proposition 8.14** (Hilbert metric on the projective cone)**.** *The function $d_{\mathcal{H}}$ defines a complete distance on the projective cone $\mathbb{R}^n_{+,*}/\sim$, where $\mathrm{u}\sim\mathrm{u}'$ means that $\mathrm{u} = s\mathrm{u}'$ for some $s>0$.*

*Proof.* The map $\mathrm{u}\mapsto\log\mathrm{u}$ identifies $\mathbb{R}^n_{+,*}/\sim$ with the quotient vector space $\mathbb{R}^n/\operatorname{Span}(\mathbb{1}_n)$, because multiplying u by $s>0$ adds the constant vector $\log(s)\mathbb{1}_n$. The variation seminorm $\|z\|_V = \max_i z_i - \min_i z_i$ vanishes exactly on constant vectors, so it induces a norm on this quotient. Symmetry, the triangle inequality and separation therefore follow from the corresponding norm properties. Completeness follows because $\mathbb{R}^n/\operatorname{Span}(\mathbb{1}_n)$ is finite-dimensional and all finite-dimensional normed spaces are complete. □

**Theorem 8.15** (Birkhoff contraction theorem)**.** *Let $\mathrm{K}\in\mathbb{R}^{n\times m}_{+,*}$. Then, for $(\mathrm{v},\mathrm{v}')\in(\mathbb{R}^m_{+,*})^2$,*

$$d_{\mathcal{H}}(\mathrm{K}\mathrm{v},\mathrm{K}\mathrm{v}') \leqslant \lambda(\mathrm{K}) d_{\mathcal{H}}(\mathrm{v},\mathrm{v}'), \qquad \lambda(\mathrm{K}) := \frac{\sqrt{\eta(\mathrm{K})}-1}{\sqrt{\eta(\mathrm{K})}+1} < 1, \qquad \eta(\mathrm{K}) := \max_{i,j,k,\ell} \frac{\mathrm{K}_{i,k}\mathrm{K}_{j,\ell}}{\mathrm{K}_{j,k}\mathrm{K}_{i,\ell}}.$$

*Proof.* Work in logarithmic coordinates and define $F(z) := \log(\mathrm{K}e^z)$, $\quad \mathsf{P}(z)_{i,k} := \frac{\mathrm{K}_{i,k}e^{z_k}}{(\mathrm{K}e^z)_i}$. Every row of $\mathsf{P}(z)$ is a probability vector and $\mathrm{d}F(z)[h] = \mathsf{P}(z)h$. We first bound its contraction in variation seminorm. If $p$ and $q$ are two probability rows, then $|\langle p-q,\,h\rangle| \leqslant \frac{1}{2}\|p-q\|_1\|h\|_V$. Indeed, subtracting $\min_k h_k$ and dividing by $\|h\|_V$ reduces this inequality to the variational characterization of total variation by functions taking values in $[0,1]$.

$$\|\mathsf{P}(z)h\|_V \leqslant \delta(\mathsf{P}(z))\|h\|_V, \qquad \delta(\mathsf{P}) := \frac{1}{2}\max_{i,j}\sum_k |\mathsf{P}_{i,k} - \mathsf{P}_{j,k}|\,. \tag{8.7}$$

Fix rows $p,q$ of $\mathsf{P}(z)$, with indices $i,j$, and put $r_k = p_k/q_k$. Then $\sum_k q_k r_k = 1$, and the definition of $\eta(\mathrm{K})$ gives $\frac{\max_k r_k}{\min_k r_k} = \max_{k,\ell}\frac{\mathrm{K}_{i,k}\mathrm{K}_{j,\ell}}{\mathrm{K}_{j,k}\mathrm{K}_{i,\ell}} \leqslant \eta(\mathrm{K})$. Write $a = \min_k r_k \leqslant 1 \leqslant b = \max_k r_k$. If $a=b=1$, then $p=q$ and the desired bound is immediate. Otherwise, $(r-1)_+ \leqslant (b-1)(r-a)/(b-a)$ on $[a,b]$, so $\frac{1}{2}\|p-q\|_1 = \sum_k q_k(r_k-1)_+ \leqslant \frac{(1-a)(b-1)}{b-a}$. Set $a = e^{-u}$ and $b = e^v$. The last quotient equals

$$\frac{2\sinh(u/2)\sinh(v/2)}{\sinh((u+v)/2)} \leqslant \tanh\left(\frac{u+v}{4}\right) \leqslant \tanh\left(\frac{\log\eta(\mathrm{K})}{4}\right) = \lambda(\mathrm{K}),$$

where the first inequality follows from $\sinh(u/2)\sinh(v/2) \leqslant \sinh^2((u+v)/4)$. Thus (8.7) holds with $\delta(\mathsf{P}(z)) \leqslant \lambda(\mathrm{K})$, uniformly in $z$.

For $z_t = (1-t)z' + tz$, the fundamental theorem of calculus and the triangle inequality for $\|\cdot\|_V$ now give $\|F(z) - F(z')\|_V \leqslant \int_0^1 \|\mathsf{P}(z_t)(z-z')\|_V \mathrm{d}t \leqslant \lambda(\mathrm{K})\|z-z'\|_V$. Taking $z = \log\mathrm{v}$ and $z' = \log\mathrm{v}'$ proves the result because $F(z) = \log(\mathrm{K}\mathrm{v})$. □

$d_{\mathcal{H}}(\mathrm{K}^\ell u^{(0)}, u^\star) \leqslant \lambda(\mathrm{K})^\ell d_{\mathcal{H}}(u^{(0)},u^\star)$. Thus Hilbert contraction supplies both the Perron–Frobenius eigenvector and the global linear convergence of power iteration. The contraction can be visualized for every initialization at once on the three-state probability simplex.

$\mathrm{K}^{\ell+1}\Sigma_3 = \mathrm{K}^\ell(\mathrm{K}\Sigma_3) \subseteq \mathrm{K}^\ell\Sigma_3$.

$$J_3 := \frac{1}{3}\mathbb{1}_3\mathbb{1}_3^\top, \qquad A := \begin{pmatrix} 0 & -1 & 1 \\ 1 & 0 & -1 \\ -1 & 1 & 0 \end{pmatrix}, \qquad \mathrm{K}_{\rho,\delta} := \rho I_3 + (1-\rho)J_3 + \delta A.$$

$$\mathrm{K}_1 = \mathrm{K}_{.90,0}, \qquad \mathrm{K}_2 = \mathrm{K}_{.90,.014}, \qquad \mathrm{K}_3 = \begin{pmatrix} .950 & .018 & .032 \\ .042 & .880 & .078 \\ .008 & .102 & .890 \end{pmatrix}.$$

**Sinkhorn contraction.**

**Theorem 8.16** (Projective linear convergence of Sinkhorn)**.** *Starting from* $\mathrm{v}^{(0)} > 0$*, use the scaling iterates of* (7.5) *and the associated primal half-step iterates already defined in* (8.2)*. Fix any optimal scaling pair* $(\mathrm{u}^\star, \mathrm{v}^\star)$*. Writing* $\lambda = \lambda(\mathrm{K})$*, one has*

$$d_{\mathcal{H}}(\mathrm{v}^{(\ell)}, \mathrm{v}^\star) \leqslant \lambda^{2\ell} d_{\mathcal{H}}(\mathrm{v}^{(0)}, \mathrm{v}^\star), \qquad d_{\mathcal{H}}(\mathrm{u}^{(\ell+1)}, \mathrm{u}^\star) \leqslant \lambda^{2\ell+1} d_{\mathcal{H}}(\mathrm{v}^{(0)}, \mathrm{v}^\star). \tag{8.8}$$

*Thus the scaling rays converge linearly. After fixing any continuous gauge, their representatives converge, and* $\mathrm{P}^{(\ell)} \to \mathrm{P}^\star$ *entrywise.*

$$d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star) \leqslant \frac{d_{\mathcal{H}}(\mathrm{P}^{(\ell)}\mathbb{1}_m, \mathrm{a})}{1-\lambda^2} \quad \textit{and} \quad d_{\mathcal{H}}(\mathrm{v}^{(\ell)}, \mathrm{v}^\star) \leqslant \frac{d_{\mathcal{H}}((\mathrm{P}^{(\ell+1/2)})^\top \mathbb{1}_n, \mathrm{b})}{1-\lambda^2}. \tag{8.9}$$

*Lastly,*

$$\|\log(\mathrm{P}^{(\ell)}) - \log(\mathrm{P}^\star)\|_\infty \leqslant d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star) + d_{\mathcal{H}}(\mathrm{v}^{(\ell)}, \mathrm{v}^\star), \tag{8.10}$$

*where* $\mathrm{P}^\star$ *is the unique solution of* (7.1)*.*

*Proof.* Entrywise multiplication by a fixed positive vector and entrywise inversion are isometries of Hilbert's metric. Therefore the row map $R(\mathrm{v}) = \mathrm{a}/(\mathrm{K}\mathrm{v})$ and the column map $C(\mathrm{u}) = \mathrm{b}/(\mathrm{K}^\top \mathrm{u})$ are both $\lambda$-Lipschitz by Theorem 8.15. Their full-cycle compositions $C \circ R$ and $R \circ C$ are $\lambda^2$-contractions. Since $\mathrm{v}^\star = C(R(\mathrm{v}^\star))$ and $\mathrm{u}^\star = R(C(\mathrm{u}^\star))$, iteration gives (8.8).
For any contraction $F$ with factor $q < 1$ and fixed point $x^\star$, the triangle inequality gives $d(x, x^\star) \leqslant d(x, Fx) + d(Fx, Fx^\star) \leqslant d(x, Fx) + q\, d(x, x^\star)$, hence $d(x, x^\star) \leqslant d(x, Fx)/(1-q)$. Apply this with $q = \lambda^2$ to the two full-cycle maps. The scaling identities $d_{\mathcal{H}}(\mathrm{u}^{(\ell+1)}, \mathrm{u}^{(\ell)}) = d_{\mathcal{H}}(\mathrm{a}, \mathrm{P}^{(\ell)}\mathbb{1}_m)$ and $d_{\mathcal{H}}(\mathrm{v}^{(\ell+1)}, \mathrm{v}^{(\ell)}) = d_{\mathcal{H}}(\mathrm{b}, (\mathrm{P}^{(\ell+1/2)})^\top \mathbb{1}_n)$ then prove (8.9).
Finally, write $\xi_i = \log(u_i^{(\ell)}/u_i^\star)$ and $\zeta_j = \log(v_j^{(\ell)}/v_j^\star)$. Then $\log(\mathrm{P}^{(\ell)}_{i,j}/\mathrm{P}^\star_{i,j}) = \xi_i + \zeta_j$ and $\mathrm{osc}(\xi \oplus \zeta) = \mathrm{osc}(\xi) + \mathrm{osc}(\zeta)$. Both plans have total mass one, so $\sum_{i,j} \mathrm{P}^\star_{i,j} e^{\xi_i + \zeta_j} = 1$. This proves (8.10). $\square$

**Nonlinear Sinkhorn images of the simplex.**

$$F_{\mathrm{u}}(\mathrm{u}) \coloneqq R(C(\mathrm{u})) = \mathrm{a} \oslash \left[\mathrm{K}\left(\mathrm{b} \oslash (\mathrm{K}^\top \mathrm{u})\right)\right].$$

$\widehat{F}_{\mathrm{u}}(p) \coloneqq \frac{F_{\mathrm{u}}(p)}{\langle F_{\mathrm{u}}(p), \mathbb{1}_3\rangle}, \qquad p \in \Sigma_3$. Thus $\widehat{\mathrm{u}}^{(\ell)} \coloneqq \mathrm{u}^{(\ell)}/\langle \mathrm{u}^{(\ell)}, \mathbb{1}_3\rangle$ follows $\widehat{\mathrm{u}}^{(\ell+1)} = \widehat{F}_{\mathrm{u}}(\widehat{\mathrm{u}}^{(\ell)})$ when $\ell$ indexes complete Sinkhorn cycles. Since $\widehat{F}_{\mathrm{u}}$ maps $\Sigma_3$ into itself, its set-valued iterates are nested:

$$\widehat{F}_{\mathrm{u}}^{\ell+1}(\Sigma_3) = \widehat{F}_{\mathrm{u}}^{\ell}\left(\widehat{F}_{\mathrm{u}}(\Sigma_3)\right) \subseteq \widehat{F}_{\mathrm{u}}^{\ell}(\Sigma_3).$$

$$\mathrm{diam}_{d_{\mathcal{H}}}\left(\widehat{F}_{\mathrm{u}}^{\ell}(\Sigma_3)\right) \leqslant \lambda(\mathrm{K})^{2(\ell-1)}\, \mathrm{diam}_{d_{\mathcal{H}}}\left(\widehat{F}_{\mathrm{u}}(\Sigma_3)\right), \qquad \ell \geqslant 1.$$

**Dual-potential form of the contraction.**

$$\|\mathrm{f}^{(\ell)} - \mathrm{f}^\star\|_V = \varepsilon d_{\mathcal{H}}(\mathrm{u}^{(\ell)}, \mathrm{u}^\star), \qquad \|\mathrm{g}^{(\ell)} - \mathrm{g}^\star\|_V = \varepsilon d_{\mathcal{H}}(\mathrm{v}^{(\ell)}, \mathrm{v}^\star).$$

Thus a complete cycle contracts both potential classes at the rate $\lambda(\mathrm{K})^2$. Moreover, the temperature factor must be retained when passing from potentials to densities:

$$\|\log \frac{\mathrm{d}\pi^{(\ell)}}{\mathrm{d}\pi^\star}\|_\infty = \frac{1}{\varepsilon}\|(\mathrm{f}^{(\ell)} - \mathrm{f}^\star) \oplus (\mathrm{g}^{(\ell)} - \mathrm{g}^\star)\|_\infty \leqslant \frac{\|\mathrm{f}^{(\ell)} - \mathrm{f}^\star\|_V + \|\mathrm{g}^{(\ell)} - \mathrm{g}^\star\|_V}{\varepsilon}.$$

$\lambda = \frac{\sqrt{\eta}-1}{\sqrt{\eta}+1} \leqslant \tanh(R/(2\varepsilon)) < 1$. $\|\mathrm{f}^{(\ell)} - \mathrm{f}^\star\|_V \leqslant \frac{\varepsilon}{1-\lambda^2}\|\log((\mathrm{P}^{(\ell)}\mathbb{1}_m) \oslash \mathrm{a})\|_V$, $\|\mathrm{g}^{(\ell)} - \mathrm{g}^\star\|_V \leqslant \frac{\varepsilon}{1-\lambda^2}\|\log(((\mathrm{P}^{(\ell+1/2)})^\top \mathbb{1}_n) \oslash \mathrm{b})\|_V$.

## 8.5 Local Convergence Analysis and Acceleration

**Conditional operator and maximal correlation.**

**Definition 8.17** (Conditional coupling operator)**.** For a coupling $\pi \in \mathcal{U}(\alpha, \beta)$, let $\pi(\mathrm{d}x, \mathrm{d}y) = \alpha(\mathrm{d}x)\pi_x(\mathrm{d}y) = \beta(\mathrm{d}y)\pi^y(\mathrm{d}x)$ be its two disintegrations. Define $T_\pi : L^2(\beta) \to L^2(\alpha)$ and $T_\pi^* : L^2(\alpha) \to L^2(\beta)$ by

$$(T_\pi k)(x) \coloneqq \int_{\mathcal{Y}} k(y) \mathrm{d}\pi_x(y), \qquad (T_\pi^* h)(y) \coloneqq \int_{\mathcal{X}} h(x) \mathrm{d}\pi^y(x). \tag{8.11}$$

The centered maximal correlation of $\pi$ is

$$\sigma(\pi) \coloneqq \sup_{\substack{k \in L_0^2(\beta) \\ \|k\|_{L^2(\beta)}=1}} \|T_\pi k\|_{L^2(\alpha)} = \sup_{\substack{h \in L_0^2(\alpha),\, k \in L_0^2(\beta) \\ \|h\|_{L^2(\alpha)} = \|k\|_{L^2(\beta)}=1}} \int h(x) k(y) \mathrm{d}\pi(x, y). \tag{8.12}$$

For the optimal entropic coupling $\pi_\varepsilon$ of (7.12), set $T_\varepsilon \coloneqq T_{\pi_\varepsilon}$ and $\sigma_\varepsilon \coloneqq \sigma(\pi_\varepsilon)$.

$$T_\varepsilon = \mathrm{diag}(\mathrm{a})^{-1}\mathrm{P}_\varepsilon, \qquad T_\varepsilon^* = \mathrm{diag}(\mathrm{b})^{-1}\mathrm{P}_\varepsilon^\top, \tag{8.13}$$

where adjointness is understood in the a- and b-weighted Euclidean inner products. Thus $\sigma_\varepsilon$ is the largest nonconstant singular value of the normalized optimal coupling.

**Dual Hessian and exact local rate.**

**Proposition 8.18** (Dual Hessian and local Sinkhorn rate)**.** *Assume the setting of Proposition 7.18, and that differentiation under the Gibbs integrals and the stated $L^2$ Taylor expansions are valid. These requirements are automatic in finite spaces and form the local regularity hypothesis in continuous settings. For bounded directions $(h,k)$ and $(h',k')$,*

$$-\varepsilon D^2\mathcal{D}_\varepsilon(f_\varepsilon,g_\varepsilon)[(h,k),(h',k')]=\int_{\mathcal{X}\times\mathcal{Y}}(h(x)+k(y))(h'(x)+k'(y))\mathrm{d}\pi_\varepsilon(x,y). \tag{8.14}$$

*Hence the Hessian form extends continuously to $L^2(\alpha)\oplus L^2(\beta)$, where the positive dual curvature operator is*

$$-\nabla^2\mathcal{D}_\varepsilon(f_\varepsilon,g_\varepsilon)=\frac{1}{\varepsilon}\begin{pmatrix}\mathrm{Id} & T_\varepsilon\\ T_\varepsilon^* & \mathrm{Id}\end{pmatrix}. \tag{8.15}$$

*After quotienting by the gauge direction $(1,-1)$ and using the product-$L^2$ quotient norm, its spectrum is contained in $[(1-\sigma_\varepsilon)/\varepsilon,2/\varepsilon]$. Both endpoints belong to the spectrum, although the lower one need not be an eigenvalue. In particular, when $\sigma_\varepsilon<1$, the spectral condition number of the full dual is $2/(1-\sigma_\varepsilon)$. The derivatives of the two soft-transform updates (7.24)–(7.25) at the optimum are*

$$D[g\mapsto g^{\bar{c},\varepsilon}](g_\varepsilon)=-T_\varepsilon,\qquad D[f\mapsto f^{c,\varepsilon}](f_\varepsilon)=-T_\varepsilon^*. \tag{8.16}$$

*$g\mapsto(g^{\bar{c},\varepsilon})^{c,\varepsilon}$ is $T_\varepsilon^*T_\varepsilon$. If the gauges are fixed so that $e^{(\ell)}:=g^{(\ell)}-g_\varepsilon\in L_0^2(\beta)$, then*

$$e^{(\ell+1)}=T_\varepsilon^*T_\varepsilon e^{(\ell)}+o\Big(\|e^{(\ell)}\|_{L^2(\beta)}\Big),\qquad \limsup_{\ell\to\infty}\frac{\|e^{(\ell+1)}\|_{L^2(\beta)}}{\|e^{(\ell)}\|_{L^2(\beta)}}\leqslant\sigma_\varepsilon^2. \tag{8.17}$$

*If $\sigma_\varepsilon^2$ is an isolated eigenvalue and the asymptotic error has a nonzero projection onto its eigenspace, the ratio in (8.17) converges to $\sigma_\varepsilon^2$.*

*Proof.* Twice differentiating the exponential term in (7.22) gives (8.14); the density law (7.23) identifies its reference measure with $\pi_\varepsilon$. Its two diagonal terms are the $L^2$ inner products of the marginals, and its cross term is represented by $T_\varepsilon$, which proves (8.15). Each quotient class has a unique representative orthogonal to the gauge direction, hence of the form $(h_0+m,k_0+m)$ with $h_0\in L_0^2(\alpha)$ and $k_0\in L_0^2(\beta)$. On the centered component, Cauchy–Schwarz and (8.12) give

$$(1-\sigma_\varepsilon)\big(\|h_0\|_{L^2(\alpha)}^2+\|k_0\|_{L^2(\beta)}^2\big)\leqslant\|h_0\|_{L^2(\alpha)}^2+\|k_0\|_{L^2(\beta)}^2+2\langle h_0,T_\varepsilon k_0\rangle_{L^2(\alpha)}\leqslant(1+\sigma_\varepsilon)\big(\|h_0\|_{L^2(\alpha)}^2+\|k_0\|_{L^2(\beta)}^2\big).$$

The common-constant direction $(1,1)$ has eigenvalue $2/\varepsilon$, whereas $(1,-1)$ is the gauge kernel. Approximate maximizers in (8.12) show that the lower spectral endpoint is $(1-\sigma_\varepsilon)/\varepsilon$, while the constant direction gives the upper endpoint $2/\varepsilon$. This proves the condition-number formula without requiring a singular-vector basis.
Differentiating either log-partition formula in Definition 7.19 produces minus the corresponding conditional expectation under $\pi_\varepsilon$, proving (8.16). The chain rule and a Taylor expansion then give (8.17). Since $\|T_\varepsilon^*T_\varepsilon\|_{L_0^2(\beta)\to L_0^2(\beta)}=\sigma_\varepsilon^2$, the rate follows. ☐

$$1-\sigma_\varepsilon^2=\inf_{\substack{k\in L_0^2(\beta)\\ \|k\|_{L^2(\beta)}=1}}\int_{\mathcal{X}}\mathrm{Var}_{\pi_{\varepsilon,x}}\big(k(Y)\big)\mathrm{d}\alpha(x). \tag{8.18}$$

Thus Sinkhorn slows down when the optimal conditional laws $Y|X=x$ become nearly deterministic. This is exactly what happens as $\varepsilon\to0$ in a smooth Monge regime.

**Proposition 8.19** (The global Hilbert factor controls the local rate)**.** *In the discrete positive-kernel setting of Theorem 8.16, let $\lambda(\mathrm{K})$ be the Birkhoff contraction factor defined in Theorem 8.15. Then*

$$\sigma_\varepsilon\leqslant\lambda(\mathrm{K})<1,\qquad\sigma_\varepsilon^2\leqslant\lambda(\mathrm{K})^2. \tag{8.19}$$

*Proof.* Let $R(\mathrm{v})=\mathrm{a}\oslash(\mathrm{Kv})$ and $C(\mathrm{u})=\mathrm{b}\oslash(\mathrm{K}^\top\mathrm{u})$ be the row and column scaling maps used in the proof of Theorem 8.16. Entrywise inversion and multiplication by a fixed positive vector are isometries of Hilbert's metric. Theorem 8.15, applied successively to K and $\mathrm{K}^\top$, therefore gives the pairwise estimate $d_{\mathcal{H}}\big(C(R(\mathrm{v})),C(R(\tilde{\mathrm{v}}))\big)\leqslant\lambda(\mathrm{K})^2d_{\mathcal{H}}(\mathrm{v},\tilde{\mathrm{v}})$, where $\lambda(\mathrm{K}^\top)=\lambda(\mathrm{K})$ follows directly from its cross-ratio definition. Under the logarithmic change of variables $g=\varepsilon\log\mathrm{v}$, Hilbert's metric becomes $\varepsilon^{-1}\|\cdot\|_V$ by (8.6). Hence the complete dual update $\mathcal{S}(g):=(g^{\bar{c},\varepsilon})^{c,\varepsilon}$ is $\lambda(\mathrm{K})^2$-Lipschitz for the variation norm on the quotient by constants: $\|\mathcal{S}(g)-\mathcal{S}(\tilde{g})\|_V\leqslant\lambda(\mathrm{K})^2\|g-\tilde{g}\|_V$. Differentiate this inequality at the fixed point $g_\varepsilon$. Proposition 8.18 gives $D\mathcal{S}(g_\varepsilon)=T_\varepsilon^*T_\varepsilon$, and therefore $\|T_\varepsilon^*T_\varepsilon h\|_V\leqslant\lambda(\mathrm{K})^2\|h\|_V$ for every $h\in L_0^2(\beta)$. In the present finite-dimensional setting, $T_\varepsilon^*T_\varepsilon$ is positive and self-adjoint for the $\beta$-weighted inner product, and its largest eigenvalue on $L_0^2(\beta)$ is $\sigma_\varepsilon^2$. If $\sigma_\varepsilon>0$, take a corresponding eigenvector $h_\star$. It is nonconstant, so $\|h_\star\|_V>0$, and the preceding bound gives $\sigma_\varepsilon^2\|h_\star\|_V=\|T_\varepsilon^*T_\varepsilon h_\star\|_V\leqslant\lambda(\mathrm{K})^2\|h_\star\|_V$. The conclusion is immediate; the case $\sigma_\varepsilon=0$ is trivial. ☐

**Blockwise over-relaxation.**

$$f^{(\ell+1)}=(1-\omega)f^{(\ell)}+\omega(g^{(\ell)})^{\bar{c},\varepsilon},\qquad g^{(\ell+1)}=(1-\omega)g^{(\ell)}+\omega(f^{(\ell+1)})^{c,\varepsilon}. \tag{8.20}$$

**Proposition 8.20** (Optimal local block relaxation)**.** *Assume the hypotheses of Proposition 8.18 and $\sigma_\varepsilon<1$. For every $0<\omega<2$, the iteration (8.20) is locally linearly convergent modulo the additive gauge. Its optimal relaxation parameter and corresponding asymptotic factor are*

$$\omega_\star=\frac{2}{1+\sqrt{1-\sigma_\varepsilon^2}},\qquad r_\star=\omega_\star-1=\frac{1-\sqrt{1-\sigma_\varepsilon^2}}{1+\sqrt{1-\sigma_\varepsilon^2}}. \tag{8.21}$$

*For $\omega=1$, the factor is $\sigma_\varepsilon^2$, as in Proposition 8.18.*

*Proof.* Use (8.16). The polar decomposition of $T_\varepsilon$ and the spectral theorem for $(T_\varepsilon^* T_\varepsilon)^{1/2}$ reduce the centered linearized iteration to scalar spectral values $s \in [0, \sigma_\varepsilon]$; in finite dimensions these are the usual singular values. On the corresponding two-dimensional mode, one relaxed cycle has matrix

$$\mathsf{M}_\omega(s) = \begin{pmatrix} 1-\omega & -\omega s \\ -\omega s(1-\omega) & 1-\omega+\omega^2 s^2 \end{pmatrix},$$

whose characteristic polynomial is $z^2 - (2(1-\omega)+\omega^2 s^2)z + (1-\omega)^2$. The supremum of the root moduli is reached, or approached, as $s \to \sigma_\varepsilon$. The two roots of this limiting mode have equal modulus precisely when $\omega^2\sigma_\varepsilon^2 = 4(\omega - 1)$. The smaller solution is $\omega_\star$ in (8.21); at this value both root moduli equal $\omega_\star - 1$. Below this threshold the dominant real root decreases with $\omega$, while above it the conjugate-root modulus $\omega - 1$ increases. This proves optimality and local convergence for $0 < \omega < 2$. The nongauge constant mode has factor $(1-\omega)^2$ and never dominates these centered modes. □

Set $\delta_\varepsilon = 1 - \sigma_\varepsilon^2$. When $\delta_\varepsilon$ is small, the exact expression in (8.21) gives the complete-cycle factor
$r_\star = \frac{1-\sqrt{\delta_\varepsilon}}{1+\sqrt{\delta_\varepsilon}} = 1 - 2\sqrt{\delta_\varepsilon} + 2\delta_\varepsilon + O(\delta_\varepsilon^{3/2}),$

**Entropic semi-dual and variable projection.** The hard semi-dual $\mathcal{E}_0$ in (5.2) eliminates one potential from the full dual $\mathcal{D}_0$. At positive temperature, the full dual is $\mathcal{D}_\varepsilon$ from (7.22), and the soft $\bar{c}$-transform $g^{\bar{c},\varepsilon}$ is defined by (7.25).

$$\mathcal{E}_\varepsilon(g) \coloneqq \mathcal{D}_\varepsilon(g^{\bar{c},\varepsilon}, g) = \int_{\mathcal{X}} g^{\bar{c},\varepsilon} \mathrm{d}\alpha + \int_{\mathcal{Y}} g \mathrm{d}\beta. \tag{8.22}$$

For a potential $g$, let $\pi_g$ be the Gibbs measure obtained from the density law (7.23) by replacing $(f_\varepsilon, g_\varepsilon)$ with $(g^{\bar{c},\varepsilon}, g)$. Write $\pi_g(\mathrm{d}x, \mathrm{d}y) = \alpha(\mathrm{d}x)\pi_{g,x}(\mathrm{d}y)$ and denote its second marginal by $\beta_g$. The first marginal is exactly $\alpha$ by construction.
$\min_{U,V} \frac{1}{2}\|U\sigma(VX) - Y\|_{\mathrm{F}}^2,$

**Proposition 8.21** (Semi-dual curvature and conditioning)**.** *For bounded directions $k, k'$,*

$$D\mathcal{E}_\varepsilon(g)[k] = \int_{\mathcal{Y}} k \mathrm{d}(\beta - \beta_g), \qquad D^2\mathcal{E}_\varepsilon(g)[k,k'] = -\frac{1}{\varepsilon}\int_{\mathcal{X}} \mathrm{Cov}_{\pi_{g,x}}\big(k(Y), k'(Y)\big)\mathrm{d}\alpha(x). \tag{8.23}$$

*At $g = g_\varepsilon$, its positive curvature on $L_0^2(\beta)$ is*

$$-\nabla^2\mathcal{E}_\varepsilon(g_\varepsilon) = \frac{1}{\varepsilon}(\mathrm{Id} - T_\varepsilon^* T_\varepsilon), \qquad \frac{1-\sigma_\varepsilon^2}{\varepsilon}\mathrm{Id} \preceq -\nabla^2\mathcal{E}_\varepsilon(g_\varepsilon) \preceq \frac{1}{\varepsilon}\mathrm{Id}. \tag{8.24}$$

*Hence its spectral condition number is at most $1/(1-\sigma_\varepsilon^2)$.*

*Proof.* The envelope theorem gives the first derivative in (8.23). Differentiating the normalized conditional Gibbs law gives the covariance formula. At the optimum, total covariance yields $\int_{\mathcal{X}} \mathrm{Cov}_{\pi_{\varepsilon,x}}(k(Y), k'(Y))\mathrm{d}\alpha(x) = \int_{\mathcal{Y}} kk'\mathrm{d}\beta - \int_{\mathcal{X}} (T_\varepsilon k)(T_\varepsilon k')\mathrm{d}\alpha$, which is (8.24). Equivalently, this operator is the Schur complement of the first identity block in (8.15). Its spectrum on centered functions lies in $[(1-\sigma_\varepsilon^2)/\varepsilon, 1/\varepsilon]$. □

### 8.6 Entropic Optimal Transport between Gaussians

**Proposition 8.22** (Quadratic closure of Sinkhorn iterates)**.** *Let $\beta = \mathcal{N}(m_\beta, \Sigma_\beta)$ on $\mathbb{R}^d$ and take $c(x,y) = \|x-y\|^2$. If $g(y)$ is a quadratic polynomial such that the Gaussian integral below is finite, then the soft transform*

$$f(x) = -\varepsilon \log \int \exp\left(\frac{g(y) - \|x-y\|^2}{\varepsilon}\right) \mathrm{d}\beta(y)$$

*is a quadratic polynomial in $x$.*

*Proof.* The exponent is the sum of a quadratic polynomial in $y$ and the logarithm of the Gaussian density of $\beta$. Completing the square in $y$ evaluates the integral as a positive constant times the exponential of a quadratic polynomial in $x$. Taking $-\varepsilon\log$ therefore gives a quadratic polynomial. □

**Proposition 8.23** (Balanced entropic OT between Gaussians)**.** *Let $\alpha = \mathcal{N}(m_\alpha, \Sigma_\alpha)$ and $\beta = \mathcal{N}(m_\beta, \Sigma_\beta)$ have positive-definite covariances, and let $\Sigma_\alpha^{1/2}\Sigma_\beta^{1/2} = U \operatorname{diag}(\sigma_i)V^\top$ be a singular-value decomposition. For the balanced entropic problem (7.12) with $c(x,y) = \|x-y\|^2$, the optimizer is Gaussian with cross-covariance $K_\varepsilon = \Sigma_\alpha^{1/2}U \operatorname{diag}(s_i)V^\top\Sigma_\beta^{1/2}$, $\quad s_i = \frac{\sqrt{\varepsilon^2+16\sigma_i^2}-\varepsilon}{4\sigma_i}$. The optimal value is*

$$\|m_\alpha - m_\beta\|^2 + \mathrm{tr}(\Sigma_\alpha) + \mathrm{tr}(\Sigma_\beta) + \sum_i \left(-2\sigma_i s_i - \frac{\varepsilon}{2}\log(1-s_i^2)\right).$$

*As $\varepsilon \downarrow 0$, $s_i \to 1$ and the full covariance contribution, including the two trace terms, converges to $\mathcal{B}(\Sigma_\alpha, \Sigma_\beta)^2$.*

*Proof.* Let $(X,Y)$ be any coupling with finite second moments and cross-covariance $K = \mathbb{E}\big[(X-m_\alpha)(Y-m_\beta)^\top\big]$. Replacing $(X,Y)$ by the Gaussian vector with the same mean and covariance leaves the quadratic cost unchanged. Since the marginals are fixed, $\mathrm{KL}(\pi|\alpha\otimes\beta) = -h(X,Y) + h(\alpha) + h(\beta)$, where $h$ denotes differential entropy when it is finite. Among laws with a fixed covariance, the Gaussian maximizes entropy; if the entropy is not finite, the relative entropy is already $+\infty$. Thus the Gaussian replacement cannot increase the objective, and it is enough to optimize over Gaussian couplings.
Any such coupling has covariance

$$\begin{pmatrix} \Sigma_\alpha & K \\ K^\top & \Sigma_\beta \end{pmatrix}.$$

Write $K = \Sigma_\alpha^{1/2} S \Sigma_\beta^{1/2}$. The block covariance constraint is equivalent to the singular values of $S$ being at most one, and finite entropy forces them to be strictly smaller than one. The cost depends on $K$ through $\|m_\alpha - m_\beta\|^2 + \mathrm{tr}(\Sigma_\alpha) + \mathrm{tr}(\Sigma_\beta) - 2\,\mathrm{tr}(K)$, while $\mathrm{KL}(\pi|\alpha\otimes\beta) = -\frac{1}{2}\log\det(\mathrm{Id} - SS^\top)$. By von Neumann's trace inequality, the minimizer aligns $S$ with the singular vectors of

$\Sigma_\alpha^{1/2}\Sigma_\beta^{1/2}$, so $S = U\,\mathrm{diag}(s_i)V^\top$. The problem separates into scalar minimizations $\min_{0\leqslant s<1} -2\sigma_i s - \frac{\varepsilon}{2}\log(1-s^2)$. The first-order condition is $2\sigma_i = \varepsilon s/(1-s^2)$, whose positive solution is the displayed $s_i$. Substitution gives the value formula. Since $s_i \to 1$ and $\varepsilon\log(1-s_i^2) \to 0$ as $\varepsilon \downarrow 0$, the spectral sum converges to $-2\sum_i \sigma_i$. The identity $\sum_i \sigma_i = \mathrm{tr}((\Sigma_\alpha^{1/2}\Sigma_\beta\Sigma_\alpha^{1/2})^{1/2})$ then gives the Bures–Wasserstein covariance contribution. ☐

**Corollary 8.24** (Gaussian Sinkhorn divergence and smoothed Bures term)**.** *Let $\alpha = \mathcal{N}(m_\alpha, \Sigma_\alpha)$ and $\beta = \mathcal{N}(m_\beta, \Sigma_\beta)$ have positive-definite covariances. For $r > 0$, define*

$$\tau_\varepsilon(r) := \frac{\sqrt{\varepsilon^2+16r^2}-\varepsilon}{4r}, \qquad \psi_\varepsilon(r) := -2r\,\tau_\varepsilon(r) - \frac{\varepsilon}{2}\log\big(1-\tau_\varepsilon(r)^2\big).$$

*If $\sigma_i(\Sigma,\Lambda)$ denotes the singular values of $\Sigma^{1/2}\Lambda^{1/2}$ and $\lambda_i(\Sigma)$ the eigenvalues of $\Sigma$, then the debiased Sinkhorn divergence (7.56) is $\bar{\mathcal{L}}^\varepsilon_{\|\cdot-\cdot\|^2}(\alpha,\beta) = \|m_\alpha - m_\beta\|^2 + \mathcal{B}_\varepsilon(\Sigma_\alpha,\Sigma_\beta)^2$, where the Gaussian covariance contribution is the debiased smoothed Bures term*

$$\mathcal{B}_\varepsilon(\Sigma,\Lambda)^2 := \sum_i \psi_\varepsilon\big(\sigma_i(\Sigma,\Lambda)\big) - \frac{1}{2}\sum_i \psi_\varepsilon\big(\lambda_i(\Sigma)\big) - \frac{1}{2}\sum_i \psi_\varepsilon\big(\lambda_i(\Lambda)\big).$$

*Moreover $\mathcal{B}_\varepsilon(\Sigma,\Lambda)^2 \to \mathcal{B}(\Sigma,\Lambda)^2$ as $\varepsilon \downarrow 0$, where $\mathcal{B}$ is the Bures–Wasserstein metric of Proposition 2.39.*

*Proof.* Proposition 8.23 writes the raw entropic value as the squared mean displacement plus trace terms and a spectral sum. With the notation above, the spectral part is exactly $\sum_i \psi_\varepsilon(\sigma_i(\Sigma_\alpha,\Sigma_\beta))$. Applying the same formula to the self-costs $(\alpha,\alpha)$ and $(\beta,\beta)$ replaces these singular values by the eigenvalues of $\Sigma_\alpha$ and $\Sigma_\beta$. In the polarization formula (7.56), the trace terms cancel because $\mathrm{tr}\,\Sigma_\alpha + \mathrm{tr}\,\Sigma_\beta - \frac{1}{2}(2\,\mathrm{tr}\,\Sigma_\alpha) - \frac{1}{2}(2\,\mathrm{tr}\,\Sigma_\beta) = 0$, while the polarization of the squared mean terms leaves exactly $\|m_\alpha - m_\beta\|^2$. Since $\tau_\varepsilon(r) \to 1$ and $\varepsilon\log(1-\tau_\varepsilon(r)^2) \to 0$, one has $\psi_\varepsilon(r) \to -2r$. The limit is therefore $\mathrm{tr}\,\Sigma + \mathrm{tr}\,\Lambda - 2\sum_i \sigma_i(\Sigma,\Lambda) = \mathcal{B}(\Sigma,\Lambda)^2$, which is the Bures formula (2.17). ☐

**Proposition 8.25** (One-dimensional Gaussian Sinkhorn rate)**.** *Consider $\alpha = \beta = \mathcal{N}(0,1)$ on $\mathbb{R}$ with $c(x,y) = (x-y)^2$. Initialize $g^{(0)} = 0$ and, after $\ell$ complete Sinkhorn cycles, write $g^{(\ell)}(y) = q^{(\ell)}y^2 + \mathrm{cst}$. One soft transform maps a quadratic coefficient $q$ to $\mathsf{Q}_\varepsilon(q) = 1 - \frac{1}{1-q+\varepsilon/2}$, $\quad q < 1+\varepsilon/2$, so that $q^{(\ell+1)} = \mathsf{Q}_\varepsilon(\mathsf{Q}_\varepsilon(q^{(\ell)}))$. Set $A_\ell := 1 - q^{(\ell)} + \frac{\varepsilon}{2}$, $\quad A_\star := 1 - q_\star + \frac{\varepsilon}{2} = \frac{\varepsilon+\sqrt{\varepsilon^2+16}}{4}$. Then $q^{(\ell)} \to q_\star = 1+\varepsilon/2 - A_\star$, and the convergence obeys the exact cross-ratio identity $\frac{A_\ell - A_\star}{A_\ell + A_\star^{-1}} = A_\star^{-4\ell}\frac{A_0-A_\star}{A_0+A_\star^{-1}}$. In particular,*

$$0 \leqslant q_\star - q^{(\ell)} \leqslant q_\star\left(\frac{4}{\varepsilon+\sqrt{\varepsilon^2+16}}\right)^{4\ell}.$$

*Proof.* Completing the square in $\int \exp((qy^2-(x-y)^2)/\varepsilon)\mathrm{d}\mathcal{N}(0,1)(y)$ gives $\mathsf{Q}_\varepsilon(q)$. Write $a = \varepsilon/2$. Applying this map twice shows that $A_\ell = 1 - q^{(\ell)} + a$ satisfies $A_{\ell+1} = M(A_\ell)$, $\quad M(A) := a + \frac{A}{1+aA}$. The fixed points of $M$ solve $A^2 - aA - 1 = 0$ and are $A_\star > 0$ and $-A_\star^{-1}$. Direct algebra for this Möbius map gives $\frac{M(A)-A_\star}{M(A)+A_\star^{-1}} = A_\star^{-4}\frac{A-A_\star}{A+A_\star^{-1}}$, which proves the exact identity by iteration. Moreover, $A_0 = 1+a > A_\star$ and $M$ leaves $[A_\star, A_0]$ invariant. On this interval, $0 < M'(A) = \frac{1}{(1+aA)^2} \leqslant \frac{1}{(1+aA_\star)^2} = A_\star^{-4}$, where $1 + aA_\star = A_\star^2$. Hence $0 \leqslant A_\ell - A_\star \leqslant A_\star^{-4\ell}(A_0 - A_\star)$. Since $A_0 - A_\star = q_\star$ and $A_\ell - A_\star = q_\star - q^{(\ell)}$, this is the announced bound. ☐

### 8.7 Continuous $\varepsilon$-Sinkhorn Flow

**Parabolic Monge–Ampère limit.** Take the quadratic torus cost $c(x,y) = d_{\mathbb{T}^d}(x,y)^2/2$. In the zero-temperature limit, the soft transforms concentrate near the map $x \mapsto x + \nabla u(x)$, while the first Laplace correction records its Jacobian determinant. Rescaling the accumulated correction in iteration time leads to the following flow.

**Definition 8.26** (Continuous $\varepsilon$-Sinkhorn flow)**.** Let $\alpha$ and $\beta$ be probability measures on the unit-volume flat torus $\mathbb{T}^d$ with smooth positive densities $\rho_\alpha = e^{-F}$ and $\rho_\beta = e^{-G}$. A gauge-fixed potential $u_t : \mathbb{T}^d \to \mathbb{R}$, with $\int_{\mathbb{T}^d} u_t\mathrm{d}x = 0$, follows the continuous $\varepsilon$-Sinkhorn flow if

$$\partial_t u_t(x) = \log\det\big(\mathrm{Id} + \nabla^2 u_t(x)\big) - G\big(x + \nabla u_t(x)\big) + F(x) - \bar{r}_t, \tag{8.25}$$

where $x + \nabla u_t(x)$ is understood modulo $\mathbb{Z}^d$, $\mathrm{Id} + \nabla^2 u_t \succ 0$, and $\bar{r}_t$ is the unique scalar preserving the mean-zero gauge.

**Proposition 8.27** (Stationary continuous Sinkhorn potentials)**.** *Assume that a smooth stationary solution of (8.25) satisfies $\mathrm{Id} + \nabla^2 u \succ 0$, and define $T(x) = x + \nabla u(x)$ modulo $\mathbb{Z}^d$. Then $T_\sharp\alpha = \beta$. Conversely, any smooth map of this form satisfying $\mathrm{Id} + \nabla^2 u \succ 0$ and $T_\sharp\alpha = \beta$ solves the stationary equation, up to the additive gauge of $u$.*

*Proof.* At stationarity, the right-hand side before subtracting $\bar{r}_t$ is a constant $r$, hence $e^{-G(T(x))}\det\nabla T(x) = e^{-F(x)}e^r$. The positive Hessian condition makes $T$ an orientation-preserving local diffeomorphism of the torus, homotopic to the identity and therefore of degree one. It is thus a diffeomorphism. Integrating the displayed identity and using that both measures have unit mass gives $e^r = 1$. The remaining equality is precisely the change-of-variables formula for $T_\sharp(e^{-F}\mathrm{d}x) = e^{-G}\mathrm{d}x$. Reversing the argument proves the converse. ☐

**Rescaled continuous Sinkhorn iterates.** Let $\mathsf{S}_\varepsilon$ denote one complete continuous dual Sinkhorn cycle (7.26), written in the sign convention $u = -f$. Using the soft transforms of Definition 7.19,

$\mathsf{S}_\varepsilon u := -\Big((-u)^{c,\varepsilon}\Big)^{\bar{c},\varepsilon}.$

**Proposition 8.28** (Rescaled continuous Sinkhorn limit)**.** *Let $u_\varepsilon^{(0)} = u_0$ be smooth and mean zero, generate $u_\varepsilon^{(\ell+1)} = \mathsf{S}_\varepsilon u_\varepsilon^{(\ell)} - \int_{\mathbb{T}^d}\mathsf{S}_\varepsilon u_\varepsilon^{(\ell)}\mathrm{d}x$, and define $u_\varepsilon(t) := u_\varepsilon^{(\lfloor t/\varepsilon\rfloor)}$. In particular, for $\varepsilon = 1/k$, one observes $u_{1/k}^{(\lfloor kt\rfloor)}$ at time $t$. Assume that, as $\varepsilon \downarrow 0$, these interpolants converge smoothly on compact time intervals to $u_t$, that $\mathrm{Id} + \nabla^2 u_t \succ 0$, and that the Laplace expansion below is uniform along the sequence. Then $u_t$ solves (8.25).*

*Proof.* The Laplace expansion of the two soft transforms gives the first-order consistency formula

$$\mathsf{S}_\varepsilon u - u = \varepsilon\left[\log\det(\mathrm{Id}+\nabla^2 u) - G(\mathrm{Id}+\nabla u) + F\right] + O(\varepsilon^2),$$

up to an $x$-independent term, which disappears under the gauge normalization. Dividing the normalized increment by $\varepsilon$ and passing formally to the smooth limit gives (8.25); the subtracted spatial mean becomes $\bar r_t$. $\square$

**One-dimensional Gaussian closure.** $\partial_t u_t(x) = \log\big(1+u_t''(x)\big) - G\big(x+u_t'(x)\big) + F(x) - \bar r_t$. Although it was introduced on the torus, the same formal equation applies on $\mathbb{R}$ for confining Gaussian densities. Let $\alpha = \mathcal{N}(m_\alpha, \sigma_\alpha^2)$ and $\beta = \mathcal{N}(m_\beta, \sigma_\beta^2)$.
$x + u_t'(x) = q_t + a_t(x - m_\alpha), \qquad a_t > 0.$

$$\dot a_t = \frac{1}{\sigma_\alpha^2} - \frac{a_t^2}{\sigma_\beta^2}, \qquad \dot q_t = -\frac{a_t}{\sigma_\beta^2}(q_t - m_\beta). \tag{8.26}$$

# 9 Statistical Optimal Transport

## 9.1 Law of Large Numbers and Central Limit Theorem

**Empirical laws.**

$$\hat\alpha_n \coloneqq \frac{1}{n}\sum_{i=1}^n \delta_{X_i}. \tag{9.1}$$

The ordinary law of large numbers says that empirical averages converge to expectations. In measure language this means that $\hat\alpha_n$ converges weakly toward $\alpha$, because testing $\hat\alpha_n$ against a bounded continuous function $\varphi$ gives the sample average $n^{-1}\sum_i \varphi(X_i)$.
**Proposition 9.1** (Empirical law of large numbers in $\mathcal{W}_p$)**.** *Let $(\mathcal{X}, d)$ be a Polish metric space, let $p \geqslant 1$, and let $\alpha \in \mathcal{P}_p(\mathcal{X})$. Let $(X_i)_{i\geqslant 1}$ be i.i.d. random variables with law $\alpha$, and define $\hat\alpha_n$ by (9.1). Then*

$$\hat\alpha_n \rightharpoonup \alpha \quad \textit{and} \quad \mathcal{W}_p(\hat\alpha_n, \alpha) \longrightarrow 0$$

*almost surely. Moreover, $\mathbb{E}\,\mathcal{W}_p(\hat\alpha_n,\alpha)^p \longrightarrow 0$.*

*Proof.* Fix a reference point $x_0 \in \mathcal{X}$, and write $r(x) = d(x, x_0)$. Since $\mathcal{X}$ is Polish, the weak topology on $\mathcal{P}(\mathcal{X})$ admits a countable convergence-determining class $(\varphi_k)_{k\geqslant 1} \subset C_b(\mathcal{X})$. For each fixed $k$, the strong law of large numbers gives $\int \varphi_k \mathrm{d}\hat\alpha_n = \frac{1}{n}\sum_{i=1}^n \varphi_k(X_i) \longrightarrow \mathbb{E}\varphi_k(X_1) = \int \varphi_k \mathrm{d}\alpha$ almost surely. Intersecting the corresponding probability-one events over the countable set of indices gives, on a single event of probability one, convergence against every $\varphi_k$. By the convergence-determining property, this is exactly $\hat\alpha_n \rightharpoonup \alpha$.
The moment condition $\alpha \in \mathcal{P}_p(\mathcal{X})$ means $\int r^p \mathrm{d}\alpha < +\infty$. Applying the strong law again to the integrable random variable $r(X_1)^p$ gives $\int r^p \mathrm{d}\hat\alpha_n = \frac{1}{n}\sum_{i=1}^n r(X_i)^p \longrightarrow \int r^p \mathrm{d}\alpha$ almost surely. Proposition **??** states that weak convergence plus convergence of the $p$-th moments is equivalent to $\mathcal{W}_p$-convergence on $\mathcal{P}_p(\mathcal{X})$. Hence $\mathcal{W}_p(\hat\alpha_n,\alpha) \to 0$ almost surely.
Set $A_n = \int r^p \mathrm{d}\hat\alpha_n$ and $M = \int r^p \mathrm{d}\alpha$. The triangle inequality through the Dirac mass $\delta_{x_0}$, followed by $(a+b)^p \leqslant 2^{p-1}(a^p + b^p)$, gives the deterministic bound $\mathcal{W}_p(\hat\alpha_n,\alpha)^p \leqslant 2^{p-1}\big(A_n + M\big)$. The family $(A_n)_n$ is uniformly integrable. Indeed, by the de la Vallée–Poussin criterion, choose a convex superlinear function $\Psi$ such that $\mathbb{E}\Psi(r(X_1)^p) < +\infty$; Jensen's inequality gives $\mathbb{E}\Psi(A_n) \leqslant \frac{1}{n}\sum_{i=1}^n \mathbb{E}\Psi(r(X_i)^p) = \mathbb{E}\Psi(r(X_1)^p)$. Thus $(A_n + M)_n$, and hence $(\mathcal{W}_p(\hat\alpha_n,\alpha)^p)_n$, is uniformly integrable. Together with almost-sure convergence to zero, this implies $\mathbb{E}\,\mathcal{W}_p(\hat\alpha_n,\alpha)^p \to 0$. $\square$

**Central-limit fluctuations.**
**Proposition 9.2** (Berry–Esseen bound in $\mathcal{W}_1$)**.** *Let $(X_i)_{i=1}^n$ be i.i.d. real random variables such that $\mathbb{E}X_i = 0$, $\mathbb{E}X_i^2 = 1$ and $\mathbb{E}|X_i|^3 < +\infty$. If $\alpha_n$ is the law of $n^{-1/2}\sum_i X_i$ and $\gamma$ is the standard Gaussian law, then $\mathcal{W}_1(\alpha_n,\gamma) \leqslant \frac{C\,\mathbb{E}|X_1|^3}{\sqrt{n}}$, where $C$ is a universal constant.*

*Proof.* By Kantorovich–Rubinstein duality,

$$\mathcal{W}_1(\alpha_n,\gamma) = \sup_{\mathrm{Lip}(h)\leqslant 1} \left|\mathbb{E}h(S_n) - \mathbb{E}h(G)\right|, \qquad S_n = n^{-1/2}\sum_i X_i,\ G\sim\gamma.$$

For each such $h$, solve Stein's equation $f_h'(x) - x f_h(x) = h(x) - \mathbb{E}h(G)$. The solution satisfies $\|f_h'\|_\infty + \|f_h''\|_\infty \leqslant C$ for a universal constant. Hence $\mathbb{E}h(S_n) - \mathbb{E}h(G) = \mathbb{E}[f_h'(S_n) - S_n f_h(S_n)]$. Write $S_n^{(i)} = S_n - X_i/\sqrt{n}$. Independence and $\mathbb{E}X_i = 0$ give

$$\mathbb{E}[S_n f_h(S_n)] = \frac{1}{\sqrt{n}}\sum_{i=1}^n \mathbb{E}\Big[X_i\big(f_h(S_n^{(i)} + X_i/\sqrt{n}) - f_h(S_n^{(i)})\big)\Big].$$

Taylor's formula with integral remainder and $\mathbb{E}X_i^2 = 1$ imply

$$\left|\mathbb{E}[S_n f_h(S_n)] - \frac{1}{n}\sum_{i=1}^n \mathbb{E}f_h'(S_n^{(i)})\right| \leqslant \frac{C\,\mathbb{E}|X_1|^3}{\sqrt{n}}.$$

Moreover, the Lipschitz bound on $f_h'$ yields

$$\left|\mathbb{E}f_h'(S_n) - \frac{1}{n}\sum_{i=1}^n \mathbb{E}f_h'(S_n^{(i)})\right| \leqslant \frac{C\,\mathbb{E}|X_1|}{\sqrt{n}} \leqslant \frac{C\,\mathbb{E}|X_1|^3}{\sqrt{n}},$$

where the last inequality uses $\mathbb{E}X_1^2 = 1$. Sharper constants and higher-order transport-distance refinements are available, but are omitted from the compact handout. $\square$

**Proposition 9.3** (Sharp symmetric lattice asymptotic)**.** *Let $X$ be centered, symmetric, of unit variance, and supported on a finite subset of $a + h\mathbb{Z}$, where $h > 0$ is the maximal lattice span. If $\alpha_n$ is the law of $n^{-1/2}\sum_{i=1}^n X_i$ and $\gamma = \mathcal{N}(0, 1)$, then $\sqrt{n}\, \mathcal{W}_1(\alpha_n, \gamma) \longrightarrow \frac{h}{4}$.*

*Proof.* Denote by $F_n$ the CDF of $\alpha_n$, and by $\Phi$ and $\varphi$ the standard Gaussian CDF and density. With the centered periodic sawtooth $\psi(u) = \frac{1}{2} - \{u\}$, the integrated lattice Edgeworth expansion gives

$$F_n(x) - \Phi(x) = \frac{h}{\sqrt{n}}\, \psi\Big(\frac{\sqrt{n}\, x - na}{h}\Big)\, \varphi(x) + r_n(x), \qquad \|r_n\|_{L^1(\mathbb{R})} = o(n^{-1/2});$$

see,, and. Vallender's one-dimensional identity states that $\mathcal{W}_1(\alpha_n, \gamma) = \int_{\mathbb{R}} |F_n - \Phi|$. Therefore $\big||u + v| - |u|\big| \leqslant |v|$ reduces the claim to

$$\int_{\mathbb{R}} \Big|\psi\Big(\frac{\sqrt{n}\, x - na}{h}\Big)\Big|\varphi(x)\mathrm{d}x \longrightarrow \int_0^1 |\psi(u)|\mathrm{d}u = \frac{1}{4}.$$

The convergence is periodic averaging: prove it first for compactly supported step functions, where it is a Riemann sum over periods, and conclude by $L^1$-approximation of $\varphi$. ☐

**Proposition 9.4** (Sharp symmetric density asymptotic)**.** *Let $X$ be centered, symmetric, of unit variance, with a density and moments of every order. Set $\kappa_4 = \mathbb{E}[X^4] - 3$ and $H_3(x) = x^3 - 3x$. If $\alpha_n$ is the law of $n^{-1/2}\sum_{i=1}^n X_i$ and $\gamma = \mathcal{N}(0, 1)$, then $\mathcal{W}_1(\alpha_n, \gamma) = \frac{|\kappa_4|}{24n}\int_{\mathbb{R}} |H_3(x)|\varphi(x)\mathrm{d}x + o(n^{-1})$.*

*Proof.* The non-lattice Edgeworth expansion through order $n^{-1}$ has an $n^{-1/2}$ term proportional to the third cumulant, and $n^{-1}$ terms proportional to the fourth cumulant and to the square of the third one. Symmetry makes the third cumulant vanish and gives $F_n(x) - \Phi(x) = -\frac{\kappa_4}{24n}H_3(x)\varphi(x) + r_n(x), \qquad \|r_n\|_{L^1(\mathbb{R})} = o(n^{-1})$; see and. Inserting this expansion into Vallender's identity and using $\big||u + v| - |u|\big| \leqslant |v|$ proves the formula. ☐

$\int_{\mathbb{R}} |H_3(x)|\varphi(x)\mathrm{d}x = 2\varphi(0) + 8\varphi(\sqrt{3}) = \frac{2+8e^{-3/2}}{\sqrt{2\pi}}$,

$$\mathcal{W}_1(\alpha_n, \gamma) \sim \begin{cases} \dfrac{1}{2\sqrt{n}}, & X \sim \frac{1}{2}(\delta_{-1} + \delta_1), \\ \dfrac{1 + 4e^{-3/2}}{10\sqrt{2\pi}}\dfrac{1}{n}, & X \sim \mathrm{Unif}[-\sqrt{3}, \sqrt{3}]. \end{cases} \tag{9.2}$$

$\alpha_n = (D_{1/\sqrt{n}})_\sharp \alpha^{*n}, \qquad n \geqslant 1,$

### 9.2 Sample Complexity Unregularized OT.

$$r_{n,p,d} := \begin{cases} n^{-1/(2p)}, & d < 2p, \\ n^{-1/(2p)}(\log(1 + n))^{1/p}, & d = 2p, \\ n^{-1/d}, & d > 2p. \end{cases} \tag{9.3}$$

**Proposition 9.5** (Empirical OT has intrinsic-dimension value rates)**.** *Let $p \geqslant 1$, let $\mathcal{X} \subset \mathbb{R}^d$ be compact, and, for each $\alpha \in \mathcal{P}(\mathcal{X})$, let $\widehat{\alpha}_n = n^{-1}\sum_{i=1}^n \delta_{X_i}$ for i.i.d. samples $X_i \sim \alpha$. Then*

$$\sup_{\alpha\in\mathcal{P}(\mathcal{X})} \mathbb{E}[\mathcal{W}_p(\widehat{\alpha}_n, \alpha)^p]^{1/p} \leqslant C_{\mathcal{X},p,d}\, r_{n,p,d}. \tag{9.4}$$

$$\mathbb{E}\Big[\Big|\mathcal{W}_p(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{W}_p(\alpha, \beta)\Big|^p\Big]^{1/p} \lesssim_{p,d} r_{n,p,d} + r_{m,p,d}.$$

**Proposition 9.6** (Dyadic partition bound for $\mathcal{W}_1$)**.** *Let $\mathcal{Q}_j$ be nested dyadic partitions of $[0, 1]^d$ into half-open cubes of side length $2^{-j}$, with cubes on the outer face closed along the boundary of $[0, 1]^d$. For all $\alpha, \beta \in \mathcal{P}([0, 1]^d)$ and every integer $J \geqslant 0$, $\mathcal{W}_1(\alpha, \beta) \leqslant \sqrt{d}\sum_{j=0}^{J-1} 2^{-j}\sum_{Q\in\mathcal{Q}_{j+1}} |\alpha(Q) - \beta(Q)| + \sqrt{d}\, 2^{-J}$.*

*Proof.* The proof is constructive: one decomposes the two measures into pieces which can be coupled at different spatial scales. Write $\Omega = [0, 1]^d$. We build positive measures $(\alpha_j, \beta_j)_{j=0}^J$ such that $\sum_{j=0}^J \alpha_j = \alpha, \qquad \sum_{j=0}^J \beta_j = \beta, \qquad \alpha_j(Q) = \beta_j(Q)$ for every $Q \in \mathcal{Q}_j$. Assuming such a decomposition for a moment, each pair $(\alpha_j, \beta_j)$ can be coupled inside the cells of $\mathcal{Q}_j$. Indeed, on every cell $Q \in \mathcal{Q}_j$ with positive mass, take the product coupling between the normalized restrictions of $\alpha_j$ and $\beta_j$ to $Q$, multiplied by the common mass $\alpha_j(Q) = \beta_j(Q)$. Summing over cells and over $j$ gives a coupling of $\alpha$ and $\beta$. Since the diameter of a cell in $\mathcal{Q}_j$ is at most $\sqrt{d}\, 2^{-j}$, this coupling has cost bounded by $\mathcal{W}_1(\alpha, \beta) \leqslant \sqrt{d}\sum_{j=0}^J 2^{-j}\alpha_j(\Omega)$.
Start at the finest scale. For $Q \in \mathcal{Q}_J$, define $\alpha_J$ and $\beta_J$ on $Q$ by keeping the common part of the two masses:

$$\alpha_J(A \cap Q) = \frac{\alpha(Q) \wedge \beta(Q)}{\alpha(Q)}\, \alpha(A \cap Q), \qquad \beta_J(A \cap Q) = \frac{\alpha(Q) \wedge \beta(Q)}{\beta(Q)}\, \beta(A \cap Q),$$

with the convention that the corresponding term is zero when the denominator vanishes. For $j = J - 1, \ldots, 1$, assume that the pieces at all finer levels $k > j$ have been constructed and define the residuals $\alpha'_j = \alpha - \sum_{k=j+1}^J \alpha_k, \qquad \beta'_j = \beta - \sum_{k=j+1}^J \beta_k$. On each $Q \in \mathcal{Q}_j$, set $\alpha_j$ and $\beta_j$ equal to the common part of $\alpha'_j$ and $\beta'_j$ inside $Q$, exactly as above. Finally set $\alpha_0 = \alpha - \sum_{k=1}^J \alpha_k$ and $\beta_0 = \beta - \sum_{k=1}^J \beta_k$. Each step removes only common residual mass, so all measures are positive, and by construction $\alpha_j(Q) = \beta_j(Q)$ on all cells $Q \in \mathcal{Q}_j$.

For notational consistency, set $\alpha'_0 = \alpha_0$, $\beta'_0 = \beta_0$, $\alpha'_J = \alpha$, and $\beta'_J = \beta$. Let $0 \leqslant j < J$ and $R \in \mathcal{Q}_j$. Since all pieces constructed at levels finer than $j+1$ have equal mass on every child $Q \in \mathcal{Q}_{j+1}$, the residual imbalance on $Q$ just before the matching at level $j+1$ is the original imbalance: $\alpha'_{j+1}(Q) - \beta'_{j+1}(Q) = \alpha(Q) - \beta(Q)$. After removing the common part at level $j+1$, the remaining source residual inside $Q$ has mass $\alpha'_j(Q) = \alpha'_{j+1}(Q) - \alpha_{j+1}(Q) = \left(\alpha'_{j+1}(Q) - \beta'_{j+1}(Q)\right)_+$. Because $\alpha_j$ is itself a common submeasure of the residual $\alpha'_j$, the mass matched at level $j$ inside $R$ satisfies

$$\alpha_j(R) = \beta_j(R) \leqslant \alpha'_j(R) = \sum_{Q \subset R} \left(\alpha'_{j+1}(Q) - \beta'_{j+1}(Q)\right)_+ \leqslant \sum_{Q \subset R} |\alpha(Q) - \beta(Q)|,$$

where the sums are over $Q \in \mathcal{Q}_{j+1}$. Summing over $R \in \mathcal{Q}_j$ gives $\alpha_j(\Omega) \leqslant \sum_{Q \in \mathcal{Q}_{j+1}} |\alpha(Q) - \beta(Q)|$. The last level satisfies $\alpha_J(\Omega) \leqslant 1$. Substituting these bounds into the cost estimate proves the claim. □

*Proof of Proposition 9.5.* General empirical Wasserstein moment estimates prove the one-sample statement for arbitrary $p$; the two-sample estimate then follows from the reverse triangle inequality and Minkowski's inequality. By the triangle inequality, $\left|\mathcal{W}_1(\hat\alpha_n, \hat\beta_m) - \mathcal{W}_1(\alpha, \beta)\right| \leqslant \mathcal{W}_1(\hat\alpha_n, \alpha) + \mathcal{W}_1(\hat\beta_m, \beta)$. It is therefore enough to bound $\mathbb{E}\,\mathcal{W}_1(\hat\alpha_n, \alpha)$. Apply Proposition 9.6 to $(\hat\alpha_n, \alpha)$. If $Q \in \mathcal{Q}_{j+1}$, then $n\hat\alpha_n(Q)$ is a binomial random variable with parameter $\alpha(Q)$, and hence $\mathbb{E}\,|\hat\alpha_n(Q) - \alpha(Q)| \leqslant \sqrt{\mathbb{E}\,|\hat\alpha_n(Q) - \alpha(Q)|^2} \leqslant \sqrt{\frac{\alpha(Q)}{n}}$. Summing over the $2^{(j+1)d}$ cells and using the Cauchy–Schwarz inequality gives

$$\sum_{Q \in \mathcal{Q}_{j+1}} \mathbb{E}\,|\hat\alpha_n(Q) - \alpha(Q)| \leqslant \frac{1}{\sqrt{n}} \sum_{Q \in \mathcal{Q}_{j+1}} \sqrt{\alpha(Q)} \leqslant \frac{2^{(j+1)d/2}}{\sqrt{n}}.$$

Therefore, for every $J \geqslant 0$, $\mathbb{E}\,\mathcal{W}_1(\hat\alpha_n, \alpha) \lesssim_d 2^{-J} + \frac{1}{\sqrt{n}} \sum_{j=0}^{J-1} 2^{j(d/2-1)}$. If $d = 1$, the sum is uniformly bounded and we choose $J$ such that $2^{-J} \lesssim n^{-1/2}$. If $d = 2$, the sum is $J$, and choosing $2^J \simeq n^{1/2}$ gives $(\log n) n^{-1/2}$. If $d \geqslant 3$, the sum is dominated by its last term; choosing $2^J \simeq n^{1/d}$ gives $2^{-J} + n^{-1/2} 2^{J(d/2-1)} \simeq n^{-1/d}$. This proves the displayed $\mathcal{W}_1$ rates. The elementary dyadic estimate is not sharp for every law in the critical case $d = 2$. The proposition deliberately retains the distribution-free $(\log n) n^{-1/2}$ bound supplied by the dyadic proof. □

**Lower bounds and minimax optimality.**

**Proposition 9.7** (Minimax lower bound for bounded densities)**.** *Let $d \geqslant 3$. Let $\mathcal{A}$ be the class of probability measures on $[0,1]^d$ admitting a density $\rho$ with respect to Lebesgue measure such that $\frac{1}{2} \leqslant \rho(x) \leqslant \frac{3}{2}$ for a.e. $x \in [0,1]^d$. For each $n \geqslant 1$, let $\mathfrak{E}_n$ be the class of measurable estimators $\tilde\alpha_n : ([0,1]^d)^n \longrightarrow \mathcal{P}([0,1]^d)$. There exists $c_d > 0$, depending only on d, such that*

$$\inf_{\tilde\alpha_n \in \mathfrak{E}_n} \sup_{\alpha \in \mathcal{A}} \mathbb{E}_{(X_1,\dots,X_n) \sim \alpha^{\otimes n}} \left[\mathcal{W}_1\left(\tilde\alpha_n(X_1, \dots, X_n), \alpha\right)\right] \geqslant c_d\, n^{-1/d}.$$

*The expectation is over i.i.d. samples $X_i \sim \alpha$. In particular, the high-dimensional rate of Proposition 9.5 is minimax sharp for $\mathcal{W}_1$, even when one allows arbitrary estimators rather than the empirical measure.*

*Proof.* We use a standard hypercube construction. Choose a smooth function $\psi$ compactly supported in $(0,1)^d$, with $\int \psi = 0$, $\|\psi\|_\infty \leqslant 1$, and whose positive and negative parts are separated in the first coordinate. There exists a compactly supported Lipschitz function $\varphi$ such that $\int \varphi\psi \mathrm{d}x > 0$. We extend both functions by zero outside $(0,1)^d$, and absorb $\mathrm{Lip}(\varphi)$ into the constants below.

Let $m$ be an integer, $h = 1/m$, and partition $[0,1]^d$ into $M = m^d$ cubes $Q_k = z_k + h[0,1)^d$. Define the rescaled bumps

$$\psi_k(x) = \psi\left(\frac{x - z_k}{h}\right) \mathbf{1}_{Q_k}(x).$$

For $\theta = (\theta_1, \dots, \theta_M) \in \{-1,1\}^M$ and a fixed small $\eta > 0$, set $\rho_\theta(x) = 1 + \eta \sum_{k=1}^M \theta_k \psi_k(x)$, $\qquad \alpha_\theta = \rho_\theta \mathrm{d}x$. For $\eta$ small enough, all these densities belong to $\mathcal{A}$, and they integrate to one because every $\psi_k$ has zero mean.

If $\theta$ and $\theta'$ differ on a set $S$ of coordinates, the dual formula for $\mathcal{W}_1$ gives an additive lower bound. Indeed, rescale $\varphi$ as $\varphi_k(x) = h\,\varphi\left(\frac{x - z_k}{h}\right)$. Each $\varphi_k$ is supported in a fixed strict subcube of $Q_k$. After dividing by this constant, Kantorovich–Rubinstein duality and the choice $s_k = \mathrm{sign}(\theta_k - \theta'_k)$ on $S$ give $\mathcal{W}_1(\alpha_\theta, \alpha_{\theta'}) \geqslant c\,\eta\, h^{d+1}\, \mathrm{Ham}(\theta, \theta')$, where Ham denotes Hamming distance and $c > 0$ depends only on the chosen bump.

Neighboring vertices of the hypercube are statistically close. If $\theta^{[k]}$ is obtained from $\theta$ by flipping only the $k$-th sign, then the one-sample Kullback–Leibler divergence is bounded by $\mathrm{KL}(\alpha_\theta | \alpha_{\theta^{[k]}}) \leqslant C\eta^2 h^d$, because the two densities differ only on $Q_k$, the perturbation has size $O(\eta)$, and the densities are uniformly bounded away from zero. For $n$ independent samples the divergence is multiplied by $n$. Choose $m$ so that $m^d \simeq n$. Then $nh^d$ is bounded, and, by taking $\eta$ sufficiently small, Pinsker's inequality (Theorem 6.13) gives a uniform bound $\mathrm{TV}\left(\alpha_\theta^{\otimes n}, \alpha_{\theta^{[k]}}^{\otimes n}\right) \leqslant q < 1$. Assouad's lemma therefore implies

$$\inf_{\tilde\alpha_n \in \mathfrak{E}_n} \sup_{\theta \in \{-1,1\}^M} \mathbb{E}_{\alpha_\theta^{\otimes n}} \left[\mathcal{W}_1\left(\tilde\alpha_n(X_1, \dots, X_n), \alpha_\theta\right)\right] \geqslant c'\eta\, M h^{d+1}(1 - q) = c'\eta(1 - q)h \simeq n^{-1/d}.$$

Decreasing the constant handles the finitely many small values of $n$, and the claim follows. □

**Leveraging smoothness.**

**Proposition 9.8** (Smoothed plug-in rates)**.** *Let $d \geqslant 3$, $0 < s \leqslant 1$, and let $\alpha = \rho_\alpha \mathrm{d}x$, $\beta = \rho_\beta \mathrm{d}x$ be probability measures on the unit-volume flat torus $\mathbb{T}^d$, with $\max\{\|\rho_\alpha\|_{H^s}, \|\rho_\beta\|_{H^s}\} \leqslant L$. Let $\kappa$ be a smooth, symmetric probability density on $\mathbb{R}^d$, with finite second moment and rapidly decaying Fourier transform; a Gaussian is a typical choice. Denote by $\kappa_h$ the periodization of $h^{-d}\kappa(\cdot/h)$. For empirical measures $\hat\alpha_n$ and $\hat\beta_m$ formed from i.i.d. samples of $\alpha$ and $\beta$, define $\tilde\alpha_n := \hat\alpha_n * \kappa_{h_n}$, $\qquad \tilde\beta_m := \hat\beta_m * \kappa_{h_m}$. Then, for every $0 < h_n, h_m \leqslant 1$,*

$$\mathbb{E}\,\mathcal{W}_1(\tilde\alpha_n, \alpha) \leqslant C\left(h_n^{s+1} + n^{-1/2} h_n^{1-d/2}\right),$$

*and the analogous bound holds for $(\tilde\beta_m,\beta)$. Choosing $h_n \simeq n^{-1/(d+2s)}$ and $h_m \simeq m^{-1/(d+2s)}$ therefore gives*

$$\mathbb{E}\left|\mathcal{W}_1(\tilde\alpha_n,\tilde\beta_m)-\mathcal{W}_1(\alpha,\beta)\right| \leqslant C\left(n^{-\frac{s+1}{d+2s}}+m^{-\frac{s+1}{d+2s}}\right).$$

*The constant depends only on $d, s, L$ and $\kappa$.*

*Proof.* It suffices to prove the one-sample bound. Set $h = h_n$ and write $\tilde\alpha_n = \rho_{\tilde\alpha_n}\mathrm{d}x$. For a zero-mean function $w$ on the torus, set $\|w\|^2_{\dot H^{-1}(\mathbb{T}^d)} := \sum_{k\in\mathbb{Z}^d\setminus\{0\}} \frac{|\widehat{w}(k)|^2}{4\pi^2\|k\|^2}$. Kantorovich–Rubinstein duality and Cauchy–Schwarz give $\mathcal{W}_1(u\mathrm{d}x, v\mathrm{d}x) \lesssim \|u-v\|_{\dot H^{-1}(\mathbb{T}^d)}$, because a 1-Lipschitz test function has $L^2$-gradient norm at most one on the unit-volume torus. Moreover, $\mathbb{E}\rho_{\tilde\alpha_n} = \rho_\alpha * \kappa_h$, so the two terms in the risk are a deterministic smoothing bias and a centered empirical fluctuation. Symmetry and the finite second moment of $\kappa$ imply $|1-\widehat{\kappa}(\xi)| \lesssim \min\{1,\|\xi\|^2\}$. Fourier estimates and Parseval's identity therefore give

$$\|\rho_\alpha * \kappa_h - \rho_\alpha\|_{\dot H^{-1}} \lesssim h^{s+1}\|\rho_\alpha\|_{H^s},$$
$$\mathbb{E}\|\rho_{\tilde\alpha_n} - \rho_\alpha * \kappa_h\|^2_{\dot H^{-1}} \lesssim \frac{1}{n}\sum_{k\neq 0}\frac{|\widehat{\kappa}(hk)|^2}{\|k\|^2} \lesssim \frac{h^{2-d}}{n}.$$

The first estimate uses $0 < s \leqslant 1$, and the last one uses $d \geqslant 3$ and the decay of $\widehat{\kappa}$. The triangle and Jensen inequalities now give the one-sample bound. Equating its two terms yields $h \simeq n^{-1/(d+2s)}$; the reverse triangle inequality for $\mathcal{W}_1$ gives the two-sample claim. □

$\tilde\alpha_{n,M} := \frac{1}{nM}\sum_{i=1}^n\sum_{j=1}^M \delta_{X_i+h_n\xi_{i,j}}$.

$$\mathbb{E}\,\mathcal{W}_1(\tilde\alpha_{n,M},\tilde\alpha_n) \leqslant h_n\,\mathbb{E}\,\mathcal{W}_1\left(\frac{1}{M}\sum_{j=1}^M\delta_{\xi_j}, \mathcal{N}(0,\mathrm{Id}_d)\right) \lesssim_d h_n M^{-1/d},$$

**MMD.** MMD contains no transport optimization: its square is a combination of expectations of kernel evaluations, estimated empirically by sums of kernel values. Ordinary Monte-Carlo averaging therefore gives the dimension-free parametric rate below for bounded kernels.

**Proposition 9.9** (MMD has a parametric value rate)**.** *Let $k$ be a bounded positive definite kernel with RKHS $\mathcal{H}_k$, and define* $\mathrm{MMD}_k(\alpha,\beta) = \|\int k(x,\cdot)\mathrm{d}(\alpha-\beta)(x)\|_{\mathcal{H}_k}$. *If $\hat\alpha_n$ and $\hat\beta_m$ are independent empirical measures, then*

$$\mathbb{E}\left|\mathrm{MMD}_k(\hat\alpha_n,\hat\beta_m)-\mathrm{MMD}_k(\alpha,\beta)\right| \leqslant \kappa\left(\frac{1}{\sqrt{n}}+\frac{1}{\sqrt{m}}\right)$$

*when $k(x,x) \leqslant \kappa^2$.*

*Proof.* Let $\Phi(x) = k(x,\cdot)$ be the feature map and $m_\alpha = \mathbb{E}\Phi(X)$. The reverse triangle inequality for the RKHS norm gives $\left|\mathrm{MMD}_k(\hat\alpha_n,\hat\beta_m)-\mathrm{MMD}_k(\alpha,\beta)\right| \leqslant \mathrm{MMD}_k(\hat\alpha_n,\alpha)+\mathrm{MMD}_k(\hat\beta_m,\beta)$. Independence cancels the cross terms after taking the squared norm and expectation, giving $\mathbb{E}\,\mathrm{MMD}_k(\hat\alpha_n,\alpha)^2 = \frac{1}{n}\mathbb{E}\|\Phi(X)-m_\alpha\|^2_{\mathcal{H}_k} = \frac{1}{n}\Big(\mathbb{E}k(X,X)-\mathbb{E}k(X,X')\Big)$. The same estimate applies to $\hat\beta_m$, and Jensen's inequality together with $k(x,x)\leqslant\kappa^2$ gives the displayed bound. □

**Entropic OT.**

**Proposition 9.10** (Fixed-temperature Sinkhorn divergence has a parametric rate)**.** *Let $c(x,y) = \|x-y\|^2/2$, and assume that $\alpha$ and $\beta$ are $\sigma^2$-subgaussian probability measures on $\mathbb{R}^d$, in the sense that $\int e^{\|x\|^2/(2d\sigma^2)}\mathrm{d}\alpha(x) \leqslant 2$, $\quad \int e^{\|y\|^2/(2d\sigma^2)}\mathrm{d}\beta(y) \leqslant 2$. Set*

$$q_d := \left\lceil\frac{5d}{2}\right\rceil + 6, \qquad b_d := \left\lceil\frac{5d}{4}\right\rceil + 3, \qquad \Lambda_{d,\sigma}(\varepsilon) := \varepsilon\left(1+\frac{\sigma^{q_d}}{\varepsilon^{b_d}}\right).$$

*Let $\hat\alpha_n$ and $\hat\beta_m$ be independent empirical measures. Then, for every $\varepsilon > 0$,*

$$\mathbb{E}\left|\bar{\mathcal{L}}^\varepsilon_c(\hat\alpha_n,\hat\beta_m)-\bar{\mathcal{L}}^\varepsilon_c(\alpha,\beta)\right| \leqslant C_d\,\Lambda_{d,\sigma}(\varepsilon)\left(\frac{1}{\sqrt{n}}+\frac{1}{\sqrt{m}}\right).$$

*Proof.* For probability measures $\alpha_0,\beta_0$, write the dual objective, following (7.22), as

$$\mathcal{D}_\varepsilon^{\alpha_0,\beta_0}(u,v) := \int u\mathrm{d}\alpha_0 + \int v\mathrm{d}\beta_0 - \varepsilon\iint\left(\exp\left(\frac{u(x)+v(y)-c(x,y)}{\varepsilon}\right)-1\right)\mathrm{d}\alpha_0(x)\mathrm{d}\beta_0(y).$$

One can choose normalized optimal potentials for which the Sinkhorn equations hold pointwise on $\mathbb{R}^d$. Thus, at an optimal pair $(u,v)$,

$$\int\exp\left(\frac{u(x)+v(y)-c(x,y)}{\varepsilon}\right)\mathrm{d}\beta_0(y) = 1 \qquad \text{for all } x.$$

Equivalently, the potentials are fixed points of the soft $c$-transform, but only the dual normalization is needed below. Let $(u,v)$ and $(u_n,v_n)$ be optimal dual potentials for $(\alpha_0,\beta_0)$ and $(\alpha_n,\beta_0)$. By optimality,

$$\mathcal{D}_\varepsilon^{\alpha_0,\beta_0}(u_n,v_n) \leqslant \mathcal{D}_\varepsilon^{\alpha_0,\beta_0}(u,v), \qquad \mathcal{D}_\varepsilon^{\alpha_n,\beta_0}(u,v) \leqslant \mathcal{D}_\varepsilon^{\alpha_n,\beta_0}(u_n,v_n).$$

Hence the difference of the two optimal values is bounded by evaluating the two objectives at either pair of potentials. For the pair $(u, v)$ associated with $(\alpha_0, \beta_0)$, the preceding Sinkhorn equation cancels the exponential term in the change of the first marginal: $\mathcal{D}_\varepsilon^{\alpha_0,\beta_0}(u, v) - \mathcal{D}_\varepsilon^{\alpha_n,\beta_0}(u, v) = \int u\mathrm{d}(\alpha_0 - \alpha_n)$. The same identity holds for $(u_n, v_n)$, using its normalization with respect to $\beta_0$.

$$|\mathcal{L}_c^\varepsilon(\alpha_n, \beta_0) - \mathcal{L}_c^\varepsilon(\alpha_0, \beta_0)| \leqslant 2 \sup_{w\in\mathcal{F}_{\varepsilon,\sigma}} \left| \int w\mathrm{d}(\alpha_n - \alpha_0) \right|,$$

where $\mathcal{F}_{\varepsilon,\sigma}$ denotes the class of first-marginal entropic potentials arising from pairs of $\sigma^2$-subgaussian marginals.
For $\varepsilon = 1$, one can show that the normalized quadratic-cost potentials belong to a polynomial-growth Hölder class. More precisely, after subtracting the quadratic part $x \mapsto \|x\|^2/2$, these potentials have derivatives up to order $s_0 = \lceil d/2 \rceil + 1$ whose local Hölder norms are bounded by polynomial functions of the subgaussian proxy. Denote by $\mathcal{F}_{\sigma,L}^s$ such a class, where $L$ is a random envelope controlled by the empirical subgaussian moment. Its empirical $L_2$-covering numbers satisfy a bound of the form $\log N(\delta, \mathcal{F}_{\sigma,L}^s, L_2(\alpha_n)) \leqslant C_d\, L^{d/(2s)} \delta^{-d/s}(1 + \sigma^{2d})$. This holds for $s > s_0 - 1$, with $\mathbb{E}L \leqslant C$ after the normalization of the subgaussian moment. Since $d/(2s_0) < 1$, Dudley's entropy integral is finite, and the standard Rademacher symmetrization and chaining bound yields $\mathbb{E}\sup_{w\in\mathcal{F}_{1,\sigma}} \left|\int w\mathrm{d}(\hat{\alpha}_n - \alpha)\right| \leqslant C_d(1+\sigma^{q_d})\, n^{-1/2}$. This is exactly the step which avoids the exponential $e^{D^2/\varepsilon}$ factor: one controls the regularity class of the dual potentials polynomially instead of bounding their sup-norm through the worst value of the Gibbs kernel.
The general $\varepsilon$ case follows by scaling. If $T_\varepsilon(x) = x/\sqrt{\varepsilon}$, $\alpha_0^\varepsilon = (T_\varepsilon)_\sharp\alpha_0$, and $\beta_0^\varepsilon = (T_\varepsilon)_\sharp\beta_0$, then $\mathcal{L}_c^\varepsilon(\alpha_0, \beta_0) = \varepsilon\, \mathcal{L}_c^1(\alpha_0^\varepsilon, \beta_0^\varepsilon)$. The push-forward of a $\sigma^2$-subgaussian law by $T_\varepsilon$ is $(\sigma^2/\varepsilon)$-subgaussian, so the standard-deviation proxy $\sigma$ is replaced by $\sigma/\sqrt{\varepsilon}$.

$$\varepsilon\, C_d \left(1 + \left(\frac{\sigma}{\sqrt{\varepsilon}}\right)^{q_d}\right) n^{-1/2} \leqslant C_d\, \varepsilon \left(1 + \frac{\sigma^{q_d}}{\varepsilon^{b_d}}\right) n^{-1/2},$$

where $b_d = \lceil q_d/2 \rceil = \lceil 5d/4 \rceil + 3$ is the rounded exponent used in. Thus $\mathbb{E}\,|\mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \beta) - \mathcal{L}_c^\varepsilon(\alpha, \beta)| \leqslant C_d\, \Lambda_{d,\sigma}(\varepsilon)\, n^{-1/2}$. Changing both marginals is handled by the triangle inequality,

$$\left|\mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{L}_c^\varepsilon(\alpha, \beta)\right| \leqslant \left|\mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{L}_c^\varepsilon(\alpha, \hat{\beta}_m)\right| + \left|\mathcal{L}_c^\varepsilon(\alpha, \hat{\beta}_m) - \mathcal{L}_c^\varepsilon(\alpha, \beta)\right|.$$

Empirical measures drawn from a subgaussian law have such a random proxy, with moments controlled by the population proxy. Integrating over this proxy gives the two-sample extension. The argument does not require the two empirical marginals to have the same sample size, so the two contributions scale as $n^{-1/2}$ and $m^{-1/2}$.
Finally, use the definition $\bar{\mathcal{L}}_c^\varepsilon(\alpha, \beta) = \mathcal{L}_c^\varepsilon(\alpha, \beta) - \frac{1}{2}\mathcal{L}_c^\varepsilon(\alpha, \alpha) - \frac{1}{2}\mathcal{L}_c^\varepsilon(\beta, \beta)$. The cross term is controlled by the two-marginal estimate. For the self term, use the telescoping inequality

$$|\mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\alpha}_n) - \mathcal{L}_c^\varepsilon(\alpha, \alpha)| \leqslant |\mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\alpha}_n) - \mathcal{L}_c^\varepsilon(\alpha, \hat{\alpha}_n)| + |\mathcal{L}_c^\varepsilon(\alpha, \hat{\alpha}_n) - \mathcal{L}_c^\varepsilon(\alpha, \alpha)|.$$

Each term is a one-marginal perturbation in the same pathwise sense: the other marginal may itself be empirical, but it is part of the common random subgaussian class. Thus the self term $\mathcal{L}_c^\varepsilon(\hat{\alpha}_n, \hat{\alpha}_n) - \mathcal{L}_c^\varepsilon(\alpha, \alpha)$ is bounded at the same order as the cross term involving $\alpha$, and similarly for $\beta$. Summing these three contributions gives the displayed estimate, up to changing $C_d$. ☐

$\sigma = \frac{R}{\sqrt{2d\log 2}} = O(R)$.

**Sample complexity of estimating OT maps.**

$$u_{n,m}^\varepsilon(x) := (\mathrm{g}_{n,m}^\varepsilon)^{\bar{c},\varepsilon}(x) = -\varepsilon \log \sum_j b_j \exp\left(\frac{\mathrm{g}_{n,m,j}^\varepsilon - c_2(x, Y_j)}{\varepsilon}\right), \qquad T_{n,m}^\varepsilon(x) := x - \nabla u_{n,m}^\varepsilon(x).$$

$$\varphi_{n,m}^\varepsilon(x) := \frac{1}{2}\|x\|^2 - u_{n,m}^\varepsilon(x) = \varepsilon \log \sum_j b_j \exp\left(\frac{\langle x, Y_j\rangle + \mathrm{g}_{n,m,j}^\varepsilon - \frac{1}{2}\|Y_j\|^2}{\varepsilon}\right), \qquad T_{n,m}^\varepsilon = \nabla\varphi_{n,m}^\varepsilon.$$

$$T_{n,m}^\varepsilon(x) = \sum_j \omega_j^\varepsilon(x) Y_j, \qquad \omega_j^\varepsilon(x) = \frac{b_j \exp\left((\mathrm{g}_{n,m,j}^\varepsilon - c_2(x, Y_j))/\varepsilon\right)}{\sum_k b_k \exp\left((\mathrm{g}_{n,m,k}^\varepsilon - c_2(x, Y_k))/\varepsilon\right)}. \tag{9.5}$$

$\omega_j^\varepsilon(X_i) = \frac{\mathrm{P}_{i,j}^\varepsilon}{a_i}$, $\quad T_{n,m}^\varepsilon(X_i) = \frac{1}{a_i}\sum_j \mathrm{P}_{i,j}^\varepsilon Y_j$. Thus the extrapolated soft-transform map coincides on the empirical support with the barycentric projection (11.43) of the optimal entropic plan. At $\varepsilon = 0$, the hard $c$-transform is generally nondifferentiable: away from Laguerre boundaries its gradient selects one target site, whereas on a boundary its subdifferential contains several sites.

**Proposition 9.11** (Consistency of empirical barycentric maps)**.** *Let $\Omega \subset \mathbb{R}^d$ be compact, let $\alpha, \beta \in \mathcal{P}_2(\Omega)$, and assume that $\alpha$ is absolutely continuous. Let $\varphi$ be a Brenier potential from $\alpha$ to $\beta$ for the cost $c_2(x, y) = \|x - y\|^2/2$, and choose a bounded Borel selection $T(x) \in \partial\varphi(x)$ on $\Omega$. At every differentiability point of $\varphi$, this selection necessarily equals $\nabla\varphi(x)$, so it is a representative of the unique Brenier map. Let $\hat{\alpha}_n = \frac{1}{n}\sum_{i=1}^n \delta_{X_i}$, $\quad \hat{\beta}_m = \frac{1}{m}\sum_{j=1}^m \delta_{Y_j}$ satisfy $\hat{\alpha}_n \to \alpha$, $\hat{\beta}_m \to \beta$ in $\mathcal{W}_2$. Let $\varepsilon_{n,m} \geqslant 0$ satisfy $\varepsilon_{n,m} \to 0$ and $\varepsilon_{n,m}\log(\min\{n, m\}) \to 0$. For each $n, m$, let $\mathrm{P}_{n,m}^\varepsilon$ be an optimal coupling between $\hat{\alpha}_n$ and $\hat{\beta}_m$, with $\varepsilon = \varepsilon_{n,m}$: unregularized if $\varepsilon = 0$, entropic if $\varepsilon > 0$. Its barycentric projection, defined on the support of $\hat{\alpha}_n$ by $\bar{T}_{n,m}^\varepsilon(X_i) := n\sum_j \mathrm{P}_{i,j}^\varepsilon Y_j$, satisfies $\int \|\bar{T}_{n,m}^\varepsilon(x) - T(x)\|^2 \mathrm{d}\hat{\alpha}_n(x) = \frac{1}{n}\sum_i \|\bar{T}_{n,m}^\varepsilon(X_i) - T(X_i)\|^2 \longrightarrow 0$ as $n, m \to +\infty$. If $\beta$ is also absolutely continuous, the analogous backward barycentric projections converge in $L^2(\hat{\beta}_m)$ to the Brenier map $S : \beta \to \alpha$.*

*Proof.* The assumptions imply $\mathcal{L}_{c_2}(\hat{\alpha}_n, \hat{\beta}_m) \to \mathcal{L}_{c_2}(\alpha, \beta)$. If $\varepsilon = 0$, $\mathrm{P}_{n,m}^\varepsilon$ is exactly optimal. If $\varepsilon > 0$, compare its entropic objective with an unregularized optimal empirical plan $Q_{n,m}$. Regarding $Q_{n,m}$ as the joint law of two uniform discrete indices, its relative entropy with respect to their product law is their mutual information. Hence $\mathrm{KL}(Q_{n,m}|\hat{\alpha}_n \otimes \hat{\beta}_m) \leqslant \min\{\log n, \log m\}$. It follows that $\int c_2\, \mathrm{dP}_{n,m}^\varepsilon \leqslant \mathcal{L}_{c_2}(\hat{\alpha}_n, \hat{\beta}_m) + \varepsilon\log(\min\{n, m\})$. Hence the transport costs of $\mathrm{P}_{n,m}^\varepsilon$ converge to the population optimum. By compactness, every weak limit point of the plans $\mathrm{P}^{\varepsilon}_{n,m}$ has marginals $\alpha, \beta$ and optimal quadratic cost. Brenier uniqueness therefore forces the whole sequence to converge weakly to $(\mathrm{Id}, T)_{\sharp}\alpha$.

Disintegrate $\mathrm{P}^{\varepsilon}_{n,m}$ with respect to its first marginal and denote its barycentric projection by $\bar{T}^{\varepsilon}_{n,m}$. Jensen's inequality gives $\int \|\bar{T}^{\varepsilon}_{n,m}(x) - T(x)\|^2 \mathrm{d}\hat{\alpha}_n(x) \leqslant \iint \|y - T(x)\|^2 \mathrm{d}\mathrm{P}^{\varepsilon}_{n,m}(x, y)$. At every differentiability point of $\varphi$, the subdifferential is the singleton $\{\nabla\varphi(x)\}$. Since the target is compact, the relevant subgradients are bounded, and the closed-graph property of $\partial\varphi$ implies continuity of the selected gradient at such a point. Convex functions are differentiable Lebesgue-almost everywhere; because $\alpha$ is absolutely continuous, the bounded function $(x, y) \mapsto \|y - T(x)\|^2$ is continuous $(\mathrm{Id}, T)_{\sharp}\alpha$-almost everywhere. The portmanteau theorem for bounded functions whose discontinuity set has zero limiting mass gives $\iint \|y - T(x)\|^2 \mathrm{d}\mathrm{P}^{\varepsilon}_{n,m}(x, y) \longrightarrow \int \|T(x) - T(x)\|^2 \mathrm{d}\alpha(x) = 0$. The proof for the backward map is identical when the optimal plan is also induced by the Brenier map from $\beta$ to $\alpha$. □

**Proposition 9.12** (Finite-sample rate for the entropic map)**.** *Assume that $\alpha$ and $\beta$ have densities on a common compact set $\Omega$, both bounded above and with $\rho_\beta$ bounded below away from zero. Let $T = \nabla\varphi$ be the quadratic Brenier map from $\alpha$ to $\beta$, and assume that $\varphi \in C^2(\Omega)$, its Legendre conjugate satisfies $\varphi^* \in C^{s+1}(\Omega)$ for some $s > 1$, and $\mu \mathrm{Id}_d \preceq \nabla^2\varphi(x) \preceq L \mathrm{Id}_d \qquad (x \in \Omega)$ for constants $0 < \mu \leqslant L$. Set $\bar{s} := s \wedge 3$, assume that $\mathcal{I}_{\mathrm{geo}}(\alpha, \beta) < +\infty$ as in Proposition 7.15, and draw $n$ independent samples from each measure. If $\varepsilon_n \simeq n^{-1/(d+\bar{s}+1)}$, then the estimator $T^{\varepsilon_n}_{n,n}$ defined in (9.5) satisfies*

$$\mathbb{E}\|T^{\varepsilon_n}_{n,n} - T\|^2_{L^2(\alpha)} \lesssim \big(1 + \mathcal{I}_{\mathrm{geo}}(\alpha, \beta)\big) n^{-\frac{\bar{s}+1}{2(d+\bar{s}+1)}} \log n.$$

**Sliced Wasserstein.** For a direction $\theta \in \mathbb{S}^{d-1}$, let $P_\theta(x) = \langle\theta, x\rangle$ be the one-dimensional projection and let $\sigma$ denote the normalized surface measure on the sphere. For $\alpha, \beta \in \mathcal{P}_p(\mathbb{R}^d)$, the sliced Wasserstein distance is $\mathcal{SW}_p(\alpha, \beta)^p := \int_{\mathbb{S}^{d-1}} \mathcal{W}_p\left((P_\theta)_\sharp\alpha, (P_\theta)_\sharp\beta\right)^p \mathrm{d}\sigma(\theta)$.

**Theorem 9.13** (Dimension-free empirical rate for $\mathcal{SW}_1$)**.** *Let $\alpha, \beta \in \mathcal{P}(\mathbb{R}^d)$ satisfy $\mathrm{supp}(\alpha) \cup \mathrm{supp}(\beta) \subset B(0, R)$, and let $\hat{\alpha}_n$ and $\hat{\beta}_m$ be their empirical measures from $n$ and $m$ i.i.d. samples, respectively. Then*

$$\mathbb{E}\left|\mathcal{SW}_1(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{SW}_1(\alpha, \beta)\right| \leqslant R\left(\frac{1}{\sqrt{n}} + \frac{1}{\sqrt{m}}\right).$$

*Proof.* The triangle inequality for $\mathcal{SW}_1$ gives $\left|\mathcal{SW}_1(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{SW}_1(\alpha, \beta)\right| \leqslant \mathcal{SW}_1(\hat{\alpha}_n, \alpha) + \mathcal{SW}_1(\hat{\beta}_m, \beta)$. Fix $\theta \in \mathbb{S}^{d-1}$, set $\alpha_\theta = (P_\theta)_\sharp\alpha$, and denote the cumulative functions of $\alpha_\theta$ and its empirical measure by $F_\theta$ and $\hat{F}_{\theta,n}$. Both measures are supported on $[-R, R]$. The one-dimensional identity (2.12) and the Cauchy–Schwarz inequality give

$$\mathbb{E}\ \mathcal{W}_1\left((P_\theta)_\sharp\hat{\alpha}_n, \alpha_\theta\right) = \int_{-R}^{R} \mathbb{E}\left|\hat{F}_{\theta,n}(s) - F_\theta(s)\right| \mathrm{d}s \leqslant \int_{-R}^{R} \sqrt{\frac{F_\theta(s)(1 - F_\theta(s))}{n}} \mathrm{d}s \leqslant \frac{R}{\sqrt{n}}.$$

Indeed, $n\hat{F}_{\theta,n}(s)$ is binomial with success probability $F_\theta(s)$. Integrating this bound over $\theta$ yields $\mathbb{E}\ \mathcal{SW}_1(\hat{\alpha}_n, \alpha) \leqslant R/\sqrt{n}$. The same argument for $\beta$, followed by the preceding triangle inequality, proves the result. □

### 9.3 Bias and Variance of OT

**Bias, variance and approximation errors.** $B^{\varepsilon}_{n,m} := \mathbb{E}\mathcal{L}^{\varepsilon}_c(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{L}^{\varepsilon}_c(\alpha, \beta)$ $Z^{\varepsilon}_{n,m} := \mathcal{L}^{\varepsilon}_c(\hat{\alpha}_n, \hat{\beta}_m) - \mathbb{E}\mathcal{L}^{\varepsilon}_c(\hat{\alpha}_n, \hat{\beta}_m)$.

$$\mathcal{L}^{\varepsilon}_c(\hat{\alpha}_n, \hat{\beta}_m) - \mathcal{L}^{0}_c(\alpha, \beta) = \underbrace{\mathcal{L}^{\varepsilon}_c(\alpha, \beta) - \mathcal{L}^{0}_c(\alpha, \beta)}_{\text{regularization bias}} + \underbrace{B^{\varepsilon}_{n,m}}_{\text{statistical bias}} + \underbrace{Z^{\varepsilon}_{n,m}}_{\text{centered fluctuation}}.$$

For the debiased Sinkhorn divergence, define $\bar{B}^{\varepsilon}_{n,m}$ and $\bar{Z}^{\varepsilon}_{n,m}$ by replacing $\mathcal{L}^{\varepsilon}_c$ with $\bar{\mathcal{L}}^{\varepsilon}_c$ above; the same decomposition then holds with bars. It is usually read after choosing a temperature $\varepsilon = \varepsilon_n$.

**Proposition 9.14** (Finite-space bias and CLT for OT)**.** *Let $\alpha = \sum_{i=1}^{N} a_i\delta_{x_i}, \qquad \beta = \sum_{j=1}^{M} b_j\delta_{y_j}$, where all weights are positive and $c$ is finite on the product support. Let $\hat{\alpha}_n$ and $\hat{\beta}_n$ be the independent empirical laws of $n$ i.i.d. samples from $\alpha$ and $\beta$, respectively. Fix $\varepsilon \geqslant 0$, with $\mathcal{L}^0_c = \mathcal{L}_c$, and let $(f^\star_\varepsilon, g^\star_\varepsilon)$ be optimal dual potentials for $\mathcal{L}^{\varepsilon}_c(\alpha, \beta)$. When $\varepsilon = 0$, assume that this pair is unique up to the additive gauge; for $\varepsilon > 0$, this uniqueness follows after fixing the gauge. Define*

$$\mathsf{v}_\varepsilon := \sum_{i=1}^{N} a_i\big(f^\star_\varepsilon(x_i) - \bar{f}_\varepsilon\big)^2 + \sum_{j=1}^{M} b_j\big(g^\star_\varepsilon(y_j) - \bar{g}_\varepsilon\big)^2,$$

*where $\bar{f}_\varepsilon = \sum_i a_i f^\star_\varepsilon(x_i)$ and $\bar{g}_\varepsilon = \sum_j b_j g^\star_\varepsilon(y_j)$. Then, as $n \to +\infty$ with $N, M$ fixed,*

$$\sqrt{n}\, B^{\varepsilon}_{n,n} \longrightarrow 0, \qquad \sqrt{n}\, Z^{\varepsilon}_{n,n} \Longrightarrow \mathcal{N}(0, \mathsf{v}_\varepsilon), \qquad n\ \mathrm{Var}\Big[\mathcal{L}^{\varepsilon}_c(\hat{\alpha}_n, \hat{\beta}_n)\Big] \longrightarrow \mathsf{v}_\varepsilon.$$

*Equivalently,*

$$\sqrt{n}\left(\mathcal{L}^{\varepsilon}_c(\hat{\alpha}_n, \hat{\beta}_n) - \mathcal{L}^{\varepsilon}_c(\alpha, \beta)\right) \Longrightarrow \mathcal{N}(0, \mathsf{v}_\varepsilon).$$

*For every fixed $\varepsilon > 0$, one moreover has $B^{\varepsilon}_{n,n} = O(n^{-1})$.*

*Proof.* Writing $\hat{\mathrm{a}}_n$ and $\hat{\mathrm{b}}_n$ for the two empirical histograms, the independent multinomial central-limit theorems give $\sqrt{n}(\hat{\mathrm{a}}_n - \mathrm{a}, \hat{\mathrm{b}}_n - \mathrm{b}) \Longrightarrow (G_{\mathrm{a}}, G_{\mathrm{b}})$, where $G_{\mathrm{a}}$ and $G_{\mathrm{b}}$ are independent centered Gaussian vectors with covariance matrices $\mathrm{diag}(\mathrm{a}) - \mathrm{a}\mathrm{a}^\top$ and $\mathrm{diag}(\mathrm{b}) - \mathrm{b}\mathrm{b}^\top$. Proposition 11.22 gives the directional derivative for every $\varepsilon \geqslant 0$; it is linear under the assumed uniqueness at $\varepsilon = 0$, and automatically at positive temperature. Thus, $D\mathcal{L}^{\varepsilon}_c(\alpha, \beta)[h, k] = \sum_i f^\star_\varepsilon(x_i)h_i + \sum_j g^\star_\varepsilon(y_j)k_j$. The delta method

therefore gives the Gaussian limit $\sum_i f_\varepsilon^\star(x_i)(G_\mathrm{a})_i+\sum_j g_\varepsilon^\star(y_j)(G_\mathrm{b})_j$. Independence and the two multinomial covariance formulas give exactly $\mathsf{v}_\varepsilon$.
Set $R\coloneqq\max_{i,j}c(x_i,y_j)-\min_{i,j}c(x_i,y_j)$. Exact dual potentials may be replaced by their $c$-transforms, and entropic potentials by their soft $c$-transforms; in either case their oscillations are at most $R$. Since differences of histograms have zero sum, integrating the corresponding subgradient bounds along segments in the two marginal simplices gives, for any histograms $(\mathrm{a}',\mathrm{b}')$ and their associated measures $(\alpha',\beta')$, $|\mathcal{L}_c^\varepsilon(\alpha',\beta')-\mathcal{L}_c^\varepsilon(\alpha,\beta)|\leqslant\frac{R}{2}\big(\|\mathrm{a}'-\mathrm{a}\|_1+\|\mathrm{b}'-\mathrm{b}\|_1\big)$. Because $N,M$ are fixed, the fourth moments of $\sqrt{n}\|\hat{\mathrm{a}}_n-\mathrm{a}\|_1$ and $\sqrt{n}\|\hat{\mathrm{b}}_n-\mathrm{b}\|_1$ are uniformly bounded. The squared rescaled OT values are therefore uniformly integrable. The Gaussian limit then yields convergence of the means and second moments, which proves the assertions for $B_{n,n}^\varepsilon$, $Z_{n,n}^\varepsilon$, and the variance.
For fixed $\varepsilon>0$, the entropic value is twice continuously differentiable in a neighborhood of the positive pair $(\mathrm{a},\mathrm{b})$. On that neighborhood, a second-order Taylor expansion has a uniformly bounded remainder; its linear term has zero expectation and $\mathbb{E}(\|\hat{\mathrm{a}}_n-\mathrm{a}\|^2+\|\hat{\mathrm{b}}_n-\mathrm{b}\|^2)=O(n^{-1})$. The probability of leaving the neighborhood is exponentially small, and the preceding global Lipschitz bound controls its contribution. Hence $B_{n,n}^\varepsilon=O(n^{-1})$. ∎

$$\sup_{(f,g)\in\mathcal{D}_c(\alpha,\beta)}\left\{\sum_i f(x_i)(G_\mathrm{a})_i+\sum_j g(y_j)(G_\mathrm{b})_j\right\},$$

**Fixed support versus a continuum.**

$$\sum_{i=1}^N f_\varepsilon^\star(x_i)(G_\mathrm{a})_i+\sum_{j=1}^M g_\varepsilon^\star(y_j)(G_\mathrm{b})_j,\qquad\begin{cases}\mathrm{Cov}(G_\mathrm{a})=\mathrm{diag}(\mathrm{a})-\mathrm{a}\mathrm{a}^\top,\\\mathrm{Cov}(G_\mathrm{b})=\mathrm{diag}(\mathrm{b})-\mathrm{b}\mathrm{b}^\top.\end{cases}$$

$\mathbb{P}[\text{some source or target atom is unobserved}]\leqslant Ne^{-na_{\min}}+Me^{-nb_{\min}}$.

### 9.4 Sketching Sinkhorn in Linear Time

**PSD kernels and ordinary feature sketches.** The RKHS/MMD section, Section 6.2, already used positive semidefinite kernels to define Hilbertian discrepancies between measures. Recall that a symmetric kernel $k:\mathcal{X}\times\mathcal{X}\to\mathbb{R}$ is positive semidefinite if, for every finite family $(x_i)_{i=1}^n$, the Gram matrix $(k(x_i,x_j))_{i,j}$ is positive semidefinite.
$k(x,y)=\int_Z\langle\varphi(x,z),\,\varphi(y,z)\rangle_{\mathbb{R}^p}\,d\xi(z),\qquad\varphi:\mathcal{X}\times Z\to\mathbb{R}^p,\ \widetilde{k}_r(x,y)=\frac{1}{r}\sum_{\ell=1}^r\langle\varphi(x,z_\ell),\,\varphi(y,z_\ell)\rangle_{\mathbb{R}^p}$.

$$(\Phi_X)_{i,(\ell,q)}=r^{-1/2}\varphi_q(x_i,z_\ell),\qquad(\Phi_Y)_{j,(\ell,q)}=r^{-1/2}\varphi_q(y_j,z_\ell),\qquad\widetilde{K}=\Phi_X\Phi_Y^\top.\tag{9.6}$$

$\widetilde{K}_{i,j}=\frac{1}{r}\sum_{\ell=1}^r\langle\varphi(x_i,z_\ell),\,\varphi(y_j,z_\ell)\rangle_{\mathbb{R}^p}=\widetilde{k}_r(x_i,y_j)$.
**Proposition 9.15** (Entrywise accuracy of rectangular feature sketches)**.** *Let $z_1,\dots,z_r$ be independent with law $\xi$, and let $K_{i,j}=k(x_i,y_j)$ and $\widetilde{K}$ be the rectangular kernel matrix and sketch defined above, with $\widetilde{K}$ given by (9.6). Assume that, for some $M<+\infty$, $|\langle\varphi(x_i,z),\,\varphi(y_j,z)\rangle_{\mathbb{R}^p}|\leqslant M$ for all $(i,j)$ and $\xi$-a.e. $z$. If $K_{\min}\coloneqq\min_{i,j}K_{i,j}>0$, then, for every $0<\delta\leqslant 1/2$,*

$$\mathbb{P}\left(\max_{i,j}\left|\frac{\widetilde{K}_{i,j}}{K_{i,j}}-1\right|>\delta\right)\leqslant 2nm\exp\left(-\frac{rK_{\min}^2\delta^2}{2M^2}\right).\tag{9.7}$$

*On the complementary event, $\widetilde{K}$ is entrywise positive and, for every $\varepsilon>0$, $\max_{i,j}\left|-\varepsilon\log\widetilde{K}_{i,j}+\varepsilon\log K_{i,j}\right|\leqslant 2\varepsilon\delta$.*

*Proof.* For each fixed $(i,j)$, the summands in the definition of $\widetilde{K}_{i,j}$ have expectation $K_{i,j}$ and lie in $[-M,M]$. Hoeffding's inequality therefore gives

$$\mathbb{P}\left(\left|\widetilde{K}_{i,j}-K_{i,j}\right|>\delta K_{i,j}\right)\leqslant 2\exp\left(-\frac{r\delta^2K_{i,j}^2}{2M^2}\right)\leqslant 2\exp\left(-\frac{r\delta^2K_{\min}^2}{2M^2}\right).$$

A union bound over the $nm$ entries proves (9.7). On its complementary event, write $\widetilde{K}_{i,j}=K_{i,j}(1+s_{i,j})$, where $|s_{i,j}|\leqslant\delta\leqslant 1/2$. Hence $\widetilde{K}_{i,j}>0$, and $|\log(1+s_{i,j})|\leqslant 2\,|s_{i,j}|$ gives the logarithmic estimate. ∎

**Application to Sinkhorn kernels.** $\alpha=\sum_{i=1}^n a_i\delta_{x_i},\qquad\beta=\sum_{j=1}^m b_j\delta_{y_j}$. Specialize the preceding construction to the Gibbs kernel $k_\varepsilon(x,y)=e^{-c(x,y)/\varepsilon}$, assumed to be PSD and to admit the displayed feature representation. Thus the rectangular matrix defined above is precisely the Sinkhorn kernel, $K_{i,j}=k_\varepsilon(x_i,y_j)$.
$u=\frac{a}{Kv},\qquad v=\frac{b}{K^\top u}$, where divisions are componentwise. Replacing $K$ by its rank-$R$ sketch (9.6), the two matrix-vector products are evaluated as
$\widetilde{K}v=\Phi_X(\Phi_Y^\top v),\qquad\widetilde{K}^\top u=\Phi_Y(\Phi_X^\top u)$,
**Proposition 9.16** (Accuracy and complexity of Gaussian Nyström–Sinkhorn)**.** *Let $\alpha=\sum_{i=1}^n a_i\delta_{x_i}$ and $\beta=\sum_{j=1}^m b_j\delta_{y_j}$ have positive weights, assume that all their support points lie in the ball $B(0,D)\subset\mathbb{R}^d$, and set $N=n+m$. For the quadratic cost $c(x,y)=\|x-y\|^2$, fix $\varepsilon>0$ and $\tau,\delta\in(0,1)$. There is a randomized Nyström construction of an entrywise-positive rank-$R$ approximation $\widetilde{K}$ of the Gaussian kernel $K_{i,j}=e^{-\|x_i-y_j\|^2/\varepsilon}$, followed by approximate Sinkhorn scaling and rounding, which returns a feasible coupling $\widehat{P}\in\mathrm{U}(a,b)$ and a value $\widehat{W}$ such that, with probability at least $1-\delta$,*

$$0\leqslant\langle C,\,\widehat{P}\rangle+\varepsilon\,\mathrm{KL}(\widehat{P}\,|\,a\otimes b)-\mathcal{L}_c^\varepsilon(\alpha,\beta)\leqslant\tau,\qquad\left|\widehat{W}-\mathcal{L}_c^\varepsilon(\alpha,\beta)\right|\leqslant\tau.$$

*The sketch rank can be chosen so that*

$$R\lesssim_d\left(1+\frac{D^2}{\varepsilon}+\log\frac{N(1+D^2)}{\tau}\right)^d\log\frac{N}{\delta},\tag{9.8}$$

*and the complete computation uses* $\widetilde{O}\left(NR\left(R+\frac{D^4}{\varepsilon\tau}\right)\right)$ *operations and* $O\left(N(R+d)\right)$ *memory. Here the constant hidden in* $\lesssim_d$ *depends only on* $d$.

*Proof.* Apply the Gaussian Nyström approximation to the Gram matrix on the union of the $N$ support points and retain its source–target block. Since $\|x-y\| \leqslant 2D$, one has $K_{\min} \geqslant e^{-4D^2/\varepsilon}$. The adaptive stopping rule of Altschuler, Bach, Rudi and Niles-Weed resolves the kernel below this scale, hence produces $\widetilde{K} > 0$ with the same logarithmic control singled out in Proposition 9.15. Their Gaussian effective-dimension estimate gives (9.8). Stability of entropic OT with respect to $\log K$, followed by approximate Sinkhorn scaling and rounding, gives the two error bounds. The low-rank factorization costs $O(NR^2)$ to construct, each kernel product costs $O(NR)$, and the quantitative scaling bound contributes $\widetilde{O}(NRD^4/(\varepsilon\tau))$, yielding the stated time and memory estimates. Their entropy convention differs from the KL-normalized value $\mathcal{L}_c^\varepsilon$ only by a constant depending on $(a,b)$, so the approximation bounds are unchanged. □

**Positive features and complete positivity.**

**Definition 9.17** (Doubly nonnegative and completely positive matrices and kernels)**.** In finite dimension, set

$$\mathrm{DNN}_n = \left\{A \in \mathbb{R}^{n\times n}\ ;\ A \succeq 0,\ A_{i,j} \geqslant 0\right\}, \qquad \mathrm{CP}_n = \left\{BB^\top\ ;\ B \in \mathbb{R}_+^{n\times q} \text{ for some } q \geqslant 1\right\}.$$

Matrices in these sets are called doubly nonnegative and completely positive, respectively. A kernel is doubly positive, respectively completely positive, if every finite Gram matrix belongs to $\mathrm{DNN}_n$, respectively $\mathrm{CP}_n$. It is a positive-feature kernel if there are a probability space $(Z,\xi)$, an integer $p \geqslant 1$, and measurable features $\varphi(x,\cdot) : Z \to \mathbb{R}_+^p$ such that $k(x,y) = \int_Z \langle \varphi(x,z), \varphi(y,z)\rangle_{\mathbb{R}^p} \mathrm{d}\xi(z) \qquad (x,y \in \mathcal{X})$.

**Proposition 9.18** (Complete positivity and positive features)**.** *Let* $k : \mathcal{X}\times\mathcal{X} \to \mathbb{R}$ *be a finite-valued PSD kernel, and assume that its canonical pseudometric* $d_k(x,y)^2 := k(x,x)+k(y,y)-2k(x,y)$ *is separable. Then* $k$ *is completely positive if and only if it is a positive-feature kernel. Moreover, scalar features* $p=1$ *suffice. For a finite set* $\mathcal{X} = \{x_1,\dots,x_n\}$*, this reduces to* $K \in \mathrm{CP}_n \iff K = BB^\top$ *for some* $B \geqslant 0$.

*Proof.* Suppose first that $k$ has a positive-feature representation. For $x_1,\dots,x_n$, let $v_q(z) := (\varphi_q(x_i,z))_{i=1}^n \geqslant 0$. Their Gram matrix is $K = \sum_{q=1}^p \int_Z v_q(z)v_q(z)^\top \mathrm{d}\xi(z)$. The integrand belongs to $\mathrm{CP}_n$, and the closed convex cone $\mathrm{CP}_n$ is stable under such integrals; hence $K \in \mathrm{CP}_n$.
Conversely, choose a $d_k$-dense sequence $(x_i)_{i\geqslant 1}$ and positive weights $(\omega_i)_{i\geqslant1}$ such that $S := \sum_i \omega_i k(x_i,x_i) < +\infty$. The case $S=0$ is trivial. For each sufficiently large $r$, complete positivity gives a factorization $K^{(r)} = B^{(r)}(B^{(r)})^\top$ of the Gram matrix on $x_1,\dots,x_r$. Write $b_\ell$ for its nonzero columns and set $t_\ell := \sum_{i=1}^r \omega_i (b_\ell)_i^2, \qquad S_r := \sum_{i=1}^r \omega_i k(x_i,x_i) = \sum_\ell t_\ell$. On the compact metrizable product $Z := \prod_{i\geqslant 1}[0,\sqrt{S/\omega_i}]$, define $z_i^{(r,\ell)} = \sqrt{S_r/t_\ell}(b_\ell)_i$ for $i \leqslant r$ and $z_i^{(r,\ell)} = 0$ otherwise. Then $\xi_r := \sum_\ell \frac{t_\ell}{S_r}\delta_{z^{(r,\ell)}}$ is a probability measure on $Z$, and $\int_Z z_i z_j \mathrm{d}\xi_r(z) = k(x_i,x_j)$ whenever $i,j \leqslant r$. Compactness yields a weakly convergent subsequence with limit $\xi$. Since every coordinate product $z_iz_j$ is continuous and bounded on $Z$, the scalar features $\varphi(x_i,z) := z_i$ satisfy $\int_Z \varphi(x_i,z)\varphi(x_j,z)\mathrm{d}\xi(z) = k(x_i,x_j) \qquad (i,j \geqslant 1)$. Finally, $\|\varphi(x_i,\cdot)-\varphi(x_j,\cdot)\|_{L^2(\xi)} = d_k(x_i,x_j)$. The feature map therefore extends by continuity to all of $\mathcal{X}$; its values remain nonnegative because $L_+^2(\xi)$ is closed. The Cauchy–Schwarz inequality for PSD kernels shows that the extended inner products equal $k(x,y)$, which proves the positive-feature representation. □

$k_\lambda(x,y) = \lambda + \cos^2\left(\frac{x-y}{2}\right), \qquad \lambda > 0.$ $\langle H,K\rangle = \frac{21}{4} - \frac{5\sqrt{5}}{2} < 0$. For a rectangular source–target matrix, Proposition 9.18 is applied to the Gram matrix on the union of the two sampled point sets, after which one extracts the rectangular cross block.
Despite this certification difficulty, completely positive kernels, equivalently positive-sketchable kernels under the assumptions of Proposition 9.18, enjoy a useful closure algebra. It provides a systematic way to construct new kernels whose low-rank features remain nonnegative.
**Proposition 9.19** (Basic closure of completely positive kernels)**.** *Let* $k_1,\dots,k_J$ *be completely positive kernels,* $a_1,\dots,a_J \geqslant 0$*, and* $(k_\theta)_{\theta\in\Theta}$ *a measurable family of completely positive kernels. Whenever they are finite, the kernels*

$$k_{\mathrm{sum}} := \sum_{j=1}^J a_j k_j, \qquad k_{\mathrm{prod}} := \prod_{j=1}^J k_j, \qquad k_{\mathrm{mix}} := \int_\Theta k_\theta \,\mathrm{d}\eta(\theta), \quad \eta \geqslant 0,$$

*are completely positive. In particular, so is every positive autocorrelation* $k(x,y) = \int g(u-x)g(u-y)\,du, \qquad g \geqslant 0$*, and every nonnegative mixture of such kernels.*

*Proof.* Restrict the kernels to arbitrary points $x_1,\dots,x_n$. If $K_1 = B_1B_1^\top$ and $K_2 = B_2B_2^\top$ with $B_1,B_2 \geqslant 0$, then a nonnegative weighted sum is factored by concatenating the correspondingly rescaled columns. Moreover, $K_1 \odot K_2 = \sum_{\ell,s}\left((B_1)_{\cdot,\ell}\odot(B_2)_{\cdot,s}\right)\left((B_1)_{\cdot,\ell}\odot(B_2)_{\cdot,s}\right)^\top$, so the pointwise product is completely positive. A measurable nonnegative mixture is an integral of matrices in the closed convex cone $\mathrm{CP}_n$, and therefore remains in that cone. Finally, the autocorrelation formula is itself a scalar positive-feature representation, so its finite Gram matrices are completely positive by the same integral argument. □

**Proposition 9.20** (Generalized Gaussian positive features)**.** *Let* $0 < p \leqslant 2$*,* $\varepsilon > 0$*, and* $k_{p,\varepsilon}(x,y) = \exp\left(-\frac{\|x-y\|^p}{\varepsilon}\right), \qquad x,y \in \mathbb{R}^d$. *Set* $a = p/2$*, let* $S_a$ *be the positive* $a$*-stable random variable normalized by* $\mathbb{E}\left(e^{-tS_a}\right) = e^{-t^a}, \qquad t \geqslant 0$*, and draw independently* $\Lambda = \varepsilon^{-1/a}S_a, \qquad Z \sim \mathcal{N}(0,\mathrm{Id}_d)$*. For any* $x_0 \in \mathbb{R}^d$*, define the positive feature* $\varphi_{p,\varepsilon}(x;\Lambda,Z) := \exp\left(\sqrt{2\Lambda}\langle Z, x-x_0\rangle - 2\Lambda\|x-x_0\|^2\right)$*. Then* $k_{p,\varepsilon}(x,y) = \mathbb{E}[\varphi_{p,\varepsilon}(x;\Lambda,Z)\varphi_{p,\varepsilon}(y;\Lambda,Z)]$*, so* $k_{p,\varepsilon}$ *is completely positive. For i.i.d. copies* $(\Lambda_\ell,Z_\ell)_{\ell=1}^r$*, define the positive sketching features* $\varphi_\ell(x) = r^{-1/2}\varphi_{p,\varepsilon}(x;\Lambda_\ell,Z_\ell), \qquad 1 \leqslant \ell \leqslant r$*. They satisfy* $\varphi_\ell \geqslant 0$ *and* $\mathbb{E}[\sum_{\ell=1}^r \varphi_\ell(x)\varphi_\ell(y)] = k_{p,\varepsilon}(x,y)$*. For* $p=2$*, one has* $S_1 = 1$ *almost surely and* $\Lambda = 1/\varepsilon$*, hence* $\varphi_{2,\varepsilon}(x;Z) = \exp\left(\sqrt{\frac{2}{\varepsilon}}\langle Z, x-x_0\rangle - \frac{2}{\varepsilon}\|x-x_0\|^2\right)$.

*Proof.* The stable-law normalization gives $\mathbb{E}e^{-t\Lambda} = \exp\left(-\frac{t^a}{\varepsilon}\right)$, while the Gaussian moment-generating function gives, conditionally on $\Lambda$, $\mathbb{E}_Z\left(\varphi_{p,\varepsilon}(x;\Lambda,Z)\varphi_{p,\varepsilon}(y;\Lambda,Z)\right) = e^{-\Lambda\|x-y\|^2}$. Taking $t = \|x-y\|^2$ in the first identity proves the feature formula. The Monte Carlo statement follows by averaging independent copies. □

**Positive sketches for Sinkhorn.** $K_{i,j} = k_{p,\varepsilon}(x_i, y_j) = \exp\left(-\frac{\|x_i-y_j\|^p}{\varepsilon}\right)$. For $p \geqslant 1$, this is the usual $p$-Wasserstein power cost; for $0 < p < 1$, it is simply a concave power transport cost. The factor $1/p$, often used in $\mathcal{W}_p^p$, only rescales $\varepsilon$. Using the sketching features $\varphi_\ell$ of Proposition 9.20, define directly
$(\Phi_X)_{i,\ell} = \varphi_\ell(x_i)$, $\quad(\Phi_Y)_{j,\ell} = \varphi_\ell(y_j)$. $P_r = \operatorname{diag}(u)\Phi_X\Phi_Y^\top\operatorname{diag}(v) = LR^\top$, $\quad L = \operatorname{diag}(u)\Phi_X$, $\quad R = \operatorname{diag}(v)\Phi_Y$.

**Connection with linear time attention.**

# 10 Generalized Wasserstein Distances

## 10.1 Unbalanced OT

**Relaxed formulation.**

**Definition 10.1** (Unbalanced optimal transport)**.** For nonnegative measures $(\alpha,\beta) \in \mathcal{M}_+(\mathcal{X}) \times \mathcal{M}_+(\mathcal{Y})$, a nonnegative measurable cost $c$, and proper lower-semicontinuous convex entropy functions $\psi_s : [0,+\infty) \to [0,+\infty]$ satisfying $\psi_s(1) = 0$, define

$$\mathrm{UW}_c(\alpha,\beta) \coloneqq \inf_{\pi\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})} \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y) + \mathcal{D}_{\psi_1}(\pi_1|\alpha) + \mathcal{D}_{\psi_2}(\pi_2|\beta), \tag{10.1}$$

where $\mathcal{D}_{\psi_s}$ is the $\varphi$-divergence of Definition 6.10.

$\mathrm{UW}_{c,\tau}(\alpha,\beta) = \inf_{\pi\geqslant 0}\int c\mathrm{d}\pi + \tau\mathcal{D}_{\bar\psi_1}(\pi_1|\alpha) + \tau\mathcal{D}_{\bar\psi_2}(\pi_2|\beta)$.

$$\mathcal{L}_c(\widetilde\alpha,\widetilde\beta) \coloneqq \begin{cases} M\mathcal{L}_c(\widetilde\alpha/M,\widetilde\beta/M), & M>0,\\ 0, & M=0.\end{cases} \tag{10.2}$$

**Proposition 10.2** (Balanced transport between relaxed marginals)**.** *Under the hypotheses of Definition 10.1,*

$$\mathrm{UW}_c(\alpha,\beta) = \inf_{\substack{\widetilde\alpha\in\mathcal{M}_+(\mathcal{X}),\ \widetilde\beta\in\mathcal{M}_+(\mathcal{Y})\\ \widetilde\alpha(\mathcal{X})=\widetilde\beta(\mathcal{Y})}} \left\{\mathcal{L}_c(\widetilde\alpha,\widetilde\beta) + \mathcal{D}_{\psi_1}(\widetilde\alpha|\alpha) + \mathcal{D}_{\psi_2}(\widetilde\beta|\beta)\right\}. \tag{10.3}$$

*A plan $\pi^*$ solves the unbalanced problem if and only if its marginals $(\pi_1^*,\pi_2^*)$ solve (10.3) and $\pi^*$ is an optimal balanced coupling between them.*

*Proof.* Every plan $\pi$ has equal-mass marginals $(\widetilde\alpha,\widetilde\beta) = (\pi_1,\pi_2)$. For fixed marginals, the divergence terms are constant and minimizing the remaining transport term gives (10.2). Grouping the plans in (10.1) by their marginals proves (10.3). The same argument shows that an optimizer must be transport-optimal conditionally on its marginals and that these marginals minimize the outer problem. Conversely, an optimal balanced coupling between any minimizing pair in (10.3) attains the unbalanced value. □

**KL mass–shape separation.** Let $A = \alpha(\mathcal{X}) > 0$, $B = \beta(\mathcal{Y}) > 0$, and write the normalized input shapes as $\widehat\alpha = \alpha/A$ and $\widehat\beta = \beta/B$.

**Proposition 10.3** (Exact mass–shape separation for KL relaxation)**.** *Assume $\psi_1 = \tau_0\varphi_{\mathrm{KL}}$ and $\psi_2 = \tau_1\varphi_{\mathrm{KL}}$, where $\varphi_{\mathrm{KL}}$ is defined in* (**??**) *and $\tau_0,\tau_1 > 0$. Set $T = \tau_0+\tau_1$, $\theta_i = \tau_i/T$, and*

$$\mathcal{E}_{c,\tau_0,\tau_1}(\widehat\alpha,\widehat\beta) \coloneqq \inf_{\substack{\widetilde\alpha\in\mathcal{P}(\mathcal{X})\\ \widetilde\beta\in\mathcal{P}(\mathcal{Y})}} \left\{\mathcal{L}_c(\widetilde\alpha,\widetilde\beta) + \tau_0\,\mathrm{KL}(\widetilde\alpha|\widehat\alpha) + \tau_1\,\mathrm{KL}(\widetilde\beta|\widehat\beta)\right\}. \tag{10.4}$$

*Then, with $e^{-\infty} = 0$,*

$$\mathrm{UW}_c(\alpha,\beta) = \tau_0 A + \tau_1 B - T A^{\theta_0}B^{\theta_1}\exp\left(-\frac{\mathcal{E}_{c,\tau_0,\tau_1}(\widehat\alpha,\widehat\beta)}{T}\right). \tag{10.5}$$

*If $(\widetilde\alpha^*,\widetilde\beta^*)$ attains the finite shape envelope and $\bar\pi^*$ is an optimal probability coupling between these marginals, then $\pi^* = M^*\bar\pi^*$ is optimal, where*

$$M^* = A^{\theta_0}B^{\theta_1}\exp\left(-\frac{\mathcal{E}_{c,\tau_0,\tau_1}(\widehat\alpha,\widehat\beta)}{T}\right) \tag{10.6}$$

*Proof.* Write every nonzero plan as $\pi = M\bar\pi$, where $M = \pi(\mathcal{X}\times\mathcal{Y}) > 0$ and $\bar\pi$ is a probability coupling with marginals $(\widetilde\alpha,\widetilde\beta)$. The KL homogeneity identity

$$\mathrm{KL}(M\widetilde\alpha|A\widehat\alpha) = M\,\mathrm{KL}(\widetilde\alpha|\widehat\alpha) + M\log(M/A) - M + A \tag{10.7}$$

holds in the extended-value sense, and similarly for the second marginal. Minimizing first over $\bar\pi$ and its normalized marginals therefore reduces the objective to $M\mathcal{E}_{c,\tau_0,\tau_1}(\widehat\alpha,\widehat\beta) + \tau_0\big(M\log(M/A) - M + A\big) + \tau_1\big(M\log(M/B) - M + B\big)$. When the shape value is finite, this strictly convex function of $M$ has the unique minimizer (10.6); substitution gives (10.5). If the shape value is infinite, every nonzero plan has infinite objective and the same formula returns the value $\tau_0 A + \tau_1 B$ of the zero plan. □

These figures should be read as pictures of a single relaxed plan $\pi$: the same nonnegative measure determines both the transported coupling and its two relaxed marginals. The small-$\tau$ result below formalizes the opposite regime, where transport becomes negligible compared with local mass variation.

**Proposition 10.4** (Small-transport-scale limit for marginal penalties)**.** *Assume that $\alpha, \beta$ are finite measures on a compact metric space $\mathcal{X}$, that $c$ is continuous, $c \geqslant 0$, and $c(x,y) = 0$ if and only if $x = y$. Assume also that the marginal divergences are nonnegative, weak-* lower semicontinuous, and have weak-* compact sublevel sets on $\mathcal{M}_+(\mathcal{X})$. Then*

$$\lim_{\tau \downarrow 0} \frac{1}{\tau} \mathrm{UW}_{c,\tau}(\alpha, \beta) = \inf_{\chi \in \mathcal{M}_+(\mathcal{X})} \mathcal{D}_{\bar{\psi}_1}(\chi|\alpha) + \mathcal{D}_{\bar{\psi}_2}(\chi|\beta).$$

*The right-hand side is the infimal gluing divergence obtained by matching the two measures through a common zero-transport marginal $\chi$. In the dominated case, if $\alpha = a\lambda$, $\beta = b\lambda$, and the minimizing common marginal is absolutely continuous, $\chi = r\lambda$, this divergence decouples pointwise as*

$$\int \mathfrak{m}_{\bar{\psi}_1,\bar{\psi}_2}(a(x), b(x)) \mathrm{d}\lambda(x), \qquad \mathfrak{m}_{\bar{\psi}_1,\bar{\psi}_2}(a,b) \coloneqq \inf_{r \geqslant 0} a\,\bar{\psi}_1(r/a) + b\,\bar{\psi}_2(r/b),$$

*with the usual recession conventions when $a = 0$ or $b = 0$. For superlinear entropies, and in particular for KL, finite energy forces this dominated form. Thus, when $\bar{\psi}_1 = \bar{\psi}_2$ is the KL entropy, $\inf_{\chi \in \mathcal{M}_+(\mathcal{X})} \mathrm{KL}(\chi|\alpha) + \mathrm{KL}(\chi|\beta) = \int (\sqrt{a} - \sqrt{b})^2 \mathrm{d}\lambda$. Thus the KL marginal relaxation contains the squared Hellinger distance as its local mass-variation limit.*

*Proof.* For the upper bound, restrict to diagonal plans $\pi = (\mathrm{Id}, \mathrm{Id})_\sharp \chi$, whose transport cost is zero and whose two marginals are both $\chi$.
For the lower bound, let $\tau_n \downarrow 0$ and let $\pi_n$ be almost minimizing plans with bounded scaled values $\tau_n^{-1} \mathrm{UW}_{c,\tau_n}(\alpha, \beta)$. Since the divergences are nonnegative, $\int c \mathrm{d}\pi_n = O(\tau_n)$, hence $\int c \mathrm{d}\pi_n \to 0$. The bounded scaled values also put the two marginals in compact divergence sublevel sets. Since a coupling has the same total mass as each marginal, the couplings are tight on $\mathcal{X} \times \mathcal{X}$. Up to subsequences, $\pi_n \rightharpoonup \pi_0$. Lower semicontinuity of the transport cost yields $\int c \mathrm{d}\pi_0 = 0$, so $\pi_0$ is concentrated on the diagonal. Its two marginals are therefore equal to a common measure $\chi$. Lower semicontinuity of the marginal divergences gives $\liminf_n \frac{1}{\tau_n} \mathrm{UW}_{c,\tau_n}(\alpha, \beta) \geqslant \mathcal{D}_{\bar{\psi}_1}(\chi|\alpha) + \mathcal{D}_{\bar{\psi}_2}(\chi|\beta)$, and optimizing over $\chi$ gives the lower bound.
In the dominated case, writing $\chi = r\lambda$ gives $\mathcal{D}_{\bar{\psi}_1}(\chi|\alpha) + \mathcal{D}_{\bar{\psi}_2}(\chi|\beta) = \int a\,\bar{\psi}_1(r/a) + b\,\bar{\psi}_2(r/b) \mathrm{d}\lambda$, so the minimization over $\chi$ decouples into the scalar envelope $\mathfrak{m}_{\bar{\psi}_1,\bar{\psi}_2}$. For KL, no singular part is admissible when $\alpha$ and $\beta$ are dominated by $\lambda$. The pointwise objective is $r\log(r/a) - r + a + r\log(r/b) - r + b$. Its optimality condition is $\log(r/a) + \log(r/b) = 0$, hence $r = \sqrt{ab}$, and the minimum is $a + b - 2\sqrt{ab} = (\sqrt{a} - \sqrt{b})^2$. □

**Proposition 10.5** (Dual of unbalanced optimal transport)**.** *Under the usual Fenchel–Rockafellar qualification assumptions, one has equality between* (10.1) *and* $\mathrm{UW}_c(\alpha, \beta) = \sup_{f \oplus g \leqslant c} -\mathcal{D}^*_{\psi_1}(-f|\alpha) - \mathcal{D}^*_{\psi_2}(-g|\beta)$.

*Proof.* Use the variational formula (6.9) for the dual of a divergence and introduce the marginal variables through continuous potentials: $\inf_{\pi \geqslant 0} \sup_{f,g} \int c \mathrm{d}\pi + \int -f \mathrm{d}\pi_1 + \int -g \mathrm{d}\pi_2 - \mathcal{D}^*_{\psi_1}(-f|\alpha) - \mathcal{D}^*_{\psi_2}(-g|\beta)$. Exchanging the infimum and the supremum gives $\sup_{f,g} -\mathcal{D}^*_{\psi_1}(-f|\alpha) - \mathcal{D}^*_{\psi_2}(-g|\beta) + \inf_{\pi \geqslant 0} \int \big(c - (f \oplus g)\big) \mathrm{d}\pi$. The last infimum is 0 when $f \oplus g \leqslant c$ and $-\infty$ otherwise, which gives the displayed dual. □

**Reverse and homogeneous formulations.**

$$\begin{aligned}\int c(x,y) \mathrm{d}\pi(x,y) + \mathcal{D}_{\psi_1}(\pi_1|\alpha) + \mathcal{D}_{\psi_2}(\pi_2|\beta) \\ = \int \left( c(x,y) + \psi_1\left(\frac{\mathrm{d}\pi_1}{\mathrm{d}\alpha}(x)\right) \frac{\mathrm{d}\alpha}{\mathrm{d}\pi_1}(x) + \psi_2\left(\frac{\mathrm{d}\pi_2}{\mathrm{d}\beta}(y)\right) \frac{\mathrm{d}\beta}{\mathrm{d}\pi_2}(y) \right) \mathrm{d}\pi(x,y).\end{aligned}$$

**Definition 10.6** (Local reverse cost and homogeneous perspective)**.** For $c, r, s \geqslant 0$, define

$$L_c(r,s) \coloneqq c + r\psi_1(1/r) + s\psi_2(1/s), \tag{10.8}$$

with the usual recession convention at $r = 0$ or $s = 0$, and

$$H_c(r,s) \coloneqq \inf_{\theta > 0} \theta L_c(r/\theta, s/\theta). \tag{10.9}$$

By construction, $H_c$ is positively 1-homogeneous. If $\alpha = F\pi_1 + \alpha^\perp$ and $\beta = G\pi_2 + \beta^\perp$ are the Lebesgue decompositions of the reference marginals with respect to the transported marginals, then the reverse formulation reads
$\mathrm{UW}_c(\alpha, \beta) = \inf_{\pi \geqslant 0} \int L_{c(x,y)}(F(x), G(y)) \mathrm{d}\pi(x,y) + \psi_1(0)\alpha^\perp(\mathcal{X}) + \psi_2(0)\beta^\perp(\mathcal{Y})$.

$$\mathrm{HW}_c(\alpha, \beta) = \inf_{\pi \geqslant 0} \int H_{c(x,y)}(F(x), G(y)) \mathrm{d}\pi(x,y) + \psi_1(0)\alpha^\perp(\mathcal{X}) + \psi_2(0)\beta^\perp(\mathcal{Y}). \tag{10.10}$$

$$\mathrm{HW}_c(\alpha, \beta) = \inf_{\substack{\lambda \in \mathcal{M}_+(\mathcal{X} \times \mathcal{Y}),\ u,v \geqslant 0 \\ (\mathrm{p}_1)_\sharp(u\lambda) = \alpha,\ (\mathrm{p}_2)_\sharp(v\lambda) = \beta}} \int H_{c(x,y)}\big(u(x,y), v(x,y)\big) \mathrm{d}\lambda(x,y). \tag{10.11}$$

**Proposition 10.7** (Homogenization does not change the unbalanced cost)**.** *Assume that the entropy functions are proper, lower semicontinuous and convex, with the usual recession extensions at zero. Then* $\mathrm{HW}_c(\alpha, \beta) = \mathrm{UW}_c(\alpha, \beta)$.

*Proof.* Since $H_c(r,s) \leqslant L_c(r,s)$ by choosing the scale equal to one, every competitor in the reverse formulation gives a homogeneous competitor of no larger cost. Hence $\mathrm{HW}_c \leqslant \mathrm{UW}_c$.
For the converse, use the semi-coupling representation (10.11) and fix a feasible triple $(\lambda, u, v)$. For a positive measurable scale $\theta(x,y)$, set $\tilde{\pi} = \theta\lambda$. Disintegrate $\lambda$ with respect to its first spatial marginal. If $\bar{u}(x)$ and $\bar{\theta}(x)$ denote the corresponding conditional

averages of $u$ and $\theta$, then $\alpha = \bar{u}\,(\mathrm{p}_1)_\sharp\lambda$ and $\tilde{\pi}_1 = \bar{\theta}\,(\mathrm{p}_1)_\sharp\lambda$. Convexity of the perspective $(a,m)\mapsto a\psi_1(m/a)$ and conditional Jensen give $\mathcal{D}_{\psi_1}(\tilde{\pi}_1|\alpha) \leqslant \int u\,\psi_1(\theta/u)\mathrm{d}\lambda$. The same argument on the second marginal yields the analogous bound with $(v,\psi_2)$. Therefore

$$\int c\mathrm{d}\tilde{\pi} + \mathcal{D}_{\psi_1}(\tilde{\pi}_1|\alpha) + \mathcal{D}_{\psi_2}(\tilde{\pi}_2|\beta) \leqslant \int \big[c\theta + u\psi_1(\theta/u) + v\psi_2(\theta/v)\big]\mathrm{d}\lambda.$$

The pointwise infimum of the integrand over $\theta > 0$ is precisely $H_c(u,v)$. A measurable $\eta$-minimizing choice of $\theta$ exists because this is a lower-semicontinuous normal integrand; the recession conventions cover $u = 0$ or $v = 0$. Letting $\eta \downarrow 0$ and then minimizing over $(\lambda, u, v)$ gives $\mathrm{UW}_c \leqslant \mathrm{HW}_c$. ☐

**Conic lifting.** Assume now that $\mathcal{X} = \mathcal{Y}$ and $\psi_1 = \psi_2 = \psi$. The homogeneous formulation naturally lifts mass to a radial coordinate.

**Definition 10.8** (Cone lift)**.** The cone over $\mathcal{X}$ is $\mathfrak{C}[\mathcal{X}] \coloneqq (\mathcal{X}\times\mathbb{R}_+)/\sim$, where all $(x,0)$ form one apex. For $p \geqslant 1$, its ground cost is $\Delta((x,r),(y,s)) \coloneqq H_{c(x,y)}(r^p,s^p)^{1/p}$.

- $\mathcal{D}_\psi = \mathrm{KL}$, $p = 2$, and $c(x,y) = -\log\cos^2(d(x,y)\wedge\pi/2)$ give the Hellinger–Kantorovich or Wasserstein–Fisher–Rao cone metric $\Delta((x,r),(y,s))^2 = r^2 + s^2 - 2rs\cos(d(x,y)\wedge\pi/2)$.
- $\mathcal{D}_\psi = \mathrm{KL}$, $p = 2$, and $c(x,y) = d(x,y)^2$ give the Gaussian Hellinger formula $\Delta((x,r),(y,s))^2 = r^2 + s^2 - 2rse^{-d(x,y)^2/2}$. It is a cone metric whenever the Gaussian kernel $k(x,y) = e^{-d(x,y)^2/2}$ is positive definite. This holds, for example, when $(\mathcal{X},d)$ is a subset of a Hilbert space: if $k(x,y) = \langle\Phi(x),\Phi(y)\rangle$ with $\|\Phi(x)\| = 1$, then the displayed quantity is $\|r\Phi(x) - s\Phi(y)\|^2$. For a general metric space, positive definiteness of this kernel is an additional hypothesis.
- $\mathcal{D}_\psi = \mathrm{TV}$, $p = 1$, and $c(x,y) = d(x,y)$ give the partial-transport cone cost $\Delta((x,r),(y,s)) = r + s - (r\wedge s)(2 - d(x,y))_+$.

For a finite measure $\eta$ on the cone, define its weighted base projection $\mathsf{P}_p\eta \in \mathcal{M}_+(\mathcal{X})$ by $\int_{\mathcal{X}}\varphi(x)\mathrm{d}(\mathsf{P}_p\eta)(x) = \int_{\mathfrak{C}[\mathcal{X}]}\varphi(x)r^p\mathrm{d}\eta(x,r)$ for every bounded Borel $\varphi$. This is well defined at the apex because the weight $r^p$ vanishes there.

$$\mathrm{CW}(\alpha,\beta) = \inf_{\gamma\in\mathcal{M}_+(\mathfrak{C}[\mathcal{X}]^2)}\left\{\int \Delta((x,r),(y,s))^p\mathrm{d}\gamma\ ;\ \mathsf{P}_p\gamma_1 = \alpha,\quad \mathsf{P}_p\gamma_2 = \beta\right\}.$$

**Definition 10.9** (Wasserstein–Fisher–Rao distance)**.** For $p = 2$ and $\Delta((x,r),(y,s))^2 = r^2 + s^2 - 2rs\cos(d(x,y)\wedge\pi/2)$, the Wasserstein–Fisher–Rao, or Hellinger–Kantorovich, distance is $\mathrm{WFR}(\alpha,\beta) \coloneqq \mathrm{CW}(\alpha,\beta)^{1/2}$.

**Theorem 10.10** (Cone formulation of unbalanced OT)**.** *Assume that $\mathcal{X}$ is a compact metric space, that $(r,s)\mapsto H_{c(x,y)}(r,s)$ is convex for every $(x,y)$, and that $(x,y,r,s)\mapsto H_{c(x,y)}(r,s)$ is lower semicontinuous. Then $\mathrm{UW}_c(\alpha,\beta) = \mathrm{HW}_c(\alpha,\beta) = \mathrm{CW}(\alpha,\beta)$. If, in addition, $\Delta$ is a continuous distance on $\mathfrak{C}[\mathcal{X}]$, then $\mathrm{CW}^{1/p}$ is a distance between nonnegative finite measures.*

*Proof.* The equality UW = HW is Proposition 10.7.
Let $(\lambda,u,v)$ be feasible in (10.11). Define the cone coupling $\gamma = \big((x,y)\mapsto((x,u(x,y)^{1/p}),(y,v(x,y)^{1/p}))\big)_\sharp\lambda$. Then $\mathsf{P}_p\gamma_1 = \alpha$ and $\mathsf{P}_p\gamma_2 = \beta$ by the weighted marginal constraints. Moreover $\int\Delta((x,r),(y,s))^p\mathrm{d}\gamma = \int H_{c(x,y)}(u(x,y),v(x,y))\mathrm{d}\lambda(x,y)$. Taking the infimum gives CW $\leqslant$ HW.
Conversely, let $\gamma$ be feasible for CW. Choose arbitrary representatives of cone points at the apex. This does not affect the weighted constraints, and it does not change the cost because, by the recession convention, $H_{c(x,y)}(0,s)$ and $H_{c(x,y)}(r,0)$ do not depend on the arbitrary spatial representative attached to the zero radius. Let $\lambda$ be the projection of $\gamma$ onto the spatial variables $(x,y)$, and disintegrate $\gamma = \gamma_{x,y}\mathrm{d}\lambda(x,y)$ with respect to this projection. Set $u(x,y) = \int r^p\mathrm{d}\gamma_{x,y}(r,s)$, $\quad v(x,y) = \int s^p\mathrm{d}\gamma_{x,y}(r,s)$. The cone marginal constraints imply $(\mathrm{p}_1)_\sharp(u\lambda) = \alpha$ and $(\mathrm{p}_2)_\sharp(v\lambda) = \beta$. Since $H_c$ is convex in its two arguments, as a perspective of the convex reverse cost $L_c$, Jensen's inequality gives, for $\lambda$-a.e. $(x,y)$, $H_{c(x,y)}(u(x,y),v(x,y)) \leqslant \int H_{c(x,y)}(r^p,s^p)\mathrm{d}\gamma_{x,y}(r,s)$. After integration and using $\Delta((x,r),(y,s))^p = H_{c(x,y)}(r^p,s^p)$, this yields HW $\leqslant$ CW. Hence UW = HW = CW.
Assume now that $\Delta$ is a distance. The nontrivial point in the triangle inequality is that two cone plans sharing the same weighted base projection need not have the same ordinary cone marginal. The radial homogeneity removes this obstruction. Indeed, common radial rescaling $((x,r),(y,s))\mapsto((x,r/\vartheta),(y,s/\vartheta))$, $\quad \gamma\mapsto\vartheta^p\gamma$, preserves both weighted projections and the $p$-action. After adding mass at the cone apex, the normalization lemma for homogeneous lifts therefore lets one rescale two almost optimal plans for $(\alpha,\beta)$ and $(\beta,\zeta)$ so that their ordinary middle cone marginal is the same lift $\eta_\beta = M\delta_{\mathfrak{o}} + (x\mapsto(x,1))_\sharp\beta$. Here $M < +\infty$ is chosen large enough for both plans; additional mass at $(\mathfrak{o},\mathfrak{o})$ changes neither the weighted projections nor the action. The ordinary gluing lemma now produces a joint cone plan $(Z_0,Z_1,Z_2)$. Since $\Delta$ is a metric, Minkowski's inequality gives

$$\left(\int\Delta(Z_0,Z_2)^p\right)^{1/p} \leqslant \left(\int\Delta(Z_0,Z_1)^p\right)^{1/p} + \left(\int\Delta(Z_1,Z_2)^p\right)^{1/p}.$$

Taking infima proves the triangle inequality. This normalization-and-gluing argument is the metric engine of the homogeneous cone construction.
For completeness, normalize each non-apex pair by taking $\vartheta = (r^p + s^p)^{1/p}$. The transformed radii then satisfy $r^p + s^p = 1$, while the transformed plan has total mass $\alpha(\mathcal{X}) + \beta(\mathcal{X})$; apex mass can again be fixed without changing the problem. Thus minimizing plans may be restricted to a weakly compact family with bounded radii. Compactness of $\mathcal{X}$ and lower semicontinuity of the cone cost give an optimal plan. If $\mathrm{CW}(\alpha,\beta) = 0$, such a plan is concentrated on the diagonal of the cone. There $x = y$ whenever the common radius is positive and $r = s$, so its two weighted base projections coincide and $\alpha = \beta$. Nonnegativity and symmetry are immediate, completing the metric proof. ☐

**Entropic KL relaxation.**

$$\inf_{\pi\in\mathcal{M}_+(\mathcal{X}\times\mathcal{Y})}\int c\mathrm{d}\pi + \mathcal{D}_{\psi_1}(\pi_1|\alpha) + \mathcal{D}_{\psi_2}(\pi_2|\beta) + \varepsilon\mathcal{D}_\varphi(\pi|\alpha\otimes\beta). \tag{10.12}$$

$$\sup_{f,g} -\mathcal{D}^*_{\psi_1}(-f|\alpha) - \mathcal{D}^*_{\psi_2}(-g|\beta) - \varepsilon\mathcal{D}^*_{\varphi}\left(\frac{f\oplus g - c}{\varepsilon}\middle|\alpha\otimes\beta\right).$$

For $\mathcal{D}_\varphi = \mathrm{KL}$, the primal-dual relation is $\mathrm{d}\pi = e^{(f\oplus g-c)/\varepsilon}\mathrm{d}\alpha\mathrm{d}\beta$. If, in addition, $\mathcal{D}_{\psi_1} = \mathcal{D}_{\psi_2} = \tau\,\mathrm{KL}$, the dual specializes to

$$\sup_{f,g} -\tau\int(e^{-f/\tau}-1)\mathrm{d}\alpha - \tau\int(e^{-g/\tau}-1)\mathrm{d}\beta - \varepsilon\iint\left(e^{(f\oplus g-c)/\varepsilon}-1\right)\mathrm{d}\alpha\mathrm{d}\beta,$$

$$f \leftarrow \omega\, g^{\bar{c},\varepsilon}, \qquad g \leftarrow \omega\, f^{c,\varepsilon}, \qquad \omega := \frac{\tau}{\tau+\varepsilon},$$

where the soft $c$-transforms are those of Definition 7.19. Equivalently,

$$f(x) = -\frac{\tau\varepsilon}{\tau+\varepsilon}\log\int_{\mathcal{Y}}\exp\left(\frac{g(y)-c(x,y)}{\varepsilon}\right)\mathrm{d}\beta(y),$$
$$g(y) = -\frac{\tau\varepsilon}{\tau+\varepsilon}\log\int_{\mathcal{X}}\exp\left(\frac{f(x)-c(x,y)}{\varepsilon}\right)\mathrm{d}\alpha(x).$$

$$u_i \leftarrow \left(\frac{\mathrm{a}_i}{(Kv)_i}\right)^{\omega}, \qquad v_j \leftarrow \left(\frac{\mathrm{b}_j}{(K^\top u)_j}\right)^{\omega}, \qquad \mathrm{P} = \mathrm{diag}(u)K\,\mathrm{diag}(v).$$

**Metric contraction of the damped updates.** For balanced entropic OT, potentials are only defined up to the gauge $(f,g)\sim(f+\lambda, g-\lambda)$, so Hilbert's projective metric is natural. The KL marginal penalties break this invariance: under the usual finite-dual assumptions, the dual potentials are unique up to $\alpha$- and $\beta$-null sets, and the natural distance is the ordinary $L^\infty$ distance on potentials.
$d_T\big((u,v),(u',v')\big) := \max\{\|\log u - \log u'\|_\infty, \|\log v - \log v'\|_\infty\}$,

**Proposition 10.11** (Linear contraction of unbalanced Sinkhorn)**.** *Assume that the damped soft-transform map sends bounded potentials to bounded potentials; for instance, this holds when c is bounded. Let $\omega = \tau/(\tau+\varepsilon) < 1$, and define $d_\infty((f,g),(f',g')) := \max\{\|f-f'\|_\infty, \|g-g'\|_\infty\}$. The damped soft-transform map*

$$\mathcal{T}_{\mathrm{GS}}(f,g) := (\tilde{f},\tilde{g}), \qquad \tilde{f} = \omega\, g^{\bar{c},\varepsilon}, \qquad \tilde{g} = \omega\, \tilde{f}^{c,\varepsilon},$$

*is a contraction: $d_\infty\big(\mathcal{T}_{\mathrm{GS}}(f,g), \mathcal{T}_{\mathrm{GS}}(f',g')\big) \leqslant \omega\, d_\infty((f,g),(f',g'))$. $(f^\star,g^\star)$ and the iterates satisfy $d_\infty((f^{(\ell)},g^{(\ell)}),(f^\star,g^\star)) \leqslant \omega^\ell d_\infty((f^{(0)},g^{(0)}),(f^\star,g^\star))$. Equivalently, for $u^{(\ell)} = e^{f^{(\ell)}/\varepsilon}$ and $v^{(\ell)} = e^{g^{(\ell)}/\varepsilon}$, $d_T((u^{(\ell)},v^{(\ell)}),(u^\star,v^\star)) \leqslant \omega^\ell d_T((u^{(0)},v^{(0)}),(u^\star,v^\star))$.*

*Proof.* The soft $c$-transform is 1-Lipschitz in $L^\infty$. Indeed, if $\|g-g'\|_\infty \leqslant \delta$, then $g-\delta \leqslant g' \leqslant g+\delta$. Since $g\mapsto g^{\bar{c},\varepsilon}$ is order reversing and satisfies $(g+\lambda)^{\bar{c},\varepsilon} = g^{\bar{c},\varepsilon} - \lambda$, this implies $g^{\bar{c},\varepsilon} - \delta \leqslant (g')^{\bar{c},\varepsilon} \leqslant g^{\bar{c},\varepsilon} + \delta$. Hence $\|g^{\bar{c},\varepsilon} - (g')^{\bar{c},\varepsilon}\|_\infty \leqslant \|g-g'\|_\infty$, and the same argument applies to $f\mapsto f^{c,\varepsilon}$.
Let $(\tilde{f},\tilde{g}) = \mathcal{T}_{\mathrm{GS}}(f,g)$ and $(\tilde{f}',\tilde{g}') = \mathcal{T}_{\mathrm{GS}}(f',g')$. With $D = d_\infty((f,g),(f',g'))$, the previous Lipschitz estimate gives $\|\tilde{f}-\tilde{f}'\|_\infty \leqslant \omega\|g-g'\|_\infty \leqslant \omega D$, and then $\|\tilde{g}-\tilde{g}'\|_\infty \leqslant \omega\|\tilde{f}-\tilde{f}'\|_\infty \leqslant \omega^2 D \leqslant \omega D$. This proves the contraction. The Banach fixed-point theorem gives existence, uniqueness and the displayed linear rate. □

**Gaussian Hellinger–Kantorovich transport.**

**Definition 10.12** (Gaussian Hellinger–Kantorovich distance)**.** For $\alpha,\beta\in\mathcal{M}_+(\mathbb{R}^d)$ and $\tau>0$, define

$$\mathrm{GHK}^2_\tau(\alpha,\beta) := \inf_{\pi\in\mathcal{M}_+(\mathbb{R}^d\times\mathbb{R}^d)}\left\{\int\|x-y\|^2\mathrm{d}\pi + \tau\,\mathrm{KL}(\pi_1|\alpha) + \tau\,\mathrm{KL}(\pi_2|\beta)\right\}. \tag{10.13}$$

$\alpha = r_0\mathcal{N}(m_0,\Sigma_\alpha), \qquad \beta = r_1\mathcal{N}(m_1,\Sigma_\beta), \qquad r_0,r_1>0, \quad \Sigma_\alpha,\Sigma_\beta\succ 0$, where $r_i$ is the total mass and $m_i\in\mathbb{R}^d$ is the mean, while $\Sigma_\alpha$ and $\Sigma_\beta$ are the respective covariances. These inputs provide one of the few continuous unbalanced problems with an explicit endpoint cost.
Write $\bar{\alpha} = \mathcal{N}(m_0,\Sigma_\alpha)$ and $\bar{\beta} = \mathcal{N}(m_1,\Sigma_\beta)$. The covariance relaxation associated with these normalized inputs is governed by a matrix Bregman divergence.

**Definition 10.13** (LogDet divergence and KL-softened Bures cost)**.** For $P,\Sigma\in\mathbb{S}^d_{++}$, define

$$D_{\mathrm{LD}}(P|\Sigma) := B_{\Phi_{\mathrm{LD}}}(P|\Sigma) = \mathrm{tr}(\Sigma^{-1}P) - \log\det(\Sigma^{-1}P) - d. \tag{10.14}$$

For $\tau>0$, define the KL-softened Bures covariance cost

$$\mathcal{B}^2_\tau(\Sigma_\alpha,\Sigma_\beta) := \min_{P,Q\in\mathbb{S}^d_{++}}\left\{\mathcal{B}^2(P,Q) + \frac{\tau}{2}D_{\mathrm{LD}}(P|\Sigma_\alpha) + \frac{\tau}{2}D_{\mathrm{LD}}(Q|\Sigma_\beta)\right\}, \qquad \mathcal{B}^2_\infty(\Sigma_\alpha,\Sigma_\beta) := \mathcal{B}^2(\Sigma_\alpha,\Sigma_\beta). \tag{10.15}$$

**Proposition 10.14** (Closed form of the KL-softened Bures cost)**.** *Set $\eta = 2/\tau$ and $R_\gamma = \Sigma_\gamma^{-1} + \eta\mathrm{Id}$ for $\gamma\in\{\alpha,\beta\}$. Define*

$$L_\tau = R_\beta^{-1/2}\left(R_\beta^{1/2}R_\alpha R_\beta^{1/2}\right)^{1/2}R_\beta^{-1/2}. \tag{10.16}$$

*Then $L_\tau R_\beta L_\tau = R_\alpha$, and the unique minimizer of (10.15) is*

$$P_\tau = \left[\Sigma_\alpha^{-1} + \eta(\mathrm{Id}-L_\tau)\right]^{-1}, \qquad Q_\tau = L_\tau P_\tau L_\tau. \tag{10.17}$$

*Its value is*

$$\mathcal{B}^2_\tau(\Sigma_\alpha,\Sigma_\beta) = \frac{\tau}{2}\log\frac{\det\Sigma_\alpha\det\Sigma_\beta}{\det P_\tau\det Q_\tau}. \tag{10.18}$$

*Proof.* For $P, Q \succ 0$, let $L$ be the Gaussian optimal map from $P$ to $Q$, so $Q = LPL$. The Frobenius gradients of the squared Bures term are $\mathrm{Id} - L$ and $\mathrm{Id} - L^{-1}$; hence stationarity of (10.15) reads

$$P^{-1} = \Sigma_\alpha^{-1} + \eta(\mathrm{Id} - L), \qquad Q^{-1} = \Sigma_\beta^{-1} + \eta(\mathrm{Id} - L^{-1}). \tag{10.19}$$

Substituting $Q^{-1} = L^{-1}P^{-1}L^{-1}$ gives $LR_\beta L = R_\alpha$, whose unique positive-definite solution is (10.16). Back-substitution gives (10.17). Positivity follows directly from $2\eta^{-1}P_\tau^{-1} = (\mathrm{Id} - L_\tau)^2 + \eta^{-1}\Sigma_\alpha^{-1} + \eta^{-1}L_\tau\Sigma_\beta^{-1}L_\tau \succ 0$. The Bures term is jointly convex by Proposition 2.42, while each LogDet term is strictly convex and coercive on $\mathbb{S}^d_{++}$; the stationary pair is therefore the unique global minimizer. Multiplying the two equations in (10.19) by $P$ and $Q$, respectively, and taking traces gives

$$\mathcal{B}^2(P, Q) + \frac{\tau}{2}\left[\mathrm{tr}(\Sigma_\alpha^{-1}P) + \mathrm{tr}(\Sigma_\beta^{-1}Q) - 2d\right] = 0.$$

At $(P, Q) = (P_\tau, Q_\tau)$, this identity cancels the Bures term against the trace parts of the two LogDet divergences. The remaining logarithms give (10.18). □

**Proposition 10.15** (Gaussian Hellinger–Kantorovich endpoint)**.** *Set $G_\tau = \Sigma_\alpha + \Sigma_\beta + \frac{\tau}{2}\mathrm{Id}$. Then*

$$\mathrm{GHK}^2_\tau(\alpha, \beta) = \tau\left\{r_0 + r_1 - 2\sqrt{r_0 r_1}\exp\left(-\frac{1}{4}\|m_0 - m_1\|^2_{G_\tau^{-1}} - \frac{\mathcal{B}^2_\tau(\Sigma_\alpha, \Sigma_\beta)}{2\tau}\right)\right\}. \tag{10.20}$$

*Proof.* Apply the equal-penalty KL mass–shape formula (10.5) to the normalized inputs $\bar\alpha$ and $\bar\beta$. Gelbrich's inequality, Theorem 15.17, and the information-projection identity for KL show that replacing each competing normalized marginal by the Gaussian with the same mean and covariance cannot increase the objective. The Gaussian formulas for $\mathcal{W}_2^2$ and KL then separate the mean and covariance variables. Direct minimization over the two means gives $\frac{\tau}{2}\|m_0 - m_1\|^2_{G_\tau^{-1}}$, while Definition 10.13 identifies the covariance minimum with $\mathcal{B}^2_\tau(\Sigma_\alpha, \Sigma_\beta)$. Substitution into (10.5) gives (10.20). □

**Partial optimal transport.**

**Definition 10.16** (Fixed-mass, penalized and TV-unbalanced transport)**.** Let $A = \alpha(\mathcal{X})$, $B = \beta(\mathcal{Y})$ and $M = \min\{A, B\}$. For $0 \leqslant m \leqslant M$, the fixed-mass partial transport value is

$$\mathrm{POT}_m(\alpha, \beta) := \inf_{\substack{\pi \in \mathcal{M}_+(\mathcal{X}\times\mathcal{Y}) \\ \pi_1 \leqslant \alpha,\ \pi_2 \leqslant \beta \\ \pi(\mathcal{X}\times\mathcal{Y}) = m}} \int c\,\mathrm{d}\pi. \tag{10.21}$$

For an unmatched-mass price $\lambda \geqslant 0$, its penalized form is

$$\mathrm{POT}^{\mathrm{TV}}_\lambda(\alpha, \beta) := \inf_{\substack{\pi \in \mathcal{M}_+(\mathcal{X}\times\mathcal{Y}) \\ \pi_1 \leqslant \alpha,\ \pi_2 \leqslant \beta}} \left\{\int c\,\mathrm{d}\pi + \lambda\big(A + B - 2\pi(\mathcal{X}\times\mathcal{Y})\big)\right\}. \tag{10.22}$$

Finally, the TV-unbalanced transport value is

$$\mathrm{UW}^{\mathrm{TV}}_{c,\lambda}(\alpha, \beta) := \inf_{\pi \in \mathcal{M}_+(\mathcal{X}\times\mathcal{Y})} \left\{\int c\,\mathrm{d}\pi + \lambda\|\alpha - \pi_1\|_{\mathrm{TV}} + \lambda\|\beta - \pi_2\|_{\mathrm{TV}}\right\}. \tag{10.23}$$

The last problem is precisely the unbalanced formulation (10.1) with $\psi_1 = \psi_2 = \lambda\varphi_{\mathrm{TV}}$, where $\varphi_{\mathrm{TV}}$ is defined in (**??**). The constraints in (10.21) and (10.22) mean that the transported marginals are submeasures of the available ones.

**Proposition 10.17** (Equivalence of partial and TV-unbalanced transport)**.** *Assume that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces and that $c$ is continuous and nonnegative. Then, for every $\lambda \geqslant 0$,*

$$\mathrm{UW}^{\mathrm{TV}}_{c,\lambda}(\alpha, \beta) = \mathrm{POT}^{\mathrm{TV}}_\lambda(\alpha, \beta) = \lambda(A + B) + \min_{0 \leqslant m \leqslant M}\{\mathrm{POT}_m(\alpha, \beta) - 2\lambda m\}. \tag{10.24}$$

*The value function $m \mapsto \mathrm{POT}_m(\alpha, \beta)$ is convex on $[0, M]$. In particular, a minimizer of the TV-unbalanced problem can be chosen with $\pi_1 \leqslant \alpha$ and $\pi_2 \leqslant \beta$. If such a minimizer has mass $m_\lambda$, then it solves $\mathrm{POT}_{m_\lambda}(\alpha, \beta)$ and $m_\lambda$ minimizes the scalar problem in (10.24). Conversely, a minimizer of $\mathrm{POT}_m(\alpha, \beta)$ solves both penalized problems whenever $2\lambda \in \partial\big[m' \mapsto \mathrm{POT}_{m'}(\alpha, \beta)\big](m)$, where the value function is extended by $+\infty$ outside $[0, M]$.*

*Proof.* Let $J_\lambda(\pi)$ denote the objective in (10.23). Starting from any $\pi$, trim its source marginal to the greatest common submeasure $\pi_1 \wedge \alpha$, obtained by taking the pointwise minimum of the two densities with respect to $\pi_1 + \alpha$. This is realized at the plan level by multiplying the disintegration of $\pi$ with respect to $\pi_1$ by the corresponding Radon–Nikodym factor. If $\delta$ units are removed, the source TV penalty decreases by $\lambda\delta$, while the target TV penalty increases by at most $\lambda\delta$ by the triangle inequality. Since $c \geqslant 0$, the transport term cannot increase. Trimming the resulting target marginal to a submeasure of $\beta$ gives, by the same argument, a plan $\widetilde\pi \leqslant \pi$ such that $\widetilde\pi_1 \leqslant \alpha$, $\widetilde\pi_2 \leqslant \beta$ and $J_\lambda(\widetilde\pi) \leqslant J_\lambda(\pi)$.
For a subcoupling $\pi$ of mass $m$, one has $\|\alpha - \pi_1\|_{\mathrm{TV}} = A - m$ and $\|\beta - \pi_2\|_{\mathrm{TV}} = B - m$. This proves the first equality in (10.24). Partitioning the admissible subcouplings according to their mass then gives the second equality. Convex combinations of admissible plans show that $m \mapsto \mathrm{POT}_m$ is convex. Compactness and continuity ensure attainment of all three minima. Equality throughout the preceding bounds yields the optimizer correspondence. Finally, a mass $m$ minimizes $m' \mapsto \mathrm{POT}_{m'} - 2\lambda m'$ exactly when the displayed subgradient condition holds. □

**Proposition 10.18** (Fixed total mass reduces partial OT to balanced OT)**.** *Let $(\mathcal{X}, d)$ be a metric space, let $p \geqslant 1$, and fix $m > 0$. If $\alpha$ and $\beta$ have total mass $m$ and finite pth moments, then*

$$\mathrm{POT}_m(\alpha, \beta) = m\,\mathcal{W}_p\left(\frac{\alpha}{m}, \frac{\beta}{m}\right)^p. \tag{10.25}$$

*Proof.* For an admissible plan in (10.21), the inequalities $\pi_1 \leqslant \alpha$ and $\pi_2 \leqslant \beta$ are equalities because all four measures have mass $m$. Hence the admissible set is the usual set of couplings between $\alpha$ and $\beta$. Dividing a coupling by $m$ identifies it with a probability coupling between $\alpha/m$ and $\beta/m$, which proves (10.25). The distance claim follows from Proposition ??. □

### 10.2 Sliced Wasserstein Distances

**One-dimensional projections.**

**Spherical averaging.**

**Definition 10.19** (Sliced Wasserstein distance)**.** Let $p \geqslant 1$, let $\alpha, \beta \in \mathcal{P}_p(\mathbb{R}^d)$, and let $\sigma$ be the uniform probability measure on the sphere $\mathbb{S}^{d-1}$. The sliced $p$-Wasserstein distance is

$$\mathcal{SW}_p(\alpha,\beta)^p \coloneqq \int_{\mathbb{S}^{d-1}} \mathcal{W}_p\left((P_\theta)_\sharp\alpha, (P_\theta)_\sharp\beta\right)^p \mathrm{d}\sigma(\theta). \tag{10.26}$$

**Proposition 10.20** (Metric properties of sliced Wasserstein)**.** *For $p \geqslant 1$, $\mathcal{SW}_p$ is a distance on $\mathcal{P}_p(\mathbb{R}^d)$. Moreover, $\mathcal{SW}_p$ metrizes weak convergence together with convergence of the pth moment. Finally,*

$$\mathcal{SW}_p(\alpha,\beta)^p \leqslant \kappa_{d,p}\, \mathcal{W}_p(\alpha,\beta)^p, \qquad \kappa_{d,p} \coloneqq \int_{\mathbb{S}^{d-1}} |\theta_1|^p \mathrm{d}\sigma(\theta) = \frac{\Gamma(d/2)\Gamma((p+1)/2)}{\sqrt{\pi}\,\Gamma((d+p)/2)}.$$

*In particular, $\kappa_{d,2} = 1/d$. Conversely, if $d \geqslant 2$, $R > 0$, and $\alpha, \beta \in \mathcal{P}_p(\mathbb{R}^d)$ are supported in $B_R \coloneqq \{x \in \mathbb{R}^d \; ; \; \|x\| \leqslant R\}$, then there is a constant $C_{d,p}$ depending only on $d$ and $p$ such that Bonnotte's historical estimate reads $\mathcal{W}_p(\alpha,\beta)^p \leqslant C_{d,p} R^{p-\frac{1}{d+1}}\, \mathcal{SW}_p(\alpha,\beta)^{\frac{1}{d+1}}$. The sharp reverse estimate for $p = 1$ also gives, for every $p \geqslant 1$, constants $\widetilde{C}_{d,p}, \widetilde{c}_{d,p} > 0$ such that*

$$\mathcal{W}_p(\alpha,\beta) \leqslant \widetilde{C}_{d,p} R^{1-\frac{1}{pd}}\, \mathcal{SW}_p(\alpha,\beta)^{\frac{1}{pd}}, \qquad \textit{equivalently} \qquad \mathcal{SW}_p(\alpha,\beta) \geqslant \widetilde{c}_{d,p} R \left(\frac{\mathcal{W}_p(\alpha,\beta)}{R}\right)^{pd}.$$

*For $p = 1$, the exponent $1/d$ in this last comparison is sharp.*

*Proof.* Non-negativity and symmetry follow from the one-dimensional Wasserstein distance. For the triangle inequality, apply the triangle inequality of $\mathcal{W}_p$ for each direction $\theta$ and then Minkowski's inequality in $L^p(\mathbb{S}^{d-1})$.
If $\mathcal{SW}_p(\alpha,\beta) = 0$, then $(P_\theta)_\sharp\alpha = (P_\theta)_\sharp\beta$ for almost every direction. By continuity of characteristic functions this holds for all directions, and the Cramér–Wold theorem implies $\alpha = \beta$. This proves separation.
Let $\pi$ be any coupling between $\alpha$ and $\beta$. Projecting $\pi$ gives an admissible coupling between the corresponding one-dimensional projections, hence $\mathcal{W}_p\left((P_\theta)_\sharp\alpha, (P_\theta)_\sharp\beta\right)^p \leqslant \int_{\mathbb{R}^d\times\mathbb{R}^d} |\langle\theta,\, x-y\rangle|^p \mathrm{d}\pi(x,y)$. Rotational invariance of $\sigma$ gives $\int_{\mathbb{S}^{d-1}} |\langle\theta,\, z\rangle|^p \mathrm{d}\sigma(\theta) = \kappa_{d,p}\|z\|^p$. Integrating the first inequality in $\theta$ and optimizing over $\pi$ proves the stated direct comparison. The beta-integral formula for the first coordinate of a uniform point on the sphere gives the displayed value of $\kappa_{d,p}$; in particular, $\kappa_{d,2} = 1/d$.
To prove the topological statement, first suppose that $\mathcal{W}_p(\alpha_n,\alpha) \to 0$. The inequality $\mathcal{SW}_p \leqslant \mathcal{W}_p$ immediately gives $\mathcal{SW}_p(\alpha_n,\alpha) \to 0$. Conversely, suppose that $\mathcal{SW}_p(\alpha_n,\alpha) \to 0$. The spherical identity

$$\int_{\mathbb{S}^{d-1}} \int_{\mathbb{R}^d} |\langle\theta, x\rangle|^p \mathrm{d}\alpha_n(x) \mathrm{d}\sigma(\theta) = \kappa_{d,p} \int_{\mathbb{R}^d} \|x\|^p \mathrm{d}\alpha_n(x), \qquad \kappa_{d,p} > 0,$$

and the triangle inequality in $L^p(\mathbb{S}^{d-1}\times(0,1))$ show that the $p$th moments of $(\alpha_n)$ are uniformly bounded. Hence the sequence is tight. From every subsequence one can extract a further subsequence for which the projected $\mathcal{W}_p$ distances converge to zero for $\sigma$-a.e. $\theta$. Every weak limit therefore has the same one-dimensional projections as $\alpha$, so Cramér–Wold identifies it with $\alpha$. Thus $\alpha_n \rightharpoonup \alpha$. Finally, let $Q_{n,\theta}$ and $Q_\theta$ be the quantile functions of the projected measures. The one-dimensional quantile formula gives $\mathcal{SW}_p(\alpha_n,\alpha) = \|Q_{n,\theta} - Q_\theta\|_{L^p(\mathbb{S}^{d-1}\times(0,1))}$. Hence the corresponding $L^p$ norms converge. By the spherical identity, their $p$th powers are respectively $\kappa_{d,p}\int\|x\|^p\mathrm{d}\alpha_n(x)$ and $\kappa_{d,p}\int\|x\|^p\mathrm{d}\alpha(x)$. Thus the $p$th moments converge. Weak convergence plus convergence of these moments is equivalent to convergence in $\mathcal{W}_p$.
The reverse estimates on compact sets are deeper and use the Radon structure of slices. Bonnotte's Lemma 5.1.4 proves $\mathcal{W}_1(\alpha,\beta) \leqslant C_d R^{\frac{d}{d+1}}\, \mathcal{SW}_1(\alpha,\beta)^{\frac{1}{d+1}}$ by mollifying a Kantorovich–Rubinstein test function, representing the regularized test through one-dimensional projections, bounding the resulting action by $\mathcal{SW}_1$, and optimizing the smoothing scale. On $B_R$, $\mathcal{W}_p^p \leqslant (2R)^{p-1}\,\mathcal{W}_1, \qquad \mathcal{SW}_1 \leqslant \mathcal{SW}_p$, where the first inequality follows by evaluating the $p$-cost on a $\mathcal{W}_1$-optimal coupling, and the second follows from the monotonicity of one-dimensional Wasserstein distances and Hölder's inequality in $\theta$. These inequalities give Bonnotte's stated general-$p$ estimate. The sharper $p = 1$ comparison theorem of Carlier, Figalli, Mérigot and Wang replaces $1/(d+1)$ by the optimal exponent $1/d$. $\mathcal{W}_p^p \leqslant C_{d,p} R^{p-\frac{1}{d}}\, \mathcal{SW}_p^{\frac{1}{d}}$, which is equivalent to the final two-sided formulation in the statement. This supplies a lower bound on $\mathcal{SW}_p$ in terms of $\mathcal{W}_p$ for every $p$, but the asserted sharpness concerns only $p = 1$. □

**Radon point of view.**

**Definition 10.21** (Measure-valued Radon transform)**.** For $\alpha \in \mathcal{P}(\mathbb{R}^d)$, define $\mathfrak{R}\alpha(\theta) \coloneqq (P_\theta)_\sharp\alpha, \qquad \theta \in \mathbb{S}^{d-1}$.

Slicing can also be understood as applying Definition 10.21 before measuring discrepancy. Thus $\mathfrak{R}\alpha$ is a collection of one-dimensional projected measures indexed by directions.
$R\rho(\theta,s) = \int_{\{x:\langle\theta,\,x\rangle=s\}} \rho(x)\,\mathrm{d}\mathcal{H}^{d-1}(x)$.

$$d_{\mathrm{Rad},p}(h_0,h_1) \coloneqq \left(\int_{\mathbb{S}^{d-1}} \mathcal{W}_p\left(h_0(\theta), h_1(\theta)\right)^p \mathrm{d}\sigma(\theta)\right)^{1/p}.$$

$\widehat{h}(\mathrm{d}\theta,\mathrm{d}s) \coloneqq h(\theta)(\mathrm{d}s)\sigma(\mathrm{d}\theta)$ $d_{\mathrm{Rad},p}(h_0,h_1) = \mathcal{W}_{p,\sigma}(\widehat{h}_0,\widehat{h}_1)$. For any map $F : E \to (Y, d_Y)$, the pull-back of the metric $d_Y$ through $F$ is the function on $E \times E$ defined by
$(F^* d_Y)(x,x') \coloneqq d_Y\big(F(x), F(x')\big)$. $\mathcal{SW}_p(\alpha,\beta) = d_{\mathrm{Rad},p}(\mathfrak{R}\alpha,\mathfrak{R}\beta) = (\mathfrak{R}^* d_{\mathrm{Rad},p})(\alpha,\beta)$.

**Infinitesimal SW geometry.**

**Proposition 10.22** (First-order sliced comparison along Brenier perturbations)**.** *Let $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$ be absolutely continuous, and let $\varphi \in C^2(\mathbb{R}^d)$ be such that $\nabla\varphi \in L^2(\alpha;\mathbb{R}^d)$ and $\nabla^2\varphi$ is bounded. For $|t|$ small enough, set $T_t(x) := x + t\nabla\varphi(x)$, $\quad \alpha_t := (T_t)_\sharp\alpha$. Then $T_t$ is the quadratic Brenier map from $\alpha$ to $\alpha_t$, and $\mathcal{W}_2(\alpha,\alpha_t)^2 = t^2 \int_{\mathbb{R}^d} \|\nabla\varphi(x)\|^2 \mathrm{d}\alpha(x)$. Moreover, $\mathcal{SW}_2(\alpha,\alpha_t)^2 \leqslant \frac{t^2}{d} \int_{\mathbb{R}^d} \|\nabla\varphi(x)\|^2 \mathrm{d}\alpha(x)$.*

*Proof.* The map $T_t$ is the gradient of $\psi_t(x) := \frac{1}{2}\|x\|^2 + t\varphi(x)$. Since $\nabla^2\varphi$ is bounded, $\nabla^2\psi_t = \mathrm{Id} + t\nabla^2\varphi$ is positive semidefinite for $|t|$ small enough. Hence $\psi_t$ is convex and $T_t = \nabla\psi_t$ is a Brenier map. Its graph is cyclically monotone, so $\mathcal{W}_2(\alpha,\alpha_t)^2 = \int \|x - T_t(x)\|^2 \mathrm{d}\alpha(x) = t^2 \int \|\nabla\varphi(x)\|^2 \mathrm{d}\alpha(x)$. For a fixed direction $\theta$, the coupling $\left(P_\theta, P_\theta \circ T_t\right)_\sharp \alpha$ is admissible between $(P_\theta)_\sharp\alpha$ and $(P_\theta)_\sharp\alpha_t$. Therefore $\mathcal{W}_2\left((P_\theta)_\sharp\alpha, (P_\theta)_\sharp\alpha_t\right)^2 \leqslant t^2 \int |\langle\theta,\, \nabla\varphi(x)\rangle|^2 \mathrm{d}\alpha(x)$. Integrating over the sphere and using $\int_{\mathbb{S}^{d-1}} |\langle\theta,\, w\rangle|^2 \mathrm{d}\sigma(\theta) = \frac{1}{d}\|w\|^2$ gives the sliced estimate. □

**Intrinsic sliced length.**

$$\ell_{\mathcal{SW}_2}(\alpha,\beta) := \inf_{\substack{\gamma_0=\alpha,\ \gamma_1=\beta \\ \gamma\ \mathcal{SW}_2\text{-absolutely continuous}}} \int_0^1 |\dot\gamma_t|_{\mathcal{SW}_2}\, \mathrm{d}t, \tag{10.27}$$

$|\dot\gamma_t|_{\mathcal{SW}_2} := \lim_{h\to 0} \frac{\mathcal{SW}_2(\gamma_{t+h},\gamma_t)}{|h|}$ $\mathcal{SW}_2 \leqslant \ell_{\mathcal{SW}_2} \leqslant d^{-1/2}\, \mathcal{W}_2$ . $\ell_{\mathcal{SW}_2}(\alpha,\alpha_t) < d^{-1/2}\, \mathcal{W}_2(\alpha,\alpha_t)$.

**Sliced Wasserstein between Gaussians.**

**Proposition 10.23** (Sliced Wasserstein between Gaussians)**.** *Let $\alpha = \mathcal{N}(m_\alpha,\Sigma_\alpha)$ and $\beta = \mathcal{N}(m_\beta,\Sigma_\beta)$ on $\mathbb{R}^d$, with positive-semidefinite covariance matrices. Then*

$$\mathcal{SW}_2(\alpha,\beta)^2 = \frac{\|m_\alpha - m_\beta\|^2}{d} + \int_{\mathbb{S}^{d-1}} \left(\sqrt{\theta^\top\Sigma_\alpha\theta} - \sqrt{\theta^\top\Sigma_\beta\theta}\right)^2 \mathrm{d}\sigma(\theta). \tag{10.28}$$

*Proof.* For every $\theta \in \mathbb{S}^{d-1}$, $(P_\theta)_\sharp\alpha = \mathcal{N}\left(\theta^\top m_\alpha, \theta^\top\Sigma_\alpha\theta\right)$, and similarly for $\beta$. The one-dimensional Gaussian formula gives

$$\mathcal{W}_2\left((P_\theta)_\sharp\alpha, (P_\theta)_\sharp\beta\right)^2 = \left|\theta^\top(m_\alpha - m_\beta)\right|^2 + \left(\sqrt{\theta^\top\Sigma_\alpha\theta} - \sqrt{\theta^\top\Sigma_\beta\theta}\right)^2.$$

Integrating over $\theta$ and using $\int_{\mathbb{S}^{d-1}} \theta\theta^\top \mathrm{d}\sigma(\theta) = \mathrm{Id}/d$ gives (10.28). □

**$L^q$-sliced, max-sliced and subspace variants.**

**Definition 10.24** ($L^q$-sliced and subspace-sliced Wasserstein)**.** Let $p \in [1,\infty)$, $q \in [1,\infty]$, and $1 \leqslant k \leqslant d$. Denote by $\mathrm{St}(d,k) := \left\{U \in \mathbb{R}^{d\times k}\ ;\ U^\top U = \mathrm{Id}_k\right\}$ the Stiefel manifold, equipped with its normalized invariant probability measure $\sigma_{d,k}$. For $q < \infty$, define

$$\mathcal{SW}_{p,q,k}(\alpha,\beta) := \left(\int_{\mathrm{St}(d,k)} \mathcal{W}_p\left((U^\top)_\sharp\alpha, (U^\top)_\sharp\beta\right)^q \mathrm{d}\sigma_{d,k}(U)\right)^{1/q}, \tag{10.29}$$

and extend the definition to $q = \infty$ by $\mathcal{SW}_{p,\infty,k}(\alpha,\beta) := \sup_{U\in\mathrm{St}(d,k)} \mathcal{W}_p\left((U^\top)_\sharp\alpha, (U^\top)_\sharp\beta\right)$. We use the abbreviations $\mathcal{SW}_{p,q} := \mathcal{SW}_{p,q,1}$, $\quad \mathcal{SW}_p = \mathcal{SW}_{p,p}$, $\quad \mathrm{MaxSW}_{p,k} := \mathcal{SW}_{p,\infty,k}$, $\quad \mathrm{MaxSW}_p = \mathcal{SW}_{p,\infty}$ . Thus $k = 1$ gives line-based slicing, $q = \infty$ gives max-slicing, and $k = d$ gives $\mathcal{SW}_{p,q,d} = \mathcal{W}_p$ for every $q$. Quadratic max-slicing is useful when only a small set of directions carries the discrepancy, for instance in generative modeling. For $p = 2$, the $k$-dimensional max-sliced case is the projection-robust Wasserstein distance, which is related to the subspace-robust distances studied later in Section 10.5.

The supremum in the max-sliced case is attained because the Stiefel manifold is compact and the projected Wasserstein distance depends continuously on $U$. Moreover, the integrand is unchanged under $U \mapsto UQ$ for $Q \in \mathrm{O}(k)$, because $Q^\top$ is an isometry of $\mathbb{R}^k$.

**Proposition 10.25** (Basic bounds for sliced variants)**.** *Let $p \in [1,\infty)$, $1 \leqslant q \leqslant r \leqslant \infty$, and $1 \leqslant k \leqslant \ell \leqslant d$. Each $\mathcal{SW}_{p,q,k}$ is a finite distance on $\mathcal{P}_p(\mathbb{R}^d)$ and metrizes the same topology as $\mathcal{W}_p$. For every $\alpha,\beta \in \mathcal{P}_p(\mathbb{R}^d)$,*

$$\mathcal{SW}_{p,q,k}(\alpha,\beta) \leqslant \mathcal{SW}_{p,r,k}(\alpha,\beta) \leqslant \mathcal{W}_p(\alpha,\beta), \tag{10.30}$$
$$\mathcal{SW}_{p,q,k}(\alpha,\beta) \leqslant \mathcal{SW}_{p,q,\ell}(\alpha,\beta) \leqslant \mathcal{W}_p(\alpha,\beta). \tag{10.31}$$

*For a unit vector $e \in \mathbb{S}^{d-1}$, set*

$$\kappa_{d,k,p} := \int_{\mathrm{St}(d,k)} \|U^\top e\|^p \mathrm{d}\sigma_{d,k}(U) = \frac{\Gamma(d/2)\Gamma((k+p)/2)}{\Gamma(k/2)\Gamma((d+p)/2)}.$$

*This constant is independent of $e$, satisfies $\kappa_{d,d,p} = 1$ and $\kappa_{d,1,p} = \kappa_{d,p}$ from Proposition 10.20, and gives the sharper bounds*

$$\mathcal{SW}_{p,q,k}(\alpha,\beta) \leqslant \begin{cases} \kappa_{d,k,p}^{1/p}\, \mathcal{W}_p(\alpha,\beta), & 1 \leqslant q \leqslant p, \\ \kappa_{d,k,p}^{1/q}\, \mathcal{W}_p(\alpha,\beta), & p \leqslant q < \infty, \\ \mathcal{W}_p(\alpha,\beta), & q = \infty. \end{cases} \tag{10.32}$$

*Moreover,*
$$\mathcal{SW}_p(\alpha,\beta) \leqslant \mathcal{SW}_{p,p,k}(\alpha,\beta) \leqslant \kappa_{d,k,p}^{1/p}\, \mathcal{W}_p(\alpha,\beta). \tag{10.33}$$

*In particular, $\kappa_{d,k,2} = k/d$ and $\mathcal{SW}_{2,2,k}^2 \leqslant (k/d)\, \mathcal{W}_2^2$. If $q \geqslant p$, the compact-support reverse estimates in Proposition 10.20 remain valid with $\mathcal{SW}_{p,q,k}$ in place of $\mathcal{SW}_p$.*

*Proof.* Write $D_U(\alpha,\beta) := \mathcal{W}_p((U^\top)_\sharp\alpha, (U^\top)_\sharp\beta)$. Non-negativity and symmetry are inherited from $\mathcal{W}_p$. Minkowski's inequality in $L^q(\sigma_{d,k})$, or the supremum triangle inequality when $q=\infty$, proves the triangle inequality. The reverse triangle inequality for $\mathcal{W}_p$ shows that $U \mapsto D_U(\alpha,\beta)$ is continuous. Indeed, for $U_n \to U$,

$$\mathcal{W}_p\left(({U_n}^\top)_\sharp\alpha, (U^\top)_\sharp\alpha\right) \leqslant \|U_n - U\|_{\mathrm{op}} \left(\int \|x\|^p \mathrm{d}\alpha(x)\right)^{1/p},$$

and likewise for $\beta$. If $\mathcal{SW}_{p,q,k}(\alpha,\beta)=0$, then $D_U = 0$ for all $U$: this is immediate for $q=\infty$, while for finite $q$ it follows from continuity and the full support of $\sigma_{d,k}$. For any $\theta \in \mathbb{S}^{d-1}$, choose a frame $U$ whose first column is $\theta$. Equality of the two $k$-dimensional projected measures then implies equality of their first-coordinate marginals, hence $(P_\theta)_\sharp\alpha = (P_\theta)_\sharp\beta$. The Cramér–Wold theorem gives $\alpha=\beta$, proving separation.
Every $U^\top$ is 1-Lipschitz, so $D_U \leqslant \mathcal{W}_p(\alpha,\beta)$. Monotonicity of $L^q$ norms on a probability space proves (10.30). To compare projection dimensions, let $V \in \mathrm{St}(d,\ell)$ and $R \in \mathrm{St}(\ell,k)$. Since $(VR)^\top = R^\top V^\top$ and $R^\top$ is 1-Lipschitz, $D_{VR}(\alpha,\beta) \leqslant D_V(\alpha,\beta)$. Under the product of the invariant measures, $VR$ is uniformly distributed on $\mathrm{St}(d,k)$. Integration proves (10.31) for finite $q$; for $q=\infty$, extend each $k$-frame to an $\ell$-frame and use the same contraction.
For the sharper constants, let $\pi \in \Gamma(\alpha,\beta)$. Then $D_U(\alpha,\beta)^p \leqslant \int \|U^\top(x-y)\|^p \mathrm{d}\pi(x,y)$. Integration in $U$, followed by minimization in $\pi$, gives $\|D_.\|^p_{L^p(\sigma_{d,k})} \leqslant \kappa_{d,k,p}\,\mathcal{W}_p(\alpha,\beta)^p$. If $q \leqslant p$, use $\|D_.\|_{L^q} \leqslant \|D_.\|_{L^p}$. If $p \leqslant q < \infty$, use $D_U \leqslant \mathcal{W}_p$ to obtain $\int D_U^q \mathrm{d}\sigma_{d,k}(U) \leqslant \mathcal{W}_p(\alpha,\beta)^{q-p}\int D_U^p \mathrm{d}\sigma_{d,k}(U)$. This proves (10.32). The beta-integral for the $p$th power of the length of a random $k$-dimensional projection gives the displayed formula for $\kappa_{d,k,p}$.
Finally, let $\omega \in \mathbb{S}^{k-1}$. Since $P_{U\omega} = P_\omega \circ U^\top$, projection contraction gives $\mathcal{W}_p\left((P_{U\omega})_\sharp\alpha, (P_{U\omega})_\sharp\beta\right) \leqslant D_U(\alpha,\beta)$. Averaging its $p$th power first over $\omega$ and then over $U$ makes $U\omega$ uniform on $\mathbb{S}^{d-1}$ and proves the first inequality in (10.33). For $q \geqslant p$, (10.30) gives $\mathcal{SW}_p \leqslant \mathcal{SW}_{p,p,k} \leqslant \mathcal{SW}_{p,q,k}$, which transfers the stated reverse estimates.
For $\eta \in \mathcal{P}_p(\mathbb{R}^d)$, set $M_p(\eta) := (\int \|x\|^p \mathrm{d}\eta(x))^{1/p}$. Since the only coupling between $(U^\top)_\sharp\eta$ and $\delta_0$ sends every point to the origin, $D_U(\eta,\delta_0)^p = \int \|U^\top x\|^p \mathrm{d}\eta(x)$, $\quad \int D_U(\eta,\delta_0)^p \mathrm{d}\sigma_{d,k}(U) = \kappa_{d,k,p} M_p(\eta)^p$. Also $D_U(\eta,\delta_0) \leqslant M_p(\eta)$.

$$\mathcal{SW}_{p,q,k}(\eta,\delta_0) \geqslant \begin{cases} \kappa_{d,k,p}^{1/q} M_p(\eta), & 1 \leqslant q \leqslant p, \\ \kappa_{d,k,p}^{1/p} M_p(\eta), & p \leqslant q \leqslant \infty. \end{cases} \tag{10.34}$$

Indeed, the second line follows from monotonicity of $L^q$ norms. For $q \leqslant p$ and $M_p(\eta) > 0$, the pointwise bound $D_U \leqslant M_p(\eta)$ gives $D_U^q \geqslant M_p(\eta)^{q-p} D_U^p$; the zero-moment case is immediate.
Suppose now that $\mathcal{SW}_{p,q,k}(\alpha_n,\alpha) \to 0$. The triangle inequality and (10.34) imply $\sup_n M_p(\alpha_n) < \infty$. Write $D_{n,U} := D_U(\alpha_n,\alpha)$. The reverse triangle inequality and the coupling of each point with itself give $|D_{n,U} - D_{n,V}| \leqslant \|U-V\|_{\mathrm{op}}\big(M_p(\alpha_n) + M_p(\alpha)\big)$, so $(D_{n,\cdot})_n$ is equicontinuous on the compact Stiefel manifold. For $q<\infty$, equicontinuity and the full support of $\sigma_{d,k}$ upgrade $\|D_{n,\cdot}\|_{L^q} \to 0$ to $\|D_{n,\cdot}\|_{L^\infty} \to 0$; for $q=\infty$ this is immediate. Given $\theta \in \mathbb{S}^{d-1}$, choose a frame $U$ whose first column is $\theta$. Projection contraction then yields $\mathcal{W}_p\left((P_\theta)_\sharp\alpha_n, (P_\theta)_\sharp\alpha\right) \leqslant D_{n,U}$, uniformly in $\theta$. Hence $\mathcal{SW}_p(\alpha_n,\alpha) \to 0$, and Proposition 10.20 gives $\mathcal{W}_p(\alpha_n,\alpha)\to 0$. The converse follows directly from $\mathcal{SW}_{p,q,k} \leqslant \mathcal{W}_p$. $\square$

**Min-SW lifted transport plans.**

$$\langle x_{\sigma_\theta(1)}, \theta\rangle < \cdots < \langle x_{\sigma_\theta(n)}, \theta\rangle, \qquad \langle y_{\tau_\theta(1)}, \theta\rangle < \cdots < \langle y_{\tau_\theta(n)}, \theta\rangle.$$

$$\pi_\theta = \frac{1}{n}\sum_{i=1}^n \delta_{(x_{\sigma_\theta(i)}, y_{\tau_\theta(i)})} \in \mathcal{U}(\alpha,\beta). \tag{10.35}$$

$\alpha_\theta = (P_\theta)_\sharp\alpha, \qquad \beta_\theta = (P_\theta)_\sharp\beta,\ \varpi_\theta = (q_{\alpha_\theta}, q_{\beta_\theta})_\sharp \mathrm{Leb}_{[0,1]}$

**Definition 10.26** (Min-SW discrepancy and lifted plans)**.**

$$\mathcal{C}_\theta(\alpha,\beta) := \{\pi \in \mathcal{U}(\alpha,\beta)\ :\ (P_\theta,P_\theta)_\sharp\pi = \varpi_\theta\}. \tag{10.36}$$

The Min-SW discrepancy is

$$\text{Min-SW}_2(\alpha,\beta)^2 := \min_{\theta\in\mathbb{S}^{d-1}} \min_{\pi\in\mathcal{C}_\theta(\alpha,\beta)} \int_{\mathbb{R}^d\times\mathbb{R}^d} \|x-y\|^2 \mathrm{d}\pi(x,y). \tag{10.37}$$

$\alpha(\mathrm{d}x) = \int_{\mathbb{R}} \alpha^s(\mathrm{d}x)\mathrm{d}\alpha_\theta(s), \qquad \beta(\mathrm{d}y) = \int_{\mathbb{R}} \beta^t(\mathrm{d}y)\mathrm{d}\beta_\theta(t)$

$$\pi^0_\theta(\mathrm{d}x,\mathrm{d}y) = \int_{\mathbb{R}^2} \alpha^s(\mathrm{d}x)\beta^t(\mathrm{d}y)\mathrm{d}\varpi_\theta(s,t) \tag{10.38}$$

**Proposition 10.27** (Min-SW lower bound, topology and metric status)**.** *For all $\alpha,\beta \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$\mathcal{W}_2(\alpha,\beta) \leqslant \text{Min-SW}_2(\alpha,\beta). \tag{10.39}$$

*The converse fails in every dimension $d \geqslant 2$: there are measures $\alpha_n$ such that*

$$\mathcal{W}_2(\alpha_n,\gamma_d) \longrightarrow 0 \qquad \textit{but} \qquad \inf_n \text{Min-SW}_2(\alpha_n,\gamma_d) > 0, \qquad \gamma_d = \mathcal{N}(0,\mathrm{Id}_d). \tag{10.40}$$

*Thus* Min-SW$_2$ *does not metrize weak convergence, or even $\mathcal{W}_2$ convergence, on $\mathcal{P}_2(\mathbb{R}^d)$, and no estimate* Min-SW$_2(\alpha,\beta) \leqslant \omega(\mathcal{W}_2(\alpha,\beta))$ *can hold for all measures with a modulus $\omega(r) \to 0$ as $r \to 0$. In particular, there is no global Hölder bound* Min-SW$_2 \leqslant C\,\mathcal{W}_2^a$ *with $a>0$. The function* Min-SW$_2$ *is non-negative, symmetric and separates measures, but it does not satisfy the triangle inequality in general. It is therefore a separating symmetric discrepancy, not a distance. In dimension one, by contrast,* Min-SW$_2 = \mathcal{W}_2$.

*Proof.* Every $\pi \in \mathcal{C}_\theta(\alpha,\beta)$ is an admissible coupling, which proves (10.39). Symmetry follows by transposing the constrained plans, and separation follows from the same lower bound.
To see that the triangle inequality can fail, let $\alpha_X, \alpha_Y, \alpha_Z$ be the uniform laws on the rows of

$$X = \begin{pmatrix} -4 & -4 \\ 0 & 3 \\ 1 & -2 \\ 4 & -4 \end{pmatrix}, \qquad Z = \begin{pmatrix} -3 & 2 \\ 2 & 2 \\ 0 & 0 \\ 2 & -4 \end{pmatrix}, \qquad Y = \frac{X+Z}{2}.$$

A direct enumeration of the 4! assignments shows that the identity assignment is quadratically optimal for both $(X,Y)$ and $(Y,Z)$. It is induced by sorting along directions proportional to $(2,-1)$ and $(3,1)$, respectively. Hence $\text{Min-SW}_2(\alpha_X,\alpha_Y)^2 = \text{Min-SW}_2(\alpha_Y,\alpha_Z)^2 = \frac{51}{16}$. For $(X,Z)$, the projected orders change only across the ten lines orthogonal to pairwise differences of rows of $X$ or $Z$. Since the cost is linear in the plan, it suffices to enumerate the resulting angular cells and, on their boundaries, the compatible extreme permutations. $\frac{1}{4}\{63, 67, 73, 79, 91, 95, 105, 131, 135\}$. Thus $\text{Min-SW}_2(\alpha_X,\alpha_Z)^2 = 63/4$, and consequently $\text{Min-SW}_2(\alpha_X,\alpha_Z) = \frac{\sqrt{63}}{2} > \frac{\sqrt{51}}{2} = \text{Min-SW}_2(\alpha_X,\alpha_Y) + \text{Min-SW}_2(\alpha_Y,\alpha_Z)$.
Write $\gamma_k = \mathcal{N}(0,\text{Id}_k)$ and let

$$e_d^2 := \inf_{z_1,\dots,z_d \in \mathbb{R}^{d-1}} \int_{\mathbb{R}^{d-1}} \min_{1\leqslant j\leqslant d} \|u - z_j\|^2 \mathrm{d}\gamma_{d-1}(u) > 0. \tag{10.41}$$

This is the optimal quadratic quantization error of a $(d-1)$-dimensional standard Gaussian by at most $d$ points. It is positive because the set of probability measures supported on at most $d$ points is closed for $\mathcal{W}_2$, whereas $\gamma_{d-1}$ is not finitely supported.
Choose a sequence $\alpha_n = n^{-1}\sum_{i=1}^n \delta_{x_i}$ converging to $\gamma_d$ in $\mathcal{W}_2$ and in affine general position, meaning that no affine hyperplane contains more than $d$ of the $x_i$. Such a deterministic sequence exists: an i.i.d. Gaussian sequence has both properties almost surely. Fix $\theta$ and $\pi \in \mathcal{C}_\theta(\alpha_n,\gamma_d)$, and let $(X,Y)\sim\pi$. Write $S = \langle\theta,\, X\rangle$, $T = \langle\theta,\, Y\rangle$ and $Z = P_{\theta^\perp}Y$, where $P_{\theta^\perp} = \text{Id}_d - \theta\theta^\top$ is identified with an orthogonal projection onto $\mathbb{R}^{d-1}$. The monotone coupling between the discrete law of $S$ and the atomless Gaussian law of $T$ makes $S$ a function of $T$. Since $Y\sim\gamma_d$, the variables $T$ and $Z\sim\gamma_{d-1}$ are independent; hence the conditional law of $Z$ given $S$ is still $\gamma_{d-1}$. For each value $s$ of $S$, affine general position gives $\#\{i : \langle\theta,\, x_i\rangle = s\} \leqslant d$. Conditionally on $S = s$, the vector $P_{\theta^\perp}X$ therefore takes at most $d$ values. By (10.41),

$$\mathbb{E}\big[\|P_{\theta^\perp}(X-Y)\|^2 \mid S = s\big] \geqslant e_d^2.$$

Averaging and discarding the non-negative displacement along $\theta$ gives $\int \|x-y\|^2\mathrm{d}\pi(x,y) \geqslant e_d^2$. This holds for every admissible $\pi$ and every $\theta$, so $\text{Min-SW}_2(\alpha_n,\gamma_d) \geqslant e_d$, which proves (10.40) and rules out every vanishing modulus $\omega$. □

**Proposition 10.28** (Min-SW between nondegenerate Gaussians)**.** *Let $\alpha = \mathcal{N}(m_\alpha,\Sigma_\alpha)$ and $\beta = \mathcal{N}(m_\beta,\Sigma_\beta)$, where $\Sigma_\alpha,\Sigma_\beta \succ 0$. Then*

$$\text{Min-SW}_2(\alpha,\beta)^2 = \mathcal{W}_2(\alpha,\beta)^2 = \|m_\alpha - m_\beta\|^2 + \text{tr}\left(\Sigma_\alpha + \Sigma_\beta - 2\left(\Sigma_\alpha^{1/2}\Sigma_\beta\Sigma_\alpha^{1/2}\right)^{1/2}\right). \tag{10.42}$$

*Proof.* Let $T(x) = m_\beta + A(x - m_\alpha)$ be the Gaussian Brenier map of Proposition 2.39, where the symmetric positive-definite matrix $A$ is given by (2.18). Choose a unit eigenvector $\theta$ of $A$, with $A\theta = \lambda\theta$ and $\lambda > 0$. If $X\sim\alpha$ and $Y = T(X)$, then $\langle\theta,\, Y - m_\beta\rangle = \langle A\theta,\, X - m_\alpha\rangle = \lambda\langle\theta,\, X - m_\alpha\rangle$. Hence the optimal Gaussian coupling $(\text{Id},T)_\sharp\alpha$ belongs to $\mathcal{C}_\theta(\alpha,\beta)$ in Definition 10.26. It is feasible for (10.37), which gives $\text{Min-SW}_2 \leqslant \mathcal{W}_2$. The reverse inequality is (10.39); formula (2.19) concludes the proof. □

## 10.3 Quotient Wasserstein and Wasserstein–Procrustes

**Quotient distances.**

**Definition 10.29** (Quotient Wasserstein distance)**.** Let a group $\mathcal{G}$ act on a metric space $(\mathcal{X},d)$ by isometries, and assume that the action preserves finite $p$th moments. Write $g_\sharp\alpha$ for the push-forward of a measure $\alpha$ by $g\in\mathcal{G}$, and $[\alpha] := \{g_\sharp\alpha \;:\; g\in\mathcal{G}\}$ for its orbit. The quotient $p$-Wasserstein distance between the orbits of $\alpha,\beta\in\mathcal{P}_p(\mathcal{X})$ is

$$\mathcal{W}_{p,\mathcal{G}}([\alpha],[\beta]) := \inf_{g,h\in\mathcal{G}} \mathcal{W}_p(g_\sharp\alpha, h_\sharp\beta) = \inf_{g\in\mathcal{G}} \mathcal{W}_p(\alpha, g_\sharp\beta).$$

**Proposition 10.30** (Metric property on the quotient)**.** *If $\mathcal{G}$ acts by isometries and preserves $\mathcal{P}_p(\mathcal{X})$, then $\mathcal{W}_{p,\mathcal{G}}$ is a pseudo-distance on $\mathcal{P}_p(\mathcal{X})$. If the infimum is attained between every pair of orbits, then it is a distance on the orbit space $\mathcal{P}_p(\mathcal{X})/\mathcal{G}$. This applies in particular to continuous actions of compact groups.*

*Proof.* Isometry of the action gives $\mathcal{W}_p(g_\sharp\alpha, g_\sharp\beta) = \mathcal{W}_p(\alpha,\beta)$, which proves the second formula in Definition 10.29. Non-negativity and symmetry are inherited from $\mathcal{W}_p$. For the triangle inequality, choose $g_1, g_2\in\mathcal{G}$. Then

$$\mathcal{W}_p(\alpha,(g_1)_\sharp\beta) + \mathcal{W}_p(\beta,(g_2)_\sharp\gamma) = \mathcal{W}_p(\alpha,(g_1)_\sharp\beta) + \mathcal{W}_p((g_1)_\sharp\beta,(g_1g_2)_\sharp\gamma).$$

Applying the triangle inequality for $\mathcal{W}_p$ and then taking infima over $g_1, g_2$ gives the result. If the infimum is attained and the quotient distance is zero, there exist $g,h\in\mathcal{G}$ such that $\mathcal{W}_p(g_\sharp\alpha,h_\sharp\beta) = 0$, hence $g_\sharp\alpha = h_\sharp\beta$ and therefore $[\alpha] = [\beta]$. For a continuous action of a compact group, the objective is continuous on a compact parameter set, so attainment follows. Without attainment, the construction is naturally a metric only after identifying distinct orbits at zero orbit distance. □

**Rigid motions and Wasserstein–Procrustes.**

**Definition 10.31** (Wasserstein–Procrustes distance)**.** For $\alpha,\beta\in\mathcal{P}_2(\mathbb{R}^d)$, define

$$\text{WP}_2(\alpha,\beta)^2 := \inf_{\substack{R\in\text{O}(d),\, t\in\mathbb{R}^d \\ \pi\in\mathcal{U}(\alpha,\beta)}} \int \|Rx + t - y\|^2 \mathrm{d}\pi(x,y).$$

Restricting $R$ to $\text{SO}(d)$ defines the orientation-preserving variant.

This is an extrinsic counterpart of the Gromov–Wasserstein viewpoint developed in Section 12.3. Procrustes alignment searches only over ambient rigid motions, whereas GW compares the metric-measure structures themselves.

$$\mathrm{P}^{(\ell)} \in \operatorname{argmin}_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \sum_{i,j} \mathrm{P}_{i,j}\|R^{(\ell)}x_i + t^{(\ell)} - y_j\|^2. \tag{10.43}$$

$$(R^{(\ell+1)}, t^{(\ell+1)}) \in \operatorname{argmin}_{R\in\mathrm{O}(d),\, t\in\mathbb{R}^d} \sum_{i,j} \mathrm{P}^{(\ell)}_{i,j}\|Rx_i + t - y_j\|^2, \tag{10.44}$$

**Proposition 10.32** (Rigid update for a fixed coupling)**.** *Let* $\alpha = \sum_i a_i\delta_{x_i}$ *and* $\beta = \sum_j b_j\delta_{y_j}$*, and fix* $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$*. Define* $\bar{x} = \sum_i a_i x_i, \qquad \bar{y} = \sum_j b_j y_j, \qquad M_{\mathrm{P}} = \sum_{i,j} \mathrm{P}_{i,j}(y_j - \bar{y})(x_i - \bar{x})^\top$*. If* $M_{\mathrm{P}} = U\Sigma V^\top$ *is a singular value decomposition, then the minimizers over the full orthogonal group of* $\min_{R\in\mathrm{O}(d),\, t\in\mathbb{R}^d} \sum_{i,j} \mathrm{P}_{i,j}\|Rx_i + t - y_j\|^2$ *are given by* $R^\star = UV^\top$ *and* $t^\star = \bar{y} - R^\star\bar{x}$*, up to the usual non-uniqueness when* $M_{\mathrm{P}}$ *is rank deficient. If one imposes* $R \in \mathrm{SO}(d)$*, set* $D = \operatorname{diag}(1,\dots,1,\det(UV^\top)), \qquad R^\star = UDV^\top, \qquad t^\star = \bar{y} - R^\star\bar{x}$*.*

*Proof.* For fixed $R$, differentiating with respect to $t$ gives $t = \bar{y} - R\bar{x}$. Substituting this value centers the variables and leaves $\sum_{i,j} \mathrm{P}_{i,j}\|R(x_i - \bar{x}) - (y_j - \bar{y})\|^2$. The two quadratic terms independent of $R$ are fixed, so the problem is equivalent to maximizing $\sum_{i,j} \mathrm{P}_{i,j}\langle R(x_i - \bar{x}),\, y_j - \bar{y}\rangle = \operatorname{tr}(R^\top M_{\mathrm{P}})$. Von Neumann's trace inequality gives $\operatorname{tr}(R^\top U\Sigma V^\top) \leqslant \operatorname{tr}(\Sigma)$ for $R \in \mathrm{O}(d)$, with equality for $R = UV^\top$. Under the constraint $R \in \mathrm{SO}(d)$, the same argument applies with the additional determinant constraint on $U^\top RV$, which gives the displayed diagonal correction. ☐

## 10.4 Vector Quantiles and Linear Optimal Transport

**Vector quantiles.** Assume that $\gamma$ is absolutely continuous. For a target law $\alpha$ with finite second moment, its vector quantile relative to $\gamma$ is the Brenier map
$T_\alpha = \nabla\varphi_\alpha, \qquad (T_\alpha)_\sharp\gamma = \alpha,\ \min_{T_\sharp\gamma=\alpha} \int \|x - T(x)\|^2 \mathrm{d}\gamma(x)$.

**Linearized Wasserstein coordinates.**

**Definition 10.33** (Linear optimal-transport embedding)**.** Fix an absolutely continuous reference measure $\gamma \in \mathcal{P}_2(\mathbb{R}^d)$, and let $T_\alpha$ denote the Brenier map from $\gamma$ to $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$. Then

$$\alpha \mapsto T_\alpha - \mathrm{Id} \in L^2(\gamma;\mathbb{R}^d), \qquad \mathrm{LOT}_\gamma(\alpha,\beta) = \|T_\alpha - T_\beta\|_{L^2(\gamma)}. \tag{10.45}$$

$\mathrm{LOT}_\gamma(\gamma,\alpha) = \|T_\alpha - \mathrm{Id}\|_{L^2(\gamma)} = \mathcal{W}_2(\gamma,\alpha)$. For two arbitrary targets, the coupling $(T_\alpha, T_\beta)_\sharp\gamma$ is admissible but not generally optimal, so $\mathrm{LOT}_\gamma$ is a tangent-space approximation of the Wasserstein geometry. Uniqueness of the Brenier maps shows that $\mathrm{LOT}_\gamma$ is itself a distance on the target measures for which these maps are defined.
$\bar{T} = \sum_s \lambda_s T_{\alpha_s}, \qquad \bar{\alpha}_{\mathrm{LOT}} = \bar{T}_\sharp\gamma$. This is exact in one dimension, where quantile functions linearize $\mathcal{W}_2$, and it is especially useful when many barycenters with changing weights must be evaluated quickly.

**Proposition 10.34** (Quantitative stability of linear OT)**.** *Let* $X, Y \subset \mathbb{R}^d$ *be compact convex sets, assume that* $X$ *has positive Lebesgue measure, and let* $\gamma$ *be normalized Lebesgue measure on* $X$*. Then there exists a constant* $C = C(d, X, Y)$ *such that, for all* $\alpha,\beta \in \mathcal{P}(Y)$*,* $\mathcal{W}_2(\alpha,\beta) \leqslant \mathrm{LOT}_\gamma(\alpha,\beta)$ *and* $\mathrm{LOT}_\gamma(\alpha,\beta) \leqslant C\,\mathcal{W}_1(\alpha,\beta)^{2/15}$*.*

*Proof.* The first inequality is immediate: $(T_\alpha, T_\beta)_\sharp\gamma$ is a feasible coupling between $\alpha$ and $\beta$. The second inequality is the quantitative stability theorem of Mérigot, Delalande and Chazal. Its proof first controls the difference of the two Kantorovich potentials in terms of $\mathcal{W}_1(\alpha,\beta)$, then combines convexity and interpolation estimates to control the gradients in $L^2(\gamma)$; the resulting dimension-independent Hölder exponent is $2/15$. In one dimension, quantile functions make the embedding isometric and the exponent improves to one. In several dimensions, the theorem is global but only Hölder; it should not be read as a global Lipschitz estimate in $\mathcal{W}_2$. ☐

**Principal components in linear OT coordinates.** $\xi_i \coloneqq T_{\alpha_i} - \mathrm{Id}, \qquad \bar{\xi} \coloneqq \frac{1}{N}\sum_{i=1}^N \xi_i$. $\mathcal{C}_{\mathrm{LOT}}h \coloneqq \frac{1}{N}\sum_{i=1}^N \langle \xi_i - \bar{\xi},\, h\rangle_{L^2(\gamma)}(\xi_i - \bar{\xi})$, $e_k = \frac{1}{\sqrt{N\lambda_k}}\sum_{i=1}^N (v_k)_i(\xi_i - \bar{\xi})$ $a_{i,k} \coloneqq \langle \xi_i - \bar{\xi},\, e_k\rangle_{L^2(\gamma)}$. $T_a(x) \coloneqq x + \bar{\xi}(x) + \sum_{k=1}^m a_k e_k(x), \qquad \alpha_a \coloneqq (T_a)_\sharp\gamma$.

## 10.5 Spectral and Robust Wasserstein Distances

**Definition 10.35** (Monotone spectral gauge)**.** A monotone spectral gauge on positive semidefinite matrices is a convex, positively 1-homogeneous map $\gamma : \mathbb{S}^d_+ \to \mathbb{R}_+$ such that $\gamma(M) = 0$ only for $M = 0$, $\gamma(QMQ^\top) = \gamma(M)$ for every orthogonal matrix $Q$, and $0 \preceq M \preceq N \implies \gamma(M) \leqslant \gamma(N)$.

**Definition 10.36** (Schatten gauge)**.** For $1 \leqslant q \leqslant +\infty$, define the Schatten gauge by

$$\gamma_q(M) \coloneqq \|M\|_{S_q} = \begin{cases} \left(\sum_{i=1}^d \lambda_i(M)^q\right)^{1/q}, & 1 \leqslant q < +\infty, \\ \lambda_{\max}(M), & q = +\infty, \end{cases}.$$

It is a monotone spectral gauge. The cases $q = 1$, $q = 2$ and $q = +\infty$ are respectively the trace, Frobenius and spectral gauges.

**Definition 10.37** (Spectral Wasserstein distance)**.** Let $\gamma$ be a monotone spectral gauge. For a coupling $\pi \in \mathcal{U}(\alpha,\beta)$, define its displacement covariance $M_\pi \coloneqq \int (x-y)(x-y)^\top \mathrm{d}\pi(x,y)$. The spectral Wasserstein distance associated with $\gamma$ is

$$\mathcal{W}_\gamma(\alpha,\beta)^2 \coloneqq \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \gamma(M_\pi). \tag{10.46}$$

The special case $\gamma(M) = \operatorname{tr}(M)$ gives the usual quadratic Wasserstein distance $\mathcal{W}_2$. The spectral gauge $\gamma(M) = \lambda_{\max}(M)$ instead measures the worst transported variance direction.

$$\mathcal{W}_{2,A}(\alpha,\beta)^2 \coloneqq \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \int (x-y)^\top A(x-y)\mathrm{d}\pi(x,y) = \mathcal{W}_2((A^{1/2})_\sharp\alpha,(A^{1/2})_\sharp\beta)^2. \tag{10.47}$$

$$\mathcal{B}_\gamma \coloneqq \{A \succeq 0\,;\ \mathrm{tr}(AM) \leqslant \gamma(M) \quad \text{for all } M\succeq 0\}, \tag{10.48}$$

$\mathcal{B}_{\gamma_q} = \left\{A\succeq 0\,;\ \|A\|_{S_{q^*}} \leqslant 1\right\}$, $\quad \frac{1}{q}+\frac{1}{q^*}=1$, with the usual endpoint conventions. Thus the trace gauge has polar set $\{A : 0 \preceq A \preceq \mathrm{Id}\}$, the Frobenius gauge is self-polar on $\mathbb{S}^d_+$, and the spectral gauge has polar set $\{A\succeq 0 : \mathrm{tr}(A)\leqslant 1\}$.

**Proposition 10.38** (Robust representation and metric equivalence)**.** *Assume, for simplicity, that the measures are compactly supported and that $\gamma$ is closed and finite on the positive semidefinite cone. Then $\mathcal{W}_\gamma(\alpha,\beta)^2 = \sup_{A\in\mathcal{B}_\gamma}\mathcal{W}_{2,A}(\alpha,\beta)^2$. If there exist constants $0<a\leqslant b<+\infty$ such that $a\mathrm{Id}\in\mathcal{B}_\gamma$ and $\mathcal{B}_\gamma\subset\{A\,;\ 0\preceq A\preceq b\mathrm{Id}\}$, equivalently $a\,\mathrm{tr}(M)\leqslant\gamma(M)\leqslant b\,\mathrm{tr}(M)$ $\quad (M\succeq 0)$, then $\sqrt{a}\,\mathcal{W}_2(\alpha,\beta)\leqslant\mathcal{W}_\gamma(\alpha,\beta)\leqslant\sqrt{b}\,\mathcal{W}_2(\alpha,\beta)$. In particular, $\mathcal{W}_\gamma$ is a distance. When $\gamma$ is the restriction of a norm to the positive semidefinite cone, these bounds hold automatically in finite dimension, so $\mathcal{W}_\gamma$ is equivalent to $\mathcal{W}_2$ on measures with finite second moments.*

*Proof.* Using the polar representation of $\gamma$, $\mathcal{W}_\gamma(\alpha,\beta)^2 = \inf_{\pi\in\mathcal{U}(\alpha,\beta)}\sup_{A\in\mathcal{B}_\gamma}\mathrm{tr}(AM_\pi)$. The coupling set is convex and compact for weak convergence under compact support. Since $\gamma$ is a finite gauge on a finite-dimensional cone and vanishes only at the origin, it is equivalent to the trace norm on the slice $\mathrm{tr}(M)=1$, so $\mathcal{B}_\gamma$ is convex and compact. The map $(\pi,A)\mapsto\mathrm{tr}(AM_\pi)$ is affine in each variable and continuous. Sion's minimax theorem gives

$$\inf_\pi\sup_{A\in\mathcal{B}_\gamma}\mathrm{tr}(AM_\pi) = \sup_{A\in\mathcal{B}_\gamma}\inf_\pi\mathrm{tr}(AM_\pi) = \sup_{A\in\mathcal{B}_\gamma}\mathcal{W}_{2,A}(\alpha,\beta)^2.$$

For fixed $A\succeq 0$, $\mathcal{W}_{2,A}$ is the Wasserstein pseudodistance associated with the seminorm $x\mapsto\|A^{1/2}x\|$. Since all terms are nonnegative, the robust identity also gives $\mathcal{W}_\gamma(\alpha,\beta)=\sup_{A\in\mathcal{B}_\gamma}\mathcal{W}_{2,A}(\alpha,\beta)$. A supremum of pseudodistances is symmetric and satisfies the triangle inequality.

If $a\mathrm{Id}\in\mathcal{B}_\gamma$ and $A\preceq b\mathrm{Id}$ for all $A\in\mathcal{B}_\gamma$, then $a\,\mathcal{W}_2(\alpha,\beta)^2=\mathcal{W}_{2,a\mathrm{Id}}(\alpha,\beta)^2\leqslant\mathcal{W}_\gamma(\alpha,\beta)^2\leqslant b\,\mathcal{W}_2(\alpha,\beta)^2$, which proves definiteness and equivalence with $\mathcal{W}_2$. The equivalence between these operator bounds and $a\,\mathrm{tr}(M)\leqslant\gamma(M)\leqslant b\,\mathrm{tr}(M)$ follows directly from the polar formula. In finite dimension, any norm restricted to the positive semidefinite cone is equivalent to the trace norm on that cone. The finite-second-moment case follows by truncation when these norm-equivalence bounds hold. ☐

**Definition 10.39** (Subspace robust Wasserstein)**.** For $1\leqslant k\leqslant d$, the Paty–Cuturi subspace robust Wasserstein distance is

$$\mathrm{SRW}_{2,k}(\alpha,\beta)\coloneqq\sup_{U^\top U=\mathrm{Id}_k}\mathcal{W}_2((U^\top)_\sharp\alpha,(U^\top)_\sharp\beta)=\sup_{P^2=P=P^\top,\ \mathrm{tr}(P)=k}\mathcal{W}_{2,P}(\alpha,\beta).$$

**Proposition 10.40** (Ky Fan relaxation of subspace robust transport)**.** *Let $\lambda_1(M)\geqslant\cdots\geqslant\lambda_d(M)\geqslant 0$ and define the Ky Fan gauge $\gamma_k(M)=\sum_{\ell=1}^k\lambda_\ell(M)$. Then*

$$\max\left\{\mathrm{SRW}_{2,k}(\alpha,\beta),\sqrt{\frac{k}{d}}\,\mathcal{W}_2(\alpha,\beta)\right\}\leqslant\mathcal{W}_{\gamma_k}(\alpha,\beta)\leqslant\mathcal{W}_2(\alpha,\beta).$$

*For $k=d$, all displayed bounds are equalities and $\mathcal{W}_{\gamma_d}=\mathcal{W}_2$.*

*Proof.* Writing $\mathcal{P}_k = \left\{P\,;\ P^2=P=P^\top,\ \mathrm{tr}(P)=k\right\}$, diagonalizing $M$ gives Ky Fan's variational formula $\gamma_k(M) = \max_{P\in\mathcal{P}_k}\mathrm{tr}(PM)=\max_{0\preceq A\preceq\mathrm{Id},\ \mathrm{tr}(A)\leqslant k}\mathrm{tr}(AM)$. Hence minimax weak duality yields

$$\mathrm{SRW}_{2,k}(\alpha,\beta)^2=\sup_{P\in\mathcal{P}_k}\inf_{\pi\in\mathcal{U}(\alpha,\beta)}\mathrm{tr}(PM_\pi)\leqslant\inf_{\pi\in\mathcal{U}(\alpha,\beta)}\sup_{P\in\mathcal{P}_k}\mathrm{tr}(PM_\pi)=\mathcal{W}_{\gamma_k}(\alpha,\beta)^2.$$

Finally, $(k/d)\,\mathrm{tr}(M)\leqslant\gamma_k(M)\leqslant\mathrm{tr}(M)$ for every $M\succeq 0$. Taking infima over couplings and square roots proves the remaining bounds. ☐

For $k=1$, $\gamma_1(M)=\lambda_{\max}(M)$. The associated spectral gradient flows and their connection with the Muon algorithm are studied in Section 14.5.

## 10.6 Conditional Wasserstein Distances

**Definition 10.41** (Conditional couplings and conditional OT)**.** Let $(S,\lambda)$ be a standard Borel probability space of conditions and let $(\Omega,d)$ be a Polish metric space. For $p\geqslant 1$, denote by $\mathcal{P}_{p,\lambda}(S\times\Omega)$ the set of probability measures on $S\times\Omega$ whose first marginal is $\lambda$ and whose disintegration $\alpha(\mathrm{d}s,\mathrm{d}x)=\alpha_s(\mathrm{d}x)\lambda(\mathrm{d}s)$ satisfies $\int_S\int_\Omega d(x,x_0)^p\,\mathrm{d}\alpha_s(x)\mathrm{d}\lambda(s)<+\infty$ for some, hence every, $x_0\in\Omega$. For $\alpha,\beta\in\mathcal{P}_{p,\lambda}(S\times\Omega)$, a conditional coupling is a measure $\Pi(\mathrm{d}s,\mathrm{d}x,\mathrm{d}y)=\pi_s(\mathrm{d}x,\mathrm{d}y)\lambda(\mathrm{d}s)$ such that $\pi_s\in\mathcal{U}(\alpha_s,\beta_s)$ for $\lambda$-a.e. $s$. The set of such couplings is denoted $\mathcal{U}_\lambda(\alpha,\beta)$. Let $(s,x,y)\mapsto c_s(x,y)$ be jointly Borel, nonnegative, and lower semicontinuous in $(x,y)$ for every $s$. The conditional OT value is

$$\begin{aligned}\mathcal{L}_c^\lambda(\alpha,\beta)&\coloneqq\inf_{\Pi\in\mathcal{U}_\lambda(\alpha,\beta)}\int_S\int_{\Omega\times\Omega}c_s(x,y)\mathrm{d}\pi_s(x,y)\mathrm{d}\lambda(s)\\&=\int_S\inf_{\pi_s\in\mathcal{U}(\alpha_s,\beta_s)}\int_{\Omega\times\Omega}c_s(x,y)\mathrm{d}\pi_s(x,y)\mathrm{d}\lambda(s).\end{aligned}\tag{10.49}$$

**Definition 10.42** (Conditional Wasserstein distance)**.** For the fiber-independent cost $c_s(x,y)=c_p(x,y)\coloneqq d(x,y)^p$, define

$$\mathcal{W}_{p,\lambda}(\alpha,\beta)\coloneqq\left(\mathcal{L}_{c_p}^\lambda(\alpha,\beta)\right)^{1/p}=\left(\int_S\mathcal{W}_p^p(\alpha_s,\beta_s)\mathrm{d}\lambda(s)\right)^{1/p}. \tag{10.50}$$

Thus $\mathcal{L}_c^\lambda$ is the general conditional Kantorovich value, whereas $\mathcal{W}_{p,\lambda}$ is its metric specialization to the constant family $c_s = c_p = d^p$, after taking the $p$th root. Equivalently,
$\mathcal{W}_{p,\lambda}(\alpha,\beta)^p = \mathcal{L}_{c_p}^\lambda(\alpha,\beta)$;

**Proposition 10.43** (Metric property)**.** *For $p \geqslant 1$, $\mathcal{W}_{p,\lambda}$ is a distance on $\mathcal{P}_{p,\lambda}(S \times \Omega)$. Moreover, this metric space is Polish.*

*Proof.* Non-negativity and symmetry follow from the corresponding properties of $\mathcal{W}_p$ on each fiber. If $\mathcal{W}_{p,\lambda}(\alpha,\beta) = 0$, then $\mathcal{W}_p(\alpha_s,\beta_s) = 0$ for $\lambda$-a.e. $s$, hence $\alpha_s = \beta_s$ for $\lambda$-a.e. $s$ and the disintegrated measures $\alpha$ and $\beta$ coincide. For the triangle inequality, let $\gamma \in \mathcal{P}_{p,\lambda}(S \times \Omega)$. Since $\mathcal{W}_p(\alpha_s,\gamma_s) \leqslant \mathcal{W}_p(\alpha_s,\beta_s) + \mathcal{W}_p(\beta_s,\gamma_s)$ for a.e. $s$, Minkowski's inequality in $L^p(S,\lambda)$ gives $\mathcal{W}_{p,\lambda}(\alpha,\gamma) \leqslant \mathcal{W}_{p,\lambda}(\alpha,\beta) + \mathcal{W}_{p,\lambda}(\beta,\gamma)$. To see completeness and separability, identify a measure with the $\lambda$-a.e. equivalence class of its measurable disintegration map $s \mapsto \alpha_s \in \mathcal{P}_p(\Omega)$. Formula (10.50) is the corresponding $L^p(S,\lambda;\mathcal{P}_p(\Omega))$ metric. Since $(\mathcal{P}_p(\Omega),\mathcal{W}_p)$ is Polish, the standard $L^p$ space of measurable metric-valued maps is Polish as well. ☐

**Proposition 10.44** (Conditional geodesics)**.** *Assume that $(\Omega,d)$ is a geodesic Polish space and let $\alpha_0,\alpha_1 \in \mathcal{P}_{p,\lambda}(S \times \Omega)$. For $\lambda$-a.e. $s$, choose a constant-speed $\mathcal{W}_p$ geodesic $t \mapsto \alpha_{t,s}$ between $\alpha_{0,s}$ and $\alpha_{1,s}$, measurably in $s$, and set $\alpha_t(\mathrm{d}s,\mathrm{d}x) = \alpha_{t,s}(\mathrm{d}x)\lambda(\mathrm{d}s)$. Then $t \mapsto \alpha_t$ is a constant-speed geodesic for $\mathcal{W}_{p,\lambda}$.*

*Proof.* For $0 \leqslant r \leqslant t \leqslant 1$, the fiberwise geodesic property gives $\mathcal{W}_p(\alpha_{r,s},\alpha_{t,s}) = (t-r)\,\mathcal{W}_p(\alpha_{0,s},\alpha_{1,s})$ for $\lambda$-a.e. $s$. Integrating the $p$th power over $S$ gives $\mathcal{W}_{p,\lambda}(\alpha_r,\alpha_t) = (t-r)\,\mathcal{W}_{p,\lambda}(\alpha_0,\alpha_1)$. Hence the conditional curve has constant speed and realizes the distance between its endpoints. In Euclidean space one may take, for each $s$, an optimal plan $\pi_s \in \mathcal{U}(\alpha_{0,s},\alpha_{1,s})$ and set $\alpha_{t,s} = ((1-t)\mathrm{p}_1 + t\mathrm{p}_2)_\sharp\pi_s$, where $\mathrm{p}_1,\mathrm{p}_2$ are the two coordinate projections. On a general geodesic space, the same construction uses a measurable family of optimal dynamical plans. Such a family can be selected measurably in the Polish setting; when geodesics are non-unique, different selections can produce different conditional geodesics. ☐

# 11 Generalized OT Problems

## 11.1 OT Barycenters

**Fréchet means.** The natural formulation is a Fréchet-mean problem on the space of probability measures: the unknown is the barycenter measure itself, and its support is not prescribed. It uses the continuous Kantorovich value $\mathcal{L}_c$ defined in (3.7).

**Definition 11.1** (Optimal-transport barycenter)**.** Given input measures $(\beta_s)_{s=1}^S$ on a space $\mathcal{X}$ and weights $\lambda \in \Sigma_S$, discard any zero-weight inputs. An *optimal-transport barycenter* of this weighted family is any solution of

$$\min_{\alpha \in \mathcal{M}_+^1(\mathcal{X})} \sum_{s=1}^S \lambda_s \mathcal{L}_c(\alpha,\beta_s). \tag{11.1}$$

**Fixed-support restriction.** $\beta_s = \sum_{j=1}^{n_s} b_{s,j}\delta_{x_{s,j}}$, $\quad \mathrm{b}_s = (b_{s,j})_{j=1}^{n_s} \in \Sigma_{n_s}$. $(\mathrm{C}_s)_{i,j} = c(y_i,x_{s,j}) \in \mathbb{R}^{n \times n_s}$. Thus its value is $\mathrm{L}_{\mathrm{C}_s}(\mathrm{a},\mathrm{b}_s)$, in the notation of the discrete Kantorovich problem (3.2).

**Definition 11.2** (Fixed-support discrete OT barycenter)**.** With the candidate sites and cost matrices above, a *fixed-support discrete OT barycenter* is a measure $\alpha^\star = \sum_i a_i^\star \delta_{y_i}$ whose weight vector $\mathrm{a}^\star = (a_i^\star)_{i=1}^n$ solves

$$\min_{\mathrm{a} \in \Sigma_n} \sum_{s=1}^S \lambda_s \mathrm{L}_{\mathrm{C}_s}(\mathrm{a},\mathrm{b}_s), \tag{11.2}$$

$$\min_{\substack{\mathrm{a} \in \Sigma_n \\ Y = (y_i)_{i=1}^n \in (\mathbb{R}^d)^n}} \sum_{s=1}^S \lambda_s \mathrm{L}_{\mathrm{C}_s(Y)}(\mathrm{a},\mathrm{b}_s), \qquad [\mathrm{C}_s(Y)]_{i,j} = \|y_i - x_{s,j}\|^2.$$

For fixed $Y$, the objective is the convex problem (11.2) in a; its dependence on $Y$ is generally nonconvex, so the joint problem is nonconvex. Thus choosing the support is itself a significant computational part of the barycenter problem, rather than a harmless preprocessing step.

**Proposition 11.3** (Two-measure barycenters are Wasserstein geodesics)**.** *Let $\beta_0,\beta_1 \in \mathcal{P}_2(\mathbb{R}^d)$ and $t \in [0,1]$. For $S = 2$, $c(x,y) = \|x-y\|^2$ and weights $(1-t,t)$, the minimizers of the barycenter problem (11.1) are exactly the time-$t$ points of constant-speed $\mathcal{W}_2$ geodesics from $\beta_0$ to $\beta_1$. The minimum value is $t(1-t)\,\mathcal{W}_2^2(\beta_0,\beta_1)$. In particular, if $\pi$ is any optimal coupling between $\beta_0$ and $\beta_1$, then $\alpha_t := \big((x,y) \mapsto (1-t)x + ty\big)_\sharp\pi$ is a barycenter. If $\beta_0$ has a density, the barycenter is unique and $\alpha_t = ((1-t)\mathrm{Id} + tT)_\sharp\beta_0$, where $T$ is the Brenier map from $\beta_0$ to $\beta_1$.*

*Proof.* The endpoint cases $t \in \{0,1\}$ are immediate, so assume $0 < t < 1$. For any candidate $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, set $A = \mathcal{W}_2(\beta_0,\alpha)$, $\quad B = \mathcal{W}_2(\alpha,\beta_1)$, $\quad D = \mathcal{W}_2(\beta_0,\beta_1)$. The identity $(1-t)A^2 + tB^2 = t(1-t)(A+B)^2 + \big((1-t)A - tB\big)^2$ and the triangle inequality $A + B \geqslant D$ give $(1-t)\,\mathcal{W}_2^2(\alpha,\beta_0) + t\,\mathcal{W}_2^2(\alpha,\beta_1) \geqslant t(1-t)D^2$. For the interpolation $\alpha_t$ induced by an optimal coupling, Proposition **??** gives $A = tD$ and $B = (1-t)D$, so the lower bound is attained.
Conversely, every minimizer must attain equality in both inequalities above. Hence $A + B = D$ and $(1-t)A = tB$, which imply $A = tD$ and $B = (1-t)D$. Parameterize a constant-speed geodesic from $\beta_0$ to $\alpha$ on $[0,t]$ and one from $\alpha$ to $\beta_1$ on $[t,1]$; both pieces then have speed $D$. Their concatenation is globally geodesic: any shorter path between points on opposite pieces, combined with the remaining subsegments, would contradict $D = \mathcal{W}_2(\beta_0,\beta_1)$. Thus the concatenated geodesic passes through $\alpha$ at time $t$. Finally, if $\beta_0$ has a density, Brenier's theorem (Theorem 2.17) gives the unique optimal coupling $(\mathrm{Id},T)_\sharp\beta_0$, and the displayed McCann interpolation follows from Definition 2.21. ☐

**Proposition 11.4** (Mean and support of quadratic Wasserstein barycenters)**.** *Let $\beta_1,\ldots,\beta_S \in \mathcal{P}_2(\mathbb{R}^d)$, let $\lambda \in \Sigma_S$, and let $\alpha^\star$ be a barycenter for the squared Euclidean cost. Define*

$$K := \overline{\mathrm{conv}}\left(\bigcup_{\{s:\,\lambda_s > 0\}} \mathrm{supp}(\beta_s)\right).$$

*Then* $\mathrm{supp}(\alpha^\star) \subset K$, $\qquad \int_{\mathbb{R}^d} x \, \mathrm{d}\alpha^\star(x) = \sum_{s=1}^S \lambda_s \int_{\mathbb{R}^d} x \, \mathrm{d}\beta_s(x)$.

*Proof.* Write $m_\alpha = \int x \, \mathrm{d}\alpha(x)$ and let $\alpha^0$ be its centered translate, as in Proposition **??**. That proposition gives, for every candidate $\alpha$, $\sum_{s=1}^S \lambda_s \mathcal{W}_2^2(\alpha, \beta_s) = \sum_{s=1}^S \lambda_s \|m_\alpha - m_{\beta_s}\|^2 + \sum_{s=1}^S \lambda_s \mathcal{W}_2^2(\alpha^0, \beta_s^0)$. For a fixed centered law $\alpha^0$, the second sum is independent of $m_\alpha$, whereas the first is uniquely minimized at $m_\alpha = \sum_s \lambda_s m_{\beta_s}$. Translating $\alpha^\star$ without changing its centered law therefore proves the mean identity.
For the support statement, let $P_K$ be the Euclidean metric projection onto the closed convex set $K$. For every $y \in K$, the projection inequality gives $\|x - y\|^2 \geqslant \|P_K(x) - y\|^2 + \|x - P_K(x)\|^2$. For each $s$ with $\lambda_s > 0$, let $\pi_s$ be an optimal coupling between $\alpha^\star$ and $\beta_s$. Since $\beta_s(K) = 1$, pushing $\pi_s$ forward by $(P_K, \mathrm{Id})$ and integrating the preceding inequality yields $\mathcal{W}_2^2\big((P_K)_\sharp \alpha^\star, \beta_s\big) \leqslant \mathcal{W}_2^2(\alpha^\star, \beta_s) - \int_{\mathbb{R}^d} \mathrm{dist}(x, K)^2 \, \mathrm{d}\alpha^\star(x)$. After multiplication by $\lambda_s$ and summation, barycenter optimality forces the last integral to vanish. Hence $\alpha^\star(K) = 1$, and the closedness of $K$ gives $\mathrm{supp}(\alpha^\star) \subset K$. ☐

If all input supports are compact, their finite union is compact and so is its convex hull; in that case the closure in the definition of $K$ is unnecessary.

**One-dimensional case.**

**Proposition 11.5** (Quantile barycenters on the line)**.** *For $\mathcal{X} = \mathbb{R}$ and $c(x, y) = |x - y|^2$, the quantile function of a Wasserstein barycenter is the weighted average of the input quantile functions:* $\mathcal{C}_{\alpha^\star}^{-1}(r) = \sum_{s=1}^S \lambda_s \mathcal{C}_{\beta_s}^{-1}(r)$, $\qquad r \in (0, 1)$.

*Proof.* The one-dimensional formula (2.11) gives $\sum_s \lambda_s \mathcal{W}_2^2(\alpha, \beta_s) = \int_0^1 \sum_s \lambda_s \left| \mathcal{C}_\alpha^{-1}(r) - \mathcal{C}_{\beta_s}^{-1}(r) \right|^2 \mathrm{d}r$. The minimization decouples pointwise in $r$. For each fixed $r$, the minimizer of $z \mapsto \sum_s \lambda_s |z - \mathcal{C}_{\beta_s}^{-1}(r)|^2$ is the weighted average $\sum_s \lambda_s \mathcal{C}_{\beta_s}^{-1}(r)$. This function is nondecreasing because it is a positive weighted sum of nondecreasing quantile functions, hence it is a valid quantile function. ☐

**Gaussian case.**

**Proposition 11.6** (Nondegenerate Gaussian inputs remain Gaussian)**.** *Let the positive-weight inputs be $\beta_s = \mathcal{N}(m_s, \Sigma_s)$, with $\Sigma_s \succeq 0$, and assume that at least one $\Sigma_s$ is positive definite. The quadratic Wasserstein barycenter is unique and has the form $\alpha^\star = \mathcal{N}(m, \Sigma)$, where $m = \sum_s \lambda_s m_s$. Its positive-definite covariance $\Sigma$ is the unique minimizer of the Bures objective*

$$\mathcal{E}(X) \coloneqq \sum_s \lambda_s \mathcal{B}(X, \Sigma_s)^2. \tag{11.3}$$

*For $X \in \mathbb{S}_{++}^d$, define*

$$R_s(X) \coloneqq \left(X^{1/2} \Sigma_s X^{1/2}\right)^{1/2}, \qquad T_s(X) \coloneqq X^{-1/2} R_s(X) X^{-1/2}, \tag{11.4}$$

$$\overline{T}(X) \coloneqq \sum_s \lambda_s T_s(X), \qquad \Psi_\lambda(X) \coloneqq \sum_s \lambda_s R_s(X). \tag{11.5}$$

*Here $T_s(X)$ is the positive-semidefinite linear optimal map from $\mathcal{N}(0, X)$ to $\mathcal{N}(0, \Sigma_s)$. The covariance $\Sigma$ is equivalently characterized by either of the stationarity equations*

$$\overline{T}(\Sigma) = \mathrm{Id} \qquad \Longleftrightarrow \qquad \Psi_\lambda(\Sigma) = \Sigma. \tag{11.6}$$

*In dimension one, if $\sigma_s = \sqrt{\Sigma_s}$ denotes the input standard deviation, then the barycenter standard deviation is $\sigma = \sum_s \lambda_s \sigma_s$.*

*Proof.* Let $\mathrm{Gauss}(\alpha)$ denote the Gaussian measure with the same mean and covariance as $\alpha$. For every competitor $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$ and every Gaussian input $\beta_s$, the Gelbrich contraction of Theorem 15.17 gives $\mathcal{W}_2^2\big(\mathrm{Gauss}(\alpha), \beta_s\big) = \mathcal{W}_2^2\big(\mathrm{Gauss}(\alpha), \mathrm{Gauss}(\beta_s)\big) \leqslant \mathcal{W}_2^2(\alpha, \beta_s)$. Summing with weights $\lambda_s$ shows that moment-matched Gaussian projection cannot increase the barycenter objective. Since a barycenter exists, projecting any minimizer therefore produces a Gaussian barycenter. The input with positive-definite covariance is absolutely continuous; the uniqueness criterion stated after Definition 11.1 then implies that the barycenter itself is this Gaussian measure.
For a Gaussian candidate, formula (2.19) separates the objective as

$$\sum_s \lambda_s \mathcal{W}_2^2\big(\mathcal{N}(m, \Sigma), \mathcal{N}(m_s, \Sigma_s)\big) = \sum_s \lambda_s \|m - m_s\|^2 + \sum_s \lambda_s \mathcal{B}(\Sigma, \Sigma_s)^2.$$

The first term is uniquely minimized at $m = \sum_s \lambda_s m_s$. The second is the Bures barycenter problem. Uniqueness of the Wasserstein barycenter makes its covariance minimizer unique, and the presence of a positive-definite input makes this minimizer positive definite.

First suppose that $\Sigma_s \succ 0$. Cyclicity of the trace gives $\mathrm{tr}\, R_s(X) = \mathrm{tr}\left(\Sigma_s^{1/2} X \Sigma_s^{1/2}\right)^{1/2}$. For a symmetric perturbation $Z$, differentiation of the trace of the matrix square root yields

$$D[\mathrm{tr}\, R_s(X)]\,[Z] = \frac{1}{2} \mathrm{tr}\big(T_s(X) Z\big).$$

Consequently,

$$D\mathcal{E}(X)[Z] = \mathrm{tr}\big((\mathrm{Id} - \overline{T}(X)) Z\big). \tag{11.7}$$

The same identity for singular $\Sigma_s$ follows by replacing $\Sigma_s$ with $\Sigma_s + \delta \mathrm{Id}$ and letting $\delta \downarrow 0$. Since the minimizer lies in $\mathbb{S}_{++}^d$, first-order optimality is $\overline{T}(\Sigma) = \mathrm{Id}$. Finally, definitions (11.4)–(11.5) give $\Psi_\lambda(X) = X^{1/2} \overline{T}(X) X^{1/2}$, so $\overline{T}(\Sigma) = \mathrm{Id}$ is equivalent to $\Psi_\lambda(\Sigma) = \Sigma$. Conversely, a positive-definite solution of this equation is stationary for the convex objective (11.3), hence is its unique minimizer. In dimension one, $\mathcal{B}(\sigma^2, \sigma_s^2)^2 = (\sigma - \sigma_s)^2$, whose minimizer is $\sigma = \sum_s \lambda_s \sigma_s$. ☐

$x \longmapsto m_s + T_s(X)(x-m)$.

$$\mathcal{K}_1(X) \coloneqq \overline{T}(X)X\overline{T}(X) = X^{-1/2}\Psi_\lambda(X)^2X^{-1/2}. \tag{11.8}$$

$B_\eta(X) \coloneqq (1-\eta)\mathrm{Id} + \eta\overline{T}(X)$

**Proposition 11.7** (Convergence of the Gaussian transport fixed-point iteration)**.** *Assume that $\lambda_s > 0$, $\Sigma_s \succeq 0$, and that at least one $\Sigma_s$ is positive definite. Fix $\eta \in (0,1]$ and $X^{(0)} \succ 0$, and iterate $X^{(\ell+1)} = \mathcal{K}_\eta(X^{(\ell)})$. Then every $X^{(\ell)}$ is positive definite and $X^{(\ell)} \to \Sigma$, where $\Sigma$ is the covariance of the Gaussian barycenter in Proposition 11.6. More precisely, with $\mathcal{R}(X) \coloneqq$ $\mathrm{tr}\big((\mathrm{Id}-\overline{T}(X))X(\mathrm{Id}-\overline{T}(X))\big)$,*

*one has*
$$\mathcal{E}(X^{(\ell+1)}) \leqslant \mathcal{E}(X^{(\ell)}) - \eta(2-\eta)\mathcal{R}(X^{(\ell)}), \tag{11.9}$$
*and therefore $\sum_{\ell\geqslant 0}\mathcal{R}(X^{(\ell)}) < +\infty$. Equivalently, $\mathcal{B}\big(X^{(\ell)}, X^{(\ell+1)}\big)^2 = \eta^2\mathcal{R}(X^{(\ell)})$.*

*Proof.* Fix $X \succ 0$, abbreviate $T_s = T_s(X)$, $\overline{T} = \overline{T}(X)$ and $B = B_\eta(X)$, and let $Z \sim \mathcal{N}(0,X)$. Since $T_sZ \sim \mathcal{N}(0,\Sigma_s)$, the pair $(BZ, T_sZ)$ is an admissible coupling between $\mathcal{N}(0,\mathcal{K}_\eta(X))$ and $\mathcal{N}(0,\Sigma_s)$. For every $z \in \mathbb{R}^d$, the Euclidean variance decomposition around $\overline{T}z = \sum_s \lambda_s T_s z$ gives

$$\sum_s \lambda_s\|Bz - T_sz\|^2 = \sum_s \lambda_s\|\overline{T}z - T_sz\|^2 + (1-\eta)^2\|z-\overline{T}z\|^2,$$
$$\sum_s \lambda_s\|z - T_sz\|^2 = \sum_s \lambda_s\|\overline{T}z - T_sz\|^2 + \|z-\overline{T}z\|^2.$$

Integrating and bounding each Wasserstein cost by the admissible coupling $(BZ, T_sZ)$ proves (11.9). Moreover, $B \succ 0$: this is immediate when $\eta < 1$, and for $\eta = 1$ it follows because the weighted sum $\overline{T}$ contains at least one positive-definite term. Hence $z \mapsto Bz$ is the optimal map from $\mathcal{N}(0,X)$ to $\mathcal{N}(0,\mathcal{K}_\eta(X))$, which proves the displayed identity for the Bures step length.

Set $\delta(X) \coloneqq \det(X)^{1/(2d)}$, $\quad c_0 \coloneqq \sum_s \lambda_s \det(\Sigma_s)^{1/(2d)} > 0$. The Minkowski determinant inequality, applied to the positive-semidefinite matrices Id and $T_s(X)$ with weights $1-\eta$ and $\eta\lambda_s$, gives

$$\begin{aligned}\delta\big(\mathcal{K}_\eta(X)\big) &= \delta(X)\det(B_\eta(X))^{1/d}\\ &\geqslant (1-\eta)\delta(X) + \eta\sum_s \lambda_s\delta(X)\det(T_s(X))^{1/d}\\ &= (1-\eta)\delta(X) + \eta c_0,\end{aligned}$$

where the last identity uses $T_s(X)XT_s(X) = \Sigma_s$. It follows that $\delta(X^{(\ell)}) \geqslant \min\{\delta(X^{(0)}), c_0\} > 0$ for every $\ell$.

The decreasing energy also bounds the traces. Indeed, the Schatten Cauchy–Schwarz inequality in the Bures formula gives $\mathcal{B}(X,\Sigma_s)^2 \geqslant \left(\sqrt{\mathrm{tr}\,X} - \sqrt{\mathrm{tr}\,\Sigma_s}\right)^2$, so (11.9) bounds $\mathrm{tr}\,X^{(\ell)}$ uniformly. A uniform trace bound together with the determinant lower bound places all iterates in a compact subset of $\mathbb{S}^d_{++}$.

Summing (11.9) gives $\mathcal{R}(X^{(\ell)}) \to 0$. Every subsequence therefore has a further subsequence converging to some $X_\infty \succ 0$. Continuity of the matrix square root and inverse on the preceding compact set yields $\mathcal{R}(X_\infty) = 0$, hence $\overline{T}(X_\infty) = \mathrm{Id}$. Proposition 11.6 identifies $X_\infty$ with the unique covariance $\Sigma$. Since every cluster point is $\Sigma$, the full sequence converges to $\Sigma$. The undamped case $\eta = 1$ is the Gaussian specialization of Theorem 4.2 of Álvarez-Esteban et al.; the argument above also proves the damped extension. □

**Sinkhorn for barycenters.**

$$\min_{\mathrm{a}\in\Sigma_n} \sum_{s=1}^S \lambda_s \mathrm{L}^\varepsilon_{\mathrm{C}_s}(\mathrm{a},\mathrm{b}_s) \tag{11.10}$$

$$\min_{(\mathrm{P}_s)_s} \left\{ \varepsilon\sum_s \lambda_s \mathrm{KL}(\mathrm{P}_s|\mathrm{K}_s)\ ;\ \forall s,\ {\mathrm{P}_s}^\top\mathbb{1}_n = \mathrm{b}_s,\quad \mathrm{P}_1\mathbb{1}_{n_1} = \ldots = \mathrm{P}_S\mathbb{1}_{n_S} \right\}, \tag{11.11}$$

where $\mathrm{K}_s \coloneqq e^{-\mathrm{C}_s/\varepsilon}$. The barycenter a is implicitly encoded in the common row marginal $\mathrm{a} = \mathrm{P}_1\mathbb{1}_{n_1} = \ldots = \mathrm{P}_S\mathbb{1}_{n_S}$.

$$\mathrm{P}_s = \mathrm{diag}(\mathrm{u}_s)\mathrm{K}_s\,\mathrm{diag}(\mathrm{v}_s), \tag{11.12}$$

$$\forall s \in [\![1,S]\!],\qquad \mathrm{v}_s^{(\ell+1)} \coloneqq \frac{\mathrm{b}_s}{{\mathrm{K}_s}^\top \mathrm{u}_s^{(\ell)}}, \tag{11.13}$$

$$\forall s \in [\![1,S]\!],\qquad \mathrm{u}_s^{(\ell+1)} \coloneqq \frac{\mathrm{a}^{(\ell+1)}}{\mathrm{K}_s\mathrm{v}_s^{(\ell+1)}}, \tag{11.14}$$

$$\text{where}\quad \mathrm{a}^{(\ell+1)} \coloneqq \prod_s (\mathrm{K}_s\mathrm{v}_s^{(\ell+1)})^{\lambda_s}. \tag{11.15}$$

$\mathrm{g}_s^+ = \varepsilon\log\left(\frac{\mathrm{b}_s}{{\mathrm{K}_s}^\top e^{\mathrm{f}_s/\varepsilon}}\right)$, $\mathrm{q}_s \coloneqq \mathrm{K}_s e^{\mathrm{g}_s^+/\varepsilon}$, $\qquad \mathrm{a}^+ = \prod_s \mathrm{q}_s^{\lambda_s}$, $\qquad \mathrm{f}_s^+ = \varepsilon\log\left(\frac{\mathrm{a}^+}{\mathrm{q}_s}\right)$. Thus a complete generalized Sinkhorn cycle is exact two-block coordinate ascent on the dual, or equivalently alternating minimization of its negative. Indeed, the constraint follows from $\log \mathrm{a}^+ = \sum_s \lambda_s \log \mathrm{q}_s$, while the first-order conditions require $e^{\mathrm{f}_s^+/\varepsilon}\mathrm{q}_s = \mathrm{a}^+$ for every $s$.

**Proposition 11.8** (Dual of entropic barycenters)**.** *Under the preceding positivity assumptions, the optimal scalings in (11.12) can be written as $(\mathrm{u}_s,\mathrm{v}_s) = (e^{\mathrm{f}_s/\varepsilon}, e^{\mathrm{g}_s/\varepsilon})$, where $(\mathrm{f}_s,\mathrm{g}_s)_s$ solve the dual problem*

$$\max_{(\mathrm{f}_s,\mathrm{g}_s)_s} \left\{ \sum_s \lambda_s \left( \langle \mathrm{g}_s,\,\mathrm{b}_s\rangle - \varepsilon\langle \mathrm{K}_s e^{\mathrm{g}_s/\varepsilon},\, e^{\mathrm{f}_s/\varepsilon}\rangle + \varepsilon\sum_{i,j}(\mathrm{K}_s)_{i,j} \right)\ ;\ \sum_s \lambda_s \mathrm{f}_s = 0 \right\}. \tag{11.16}$$

*Proof.* Introduce Lagrange multipliers in (11.11):

$$\min_{(\mathrm{P}_s)_s,\mathrm{a}}\max_{(\mathrm{f}_s,\mathrm{g}_s)_s}\sum_s \lambda_s\Big(\varepsilon\mathrm{KL}(\mathrm{P}_s|\mathrm{K}_s)+\langle \mathrm{a}-\mathrm{P}_s\mathbb{1}_{n_s},\,\mathrm{f}_s\rangle+\langle \mathrm{b}_s-\mathrm{P}_s{}^\top\mathbb{1}_n,\,\mathrm{g}_s\rangle\Big).$$

The explicit constraint $\mathrm{a}\in\Sigma_n$ may be dropped here: nonnegativity of the couplings, together with $\mathrm{P}_s\mathbb{1}_{n_s}=\mathrm{a}$ and $\mathrm{P}_s{}^\top\mathbb{1}_n=\mathrm{b}_s$, already forces $\mathrm{a}\in\Sigma_n$. We may therefore minimize the Lagrangian over $\mathrm{a}\in\mathbb{R}^n$. Strong duality holds, so one can exchange the minimum and maximum. Finiteness of the minimum with respect to a gives the vector constraint $\sum_s\lambda_s\mathrm{f}_s=0$, while minimization with respect to $\mathrm{P}_s$ gives the Legendre transform of $\mathrm{KL}(\cdot|\mathrm{K}_s)$:

$$\max_{(\mathrm{f}_s,\mathrm{g}_s)_s}\sum_s\lambda_s\left[\langle\mathrm{g}_s,\,\mathrm{b}_s\rangle-\varepsilon\mathrm{KL}^*\left(\frac{\mathrm{f}_s\oplus\mathrm{g}_s}{\varepsilon}\middle|\mathrm{K}_s\right)\right],\qquad \sum_s\lambda_s\mathrm{f}_s=0.$$

The separable conjugate is
$$\mathrm{KL}^*(\mathrm{U}|\mathrm{K})=\sum_{i,j}\mathrm{K}_{i,j}\left(e^{\mathrm{U}_{i,j}}-1\right),\tag{11.17}$$

because for $k>0$, $\sup_{r\geqslant 0} ur-\big(r\log(r/k)-r+k\big)=k(e^u-1)$, and the case $k=0$ follows by lower semicontinuity. Substituting this conjugate gives (11.16), including its additive constant. The coordinate maximization in $\mathrm{g}_s$ gives (11.13); the block maximization in all $(\mathrm{f}_s)_s$ under $\sum_s\lambda_s\mathrm{f}_s=0$ gives the weighted geometric mean (11.15) and then (11.14). □

**Wasserstein-over-Wasserstein and barycenters.** $\int_{\mathcal{P}(\Omega)}\mathcal{L}_c(\beta_0,\alpha)\mathrm{d}\mathfrak{A}(\alpha)<+\infty$, $\mathcal{B}_c(\mathfrak{A}):=\mathrm{Argmin}_{\beta\in\mathcal{P}(\Omega)}\int_{\mathcal{P}(\Omega)}\mathcal{L}_c(\beta,\alpha)\mathrm{d}\mathfrak{A}(\alpha)$. $\widetilde{\alpha}_{\mathfrak{A}}:=\underset{\beta\in\mathcal{P}(\Omega)}{\mathrm{argmin}}\int_{\mathcal{P}(\Omega)}\mathcal{L}_c(\beta,\alpha)\mathrm{d}\mathfrak{A}(\alpha)$,

**Proposition 11.9** (Law of large numbers for barycenters over measures)**.** *Let $\Omega$ be a compact metric space, let $c:\Omega\times\Omega\to\mathbb{R}$ be continuous, and let $\mathfrak{A}\in\mathcal{P}(\mathcal{P}(\Omega))$. Let $\alpha_1,\alpha_2,\dots$ be independent random probability measures with common law $\mathfrak{A}$, and set $\widehat{\mathfrak{A}}_p:=\frac{1}{p}\sum_{i=1}^p\delta_{\alpha_i}\in\mathcal{P}(\mathcal{P}(\Omega))$. Then, almost surely,*

$$\widehat{\mathfrak{A}}_p\rightharpoonup\mathfrak{A}\quad\textit{in } \mathcal{P}(\mathcal{P}(\Omega)),\qquad \bar{\alpha}_{\widehat{\mathfrak{A}}_p}\rightharpoonup\bar{\alpha}_{\mathfrak{A}}\quad\textit{in } \mathcal{P}(\Omega).\tag{11.18}$$

*Moreover, if $\mathcal{B}_c(\mathfrak{A})=\{\widetilde{\alpha}_{\mathfrak{A}}\}$ is a singleton and if $\widetilde{\alpha}_{\widehat{\mathfrak{A}}_p}\in\mathcal{B}_c(\widehat{\mathfrak{A}}_p)$ is any empirical barycenter, then, almost surely,*

$$\widetilde{\alpha}_{\widehat{\mathfrak{A}}_p}\rightharpoonup\widetilde{\alpha}_{\mathfrak{A}}\quad\textit{in } \mathcal{P}(\Omega).\tag{11.19}$$

*Proof.* Set $K=\mathcal{P}(\Omega)$. Since $\Omega$ is compact metric, $K$ is compact metric for weak convergence, and so is $\mathcal{P}(K)$. The space $\mathcal{C}(K)$ is separable for the uniform norm. Applying the scalar strong law to a countable dense family of test functions and then using uniform approximation gives, almost surely, for every $\Phi\in\mathcal{C}(K)$, $\int\Phi(\alpha)\mathrm{d}\widehat{\mathfrak{A}}_p(\alpha)=\frac{1}{p}\sum_{i=1}^p\Phi(\alpha_i)\longrightarrow\int\Phi(\alpha)\mathrm{d}\mathfrak{A}(\alpha)$ This is exactly (11.18).
For the collapsed mixtures, take $f\in\mathcal{C}(\Omega)$ and define $\Phi_f(\alpha)=\int_\Omega f\mathrm{d}\alpha$. This function is continuous on $\mathcal{P}(\Omega)$. Hence

$$\int_\Omega f\mathrm{d}\bar{\alpha}_{\widehat{\mathfrak{A}}_p}=\int_{\mathcal{P}(\Omega)}\Phi_f(\alpha)\mathrm{d}\widehat{\mathfrak{A}}_p(\alpha)\longrightarrow\int_{\mathcal{P}(\Omega)}\Phi_f(\alpha)\mathrm{d}\mathfrak{A}(\alpha)=\int_\Omega f\mathrm{d}\bar{\alpha}_{\mathfrak{A}},$$

which proves the second convergence in (11.18).
Since $c$ is continuous on the compact product $\Omega\times\Omega$, the value $(\beta,\alpha)\mapsto\mathcal{L}_c(\beta,\alpha)$ is continuous on $K^2$. Therefore the map $\beta\mapsto h_\beta$, where $h_\beta(\alpha)=\mathcal{L}_c(\beta,\alpha)$, is continuous from the compact space $K$ to $\mathcal{C}(K)$. Its image $\mathcal{H}:=\{h_\beta:\beta\in K\}$ is compact in $\mathcal{C}(K)$. The convergence of $\widehat{\mathfrak{A}}_p$ to $\mathfrak{A}$ is uniform over $\mathcal{H}$: given $\eta>0$, cover $\mathcal{H}$ by finitely many $\eta$-balls in $\|\cdot\|_\infty$, use weak convergence for the centers, and bound the error on the balls by the total variation of $\widehat{\mathfrak{A}}_p-\mathfrak{A}$. Hence the empirical objectives $F_p(\beta):=\int\mathcal{L}_c(\beta,\alpha)\mathrm{d}\widehat{\mathfrak{A}}_p(\alpha)$ converge uniformly on $K$ to $F(\beta):=\int\mathcal{L}_c(\beta,\alpha)\mathrm{d}\mathfrak{A}(\alpha)$. Let $\beta_p\in\mathcal{B}_c(\widehat{\mathfrak{A}}_p)$. Compactness gives a subsequence $\beta_{p_k}\rightharpoonup\beta$. Uniform convergence, continuity of $F$, and optimality of $\beta_{p_k}$ give, for any $\gamma\in K$, $F(\beta)=\lim_k F_{p_k}(\beta_{p_k})\leqslant\lim_k F_{p_k}(\gamma)=F(\gamma)$, so $\beta\in\mathcal{B}_c(\mathfrak{A})$. If this set is the singleton $\{\widetilde{\alpha}_{\mathfrak{A}}\}$, every converging subsequence has the same limit, and therefore the whole sequence $\widetilde{\alpha}_{\widehat{\mathfrak{A}}_p}$ converges to $\widetilde{\alpha}_{\mathfrak{A}}$, proving (11.19). □

**Toward central limit theorems on Wasserstein space.** There are nevertheless important settings where such results can be proved. In one dimension, the quantile representation linearizes $\mathcal{W}_2$, so barycenter fluctuations can be studied through empirical averages of quantile functions.

**Sliced and Radon barycenters.** Slicing gives a scalable surrogate for high-dimensional barycenters by applying the one-dimensional quantile formula of Proposition 11.5 in every projection direction. For measures on $\mathbb{R}^d$, define the sliced barycenter by replacing $\mathcal{W}_2$ with $\mathcal{SW}_2$, introduced in Definition 10.19 and interpreted through the Radon transform in Section 10.2:

$$\min_{\alpha\in\mathcal{M}_+^1(\mathbb{R}^d)}\sum_{s=1}^S\lambda_s\,\mathcal{SW}_2(\alpha,\beta_s)^2=\min_{\alpha\in\mathcal{M}_+^1(\mathbb{R}^d)}\int_{\mathbb{S}^{d-1}}\sum_{s=1}^S\lambda_s\,\mathcal{W}_2\left((P_\theta)_\sharp\alpha,(P_\theta)_\sharp\beta_s\right)^2\mathrm{d}\sigma(\theta).$$

$$\min_{(\gamma_\theta)_\theta}\int_{\mathbb{S}^{d-1}}\sum_{s=1}^S\lambda_s\,\mathcal{W}_2(\gamma_\theta,\beta_{s,\theta})^2\mathrm{d}\sigma(\theta),\qquad \beta_{s,\theta}:=(P_\theta)_\sharp\beta_s.$$

$\mathcal{C}_{\gamma_\theta}^{-1}(r)=\sum_{s=1}^S\lambda_s\mathcal{C}_{\beta_{s,\theta}}^{-1}(r),\qquad r\in[0,1]$. For two inputs $\beta_0,\beta_1$ and interpolation weight $t\in[0,1]$, it is useful to make this directionwise construction explicit. Define

$$Q_i(\theta,r):=\mathcal{C}_{\beta_{i,\theta}}^{-1}(r),\qquad Q_t(\theta,r):=(1-t)Q_0(\theta,r)+tQ_1(\theta,r),\qquad \gamma_{t,\theta}:=\big(Q_t(\theta,\cdot)\big)_\sharp\mathrm{Leb}_{[0,1]}.\tag{11.20}$$

Thus $Q_t$ is the quantile field of the independently interpolated projected laws $(\gamma_{t,\theta})_\theta$. When these laws have densities, we denote them by $h_t(\theta,\cdot)$.
Let $h(\theta,t)$ denote a density of $\gamma_\theta$. We use the one-dimensional Fourier transform in the variable $t$,

$$\widehat{h}(\theta,\omega) := \int_{\mathbb{R}} e^{-\mathrm{i}\omega t} h(\theta,t)\mathrm{d}t, \qquad h(\theta,t) = \frac{1}{2\pi}\int_{\mathbb{R}} e^{\mathrm{i}\omega t}\widehat{h}(\theta,\omega)\mathrm{d}\omega, \tag{11.21}$$

**Proposition 11.10** (Radon least-squares pseudoinverse)**.** *Let $d \geqslant 2$, let $\sigma$ be the uniform probability measure on $\mathbb{S}^{d-1}$, and let $R$ be the density Radon transform introduced in the Radon point of view paragraph. Write $\mathcal{D}(R) := \{\rho \in L^2(\mathbb{R}^d)\ ;\ R\rho \in L^2(\mathbb{S}^{d-1}\times\mathbb{R}, \mathrm{d}\sigma\,\mathrm{d}t)\}$. Let $h \in L^2(\mathbb{S}^{d-1}\times\mathbb{R}, \mathrm{d}\sigma\,\mathrm{d}t)$ and assume that the Fourier expressions below define an element of $\mathcal{D}(R)$. Then the unique solution of*

$$\min_{\rho\in\mathcal{D}(R)} \int_{\mathbb{S}^{d-1}}\int_{\mathbb{R}} |R\rho(\theta,t)-h(\theta,t)|^2\,\mathrm{d}t\,\mathrm{d}\sigma(\theta) \tag{11.22}$$

*is $\rho^\dagger = R^\dagger h$, where*

$$R^\dagger h(x) = \frac{|\mathbb{S}^{d-1}|}{2(2\pi)^d}\int_{\mathbb{S}^{d-1}}\int_{\mathbb{R}} e^{\mathrm{i}\omega\langle\theta,\,x\rangle}|\omega|^{d-1}\widehat{h}(\theta,\omega)\mathrm{d}\omega\,\mathrm{d}\sigma(\theta). \tag{11.23}$$

*Thus $R^\dagger h$ is the density of the pseudoinverse reconstruction, which is a priori a signed measure. If $h = R\rho$ is Radon-consistent and $\rho$ is sufficiently regular, then $R^\dagger R\rho = \rho$.*

*Proof.* For the $d$-dimensional Fourier transform, use the compatible convention

$$\widehat{\rho}(\xi) = \int_{\mathbb{R}^d} e^{-\mathrm{i}\langle\xi,\,x\rangle}\rho(x)\mathrm{d}x, \qquad \rho(x) = \frac{1}{(2\pi)^d}\int_{\mathbb{R}^d} e^{\mathrm{i}\langle\xi,\,x\rangle}\widehat{\rho}(\xi)\mathrm{d}\xi.$$

The Fourier-slice theorem gives

$$\widehat{R\rho}(\theta,\omega) = \widehat{\rho}(\omega\theta). \tag{11.24}$$

Applying the one-dimensional Plancherel identity in $t$ rewrites the objective in (11.22), up to the positive factor $1/(2\pi)$, as $\int_{\mathbb{S}^{d-1}}\int_{\mathbb{R}}\left|\widehat{\rho}(\omega\theta)-\widehat{h}(\theta,\omega)\right|^2\mathrm{d}\omega\,\mathrm{d}\sigma(\theta)$. Every $\xi \neq 0$ has the two signed-polar representations $(\theta,\omega) = (\xi/\|\xi\|, \|\xi\|)$ and $(-\xi/\|\xi\|, -\|\xi\|)$. The pointwise least-squares minimizer is therefore

$$\widehat{\rho^\dagger}(\xi) = \frac{1}{2}\left(\widehat{h}\left(\frac{\xi}{\|\xi\|}, \|\xi\|\right) + \widehat{h}\left(-\frac{\xi}{\|\xi\|}, -\|\xi\|\right)\right). \tag{11.25}$$

This also proves uniqueness under the stated existence assumptions. Applying the inverse $d$-dimensional Fourier transform to (11.25) and using signed polar coordinates yields (11.23); the factor $|\mathbb{S}^{d-1}|/2$ accounts for the normalization of $\sigma$ and the two signed representations of each nonzero frequency. Finally, if $h = R\rho$, then (11.24) makes both terms in (11.25) equal to $\widehat{\rho}(\xi)$, proving $R^\dagger R\rho = \rho$. $\square$

$$m_\Omega(\omega) := |\omega|^{d-1}\chi(\omega/\Omega), \qquad R_\Omega^\dagger h(x) := \frac{|\mathbb{S}^{d-1}|}{2(2\pi)^d}\int_{\mathbb{S}^{d-1}}\int_{\mathbb{R}} e^{\mathrm{i}\omega\langle\theta,\,x\rangle}m_\Omega(\omega)\widehat{h}(\theta,\omega)\mathrm{d}\omega\,\mathrm{d}\sigma(\theta), \tag{11.26}$$

where $\chi$ is an even low-pass window with $\chi(0) = 1$. For the two-input path (11.20), choose $\eta_t \geqslant 0$ so that the positive part below has nonzero mass, and define the nonnegative, unit-mass reconstruction

$$A_t(x) := \frac{\left((R_\Omega^\dagger h_t)(x) - \eta_t\right)_+}{\displaystyle\int_{\mathbb{R}^d}\left((R_\Omega^\dagger h_t)(z) - \eta_t\right)_+\mathrm{d}z}. \tag{11.27}$$

For the endpoints, set $A_i = \rho_i$ when $\beta_i = \rho_i\mathrm{d}x$, $i \in \{0,1\}$. The resulting regularized density is generally only a least-squares approximation to the independently averaged slices. With finitely many directions, the sampled Radon operator is also non-injective, so the pseudoinverse reconstruction and the constrained sliced barycenter may differ.

## 11.2 Multimarginal OT

### Definition and basic structure.

**Definition 11.11** (Multimarginal optimal transport)**.** Let $\alpha_s \in \mathcal{P}(\mathcal{X}_s)$ for $s = 1,\dots,S$, and denote by $\mathcal{U}(\alpha_1,\dots,\alpha_S)$ the probability measures whose $s$-th marginal is $\alpha_s$. For a measurable cost $c$ bounded from below, define

$$\mathrm{MMOT}_c(\alpha_1,\dots,\alpha_S) := \inf_{\pi\in\mathcal{U}(\alpha_1,\dots,\alpha_S)}\int_{\mathcal{X}_1\times\cdots\times\mathcal{X}_S} c(x_1,\dots,x_S)\mathrm{d}\pi(x_1,\dots,x_S). \tag{11.28}$$

### Monge structure and splitting-set twist.

**Definition 11.12** (Twist on splitting sets)**.** Fix $x_1 \in \mathcal{X}_1$. A set $M \subset \mathcal{X}_2\times\cdots\times\mathcal{X}_S$ is a $c$-splitting set at $x_1$ if there exist functions $u_s : \mathcal{X}_s \to \mathbb{R}\cup\{-\infty\}$, for $s = 2,\dots,S$, such that $\sum_{s=2}^S u_s(x_s) \leqslant c(x_1,x_2,\dots,x_S)$ for all $(x_2,\dots,x_S)$, with equality on $M$. Assume $c$ is differentiable in $x_1$. The cost is twisted on splitting sets if, for every $x_1$ and every $c$-splitting set $M$ at $x_1$, the map $(x_2,\dots,x_S) \longmapsto \nabla_{x_1}c(x_1,x_2,\dots,x_S)$ is injective on $M$.

**Proposition 11.13** (Multi-marginal Monge structure)**.** *Assume that $\mathcal{X}_s \subset \mathbb{R}^d$, that $c$ is continuous and differentiable with respect to $x_1$, and that $c$ is twisted on splitting sets. Suppose that Kantorovich dual maximizers $(\varphi_s)_{s=1}^S$ exist, that $\alpha_1$ is absolutely continuous, and that $\varphi_1$ is differentiable $\alpha_1$-a.e. Then every optimal plan $\pi^\star \in \mathcal{U}(\alpha_1,\dots,\alpha_S)$ is concentrated on the graph of maps $\pi^\star = (\mathrm{Id}, T_2,\dots,T_S)_\sharp\alpha_1$, $\qquad (T_s)_\sharp\alpha_1 = \alpha_s$. In particular, under these hypotheses the optimizer is unique.*

*Proof.* Let $(\varphi_s)_{s=1}^S$ be optimal dual potentials. Complementary slackness gives a Borel contact set $\Gamma$ of full $\pi^\star$-measure on which $\sum_s \varphi_s(x_s) = c(x_1,\dots,x_S)$. After disintegrating with respect to the first marginal, fix a point $x_1$ where $\varphi_1$ is differentiable and where the conditional plan is concentrated on the fiber $M(x_1) = \{(x_2,\dots,x_S) : (x_1,x_2,\dots,x_S) \in \Gamma\}$. Indeed, set $u_2 = \varphi_2 + \varphi_1(x_1)$ and $u_s = \varphi_s$ for $s \geqslant 3$. Dual feasibility gives $\sum_{s=2}^S u_s(x_s) \leqslant c(x_1,x_2,\dots,x_S)$, and equality holds on $M(x_1)$. If $(x_2,\dots,x_S) \in M(x_1)$, the function $z \longmapsto c(z,x_2,\dots,x_S) - \sum_{s=2}^S \varphi_s(x_s)$ touches $\varphi_1$ from above at $z = x_1$. Differentiating at this contact point gives $\nabla\varphi_1(x_1) = \nabla_{x_1} c(x_1,x_2,\dots,x_S)$. All points in the fiber therefore have the same value of $\nabla_{x_1} c$. Twist on splitting sets makes the fiber a singleton for $\alpha_1$-a.e. $x_1$. Disintegrating $\pi^\star$ with respect to its first marginal gives Dirac conditional measures, hence measurable maps $(T_2,\dots,T_S)$. If $\pi^1$ and $\pi^2$ are two optimal plans, their average is also optimal. The conditional measure of this average over $x_1$ is the average of the two Dirac conditionals, and it must again be a Dirac mass by the preceding argument. Hence the two Dirac masses coincide for $\alpha_1$-a.e. $x_1$, proving uniqueness. ☐

**Coulomb cost and density-functional theory.** $c_{\mathrm{Coul}}(x_1,\dots,x_N) := \sum_{1\leqslant i<j\leqslant N} \frac{1}{\|x_i-x_j\|}$, $\nabla_{x_1} c_{\mathrm{Coul}}(x_1,\dots,x_N) = -\sum_{j=2}^N \frac{x_1-x_j}{\|x_1-x_j\|^3}$. For $N=2$, the map from $x_2$ to this vector is injective, so the ordinary two-marginal twist argument can be recovered under the required existence, duality and differentiability hypotheses. For $N \geqslant 3$, however, the displayed total force does not by itself determine the entire tuple $(x_2,\dots,x_N)$; twist on splitting sets, and hence a Monge representation, is not automatic.

If $\rho$ is an electron density with $\int_{\mathbb{R}^3} \rho(x)\mathrm{d}x = N$ and $\alpha = \rho/N$ is the associated probability density, the strictly-correlated-electrons relaxation of density-functional theory is the equal-marginal problem

$$V_{\mathrm{ee}}^{\mathrm{SCE}}[\rho] := \inf_{\pi\in\mathcal{U}(\alpha,\dots,\alpha)} \int_{(\mathbb{R}^3)^N} c_{\mathrm{Coul}}(x_1,\dots,x_N)\mathrm{d}\pi(x_1,\dots,x_N).$$

$\pi = (\mathrm{Id}, T_2,\dots,T_N)_\sharp\alpha, \qquad (T_i)_\sharp\alpha = \alpha,$

**Proposition 11.14** (Cyclic co-motion plans)**.** *Let $\alpha \in \mathcal{P}(\mathbb{R}^3)$ and let $T : \mathbb{R}^3 \to \mathbb{R}^3$ be measurable with $T_\sharp\alpha = \alpha$ and $T^N = \mathrm{Id}$ $\alpha$-a.e. Set $T^0 = \mathrm{Id}$ and $\pi_T = (T^0, T^1,\dots,T^{N-1})_\sharp\alpha$. Then $\pi_T \in \mathcal{U}(\alpha,\dots,\alpha)$. If $R_\sigma(x_1,\dots,x_N) = (x_{\sigma(1)},\dots,x_{\sigma(N)})$ for $\sigma \in \mathrm{Perm}(N)$, then $\bar\pi_T := \frac{1}{N!}\sum_{\sigma\in\mathrm{Perm}(N)} (R_\sigma)_\sharp\pi_T$ is a symmetric admissible plan and*

$$\int c_{\mathrm{Coul}}\mathrm{d}\bar\pi_T = \int c_{\mathrm{Coul}}\mathrm{d}\pi_T = \int_{\mathbb{R}^3} \sum_{0\leqslant i<j\leqslant N-1} \frac{1}{\|T^i(x) - T^j(x)\|}\mathrm{d}\alpha(x).$$

*Proof.* Since $T_\sharp\alpha = \alpha$, all iterates $T^i$ preserve $\alpha$, so every marginal of $\pi_T$ is $\alpha$. The plan $\bar\pi_T$ is an average of coordinate permutations of $\pi_T$, hence has the same marginals and is invariant under coordinate permutations. The Coulomb cost is symmetric in its arguments, so its integral is unchanged by each $R_\sigma$. The last identity follows by evaluating $c_{\mathrm{Coul}}$ on the graph $(x, T(x),\dots,T^{N-1}(x))$. ☐

Proposition 11.13 explains the general mechanism that can force graph solutions, while Proposition 11.14 records the additional equal-marginal symmetry used by co-motion maps. Thus the DFT problem is both a central application and a warning that multi-marginal OT is richer than a naive deterministic matching problem.

**Multi-marginal formulation of barycenters.** $c_{\mathrm{bar}}(x_1,\dots,x_S) = \min_{x\in\mathbb{R}^d} \sum_{s=1}^S \lambda_s\|x - x_s\|^2$.

**Proposition 11.15** (Multi-marginal formula for quadratic barycenters)**.** *Let $\beta_1,\dots,\beta_S \in \mathcal{P}_2(\mathbb{R}^d)$ and $\lambda \in \Sigma_S$. Define $B(x_1,\dots,x_S) = \sum_{s=1}^S \lambda_s x_s$, $\quad c_{\mathrm{bar}}(x_1,\dots,x_S) = \min_x \sum_s \lambda_s\|x - x_s\|^2$. If $\pi^\star$ solves the multi-marginal OT problem with marginals $(\beta_s)_s$ and cost $c_{\mathrm{bar}}$, then $\alpha^\star = B_\sharp\pi^\star$ is a Wasserstein barycenter. Conversely, every barycenter is obtained this way from an optimal multi-marginal plan.*

*Proof.* For any candidate barycenter $\alpha$ and couplings $\pi_s \in \mathcal{U}(\alpha,\beta_s)$, glue the couplings along their common $\alpha$ marginal to obtain a joint law of $(X, Y_1,\dots,Y_S)$. Conditioning on $(Y_s)_s$ and minimizing over $X$ gives $\sum_s \lambda_s\mathbb{E}\|X - Y_s\|^2 \geqslant \mathbb{E}\min_x \sum_s \lambda_s\|x - Y_s\|^2 = \mathbb{E}c_{\mathrm{bar}}(Y_1,\dots,Y_S)$. Taking the infimum over the couplings gives that the barycenter value is at least the multi-marginal value. Conversely, from an optimal multi-marginal plan $\pi^\star$, set $X = B(Y_1,\dots,Y_S)$. The couplings between $X$ and each $Y_s$ are feasible for the barycenter problem and attain exactly the multi-marginal cost, proving equality and the formula. If $\alpha^\star$ is any barycenter, choose optimal couplings between $\alpha^\star$ and each $\beta_s$ and glue them along the common $\alpha^\star$ marginal. Since the barycenter and multi-marginal values are equal, the conditional minimization inequality above must be an equality. Thus $X = B(Y_1,\dots,Y_S)$ almost surely for the induced optimal multi-marginal plan, and $\alpha^\star = B_\sharp\pi^\star$. ☐

**Corollary 11.16** (Sparse discrete barycenters)**.** *Let $\beta_s = \sum_{i_s=1}^{n_s} b_{s,i_s}\delta_{x_{s,i_s}}$, $\qquad s = 1,\dots,S$, be discrete input measures. There exists a quadratic Wasserstein barycenter $\alpha^\star$ supported on weighted averages $\sum_s \lambda_s x_{s,i_s}$ of one support point from each input and satisfying $\#\,\mathrm{supp}(\alpha^\star) \leqslant \sum_{s=1}^S n_s - S + 1$.*

*Proof.* Write a discrete multi-marginal plan as a nonnegative tensor $\mathrm{P} = (\mathrm{P}_{i_1,\dots,i_S})$ and let $\mathcal{A}_{\mathrm{marg}}$ collect its $S$ marginals: $(\mathcal{A}_{\mathrm{marg}}\mathrm{P})_{s,i_s} = \sum_{(i_r)_{r\neq s}} \mathrm{P}_{i_1,\dots,i_S}$. After vectorizing $\mathrm{P}$, Proposition 3.6 applies to the multi-marginal linear program. Its constraint operator has rank $\mathrm{rank}(\mathcal{A}_{\mathrm{marg}}) = \sum_{s=1}^S n_s - S + 1$. Indeed, a family of vectors $u_s \in \mathbb{R}^{n_s}$ belongs to the annihilator of its image precisely when $\sum_{s=1}^S u_{s,i_s} = 0 \qquad$ for every $(i_1,\dots,i_S)$. Varying one index at a time shows that every $u_s$ is constant, say $u_s = c_s\mathbb{1}_{n_s}$, and the remaining condition is $\sum_s c_s = 0$. The annihilator therefore has dimension $S-1$, which proves the rank formula.

The rank-controlled sparsity result therefore supplies an optimal multi-marginal tensor $\mathrm{P}^\star$ with at most $\sum_s n_s - S + 1$ positive entries. Proposition 11.15 gives the barycenter $\alpha^\star = B_\sharp\mathrm{P}^\star$. Each positive tensor entry produces at most one atom under $B$, and collisions can only reduce the support size. This proves the claim. ☐

**Entropic regularization of multi-marginal OT.** $\inf_{\pi\in\mathcal{U}(\alpha_1,\dots,\alpha_S)} \int c\mathrm{d}\pi + \varepsilon\,\mathrm{KL}(\pi|\alpha_1\otimes\cdots\otimes\alpha_S)$.

$$\mathrm{d}\pi^\star(x_1,\dots,x_S) = \exp\left(\frac{\sum_s f_s(x_s) - c(x_1,\dots,x_S)}{\varepsilon}\right)\prod_s \mathrm{d}\alpha_s(x_s),$$

**Treewidth and graphical structure.** The important exception is when the cost factors over a sparse interaction graph. Let $G = (V, E)$ be a finite undirected graph with $V = \{1, \dots, S\}$ and suppose, for simplicity, that

$$c(x_1, \dots, x_S) = \sum_{(r,s)\in E} c_{r,s}(x_r, x_s). \tag{11.29}$$

**Definition 11.17** (Tree decomposition and treewidth)**.** A *tree decomposition* of a finite graph $G = (V, E)$ is a tree $\mathcal{T} = (Q, F)$ together with bags $B_q \subseteq V$, $q \in Q$, such that
1. $\bigcup_{q\in Q} B_q = V$;
2. for every edge $(r, s) \in E$, some bag contains both $r$ and $s$;
3. for every vertex $s \in V$, the nodes $\{q \in Q\ ;\ s \in B_q\}$ form a connected subtree of $\mathcal{T}$.

The width of this decomposition is $\max_{q\in Q} |B_q| - 1$, and the *treewidth* $\mathrm{tw}(G)$ is the minimum width over all tree decompositions.

**Junction-tree contractions inside Sinkhorn.**

$$\mathrm{K}^{r,s}_{i_r,i_s} = \exp\left(-\frac{\mathrm{C}^{r,s}_{i_r,i_s}}{\varepsilon}\right), \qquad h_s(i_s) = (\mathrm{a}_s)_{i_s}(u_s)_{i_s}.$$

$$\mathrm{P}_{i_1,\dots,i_S} = \prod_{s\in V} h_s(i_s) \prod_{(r,s)\in E} \mathrm{K}^{r,s}_{i_r,i_s}. \tag{11.30}$$

When $G$ is a tree, let $N_G(r)$ denote the neighbors of $r$. The directed sum-product messages satisfy

$$m_{r\to s}(i_s) = \sum_{i_r=1}^{n_r} \mathrm{K}^{r,s}_{i_r,i_s} h_r(i_r) \prod_{q\in N_G(r)\setminus\{s\}} m_{q\to r}(i_r). \tag{11.31}$$

$$(\widehat{\mathrm{a}}_s)_{i_s} = h_s(i_s) \prod_{r\in N_G(s)} m_{r\to s}(i_s). \tag{11.32}$$

For the block $s$ currently selected by the cyclic scaling algorithm, the exact coordinate update is then

$$u_s \leftarrow u_s \odot \frac{\mathrm{a}_s}{\widehat{\mathrm{a}}_s},$$

with componentwise products and quotients. This changes only the unary factor $h_s$.
For a general tree decomposition, choose one host bag for each unary factor, assign each edge factor in (11.30) to a bag containing both endpoints, and denote the product assigned to bag $B_q$ by $\psi_q(i_{B_q})$. For adjacent bags $q, q' \in Q$, set $\mathcal{S}_{q,q'} = B_q \cap B_{q'}$. Junction-tree messages take the form

$$M_{q\to q'}(i_{\mathcal{S}_{q,q'}}) = \sum_{i_{B_q\setminus\mathcal{S}_{q,q'}}} \psi_q(i_{B_q}) \prod_{\ell\in N_{\mathcal{T}}(q)\setminus\{q'\}} M_{\ell\to q}(i_{\mathcal{S}_{\ell,q}}). \tag{11.33}$$

$$\mathfrak{b}_q(i_{B_q}) = \psi_q(i_{B_q}) \prod_{\ell\in N_{\mathcal{T}}(q)} M_{\ell\to q}(i_{\mathcal{S}_{\ell,q}}). \tag{11.34}$$

For any $s \in B_q$, summing $\mathfrak{b}_q$ over $B_q \setminus \{s\}$ gives $\widehat{\mathrm{a}}_s$; the running-intersection property ensures that every bag containing $s$ gives the same result.
Redundant adjacent bags may be contracted, so one can assume that the decomposition is reduced. If its width is $w$, every bag contains at most $w + 1$ variables and every separator contains at most $w$.

$$O\big(|Q|\, n^{w+1}\big) \quad \text{time}, \qquad O\big(|Q|\, n^{w}\big) \quad \text{message memory.}$$

$$O\left(\sum_{q\in Q} \big(1 + \deg_{\mathcal{T}}(q)\big) \prod_{s\in B_q} n_s\right),$$

$$O\left(\sum_{(q,q')\in F} \prod_{s\in\mathcal{S}_{q,q'}} n_s\right)$$

## 11.3 Low-Rank Optimal Transport

**Definition 11.18** (Low-rank factored couplings)**.** Let $\mathrm{a} \in \Sigma_n$, $\mathrm{b} \in \Sigma_m$ and let $r \geqslant 1$. A rank-$r$ factored coupling is a triple $(\mathrm{Q}, \mathrm{R}, g)$ such that $\mathrm{Q} \in \mathbb{R}_+^{n\times r}$, $\mathrm{R} \in \mathbb{R}_+^{m\times r}$, $g \in \Sigma_r$, with $\mathrm{Q}\mathbb{1}_r = \mathrm{a}$, $\mathrm{R}\mathbb{1}_r = \mathrm{b}$, $\mathrm{Q}^\top\mathbb{1}_n = \mathrm{R}^\top\mathbb{1}_m = g$. It induces the coupling

$$\mathrm{P}(\mathrm{Q}, \mathrm{R}, g) = \mathrm{Q}\,\mathrm{diag}(g)^{-1}\mathrm{R}^\top, \qquad \mathrm{P}_{i,j} = \sum_{k=1}^{r} \frac{\mathrm{Q}_{i,k}\mathrm{R}_{j,k}}{g_k}, \tag{11.35}$$

where columns with $g_k = 0$ are discarded before applying the formula.

**Proposition 11.19** (Factored couplings and nonnegative rank)**.** *For every feasible triple* $(\mathrm{Q}, \mathrm{R}, g)$ *in Definition 11.18, the matrix* $\mathrm{P}(\mathrm{Q}, \mathrm{R}, g)$ *belongs to* $\mathrm{U}(\mathrm{a}, \mathrm{b})$ *and has nonnegative rank at most* $r$*. Conversely, every coupling* $\mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$ *with nonnegative rank at most* $r$ *admits a representation of the form* (11.35)*, after possibly deleting zero latent components.*

*Proof.* The marginal constraints follow by direct summation: $\mathrm{P}\mathbb{1}_m = \mathrm{Q}\,\mathrm{diag}(g)^{-1}\mathrm{R}^\top\mathbb{1}_m = \mathrm{Q}\,\mathrm{diag}(g)^{-1}g = \mathrm{Q}\mathbb{1}_r = \mathrm{a}$, and similarly $\mathrm{P}^\top\mathbb{1}_n = \mathrm{b}$. Since P is a sum of $r$ nonnegative rank-one matrices $\mathrm{Q}_{:,k}\mathrm{R}_{:,k}^\top/g_k$, its nonnegative rank is at most $r$.
Conversely, suppose $\mathrm{P} = UV^\top$ with $U \in \mathbb{R}_+^{n\times r}$ and $V \in \mathbb{R}_+^{m\times r}$. Set $q_k = \sum_i U_{i,k}$ and $s_k = \sum_j V_{j,k}$. Components with $q_k s_k = 0$ do not contribute and can be removed. Define $g_k = q_k s_k$, $\mathrm{Q}_{i,k} = U_{i,k}s_k$ and $\mathrm{R}_{j,k} = V_{j,k}q_k$. Since P has total mass one, $g \in \Sigma_r$. Moreover $\mathrm{Q}\mathbb{1}_r = \mathrm{P}\mathbb{1}_m = \mathrm{a}$, $\mathrm{R}\mathbb{1}_r = \mathrm{P}^\top\mathbb{1}_n = \mathrm{b}$, and both column marginals are equal to $g$. Finally, $\sum_k \frac{\mathrm{Q}_{i,k}\mathrm{R}_{j,k}}{g_k} = \sum_k \frac{U_{i,k}s_kV_{j,k}q_k}{q_ks_k} = \mathrm{P}_{i,j}$. □

For a cost matrix $\mathrm{C} \in \mathbb{R}^{n\times m}$, the low-rank constrained OT value is therefore
$\min_{(\mathrm{Q},\mathrm{R},g)}\langle \mathrm{C}, \mathrm{Q}\,\mathrm{diag}(g)^{-1}\mathrm{R}^\top\rangle$ subject to Definition 11.18.

$$\min_{\substack{\mathrm{Q}\mathbb{1}_r=\mathrm{a},\ \mathrm{Q}^\top\mathbb{1}_n=g\\ \mathrm{R}\mathbb{1}_r=\mathrm{b},\ \mathrm{R}^\top\mathbb{1}_m=g}} \langle \mathrm{C}, \mathrm{Q}\,\mathrm{diag}(g)^{-1}\mathrm{R}^\top\rangle + \varepsilon\mathrm{KL}(\mathrm{Q}|\mathrm{a}\otimes g) + \varepsilon\mathrm{KL}(\mathrm{R}|\mathrm{b}\otimes g), \tag{11.36}$$

where Q, R are nonnegative. For fixed $g$, this differs only by constants from adding $-\varepsilon(\mathrm{H}(\mathrm{Q})+\mathrm{H}(\mathrm{R}))$ to the factorized transport cost.

## 11.4 Capacity-Constrained Optimal Transport

**Definition 11.20** (Capacity-constrained optimal transport)**.** Let $\alpha \in \mathcal{M}_+^1(\mathcal{X})$, $\beta \in \mathcal{M}_+^1(\mathcal{Y})$, let $c : \mathcal{X}\times\mathcal{Y} \to [0,+\infty]$ be measurable, and let $u : \mathcal{X}\times\mathcal{Y} \to [0,+\infty]$ be a measurable capacity. Define

$$\mathcal{L}_c^u(\alpha,\beta) := \inf_{\pi\in\mathcal{U}(\alpha,\beta)} \left\{ \int_{\mathcal{X}\times\mathcal{Y}} c(x,y)\mathrm{d}\pi(x,y)\ ;\ \pi \ll \alpha\otimes\beta,\quad \frac{\mathrm{d}\pi}{\mathrm{d}(\alpha\otimes\beta)}(x,y) \leqslant u(x,y) \quad (\alpha\otimes\beta)\text{-a.e.} \right\}. \tag{11.37}$$

By convention, $u \equiv +\infty$ removes both the density bound and the absolute-continuity requirement, so that $\mathcal{L}_c^{+\infty}(\alpha,\beta) = \mathcal{L}_c(\alpha,\beta)$.

The product coupling $\alpha\otimes\beta$ is feasible whenever $u \geqslant 1$. Thus the constraint is not meant to promote the product plan; rather, it prevents the optimizer from using any pair more than the prescribed density ratio.
For discrete measures $\alpha = \sum_i a_i\delta_{x_i}$ and $\beta = \sum_j b_j\delta_{y_j}$, a capacity is an upper matrix $\overline{\mathrm{P}} \in \mathbb{R}_+^{n\times m}$. Given a cost matrix C, the density-ratio discretization of Definition 11.20, with $\overline{\mathrm{P}}_{i,j} = u_{i,j}a_ib_j$, is the capped linear program

$$\min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle \mathrm{C}, \mathrm{P}\rangle \quad \text{subject to} \quad 0 \leqslant \mathrm{P}_{i,j} \leqslant \overline{\mathrm{P}}_{i,j} \quad \forall (i,j). \tag{11.38}$$

**Proposition 11.21** (Feasibility of a capped transport polytope)**.** *The constraints in* (11.38) *are feasible if and only if*

$$a(I) + b(J) - 1 \leqslant \overline{\mathrm{P}}(I,J) \qquad \text{for every } I \subset \{1,\dots,n\},\ J \subset \{1,\dots,m\}. \tag{11.39}$$

*Proof.* If P is feasible, then $\mathrm{P}(I,J) = a(I) - \mathrm{P}(I,J^c) \geqslant a(I) - b(J^c) = a(I)+b(J)-1$. Since $\mathrm{P}(I,J) \leqslant \overline{\mathrm{P}}(I,J)$, condition (11.39) is necessary.
For sufficiency, build a flow network with capacity $a_i$ from the source to row vertex $i$, capacity $\overline{\mathrm{P}}_{i,j}$ from row $i$ to column $j$, and capacity $b_j$ from column $j$ to the sink. A cut containing row set $I$ and column set $J^c$ has capacity $1 - a(I) + \overline{\mathrm{P}}(I,J) + 1 - b(J)$. Condition (11.39) says that every such cut has capacity at least one. The max-flow/min-cut theorem therefore gives a unit flow, and its row-to-column edge values form the required matrix P. □

$$\min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b}),\,0\leqslant\mathrm{P}\leqslant\overline{\mathrm{P}}} \langle \mathrm{C}, \mathrm{P}\rangle + \varepsilon\mathrm{KL}(\mathrm{P}|\mathrm{a}\otimes\mathrm{b}). \tag{11.40}$$

On the active support, all iterates and correction factors are positive, so every division in the KL–Dykstra iteration is well defined. KL-Dykstra convergence on finite-dimensional affine and box constraints then shows that, when the capped polytope is nonempty, the iterates converge to the unique minimizer of (11.40).

## 11.5 Metric Learning and Inverse OT

### *11.5.1 Differentiating OT losses*

**First variations of OT values.**

**Proposition 11.22** (First variations of OT values)**.** *Let $\alpha$ and $\beta$ be probability measures with compact supports $\mathcal{X} = \mathrm{supp}(\alpha)$ and $\mathcal{Y} = \mathrm{supp}(\beta)$, let $c \in \mathcal{C}(\mathcal{X}\times\mathcal{Y})$, and let $\varepsilon \geqslant 0$. Use the convention $\mathcal{L}_c^0 = \mathcal{L}_c$, where the unregularized and positive-temperature values are defined in Definitions 3.23 and 7.10. Denote by $\mathcal{O}_c^\varepsilon(\alpha,\beta)$ and $\mathcal{D}_c^\varepsilon(\alpha,\beta)$ the sets of primal and dual optimizers of the corresponding problem. If $\chi$ is a signed measure with $\chi(\mathcal{X}) = 0$ and $\alpha_t = \alpha + t\chi$ is a probability measure for $0 \leqslant t \leqslant t_0$, then*

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0^+} \mathcal{L}_c^\varepsilon(\alpha_t,\beta) = \sup_{(f,g)\in\mathcal{D}_c^\varepsilon(\alpha,\beta)} \int_{\mathcal{X}} f(x)\mathrm{d}\chi(x).$$

*If $h \in \mathcal{C}(\mathcal{X}\times\mathcal{Y})$ and $c_t = c + th$, then*

$$\left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0^+} \mathcal{L}_{c_t}^\varepsilon(\alpha,\beta) = \inf_{\pi\in\mathcal{O}_c^\varepsilon(\alpha,\beta)} \int_{\mathcal{X}\times\mathcal{Y}} h(x,y)\mathrm{d}\pi(x,y).$$

*For fixed $(\alpha,\beta)$, a finite signed Radon measure $\eta$ on $\mathcal{X}\times\mathcal{Y}$ represents the first variation of the functional $c \mapsto \mathcal{L}_c^\varepsilon(\alpha,\beta)$ if, for every $h \in \mathcal{C}(\mathcal{X}\times\mathcal{Y})$, $\mathcal{L}_{c+th}^\varepsilon(\alpha,\beta) = \mathcal{L}_c^\varepsilon(\alpha,\beta) + t\int_{\mathcal{X}\times\mathcal{Y}} h\mathrm{d}\eta + o(t) \qquad (t \to 0)$. In particular, if the normalized optimal potential $f_\varepsilon^\star$ and the optimal plan $\pi_\varepsilon^\star$ are unique (so in particular when $\varepsilon > 0$), then, with the marginal first variation understood as in Definition 14.2,*

$$\frac{\delta\mathcal{L}_c^\varepsilon}{\delta\alpha}(\alpha,\beta) = f_\varepsilon^\star, \qquad \frac{\delta\mathcal{L}_c^\varepsilon}{\delta c}(\alpha,\beta) = \pi_\varepsilon^\star.$$

*Proof.* For $\varepsilon = 0$, the Kantorovich duality theorem is stated in Proposition 4.5. Its dual value is a supremum of affine functions of $\alpha$, so Danskin's theorem gives the first formula as the supremum over active dual potentials. The condition $\chi(\mathcal{X}) = 0$ makes this expression independent of the additive gauge $(f, g) \mapsto (f + \lambda, g - \lambda)$.
For $\varepsilon > 0$, use the continuous entropic duality and primal–dual density law of Proposition 7.18. At an optimal pair, the marginal constraint makes the density in (7.23) integrate to one with respect to $\beta$ for every $x$ on the source support. Therefore the derivative of the dual objective (7.22) with respect to $\alpha$ at fixed optimal potentials is $\int f_\varepsilon^\star \mathrm{d}\chi$: the derivative of its exponential term vanishes after the row normalization. Danskin's theorem then gives the same first formula, now with a singleton set of normalized active potentials.
For every $\varepsilon \geqslant 0$, the primal objective depends on $c$ only through the affine term $\int c\mathrm{d}\pi$. The minimum form of Danskin's theorem therefore gives the second formula as the infimum of $\int h\mathrm{d}\pi$ over active primal optimizers. When the normalized potential or plan is unique, the corresponding directional derivative is linear in the perturbation, which gives the two first variations. $\square$

$f^\star \in \partial_{\mathrm{a}} \mathrm{L}_{\mathrm{C}}(\mathrm{a}, \mathrm{b}), \qquad \mathrm{P}^\star \in \partial_{\mathrm{C}}^{\sup} \mathrm{L}_{\mathrm{C}}(\mathrm{a}, \mathrm{b})$. For $\varepsilon > 0$, these are ordinary derivatives on the relative interiors of finite-dimensional simplices. For a finite-dimensional parametrization $c_\theta$, the entropic formula gives the backpropagation rule
$\partial_\theta \mathcal{L}^\varepsilon_{c_\theta}(\alpha, \beta) = \int \partial_\theta c_\theta(x, y) \mathrm{d}\pi^\star_{\theta,\varepsilon}(x, y)$, where $\pi^\star_{\theta,\varepsilon}$ is the entropic optimizer between $(\alpha, \beta)$ for the cost $c_\theta$. For every perturbation $H \in \mathbb{R}^{d\times d}$,

$$D_A \mathcal{L}^\varepsilon_{c_A}(\alpha, \beta)[H] = \int \langle Hx,\, y\rangle \mathrm{d}\pi^\star_{A,\varepsilon}(x, y) = \left\langle H,\, \int yx^\top \mathrm{d}\pi^\star_{A,\varepsilon}(x, y) \right\rangle_{\mathrm{F}}.$$

$$\nabla_A \mathcal{L}^\varepsilon_{c_A}(\alpha, \beta) = \int yx^\top \mathrm{d}\pi^\star_{A,\varepsilon}(x, y) = \mathrm{Cov}_{\pi^\star_{A,\varepsilon}}(Y, X) + m_\beta m_\alpha^\top,$$

where $(X, Y) \sim \pi^\star_{A,\varepsilon}$, $m_\alpha = \int x\mathrm{d}\alpha(x)$ and $m_\beta = \int y\mathrm{d}\beta(y)$. Thus the gradient is the raw cross-moment of the optimal coupling, and it is its cross-covariance when both marginals are centered.

*11.5.2 Inverse Optimal Transport*

**Definition 11.23** (Regularized inverse-OT loss)**.** Let $\varepsilon \geqslant 0$ and $\widehat{\pi} \in \mathcal{U}(\alpha, \beta)$, with $\mathrm{KL}(\widehat{\pi}|\alpha \otimes \beta) < +\infty$ when $\varepsilon > 0$. The regularized inverse-OT loss is

$$\mathcal{F}_\varepsilon(c \mid \widehat{\pi}) := \int c\mathrm{d}\widehat{\pi} + \varepsilon\, \mathrm{KL}(\widehat{\pi}|\alpha \otimes \beta) - \mathcal{L}^\varepsilon_c(\alpha, \beta), \tag{11.41}$$

where the entropy term is set to zero when $\varepsilon = 0$.

As shown next, this loss is convex in $c$. Minimizing it over a convex class of costs is therefore a convex problem that avoids non-convex bilevel optimization through a forward OT solver.

**Proposition 11.24** (Convexity and calibration of the inverse-OT loss)**.** *For fixed $(\alpha, \beta, \widehat{\pi})$, the map $c \mapsto \mathcal{F}_\varepsilon(c \mid \widehat{\pi})$ is convex and nonnegative. Moreover, $\mathcal{F}_\varepsilon(c \mid \widehat{\pi}) = 0 \iff \widehat{\pi}$ solves (7.12) for the cost $c$. When $\varepsilon > 0$, the regularized optimizer is unique whenever the value is finite, so the zero set identifies the forward coupling for a fixed cost.*

*Proof.* The first two terms in (11.41) are respectively linear and constant in $c$, whereas $c \mapsto \mathcal{L}^\varepsilon_c(\alpha, \beta)$ is concave because it is the infimum of affine functions of $c$, as in Proposition 3.24. Testing (7.12) with $\widehat{\pi}$ proves nonnegativity, and equality holds precisely when this test plan attains the infimum. Uniqueness for $\varepsilon > 0$ follows from the strict convexity of relative entropy on finite-entropy couplings. $\square$

**Bilinear cost learning.** For concreteness, consider the bilinear model on $\mathbb{R}^d$,
$c_A(x, y) = \langle Ax,\, y\rangle, \qquad A \in \mathbb{R}^{d\times d}$.

$$A^\star \in \operatorname*{argmin}_{A\in\mathcal{A}} \mathcal{F}_\varepsilon(c_A \mid \widehat{\pi}). \tag{11.42}$$

**Finite-sample and population geometry.** Let $A_0 = -\mathrm{Id}$, let $\pi_0$ be the associated quadratic $\mathcal{W}_2$ plan, and set $\widehat{\pi}_n = n^{-1} \sum_{i=1}^n \delta_{(X_i, Y_i)}$ for i.i.d. $(X_i, Y_i) \sim \pi_0$. Along any affine matrix path $A_t$, the empirical loss $t \mapsto \mathcal{F}_0(c_{A_t} \mid \widehat{\pi}_n)$ is convex and polyhedral. Under the smoothness, invertibility and Hessian-spanning assumptions of Peyré, Poon and Tron (2026), its population limit instead has local quadratic curvature transverse to the positive-scaling ray of $A_0$.

**Statistical estimation.** $\|A_n^\star - A_0\|_{\mathrm{F}} = O_{\mathbb{P}}(n^{-1/2})$

## 11.6 Weak Optimal Transport

**Barycentric projection of a coupling.**

**Definition 11.25** (Barycentric projection of a coupling)**.** Let $\alpha, \beta \in \mathcal{P}_1(\mathbb{R}^d)$ and let $\pi \in \mathcal{U}(\alpha, \beta)$. Disintegrate $\pi$ with respect to its first marginal as $\pi(\mathrm{d}x, \mathrm{d}y) = \pi_x(\mathrm{d}y)\alpha(\mathrm{d}x)$. Since $\beta$ has finite first moment, the conditional mean is finite for $\alpha$-a.e. $x$, and the barycentric projection of $\pi$ is the map

$$\bar{T}_\pi(x) := \int_{\mathbb{R}^d} y\mathrm{d}\pi_x(y), \qquad \bar{\beta}_\pi := (\bar{T}_\pi)_\sharp \alpha. \tag{11.43}$$

For discrete measures $\alpha = \sum_{i=1}^n a_i \delta_{x_i}$ and $\beta = \sum_{j=1}^m b_j \delta_{y_j}$, write $\pi = \sum_{i,j} \mathrm{P}_{i,j} \delta_{(x_i, y_j)}$ with $\mathrm{P} \in \mathrm{U}(\mathrm{a}, \mathrm{b})$. Then $\pi_{x_i} = \sum_{j=1}^m \frac{\mathrm{P}_{i,j}}{a_i} \delta_{y_j}$ and $\bar{T}_\pi(x_i) = \frac{1}{a_i} \sum_{j=1}^m \mathrm{P}_{i,j} y_j$.

**Proposition 11.26** (Barycentric projection of a quadratic optimal plan)**.** *Let $\pi \in \mathcal{U}(\alpha, \beta)$ be optimal for the quadratic cost $\|x - y\|^2$ between $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$, and define $\bar{T}_\pi$ and $\bar{\beta}_\pi$ by (11.43). Then $(\mathrm{Id}, \bar{T}_\pi)_\sharp \alpha$ is an optimal coupling between $\alpha$ and $\bar{\beta}_\pi$. Equivalently, $\bar{T}_\pi$ is a quadratic optimal transport map from $\alpha$ to the projected target $\bar{\beta}_\pi$.*

*Proof.* By Theorem 3.35, $\pi$ is concentrated on a $c$-cyclically monotone set $\Gamma$ for $c(x, y) = \|x - y\|^2$. For the quadratic cost, and since it is enough to check cyclic permutations, this means that every finite cycle $(x_i, y_i)_{i=1}^m \subset \Gamma$ satisfies $\sum_{i=1}^m \langle x_i,\, y_i\rangle \geqslant \sum_{i=1}^m \langle x_i,\, y_{i+1}\rangle, \qquad y_{m+1} = y_1$. After changing the disintegration on an $\alpha$-negligible set, $\pi_x$ is supported on the section $\Gamma_x = \{y\,;\, (x, y) \in \Gamma\}$ for $\alpha$-a.e. $x$. Choose $x_1, \ldots, x_m$ in this full-measure set and independently sample $Y_i \sim \pi_{x_i}$. Applying the cyclic inequality to $(x_i, Y_i)$ and taking expectations gives $\sum_{i=1}^m \langle x_i,\, \bar{T}_\pi(x_i)\rangle \geqslant \sum_{i=1}^m \langle x_i,\, \bar{T}_\pi(x_{i+1})\rangle$. Thus $(\mathrm{Id}, \bar{T}_\pi)_\sharp \alpha$ is concentrated on a cyclically monotone graph. By the cyclic-monotonicity characterization of quadratic optimality, this plan is optimal between its two marginals, namely $\alpha$ and $\bar{\beta}_\pi$. $\square$

**Weak transport costs.**

**Definition 11.27** (Weak optimal transport)**.** Let $\mathcal{X}$ and $\mathcal{Y}$ be Polish spaces, let $\alpha \in \mathcal{P}(\mathcal{X})$ and $\beta \in \mathcal{P}(\mathcal{Y})$, and let $C : \mathcal{X} \times \mathcal{P}(\mathcal{Y}) \to \mathbb{R} \cup \{+\infty\}$ be measurable. For each $\pi \in \mathcal{U}(\alpha, \beta)$, let $\pi(\mathrm{d}x, \mathrm{d}y) = \alpha(\mathrm{d}x)\pi_x(\mathrm{d}y)$ be a disintegration. Then

$$\mathrm{WOT}_C(\alpha, \beta) := \inf_{\pi \in \mathcal{U}(\alpha,\beta)} \int C(x, \pi_x)\mathrm{d}\alpha(x). \tag{11.44}$$

If $C(x, \cdot)$ is convex for $\alpha$-a.e. $x$, then the objective in (11.44) is convex in $\pi$: the conditional law of $(1-t)\pi^0 + t\pi^1$ is $(1-t)\pi^0_x + t\pi^1_x$. Since $\mathcal{U}(\alpha, \beta)$ is convex, weak OT is then a convex optimization problem; without this hypothesis it need not be.

**Proposition 11.28** (Weak Kantorovich duality)**.** *Let $\mathcal{X}$ and $\mathcal{Y}$ be compact metric spaces, and let $C : \mathcal{X} \times \mathcal{P}(\mathcal{Y}) \to \mathbb{R} \cup \{+\infty\}$ be proper, jointly lower semicontinuous, bounded from below and convex in its second argument. Fix $\alpha \in \mathcal{P}(\mathcal{X})$ and $\beta \in \mathcal{P}(\mathcal{Y})$, and assume that $\mathrm{WOT}_C(\alpha, \beta) < +\infty$. For $g \in \mathcal{C}(\mathcal{Y})$, define the weak $C$-transform*

$$g^C(x) := \inf_{\nu \in \mathcal{P}(\mathcal{Y})} \left\{ C(x, \nu) - \int g(y)\mathrm{d}\nu(y) \right\}.$$

*Then*

$$\mathrm{WOT}_C(\alpha, \beta) = \sup_{g \in \mathcal{C}(\mathcal{Y})} \left\{ \int g^C(x)\mathrm{d}\alpha(x) + \int g(y)\mathrm{d}\beta(y) \right\}.$$

*When $C(x, \nu) = \int c(x, y)\mathrm{d}\nu(y)$, this reduces to the usual Kantorovich dual with $g^C(x) = \inf_y(c(x, y) - g(y))$.*

*Proof.* For any coupling $\pi$ and any $g \in \mathcal{C}(\mathcal{Y})$, the definition of $g^C$ gives $C(x, \pi_x) \geqslant g^C(x) + \int g(y)\mathrm{d}\pi_x(y)$. After integration with respect to $\alpha$, the second term becomes $\int g\mathrm{d}\beta$ because the second marginal of $\pi$ is $\beta$. This proves weak duality.
For the converse, lift a kernel $x \mapsto \pi_x$ to the measure $\alpha(\mathrm{d}x)\delta_{\pi_x}(\mathrm{d}\nu)$ on $\mathcal{X} \times \mathcal{P}(\mathcal{Y})$. Relaxing this graph constraint gives probability measures $P$ whose first marginal is $\alpha$ and whose intensity satisfies $\int_{\mathcal{X} \times \mathcal{P}(\mathcal{Y})} \nu \, \mathrm{d}P(x, \nu) = \beta$. This relaxation does not change the value: disintegrating $P(\mathrm{d}x, \mathrm{d}\nu) = P_x(\mathrm{d}\nu)\alpha(\mathrm{d}x)$ and replacing $P_x$ by its barycenter $\bar{\nu}_x = \int \nu \, \mathrm{d}P_x(\nu)$ preserves the intensity constraint and cannot increase the objective, by convexity of $C(x, \cdot)$. The relaxed feasible set is compact, and the integral functional is lower semicontinuous. Fenchel–Rockafellar duality for the affine intensity constraint therefore has no gap; its continuous multiplier is $g \in \mathcal{C}(\mathcal{Y})$. Minimization in $\nu$ gives $g^C(x)$, while the constraint contributes $\int g\mathrm{d}\beta$. This is the weak-cost analogue of Proposition 4.2; see for the Polish-space formulation. $\square$

The dual objective is concave in $g$, because $g^C(x)$ is a pointwise infimum of affine functions of $g$; it is generally non-smooth. The most common weak transport model on $\mathbb{R}^d$ retains only the conditional barycenter.

**Proposition 11.29** (Barycentric weak transport is weaker than $\mathcal{W}_2$)**.** *Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ and define $C_{\mathrm{bar}}(x, \nu) = \|x - \int y\mathrm{d}\nu(y)\|^2$. Equivalently,*

$$\mathrm{WOT}_{C_{\mathrm{bar}}}(\alpha, \beta) = \inf_{\pi \in \mathcal{U}(\alpha,\beta)} \int \|x - \bar{T}_\pi(x)\|^2\mathrm{d}\alpha(x). \tag{11.45}$$

*Then $\mathrm{WOT}_{C_{\mathrm{bar}}}(\alpha, \beta) \leqslant \mathcal{W}_2^2(\alpha, \beta)$.*

*Proof.* Let $\pi$ be any coupling and disintegrate it as $\pi_x\alpha$. By Jensen's inequality, $\|x - \bar{T}_\pi(x)\|^2 \leqslant \int \|x - y\|^2\mathrm{d}\pi_x(y)$. Integrating in $x$ gives $\int C_{\mathrm{bar}}(x, \pi_x)\mathrm{d}\alpha(x) \leqslant \int \|x - y\|^2\mathrm{d}\pi(x, y)$. Taking the infimum over $\pi$ proves the claim. $\square$

For $\alpha = \sum_i a_i\delta_{x_i}$ and $\beta = \sum_j b_j\delta_{y_j}$, this is the convex quadratic program

$$\min_{\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})} \sum_i a_i \|x_i - \frac{1}{a_i} \sum_j \mathrm{P}_{i,j} y_j\|^2. \tag{11.46}$$

Unlike the generally non-convex quadratic GW problem of Section 12.3, every row barycenter here is affine in P. Adding $\varepsilon\,\mathrm{KL}(\pi|\alpha \otimes \beta)$ gives a convex entropic problem. If $F_{\mathrm{bar}}$ denotes the objective in (11.45), its linearized cost, modulo a source-only gauge, is

$$c^{\mathrm{bar}}_\pi(x, y) := 2\langle x - \bar{T}_\pi(x),\, x - y\rangle, \qquad \left.\frac{\mathrm{d}}{\mathrm{d}t}\right|_{t=0} F_{\mathrm{bar}}((1-t)\pi + t\widehat{\pi}) = \int c^{\mathrm{bar}}_\pi \mathrm{d}(\widehat{\pi} - \pi). \tag{11.47}$$

Hence a regularized minimizer $\pi^\star$ solves the entropic OT problem with frozen cost $c^{\mathrm{bar}}_{\pi^\star}$, and convexity makes this condition globally sufficient. This suggests Sinkhorn subproblems as in the iterative GW solver of Section 12.3, but the tangent is a lower model, so a damped or proximal update is needed for a generic descent guarantee. The barycentric cost is the canonical example to keep in mind: admissibility still constrains the full conditional laws to have second marginal $\beta$, but the objective only charges the displacement from $x$ to $\bar{T}_\pi(x)$ and ignores the conditional variance around this barycenter. Vanishing cost means $\bar{T}_\pi = \mathrm{Id}$, which is exactly the martingale constraint studied next.

### *11.6.1 Martingale Optimal Transport*

**Definition 11.30** (Martingale couplings and martingale OT)**.** Let $\alpha, \beta \in \mathcal{P}_1(\mathbb{R}^d)$. A coupling $\pi \in \mathcal{U}(\alpha, \beta)$ is a martingale coupling if, for the disintegration $\pi(\mathrm{d}x, \mathrm{d}y) = \pi_x(\mathrm{d}y)\alpha(\mathrm{d}x)$,

$$\int_{\mathbb{R}^d} y\mathrm{d}\pi_x(y) = x \qquad \text{for } \alpha\text{-a.e. } x. \tag{11.48}$$

Equivalently, $\bar{T}_\pi = \mathrm{Id}$ in (11.43). The set of such couplings is denoted $\mathcal{U}_{\mathrm{mart}}(\alpha, \beta) := \left\{\pi \in \mathcal{U}(\alpha, \beta) \,;\, \bar{T}_\pi = \mathrm{Id} \quad \alpha\text{-a.e.}\right\}$.

**Remark 11.31 (Zero barycentric cost and martingale feasibility).** For $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$,

$$\mathrm{WOT}_{C_{\mathrm{bar}}}(\alpha, \beta) = 0 \quad \Longleftrightarrow \quad \mathcal{U}_{\mathrm{mart}}(\alpha, \beta) \neq \emptyset. \tag{11.49}$$

A martingale coupling is a zero-cost competitor. Conversely, existence for the barycentric weak problem and nonnegativity of (11.45) show that a zero value yields an optimizer with $\bar{T}_\pi = \mathrm{Id}$. By Theorem 11.35, this is also equivalent to $\alpha \preceq_{\mathrm{cx}} \beta$.

For a cost $c$, the martingale OT problem minimizes $\int c\mathrm{d}\pi$ over $\mathcal{U}_{\mathrm{mart}}(\alpha, \beta)$, with value $+\infty$ when the admissible set is empty. The terminology comes from probability: if $(X, Y) \sim \pi$, then (11.48) is exactly $\mathbb{E}[Y|X] = X$. Hence martingale OT is a Kantorovich problem with the usual two marginal constraints plus a barycentric constraint on the conditional laws.

**Stochastic orders.**

**Definition 11.32** (Classical stochastic order)**.** For $\alpha, \beta \in \mathcal{P}(\mathbb{R})$,

$$\alpha \preceq_{\mathrm{st}} \beta \quad \Longleftrightarrow \quad \int \varphi \, \mathrm{d}\alpha \leqslant \int \varphi \, \mathrm{d}\beta \quad \text{for every bounded increasing } \varphi.$$

**Proposition 11.33** (Strassen's theorem for stochastic order)**.** *Let $\alpha, \beta \in \mathcal{P}(\mathbb{R})$. Then $\alpha \preceq_{\mathrm{st}} \beta$ if and only if there exists $\pi \in \mathcal{U}(\alpha, \beta)$ concentrated on $\{(x, y) \,;\, x \leqslant y\}$.*

*Proof.* A coupling supported on $x \leqslant y$ immediately gives the integral inequality for every increasing $\varphi$. Conversely, testing approximate indicators of intervals $(t, +\infty)$ gives $F_\alpha(t) \geqslant F_\beta(t)$ for all continuity points of the distribution functions. If $U$ is uniform on $(0, 1)$ and $F_\alpha^{-1}, F_\beta^{-1}$ are the generalized quantiles, then $X = F_\alpha^{-1}(U)$ and $Y = F_\beta^{-1}(U)$ have laws $\alpha$ and $\beta$, and the inequality between distribution functions gives $X \leqslant Y$ almost surely. ☐

**Convex order and martingale feasibility.**

**Definition 11.34** (Convex order)**.** For $\alpha, \beta \in \mathcal{P}_1(\mathbb{R}^d)$, write $\alpha \preceq_{\mathrm{cx}} \beta$ when $\int \varphi \, \mathrm{d}\alpha \leqslant \int \varphi \, \mathrm{d}\beta$ for every continuous convex $\varphi : \mathbb{R}^d \to \mathbb{R}$ with at most linear growth.

**Theorem 11.35** (Strassen's martingale theorem)**.** *Let $\alpha, \beta \in \mathcal{P}_1(\mathbb{R}^d)$. Then $\alpha \preceq_{\mathrm{cx}} \beta \iff \mathcal{U}_{\mathrm{mart}}(\alpha, \beta) \neq \emptyset$. Equivalently, $\beta$ is above $\alpha$ in convex order if and only if there exists a coupling $\pi(\mathrm{d}x, \mathrm{d}y) = \pi_x(\mathrm{d}y)\alpha(\mathrm{d}x)$ such that $\int y \mathrm{d}\pi_x(y) = x$ for $\alpha$-almost every $x$.*

*Proof.* If $\pi$ is a martingale coupling, then Jensen's inequality gives, for every convex $\varphi$,

$$\varphi(x) = \varphi\left(\int y \mathrm{d}\pi_x(y)\right) \leqslant \int \varphi(y) \mathrm{d}\pi_x(y),$$

and integration in $x$ gives $\int \varphi \mathrm{d}\alpha \leqslant \int \varphi \mathrm{d}\beta$.
Conversely, assume $\alpha \preceq_{\mathrm{cx}} \beta$. First suppose that both measures are supported in a compact convex set $K$. Separate $\beta$ from the closed convex set of probability measures on $K$ that are reachable from $\alpha$ by martingale kernels supported in $K$. If $\beta$ were not in this set, a continuous separating function $\psi : K \to \mathbb{R}$ would satisfy

$$\int \psi \mathrm{d}\beta > \sup \left\{ \int \psi \mathrm{d}\eta \;:\; \eta \in \mathcal{P}(K),\, \mathcal{U}_{\mathrm{mart}}(\alpha, \eta) \neq \emptyset \right\}.$$

For a fixed $x \in K$, optimizing over probability measures on $K$ with barycenter $x$ gives the concave envelope $\mathrm{conc}_K \psi(x)$; hence the right-hand side is $\int \mathrm{conc}_K \psi \mathrm{d}\alpha$. Since convex order is equivalently the reverse inequality for concave functions, $\int \mathrm{conc}_K \psi \mathrm{d}\alpha \geqslant \int \mathrm{conc}_K \psi \mathrm{d}\beta \geqslant \int \psi \mathrm{d}\beta$, a contradiction. For general measures in $\mathcal{P}_1(\mathbb{R}^d)$, the same separation argument is carried out in the $\mathcal{W}_1$ topology, whose continuous test functions have at most linear growth. Compactness is replaced by tightness and uniform integrability of first moments; these properties also make the set of attainable second marginals closed and preserve the martingale constraint under limits. This is the standard extension in Strassen's theorem. ☐

**Proposition 11.36** (Brenier–Strassen projection formula)**.** *Let $\alpha, \beta \in \mathcal{P}_2(\mathbb{R}^d)$ and let $C_{\mathrm{bar}}$ be the barycentric quadratic cost of Proposition 11.29. Then*

$$\mathrm{WOT}_{C_{\mathrm{bar}}}(\alpha, \beta) = \inf_{\eta \in \mathcal{P}_2(\mathbb{R}^d) :\, \eta \preceq_{\mathrm{cx}} \beta} \mathcal{W}_2^2(\alpha, \eta). \tag{11.50}$$

*Proof.* Let $(X, Y)$ have law $\pi \in \mathcal{U}(\alpha, \beta)$, set $Z = \mathbb{E}[Y \mid X]$, and denote its law by $\eta$. Conditional Jensen gives $\mathbb{E}[\varphi(Z)] \leqslant \mathbb{E}[\varphi(Y)]$ for every convex $\varphi$, so $\eta \preceq_{\mathrm{cx}} \beta$. Since $(X, Z)$ couples $\alpha$ and $\eta$, $\mathcal{W}_2^2(\alpha, \eta) \leqslant \mathbb{E}\|X - Z\|^2 = \int C_{\mathrm{bar}}(x, \pi_x) \mathrm{d}\alpha(x)$. Taking the infimum over $\pi$ proves that the left-hand side of (11.50) is no smaller than the right-hand side.
Conversely, fix $\eta \preceq_{\mathrm{cx}} \beta$. By Theorem 11.35, there is a martingale coupling from $\eta$ to $\beta$. Glue it to an optimal coupling $(X, Z)$ between $\alpha$ and $\eta$, conditionally independently given $Z$. The resulting variables satisfy $\mathbb{E}[Y \mid Z] = Z$, hence $\mathbb{E}[Y \mid X] = \mathbb{E}[Z \mid X]$. Conditional Jensen therefore gives $\mathbb{E}\|X - \mathbb{E}[Y \mid X]\|^2 = \mathbb{E}\|X - \mathbb{E}[Z \mid X]\|^2 \leqslant \mathbb{E}\|X - Z\|^2 = \mathcal{W}_2^2(\alpha, \eta)$. Taking the infimum over $\eta$ proves the reverse inequality. ☐

$\mathcal{N}(m, \Sigma_0) \preceq_{\mathrm{cx}} \mathcal{N}(m, \Sigma_1) \iff \Sigma_1 - \Sigma_0 \succeq 0.$

# 12 Beyond Comparing Measures

## 12.1 Vector and Matrix-Valued Measures

**Positive vector-valued measures.**

**Definition 12.1** (Positive vector-valued measure)**.** A positive $\mathbb{R}_+^m$-valued measure on $\mathcal{X}$ is a tuple $\alpha = (\alpha^1, \ldots, \alpha^m) \in \mathcal{M}_+(\mathcal{X}; \mathbb{R}_+^m)$, where each component $\alpha^k$ is a nonnegative finite measure.

For the dynamic formulas below, assume that $\mathcal{X}$ is either $\mathbb{R}^d$, the flat torus, or a bounded convex domain in $\mathbb{R}^d$ equipped with no-flux boundary conditions. First suppose that the endpoints and the curve have densities $u_t(x) \in \mathbb{R}_+^m$, and write $V_{t,\ell}^k$ for the velocity of channel $1 \leqslant k \leqslant m$ in spatial direction $1 \leqslant \ell \leqslant d$.

**Definition 12.2** (Vector-valued dynamic transport)**.** For a nonnegative extended-valued action density $\Phi : \mathbb{R}_+^m \times (\mathbb{R}^m)^d \to [0, +\infty]$, define

$$\mathcal{W}_\Phi^2(\alpha_0, \alpha_1) := \inf_{u,V} \int_0^1 \int_{\mathcal{X}} \Phi(u_t(x), V_t(x))\, \mathrm{d}x\, \mathrm{d}t \tag{12.1}$$

subject to the endpoint constraints $u_0 \mathrm{d}x = \alpha_0$, $u_1 \mathrm{d}x = \alpha_1$ and the componentwise continuity equation

$$\partial_t u_t + \nabla_x \cdot V_t = 0, \qquad (\nabla_x \cdot V_t)^k = \sum_{\ell=1}^d \partial_{x_\ell} V_{t,\ell}^k. \tag{12.2}$$

The notation $\mathcal{W}_\Phi$ does not automatically define a distance. It is an extended distance on each finite-action component when the action is lower semicontinuous, even and quadratic in $V$, and coercively nondegenerate. These conditions hold for the linear mobility action $\Phi_{\mathsf{M}}(u, V) = \sum_\ell V_\ell^\top \mathsf{M}(u)^\dagger V_\ell$ with the range convention; the diagonal case below is the weighted product of scalar Wasserstein distances on a fixed channel-mass class. For the diagonal action $\Phi_{\mathrm{diag}}(u, V) = \sum_{k=1}^m |V^k|^2/u^k$, the channels move independently.

**Positive matrix-valued measures.**

**Definition 12.3** (Positive matrix-valued measure)**.** Write $\mathbb{S}^m$ for real symmetric matrices and $\mathbb{S}_+^m$ for the positive semidefinite cone. A positive $\mathbb{S}_+^m$-valued measure is a countably additive map from Borel sets of $\mathcal{X}$ to $\mathbb{S}_+^m$; we denote the cone of such measures by $\mathcal{A} \in \mathcal{M}_+(\mathcal{X}; \mathbb{S}_+^m)$.

**Definition 12.4** (Matrix-valued dynamic transport)**.**

$$\mathcal{W}_{\mathrm{mat}}^2(\mathcal{A}_0, \mathcal{A}_1) := \inf_{A,P} \int_0^1 \int_{\mathcal{X}} \sum_{\ell=1}^d \mathrm{tr}\left(P_{t,\ell}^\top A_t^\dagger P_{t,\ell}\right) \mathrm{d}x \mathrm{d}t \tag{12.3}$$

subject to $A_0 \mathrm{d}x = \mathcal{A}_0$, $A_1 \mathrm{d}x = \mathcal{A}_1$ and to the matrix-valued continuity equation

$$\partial_t A_t + \nabla_x \cdot P_t = 0, \qquad \nabla_x \cdot P_t = \sum_{\ell=1}^d \partial_{x_\ell} P_{t,\ell}. \tag{12.4}$$

Here $A_t^\dagger$ is the Moore–Penrose inverse, and the integrand is $+\infty$ unless $\mathrm{Ran}(P_{t,\ell}) \subseteq \mathrm{Ran}(A_t)$ for every $\ell$.

The action in (12.3) is reversible, quadratic under time rescaling and nondegenerate on its effective domain. Hence $\mathcal{W}_{\mathrm{mat}}$ is an extended distance, and a genuine distance on each finite-action component. Equality of the total matrices is necessary for finiteness but need not be sufficient.

**Proposition 12.5** (Diagonal matrix subproblem)**.** *Assume that the endpoints are diagonal in a fixed orthonormal basis,* $\mathcal{A}_i = \mathrm{diag}(\alpha_i^1, \ldots, \alpha_i^m)$, $\quad i = 0, 1$, *and that* $\alpha_0^k(\mathcal{X}) = \alpha_1^k(\mathcal{X}) = m_k$ *for every* $k$. *If one restricts the admissible curves in* (12.3) *to remain diagonal in that basis,* $A_t = \mathrm{diag}(u_t^1, \ldots, u_t^m)$, $\quad P_{t,\ell} = \mathrm{diag}(V_{t,\ell}^1, \ldots, V_{t,\ell}^m)$, *then the value of this restricted matrix problem is*

$$\sum_{k:m_k>0} m_k\, \mathcal{W}_2^2\left(\frac{\alpha_0^k}{m_k}, \frac{\alpha_1^k}{m_k}\right),$$

*with zero contribution from zero-mass channels. Thus the commuting matrix submodel is exactly the diagonal positive vector-valued Benamou–Brenier model above.*

*Proof.* The continuity equation (12.4) is diagonal entry by diagonal entry and gives $\partial_t u^k + \nabla \cdot V^k = 0$. Moreover, $\sum_{\ell=1}^d \mathrm{tr}\left(P_{t,\ell}^\top A_t^\dagger P_{t,\ell}\right) = \sum_{k=1}^m \frac{|V_t^k|^2}{u_t^k}$ with the same scalar perspective convention as before. The admissible set and the action are therefore exactly those of the diagonal vector model. □

### 12.2 Wasserstein over Wasserstein

The Wasserstein construction can itself be iterated: $(\mathcal{P}_p(\mathcal{X}), \mathcal{W}_p)$ becomes the ground space for a second Wasserstein distance. We first record its metric structure, then introduce random measures and collapsed mixtures, and finally transport laws over measures.

**Wasserstein spaces as ground spaces.**

**Proposition 12.6** (Wasserstein spaces as ground spaces)**.** *Let* $1 \leqslant p < \infty$. *If* $(\mathcal{X}, d)$ *is Polish, then* $(\mathcal{P}_p(\mathcal{X}), \mathcal{W}_p)$ *is Polish. If* $\mathcal{X}$ *is compact, then* $\mathcal{P}_p(\mathcal{X}) = \mathcal{P}(\mathcal{X})$ *and this space is compact for* $\mathcal{W}_p$.

*Proof.* This is a standard structural theorem for Wasserstein spaces. For completeness, extract from a $\mathcal{W}_p$-Cauchy sequence a subsequence $(\alpha_k)_k$ such that $\sum_k \mathcal{W}_p(\alpha_k, \alpha_{k+1}) < \infty$. Gluing optimal couplings of consecutive terms produces random variables $(X_k)_k$ with laws $\alpha_k$ and $\sum_k \|d(X_k, X_{k+1})\|_{L^p} < \infty$. Hence $X_k$ converges almost surely and in $L^p$ to an $\mathcal{X}$-valued random variable; its law is the $\mathcal{W}_p$ limit of the subsequence, and the Cauchy property gives convergence of the original sequence. Separability follows by approximating measures with finitely supported measures on a countable dense subset and rational weights. If $\mathcal{X}$ is compact, Prokhorov's theorem gives compactness of $\mathcal{P}(\mathcal{X})$ for weak convergence, and Proposition **??** identifies this topology with the $\mathcal{W}_p$ topology. □

**Random measures and collapsed mixtures.** Fix $1 \leqslant p < \infty$. We denote elements of $\mathcal{P}_p(\mathcal{X})$ by $\alpha, \beta, \ldots$. Elements of $\mathcal{P}_p(\mathcal{P}_p(\mathcal{X}))$ are denoted by fraktur letters, for instance $\mathfrak{A}, \mathfrak{B}$; they are probability laws over probability measures, or random probability measures. A basic parametric example is obtained from a measurable family $(\alpha_\zeta)_{\zeta \in Z}$ and a probability law $\gamma$ on the parameter space:

$$\mathfrak{A} = (\zeta \mapsto \alpha_\zeta)_\sharp \gamma. \tag{12.5}$$

$\int_Z \mathcal{W}_p(\alpha_\zeta, \delta_{x_0})^p \, \mathrm{d}\gamma(\zeta) < \infty$. If $\gamma = \sum_{i=1}^K a_i \delta_{\zeta_i}$, then
$\mathfrak{A} = \sum_{i=1}^K a_i \delta_{\alpha_{\zeta_i}}$.

**Definition 12.7** (Collapsed, or barycentric, mixture)**.** For $\mathfrak{A} \in \mathcal{P}(\mathcal{P}(\mathcal{X}))$, the collapsed, or barycentric, mixture associated with $\mathfrak{A}$ is the measure $\bar{\alpha}_{\mathfrak{A}}$ defined by

$$\int_{\mathcal{X}} f(x) \mathrm{d}\bar{\alpha}_{\mathfrak{A}}(x) = \int_{\mathcal{P}(\mathcal{X})} \left( \int_{\mathcal{X}} f(x) \mathrm{d}\alpha(x) \right) \mathrm{d}\mathfrak{A}(\alpha), \tag{12.6}$$

for bounded continuous $f$.

Unlike this linear collapse, the Wasserstein barycenter of Definition 11.1 in Section 11.1 is nonlinear. $\int_{\mathcal{X}} d(x, x_0)^p \, \mathrm{d}\bar{\alpha}_{\mathfrak{A}}(x) = \int_{\mathcal{P}_p(\mathcal{X})} \mathcal{W}_p(\alpha, \delta_{x_0})^p \, \mathrm{d}\mathfrak{A}(\alpha)$.

**Transport between random measures.** Equipping $\mathcal{P}_p(\mathcal{X})$ with $\mathcal{W}_p$ gives the following ordinary Wasserstein distance between laws over measures.

**Definition 12.8** (Wasserstein-over-Wasserstein distance)**.** Let $1 \leqslant p < +\infty$ and let $\mathcal{X}$ be a Polish metric space. For $\mathfrak{A}, \mathfrak{B} \in \mathcal{P}_p(\mathcal{P}_p(\mathcal{X}))$, define

$$\mathbb{W}_p^p(\mathfrak{A}, \mathfrak{B}) := \inf_{\Pi \in \mathcal{U}(\mathfrak{A}, \mathfrak{B})} \int_{\mathcal{P}_p(\mathcal{X}) \times \mathcal{P}_p(\mathcal{X})} \mathcal{W}_p^p(\alpha, \beta) \mathrm{d}\Pi(\alpha, \beta). \tag{12.7}$$

For $p = 2$, Gaussian mixtures provide a particularly explicit example with two levels of geometry. A mixture $\sum_i a_i \mathcal{N}(m_i, \Sigma_i)$ can either be viewed as the collapsed measure on $\mathcal{X}$, or as the component law
$\mathfrak{A} = \sum_i a_i \delta_{\mathcal{N}(m_i, \Sigma_i)}$ $\mathfrak{A} = \sum_i a_i \delta_{\mathcal{N}(m_i, \Sigma_i)}$, $\quad \mathfrak{B} = \sum_j b_j \delta_{\mathcal{N}(n_j, \Lambda_j)}$, $\mathrm{C}_{i,j} = \|m_i - n_j\|^2 + \mathcal{B}(\Sigma_i, \Lambda_j)^2$. Let $\mathrm{P}^\star$ be an optimal coupling between the weights $a$ and $b$. If $A_{i,j}$ denotes the Brenier linear part from $\Sigma_i$ to $\Lambda_j$, then each active component pair is interpolated by
$m_{ij,t} = (1-t)m_i + t n_j, \qquad \Sigma_{ij,t} = \big((1-t)\mathrm{Id} + tA_{i,j}\big)\Sigma_i\big((1-t)\mathrm{Id} + tA_{i,j}\big)^\top$, $\bar{\alpha}_t = \sum_{i,j} \mathrm{P}^\star_{i,j} \mathcal{N}(m_{ij,t}, \Sigma_{ij,t})$.

**Proposition 12.9** (Collapsing is non-expansive)**.** *Let $(\mathcal{X}, d)$ be Polish, let $1 \leqslant p < \infty$, and let $\mathfrak{A}, \mathfrak{B} \in \mathcal{P}_p(\mathcal{P}_p(\mathcal{X}))$. For the collapsed mixtures defined by (12.6), $\mathcal{W}_p(\bar{\alpha}_{\mathfrak{A}}, \bar{\beta}_{\mathfrak{B}}) \leqslant \mathbb{W}_p(\mathfrak{A}, \mathfrak{B})$.*

*Proof.* Fix $\Pi \in \mathcal{U}(\mathfrak{A}, \mathfrak{B})$ and $\eta > 0$. A standard measurable-selection theorem for optimal transport on Polish spaces gives a measurable kernel $(\alpha, \beta) \mapsto \pi_{\alpha,\beta} \in \mathcal{U}(\alpha, \beta)$ such that $\int_{\mathcal{X} \times \mathcal{X}} d(x, y)^p \, \mathrm{d}\pi_{\alpha,\beta}(x, y) \leqslant \mathcal{W}_p(\alpha, \beta)^p + \eta$. Integrating this kernel against $\Pi$ gives a coupling $\bar{\pi}$ between $\bar{\alpha}_{\mathfrak{A}}$ and $\bar{\beta}_{\mathfrak{B}}$. Its cost satisfies $\int_{\mathcal{X} \times \mathcal{X}} d(x, y)^p \mathrm{d}\bar{\pi}(x, y) \leqslant \int_{\mathcal{P}_p(\mathcal{X})^2} \mathcal{W}_p^p(\alpha, \beta) \mathrm{d}\Pi(\alpha, \beta) + \eta$. Letting $\eta \downarrow 0$, taking the infimum over $\Pi$, and then taking the $p$-th root proves the claim. $\square$

## 12.3 Gromov–Wasserstein

**Discrete formulation.** Optimal transport needs a ground cost C to compare histograms (a, b), and thus cannot be used directly if the histograms are not defined on the same underlying space, or if one cannot pre-register these spaces to define a ground cost. To address this issue, one can instead use a weaker requirement: two matrices $\mathrm{D} \in \mathbb{R}^{n \times n}$ and $\mathrm{D}' \in \mathbb{R}^{m \times m}$ are available and represent relationships between the points on which the histograms are defined. A typical scenario is when these matrices are powers of distance matrices.

$$\mathcal{E}_{\mathrm{D},\mathrm{D}'}(\mathrm{P}) := \sum_{i,j,i',j'} \Delta(\mathrm{D}_{i,i'}, \mathrm{D}'_{j,j'})^p \mathrm{P}_{i,j} \mathrm{P}_{i',j'}, \qquad \mathrm{GW}((\mathrm{a}, \mathrm{D}), (\mathrm{b}, \mathrm{D}'))^p := \min_{\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})} \mathcal{E}_{\mathrm{D},\mathrm{D}'}(\mathrm{P}). \tag{12.8}$$

where $p \geqslant 1$ and $\Delta$ is a distance on $\mathbb{R}$, typically $\Delta(u, v) = |u - v|$.
When D and D′ satisfy the metric axioms on $\{1, \ldots, m\}$ and $\{1, \ldots, n\}$, respectively, these weighted sets define finite metric-measure spaces with measures $\sum_i \mathrm{a}_i \delta_i$ and $\sum_j \mathrm{b}_j \delta_j$. Theorem 12.16 therefore applies directly to (12.8): GW satisfies the triangle inequality and defines a distance modulo measure-preserving isometries.

**General setting.**

**Definition 12.10** (Metric-measure space)**.** A metric-measure space is a triple $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$, where $(\mathcal{X}, d_{\mathcal{X}})$ is Polish and $\alpha$ is a Borel probability measure on $\mathcal{X}$. For a finite $p$-Gromov–Wasserstein distance, one additionally assumes the finite-size condition $\int_{\mathcal{X} \times \mathcal{X}} d_{\mathcal{X}}(x, x')^p \, \mathrm{d}\alpha(x) \mathrm{d}\alpha(x') < \infty$.

The general setting corresponds to computing couplings between metric-measure spaces $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$ and $\mathbb{Y} = (\mathcal{Y}, d_{\mathcal{Y}}, \beta)$, where the distance and the measure are both part of the data. Compactness is often assumed below to avoid additional tightness and integrability arguments.

**Definition 12.11** (Gromov–Wasserstein distance)**.** Fix $p \geqslant 1$ and a distance $\Delta$ on $[0, +\infty)$. Then

$$\mathcal{GW}(\mathbb{X}, \mathbb{Y})^p := \min_{\pi \in \mathcal{U}(\alpha, \beta)} \int_{\mathcal{X}^2 \times \mathcal{Y}^2} \Delta(d_{\mathcal{X}}(x, x'), d_{\mathcal{Y}}(y, y'))^p \mathrm{d}\pi(x, y) \mathrm{d}\pi(x', y'). \tag{12.9}$$

**Definition 12.12** (Infinite-order Gromov–Wasserstein distance)**.** For a distance $\Delta$ on $[0, +\infty)$, the infinite-order Gromov–Wasserstein distance is

$$\mathcal{GW}_\infty(\mathbb{X}, \mathbb{Y}) := \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \operatorname*{ess\,sup}_{\pi \otimes \pi} \Delta\big(d_{\mathcal{X}}(x, x'), d_{\mathcal{Y}}(y, y')\big). \tag{12.10}$$

**Proposition 12.13** (Infinite-order limit of Gromov–Wasserstein)**.** *Let $\mathbb{X}$ and $\mathbb{Y}$ be compact metric-measure spaces, and assume that $\Delta$ is continuous. Then*

$$\mathcal{GW}_\infty(\mathbb{X}, \mathbb{Y}) = \inf_{\pi \in \mathcal{U}(\alpha, \beta)} \sup_{\big((x,y),(x',y')\big) \in \mathrm{supp}(\pi)^2} \Delta\big(d_{\mathcal{X}}(x, x'), d_{\mathcal{Y}}(y, y')\big). \tag{12.11}$$

*Moreover, $\mathcal{GW}_p(\mathbb{X},\mathbb{Y})$ is nondecreasing in $p$ and*

$$\lim_{p\to\infty} \mathcal{GW}_p(\mathbb{X},\mathbb{Y}) = \mathcal{GW}_\infty(\mathbb{X},\mathbb{Y}). \tag{12.12}$$

*Proof.* For each coupling $\pi$, continuity of the distortion integrand identifies its essential supremum under $\pi\otimes\pi$ with its supremum over $\operatorname{supp}(\pi)^2$, which proves (12.11). Monotonicity follows from monotonicity of probability-space $L^p$ norms, and gives a limit $L \leqslant \mathcal{GW}_\infty(\mathbb{X},\mathbb{Y})$.
Let $p_k\to\infty$ and let $\pi_k$ minimize the order-$p_k$ problem. Compactness of $\mathcal{U}(\alpha,\beta)$ gives, after extraction, $\pi_k \rightharpoonup \pi_\star$. If the distortion vanishes on $\operatorname{supp}(\pi_\star)^2$, the result is immediate. Otherwise, fix $(x,y),(x',y')\in\operatorname{supp}(\pi_\star)$ with positive distortion and a number $0<r<\Delta\big(d_\mathcal{X}(x,x'),d_\mathcal{Y}(y,y')\big)$. By continuity, there are neighborhoods $U$ and $V$ of these points on whose product the distortion exceeds $r$. The Portmanteau theorem gives $\liminf_k \pi_k(U)>0$ and $\liminf_k \pi_k(V)>0$. Hence, for some $c>0$ and all sufficiently large $k$, $\mathcal{GW}_{p_k}(\mathbb{X},\mathbb{Y}) \geqslant r\big(\pi_k(U)\pi_k(V)\big)^{1/p_k} \geqslant rc^{1/p_k}$. Letting first $k\to\infty$ and then $r$ increase to the distortion of the selected pair shows that $L$ dominates every distortion on $\operatorname{supp}(\pi_\star)^2$. Therefore $L \geqslant \sup_{((x,y),(x',y'))\in\operatorname{supp}(\pi_\star)^2} \Delta\big(d_\mathcal{X}(x,x'),d_\mathcal{Y}(y,y')\big) \geqslant \mathcal{GW}_\infty(\mathbb{X},\mathbb{Y})$, which proves (12.12). □

Taking $\mathcal{X}=\{1,\dots,m\}$, $\mathcal{Y}=\{1,\dots,n\}$, $\alpha=\sum_i \mathrm{a}_i\delta_i$ and $\beta=\sum_j \mathrm{b}_j\delta_j$ reduces (12.9) exactly to the discrete problem (12.8), since every coupling is represented by a matrix $\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})$. Thus the construction above is precisely the finite-space, or equivalently finitely supported, specialization of the metric-measure formulation.

**Comparison with Wasserstein and Wasserstein–Procrustes.**

**Proposition 12.14** (GW, Wasserstein–Procrustes and Wasserstein)**.** *Let $p\geqslant 1$, let $\alpha,\beta\in\mathcal{P}_p(\mathcal{X})$ be probability measures on the same metric space $(\mathcal{X},d_\mathcal{X})$, and take $\Delta(u,v)=|u-v|$ in (12.9). Then*

$$\mathcal{GW}((\mathcal{X},d_\mathcal{X},\alpha),(\mathcal{X},d_\mathcal{X},\beta)) \leqslant 2\,\mathcal{W}_p(\alpha,\beta), \tag{12.13}$$

*where $\mathcal{W}_p$ uses the same ground metric $d_\mathcal{X}$. In particular, when $\mathcal{X}=\mathbb{R}^d$ is equipped with the Euclidean distance, write $\mathbb{X}_\alpha = (\mathbb{R}^d,\|\cdot\|,\alpha)$ and $\mathbb{X}_\beta=(\mathbb{R}^d,\|\cdot\|,\beta)$. Then*

$$\frac{1}{2}\,\mathcal{GW}(\mathbb{X}_\alpha,\mathbb{X}_\beta) \leqslant \mathcal{W}_{p,\mathrm{E}(d)}([\alpha],[\beta]) \leqslant \mathcal{W}_p(\alpha,\beta). \tag{12.14}$$

*For $p=2$, the middle term is the Wasserstein–Procrustes distance of Definition 10.31. The factor $1/2$ reflects the normalization in (12.9); it disappears under the alternative convention $\Delta(u,v)=|u-v|/2$.*

*Proof.* Fix $\pi\in\mathcal{U}(\alpha,\beta)$, and draw $(x,y)$ and $(x',y')$ independently from $\pi$. The reverse triangle inequality gives $|d_\mathcal{X}(x,x')-d_\mathcal{X}(y,y')| \leqslant d_\mathcal{X}(x,y)+d_\mathcal{X}(x',y')$. Taking the $L^p(\pi\otimes\pi)$ norm, applying Minkowski's inequality, and optimizing over $\pi$ proves (12.13).
Now specialize to $\mathbb{R}^d$. Choosing the identity in Definition 10.29 proves the right inequality in (12.14). Rigid motions preserve pairwise Euclidean distances, so the GW value is unchanged when $\alpha$ and $\beta$ are pushed forward by arbitrary $g,h\in\mathrm{E}(d)$. Applying (12.13) to these aligned measures gives $\mathcal{GW}(\mathbb{X}_\alpha,\mathbb{X}_\beta) \leqslant 2\,\mathcal{W}_p(g_\sharp\alpha,h_\sharp\beta)$. Taking the infimum over $g,h$ proves the left inequality. □

**Definition 12.15** (Isometric metric-measure spaces)**.** Two metric-measure spaces $\mathbb{X}=(\mathcal{X},d_\mathcal{X},\alpha)$ and $\mathbb{Y}=(\mathcal{Y},d_\mathcal{Y},\beta)$ are isometric if there exists a bijection $\varphi:\operatorname{supp}(\alpha)\to\operatorname{supp}(\beta)$ such that $\varphi_\sharp\alpha=\beta$ and $d_\mathcal{Y}(\varphi(x),\varphi(x'))=d_\mathcal{X}(x,x')$ for all $x,x'\in\operatorname{supp}(\alpha)$.

**Theorem 12.16** (Gromov–Wasserstein metric modulo isometries)**.** *For compact metric-measure spaces, $p\geqslant 1$ and $\Delta(u,v)=|u-v|$, $\mathcal{GW}$ defines a distance up to measure-preserving isometries.*

*Proof.* Compactness of the supports makes the coupling set compact and the distortion functional continuous, so an optimal coupling exists. If $\varphi$ is a measure-preserving isometry, the graph coupling $(\mathrm{Id},\varphi)_\sharp\alpha$ has zero distortion; in particular, $\mathcal{GW}(\mathbb{X},\mathbb{X})=0$. Symmetry and non-negativity are immediate.
If $\mathcal{GW}(\mathbb{X},\mathbb{Y})=0$ and $\pi$ is an optimal plan, then $d_\mathcal{X}(x,x')=d_\mathcal{Y}(y,y')$ holds $\pi\otimes\pi$-almost everywhere. By continuity, this equality holds on $\operatorname{supp}(\pi)^2$. $d_\pi((x,y),(x',y')) \coloneqq \frac{1}{2}d_\mathcal{X}(x,x')+\frac{1}{2}d_\mathcal{Y}(y,y')$. The first projection $\psi:(x,y)\mapsto x$ is measure-preserving. For $((x,y),(x',y'))\in\operatorname{supp}(\pi)^2$, $d_\mathcal{X}(\psi(x,y),\psi(x',y'))=d_\mathcal{X}(x,x')=d_\mathcal{Y}(y,y')=d_\pi((x,y),(x',y'))$, so $\psi$ is an isometry and therefore injective. To see surjectivity onto $\operatorname{supp}(\alpha)$, take $x\in\operatorname{supp}(\alpha)$. Since $\psi_\sharp\pi=\alpha$, there is a sequence $(x_k,y_k)\in\operatorname{supp}(\pi)$ with $x_k\to x$. The equality of distances on $\operatorname{supp}(\pi)$ makes $(y_k)_k$ Cauchy, and compactness gives a convergent subsequence with limit $(x,y)\in\operatorname{supp}(\pi)$. The same argument for the second projection shows that the support space is also isometric to $\mathbb{Y}$.
For the triangle inequality, let $\pi$ be an optimal coupling between $\mathbb{X}$ and $\mathbb{Y}$, and $\xi$ an optimal coupling between $\mathbb{Y}$ and $\mathbb{Z}=(Z,d_Z,\gamma)$. By the gluing lemma, take $\sigma$ on $\mathcal{X}\times\mathcal{Y}\times Z$ whose $(\mathcal{X},\mathcal{Y})$ and $(\mathcal{Y},Z)$ marginals are $\pi$ and $\xi$. Let $\rho=(P_{\mathcal{X},Z})_\sharp\sigma$, and write $\bar\sigma=\sigma\otimes\sigma$ for the product law of two independent triples $(x,y,z)$ and $(x',y',z')$. Then $\rho$ is feasible between $\mathbb{X}$ and $\mathbb{Z}$, and

$$\begin{aligned}
\mathcal{GW}(\mathbb{X},\mathbb{Z}) &\leqslant \left(\int |d_\mathcal{X}(x,x')-d_Z(z,z')|^p\,\mathrm{d}\bar\sigma\right)^{1/p}\\
&\leqslant \left(\int |d_\mathcal{X}(x,x')-d_\mathcal{Y}(y,y')|^p\,\mathrm{d}\bar\sigma\right)^{1/p} + \left(\int |d_\mathcal{Y}(y,y')-d_Z(z,z')|^p\,\mathrm{d}\bar\sigma\right)^{1/p}\\
&= \mathcal{GW}(\mathbb{X},\mathbb{Y})+\mathcal{GW}(\mathbb{Y},\mathbb{Z}),
\end{aligned}$$

where the second inequality uses the pointwise triangle inequality followed by Minkowski's inequality. This proves the triangle inequality and completes the metric proof. □

**Marginal stability and geodesics.**

**Proposition 12.17** (Marginal stability and empirical GW rates)**.** *Let $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$ and $\mathbb{Y} = (\mathcal{Y}, d_{\mathcal{Y}}, \beta)$ be compact metric-measure spaces, with $\Delta(u,v) = |u-v|$ and exponent $p \geqslant 1$. For any probability measures $\tilde\alpha$ on $\mathcal{X}$ and $\tilde\beta$ on $\mathcal{Y}$, set $\widetilde{\mathbb{X}} = (\mathcal{X}, d_{\mathcal{X}}, \tilde\alpha)$, $\quad \widetilde{\mathbb{Y}} = (\mathcal{Y}, d_{\mathcal{Y}}, \tilde\beta)$. Then, deterministically, $\left|\mathcal{GW}(\widetilde{\mathbb{X}}, \widetilde{\mathbb{Y}}) - \mathcal{GW}(\mathbb{X}, \mathbb{Y})\right| \leqslant 2\,\mathcal{W}_p^{\mathcal{X}}(\tilde\alpha, \alpha) + 2\,\mathcal{W}_p^{\mathcal{Y}}(\tilde\beta, \beta)$, where $\mathcal{W}_p^{\mathcal{X}}$ and $\mathcal{W}_p^{\mathcal{Y}}$ denote Wasserstein distances computed with $d_{\mathcal{X}}$ and $d_{\mathcal{Y}}$. If $\hat\alpha_n$ and $\hat\beta_m$ are independent empirical measures built from iid samples with laws $\alpha$ and $\beta$, write $\widehat{\mathbb{X}}_n = (\mathcal{X}, d_{\mathcal{X}}, \hat\alpha_n)$ and $\widehat{\mathbb{Y}}_m = (\mathcal{Y}, d_{\mathcal{Y}}, \hat\beta_m)$. If, in addition, $\mathcal{X}$ and $\mathcal{Y}$ are regular compact subsets of Euclidean spaces of intrinsic dimensions $d_{\mathcal{X}}$ and $d_{\mathcal{Y}}$, and the high-dimensional Wasserstein regime $d_{\mathcal{X}}, d_{\mathcal{Y}} > 2p$ applies, then $\mathbb{E}\left|\mathcal{GW}(\widehat{\mathbb{X}}_n, \widehat{\mathbb{Y}}_m) - \mathcal{GW}(\mathbb{X}, \mathbb{Y})\right| = O(n^{-1/d_{\mathcal{X}}} + m^{-1/d_{\mathcal{Y}}})$. For $p = 1$ on subsets of $[0,1]^d$, the rates, including the low-dimensional logarithmic cases, are those of Proposition 9.5.*

*Proof.* The triangle inequality of Theorem 12.16 gives $\mathcal{GW}(\widetilde{\mathbb{X}}, \widetilde{\mathbb{Y}}) - \mathcal{GW}(\mathbb{X}, \mathbb{Y}) \leqslant \mathcal{GW}(\widetilde{\mathbb{X}}, \mathbb{X}) + \mathcal{GW}(\mathbb{Y}, \widetilde{\mathbb{Y}})$. Exchanging the two pairs gives the same bound for the opposite difference, hence the absolute value. The two terms on the right compare metric-measure spaces with the same underlying metric and only different measures, so Proposition 12.14 bounds them by $2\,\mathcal{W}_p^{\mathcal{X}}(\tilde\alpha, \alpha)$ and $2\,\mathcal{W}_p^{\mathcal{Y}}(\tilde\beta, \beta)$. Substituting $\tilde\alpha = \hat\alpha_n$ and $\tilde\beta = \hat\beta_m$, taking expectations and applying the empirical Wasserstein rates recalled in Section 9.2 gives the stated Euclidean rate. □

**Proposition 12.18** (Gromov–Wasserstein geodesics)**.** *Let $\mathbb{X}_0 = (\mathcal{X}_0, d_{\mathcal{X}_0}, \alpha_0)$ and $\mathbb{X}_1 = (\mathcal{X}_1, d_{\mathcal{X}_1}, \alpha_1)$ be compact metric-measure spaces, take $\Delta(u,v) = |u-v|$ in (12.9), and let $\pi^\star$ be an optimal coupling. Define, on $Z = \mathcal{X}_0 \times \mathcal{X}_1$, $d_t((x_0,x_1),(x_0',x_1')) := (1-t)d_{\mathcal{X}_0}(x_0,x_0') + t d_{\mathcal{X}_1}(x_1,x_1')$, $\quad \mathbb{X}_t = (Z, d_t, \pi^\star)$. For $0<t<1$, $d_t$ is a metric. At $t=0$ and $t=1$, one quotients $Z$ by the zero-distance relation associated with $d_t$. Then $t \mapsto \mathbb{X}_t$ is a constant-speed geodesic: $\mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) = |t-s|\,\mathcal{GW}(\mathbb{X}_0, \mathbb{X}_1) \quad \forall s,t \in [0,1]$.*

*Proof.* Write $D = \mathcal{GW}(\mathbb{X}_0, \mathbb{X}_1)$. For $s<t$, couple $\mathbb{X}_s$ and $\mathbb{X}_t$ by the diagonal coupling induced by the identity on $Z$ and the measure $\pi^\star$. For two independent points $z = (x_0,x_1)$ and $z' = (x_0',x_1')$ sampled from $\pi^\star$, $d_t(z,z') - d_s(z,z') = (t-s)\big(d_{\mathcal{X}_1}(x_1,x_1') - d_{\mathcal{X}_0}(x_0,x_0')\big)$. Using this feasible coupling gives $\mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) \leqslant (t-s)D$. The same construction with the projections from $Z$ to $\mathcal{X}_0$ and $\mathcal{X}_1$ gives $\mathcal{GW}(\mathbb{X}_0, \mathbb{X}_t) \leqslant tD$ and $\mathcal{GW}(\mathbb{X}_t, \mathbb{X}_1) \leqslant (1-t)D$. The triangle inequality for $\mathcal{GW}$ then yields, for $0 \leqslant s \leqslant t \leqslant 1$, $D \leqslant \mathcal{GW}(\mathbb{X}_0, \mathbb{X}_s) + \mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) + \mathcal{GW}(\mathbb{X}_t, \mathbb{X}_1) \leqslant sD + \mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) + (1-t)D$, so $\mathcal{GW}(\mathbb{X}_s, \mathbb{X}_t) \geqslant (t-s)D$. This proves equality. □

**Distance-profile lower bound.** This law of distance-profile laws is a Wasserstein-over-Wasserstein construction; see Section 12.2.

**Proposition 12.19** (Mémoli profile lower bound)**.** *Let $\mathbb{X} = (\mathcal{X}, d_{\mathcal{X}}, \alpha)$ and $\mathbb{Y} = (\mathcal{Y}, d_{\mathcal{Y}}, \beta)$ be compact metric-measure spaces and take $\Delta(u,v) = |u-v|$ in (12.9), with the same exponent $p \geqslant 1$. For each $x \in \mathcal{X}$ and $y \in \mathcal{Y}$, define the distance-profile measures on $\mathbb{R}_+$ by $\alpha_x := (d_{\mathcal{X}}(x,\cdot))_\sharp \alpha$, $\quad \beta_y := (d_{\mathcal{Y}}(y,\cdot))_\sharp \beta$. Let $\mathfrak{D}_{\mathbb{X}} = (x \mapsto \alpha_x)_\sharp \alpha$ and $\mathfrak{D}_{\mathbb{Y}} = (y \mapsto \beta_y)_\sharp \beta$, which are probability measures on $\mathcal{P}_p(\mathbb{R}_+)$. Then $\mathbb{W}_p\big(\mathfrak{D}_{\mathbb{X}}, \mathfrak{D}_{\mathbb{Y}}\big) \leqslant \mathcal{GW}(\mathbb{X}, \mathbb{Y})$. Here $\mathbb{W}_p$ is the Wasserstein-over-Wasserstein distance (12.7): its ground space is $(\mathcal{P}_p(\mathbb{R}_+), \mathcal{W}_p)$.*

*Proof.* Fix any $\pi \in \mathcal{U}(\alpha, \beta)$. It induces a coupling $(x,y) \mapsto (\alpha_x, \beta_y)$ between $\mathfrak{D}_{\mathbb{X}}$ and $\mathfrak{D}_{\mathbb{Y}}$, hence $\mathbb{W}_p\big(\mathfrak{D}_{\mathbb{X}}, \mathfrak{D}_{\mathbb{Y}}\big)^p \leqslant \int_{\mathcal{X}\times\mathcal{Y}} \mathcal{W}_p(\alpha_x, \beta_y)^p\, \mathrm{d}\pi(x,y)$. For fixed $(x,y)$, the map $(x',y') \mapsto (d_{\mathcal{X}}(x,x'), d_{\mathcal{Y}}(y,y'))$ pushes the same coupling $\pi$ to a coupling between $\alpha_x$ and $\beta_y$. Therefore $\mathcal{W}_p(\alpha_x, \beta_y)^p \leqslant \int_{\mathcal{X}\times\mathcal{Y}} |d_{\mathcal{X}}(x,x') - d_{\mathcal{Y}}(y,y')|^p\, \mathrm{d}\pi(x',y')$. Integrating in $(x,y)$ gives $\mathbb{W}_p\big(\mathfrak{D}_{\mathbb{X}}, \mathfrak{D}_{\mathbb{Y}}\big)^p \leqslant \int_{\mathcal{X}^2\times\mathcal{Y}^2} |d_{\mathcal{X}}(x,x') - d_{\mathcal{Y}}(y,y')|^p\, \mathrm{d}\pi(x,y)\mathrm{d}\pi(x',y')$. Taking the infimum over $\pi$ and then the $p$-th root proves the claim. □

$$\mathbb{W}_p(\mathfrak{D}_{\mathbb{X}}, \mathfrak{D}_{\mathbb{Y}}) \leqslant \mathcal{GW}(\mathbb{X}, \mathbb{Y}) \leqslant 2\,\mathcal{W}_{p,\mathrm{E}(d)}([\alpha],[\beta]) \leqslant 2\,\mathcal{W}_p(\alpha,\beta), \tag{12.15}$$

where $\mathbb{X} = (\mathbb{R}^d, \|\cdot\|, \alpha)$ and $\mathbb{Y} = (\mathbb{R}^d, \|\cdot\|, \beta)$. The profile term is an inexpensive intrinsic certificate, while the Wasserstein–Procrustes term gives an extrinsic certificate after optimal rigid registration.

**First-order optimality condition as self-consistent OT.** $L\big((x,y),(x',y')\big) := \Delta\big(d_{\mathcal{X}}(x,x'), d_{\mathcal{Y}}(y,y')\big)^p$. For finite signed measures $\xi, \eta \in \mathcal{M}(\mathcal{Z})$ for which the integrals are absolutely convergent, define the symmetric bilinear form and its associated quadratic form by

$$\mathcal{B}_L(\xi,\eta) := \iint_{\mathcal{Z}^2} L(z,z')\mathrm{d}\xi(z)\mathrm{d}\eta(z'), \qquad \mathcal{Q}_L(\xi) := \mathcal{B}_L(\xi,\xi). \tag{12.16}$$

Thus (12.9) minimizes $\mathcal{Q}_L(\pi)$ over $\pi \in \mathcal{U}(\alpha,\beta)$. At a coupling $\pi$, define the linearized ground cost

$$c_\pi(z) := 2\int_{\mathcal{Z}} L(z,z')\mathrm{d}\pi(z'). \tag{12.17}$$

For any $\widehat\pi \in \mathcal{U}(\alpha,\beta)$ and $\xi = \widehat\pi - \pi$, the exact quadratic expansion is

$$\mathcal{Q}_L(\pi + t\xi) = \mathcal{Q}_L(\pi) + t\int_{\mathcal{Z}} c_\pi \mathrm{d}\xi + t^2 \mathcal{Q}_L(\xi). \tag{12.18}$$

**Definition 12.20** (First-order stationary GW coupling)**.** A coupling $\pi \in \mathcal{U}(\alpha,\beta)$ is *first-order stationary for GW* if it satisfies the variational inequality

$$\int_{\mathcal{Z}} c_\pi \mathrm{d}(\widehat\pi - \pi) \geqslant 0 \qquad \text{for every } \widehat\pi \in \mathcal{U}(\alpha,\beta). \tag{12.19}$$

**Proposition 12.21** (Frozen-OT characterization of stationarity)**.** *A coupling $\pi$ is first-order stationary in the sense of Definition 12.20 if and only if*

$$\pi \in \operatorname{argmin}_{\widehat\pi \in \mathcal{U}(\alpha,\beta)} \int_{\mathcal{Z}} c_\pi \mathrm{d}\widehat\pi. \tag{12.20}$$

*In particular, every local minimizer for the narrow topology, and hence every global minimizer, of $\mathcal{Q}_L$ is an optimal Kantorovich coupling for its own linearized cost $c_\pi$.*

*Proof.* The variational inequality (12.19) is exactly the assertion that $\pi$ minimizes the linear functional $\widehat{\pi} \mapsto \int c_\pi \mathrm{d}\widehat{\pi}$, proving the equivalence. If $\pi$ is a local minimizer and $\widehat{\pi}$ is arbitrary, the segment $\pi_t = (1-t)\pi + t\widehat{\pi}$ remains feasible. The right derivative at $t = 0$ in (12.18) must be nonnegative, which gives (12.19). □

**Conditional concavity of GW in Euclidean spaces.**

$$\mathcal{T}_{\mathcal{X},\mathcal{Y}} \coloneqq \{\xi \in \mathcal{M}(\mathcal{X} \times \mathcal{Y})\ ;\ (P_\mathcal{X})_\sharp \xi = 0,\ (P_\mathcal{Y})_\sharp \xi = 0\}. \tag{12.21}$$

$\mathcal{T}_{\mathcal{X},\mathcal{Y}} = \{\Xi \in \mathbb{R}^{n\times m}\ ;\ \Xi \mathbb{1}_m = 0,\ \Xi^\top \mathbb{1}_n = 0\}$, $\quad \dim \mathcal{T}_{\mathcal{X},\mathcal{Y}} = (m-1)(n-1)$. $\frac{\mathrm{d}^2}{\mathrm{d}t^2}\mathcal{Q}_L(\pi_t) = 2\mathcal{Q}_L(\xi)$.

**Proposition 12.22** (Conditional concavity of squared GW)**.** *Let $\mathcal{X}$ and $\mathcal{Y}$ be compact metric spaces, and let $\alpha \in \mathcal{M}_+^1(\mathcal{X})$ and $\beta \in \mathcal{M}_+^1(\mathcal{Y})$. In (12.9), take $p = 2$ and $\Delta(u,v) = |u - v|$. Assume that the two distance kernels $d_\mathcal{X}$ and $d_\mathcal{Y}$ are either both conditionally positive definite or both conditionally negative definite in the sense of Definition 6.4. Then*

$$\mathcal{Q}_L(\xi) \leqslant 0 \qquad \text{for every } \xi \in \mathcal{T}_{\mathcal{X},\mathcal{Y}}. \tag{12.22}$$

*Proof.* The zero marginal conditions make the two pure terms in the squared loss vanish, and therefore $\mathcal{Q}_L(\xi) = -2\iint d_\mathcal{X}(x,x')d_\mathcal{Y}(y,y')\mathrm{d}\xi(x,y)\mathrm{d}\xi(x',y')$. Choose $s \in \{-1,1\}$ so that $sd_\mathcal{X}$ and $sd_\mathcal{Y}$ are conditionally positive definite. Fix base points $x_0 \in \mathcal{X}$ and $y_0 \in \mathcal{Y}$, and define the centered kernels

$$\begin{aligned}\widetilde{d}_\mathcal{X}(x,x') &= s\big(d_\mathcal{X}(x,x') - d_\mathcal{X}(x,x_0) - d_\mathcal{X}(x_0,x') + d_\mathcal{X}(x_0,x_0)\big),\\ \widetilde{d}_\mathcal{Y}(y,y') &= s\big(d_\mathcal{Y}(y,y') - d_\mathcal{Y}(y,y_0) - d_\mathcal{Y}(y_0,y') + d_\mathcal{Y}(y_0,y_0)\big).\end{aligned}$$

These kernels are positive definite. Since $\xi$ has zero marginals and $s^2 = 1$, every centering term vanishes after integration. Let $\Phi_\mathcal{X}$ and $\Phi_\mathcal{Y}$ be Hilbert feature maps for $\widetilde{d}_\mathcal{X}$ and $\widetilde{d}_\mathcal{Y}$. Compactness and continuity make these maps bounded. Writing $z = (x,y)$ and $z' = (x',y')$, one obtains $\iint_{\mathcal{Z}^2} d_\mathcal{X}(x,x')d_\mathcal{Y}(y,y')\mathrm{d}\xi(z)\mathrm{d}\xi(z') = \|\int_\mathcal{Z} \Phi_\mathcal{X}(x) \otimes \Phi_\mathcal{Y}(y)\mathrm{d}\xi(z)\|^2 \geqslant 0$. This proves (12.22); the second-derivative identity above then proves concavity on every feasible segment. □

**Proposition 12.23** (Extreme-point and permutation optimizers in the concave regime)**.** *Assume the hypotheses of Proposition 12.22. Then the squared GW problem has an optimal coupling that is an extreme point of the convex set $\mathcal{U}(\alpha,\beta)$. If, in addition, $\alpha$ and $\beta$ are supported on respectively $n$ and $m$ points, such an optimizer can be chosen with at most $n+m-1$ nonzero entries. If $n = m$ and both measures are uniform, it can be chosen as a permutation coupling $\mathrm{P}_\sigma/n$.*

*Proof.* Set $K = \mathcal{U}(\alpha,\beta)$. Because $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces, $K$ is weakly compact, metrizable and convex in the locally convex space of finite signed measures on $\mathcal{X} \times \mathcal{Y}$. The kernel $L$ is continuous and bounded, so $\mathcal{Q}_L$ is weakly continuous and admits a minimizer $\pi_\star \in K$. The Choquet representation theorem for metrizable compact convex sets provides a probability measure $\nu$ concentrated on the extreme points of $K$ whose barycenter is $\pi_\star$. Concavity of $\mathcal{Q}_L$ and minimality of $\pi_\star$ give $\mathcal{Q}_L(\pi_\star) \geqslant \int_K \mathcal{Q}_L(\eta)\mathrm{d}\nu(\eta) \geqslant \mathcal{Q}_L(\pi_\star)$. Both inequalities are therefore equalities. Since $\mathcal{Q}_L(\eta) \geqslant \mathcal{Q}_L(\pi_\star)$ throughout $K$, equality of the integral implies that $\mathcal{Q}_L(\eta) = \mathcal{Q}_L(\pi_\star)$ for $\nu$-almost every $\eta$. Because $\nu$ is concentrated on extreme couplings, at least one extreme coupling is optimal. This proves directly the instance of the Bauer maximum principle needed here.
For finitely supported measures, the coupling set is a transport polytope. Its marginal operator has rank $n + m - 1$, as computed in the proof of Proposition 3.7. If an extreme coupling had more than $n + m - 1$ positive entries, the restriction of this operator to its support would have a nonzero kernel direction $H$; for sufficiently small $t > 0$, both $\mathrm{P} + tH$ and $\mathrm{P} - tH$ would be distinct feasible couplings with midpoint $\mathrm{P}$, contradicting extremality. This proves the support bound. In the equal-size uniform case, Theorem 3.15 identifies the extreme points of the scaled Birkhoff polytope as $\mathrm{P}_\sigma/n$. □

**Proposition 12.24** (Tangent majorization and frozen-OT replacement)**.** *Assume that $\mathcal{Q}_L$ is concave on $\mathcal{U}(\alpha,\beta)$. Then, for every $\pi, \widehat{\pi} \in \mathcal{U}(\alpha,\beta)$,*

$$\mathcal{Q}_L(\widehat{\pi}) \leqslant \mathcal{Q}_L(\pi) + \int_\mathcal{Z} c_\pi \mathrm{d}(\widehat{\pi} - \pi) = 2\mathcal{B}_L(\pi,\widehat{\pi}) - \mathcal{Q}_L(\pi). \tag{12.23}$$

1. *if $\pi^+ \in \operatorname{argmin}_{\eta\in\mathcal{U}(\alpha,\beta)} \int c_\pi \mathrm{d}\eta$, then $\mathcal{Q}_L(\pi^+) \leqslant \mathcal{Q}_L(\pi)$;*
2. *if $\pi$ is a local minimizer of $\mathcal{Q}_L$ and $\pi^+$ minimizes its frozen OT problem, then $\mathcal{Q}_L\big((1-t)\pi + t\pi^+\big) = \mathcal{Q}_L(\pi)$ for every $t \in [0,1]$;*
3. *if $\pi_\star$ is globally GW-optimal, then every minimizer of the frozen OT problem with cost $c_{\pi_\star}$ is also globally GW-optimal.*

*Proof.* Apply (12.18) at $t = 1$ with $\xi = \widehat{\pi} - \pi$. Concavity gives $\mathcal{Q}_L(\xi) \leqslant 0$, which proves (12.23). For the first conclusion, frozen optimality gives $\int c_\pi \mathrm{d}(\pi^+ - \pi) \leqslant 0$.
Now assume that $\pi$ is a local minimizer. Proposition 12.21 shows that $\pi$ itself minimizes its frozen objective. Hence every other frozen minimizer $\pi^+$ satisfies $\int c_\pi \mathrm{d}(\pi^+ - \pi) = 0$. With $\xi = \pi^+ - \pi$, the exact expansion (12.18) becomes $\mathcal{Q}_L(\pi + t\xi) = \mathcal{Q}_L(\pi) + t^2\mathcal{Q}_L(\xi)$. Local minimality for sufficiently small $t > 0$ gives $\mathcal{Q}_L(\xi) \geqslant 0$, while concavity gives the reverse inequality. Thus $\mathcal{Q}_L(\xi) = 0$, proving the second conclusion for every $t \in [0,1]$. If $\pi = \pi_\star$ is globally optimal, the common value is the global minimum, which proves the third conclusion. □

Thus freezing one occurrence of the coupling produces a global upper bound, not merely a formal linearization. The second conclusion is a flat-replacement theorem for a coupling already known to be locally minimizing; it is not a converse to Proposition 12.21.

**GW as cost-regularized robust Wasserstein transport.**

**Proposition 12.25** (Concave coupling energies as cost-regularized OT)**.** *Let $\mathcal{X}$ and $\mathcal{Y}$ be compact, set $\mathcal{Z} = \mathcal{X} \times \mathcal{Y}$, and let $Q : \mathcal{U}(\alpha,\beta) \to \mathbb{R} \cup \{-\infty\}$ be proper, upper semicontinuous and concave. Let $\mathfrak{F}_Q$ be any proper convex weak-* lower-semicontinuous extension of $-Q$ from $\mathcal{U}(\alpha,\beta)$ to finite signed measures on $\mathcal{Z}$. Such an extension always exists: one possible choice is the canonical constrained extension*

$$\mathfrak{F}_Q^{\mathrm{can}}(\xi) \coloneqq \begin{cases} -Q(\xi), & \xi \in \mathcal{U}(\alpha,\beta), \\ +\infty, & \textit{otherwise}, \end{cases}.$$

*Define the convex conjugate of the chosen extension by*

$$\mathfrak{F}_Q^*(h) := \sup_{\xi\in\mathcal{M}(\mathcal{Z})} \left\{ \int_{\mathcal{Z}} h\mathrm{d}\xi - \mathfrak{F}_Q(\xi) \right\}, \qquad h \in \mathcal{C}(\mathcal{Z}).$$

*Then the convex cost penalty $\mathfrak{R}_Q(c) := \mathfrak{F}_Q^*(-c)$ satisfies*

$$\inf_{\pi\in\mathcal{U}(\alpha,\beta)} Q(\pi) = \inf_{c\in\mathcal{C}(\mathcal{X}\times\mathcal{Y})} \left\{ \mathcal{L}_c(\alpha,\beta) + \mathfrak{R}_Q(c) \right\}. \tag{12.24}$$

*Proof.* Fenchel–Moreau gives, for every feasible $\pi$,

$$Q(\pi) = \inf_{c\in\mathcal{C}(\mathcal{Z})} \left\{ \int_{\mathcal{Z}} c\mathrm{d}\pi + \mathfrak{F}_Q^*(-c) \right\}.$$

Taking the infimum in $\pi$ and exchanging the two infima yields (12.24); the inner infimum in $\pi$ is exactly $\mathcal{L}_c(\alpha,\beta)$. □

$$\mathcal{Q}_L(\pi) = C(\alpha,\beta) + \int_{\mathcal{Z}} c_0\mathrm{d}\pi - 2\|m_\pi\|_{\mathcal{H}_\mathcal{Z}}^2, \qquad m_\pi := \int_{\mathcal{Z}} \Phi(z)\mathrm{d}\pi(z).$$

$$\mathcal{Q}_L(\pi) = C(\alpha,\beta) + \inf_{c\in c_0+\mathcal{H}_\mathcal{Z}} \left\{ \int_{\mathcal{Z}} c\mathrm{d}\pi + \frac{1}{8}\|c-c_0\|_{\mathcal{H}_\mathcal{Z}}^2 \right\}. \tag{12.25}$$

Thus, after quotienting out separable costs that integrate to constants on $\mathcal{U}(\alpha,\beta)$, the quadratic extension in Proposition 12.25 makes $\mathfrak{R}_Q$, up to the additive marginal constant, a squared RKHS norm, with value $+\infty$ outside the admissible affine cost space.

**Proposition 12.26** (Finite-dimensional cost learning for Euclidean GW)**.** *Let $\mathcal{X}\subset\mathbb{R}^{d_x}$ and $\mathcal{Y}\subset\mathbb{R}^{d_y}$ be compact, and use $p=2$ and $\Delta(u,v)=|u-v|$ throughout. Retain the notation $\mathcal{GW}$ of (12.9) when the within-space functions are symmetric dissimilarities rather than metrics. For each of the two choices below, set $\mathbb{X}=(\mathcal{X},d_\mathcal{X},\alpha)$ and $\mathbb{Y}=(\mathcal{Y},d_\mathcal{Y},\beta)$. There is a constant $C(\alpha,\beta)$, depending only on the marginals and on the chosen dissimilarities, such that*

$$\mathcal{GW}(\mathbb{X},\mathbb{Y})^2 = C(\alpha,\beta) + \inf_{M\in\mathbb{R}^{d_y\times d_x}} \left\{ \mathcal{L}_{c_M}(\alpha,\beta) + \frac{1}{2}\|M\|_\mathrm{F}^2 \right\}. \tag{12.26}$$

*The constant $C(\alpha,\beta)$ is different in the two cases, while the learned cost is as follows.*

1. *For the inner-product dissimilarities*

$$d_\mathcal{X}(x,x') = -\langle x,\, x'\rangle, \qquad d_\mathcal{Y}(y,y') = -\langle y,\, y'\rangle, \tag{12.27}$$

*the learned costs form the linear family $c_M(x,y) = -2\langle Mx,\, y\rangle$.*

2. *For the squared-distance dissimilarities*

$$d_\mathcal{X}(x,x') = \|x-x'\|^2, \qquad d_\mathcal{Y}(y,y') = \|y-y'\|^2, \tag{12.28}$$

*translate $\alpha$ and $\beta$ independently so that both have zero mean, which leaves $\mathcal{GW}$ unchanged. The learned costs then form the affine family $c_M(x,y) = -4\|x\|^2\|y\|^2 - 4\langle Mx,\, y\rangle$.*

*Proof.* For $\pi\in\mathcal{U}(\alpha,\beta)$, write $S_\pi := \int yx^\top \mathrm{d}\pi(x,y)$. For two independent pairs $(x,y),(x',y')\sim\pi$, $\iint\langle x,\, x'\rangle\langle y,\, y'\rangle\mathrm{d}\pi\mathrm{d}\pi = \|S_\pi\|_\mathrm{F}^2$. For (12.27), expanding the square therefore gives $\mathcal{Q}_L(\pi) = C(\alpha,\beta) - 2\|S_\pi\|_\mathrm{F}^2$. For fixed $\pi$, completing the square gives

$$\inf_M \left\{ -2\langle M,\, S_\pi\rangle_\mathrm{F} + \frac{1}{2}\|M\|_\mathrm{F}^2 \right\} = -2\|S_\pi\|_\mathrm{F}^2, \qquad M = 2S_\pi,$$

which proves (12.26) in the first case after minimizing over $\pi$.
For centered marginals, direct expansion yields

$$\iint \|x-x'\|^2\|y-y'\|^2\mathrm{d}\pi\mathrm{d}\pi = 2\int \|x\|^2\|y\|^2\mathrm{d}\pi + 2\left(\int_\mathcal{X}\|x\|^2\mathrm{d}\alpha(x)\right)\left(\int_\mathcal{Y}\|y\|^2\mathrm{d}\beta(y)\right) + 4\|S_\pi\|_\mathrm{F}^2.$$

Substitution into the squared-distance plan objective gives

$$\mathcal{Q}_L(\pi) = C(\alpha,\beta) - 4\int \|x\|^2\|y\|^2\mathrm{d}\pi(x,y) - 8\|S_\pi\|_\mathrm{F}^2. \tag{12.29}$$

Finally,

$$\inf_M \left\{ -4\langle M,\, S_\pi\rangle_\mathrm{F} + \frac{1}{2}\|M\|_\mathrm{F}^2 \right\} = -8\|S_\pi\|_\mathrm{F}^2, \qquad M = 4S_\pi,$$

which proves (12.26) in the second case. □

**GW between Gaussian measures.** $\alpha = \mathcal{N}(m_\alpha, \Sigma_\alpha) \in \mathcal{P}_2(\mathbb{R}^{d_x})$, $\beta = \mathcal{N}(m_\beta, \Sigma_\beta) \in \mathcal{P}_2(\mathbb{R}^{d_y})$,

**Proposition 12.27** (Exact inner-product and Gaussian-constrained squared GW)**.** *For the inner-product dissimilarities (12.27), the unrestricted problem admits a jointly Gaussian minimizer for arbitrary $m_\alpha$ and $m_\beta$. If both means vanish, then*

$$\mathcal{GW}(\mathbb{X}, \mathbb{Y})^2 = \sum_{i=1}^{D} (\lambda_i - \eta_i)^2. \tag{12.30}$$

*For the squared-distance dissimilarities (12.28), define the Gaussian-constrained squared distortion*

$$\mathcal{GW}^{\mathrm{Gau}}(\mathbb{X}, \mathbb{Y})^2 := \inf \left\{ \mathcal{Q}_L(\pi) : \pi \in \mathcal{U}(\alpha, \beta), \pi \text{ is jointly Gaussian} \right\}. \tag{12.31}$$

*Then*

$$\mathcal{GW}^{\mathrm{Gau}}(\mathbb{X}, \mathbb{Y})^2 = 4 \left( \sum_{i=1}^{D} \lambda_i - \sum_{i=1}^{D} \eta_i \right)^2 + 8 \sum_{i=1}^{D} (\lambda_i - \eta_i)^2. \tag{12.32}$$

*If $d_x \geqslant d_y$ and $\Sigma_\alpha \succ 0$, a minimizer in (12.31) is induced by a deterministic affine map that aligns the principal covariance directions, rescales the first $d_y$ coordinates and discards the remaining ones.*

*Proof.* For a coupling $\pi$, set $K_\pi := \int (y - m_\beta)(x - m_\alpha)^\top \mathrm{d}\pi(x, y)$. Its block covariance is positive semidefinite, $\begin{pmatrix} \Sigma_\alpha & K_\pi^\top \\ K_\pi & \Sigma_\beta \end{pmatrix} \succeq 0$. Conversely, every matrix $K$ satisfying this constraint is the cross-covariance of a jointly Gaussian coupling. The expansion in the proof of Proposition 12.26 shows that the inner-product objective $\mathcal{Q}_L(\pi)$ depends on $\pi$ only through the cross-moment $\int yx^\top \mathrm{d}\pi = m_\beta m_\alpha^\top + K_\pi$. Replacing any coupling by this Gaussian coupling therefore leaves the objective unchanged. This proves Gaussian optimality, including for nonzero means.

When the means vanish, the positive-semidefinite block constraint is equivalent to the exact factorization $K = \Sigma_\beta^{1/2} R^\top \Sigma_\alpha^{1/2}$ with $R \in \mathbb{R}^{d_x \times d_y}$ and $\|R\|_{\mathrm{op}} \leqslant 1$; this remains valid for singular covariances. Von Neumann's trace inequality then gives $\max_K \|K\|_{\mathrm{F}}^2 = \sum_{i=1}^{\min(d_x, d_y)} \lambda_i \eta_i$. Substitution into the inner-product instance of (12.26) yields (12.30).

For a centered jointly Gaussian pair $(X, Y)$ with cross-covariance $K$, Isserlis' formula gives

$$\mathcal{Q}_L(\pi) = 4\big(\operatorname{tr} \Sigma_\alpha - \operatorname{tr} \Sigma_\beta\big)^2 + 8 \left( \operatorname{tr} \Sigma_\alpha^2 + \operatorname{tr} \Sigma_\beta^2 - 2\|K\|_{\mathrm{F}}^2 \right).$$

Squared distances are translation invariant, so the same expression applies for arbitrary means. Maximizing $\|K\|_{\mathrm{F}}^2$ as above proves (12.32). More explicitly, write $\Sigma_\alpha = U \operatorname{diag}(\lambda_i) U^\top$ and $\Sigma_\beta = V \operatorname{diag}(\eta_j) V^\top$. For arbitrary signs $s_j \in \{-1, 1\}$, the map

$$T(x) = m_\beta + V \left[ \operatorname{diag}\Big( s_j \sqrt{\eta_j / \lambda_j} \Big)_{j=1}^{d_y} \quad 0 \right] U^\top (x - m_\alpha)$$

pushes $\alpha$ to $\beta$ and attains the maximizing cross-covariance. This is the claimed principal-component alignment. □

For the squared-distance dissimilarities (12.28), the unrestricted problem is subtler because (12.29) also contains the mixed fourth moment $\int \|x\|^2 \|y\|^2 \mathrm{d}\pi$, which is not fixed by the cross-moment $\int yx^\top \mathrm{d}\pi$ outside the Gaussian class.

**Proposition 12.28** (Failure of Gaussian closure for squared-distance GW)**.** *For all sufficiently large integers $n$, consider $\alpha = \mathcal{N}(0, 1)$ on $\mathbb{R}$, $\beta_n = \mathcal{N}(0, \mathrm{Id}_n)$ on $\mathbb{R}^n$, with the squared-distance dissimilarities (12.28), and write $\mathbb{X}_n = (\mathbb{R}, d_{\mathcal{X}}, \alpha)$ and $\mathbb{Y}_n = (\mathbb{R}^n, d_{\mathcal{Y}}, \beta_n)$. No jointly Gaussian coupling is optimal for the unrestricted GW problem between $\alpha$ and $\beta_n$. In particular, $\mathcal{GW}(\mathbb{X}_n, \mathbb{Y}_n)^2 < \mathcal{GW}^{\mathrm{Gau}}(\mathbb{X}_n, \mathbb{Y}_n)^2$.*

*Proof.* For $(X, Y) \sim \pi$, equation (12.29) shows that minimizing the GW objective is equivalent to maximizing $\mathcal{G}(\pi) := \mathbb{E}[X^2 \|Y\|^2] + 2\|\mathbb{E}[YX]\|^2$. If $(X, Y)$ is jointly Gaussian, write $k = \mathbb{E}[YX] \in \mathbb{R}^n$. Positivity of its block covariance gives $\|k\|^2 \leqslant 1$, while Isserlis' formula gives $\mathbb{E}[X^2 \|Y\|^2] = n + 2\|k\|^2$.

We construct a better non-Gaussian coupling. Let $Y \sim \mathcal{N}(0, \mathrm{Id}_n)$, set $R = \|Y\|^2$, and denote by $G_j$ the cumulative distribution function of a $\chi_j^2$ variable. Then $U = G_n(R)$ is uniform on $(0, 1)$ and $A = G_1^{-1}(U)$ has law $\chi_1^2$. If $\varepsilon$ is an independent Rademacher variable and $X = \varepsilon\sqrt{A}$, then $X \sim \mathcal{N}(0, 1)$ and $\mathbb{E}[YX] = 0$. This coupling therefore satisfies $\mathcal{G}(\pi) = \mathbb{E}[AR] = \int_0^1 G_1^{-1}(u) G_n^{-1}(u) \mathrm{d}u$. The central limit theorem gives weak convergence of the standardized $\chi_n^2$ variable to a standard Gaussian. Since their second moments also converge, this convergence holds in $\mathcal{W}_2$; the quantile formula of Proposition 2.32 therefore gives $\frac{G_n^{-1}(u) - n}{\sqrt{2n}} \longrightarrow \Phi^{-1}(u)$ in $L^2(0, 1)$, where $\Phi$ is the standard Gaussian cumulative distribution function. Since $G_1^{-1} \in L^2(0, 1)$, $\mathcal{G}(\pi) = n + \sqrt{2n}\big(c + o(1)\big)$, $c := \int_0^1 G_1^{-1}(u) \Phi^{-1}(u) \mathrm{d}u > 0$. Here $c$ is the covariance of the two strictly increasing, nonconstant functions $G_1^{-1}$ and $\Phi^{-1}$ of a uniform random variable, so $c > 0$. Thus $\mathcal{G}(\pi) > n + 4$ for all sufficiently large $n$, which is strictly better than every jointly Gaussian coupling. □

**Monge–GW maps.**

**Definition 12.29** (Gromov–Monge problem)**.** Fix $p \geqslant 1$ and the discrepancy $\Delta$ used in Definition 12.11. The directed Gromov–Monge value is

$$\mathcal{GW}_p^{\mathrm{M}}(\mathbb{X}, \mathbb{Y}) := \inf_{\substack{T : \mathcal{X} \to \mathcal{Y} \text{ measurable} \\ T_\sharp \alpha = \beta}} \left( \iint_{\mathcal{X}^2} \Delta\big(d_{\mathcal{X}}(x, x'), d_{\mathcal{Y}}(T(x), T(x'))\big)^p \mathrm{d}\alpha(x) \mathrm{d}\alpha(x') \right)^{1/p}, \tag{12.33}$$

with value $+\infty$ when no such measure-preserving map exists.

$$\mathcal{GW}_\infty^{\mathrm{M}}(\mathbb{X}, \mathbb{Y}) := \inf_{\substack{T : \mathcal{X} \to \mathcal{Y} \text{ measurable} \\ T_\sharp \alpha = \beta}} \operatorname*{ess\,sup}_{\alpha \otimes \alpha} \Delta\big(d_{\mathcal{X}}(x, x'), d_{\mathcal{Y}}(T(x), T(x'))\big). \tag{12.34}$$

$$\begin{gathered} \mathcal{GW}_p^{\mathrm{M}}(\mathbb{X}, \mathbb{Y}) \leqslant \mathcal{GW}_q^{\mathrm{M}}(\mathbb{X}, \mathbb{Y}) \leqslant \mathcal{GW}_\infty^{\mathrm{M}}(\mathbb{X}, \mathbb{Y}), \\ \mathcal{GW}_p(\mathbb{X}, \mathbb{Y}) \leqslant \mathcal{GW}_p^{\mathrm{M}}(\mathbb{X}, \mathbb{Y}), \qquad \mathcal{GW}_\infty(\mathbb{X}, \mathbb{Y}) \leqslant \mathcal{GW}_\infty^{\mathrm{M}}(\mathbb{X}, \mathbb{Y}). \end{gathered} \tag{12.35}$$

**Proposition 12.30** (Finite uniform Gromov–Monge limits)**.** *Let $\mathcal{X} = \{x_i\}_{i=1}^n$ and $\mathcal{Y} = \{y_j\}_{j=1}^n$ carry uniform measures. Then*

$$\mathcal{GW}_p^{\mathrm{M}}(\mathbb{X}, \mathbb{Y}) = \min_{\sigma \in \mathfrak{S}_n} \left( \frac{1}{n^2} \sum_{i,i'=1}^n \Delta\big(d_{\mathcal{X}}(x_i, x_{i'}), d_{\mathcal{Y}}(y_{\sigma(i)}, y_{\sigma(i')})\big)^p \right)^{1/p}, \tag{12.36}$$

$$\lim_{p\to\infty} \mathcal{GW}_p^{\mathrm{M}}(\mathbb{X}, \mathbb{Y}) = \mathcal{GW}_\infty^{\mathrm{M}}(\mathbb{X}, \mathbb{Y}) = \mathcal{GW}_\infty(\mathbb{X}, \mathbb{Y}). \tag{12.37}$$

*Proof.* A measure-preserving map between equal-size uniform spaces is a permutation; this proves (12.36). Since $\mathfrak{S}_n$ is finite, the minimum commutes with $p \to \infty$.
For the last equality, let $\mathrm{P}_\infty$ minimize $\mathcal{GW}_\infty$. The scaled matrix $n\mathrm{P}_\infty$ is doubly stochastic, so Theorem 3.15 provides a permutation whose graph is contained in $\operatorname{supp}(\mathrm{P}_\infty)$. Its distortion cannot exceed the maximal distortion on $\operatorname{supp}(\mathrm{P}_\infty)^2$, and hence $\mathcal{GW}_\infty^{\mathrm{M}} \leqslant \mathcal{GW}_\infty$. The reverse inequality is (12.35). □

For the inner-product dissimilarities (12.27), the learned-cost representation yields a Monge optimizer; the stronger transformed-Brenier description obtained from the twist condition has a precise rank limitation.
**Proposition 12.31** (Monge–GW map for inner-product dissimilarities)**.** *Let $\mathcal{X} \subset \mathbb{R}^{d_x}$ and $\mathcal{Y} \subset \mathbb{R}^{d_y}$ be compact, assume $d_x \geqslant d_y$, and let $\alpha$ be absolutely continuous with respect to Lebesgue measure. Then the GW problem with dissimilarities (12.27) admits an optimal coupling of the form $(\mathrm{Id}, T)_\sharp\alpha$. In the inner-product instance of (12.26), let $M^\star$ be an optimal learned matrix. If* $\operatorname{rank}(M^\star) = d_y$*, then $c_{M^\star}$ satisfies the twist condition of Definition 2.25; the optimal coupling for this fixed cost is unique and*

$$T(x) = \nabla\varphi(M^\star x) \tag{12.38}$$

*for the Brenier potential $\varphi$ transporting $(M^\star)_\sharp\alpha$ to $\beta$.*

*Proof.* Compactness of the coupling set and coercivity of the quadratic penalty in $M$ give an optimal pair $(\pi^\star, M^\star)$ in the inner-product instance of (12.26). With $M^\star$ frozen, $\pi^\star$ solves linear OT for $c_{M^\star}$. The linear-cost Monge-existence theorem of Dumont, Lacombe and Vialard, also used in the cost-regularized setting in, gives a graph-supported optimizer for every matrix $M^\star$ under the stated dimension and absolute-continuity assumptions. Its rank-deficient case combines transport on the image of $M^\star$ with measurable transport along its fibers. Replacing $\pi^\star$ by this optimizer leaves the joint objective unchanged.
If $M^\star$ has full row rank, then $y \longmapsto \nabla_x c_{M^\star}(x,y) = -2(M^\star)^\top y$ is injective, so the cost is twisted. Moreover, $(M^\star)_\sharp\alpha$ is absolutely continuous on $\mathbb{R}^{d_y}$, and minimizing $-2\langle z, y\rangle$ is equivalent, up to marginal-only terms, to quadratic transport between $z = M^\star x$ and $y$. Brenier's theorem, Theorem 2.17, therefore gives the unique map $z \mapsto \nabla\varphi(z)$, proving (12.38). This is precisely the full-rank construction of. □

If $M^\star$ is rank deficient, the elementary twist argument fails and uniqueness need not hold; the existence assertion nevertheless remains valid by the linear-cost analysis cited in the proof. This distinction is essential: a Monge optimizer always exists here under the hypotheses of Proposition 12.31, but a literal full-rank Brenier characterization does not always apply.
**Entropic regularization and biconvex alternating minimization.** Assume in this paragraph that $\mathcal{X}$ and $\mathcal{Y}$ are compact metric spaces, that $L$ in (12.16) is continuous, and set $\mathfrak{m} = \alpha \otimes \beta$. With the relative entropy of Definition 7.9, define

$$\mathcal{R}_\varepsilon(\pi) := \begin{cases} 0, & \varepsilon = 0, \\ \varepsilon\,\mathrm{KL}(\pi|\mathfrak{m}), & \varepsilon > 0. \end{cases}$$

**Definition 12.32** (Entropic GW and its biconvex relaxation)**.** For $\varepsilon \geqslant 0$, set

$$\mathcal{F}_\varepsilon(\pi) := \mathcal{Q}_L(\pi) + \mathcal{R}_\varepsilon(\pi), \qquad \mathcal{GW}_\varepsilon(\mathbb{X}, \mathbb{Y})^p := \min_{\pi \in \mathcal{U}(\alpha,\beta)} \mathcal{F}_\varepsilon(\pi), \tag{12.39}$$

$$\mathcal{F}_\varepsilon^{\mathrm{bi}}(\pi, \xi) := \mathcal{B}_L(\pi, \xi) + \frac{1}{2}\mathcal{R}_\varepsilon(\pi) + \frac{1}{2}\mathcal{R}_\varepsilon(\xi), \qquad \mathcal{GW}_{\varepsilon,\mathrm{bi}}(\mathbb{X}, \mathbb{Y})^p := \min_{\pi,\xi \in \mathcal{U}(\alpha,\beta)} \mathcal{F}_\varepsilon^{\mathrm{bi}}(\pi, \xi). \tag{12.40}$$

$$\mathcal{GW}_{\varepsilon,\mathrm{bi}}(\mathbb{X}, \mathbb{Y})^p \leqslant \mathcal{GW}_\varepsilon(\mathbb{X}, \mathbb{Y})^p. \tag{12.41}$$

Block optimality explains both the usefulness and the limitation of the relaxation. If $(\pi_\star, \xi_\star)$ minimizes (12.40), then each component minimizes an OT problem with the other component frozen.

$$\pi_\star \in \operatorname*{argmin}_{\pi \in \mathcal{U}(\alpha,\beta)} \left\{ \int c_{\xi_\star}\,\mathrm{d}\pi + \mathcal{R}_\varepsilon(\pi) \right\},$$

and symmetrically for $\xi_\star$. Thus the relaxation replaces the quadratic problem by two OT problems, but their ground costs are themselves unknown.
In the unregularized discrete case, these block problems are linear programs. If $\alpha$ and $\beta$ are both uniform on $n$ points, at least one *global minimizer of the lifted problem* has the form $(\mathrm{P}_\sigma/n, \mathrm{P}_\tau/n)$.
**Proposition 12.33** (Tightness of the biconvex GW relaxation)**.** *Assume that $\mathcal{Q}_L$ is concave on $\mathcal{U}(\alpha, \beta)$. Since $\mathcal{Q}_L$ is quadratic, this is equivalent to*

$$\mathcal{Q}_L(\pi - \xi) \leqslant 0 \qquad \text{for every } \pi, \xi \in \mathcal{U}(\alpha, \beta). \tag{12.42}$$

*Then (12.41) is an equality for every $\varepsilon \geqslant 0$. Moreover, every minimizing pair $(\pi_\star, \xi_\star)$ satisfies $\mathcal{F}_\varepsilon(\pi_\star) = \mathcal{F}_\varepsilon(\xi_\star) = \mathcal{GW}_\varepsilon(\mathbb{X}, \mathbb{Y})^p$, $\mathcal{Q}_L(\pi_\star - \xi_\star) = 0$. If $\varepsilon > 0$, then $\pi_\star = \xi_\star$. The same conclusion holds for $\varepsilon = 0$ when the inequality in (12.42) is strict for every nonzero feasible difference.*

*Proof.* Symmetry and bilinearity give the polarization identity

$$\mathcal{F}_\varepsilon^{\mathrm{bi}}(\pi, \xi) = \frac{1}{2}\mathcal{F}_\varepsilon(\pi) + \frac{1}{2}\mathcal{F}_\varepsilon(\xi) - \frac{1}{2}\mathcal{Q}_L(\pi - \xi). \tag{12.43}$$

Under (12.42), the right-hand side is at least $\mathcal{GW}_\varepsilon(\mathbb{X},\mathbb{Y})^p$. Conversely, diagonal pairs recover $\mathcal{F}_\varepsilon$, proving equality of the optimal values. Equality in the preceding bounds gives the two asserted identities. Strict negativity immediately forces $\pi_\star = \xi_\star$. Suppose now that $\varepsilon > 0$. The two diagonal objective values are finite, so both relative entropies are finite. If $\pi_\star \neq \xi_\star$, then $\mathcal{Q}_L(\pi_\star - \xi_\star) = 0$ makes the quadratic term affine on their joining segment, while strict convexity of relative entropy makes its value at the midpoint strictly smaller than the average of its endpoint values. The midpoint would therefore have a smaller diagonal objective, a contradiction. □

Starting from $(\pi^{(0)}, \xi^{(0)}) \in \mathcal{U}(\alpha,\beta)^2$, exact alternating minimization of the general-measure relaxation reads

$$\pi^{(\ell+1)} \in \operatorname*{argmin}_{\pi\in\mathcal{U}(\alpha,\beta)} \left\{ \int c_{\xi^{(\ell)}} \mathrm{d}\pi + \mathcal{R}_\varepsilon(\pi) \right\}, \tag{12.44}$$

$$\xi^{(\ell+1)} \in \operatorname*{argmin}_{\xi\in\mathcal{U}(\alpha,\beta)} \left\{ \int c_{\pi^{(\ell+1)}} \mathrm{d}\xi + \mathcal{R}_\varepsilon(\xi) \right\}. \tag{12.45}$$

For discrete measures, let H denote the discrete Shannon–Boltzmann entropy of Definition 7.1. Given $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$, define its contracted GW cost by

$$\mathrm{C}(\mathrm{P})_{i,j} \coloneqq \sum_{i'=1}^{n}\sum_{j'=1}^{m} \Delta\big(\mathrm{D}_{i,i'}, \mathrm{D}'_{j,j'}\big)^p \mathrm{P}_{i',j'}. \tag{12.46}$$

$$\min_{\mathrm{P},\mathrm{Q}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \langle \mathrm{C}(\mathrm{P}), \mathrm{Q}\rangle - \frac{\varepsilon}{2}\big(\mathrm{H}(\mathrm{P}) + \mathrm{H}(\mathrm{Q})\big), \tag{12.47}$$

and the original lifted optimal value is obtained by adding this constant back. There is no factor $1/2$ in front of the bilinear term under the convention (12.46): $\mathrm{C}(\mathrm{P})$ is half of the full linearized cost in (12.17).

$$\min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} \mathcal{E}_{\mathrm{D},\mathrm{D}'}(\mathrm{P}) - \varepsilon\mathrm{H}(\mathrm{P}). \tag{12.48}$$

For an unstructured distortion kernel, evaluating one entry of (12.46) requires $nm$ terms. Forming all $nm$ entries therefore costs $O(n^2m^2)$ dense arithmetic operations; explicitly storing the kernel would likewise require $O(n^2m^2)$ memory. The squared loss admits a substantially cheaper contraction without forming this four-index tensor.

**Proposition 12.34** (Fast contracted cost for squared GW)**.** *Assume that* $\Delta\big(\mathrm{D}_{i,i'}, \mathrm{D}'_{j,j'}\big)^p = \big(\mathrm{D}_{i,i'} - \mathrm{D}'_{j,j'}\big)^2$. *Then, for every* $\mathrm{P} \in \mathrm{U}(\mathrm{a},\mathrm{b})$,

$$\mathrm{C}(\mathrm{P}) = \mathrm{D}^{\odot 2}\mathrm{a}\,\mathbb{1}_m^\top + \mathbb{1}_n(\mathrm{D}'^{\odot 2}\mathrm{b})^\top - 2\mathrm{D}\,\mathrm{P}\,\mathrm{D}'^\top. \tag{12.49}$$

*Its ambient Frobenius gradient satisfies* $\nabla\mathcal{E}_{\mathrm{D},\mathrm{D}'}(\mathrm{P}) = 2\mathrm{C}(\mathrm{P})$.

*Proof.* Expanding the square in (12.46) gives

$$\mathrm{C}(\mathrm{P})_{i,j} = \sum_{i',j'} \mathrm{D}_{i,i'}^2 \mathrm{P}_{i',j'} + \sum_{i',j'} (\mathrm{D}'_{j,j'})^2 \mathrm{P}_{i',j'} - 2\sum_{i',j'} \mathrm{D}_{i,i'}\mathrm{D}'_{j,j'}\mathrm{P}_{i',j'}.$$

The marginal identities $\mathrm{P}\mathbb{1}_m = \mathrm{a}$ and $\mathrm{P}^\top\mathbb{1}_n = \mathrm{b}$ turn the first two sums into $(\mathrm{D}^{\odot 2}\mathrm{a})_i$ and $(\mathrm{D}'^{\odot 2}\mathrm{b})_j$, while the last sum is $(\mathrm{DPD}'^\top)_{i,j}$. This proves (12.49). The underlying four-index kernel is symmetric, so differentiating $\mathcal{E}_{\mathrm{D},\mathrm{D}'}(\mathrm{P}) = \langle \mathrm{C}(\mathrm{P}), \mathrm{P}\rangle$ gives the factor two in the gradient. Finally, computing DP and then $(\mathrm{DP})\mathrm{D}'^\top$ costs respectively $O(n^2m)$ and $O(nm^2)$ operations; the marginal terms are cheaper. □

$\mathrm{Q}^{(0)},\ \mathrm{P}^{(1)},\ \mathrm{Q}^{(1)},\ \mathrm{P}^{(2)}, \ldots$

**Fused GW: adding features.** $\varphi_{\mathcal{X}} : \mathcal{X} \to \mathcal{Z}_\mathrm{f}, \qquad \varphi_{\mathcal{Y}} : \mathcal{Y} \to \mathcal{Z}_\mathrm{f},$

**Definition 12.35** (Fused Gromov–Wasserstein problem)**.** Under the notation of (12.16), let $c_{\mathcal{X},\mathcal{Y}} : \mathcal{X}\times\mathcal{Y} \to \mathbb{R}_+$ be a measurable cross-space cost and let $0 \leqslant \lambda \leqslant 1$. The fused GW value is

$$\mathrm{FGW}_{\lambda,p}(\mathbb{X},\mathbb{Y}; c_{\mathcal{X},\mathcal{Y}})^p \coloneqq \min_{\pi\in\mathcal{U}(\alpha,\beta)} (1-\lambda)\int_{\mathcal{X}\times\mathcal{Y}} c_{\mathcal{X},\mathcal{Y}}\mathrm{d}\pi + \lambda\mathcal{Q}_L(\pi). \tag{12.50}$$

The first term compares features by ordinary Kantorovich transport, while the second compares intrinsic geometry. Thus $\lambda = 0$ gives feature-only OT and $\lambda = 1$ gives pure GW; intermediate values balance pointwise feature fidelity against pairwise geometric fidelity.

For discrete measures, setting $M_{i,j} = c_{\mathcal{X},\mathcal{Y}}(x_i, y_j)$ reduces (12.50) to $\min_{\mathrm{P}\in\mathrm{U}(\mathrm{a},\mathrm{b})} (1-\lambda)\langle M, \mathrm{P}\rangle + \lambda\mathcal{E}_{\mathrm{D},\mathrm{D}'}(\mathrm{P})$, The algorithms above extend by the same linearization principle. With the coupling regularizer $\mathcal{R}_\varepsilon$ defined before (12.39), the regularized fused objective is

$$\mathcal{F}_{\varepsilon,\lambda}^{\mathrm{FGW}}(\pi) \coloneqq (1-\lambda)\int_{\mathcal{X}\times\mathcal{Y}} c_{\mathcal{X},\mathcal{Y}}\mathrm{d}\pi + \lambda\mathcal{Q}_L(\pi) + \mathcal{R}_\varepsilon(\pi).$$

$$c_{\pi,\lambda}^{\mathrm{FGW}}(x,y) \coloneqq (1-\lambda)c_{\mathcal{X},\mathcal{Y}}(x,y) + 2\lambda\int_{\mathcal{X}\times\mathcal{Y}} L\big((x,y),(x',y')\big)\mathrm{d}\pi(x',y'). \tag{12.51}$$

$$\mathcal{F}_{\varepsilon,\lambda}^{\mathrm{FGW,bi}}(\pi,\xi) \coloneqq \frac{1-\lambda}{2}\int_{\mathcal{X}\times\mathcal{Y}} c_{\mathcal{X},\mathcal{Y}}\mathrm{d}(\pi+\xi) + \lambda\mathcal{B}_L(\pi,\xi) + \frac{1}{2}\mathcal{R}_\varepsilon(\pi) + \frac{1}{2}\mathcal{R}_\varepsilon(\xi).$$

$$\operatorname*{argmin}_{\pi\in\mathcal{U}(\alpha,\beta)} \left\{ \int_{\mathcal{X}\times\mathcal{Y}} c_{\xi,\lambda}^{\mathrm{FGW}}\mathrm{d}\pi + \mathcal{R}_\varepsilon(\pi) \right\}.$$

**Hausdorff and Gromov–Hausdorff viewpoints.**

**Definition 12.36** (Hausdorff distance)**.** If $A, B$ are nonempty compact subsets of a metric space $(Z, d_Z)$, their Hausdorff distance is

$$d_{\mathrm{H}}^{Z}(A,B) := \max\left\{\sup_{a\in A}\inf_{b\in B} d_Z(a,b), \sup_{b\in B}\inf_{a\in A} d_Z(a,b)\right\}.$$

**Definition 12.37** (Gromov–Hausdorff distance)**.** For compact metric spaces $(\mathcal{X}, d_{\mathcal{X}})$ and $(\mathcal{Y}, d_{\mathcal{Y}})$, their Gromov–Hausdorff distance is $d_{\mathrm{GH}}(\mathcal{X},\mathcal{Y}) := \inf_{Z,\varphi,\psi} d_{\mathrm{H}}^{Z}(\varphi(\mathcal{X}), \psi(\mathcal{Y}))$.

One has $d_{\mathrm{GH}}(\mathcal{X},\mathcal{Y}) = 0$ exactly when $\mathcal{X}$ and $\mathcal{Y}$ are isometric. Hence $d_{\mathrm{GH}}$ is a genuine metric on isometry classes of compact metric spaces.
$\mathrm{dis}(R) := \sup_{(x,y),(x',y')\in R} |d_{\mathcal{X}}(x,x') - d_{\mathcal{Y}}(y,y')|$.

$$2d_{\mathrm{GH}}(\mathcal{X},\mathcal{Y}) = \inf_{R\in\mathfrak{R}(\mathcal{X},\mathcal{Y})} \mathrm{dis}(R). \tag{12.52}$$

**Proposition 12.38** (Gromov–Hausdorff as an infinite-order intrinsic matching)**.** *Let $\mathcal{X}$ and $\mathcal{Y}$ be compact metric spaces and let $\alpha, \beta$ have full support. Then*

$$2d_{\mathrm{GH}}(\mathcal{X},\mathcal{Y}) \leqslant \mathcal{GW}_\infty(\mathbb{X},\mathbb{Y}). \tag{12.53}$$

*Moreover, removing the prescribed weights recovers Gromov–Hausdorff exactly:*

$$2d_{\mathrm{GH}}(\mathcal{X},\mathcal{Y}) = \inf_{\substack{\alpha\in\mathcal{M}_+^1(\mathcal{X}),\ \mathrm{supp}(\alpha)=\mathcal{X}\\ \beta\in\mathcal{M}_+^1(\mathcal{Y}),\ \mathrm{supp}(\beta)=\mathcal{Y}}} \mathcal{GW}_\infty\big((\mathcal{X},d_{\mathcal{X}},\alpha),(\mathcal{Y},d_{\mathcal{Y}},\beta)\big). \tag{12.54}$$

*Proof.* The support of any $\pi \in \mathcal{U}(\alpha,\beta)$ is compact. The support of each marginal is the closure of the corresponding coordinate projection of $\mathrm{supp}(\pi)$. These projections are compact, hence already closed; because $\alpha$ and $\beta$ have full support, they are therefore equal to $\mathcal{X}$ and $\mathcal{Y}$. Thus $\mathrm{supp}(\pi)$ is a correspondence, and (12.52) gives (12.53).
For (12.54), the preceding lower bound applies to every pair of full-support weights. Conversely, choose a correspondence $R$ with distortion arbitrarily close to the right-hand side of (12.52). Replacing $R$ by its closure preserves both the correspondence property and its distortion, so $R$ may be assumed compact. It then carries a probability $\pi$ with support exactly $R$. Let $\alpha$ and $\beta$ be its marginals. They have full support, while (12.11) gives $\mathcal{GW}_\infty(\mathbb{X},\mathbb{Y}) \leqslant \mathrm{dis}(R)$. Taking the infimum over $R$ proves the identity. ☐

Thus Gromov–Hausdorff is the mass-free, hard-support counterpart of GW: it minimizes worst-case distortion over correspondences between the two supports. By contrast, finite-$p$ GW fixes the masses and averages distortion under a coupling; optimizing these masses and taking $p = \infty$ recovers $2d_{\mathrm{GH}}$ through (12.54).

## 12.4 Quantum Optimal Transport

**Finite-dimensional states and couplings.**

**Definition 12.39** (Hermitian and density matrices)**.** Let $\mathbb{H}_n$ be the real vector space of $n \times n$ Hermitian matrices, $\mathbb{H}_n^+ = \{A \in \mathbb{H}_n\ ;\ A \succeq 0\}$, $\mathbb{H}_n^{+,1} = \{A \in \mathbb{H}_n^+\ ;\ \mathrm{tr}(A) = 1\}$. Here $A \succeq 0$ means $z^*Az \geqslant 0$ for every $z \in \mathbb{C}^n$, equivalently that all eigenvalues of $A$ are nonnegative. Elements of $\mathbb{H}_n^{+,1}$ are density matrices.

For $A \in \mathbb{H}_n$ and $B \in \mathbb{H}_m$, the tensor product is characterized in operator and matrix form by

$$(A\otimes B)(x\otimes y) = (Ax)\otimes(By), \qquad A\otimes B = [A_{i,k}B]_{i,k=1}^n, \qquad (A\otimes B)_{(i,j),(k,\ell)} = A_{i,k}B_{j,\ell}. \tag{12.55}$$

**Definition 12.40** (Joint quantum states and partial traces)**.** A joint quantum state between $\mathbb{C}^n$ and $\mathbb{C}^m$ is a density matrix $T \in \mathbb{H}_{nm}^{+,1}$ acting on $\mathbb{C}^n \otimes \mathbb{C}^m$. Its marginals are the partial traces $\mathrm{Tr}_B(T) \in \mathbb{H}_n^{+,1}$ and $\mathrm{Tr}_A(T) \in \mathbb{H}_m^{+,1}$ defined, for all $F \in \mathbb{H}_n$ and $G \in \mathbb{H}_m$, by

$$\mathrm{tr}(F\ \mathrm{Tr}_B\,T) = \mathrm{tr}((F\otimes \mathrm{Id}_m)T), \qquad \mathrm{tr}(G\ \mathrm{Tr}_A\,T) = \mathrm{tr}((\mathrm{Id}_n\otimes G)T), \tag{12.56}$$

and they play the role of the two marginals of a classical coupling.

In product bases, $T$ is the four-index tensor $T_{(i,j),(k,\ell)} = \langle e_i\otimes f_j, T(e_k\otimes f_\ell)\rangle$, and

$$\mathrm{tr}(T) = \sum_{i=1}^n\sum_{j=1}^m T_{(i,j),(i,j)}, \quad [\mathrm{Tr}_B\,T]_{i,k} = \sum_{j=1}^m T_{(i,j),(k,j)}, \quad [\mathrm{Tr}_A\,T]_{j,\ell} = \sum_{i=1}^n T_{(i,j),(i,\ell)}. \tag{12.57}$$

Thus $\mathrm{tr}(T) = \mathrm{tr}(\mathrm{Tr}_B\,T) = \mathrm{tr}(\mathrm{Tr}_A\,T)$, and $A \otimes B$ has partial traces $A$ and $B$ for density matrices $A, B$.

**Definition 12.41** (Finite-dimensional quantum OT)**.** Let $A \in \mathbb{H}_n^{+,1}$, $B \in \mathbb{H}_m^{+,1}$ and let $C \in \mathbb{H}_{nm}$ be a Hermitian cost observable. The quantum OT value is the semidefinite program

$$\mathrm{QOT}_C(A,B) := \min_{T\in\mathbb{H}_{nm}^+} \{\mathrm{tr}(CT)\ ;\ \mathrm{Tr}_B(T) = A,\ \mathrm{Tr}_A(T) = B\}. \tag{12.58}$$

**Proposition 12.42** (Quantum Kantorovich duality)**.** *For $A \in \mathbb{H}_n^{+,1}$ and $B \in \mathbb{H}_m^{+,1}$, the dual of* (12.58) *is*

$$\mathrm{QOT}_C(A,B) = \sup_{F\in\mathbb{H}_n,\ G\in\mathbb{H}_m} \{\mathrm{tr}(FA) + \mathrm{tr}(GB) :\ F\otimes\mathrm{Id}_m + \mathrm{Id}_n\otimes G \preceq C\}. \tag{12.59}$$

*There is no duality gap. If $A$ and $B$ are positive definite, the supremum is attained. For singular marginals, the primal reduces to the tensor product of their supports; on this reduced space the corresponding dual maximum is attained, while the full-space formula above is always valid as a supremum.*

*Proof.* Introduce Hermitian Lagrange multipliers $F$ and $G$ for the two marginal constraints. The Lagrangian is

$$\mathrm{tr}(CT) + \mathrm{tr}(F(A - \mathrm{Tr}_B\, T)) + \mathrm{tr}(G(B - \mathrm{Tr}_A\, T)) = \mathrm{tr}(FA) + \mathrm{tr}(GB) + \mathrm{tr}((C - F \otimes \mathrm{Id}_m - \mathrm{Id}_n \otimes G)T)\,,$$

where (12.56) was used in the last equality. Minimizing over $T \succeq 0$ gives a finite lower bound if and only if $C - F\otimes\mathrm{Id}_m - \mathrm{Id}_n \otimes G \succeq 0$, in which case the infimum in $T$ is 0. When $A, B \succ 0$, the coupling $A \otimes B$ is strictly feasible, so Slater's theorem gives equality of primal and dual values and dual attainment.
For singular marginals, let $P_A$ and $P_B$ be their support projections. If $T$ is feasible, then $\mathrm{tr}\big(((\mathrm{Id}_n - P_A) \otimes \mathrm{Id}_m)T\big) = \mathrm{tr}((\mathrm{Id}_n - P_A)A) = 0$. Positivity implies that $T$ vanishes on $(\mathrm{supp}\, A)^\perp \otimes \mathbb{C}^m$; the analogous argument for $B$ yields $T = (P_A \otimes P_B)T(P_A \otimes P_B)$. Thus the primal is exactly the problem obtained by compressing $A$, $B$ and $C$ to $\mathrm{supp}\, A \otimes \mathrm{supp}\, B$. The reduced marginals are positive definite, so Slater applies there. Equivalently, the unreduced dual values approach the reduced maximum after assigning sufficiently negative potentials on the orthogonal complements. This proves the stated supremum formula without a gap. ☐

**Entropic regularization and Bregman iterations.**

**Definition 12.43** (von Neumann quantum entropy)**.** For a density matrix or positive semidefinite matrix $T$, the shifted von Neumann entropy functional used here is

$$H(T) = \mathrm{tr}(T(\log T - \mathrm{Id}))\,, \qquad \nabla H(T) = \log T,$$

with the convention $0 \log 0 = 0$ on eigenvalues. This is the convex negative quantum entropy; on trace-one states it differs from the physical entropy $-\mathrm{tr}(T \log T)$ by a sign and an additive constant.

**Proposition 12.44** (Entropic quantum OT duality)**.** *Assume $A \succ 0$, $B \succ 0$ and $\varepsilon > 0$. Then (**??**) has a unique positive minimizer. Its dual is*

$$\mathrm{QOT}^{\varepsilon}_C(A, B) = \max_{F\in\mathbb{H}_n,\ G\in\mathbb{H}_m} \left\{\mathrm{tr}(FA) + \mathrm{tr}(GB) - \varepsilon\, \mathrm{tr}\exp\left(\frac{F \otimes \mathrm{Id}_m + \mathrm{Id}_n \otimes G - C}{\varepsilon}\right)\right\}. \tag{12.60}$$

*At optimality, primal and dual variables are linked by the Gibbs formula*

$$T_e(F, G) = \exp\left(\frac{F \otimes \mathrm{Id}_m + \mathrm{Id}_n \otimes G - C}{\varepsilon}\right), \tag{12.61}$$

*with $\mathrm{Tr}_B(T_e) = A$ and $\mathrm{Tr}_A(T_e) = B$.*

*Proof.* The feasible set is compact and nonempty, and it contains the positive definite point $A \otimes B$. The trace entropy is strictly convex on positive semidefinite matrices, hence the regularized primal has a unique minimizer. This minimizer is positive definite: if it had a non-trivial kernel, moving toward $A \otimes B$ would have entropy directional derivative $-\infty$ at the boundary while changing the linear cost only at finite rate. Slater's condition justifies the Lagrange dual computation. The Fenchel identity $\sup_{T\succeq 0}\ \mathrm{tr}(YT) - \varepsilon H(T) = \varepsilon\, \mathrm{tr}\exp(Y/\varepsilon)$ is the matrix analogue of the scalar exponential conjugacy. Applying it to the Lagrangian of (**??**), with $Y = F \otimes \mathrm{Id}_m + \mathrm{Id}_n \otimes G - C$, gives (12.60). The stationarity condition of this Fenchel identity gives (12.61); differentiating the dual objective with respect to $F$ and $G$ yields the two marginal equations. ☐

$$D_H(T|K) = \mathrm{tr}(T(\log T - \log K) - T + K)\,.$$

$\mathcal{M}_A = \{T \succeq 0 : \mathrm{Tr}_B(T) = A\}, \qquad \mathcal{M}_B = \{T \succeq 0 : \mathrm{Tr}_A(T) = B\}.$

**Proposition 12.45** (Exact Bregman projections)**.** *Assume $A, B \succ 0$ and let $K = \exp(-C/\varepsilon)$. The minimizer of (**??**) is equivalently the minimizer of $D_H(T|K)$ over $\mathcal{M}_A \cap \mathcal{M}_B$. Moreover, if a current positive definite matrix has Gibbs form $T_e(F, G)$, then its Bregman projection onto $\mathcal{M}_A$ has the form $T_e(F^+, G)$, where $F^+$ is chosen so that $\mathrm{Tr}_B\, T_e(F^+, G) = A$. The projection onto $\mathcal{M}_B$ is analogous. Thus, when each one-block marginal equation is solved exactly, alternating Bregman projections are equivalent to alternating block maximization of the dual* (12.60)*.*

*Proof.* Since $\log K = -C/\varepsilon$, the identity $\mathrm{tr}(CT) + \varepsilon H(T) = \varepsilon D_H(T|K) - \varepsilon\, \mathrm{tr}(K)$ holds, so the primal minimizer is the constrained Bregman projection of $K$ up to an additive constant. For the projection of a positive definite matrix $S$ onto $\mathcal{M}_A$, the affine set contains the positive definite point $A \otimes \mathrm{Id}_m/m$; the entropy derivative $\log T - \log S$ is singular at the boundary, so the projection lies in the interior of the positive cone. Its Lagrangian first variation is $\log T - \log S - \Lambda \otimes \mathrm{Id}_m = 0$ for a Hermitian multiplier $\Lambda$. Hence $T = \exp(\log S + \Lambda \otimes \mathrm{Id}_m)$. If $S = T_e(F, G)$, this is again of the form $T_e(F + \varepsilon\Lambda, G)$. The multiplier is fixed by the marginal equation $\mathrm{Tr}_B(T) = A$. The same argument applies to $\mathcal{M}_B$. Finally, the first-order optimality condition for maximizing (12.60) over one block is exactly the corresponding marginal equation, so the Bregman and block-dual views coincide. ☐

$\mathrm{Tr}_B\, T_e(F, G) = A, \qquad \mathrm{Tr}_A\, T_e(F, G) = B$

**Gurvits scaling and quantum Sinkhorn.** Exact Bregman projection onto either non-commutative marginal constraint requires solving a nonlinear matrix equation and has no closed form in general. Gurvits scaling instead gives an explicit fixed-point construction of a feasible quantum coupling, used as a surrogate for entropic QOT. It does not, in general, minimize the entropic QOT objective or arise as block minimization of an underlying energy; in the diagonal case it coincides with classical Sinkhorn and recovers its variational interpretation.

$$T_s(F, G) = \exp\left(\frac{Z}{2\varepsilon}\right) \exp(-C/\varepsilon) \exp\left(\frac{Z}{2\varepsilon}\right) = (U \otimes V)K(U \otimes V), \qquad Z = F \otimes \mathrm{Id}_m + \mathrm{Id}_n \otimes G, \tag{12.62}$$

where $U = \exp(F/(2\varepsilon))$, $V = \exp(G/(2\varepsilon))$ and $K = \exp(-C/\varepsilon)$. For matrices $M, N$, the commutator is $[M, N] \coloneqq MN - NM$. Thus $[Z, C] = 0$ means that $Z$ and $C$ commute; then $T_s(F, G) = T_e(F, G)$, whereas otherwise $T_s$ is a Strang-type symmetric surrogate.
Using the product bases fixed above, associate with $K$ the positive map $\mathcal{K} : \mathbb{H}_m \to \mathbb{H}_n$ and its Hilbert–Schmidt adjoint through

$$\mathcal{K}(X) \coloneqq \mathrm{Tr}_B\big(K(\mathrm{Id}_n \otimes X)\big), \qquad \mathcal{K}^\star(Y) \coloneqq \mathrm{Tr}_A\big(K(Y \otimes \mathrm{Id}_m)\big). \tag{12.63}$$

In indices, $[\mathcal{K}(X)]_{i,k} = \sum_{j,\ell} K_{(i,j),(k,\ell)} X_{\ell,j}$ and $[\mathcal{K}^\star(Y)]_{j,\ell} = \sum_{i,k} K_{(i,j),(k,\ell)} Y_{k,i}$. Thus $\operatorname{tr}(Y\mathcal{K}(X)) = \operatorname{tr}(\mathcal{K}^\star(Y)X)$. The standard completely positive Choi map is $X \mapsto \mathcal{K}(X^\top)$; convention (12.63) moves this basis-dependent transpose out of the marginal equations, which become exactly
$U\,\mathcal{K}(V^2)\,U = A, \qquad V\,\mathcal{K}^\star(U^2)\,V = B.$

$$\begin{aligned} R_V &= \mathcal{K}(V^2), \quad U \leftarrow R_V^{-1/2}\big(R_V^{1/2} A R_V^{1/2}\big)^{1/2} R_V^{-1/2}, \\ S_U &= \mathcal{K}^\star(U^2), \quad V \leftarrow S_U^{-1/2}\big(S_U^{1/2} B S_U^{1/2}\big)^{1/2} S_U^{-1/2}. \end{aligned} \tag{12.64}$$

These inverse square roots are well-defined when $K \succ 0$ and $U, V, A, B \succ 0$. This is Gurvits/operator scaling with prescribed targets.

## 12.5 Dynamic Time Warping

**Ordered alignments versus transport couplings.**

**Discrete variational problem.** Let $x = (x_i)_{i=1}^n$ and $y = (y_j)_{j=1}^m$ be two sequences in a feature space $\mathcal{Z}$, and set $\mathrm{C}_{i,j} = c(x_i, y_j)$ for a nonnegative cost $c : \mathcal{Z} \times \mathcal{Z} \to \mathbb{R}_+$. A warping path is a sequence
$\omega = ((i_\ell, j_\ell))_{\ell=1}^L$ $(i_{\ell+1} - i_\ell, j_{\ell+1} - j_\ell) \in \{(1,0),(0,1),(1,1)\}$. Denote the set of such paths by $\Omega_{n,m}$ and the corresponding incidence matrix by $(A_\omega)_{i,j} = \mathbf{1}_{\{(i,j)\in\omega\}}$. Its length satisfies $\max\{n,m\} \leqslant L \leqslant n+m-1$, its total mass is $\sum_{i,j}(A_\omega)_{i,j} = L$, and its two marginals are precisely the row and column visit counts.

**Definition 12.46** (Dynamic time warping)**.** The DTW value associated with the feature cost $c$ is

$$\mathrm{DTW}_c(x,y) \coloneqq \min_{\omega\in\Omega_{n,m}} \sum_{\ell=1}^{L} c(x_{i_\ell}, y_{j_\ell}) = \min_{\omega\in\Omega_{n,m}} \langle A_\omega,\, \mathrm{C}\rangle. \tag{12.65}$$

**Dynamic programming.**

**Proposition 12.47** (DTW dynamic-programming recurrence)**.** *Set $D_{0,0} = 0$, $D_{i,0} = D_{0,j} = +\infty$ for $i, j > 0$, and, for $1 \leqslant i \leqslant n$, $1 \leqslant j \leqslant m$, define*

$$D_{i,j} = \mathrm{C}_{i,j} + \min\{D_{i-1,j}, D_{i,j-1}, D_{i-1,j-1}\}. \tag{12.66}$$

*Then $D_{n,m} = \mathrm{DTW}_c(x,y)$. Storing one minimizing predecessor at every cell and backtracking from $(n,m)$ recovers an optimal path. The computation takes $O(nm)$ time and $O(nm)$ memory; if only the value is required, two rows give $O(\min\{n,m\})$ memory.*

*Proof.* Every admissible path reaching $(i,j)$ enters it from exactly one of $(i-1,j)$, $(i,j-1)$ or $(i-1,j-1)$. Removing the last cell therefore leaves an admissible path to one of these predecessors, while appending $(i,j)$ to any predecessor path gives an admissible path to $(i,j)$. Minimizing over the three mutually exhaustive last steps proves (12.66) by induction on $i+j$. The complexity follows because each of the $nm$ cells performs constant work. □

**Continuous time warping.** For simplicity, let $x, y : [0,1] \to \mathcal{Z}$ use normalized clocks. The symmetric continuous registration problem uses two monotone clocks.

**Definition 12.48** (Continuous dynamic time warping)**.** Let $\Gamma_\uparrow$ contain pairs $(\varphi,\psi)$ of absolutely continuous, nondecreasing surjections of $[0,1]$ onto itself. Define

$$\mathrm{CDTW}_c(x,y) \coloneqq \inf_{(\varphi,\psi)\in\Gamma_\uparrow} \int_0^1 c\big(x(\varphi(s)), y(\psi(s))\big)\big(\dot\varphi(s) + \dot\psi(s)\big)\mathrm{d}s. \tag{12.67}$$

**Soft-DTW and the Sinkhorn analogy.**

$$\mathrm{softmin}_\varepsilon(r_1,\dots,r_q) = -\varepsilon\log\left(\sum_{k=1}^q e^{-r_k/\varepsilon}\right), \tag{12.68}$$

$$D^\varepsilon_{i,j} = \mathrm{C}_{i,j} + \mathrm{softmin}_\varepsilon\big(D^\varepsilon_{i-1,j}, D^\varepsilon_{i,j-1}, D^\varepsilon_{i-1,j-1}\big), \qquad \mathrm{sDTW}_{c,\varepsilon}(x,y) = D^\varepsilon_{n,m}. \tag{12.69}$$

$$\mathrm{sDTW}_{c,\varepsilon}(x,y) = -\varepsilon\log\sum_{\omega\in\Omega_{n,m}}\exp\left(-\frac{\langle A_\omega,\,\mathrm{C}\rangle}{\varepsilon}\right). \tag{12.70}$$

$$\mathrm{sDTW}_{c,\varepsilon}(x,y) = \min_{q\in\Delta(\Omega_{n,m})}\left\{\sum_\omega q_\omega\langle A_\omega,\,\mathrm{C}\rangle - \varepsilon H(q)\right\}. \tag{12.71}$$

$$\mathbb{P}_\varepsilon(\omega) = \frac{\exp(-\langle A_\omega,\,\mathrm{C}\rangle/\varepsilon)}{\sum_{\omega'}\exp(-\langle A_{\omega'},\,\mathrm{C}\rangle/\varepsilon)}, \qquad E_\varepsilon \coloneqq \nabla_{\mathrm{C}}\mathrm{sDTW}_{c,\varepsilon}(x,y). \tag{12.72}$$

$$\mathrm{DTW}_c(x,y) - \varepsilon\log|\Omega_{n,m}| \leqslant \mathrm{sDTW}_{c,\varepsilon}(x,y) \leqslant \mathrm{DTW}_c(x,y), \tag{12.73}$$

because the partition sum is bounded below by its largest term and above by $|\Omega_{n,m}|$ times that term. Hence $\mathrm{sDTW}_{c,\varepsilon} \to \mathrm{DTW}_c$ as $\varepsilon \to 0$.

$$E_\varepsilon = \mathbb{E}_{\omega\sim\mathbb{P}_\varepsilon}[A_\omega]. \tag{12.74}$$

Thus $(E_\varepsilon)_{i,j}$ is the probability that a Gibbs path visits cell $(i,j)$. The matrix $E_\varepsilon$ is a diffuse expected alignment and converges, when the hard optimum is unique, to its path-incidence matrix.

$$\overline{\mathrm{sDTW}}_{c,\varepsilon}(x,y) = \mathrm{sDTW}_{c,\varepsilon}(x,y) - \frac{1}{2}\mathrm{sDTW}_{c,\varepsilon}(x,x) - \frac{1}{2}\mathrm{sDTW}_{c,\varepsilon}(y,y). \tag{12.75}$$

# 13 Dynamic Optimal Transport

## 13.1 Evolutions over the Space of Measures

**Lagrangian and Eulerian descriptions.**

$$\frac{\mathrm{d}x(t)}{\mathrm{d}t} = v_t(x(t)), \tag{13.1}$$

$$\frac{\partial \alpha_t}{\partial t} + \operatorname{div}(v_t \alpha_t) = 0. \tag{13.2}$$

$$\int_0^1 \int_{\mathbb{R}^d} \left(\partial_t \varphi(t,x) + \langle v_t(x),\, \nabla_x \varphi(t,x)\rangle\right) \mathrm{d}\alpha_t(x)\mathrm{d}t = 0. \tag{13.3}$$

This equation is obtained from (13.2) by integration by parts. Hence, for smooth positive densities, the classical and weak formulations are equivalent; the weak formulation is useful because it still makes sense for discrete measures whose particles evolve according to (13.1).

**Definition 13.1** (Admissible continuity-equation evolution)**.** A narrowly continuous curve $(\alpha_t)_{t\in[0,1]}$ and a Borel velocity field $v$ are admissible between $\alpha_0$ and $\alpha_1$ if $\int_0^1 \int \|v_t(x)\|^2 \mathrm{d}\alpha_t(x)\mathrm{d}t < +\infty$ and $\partial_t \alpha_t + \operatorname{div}(\alpha_t v_t) = 0$ in distributions, with the prescribed endpoints and no flux through the boundary when the domain is bounded.

**Proposition 13.2** (Lagrangian flows solve the continuity equation)**.** *Let $(T_t)_{t\in[0,1]}$ be a $C^1$ family of diffeomorphisms of $\mathbb{R}^d$ and define $\alpha_t = (T_t)_\sharp \alpha_0$. Assume that the derivatives below are integrable with respect to $\alpha_0$, and define the Eulerian velocity field by $v_t(T_t(y)) = \partial_t T_t(y)$. Then $(\alpha_t, v_t)$ solves the continuity equation in the weak sense (13.3). In particular, if $\alpha_0 = \frac{1}{n}\sum_i \delta_{x_i(0)}$ is empirical, then $\alpha_t = \frac{1}{n}\sum_i \delta_{x_i(t)}$ is empirical as well, with particle velocities $\dot{x}_i(t) = v_t(x_i(t))$.*

*Proof.* Let $\varphi \in C_c^1((0,1)\times\mathbb{R}^d)$. Since $\alpha_t = (T_t)_\sharp \alpha_0$, $\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(t,x)\mathrm{d}\alpha_t(x) = \frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(t,T_t(y))\mathrm{d}\alpha_0(y)$. The chain rule gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(t,T_t(y))\mathrm{d}\alpha_0(y) = \int \left(\partial_t\varphi(t,T_t(y)) + \langle \nabla_x \varphi(t,T_t(y)),\, \partial_t T_t(y)\rangle\right)\mathrm{d}\alpha_0(y).$$

Using the definition of $v_t$ and the push-forward relation, this equals

$$\int \left(\partial_t \varphi(t,x) + \langle \nabla_x\varphi(t,x),\, v_t(x)\rangle\right)\mathrm{d}\alpha_t(x).$$

Integrating in time and using the boundary values of $\varphi$ gives (13.3). □

**From measure evolutions to vector fields.** For a given evolution $(\alpha_t)_t$, there are typically infinitely many velocity fields $v_t$ satisfying

$$\partial_t \alpha_t + \operatorname{div}(\alpha_t v_t) = 0. \tag{13.4}$$

$\mathcal{H}_\alpha = \left\{ v \in L^2(\alpha;\mathbb{R}^d)\ ;\ \operatorname{div}(\alpha v) = 0 \text{ in distributions}\right\}.$

**Dacorogna–Moser inversion.**

$$v_t = -\frac{1}{\rho_t}\nabla\Delta^{-1}(\partial_t\rho_t), \tag{13.5}$$

$\alpha_t = (1-t)\alpha_0 + t\alpha_1 = \rho_t\,\mathrm{d}x, \qquad \rho_t = (1-t)\rho_0 + t\rho_1.$ $\operatorname{div} w = \rho_0 - \rho_1, \qquad w\cdot n = 0$ on $\partial\Omega$, for instance $w = -\nabla\varphi$ with $\Delta\varphi = \rho_1 - \rho_0$ and Neumann boundary condition. Then
$v_t = \frac{w}{\rho_t}$ satisfies $\partial_t\rho_t + \operatorname{div}(\rho_t v_t) = 0$. The flow $\partial_t T_t = v_t \circ T_t$, $T_0 = \mathrm{Id}$, therefore transports $\rho_0 \mathrm{d}x$ onto $\rho_t\mathrm{d}x$, and $T_1$ solves the prescribed-Jacobian problem $\rho_1(T_1(x))\det(\nabla T_1(x)) = \rho_0(x)$.

**Least-square inversion and gradient structure.**

$$\min_v \frac{1}{2}\int_0^1\int_{\mathbb{R}^d} \|v_t(x)\|^2\,\mathrm{d}\alpha_t(x)\mathrm{d}t \quad \text{subject to} \quad \partial_t\alpha_t + \operatorname{div}(\alpha_t v_t) = 0. \tag{13.6}$$

**Proposition 13.3** (Least-square velocities are gradients)**.** *Assume that $\alpha_t = \rho_t\,\mathrm{d}x$ is a smooth positive density curve, that $\partial_t\rho_t$ has zero integral, and that boundary terms vanish. The minimizer of (13.6), if it exists, is a gradient field $v_t = \nabla\varphi_t$, where $\varphi_t$, unique up to an additive constant on each connected component, solves the weighted Poisson equation*

$$-\operatorname{div}(\rho_t\nabla\varphi_t) = \partial_t\rho_t, \qquad v_t = -\nabla\Delta_{\alpha_t}^{-1}(\partial_t\alpha_t), \quad \Delta_{\alpha_t}\varphi = \operatorname{div}(\alpha_t\nabla\varphi). \tag{13.7}$$

*Proof.* Introduce a Lagrange multiplier $\varphi_t$ for the continuity equation. The constrained problem has the formal saddle formulation

$$\min_v \max_\varphi \int_0^1 \left[\frac{1}{2}\int_{\mathbb{R}^d}\|v_t(x)\|^2\,\mathrm{d}\alpha_t(x) + \int_{\mathbb{R}^d}\varphi_t(x)\big(\operatorname{div}(\alpha_t v_t)(x) + \partial_t\alpha_t(x)\big)\mathrm{d}x\right]\mathrm{d}t.$$

Integrating by parts in the divergence term gives, for each $t$,

$$\int\left(\frac{1}{2}\|v_t\|^2 - \langle\nabla\varphi_t,\, v_t\rangle\right)\mathrm{d}\alpha_t + \int \varphi_t\,\partial_t\alpha_t.$$

The pointwise minimizer in $v_t$ is therefore $v_t = \nabla\varphi_t$. Substituting this into the constraint $\partial_t\rho_t + \operatorname{div}(\rho_t v_t) = 0$ gives the weighted Poisson equation in (13.7). The inverse notation is just a shorthand for solving this equation on zero-mean right-hand sides, modulo additive constants. □

## 13.2 Benamou–Brenier dynamic formulation of OT

**Benamou–Brenier formulation.**

**Definition 13.4** (Benamou–Brenier action)**.** For an admissible pair $(\alpha_t, v_t)$, define $\mathrm{Act}_2(\alpha, v) := \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 \mathrm{d}\alpha_t(x)\mathrm{d}t$.

**Theorem 13.5** (Benamou–Brenier)**.** *For probability measures $\alpha_0, \alpha_1 \in \mathcal{P}_2(\mathbb{R}^d)$,*

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{(\alpha_t, v_t)} \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 \, \mathrm{d}\alpha_t(x)\mathrm{d}t, \tag{13.8}$$

*where the infimum is over $(\alpha_t, v_t)$ solving $\partial_t \alpha_t + \nabla \cdot (\alpha_t v_t) = 0$ with $\alpha_{t=0} = \alpha_0$ and $\alpha_{t=1} = \alpha_1$. If $\alpha_0$ has a density and $T$ is the optimal Monge map $T_\sharp \alpha_0 = \alpha_1$, a minimizing curve is*

$$\alpha_t = ((1-t)\mathrm{Id} + tT)_\sharp \alpha_0, \qquad v_t\big((1-t)x + tT(x)\big) = T(x) - x \quad \textit{for } \alpha_0\textit{-a.e. } x \textit{ and a.e. } t \in (0,1). \tag{13.9}$$

*Proof.* For the inequality "dynamic $\leqslant$ static", assume first that a Monge map $T$ exists and define $(\alpha_t, v_t)$ by (13.9). Since the Lagrangian velocity $T(x) - x$ is independent of $t$, $\int_0^1 \int \|v_t\|^2 \, \mathrm{d}\alpha_t \mathrm{d}t = \int \|T(x) - x\|^2 \, \mathrm{d}\alpha_0(x)$, so the dynamic cost is no larger than the static Monge cost. Without a Monge map, the same construction is made with an optimal coupling $\pi$: sample $(X, Y) \sim \pi$ and move along the straight path $\gamma_{X,Y}(t) = (1-t)X + tY$. This path measure has action $\int \|x - y\|^2 \mathrm{d}\pi(x, y)$; projecting path velocities onto their conditional mean at time $t$ gives an admissible Eulerian velocity with no larger action, so the dynamic value is no larger than the Kantorovich value.
Conversely, for a smooth deterministic path, take the flow $T_t$ defined by $\dot{T}_t = v_t \circ T_t$ and $T_0 = \mathrm{Id}$. Then $\alpha_t = (T_t)_\sharp \alpha_0$ and $(T_1)_\sharp \alpha_0 = \alpha_1$. Jensen's inequality gives $\|T_1(x) - x\|^2 \leqslant \int_0^1 \|v_t(T_t(x))\|^2 \mathrm{d}t$. After integration with respect to $\alpha_0$, the Monge cost is bounded above by the dynamic action. For general finite-energy solutions of the continuity equation, the superposition principle lifts the curve to a probability measure on absolutely continuous paths; applying Jensen's inequality pathwise gives a coupling of the endpoints whose quadratic cost is no larger than the action. Thus the Kantorovich value is bounded above by the dynamic value. ☐

**Convex moment-based reformulation.** Although (13.8) is not jointly convex in $(\alpha_t, v_t)$, it becomes convex after replacing velocities by momenta. Given a velocity $v \in L^2(\alpha; \mathbb{R}^d)$, define the momentum
$\omega := \alpha v, \qquad \omega(A) = \int_A v(x) \, \mathrm{d}\alpha(x),$

**Definition 13.6** (Quadratic perspective action)**.**

$$J(a, m) := \begin{cases} \|m\|^2 / a, & a > 0, \\ 0, & a = 0 \text{ and } m = 0, \\ +\infty, & a = 0 \text{ and } m \neq 0, \end{cases} \qquad (a, m) \in [0, +\infty) \times \mathbb{R}^d. \tag{13.10}$$

If $\lambda$ dominates $\alpha$ and $|\omega|$, define

$$\mathbb{J}(\alpha, \omega) := \int J\left(\frac{\mathrm{d}\alpha}{\mathrm{d}\lambda}(x), \frac{\mathrm{d}\omega}{\mathrm{d}\lambda}(x)\right) \mathrm{d}\lambda(x). \tag{13.11}$$

$$\mathbb{J}(\alpha, \omega) < +\infty \quad \iff \quad \omega = v\alpha \text{ with } v \in L^2(\alpha; \mathbb{R}^d), \qquad \mathbb{J}(\alpha, \omega) = \int \|v\|^2 \, \mathrm{d}\alpha.$$

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{\substack{\partial_t \alpha_t + \mathrm{div}\, \omega_t = 0 \\ \alpha_{t=0} = \alpha_0, \ \alpha_{t=1} = \alpha_1}} \int_0^1 \mathbb{J}(\alpha_t, \omega_t) \, \mathrm{d}t. \tag{13.12}$$

$$\inf_{\substack{\partial_t \rho_t + \mathrm{div}\, m_t = 0 \\ \rho_{t=0} \mathrm{d}x = \alpha_0, \ \rho_{t=1} \mathrm{d}x = \alpha_1}} \int_0^1 \int_{\mathbb{R}^d} J(\rho_t(x), m_t(x)) \, \mathrm{d}x \, \mathrm{d}t = \inf_{\substack{\partial_t \rho_t + \mathrm{div}\, m_t = 0 \\ \rho_{t=0} \mathrm{d}x = \alpha_0, \ \rho_{t=1} \mathrm{d}x = \alpha_1}} \int_0^1 \int_{\mathbb{R}^d} \frac{\|m_t(x)\|^2}{\rho_t(x)} \, \mathrm{d}x \, \mathrm{d}t,$$

**Dual Hamilton–Jacobi formulation.** The momentum formulation also has a useful dual. It turns the least-action problem into a Hamilton–Jacobi subsolution inequality for a scalar potential, with equality on the part of space-time actually visited by the optimal curve.
**Proposition 13.7** (Dual Benamou–Brenier problem)**.** *Assume, for simplicity, that the densities are smooth, compactly supported, and that boundary terms vanish. Then the convex dynamic value* (13.12) *has the dual formulation*

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \sup_{\varphi} \left\{ \int_{\mathbb{R}^d} \varphi_1 \, \mathrm{d}\alpha_1 - \int_{\mathbb{R}^d} \varphi_0 \, \mathrm{d}\alpha_0 \; ; \; \partial_t \varphi_t + \frac{1}{4} \|\nabla \varphi_t\|^2 \leqslant 0 \right\}, \tag{13.13}$$

*where the supremum is over smooth test potentials $\varphi : [0,1] \times \mathbb{R}^d \to \mathbb{R}$. In the general metric-measure statement, the same formula is understood after relaxation: $\varphi$ is a Hamilton–Jacobi subsolution, for instance in the viscosity sense, and finite-dimensional discretizations follow directly from Fenchel–Rockafellar duality. If $(\rho, m)$ and $\varphi$ are smooth primal and dual optimizers, then*

$$m_t = \frac{\rho_t}{2} \nabla \varphi_t, \qquad \partial_t \varphi_t + \frac{1}{4} \|\nabla \varphi_t\|^2 = 0 \quad \textit{on } \{\rho_t > 0\}. \tag{13.14}$$

*Equivalently, the optimal Eulerian velocity is $v_t = m_t / \rho_t = \nabla \varphi_t / 2$.*

*Proof.* Let $(\rho, m)$ satisfy the continuity equation and let $\varphi$ be smooth. Multiplying the constraint by $\varphi$ and integrating by parts gives

$$\int_{\mathbb{R}^d} \varphi_1 \, \mathrm{d}\alpha_1 - \int_{\mathbb{R}^d} \varphi_0 \, \mathrm{d}\alpha_0 = \int_0^1 \int_{\mathbb{R}^d} (\rho_t \, \partial_t \varphi_t + \langle m_t, \nabla \varphi_t \rangle) \, \mathrm{d}x\mathrm{d}t.$$

If $\partial_t\varphi_t + \|\nabla\varphi_t\|^2/4 \leqslant 0$, Young's inequality yields, pointwise, $\rho\,\partial_t\varphi + \langle m, \nabla\varphi\rangle \leqslant -\frac{\rho}{4}\|\nabla\varphi\|^2 + \langle m, \nabla\varphi\rangle \leqslant J(\rho, m)$, with the perspective convention (13.10). Hence the dual objective of every feasible $\varphi$ is no larger than the action of every feasible primal pair.

Conversely, introduce $\varphi$ as a Lagrange multiplier for $\partial_t\rho + \operatorname{div} m = 0$. After integration by parts, and up to the fixed endpoint contribution in (13.13), the pointwise minimization over $(\rho, m)$ contains $J(\rho,m) - \rho\,\partial_t\varphi - \langle m, \nabla\varphi\rangle$. For fixed $\rho > 0$, its minimum over $m$ is attained at $m = \rho\nabla\varphi/2$ and equals $-\rho(\partial_t\varphi + \|\nabla\varphi\|^2/4)$. The recession cases in (13.10) give the same conclusion when $\rho = 0$. Thus the infimum over $\rho \geqslant 0$ is finite exactly when the Hamilton–Jacobi inequality holds, in which case the pointwise infimum is 0. Fenchel–Rockafellar duality then gives no duality gap. Equality in the two pointwise inequalities gives (13.14). ☐

$\frac{\mathrm{d}}{\mathrm{d}t}\varphi_t(\gamma(t)) = \partial_t\varphi_t(\gamma(t)) + \langle\nabla\varphi_t(\gamma(t)), \dot\gamma(t)\rangle \leqslant \|\dot\gamma(t)\|^2$, where the last inequality uses $\partial_t\varphi + \|\nabla\varphi\|^2/4 \leqslant 0$ and Young's inequality. Integrating and minimizing over curves gives
$\varphi_1(y) - \varphi_0(x) \leqslant \|x-y\|^2$. Thus $(-\varphi_0, \varphi_1)$ is a feasible static Kantorovich dual pair for the quadratic cost. At optimality the inequality is saturated on the endpoint pairs connected by the primal characteristics, which is the Eulerian version of complementary slackness.

**Proximal splitting.** The convex momentum formulation also explains the original Benamou–Brenier solver. Suppressing discretization indices, write $U = (\rho, m)$, where $m$ is the density of the momentum measure.

$$\mathcal{F}(U) = \int_0^1\int_{\mathbb{R}^d} J(\rho_t(x), m_t(x))\,\mathrm{d}x\,\mathrm{d}t, \qquad \mathcal{C} = \{(\rho, m)\ ;\ \partial_t\rho + \operatorname{div} m = 0,\ \rho_0\mathrm{d}x = \alpha_0,\ \rho_1\mathrm{d}x = \alpha_1\},$$

$\min_U\ \mathcal{F}(U) + \mathcal{G}(U)$.

$$\operatorname{prox}_{\tau\mathcal{F}}(\bar U) = \operatorname{argmin}_U \frac{1}{2}\|U - \bar U\|_{\mathsf{H}}^2 + \tau\mathcal{F}(U), \qquad \operatorname{prox}_{\tau\mathcal{G}}(\bar U) = \operatorname{argmin}_{U\in\mathcal{C}} \frac{1}{2}\|U - \bar U\|_{\mathsf{H}}^2.$$

For the standard product metric, the first map is local in $(t, x)$: it is the proximal operator of the convex perspective $J$. The second map is the orthogonal projection onto the affine set defined by the divergence equation and endpoint constraints. Douglas–Rachford then alternates these two simple operations:

$$\begin{aligned} U^{(\ell+1)} &= \operatorname{prox}_{\tau\mathcal{F}}(Z^{(\ell)}),\\ \widetilde U^{(\ell+1)} &= \operatorname{prox}_{\tau\mathcal{G}}(2U^{(\ell+1)} - Z^{(\ell)}),\\ Z^{(\ell+1)} &= Z^{(\ell)} + \widetilde U^{(\ell+1)} - U^{(\ell+1)}. \end{aligned}$$

**Path-space formulation.** Let $\mathcal{S} = C([0,1];\mathbb{R}^d)$ be the space of continuous paths endowed with the uniform topology. For $t \in [0,1]$ define the evaluation map
$e_t : \mathcal{S} \to \mathbb{R}^d, \qquad e_t(\gamma) = \gamma(t)$.

$$\mathcal{W}_2^2(\alpha_0, \alpha_1) = \inf_{M\in\mathcal{P}(\mathcal{S})}\left\{\int_{\mathcal{S}}\int_0^1 \|\dot\gamma(t)\|^2\mathrm{d}t\,\mathrm{d}M(\gamma)\ ;\ (e_0)_\sharp M = \alpha_0,\ (e_1)_\sharp M = \alpha_1\right\}.$$

$(e_t, \dot e_t)_\sharp M^*(\mathrm{d}x, \mathrm{d}q) = \alpha_t(\mathrm{d}x)\delta_{v_t^*(x)}(\mathrm{d}q)$, where $v_t^*$ is the optimal velocity field in the Benamou–Brenier formulation. Hence $M^*$ concentrates on straight-line geodesics and, for a.e. $t$, assigns exactly one direction at $\alpha_t$-a.e. spatial point.

### 13.3 Generalized Dynamic Wasserstein Distances

The quadratic Benamou–Brenier formula is only one instance of a broader fixed-mass dynamic language. The objects introduced here are metric: they specify admissible curves, tangent variables and path energies.

**Path actions.** In the mass-preserving Euclidean setting, the basic input is an instantaneous action $\mathbb{A}(\alpha, w)$, where $\alpha$ is the current measure and $w$ is an admissible velocity representative. Its endpoint geometry is defined dynamically.

**Definition 13.8** (Generalized dynamic action distance)**.** Let $\mathbb{A}$ be a nonnegative extended-valued action on measure–velocity pairs. Define

$$\mathsf{D}_{\mathbb{A}}^2(\alpha_0, \alpha_1) := \inf_{\alpha_t, v_t}\left\{\int_0^1 \mathbb{A}(\alpha_t, v_t)\,\mathrm{d}t : \partial_t\alpha_t + \operatorname{div}(\alpha_t v_t) = 0,\ \alpha_{t=0} = \alpha_0,\ \alpha_{t=1} = \alpha_1\right\}. \tag{13.15}$$

**Quadratic, or Riemannian, tangent actions.**

$$\mathbb{A}(\alpha; w, z) = \langle Q_\alpha w, z\rangle_{L^2(\alpha)}, \qquad \mathbb{A}(\alpha, w) = \langle Q_\alpha w, w\rangle_{L^2(\alpha)},$$

$$\mathsf{D}_Q^2(\alpha_0, \alpha_1) = \mathsf{D}_{\mathbb{A}}^2(\alpha_0, \alpha_1) = \inf_{\substack{\partial_t\alpha_t + \operatorname{div}(\alpha_t v_t) = 0\\ \alpha_{t=0}=\alpha_0,\ \alpha_{t=1}=\alpha_1}} \int_0^1 \langle Q_{\alpha_t} v_t, v_t\rangle_{L^2(\alpha_t)}\,\mathrm{d}t. \tag{13.16}$$

The usual $\mathcal{W}_2$ geometry corresponds to $Q_\alpha = \mathrm{Id}$ in this simplified notation. Thus $Q_\alpha$ records how the chosen geometry deforms the Euclidean $L^2(\alpha)$ tangent norm: no deformation for $\mathcal{W}_2$, and a nontrivial tensor for generalized Riemannian geometries.

**Local velocity actions.** $A : [0, +\infty) \times \mathbb{R}^d \to [0, +\infty], \qquad (a, w) \mapsto A(a, w)$, where $a \in \mathbb{R}_+$ is a density value and $w \in \mathbb{R}^d$ is a velocity value, and defines

$$\mathbb{A}(\alpha, w) = \int A\left(\frac{\mathrm{d}\alpha}{\mathrm{d}\lambda}(x), w(x)\right)\mathrm{d}\lambda(x). \tag{13.17}$$

For a fixed reference $\lambda$, this covers density-dependent mobilities and congestion constraints. If, in addition, $A$ is positively 1-homogeneous in its first variable, $A(\eta a, w) = \eta A(a, w)$ for $\eta \geqslant 0$, then the same formula is intrinsic: replacing $\lambda$ by another dominating measure gives the same value. The usual Benamou–Brenier action is the model case

$$A_2(a, w) = a\|w\|^2, \qquad \mathbb{A}(\alpha, w) = \int \|w\|^2\mathrm{d}\alpha. \tag{13.18}$$

**Homogeneous momentum actions.**

$$J_A(a,m) := \begin{cases} A(a,m/a), & a>0, \\ 0, & a=0 \text{ and } m=0, \\ +\infty, & a=0 \text{ and } m\neq 0, \end{cases} \tag{13.19}$$

$$\mathbb{J}_{A,\lambda}(\alpha,\omega) := \int J_A\left(\frac{\mathrm{d}\alpha}{\mathrm{d}\lambda}, \frac{\mathrm{d}\omega}{\mathrm{d}\lambda}\right) \mathrm{d}\lambda, \tag{13.20}$$

$J_2(a,m) = \frac{\|m\|^2}{a}$,

**Proposition 13.9** (Concave mobilities give convex momentum actions)**.** *Let $I \subset [0,+\infty)$ be a convex interval, let $\theta : I \to [0,+\infty)$ be concave, and let $L : \mathbb{R}^d \to [0,+\infty]$ be convex with $L(0)=0$. Define, on the set where $\theta(a)>0$,*

$$J_{\theta,L}(a,m) := \theta(a) L\left(\frac{m}{\theta(a)}\right), \qquad A_{\theta,L}(a,w) := J_{\theta,L}(a,aw) = \theta(a) L\left(\frac{aw}{\theta(a)}\right), \tag{13.21}$$

*and extend $J_{\theta,L}$ to the boundary by lower semicontinuity. Then $J_{\theta,L}$ is convex in $(a,m)$. In particular, this single construction contains $A(a,w) = aL(w)$ by taking $\theta(a)=a$, and the concave-mobility quadratic action $A(a,w) = \frac{a^2\|w\|^2}{\theta(a)}$ by taking $L(u) = \|u\|^2$.*

*Proof.* The perspective $P(s,m) = sL(m/s)$ of a convex function is convex on $s>0$. Since $L(0)=0$, it is also nonincreasing in the scalar $s$ for fixed $m$: if $s_2 \geqslant s_1 > 0$, then $L(m/s_2) = L\big((s_1/s_2)(m/s_1)\big) \leqslant (s_1/s_2)L(m/s_1)$, hence $P(s_2,m) \leqslant P(s_1,m)$. Let $a_\zeta = (1-\zeta)a_0 + \zeta a_1$ and $m_\zeta = (1-\zeta)m_0 + \zeta m_1$. By concavity of $\theta$, $\theta(a_\zeta) \geqslant (1-\zeta)\theta(a_0) + \zeta\theta(a_1)$. Using the monotonicity in $s$, and then the convexity of $P$, gives

$$J_{\theta,L}(a_\zeta,m_\zeta) = P(\theta(a_\zeta),m_\zeta) \leqslant P((1-\zeta)\theta(a_0)+\zeta\theta(a_1),m_\zeta) \leqslant (1-\zeta)J_{\theta,L}(a_0,m_0) + \zeta J_{\theta,L}(a_1,m_1).$$

The boundary cases follow by the chosen lower-semicontinuous extension. □

**Proposition 13.10** (Homogeneous dynamic actions define distances)**.** *Fix a reference measure $\lambda$, omitted from the notation only in the intrinsic case where $\mathbb{J}_{A,\lambda}$ does not depend on $\lambda$. Assume that the momentum perspective $J_A$ defined in (13.19) is lower semicontinuous, convex in $(a,m)$, even in $m$, and satisfies $J_A(a,0)=0$. Assume moreover that, for some $r>1$, $J_A(a,\xi m) = |\xi|^r J_A(a,m)$ for all admissible $a$, all $m$, and all $\xi \in \mathbb{R}$, and that $J_A(a,m)=0$ if and only if $m=0$. Equivalently, the evenness, homogeneity and nondegeneracy assumptions translate, away from zero density, into $A(a,-w) = A(a,w)$, $\quad A(a,\xi w) = |\xi|^r A(a,w)$, $\quad A(a,w)=0 \Longleftrightarrow w=0 \quad (a>0)$. Define, on every fixed-mass class,*

$$\mathsf{D}_{A,\lambda}(\alpha_0,\alpha_1) := \inf_{\substack{\partial_t\alpha_t + \operatorname{div}\omega_t = 0 \\ \alpha_{t=0}=\alpha_0,\ \alpha_{t=1}=\alpha_1}} \left(\int_0^1 \mathbb{J}_{A,\lambda}(\alpha_t,\omega_t)\,\mathrm{d}t\right)^{1/r}.$$

*Assume finally that the relaxed dynamic problem is sequentially closed and attains its infimum whenever the value is finite. Then $\mathsf{D}_{A,\lambda}$ is an extended distance on each finite-action component: it is symmetric, satisfies the triangle inequality, and $\mathsf{D}_{A,\lambda}(\alpha_0,\alpha_1) = 0$ only when $\alpha_0=\alpha_1$. In the intrinsic case, we write $\mathsf{D}_A$.*

*Proof.* The constant curve $\alpha_t = \alpha_0$, $\omega_t = 0$, gives $\mathsf{D}_{A,\lambda}(\alpha_0,\alpha_0)=0$. Conversely, if $\mathsf{D}_{A,\lambda}(\alpha_0,\alpha_1)=0$, attainment provides a relaxed zero-action minimizer, which satisfies $\int_0^1 \mathbb{J}_{A,\lambda}(\alpha_t,\omega_t)\,\mathrm{d}t = 0$. Since $\mathbb{J}_{A,\lambda} \geqslant 0$, this implies $\omega_t = 0$ for a.e. $t$. The continuity equation then gives $\partial_t\alpha_t = 0$ in the sense of distributions, so $\alpha_0=\alpha_1$.
Symmetry follows by time reversal. If $(\alpha_t,\omega_t)_{t\in[0,1]}$ connects $\alpha_0$ to $\alpha_1$, then $\widetilde{\alpha}_t = \alpha_{1-t}$, $\quad \widetilde{\omega}_t = -\omega_{1-t}$ connects $\alpha_1$ to $\alpha_0$, satisfies the same continuity equation, and has the same action because $J_A$ is even in $m$.
Let $(\alpha^1,\omega^1)$ connect $\alpha_0$ to $\alpha_1$ with action $E_1$, and let $(\alpha^2,\omega^2)$ connect $\alpha_1$ to $\alpha_2$ with action $E_2$. For $\tau \in (0,1)$, concatenate the two curves after the time changes $s = t/\tau$ and $s = (t-\tau)/(1-\tau)$:

$$(\alpha_t,\omega_t) = \begin{cases} \big(\alpha^1_{t/\tau},\ \tau^{-1}\omega^1_{t/\tau}\big), & 0 \leqslant t \leqslant \tau, \\ \big(\alpha^2_{(t-\tau)/(1-\tau)},\ (1-\tau)^{-1}\omega^2_{(t-\tau)/(1-\tau)}\big), & \tau \leqslant t \leqslant 1. \end{cases}$$

The rescaled curve is admissible from $\alpha_0$ to $\alpha_2$. By $r$-homogeneity in the momentum variable, its action is $\tau^{1-r}E_1 + (1-\tau)^{1-r}E_2$. Writing $X = E_1^{1/r}$ and $Y = E_2^{1/r}$, the optimal allocation is $\tau = X/(X+Y)$, with the evident limiting convention if $X+Y=0$, and gives

$$\inf_{\tau\in(0,1)} \left\{\tau^{1-r}E_1 + (1-\tau)^{1-r}E_2\right\} = (X+Y)^r.$$

Hence $\mathsf{D}_{A,\lambda}(\alpha_0,\alpha_2) \leqslant E_1^{1/r} + E_2^{1/r}$. Taking $E_1$ and $E_2$ arbitrarily close to the two optimal action values yields $\mathsf{D}_{A,\lambda}(\alpha_0,\alpha_2) \leqslant \mathsf{D}_{A,\lambda}(\alpha_0,\alpha_1) + \mathsf{D}_{A,\lambda}(\alpha_1,\alpha_2)$, with the usual extended-real convention when one of the distances is infinite. □

**Concave-mobility actions.**

**Definition 13.11** (Concave-mobility dynamic distance)**.** Let $I \subset [0,+\infty)$ be a convex interval and let $\theta : I \to [0,+\infty)$ be concave. Define

$$J_\theta(a,m) := \begin{cases} \|m\|^2/\theta(a), & \theta(a)>0, \\ 0, & \theta(a)=0 \text{ and } m=0, \\ +\infty, & \theta(a)=0 \text{ and } m\neq 0. \end{cases}$$

The corresponding velocity action is $A_\theta(a,w) = J_\theta(a,aw) = \frac{a^2\|w\|^2}{\theta(a)}$. For a fixed reference measure $\lambda$, set $\mathsf{W}_{\theta,\lambda}(\alpha_0,\alpha_1) := \mathsf{D}_{A_\theta,\lambda}(\alpha_0,\alpha_1)$.

$\mathbb{A}_{\theta,\lambda}(\alpha,w) = \int A_\theta(a(x),w(x))\,\mathrm{d}\lambda(x) = \int \frac{a(x)^2}{\theta(a(x))}\|w(x)\|^2\,\mathrm{d}\lambda(x)$, $\mathbb{A}_{\theta,\lambda}(\alpha,w) = \int \frac{a(x)}{\theta(a(x))}\|w(x)\|^2\,\mathrm{d}\alpha(x)$. Hence this is a local Riemannian case, in the sense of (13.16), whenever the multiplier $a(x)/\theta(a(x))$ is finite and positive. The associated tensor is the multiplication operator

$$(Q_{\theta,\lambda,\alpha}w)(x) = \frac{a(x)}{\theta(a(x))}w(x), \qquad \alpha = a\lambda, \tag{13.22}$$

defined $\alpha$-a.e. on the set where $\theta(a) > 0$. Except for linear mobilities $\theta(a) = ca$, and in particular the normalized case $\theta(a) = a$ which recovers $\mathcal{W}_2$, the pointwise velocity action $A_\theta(a,w)$ is not positively 1-homogeneous in the density variable $a$.

**Proposition 13.12** (Concave-mobility dynamic distances)**.** *For a fixed reference measure $\lambda$, and under the standard compactness and lower-semicontinuity hypotheses ensuring existence of relaxed minimizers, $\mathsf{W}_{\theta,\lambda} = \mathsf{D}_{A_\theta,\lambda}$ is an extended distance on each finite-action component contained in $\{\alpha : \alpha \ll \lambda\}$. In particular, whenever the relevant component has finite pairwise distances, $\mathsf{W}_{\theta,\lambda}$ is a genuine metric there.*

*Proof.* Proposition 13.9, applied with $L(u) = \|u\|^2$, gives the lower-semicontinuous convex momentum density $J_\theta$. It is even in $m$, satisfies $J_\theta(a,0) = 0$, and obeys $J_\theta(a,\xi m) = |\xi|^2 J_\theta(a,m)$. Moreover $J_\theta(a,m) = 0$ if and only if $m = 0$, with the boundary convention used in its definition. For the fixed reference $\lambda$, the hypotheses of Proposition 13.10 therefore hold with $r = 2$; the existence assumption is supplied by the compactness hypothesis in the statement. That proposition gives symmetry, separation and the triangle inequality for $\mathsf{D}_{A_\theta,\lambda}$, hence for $\mathsf{W}_{\theta,\lambda}$. □

**Dynamic spectral Wasserstein distances.**

**Definition 13.13** (Dynamic spectral Wasserstein distance)**.** Let $\gamma$ be a monotone spectral gauge on $\mathbb{S}^d_+$. Define

$$\mathbb{A}_\gamma(\alpha,v) := \gamma\left(\int v(x)v(x)^\top\mathrm{d}\alpha(x)\right). \tag{13.23}$$

The associated distance is $$\mathsf{W}^2_{\gamma,\mathrm{dyn}}(\alpha_0,\alpha_1) := \mathsf{D}^2_{\mathbb{A}_\gamma}(\alpha_0,\alpha_1). \tag{13.24}$$

$$\mathbb{J}_\gamma(\alpha,\omega) = \gamma\left(\int\left(\frac{\mathrm{d}\omega}{\mathrm{d}\alpha}\right)\left(\frac{\mathrm{d}\omega}{\mathrm{d}\alpha}\right)^\top\mathrm{d}\alpha\right),$$

$$\mathbb{J}_\gamma(\rho,m) = \gamma\left(\int\frac{m(x)m(x)^\top}{\rho(x)}\,\mathrm{d}x\right).$$

$A_{\mathrm{lin}}(a,w) = a\,w^\top G w, \qquad J_{\mathrm{lin}}(a,m) = \frac{m^\top G m}{a},$

**Proposition 13.14** (Static/dynamic equality for spectral Wasserstein)**.** *Let $\gamma$ be a monotone spectral gauge and let $\alpha_0,\alpha_1 \in \mathcal{P}_2(\mathbb{R}^d)$. Then $\mathsf{W}^2_{\gamma,\mathrm{dyn}}(\alpha_0,\alpha_1) = \mathcal{W}^2_\gamma(\alpha_0,\alpha_1)$. The common value defines a distance on $\mathcal{P}_2(\mathbb{R}^d)$; the triangle inequality is established by the robust representation in Proposition 10.38.*

*Proof.* We first prove $\mathsf{W}^2_{\gamma,\mathrm{dyn}} \leqslant \mathcal{W}^2_\gamma$. Let $\pi \in \mathcal{U}(\alpha_0,\alpha_1)$ and let $(X,Y) \sim \pi$. Set $Z_t = (1-t)X + tY$ and $\alpha_t = (Z_t)_\sharp\pi$. Define the Eulerian velocity as the conditional mean $v_t(z) := \mathbb{E}[Y - X \mid Z_t = z]$. Then $(\alpha_t, v_t)$ solves the continuity equation. If $M_\pi := \int(x-y)(x-y)^\top\mathrm{d}\pi(x,y), \qquad C_t := \int v_t(z)v_t(z)^\top\mathrm{d}\alpha_t(z)$, conditional Jensen gives $C_t \preceq M_\pi$. Indeed, for every direction $u \in \mathbb{R}^d$,

$$u^\top C_t u = \mathbb{E}\big[\mathbb{E}[\langle u, Y-X\rangle \mid Z_t]^2\big] \leqslant \mathbb{E}[(\langle u, Y-X\rangle)^2] = u^\top M_\pi u.$$

By the Loewner monotonicity of $\gamma$, $\int_0^1\gamma(C_t)\mathrm{d}t \leqslant \gamma(M_\pi)$. Taking the infimum over $\pi$ gives the first inequality.

Conversely, let $(\alpha_t, v_t)$ be an admissible finite-action competitor. Since $\gamma$ is a finite positive gauge on the finite-dimensional cone $\mathbb{S}^d_+$, it is equivalent to the trace on this cone; hence finite spectral action implies the usual finite kinetic energy needed for the superposition principle. Thus there exists a probability law $\eta$ on absolutely continuous paths $\omega$. such that $(e_t)_\sharp\eta = \alpha_t$ and $\dot\omega_t = v_t(\omega_t)$ for a.e. $t$. Let $\pi = (e_0,e_1)_\sharp\eta$ be the induced endpoint coupling. With $C_t = \int v_t v_t^\top\mathrm{d}\alpha_t$, Jensen's inequality along each path gives

$$(\omega_1-\omega_0)(\omega_1-\omega_0)^\top = \left(\int_0^1\dot\omega_t\mathrm{d}t\right)\left(\int_0^1\dot\omega_t\mathrm{d}t\right)^\top \preceq \int_0^1\dot\omega_t\dot\omega_t^\top\mathrm{d}t.$$

After integration over $\eta$, $M_\pi \preceq \int_0^1 C_t\mathrm{d}t$. Monotonicity of $\gamma$, followed by convexity, yields

$$\gamma(M_\pi) \leqslant \gamma\left(\int_0^1 C_t\mathrm{d}t\right) \leqslant \int_0^1\gamma(C_t)\mathrm{d}t.$$

Since $\pi$ is admissible for the static problem, $\mathcal{W}^2_\gamma(\alpha_0,\alpha_1)$ is bounded by the action of the dynamic competitor. Taking the infimum over all competitors gives the reverse inequality.

The crucial assumption is the monotonicity of $\gamma$: both parts of the proof produce Loewner-order comparisons between covariance matrices, and these comparisons imply inequalities between actions only for monotone gauges. □

**Kernelized Benamou–Brenier distances.**

**Definition 13.15** (Kernelized Benamou–Brenier distance)**.** Let $k$ be a positive definite kernel on $\mathbb{R}^d$ with scalar RKHS $\mathcal{H}_k$. Set $\mathcal{H}^d_k := \mathcal{H}_k \times\cdots\times\mathcal{H}_k, \qquad \|v\|^2_{\mathcal{H}^d_k} := \sum_{\ell=1}^d\|v_\ell\|^2_{\mathcal{H}_k}$.

$$\mathbb{A}_k(\alpha,v) := \|v\|^2_{\mathcal{H}^d_k}, \qquad \mathcal{W}^2_k(\alpha_0,\alpha_1) := \mathsf{D}^2_{\mathbb{A}_k}(\alpha_0,\alpha_1). \tag{13.25}$$

**Proposition 13.16** (Kernelized dynamic distance)**.** *Assume that the kernel has uniformly bounded diagonal, $\kappa_k^2 \coloneqq \sup_{x\in\mathbb{R}^d} k(x,x) < +\infty$. Then the quantity $\mathcal{W}_k$ defined by (13.25) is an extended distance on probability measures. More precisely, it is symmetric, satisfies the triangle inequality, and $\mathcal{W}_k(\alpha_0,\alpha_1)=0$ implies $\alpha_0=\alpha_1$. It can take the value $+\infty$ between different finite-action components.*

*Proof.* The constant curve gives zero self-distance. To prove separation without assuming existence of a minimizing curve, use the RKHS evaluation bound $\|v(x)\| \leqslant \kappa_k \|v\|_{\mathcal{H}_k^d}$. For every $\varphi \in C_c^1(\mathbb{R}^d)$ and every admissible curve of action $E$, the weak continuity equation and Cauchy–Schwarz give

$$\begin{aligned}\left|\int \varphi \, \mathrm{d}(\alpha_1-\alpha_0)\right| &\leqslant \int_0^1 \int \|\nabla\varphi(x)\| \, \|v_t(x)\| \, \mathrm{d}\alpha_t(x)\mathrm{d}t \\ &\leqslant \kappa_k \|\nabla\varphi\|_\infty \left(\int_0^1 \|v_t\|^2_{\mathcal{H}_k^d}\mathrm{d}t\right)^{1/2} = \kappa_k\|\nabla\varphi\|_\infty\sqrt{E}.\end{aligned}$$

Taking the infimum over admissible curves shows that $\mathcal{W}_k(\alpha_0,\alpha_1)=0$ forces equality of the integrals of every compactly supported smooth test function, hence $\alpha_0=\alpha_1$.
Symmetry follows by time reversal and by replacing $v_t$ with $-v_{1-t}$. For the triangle inequality, concatenate an almost optimal curve from $\alpha_0$ to $\alpha_1$ of action $E_1$ with one from $\alpha_1$ to $\alpha_2$ of action $E_2$. Compressing the first curve into a time interval of length $\tau$ multiplies its action by $1/\tau$, and compressing the second into length $1-\tau$ multiplies its action by $1/(1-\tau)$. Optimizing over $\tau\in(0,1)$ gives $(\sqrt{E_1}+\sqrt{E_2})^2$, and taking infima yields the triangle inequality. ☐

One should read $\mathcal{W}_k$ as an extended distance on finite-action components, not as a replacement for $\mathcal{W}_2$ on all of $\mathcal{P}_2(\mathbb{R}^d)$.

## 13.4 Nonlocal Wasserstein Distances

**Definition 13.17** (Logarithmic mean)**.**

$$\theta(a,b) \coloneqq \begin{cases} \dfrac{a-b}{\log a - \log b}, & a,b>0 \text{ and } a\neq b,\\ a, & a=b,\\ 0, & ab=0,\end{cases} \tag{13.26}$$

for $a,b\geqslant 0$.

**Continuum jump kernels.**

$$\int_{\mathcal{X}\times\mathcal{X}} \Phi(x,y)\,\mathsf{J}(\mathrm{d}x,\mathrm{d}y) \coloneqq \int_{\mathcal{X}}\left(\int_{\mathcal{X}}\Phi(x,y)\,K(x,\mathrm{d}y)\right)\mathfrak{m}(\mathrm{d}x). \tag{13.27}$$

$\bar\nabla\varphi(x,y) \coloneqq \varphi(y)-\varphi(x)$ for the nonlocal gradient, and use the logarithmic mean $\theta$ defined in (13.26). These objects define the nonlocal continuity equation and its least-action distance.

**Definition 13.18** (Continuum nonlocal Wasserstein distance)**.** For $\alpha_t=\rho_t\mathfrak{m}$ and antisymmetric $v_t(x,y)$, require, for every test function $\varphi$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int\varphi\,\mathrm{d}\alpha_t = \frac12\iint \bar\nabla\varphi(x,y)\,v_t(x,y)\,\theta(\rho_t(x),\rho_t(y))\,\mathsf{J}(\mathrm{d}x,\mathrm{d}y). \tag{13.28}$$

Define $\mathbb{A}_K(\alpha,v) \coloneqq \frac12\iint |v(x,y)|^2\theta(\rho(x),\rho(y))\,\mathsf{J}(\mathrm{d}x,\mathrm{d}y)$,

and
$$\mathcal{W}_K^2(\alpha_0,\alpha_1) \coloneqq \inf_{\rho_t,v_t}\int_0^1 \mathbb{A}_K(\alpha_t,v_t)\,\mathrm{d}t, \tag{13.29}$$

where the infimum is over all curves solving (13.28) with endpoints $\alpha_0,\alpha_1$.

**Proposition 13.19** (Metric properties of nonlocal transport)**.** *Under standard regularity and irreducibility assumptions on the jump kernel, $\mathcal{W}_K$ is an extended distance on the set of probability measures absolutely continuous with respect to $\mathfrak{m}$, and finite-distance pairs are connected by constant-speed geodesics.*

*Proof.* We use the analytic compactness and lower-semicontinuity theorem for the logarithmic-mean action. In particular, action-bounded sequences of admissible curves are compact for the narrow topology, the weak nonlocal continuity equation is closed under this convergence, and the action is lower semicontinuous. These facts are the nonlocal analogues of the compactness theorem used in the Benamou–Brenier formulation.
Nonnegativity is immediate from the definition of $\mathbb{A}_K(\alpha,v)$. If $\alpha_0=\alpha_1$, the constant curve $\rho_t=\rho_0$, $v_t=0$, is admissible and has zero action, hence $\mathcal{W}_K(\alpha_0,\alpha_0)=0$.
Symmetry follows by time reversal. If $(\rho_t,v_t)$ transports $\alpha_0$ to $\alpha_1$, set $\tilde\rho_t=\rho_{1-t}$, $\tilde v_t=-v_{1-t}$. For every test function $\varphi$, differentiating $\int\varphi\tilde\rho_t\,\mathrm{d}\mathfrak{m}$ and using (13.28) shows that $(\tilde\rho_t,\tilde v_t)$ solves the same continuity equation from $\alpha_1$ to $\alpha_0$. Since the action is quadratic in $v$, $\mathbb{A}_K(\tilde\alpha_t,\tilde v_t)=\mathbb{A}_K(\alpha_{1-t},v_{1-t})$, so the action is unchanged.
For the triangle inequality, let $(\rho_t^0,v_t^0)$ connect $\alpha_0$ to $\alpha_1$ with action $E_0$, and let $(\rho_t^1,v_t^1)$ connect $\alpha_1$ to $\alpha_2$ with action $E_1$. For $0<\zeta<1$, concatenate the two curves after the rescaling

$$(\rho_t,v_t) = \begin{cases}\left(\rho^0_{t/\zeta},\,\zeta^{-1}v^0_{t/\zeta}\right), & 0\leqslant t\leqslant\zeta,\\ \left(\rho^1_{(t-\zeta)/(1-\zeta)},\,(1-\zeta)^{-1}v^1_{(t-\zeta)/(1-\zeta)}\right), & \zeta<t\leqslant 1.\end{cases}$$

The factors $\zeta^{-1}$ and $(1-\zeta)^{-1}$ are exactly those needed for the weak continuity equation after changing time. Since $v\mapsto\mathbb{A}_K(\alpha,v)$ is quadratic, $\int_0^1\mathbb{A}_K(\alpha_t,v_t)\,\mathrm{d}t = \frac{E_0}{\zeta}+\frac{E_1}{1-\zeta}$. Optimizing in $\zeta$, or taking $\zeta=\sqrt{E_0}/(\sqrt{E_0}+\sqrt{E_1})$ when both actions are positive, gives a concatenated action $(\sqrt{E_0}+\sqrt{E_1})^2$. Taking infima over the two curves yields $\mathcal{W}_K(\alpha_0,\alpha_2)\leqslant\mathcal{W}_K(\alpha_0,\alpha_1)+\mathcal{W}_K(\alpha_1,\alpha_2)$.

If $\mathcal{W}_K(\alpha_0,\alpha_1)=0$, choose admissible curves with actions tending to zero. By the compactness and lower-semicontinuity theorem quoted above, a subsequence converges to an admissible curve $(\rho_t,v_t)$ of zero action. Hence $v_t=0$ for $\theta(\rho_t(x),\rho_t(y))\mathsf{J}(\mathrm{d}x,\mathrm{d}y)\mathrm{d}t$-a.e. $(t,x,y)$. Substituting in the weak continuity equation gives $\frac{\mathrm{d}}{\mathrm{d}t}\int\varphi\,\mathrm{d}\alpha_t=0$ for every admissible test function $\varphi$. The irreducibility/separation assumption in ensures that these test functions determine the measure, so $\alpha_t$ is constant and $\alpha_0=\alpha_1$. Finally, if $\mathcal{W}_K(\alpha_0,\alpha_1)<+\infty$, the same direct-method compactness applied to a minimizing sequence gives a minimizer of (13.29). Reparametrizing this minimizing curve by metric arclength gives a constant-speed curve. Equivalently, for every $0\leqslant s<t\leqslant 1$, the restriction of the minimizer to $[s,t]$, rescaled to unit time, is admissible between $\alpha_s$ and $\alpha_t$; the triangle inequality and equality of the total minimal action force $\mathcal{W}_K(\alpha_s,\alpha_t)=(t-s)\mathcal{W}_K(\alpha_0,\alpha_1)$ after this constant-speed parametrization. Thus finite-distance pairs are joined by constant-speed geodesics. The consequences for entropy dynamics and fractional PDE examples are developed in Section 14.6. □

For a fixed jump kernel, this geometry is genuinely nonlocal and does not coincide with ordinary $\mathcal{W}_2$. The local metric is nevertheless recovered in a small-jump limit. More explicitly, on $\mathcal{X}=\mathbb{R}^d$, let $\eta(z)=\bar\eta(\|z\|)$ be a nonnegative radial profile with $M_2(\eta)\coloneqq\int_{\mathbb{R}^d}\|z\|^2\eta(z)\,\mathrm{d}z\in(0,+\infty)$,

$$\int\|y-x\|^2K_\varepsilon(x,\mathrm{d}y)=\varepsilon^2M_2(\eta),\qquad \int(y-x)(y-x)^\top K_\varepsilon(x,\mathrm{d}y)=\varepsilon^2\frac{M_2(\eta)}{d}\mathrm{Id}. \tag{13.30}$$

$\widehat{K}_\varepsilon\coloneqq\frac{2d}{\varepsilon^2M_2(\eta)}K_\varepsilon,\qquad \int(y-x)(y-x)^\top\widehat{K}_\varepsilon(x,\mathrm{d}y)=2\mathrm{Id}.$

$$\mathcal{W}_{\widehat{K}_\varepsilon}=\varepsilon\sqrt{\frac{M_2(\eta)}{2d}}\,\mathcal{W}_{K_\varepsilon}\underset{\varepsilon\to0}{\longrightarrow}\mathcal{W}_2\,.$$

**Discrete Wasserstein distances on Markov chains.** $\mathsf{J}_{i,j}\coloneqq\pi_iK_{i,j}=\pi_jK_{j,i}$

$$\Sigma_n\coloneqq\left\{a\in\mathbb{R}^n_+:\sum_i a_i=1\right\}.$$

$\rho_i(a)\coloneqq\frac{a_i}{\pi_i},\qquad a_i=\pi_i\rho_i(a)$. The logarithmic mean $\theta$ defined in (13.26) is the mobility selected so that the entropy calculus later recovers exactly the Markov evolution; see Proposition 14.41. The identity $\theta(a,b)(\log a-\log b)=a-b$ converts entropy gradients into density differences along graph edges.
$\bar\nabla\psi(i,j)\coloneqq\psi_j-\psi_i$.

**Definition 13.20** (Discrete Markov-chain Wasserstein distance)**.**

$$(\mathcal{K}_\rho\psi)_i\coloneqq\sum_j K_{i,j}\theta(\rho_i,\rho_j)(\psi_i-\psi_j),\qquad(\mathcal{L}_a\psi)_i\coloneqq\pi_i(\mathcal{K}_{\rho(a)}\psi)_i. \tag{13.31}$$

Its associated tangent action is

$$\mathbb{A}_K(a,\psi)\coloneqq\frac12\sum_{i,j}\mathsf{J}_{i,j}\theta(\rho_i(a),\rho_j(a))(\bar\nabla\psi(i,j))^2. \tag{13.32}$$

$$\mathcal{W}_K^2(a_0,a_1)\coloneqq\inf_{a_t,\psi_t}\int_0^1\mathbb{A}_K(a_t,\psi_t)\,\mathrm{d}t,\qquad \dot a_t+\mathcal{L}_{a_t}\psi_t=0, \tag{13.33}$$

with endpoint conditions $a_{t=0}=a_0$ and $a_{t=1}=a_1$.

### 13.5 Dynamic Unbalanced Wasserstein Distances

**Balance equation and tangent variables.**

$$\partial_t\rho_t+\nabla\cdot m_t=s_t,\qquad\int_0^1\int\left(\frac{\|m_t\|^2}{\rho_t}+\kappa^2\frac{s_t^2}{\rho_t}\right)\mathrm{d}x\,\mathrm{d}t,$$

**Reaction–transport action.**

**Definition 13.21** (Dynamic Wasserstein–Fisher–Rao distance)**.**

For $\kappa>0$, define

$$A_\kappa(a,w,g)\coloneqq a\big(\|w\|^2+\kappa^2g^2\big), \tag{13.34}$$

and

$$J_\kappa(a,m,r)\coloneqq\begin{cases}(\|m\|^2+\kappa^2r^2)/a, & a>0,\\ 0, & (a,m,r)=(0,0,0),\\ +\infty, & a=0,\ (m,r)\neq(0,0).\end{cases} \tag{13.35}$$

If $\lambda$ dominates $\alpha$, $|\omega|$, and $|\sigma|$, set

$$\mathbb{J}_\kappa(\alpha,\omega,\sigma)\coloneqq\int J_\kappa\left(\frac{\mathrm{d}\alpha}{\mathrm{d}\lambda},\frac{\mathrm{d}\omega}{\mathrm{d}\lambda},\frac{\mathrm{d}\sigma}{\mathrm{d}\lambda}\right)\mathrm{d}\lambda. \tag{13.36}$$

The induced dynamic Wasserstein–Fisher–Rao distance is

$$\mathrm{WFR}_\kappa^2(\alpha_0,\alpha_1)\coloneqq\inf_{\substack{\partial_t\alpha_t+\nabla\cdot\omega_t=\sigma_t\\ \alpha_{t=0}=\alpha_0,\ \alpha_{t=1}=\alpha_1}}\int_0^1\mathbb{J}_\kappa(\alpha_t,\omega_t,\sigma_t)\mathrm{d}t. \tag{13.37}$$

**Static and dynamic viewpoints.** The dynamic balance-equation formula is not a separate object from static unbalanced OT: it is the least-action representation of the same cone distance. To make the normalization explicit, define on the cone $\mathfrak{C}[\mathbb{R}^d]$ the squared cost

$$\Delta_\kappa\big((x,r),(y,s)\big)^2 := 4\kappa^2\left[r^2+s^2-2rs\cos\left(\frac{\|x-y\|}{2\kappa}\wedge\frac{\pi}{2}\right)\right]. \tag{13.38}$$

The radii encode masses through the weighted projection $\mathsf{P}_2$ defined in Section 10.1. Accordingly, set

$$\mathrm{CW}_\kappa(\alpha_0,\alpha_1) := \inf_{\substack{\gamma\in\mathcal{M}_+(\mathfrak{C}[\mathbb{R}^d]^2)\\ \mathsf{P}_2\gamma_1=\alpha_0,\ \mathsf{P}_2\gamma_2=\alpha_1}} \int \Delta_\kappa\big((x,r),(y,s)\big)^2 \mathrm{d}\gamma(x,r,y,s). \tag{13.39}$$

For $\kappa=1/2$, this is exactly the normalization of the Hellinger–Kantorovich cone cost used in Section 10.1.
Thus Definition 13.21 gives both the reaction–transport action and the induced dynamic WFR metric.
**Proposition 13.22** (Static/dynamic equivalence for unbalanced OT)**.** *For nonnegative finite measures $\alpha_0,\alpha_1$ on $\mathbb{R}^d$, the dynamic value (13.37) equals the static cone value (13.39). Hence $\mathrm{WFR}_\kappa=\mathrm{CW}_\kappa^{1/2}$ is the geodesic distance generated by the balance-equation least-action problem.*

*Proof.* The cone construction turns variation of mass into radial motion and spatial transport into angular motion on $\mathfrak{C}[\mathbb{R}^d]$. The normalization can be checked on a smooth cone path. If its base position is $x_t$, its cone radius is $r_t$, and the projected mass is $a_t=r_t^2$, then its relative growth rate is $g_t=\dot a_t/a_t=2\dot r_t/r_t$. The infinitesimal energy induced by (13.38) is therefore

$$4\kappa^2\dot r_t^2+r_t^2\|\dot x_t\|^2=a_t\left(\|\dot x_t\|^2+\kappa^2g_t^2\right),$$

which is precisely the velocity action (13.34).
Applying the Benamou–Brenier theorem on the cone to the lifted endpoint measures now gives a dynamic least-action problem whose static value is $\mathrm{CW}_\kappa$. Projecting a cone curve back to the base space with the squared cone radius as weight produces a measure curve $\alpha_t$, a spatial flux measure $\omega_t$, and a source measure $\sigma_t$ satisfying the balance equation. The cone kinetic energy decomposes exactly into the three-variable perspective action $\mathbb{J}_\kappa$ in (13.37). Conversely, any finite-action triple $(\alpha_t,\omega_t,\sigma_t)$ can be lifted, after relaxation, to a cone curve whose radial and spatial velocities realize the source and flux with the same action. This is the standard static/dynamic identification for the Hellinger–Kantorovich and Wasserstein–Fisher–Rao metrics.
The two infima are therefore equal; lower semicontinuity gives the general finite-measure statement from the smooth positive case. □

**Balanced versus unbalanced interpolations.**
### 13.6 Variational Mean Field Games
**From individual control to a population equilibrium.**

$$\inf_{\gamma(0)=x}\left\{\int_0^1\left(\frac{1}{2}\|\dot\gamma(t)\|^2+g\big(\rho_t(\gamma(t))\big)\right)\mathrm{d}t+\Psi(\gamma(1))\right\}. \tag{13.40}$$

**A Benamou–Brenier planning problem.**

$$g(r)=G'(r) \tag{13.41}$$

**Definition 13.23** (Variational mean-field-game planning problem)**.**

$$\inf_{(\rho_t,v_t)}\left\{\int_0^1\int_{\mathbb{R}^d}\left(\frac{1}{2}\rho_t(x)\|v_t(x)\|^2+G(\rho_t(x))\right)\mathrm{d}x\,\mathrm{d}t+\int_{\mathbb{R}^d}\Psi(x)\rho_1(x)\mathrm{d}x\right\}, \tag{13.42}$$

subject to

$$\partial_t\rho_t+\mathrm{div}(\rho_tv_t)=0,\qquad \rho_{t=0}\,\mathrm{d}x=\alpha_0. \tag{13.43}$$

Unlike in (13.8), the final measure is not prescribed. The terminal potential $\Psi$ selects desirable final states softly, while the integral of $G(\rho_t)$ penalizes crowded intermediate configurations.
**Convex momentum formulation.** As in the Benamou–Brenier problem, the density–velocity integrand and the constraint in (13.42)–(13.43) are not jointly convex. Set $\omega_t=\alpha_tv_t$, or $m_t=\rho_tv_t$ when densities exist, and define the congestion functional

$$\mathcal{C}_G(\alpha) := \begin{cases}\displaystyle\int_{\mathbb{R}^d}G(\rho(x))\mathrm{d}x, & \alpha=\rho\,\mathrm{d}x,\\ +\infty, & \text{otherwise.}\end{cases} \tag{13.44}$$

$$\inf_{(\alpha_t,\omega_t)}\left\{\int_0^1\left(\frac{1}{2}\mathbb{J}(\alpha_t,\omega_t)+\mathcal{C}_G(\alpha_t)\right)\mathrm{d}t+\int_{\mathbb{R}^d}\Psi\,\mathrm{d}\alpha_1\right\},\qquad \partial_t\alpha_t+\mathrm{div}\,\omega_t=0,\quad \alpha_{t=0}=\alpha_0. \tag{13.45}$$

$$\frac{1}{2}J(\rho_t,m_t)+G(\rho_t)=\frac{\|m_t\|^2}{2\rho_t}+G(\rho_t), \tag{13.46}$$

**Optimality system and game interpretation.**
**Proposition 13.24** (Potential MFG system)**.** *Assume that an optimizer of (13.45) has a smooth positive density $\rho$, that $G$ is differentiable, and set $g=G'$. Then there is a value function $u$ such that*

$$\begin{cases}-\partial_tu+\dfrac{1}{2}\|\nabla u\|^2=g(\rho), & u_{t=1}=\Psi,\\ \partial_t\rho-\mathrm{div}(\rho\nabla u)=0, & \rho_{t=0}\,\mathrm{d}x=\alpha_0,\end{cases}\qquad m=-\rho\nabla u. \tag{13.47}$$

*Proof.* Use $-u$ as the multiplier for $\partial_t\rho + \operatorname{div} m = 0$. After integration by parts, the space–time Lagrangian density is $\frac{\|m\|^2}{2\rho} + G(\rho) + \rho\,\partial_t u + m\cdot\nabla u$. Stationarity in $m$ gives $m = -\rho\nabla u$. Stationarity in $\rho$, followed by this substitution, gives the first equation in (13.47). The free terminal density contributes $(\Psi - u_1)\rho_1$, hence $u_1 = \Psi$ under the positivity assumption. Primal feasibility gives the second equation. $\square$

**Hard congestion through a bottleneck.**

$$G_\kappa(r) = \iota_{[0,\kappa]}(r) = \begin{cases} 0, & 0 \leqslant r \leqslant \kappa, \\ +\infty, & \text{otherwise.} \end{cases} \tag{13.48}$$

$$\Gamma_\eta(\rho_1) = \frac{\eta}{2}\int_\Omega |\rho_1(x) - \rho_\star(x)|^2 \,\mathrm{d}x, \tag{13.49}$$

where $\rho_\star$ is a desired terminal density.

The role of endpoint relaxation depends on the geometry. If two fixed endpoint densities on $\mathbb{R}^d$ satisfy a spatially constant cap, their ordinary $\mathcal{W}_2$ geodesic satisfies the same cap by Proposition 14.14, so the cap does not alter that geodesic.

**Computational consequence.**

# 14 Wasserstein Gradient Flows

## 14.1 Minimizing Movements and Wasserstein Gradients

**Definition 14.1** (JKO minimizing movement)**.** Given $\tau > 0$ and $\alpha^{(0)}$, define recursively

$$\alpha^{(\ell+1)} \in \underset{\alpha\in\mathcal{P}_2(\mathbb{R}^d)}{\operatorname{argmin}} \left\{ \frac{1}{2\tau}\mathcal{W}_2(\alpha^{(\ell)},\alpha)^2 + f(\alpha) \right\}. \tag{14.1}$$

**Euclidean gradient flows.** If we restrict (14.1) to finite dimensions and assume $\alpha_t = \delta_{x(t)}$ and $\alpha = \delta_x$ (single Dirac measures), this matches the implicit Euler scheme:

$$x(t+\tau) \in \underset{x}{\operatorname{argmin}} \left\{ \frac{1}{2\tau}\|x - x(t)\|^2 + h(x) \right\},$$

where $h(x) = f(\delta_x)$. When $h$ is differentiable and the minimizer is unique, the solution is characterized by the implicit Euler formula
$x(t+\tau) = (\mathrm{Id} + \tau\nabla h)^{-1}(x(t))$. $x(t+\tau) = (\mathrm{Id} - \tau\nabla h)(x(t)) = x(t) - \tau\nabla h(x(t))$.

$$\dot{x}(t) = -\nabla h(x(t)). \tag{14.2}$$

**Vertical derivatives (first- and higher-order variations).**

**Definition 14.2** (First variation)**.** Assume that $f(\alpha) < +\infty$. A measurable function $\delta f(\alpha)$ is a first variation of $f$ at $\alpha$ if, for every $\beta \in \mathcal{P}(\mathbb{R}^d)$ such that $\delta f(\alpha)$ is integrable with respect to $|\beta - \alpha|$ and the mixture $\alpha_\vartheta$ remains in the domain of $f$ for small $\vartheta > 0$, $f(\alpha_\vartheta) = f(\alpha) + \vartheta\int_{\mathbb{R}^d}[\delta f(\alpha)](x)\mathrm{d}(\beta-\alpha)(x) + o(\vartheta) \qquad (\vartheta \downarrow 0)$.

$Df(\alpha)[\eta] := \int_{\mathbb{R}^d}[\delta f(\alpha)](x)\mathrm{d}\eta(x)$

**Definition 14.3** (Higher-order vertical derivatives)**.** Set $D^0 f = f$. The functional $f$ admits vertical derivatives through order $r$ if there are symmetric multilinear forms $D^j f(\alpha) : \mathcal{M}_0(\mathcal{X})^j \longrightarrow \mathbb{R}, \qquad 1 \leqslant j \leqslant r$, such that, along every mixture segment in $\mathcal{P}(\mathcal{X})$ and for fixed $\eta_1,\dots,\eta_{j-1} \in \mathcal{M}_0(\mathcal{X})$, $\frac{\mathrm{d}}{\mathrm{d}\vartheta}D^{j-1}f(\alpha_\vartheta)[\eta_1,\dots,\eta_{j-1}] = D^j f(\alpha_\vartheta)[\beta-\alpha,\eta_1,\dots,\eta_{j-1}]$. At the endpoints, derivatives are understood one-sided.

$D^j f(\alpha)[\eta_1,\dots,\eta_j] = \int_{\mathcal{X}^j}\frac{\delta^j f}{\delta\alpha^j}(\alpha,x_1,\dots,x_j)\mathrm{d}\eta_1(x_1)\cdots\mathrm{d}\eta_j(x_j)$

**Horizontal derivative (Wasserstein gradient).**

**Definition 14.4** (Wasserstein gradient)**.** Assume that $f$ admits a smooth first variation $\delta f(\alpha)$. In the smooth formal calculus on $\mathcal{P}_2(\mathbb{R}^d)$, the Wasserstein gradient of $f$ at $\alpha$ is the gradient vector field $\nabla\!\!\!\!\nabla f(\alpha) = \nabla_x\delta f(\alpha)$.

$$\frac{\partial\alpha_t}{\partial t} + \operatorname{div}(-\nabla\!\!\!\!\nabla f(\alpha_t)\alpha_t) = 0. \tag{14.3}$$

**Proposition 14.5** (Formal Wasserstein gradient)**.** *Assume that $f$ admits a smooth first variation $\delta f(\alpha)$ and that $\alpha$ has a smooth positive density. Let $v$ be a sufficiently regular velocity field, compactly supported or satisfying a no-flux boundary condition. For perturbations generated by $(\mathrm{Id}+\tau v)_\sharp\alpha$, the differential of $f$ is $\frac{\mathrm{d}}{\mathrm{d}\tau}_{|\tau=0} f\big((\mathrm{Id}+\tau v)_\sharp\alpha\big) = \int\langle\nabla\delta f(\alpha)(x),\, v(x)\rangle\mathrm{d}\alpha(x)$. Hence, for the Riemannian metric $\|v\|^2_{L^2(\alpha)} = \int\|v\|^2\mathrm{d}\alpha$, the Wasserstein gradient is the vector field $\nabla\!\!\!\!\nabla f(\alpha) = \nabla\delta f(\alpha)$.*

*Proof.* The push-forward expansion gives, in the sense of distributions, $(\mathrm{Id}+\tau v)_\sharp\alpha = \alpha - \tau\operatorname{div}(\alpha v) + o(\tau)$. Using the definition of the first variation, $f((\mathrm{Id}+\tau v)_\sharp\alpha) = f(\alpha) - \tau\int\delta f(\alpha)\operatorname{div}(\alpha v)\mathrm{d}x + o(\tau)$. An integration by parts, with either compact support or vanishing boundary flux, gives $-\int\delta f(\alpha)\operatorname{div}(\alpha v)\mathrm{d}x = \int\langle\nabla\delta f(\alpha),\, v\rangle\mathrm{d}\alpha$. By definition of the Riesz representative for the $L^2(\alpha)$ metric, this representative is $\nabla\delta f(\alpha)$. $\square$

**From the JKO step to the velocity field.** A first-order expansion of the JKO step explains why (14.3) uses the vector field $\nabla\!\!\!\!\nabla f(\alpha)$. This is a formal local argument: assume that $\alpha_t$ is absolutely continuous and that, for the nearby competitors selected by the JKO step, the optimal map from $\alpha_t$ can be written as $\mathrm{Id}+\tau v$.
$\mathcal{W}_2^2\big(\alpha_t,(\mathrm{Id}+\tau v)_\sharp\alpha_t\big) = \tau^2\|v\|^2_{L^2(\alpha_t)}$. $\min_v \frac{\tau}{2}\|v\|^2_{L^2(\alpha_t)} + f\big((\mathrm{Id}+\tau v)_\sharp\alpha_t\big)$,

$$(\mathrm{Id}+\tau v)_\sharp\alpha_t = \alpha_t - \tau\operatorname{div}(v\alpha_t) + o(\tau),$$
$$f\big((\mathrm{Id}+\tau v)_\sharp\alpha_t\big) = f(\alpha_t) + \tau\int\langle\nabla_x\delta f(\alpha_t)(x),\, v(x)\rangle\mathrm{d}\alpha_t(x) + o(\tau)$$

$$\min_v f(\alpha_t) + \tau \int \left[\frac{1}{2}\|v(x)\|^2 + \langle \nabla\!\!\!\!\nabla f(\alpha_t)(x),\, v(x)\rangle\right] \mathrm{d}\alpha_t(x) + o(\tau).$$

**Metric derivative and curves of maximal slope.**

**Definition 14.6** (Metric derivative and absolutely continuous curve)**.** For $1 \leqslant p \leqslant +\infty$, a curve $x : [0,T] \to (\mathcal{X}, d)$ belongs to $AC^p([0,T];\mathcal{X})$ if some $g \in L^p(0,T)$ satisfies $d(x_s, x_t) \leqslant \int_s^t g(r)\mathrm{d}r$. Absolutely continuous means $AC^1$. Its metric derivative is $|\dot{x}_t| := \lim_{h\to 0} \frac{d(x_{t+h}, x_t)}{|h|}$ for a.e. $t$.

**Definition 14.7** (Metric slope and curve of maximal slope)**.** For lower-semicontinuous $f : \mathcal{X} \to (-\infty, +\infty]$, define

$$|\partial f|(x) := \begin{cases} \limsup_{\substack{y\to x\\ y\neq x}} \dfrac{(f(x) - f(y))_+}{d(x,y)}, & f(x) < +\infty, \\ +\infty, & f(x) = +\infty. \end{cases}$$

It is a strong upper gradient if $|f(y_t) - f(y_s)| \leqslant \int_s^t |\partial f|(y_r)|\dot{y}_r|\mathrm{d}r$ along every absolutely continuous $y$. When this property holds, a curve $x_t$ is a curve of maximal slope for $f$ if $f(x_t) + \frac{1}{2}\int_s^t |\dot{x}_r|^2\mathrm{d}r + \frac{1}{2}\int_s^t |\partial f|^2(x_r)\mathrm{d}r \leqslant f(x_s), \qquad 0 \leqslant s < t.$

**Discrete evolutions.** If $f$ admits a first variation whose spatial gradient is well defined at empirical measures, the characteristic form of (14.3) preserves the number and weights of the atoms. For equal weights, write $\alpha_t = \frac{1}{n}\sum_i \delta_{x_i(t)}$. The particles $X(t) := (x_i(t))_i$ evolve according to the coupled ODEs

$$\dot{x}_i(t) = -n\nabla_{x_i} F(X(t)), \tag{14.4}$$

where $F(X) := f\left(\frac{1}{n}\sum_i \delta_{x_i}\right)$ and the factor $n$ comes from the empirical Wasserstein metric $\frac{1}{n}\sum_i \|\dot{x}_i\|^2$.

**Linear functionals.**

**Shannon neg-entropy.**

**Definition 14.8** (Score function)**.** For $\alpha = \rho\,\mathrm{d}x$ with differentiable positive density $\rho$, its score is $\nabla \log\rho$. Relative to $\beta$, its relative score is $\nabla\log(\mathrm{d}\alpha/\mathrm{d}\beta)$ whenever this logarithmic density has a weak gradient.

**Interaction energies.**

$$f(\alpha) := \iint k(x,y)\mathrm{d}\alpha(x)\mathrm{d}\alpha(y). \tag{14.5}$$

For a symmetric kernel $k$:
$\delta f(\alpha)(x) = 2\int k(x,y)\mathrm{d}\alpha(y), \quad \nabla\!\!\!\!\nabla f(\alpha)(x) = 2\int \nabla_x k(x,y)\mathrm{d}\alpha(y)$. For $\alpha_0 = \frac{1}{n}\sum_i \delta_{x_i}$, the flow (14.3) implies particles $(x_i(t))_i$ obey: $\dot{x}_i(t) = -\frac{2}{n}\sum_j \nabla k(x_i(t), x_j(t))$. If $k$ is positive definite, or more generally conditionally positive definite on signed measures of zero total mass as for the energy-distance kernel $k(x,y) = -\|x-y\|$, and one minimizes the squared kernel discrepancy to a teacher distribution $\beta$, then

$$\|\alpha - \beta\|_k^2 = \iint k\mathrm{d}\alpha\mathrm{d}\alpha - 2\int\left(\int k(x,y)\mathrm{d}\beta(y)\right)\mathrm{d}\alpha(x) + \text{constant}.$$

Thus MMD-type training energies are exactly an interaction energy plus a linear potential; the teacher distribution appears through the potential $x \mapsto -2\int k(x,y)\mathrm{d}\beta(y)$. The corresponding empirical Wasserstein gradient flow is
$\dot{x}_i(t) = -\frac{2}{n}\sum_j \nabla_x k(x_i(t), x_j(t)) + 2\int \nabla_x k(x_i(t), y)\mathrm{d}\beta(y)$.

**Stochastic particles and McKean–Vlasov limits.** $\mathrm{d}X_t = b(X_t)\mathrm{d}t + \sqrt{2}\sigma\mathrm{d}B_t$, and the one-particle law $\alpha_t = \rho_t\mathrm{d}x$ directly satisfies the linear Fokker–Planck equation
$\partial_t\rho_t = -\operatorname{div}(b\rho_t) + \sigma^2\Delta\rho_t$. $\mathrm{d}X_i^n(t) = b(X_i^n(t), \alpha_t^n)\mathrm{d}t + \sqrt{2}\sigma\mathrm{d}B_i(t), \qquad \alpha_t^n = \frac{1}{n}\sum_{i=1}^n \delta_{X_i^n(t)}$. For finite $n$, the empirical law $\alpha_t^n$ is itself random. Under suitable Lipschitz, growth and chaotic-initialization assumptions, propagation of chaos states that finitely many particles become asymptotically independent as $n \to \infty$, all with the same deterministic law $\rho_t\mathrm{d}x$; equivalently, the empirical measure $\alpha_t^n$ converges in probability to this law.
$\partial_t\rho_t = -\operatorname{div}\left(b(x,\rho_t)\rho_t\right) + \sigma^2\Delta\rho_t$. $b(x,\rho) = -\nabla\frac{\delta\mathcal{E}}{\delta\rho}(x)$, $\mathcal{E}(\rho) + \sigma^2\int \rho\log\rho\,\mathrm{d}x$. For a prescribed target $\beta = \rho_\beta\mathrm{d}x$, take $b = \sigma^2\nabla\log\rho_\beta$. The SDE, the Fokker–Planck equation, and the deterministic continuity equation with score velocity are then three representations of the same gradient flow:

$$\partial_t\rho_t = \sigma^2\operatorname{div}\left(\rho_t\nabla\log\frac{\rho_t}{\rho_\beta}\right), \qquad v_t = \sigma^2\left(\nabla\log\rho_\beta - \nabla\log\rho_t\right). \tag{14.6}$$

**Gradient flows under density constraints.**

$$f_{\rho_{\max}}(\rho\mathrm{d}x) = \int_\Omega h(x)\rho(x)\mathrm{d}x + \iota_{\mathcal{K}_{\rho_{\max}}}(\rho\mathrm{d}x), \qquad \mathcal{K}_{\rho_{\max}} = \{\rho\mathrm{d}x \in \mathcal{P}_2(\Omega)\,;\, 0 \leqslant \rho \leqslant \rho_{\max} \text{ a.e.}\}, \tag{14.7}$$

where $\Omega \subset \mathbb{R}^d$ is a convex domain and $\mathcal{K}_{\rho_{\max}}$ is assumed nonempty; if $\Omega$ has finite volume, this requires $\rho_{\max}|\Omega| \geqslant 1$. The indicator term has no classical first variation; it contributes the normal cone to $\mathcal{K}_{\rho_{\max}}$ in the Wasserstein first-order condition.

$$\alpha^{(\ell+1)} \in \operatorname*{argmin}_{\alpha\in\mathcal{K}_{\rho_{\max}}} \frac{1}{2\tau}\mathcal{W}_2^2(\alpha^{(\ell)}, \alpha) + \int_\Omega h\,\mathrm{d}\alpha. \tag{14.8}$$

$$\partial_t\rho_t = \operatorname{div}\left(\rho_t\nabla(h + p_t)\right), \qquad 0 \leqslant \rho_t \leqslant \rho_{\max}, \qquad p_t \geqslant 0, \qquad p_t(\rho_{\max} - \rho_t) = 0. \tag{14.9}$$

The pressure vanishes in the unsaturated region and acts only where $\rho_t = \rho_{\max}$, pushing mass away from congested zones so that the density cap remains satisfied.

**Multi-species gradient flows.** Several applications involve several densities living on the same physical space: chemical species, color channels, cell populations or competing phases. Let $\Omega \subset \mathbb{R}^d$ be the physical domain and assume that the component masses $\mathbf{m} = (m_1, \dots, m_p)$ are positive and fixed.

$$\mathcal{P}_{2,\mathbf{m}}(\Omega; \mathbb{R}^p_+) = \left\{ \alpha = (\alpha_1, \dots, \alpha_p) \ ; \ \alpha_i \in \mathcal{M}_+(\Omega),\ \alpha_i(\Omega) = m_i,\ \int_\Omega \|x\|^2 \mathrm{d}\alpha_i(x) < +\infty \right\}. \tag{14.10}$$

$$\mathcal{W}^2_{2,\oplus}(\alpha, \eta) = \sum_{i=1}^p m_i\, \mathcal{W}_2^2\left( \frac{\alpha_i}{m_i}, \frac{\eta_i}{m_i} \right). \tag{14.11}$$

$$\alpha^{(\ell+1)} \in \underset{\alpha \in \mathcal{P}_{2,\mathbf{m}}(\Omega;\mathbb{R}^p_+)}{\operatorname{argmin}} \ \frac{1}{2\tau} \mathcal{W}^2_{2,\oplus}(\alpha^{(\ell)}, \alpha) + f(\alpha). \tag{14.12}$$

For smooth densities and first variations $\varphi_i = \delta f / \delta \rho_i$, this yields

$$\partial_t \rho_i = \operatorname{div}\left(\rho_i \nabla \varphi_i\right), \qquad \varphi_i = \frac{\delta f}{\delta \rho_i}, \qquad i = 1, \dots, p. \tag{14.13}$$

$$\mathcal{C}_\beta = \left\{ \alpha \in \mathcal{P}_{2,\mathbf{m}}(\Omega; \mathbb{R}^p_+) \ ; \ \sum_{i=1}^p \alpha_i = \beta \right\}. \tag{14.14}$$

$$\alpha^{(\ell+1)} \in \underset{\alpha \in \mathcal{C}_\beta}{\operatorname{argmin}} \ \frac{1}{2\tau} \mathcal{W}^2_{2,\oplus}(\alpha^{(\ell)}, \alpha) + f(\alpha). \tag{14.15}$$

$$\partial_t \rho_i = \operatorname{div}\left(\rho_i \nabla(\varphi_i - \lambda_t)\right), \qquad \operatorname{div}(\rho_\beta \nabla \lambda_t) = \operatorname{div}\left( \sum_{i=1}^p \rho_i \nabla \varphi_i \right). \tag{14.16}$$

For the separable Shannon entropy $f(\alpha) = \sum_i \int_\Omega \rho_i \log \rho_i \mathrm{d}x$, this becomes

$$\partial_t \rho_i = \Delta \rho_i - \operatorname{div}(\rho_i \nabla \lambda_t), \qquad \operatorname{div}(b \nabla \lambda_t) = \Delta b. \tag{14.17}$$

When $b$ is constant, the pressure can be chosen constant and the system reduces to independent heat equations which nevertheless preserve the pointwise sum $\sum_i \rho_i = b$. This product geometry should be contrasted with the vector-valued Benamou–Brenier distances of Section 12.1, where the metric itself can couple the components through a non-diagonal mobility.

## 14.2 Geodesic Convexity and Convergence

**Geodesics and convexity.** $\alpha_t = ((1-t)P_0 + tP_1)_\sharp \pi^\star, \qquad t \in [0,1]$, where $P_0(x,y) = x$ and $P_1(x,y) = y$. If the optimal plan is induced by a Brenier map $T$, this reduces to $((1-t)\mathrm{Id} + tT)_\sharp \alpha_0$.

**Definition 14.9** (Geodesic convexity)**.** A functional $f$ on $\mathcal{P}_2(\mathbb{R}^d)$ is geodesically convex if for every $\mathcal{W}_2$ geodesic $(\alpha_t)_t$, $f(\alpha_t) \leqslant (1-t) f(\alpha_0) + t f(\alpha_1)$. It is $\lambda$-geodesically convex if the right-hand side is improved by $-\frac{\lambda}{2} t(1-t)\, \mathcal{W}_2^2(\alpha_0, \alpha_1)$.

**Proposition 14.10** (Geodesic convexity of linear and quadratic energies)**.** *1. If $h$ is convex, then the linear energy $\alpha \mapsto \int h \mathrm{d}\alpha$ is geodesically convex; if $h$ is $\lambda$-strongly convex, it is $\lambda$-geodesically convex. Conversely, if this linear energy is geodesically convex for all Dirac endpoints, then $h$ is convex.*

*2. More generally, let $k \geqslant 2$ and let $H_k : (\mathbb{R}^d)^k \to \mathbb{R} \cup \{+\infty\}$ be convex. Then the polynomial energy $\alpha \mapsto \int_{(\mathbb{R}^d)^k} H_k(x_1, \dots, x_k)\, \mathrm{d}\alpha(x_1) \cdots \mathrm{d}\alpha(x_k)$ is geodesically convex whenever the integral is well defined. If $H_k$ is $\lambda$-strongly convex on the product space, this energy is $k\lambda$-geodesically convex. In particular, the quadratic interaction $\alpha \mapsto \frac{1}{2} \iint W(x,y) \mathrm{d}\alpha(x) \mathrm{d}\alpha(y)$ is geodesically convex when $W$ is convex on $\mathbb{R}^d \times \mathbb{R}^d$, and it is $\lambda$-geodesically convex when $W$ is $\lambda$-strongly convex there.*

*Proof.* Let $\pi$ be an optimal coupling between $\alpha_0$ and $\alpha_1$, and set $X_t = (1-t)X_0 + tX_1$ under $(X_0, X_1) \sim \pi$. Convexity of $h$ gives $h(X_t) \leqslant (1-t)h(X_0) + th(X_1)$, and strong convexity gives the additional term $-\frac{\lambda}{2} t(1-t) \|X_0 - X_1\|^2$. Integrating proves the first claim. Necessity follows by testing on Dirac masses: the geodesic between $\delta_x$ and $\delta_y$ is $\delta_{(1-t)x+ty}$, so geodesic convexity of $\int h \mathrm{d}\alpha$ gives the usual convexity inequality for $h$.

For the polynomial statement, take $k$ independent copies $(X_0^\ell, X_1^\ell)_{\ell=1}^k$ of the same optimal coupling and define $X_t^\ell = (1-t)X_0^\ell + tX_1^\ell$. Convexity of $H_k$ on the product space gives $H_k(X_t^1, \dots, X_t^k) \leqslant (1-t) H_k(X_0^1, \dots, X_0^k) + t H_k(X_1^1, \dots, X_1^k)$. Integrating over the product coupling proves the claim. If $H_k$ is $\lambda$-strongly convex, the additional term is $-\frac{\lambda}{2} t(1-t) \sum_{\ell=1}^k \mathbb{E}\|X_0^\ell - X_1^\ell\|^2 = -\frac{k\lambda}{2} t(1-t)\, \mathcal{W}_2^2(\alpha_0, \alpha_1)$, which proves the stated constant. For the quadratic interaction, the prefactor $1/2$ cancels the factor $k = 2$, leaving the constant $\lambda$. □

**Proposition 14.11** (One-dimensional interactions and energy-distance MMD)**.** *Let $\varphi : \mathbb{R}_+ \to \mathbb{R}$ be continuous with at most quadratic growth, and define on $\mathcal{P}_2(\mathbb{R})$ $\mathcal{I}_\varphi(\alpha) := \frac{1}{2} \iint \varphi(|x-y|) \mathrm{d}\alpha(x) \mathrm{d}\alpha(y)$. Then $\mathcal{I}_\varphi$ is geodesically convex for $\mathcal{W}_2$ on $\mathcal{P}_2(\mathbb{R})$ if and only if $\varphi$ is convex on $\mathbb{R}_+$. Fix also $\beta \in \mathcal{P}_2(\mathbb{R})$. For the conditionally positive kernel $k(x,y) = -|x-y|$ from Definition 6.4, the squared MMD of Definition 6.5, equivalently the squared energy distance, $\mathcal{E}_\beta(\alpha) := \mathrm{MMD}_k^2(\alpha, \beta) = \iint -|x-y|\, \mathrm{d}(\alpha - \beta)(x) \mathrm{d}(\alpha - \beta)(y)$, is geodesically convex as a function of $\alpha$.*

*Proof.* Let $Q_0, Q_1$ be the quantile functions of $\alpha_0, \alpha_1$, and let $Q_t = (1-t)Q_0 + tQ_1$ be the quantile function of the one-dimensional $\mathcal{W}_2$ geodesic $\alpha_t$. For $r > s$, monotonicity gives $Q_i(r) - Q_i(s) \geqslant 0$ for $i = 0, 1$, and hence $Q_t(r) - Q_t(s) = (1-t)(Q_0(r) - Q_0(s)) + t(Q_1(r) - Q_1(s))$. Using the symmetry of the double integral and the quantile parametrization, one can write $\mathcal{I}_\varphi(\alpha_t) = \int_0^1 \int_0^r \varphi(Q_t(r) - Q_t(s))\, \mathrm{d}s\, \mathrm{d}r$. Convexity of $\varphi$ on $\mathbb{R}_+$ gives the desired convexity after integration.

Conversely, test geodesic convexity on two-point measures $\alpha_i = \frac{1}{2}(\delta_0 + \delta_{a_i})$, with $a_i \geqslant 0$. Their monotone geodesic is $\alpha_t = \frac{1}{2}(\delta_0 + \delta_{(1-t)a_0 + ta_1})$. Since $\mathcal{I}_\varphi(\alpha_t) = \frac{1}{4}\varphi(0) + \frac{1}{4}\varphi((1-t)a_0 + ta_1)$, and the identical $\frac{1}{4}\varphi(0)$ terms cancel from the geodesic-convexity inequality, one obtains the ordinary convexity inequality for $\varphi$ on $\mathbb{R}_+$.

For the energy distance, the function $\varphi(r) = -r$ is affine on $\mathbb{R}_+$, so the self-interaction $-\iint |x-y| \mathrm{d}\alpha(x)\mathrm{d}\alpha(y)$ is affine along one-dimensional Wasserstein geodesics even though $z \mapsto -|z|$ is not convex on $\mathbb{R}$. Write $Q_\beta$ for the quantile function of $\beta$. Expanding the MMD and dropping the term depending only on $\beta$ gives

$$\mathcal{E}_\beta(\alpha) = 2\int_0^1\int_0^1 |Q_\alpha(r) - Q_\beta(s)| \,\mathrm{d}r\,\mathrm{d}s - \int_0^1\int_0^1 |Q_\alpha(r) - Q_\alpha(s)| \,\mathrm{d}r\,\mathrm{d}s + \text{constant}.$$

Here the constant is independent of $\alpha$. The positive cross-distance term is convex in $Q_\alpha$, since the absolute value is convex and $Q_t$ is affine in $t$. The self-distance term is affine along quantile geodesics because $\int_0^1\int_0^1 |Q_t(r)-Q_t(s)| \,\mathrm{d}r\,\mathrm{d}s = 2\int_0^1\int_0^r (Q_t(r)-Q_t(s)) \,\mathrm{d}s\,\mathrm{d}r$. Thus $\alpha \mapsto \mathcal{E}_\beta(\alpha)$ is geodesically convex. □

$\mathrm{MMD}_k^2(\alpha,\beta) = \iint k \,\mathrm{d}\alpha\mathrm{d}\alpha - 2\iint k \,\mathrm{d}\alpha\mathrm{d}\beta + \text{constant}$

**Theorem 14.12** (McCann displacement convexity for internal energies)**.** *Let $g : [0,+\infty) \to \mathbb{R}\cup\{+\infty\}$ be convex with $g(0)=0$, and define $\psi(r) = r^d g(r^{-d})$, $\quad r > 0$. If $\psi$ is convex and nonincreasing on $(0,+\infty)$, then the internal energy*

$$\mathcal{U}_g(\alpha) = \begin{cases} \displaystyle\int_{\mathbb{R}^d} g(\rho(x))\,\mathrm{d}x, & \alpha = \rho\mathrm{d}x,\\ +\infty, & \textit{otherwise}, \end{cases}$$

*is geodesically convex on $\mathcal{P}_2(\mathbb{R}^d)$.*

*Proof.* It is enough to prove the inequality for smooth positive compactly supported densities; the general case follows by approximation and lower semicontinuity. Let $T = \nabla\varphi$ be the Brenier map from $\alpha_0 = \rho_0\mathrm{d}x$ to $\alpha_1 = \rho_1\mathrm{d}x$, set $T_t = (1-t)\mathrm{Id} + tT$, and denote $J_t(x) = \det\big((1-t)\mathrm{Id} + t\nabla T(x)\big)$. The interpolation satisfies $\alpha_t = (T_t)_\sharp\alpha_0$ and $\rho_t(T_t(x))J_t(x) = \rho_0(x)$. By concavity of $\det^{1/d}$ on positive semidefinite matrices, $J_t(x)^{1/d} \geqslant (1-t) + t\det(\nabla T(x))^{1/d}$. Introduce $s_t(x) = (J_t(x)/\rho_0(x))^{1/d}$. Since $\rho_1(T(x))\det\nabla T(x) = \rho_0(x)$, this gives $s_t(x) \geqslant (1-t)\rho_0(x)^{-1/d} + t\rho_1(T(x))^{-1/d}$. Using the change of variables $y = T_t(x)$, one obtains $\int g(\rho_t(y))\,\mathrm{d}y = \int \rho_0(x)\,\psi(s_t(x))\,\mathrm{d}x$. Because $\psi$ is nonincreasing and convex, $\psi(s_t(x)) \leqslant (1-t)\psi(\rho_0(x)^{-1/d}) + t\psi(\rho_1(T(x))^{-1/d})$. Multiplying by $\rho_0$ and integrating gives $\mathcal{U}_g(\alpha_t) \leqslant (1-t)\mathcal{U}_g(\alpha_0) + t\mathcal{U}_g(\alpha_1)$. □

**Proposition 14.13** (Geodesic convexity of power divergences)**.** *For $m > 0$, define*

$$\varphi_m(r) = \begin{cases} \dfrac{r^m - mr + m - 1}{m(m-1)}, & m \neq 1,\\ r\log r - r + 1, & m = 1. \end{cases}$$

*Then the following geodesic-convexity statements hold.*

*1. If $\Omega \subset \mathbb{R}^d$ is convex, $\beta = \rho_\beta\mathrm{d}x$ is uniform on $\Omega$, and $m \geqslant 1 - 1/d$, then $\alpha \mapsto \mathcal{D}_{\varphi_m}(\alpha|\beta)$ is geodesically convex on $\mathcal{P}_2(\Omega)$.*

*2. If $\beta = Z^{-1}e^{-V}\mathrm{d}x$ on $\mathbb{R}^d$, $V$ is convex, and $m \geqslant 1$, then $\alpha \mapsto \mathcal{D}_{\varphi_m}(\alpha|\beta)$ is geodesically convex on $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$.*

*3. In the KL case $m = 1$, if $V$ is $\lambda$-strongly convex, then $\alpha \mapsto \mathrm{KL}(\alpha|\beta) = \mathcal{D}_{\varphi_1}(\alpha|\beta)$ is $\lambda$-geodesically convex.*

*All statements use the convention that the divergence is $+\infty$ when $\alpha$ is not absolutely continuous with respect to $\beta$.*

*Proof.* For the flat reference case, write $\alpha = \rho\mathrm{d}x$ and let $\rho_\beta = b\,\mathbb{1}_\Omega$, where $b = 1/|\Omega|$. Up to affine terms in $\rho$, which only add constants on probability measures, the integrand $b\,\varphi_m(\rho/b)$ has non-affine part $b^{1-m}\rho^m/(m(m-1))$ for $m \neq 1$, and $\rho\log\rho$ for $m = 1$. For $m = 1$, McCann's criterion gives $\psi(r) = -d\log r$. For $m \neq 1$, the transformed non-affine part is, up to the irrelevant factor $b^{1-m}$, $\psi_m(r) = \frac{r^{d(1-m)}}{m(m-1)}$. If $m > 1$, this function is convex and nonincreasing. If $0 < m < 1$, it is convex and nonincreasing exactly when $d(1-m) \leqslant 1$, i.e. $m \geqslant 1 - 1/d$. This proves the first claim by Theorem 14.12.

Assume next that $m > 1$ and $V$ is convex. Up to constants, $\mathcal{D}_{\varphi_m}(\alpha|\beta) = \frac{Z^{m-1}}{m(m-1)}\int \rho(x)^m e^{(m-1)V(x)}\,\mathrm{d}x$. Let $T = \nabla\varphi$ be the Brenier map from $\alpha_0 = \rho_0\mathrm{d}x$ to $\alpha_1$, set $T_t = (1-t)\mathrm{Id} + tT$, and write $J_t(x) = \det((1-t)\mathrm{Id} + t\nabla T(x))$. Along the geodesic $\alpha_t = (T_t)_\sharp\alpha_0$,

$$\int \rho_t^m e^{(m-1)V}\,\mathrm{d}x = \int \rho_0(x)^m \exp\big((m-1)\big(V(T_t(x)) - \log J_t(x)\big)\big)\,\mathrm{d}x.$$

For each $x$, the function $t \mapsto V(T_t(x))$ is convex, and $t \mapsto -\log J_t(x)$ is convex because $\log\det$ is concave on positive definite matrices. Since $m - 1 > 0$, the exponential of their sum is convex in $t$. Integrating gives geodesic convexity. The nonsmooth case follows by approximation.

Finally, for $m = 1$, $\mathrm{KL}(\alpha|\beta) = \int \rho\log\rho\,\mathrm{d}x + \int V\mathrm{d}\alpha + \log Z$. The entropy term is 0-geodesically convex by Theorem 14.12. The linear potential term is $\lambda$-geodesically convex when $V$ is $\lambda$-strongly convex, by Proposition 14.10. Taking $\lambda = 0$ also covers the $m = 1$ case in the second claim. □

**Geodesically convex constraints.**

**Proposition 14.14** (Geodesic convexity of density caps)**.** *Assume that $\Omega \subset \mathbb{R}^d$ is convex and let $\rho_{\max} > 0$. The set $\mathcal{K}_{\rho_{\max}}$ in (14.7) is geodesically convex in $\mathcal{P}_2(\Omega)$: if $\alpha_0, \alpha_1 \in \mathcal{K}_{\rho_{\max}}$, then the constant-speed $\mathcal{W}_2$ geodesic $(\alpha_t)_{t\in[0,1]}$ between them also belongs to $\mathcal{K}_{\rho_{\max}}$.*

*Proof.* Assume first that the endpoints have smooth positive densities $\alpha_i = \rho_i\mathrm{d}x$ with $\rho_i \leqslant \rho_{\max}$, and let $T = \nabla\varphi$ be the Brenier map from $\alpha_0$ to $\alpha_1$. Since $\Omega$ is convex, $T_t = (1-t)\mathrm{Id} + tT$ maps $\Omega$ into $\Omega$. The interpolation is $\alpha_t = (T_t)_\sharp\alpha_0$ and $\rho_t(T_t(x)) = \frac{\rho_0(x)}{\det((1-t)\mathrm{Id}+t\nabla T(x))}$. The concavity of $\det^{1/d}$ on positive semidefinite matrices and the identity $\rho_1(T(x))\det\nabla T(x) = \rho_0(x)$ give $\rho_t(T_t(x))^{-1/d} \geqslant (1-t)\rho_0(x)^{-1/d} + t\rho_1(T(x))^{-1/d} \geqslant \rho_{\max}^{-1/d}$. Hence $\rho_t \leqslant \rho_{\max}$ for all $t$. The general case follows by approximation and lower semicontinuity of the $L^\infty$ bound, as in McCann's displacement-convexity theory. □

**Dissipation and finite metric length.** $\frac{\mathrm{d}}{\mathrm{d}t}h(x_t) = -\|\nabla h(x_t)\|^2 = -\|\dot x_t\|^2$. Thus, for $0 \leqslant s < t$, Cauchy's inequality gives

$$\int_s^t \|\dot x_r\| \,\mathrm{d}r \leqslant \sqrt{t-s}\left(\int_s^t \|\dot x_r\|^2 \mathrm{d}r\right)^{1/2} = \sqrt{t-s}\,\big(h(x_s) - h(x_t)\big)^{1/2}. \tag{14.18}$$

**Proposition 14.15** (Finite metric length from dissipation)**.** *Let $(\alpha_r)_{r\in[0,T]}$ be an absolutely continuous curve in $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$. Assume that, for all $0 \leqslant s < t \leqslant T$, it satisfies the exact energy-dissipation identity $f(\alpha_s) - f(\alpha_t) = \int_s^t |\dot\alpha_r|^2_{\mathcal{W}_2}\mathrm{d}r$. Then*

$$\mathrm{Length}_{\mathcal{W}_2}\big((\alpha_r)_{r\in[s,t]}\big) = \int_s^t |\dot\alpha_r|_{\mathcal{W}_2}\,\mathrm{d}r \leqslant \sqrt{t-s}\,\big(f(\alpha_s) - f(\alpha_t)\big)^{1/2}. \tag{14.19}$$

*In the smooth Wasserstein gradient-flow setting with velocity $v_r = -\nabla\!\!\!\!\nabla f(\alpha_r)$, the dissipation identity reads equivalently $f(\alpha_s) - f(\alpha_t) = \int_s^t |\dot\alpha_r|^2_{\mathcal{W}_2}\mathrm{d}r = \int_s^t \int \|v_r(x)\|^2 \mathrm{d}\alpha_r(x)\mathrm{d}r$. If one assumes only the metric energy-dissipation inequality for a curve of maximal slope, then the same length estimate holds with the right-hand side multiplied by $\sqrt{2}$.*

*Proof.* Cauchy's inequality gives

$$\int_s^t |\dot\alpha_r|_{\mathcal{W}_2}\,\mathrm{d}r \leqslant \sqrt{t-s}\left(\int_s^t |\dot\alpha_r|^2_{\mathcal{W}_2}\mathrm{d}r\right)^{1/2}.$$

The exact energy-dissipation identity substitutes the last integral by $f(\alpha_s) - f(\alpha_t)$, which proves (14.19). In the metric theory of curves of maximal slope, the energy-dissipation inequality gives instead $\frac{1}{2}\int_s^t |\dot\alpha_r|^2_{\mathcal{W}_2}\mathrm{d}r \leqslant f(\alpha_s) - f(\alpha_t)$, and the same argument yields the additional factor $\sqrt{2}$. ◻

$\mathcal{W}_2(\alpha_s, \alpha_t) \leqslant \mathrm{Length}_{\mathcal{W}_2}\big((\alpha_r)_{r\in[s,t]}\big) \leqslant \sqrt{f(\alpha_0) - m_T}\,\sqrt{t-s}$, with an additional factor $\sqrt{2}$ under the energy-dissipation inequality. Thus an energy-dissipating trajectory is 1/2-Hölder continuous as a curve in Wasserstein space on every time interval where the energy remains bounded from below.

**Proposition 14.16** (Energy decay for convex Wasserstein flows)**.** *Assume formally that $f$ is geodesically convex, admits a smooth first variation, and has a minimizer $\alpha^\star$. Let $(\alpha_t)_t$ be a smooth solution of the Wasserstein gradient flow $\partial_t\alpha_t + \mathrm{div}(\alpha_t v_t) = 0, \quad v_t = -\nabla\!\!\!\!\nabla f(\alpha_t)$. Then $\frac{\mathrm{d}}{\mathrm{d}t}f(\alpha_t) = -\int \|\nabla\!\!\!\!\nabla f(\alpha_t)(x)\|^2\,\mathrm{d}\alpha_t(x) \leqslant 0$. If $T_t$ is the optimal map from $\alpha_t$ to $\alpha^\star$, then $f(\alpha_t) - f(\alpha^\star) \leqslant -\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\mathcal{W}_2^2(\alpha_t, \alpha^\star)$, and consequently $f(\alpha_t) - f(\alpha^\star) \leqslant \frac{\mathcal{W}_2^2(\alpha_0,\alpha^\star)}{2t}$.*

*Proof.* The chain rule and Proposition 14.5 give $\frac{\mathrm{d}}{\mathrm{d}t}f(\alpha_t) = \int \langle \nabla\!\!\!\!\nabla f(\alpha_t)(x), v_t(x)\rangle\,\mathrm{d}\alpha_t(x) = -\int \|\nabla\!\!\!\!\nabla f(\alpha_t)(x)\|^2\,\mathrm{d}\alpha_t(x)$. Geodesic convexity along the geodesic $((1-s)\mathrm{Id} + sT_t)_\sharp\alpha_t$ gives $f(\alpha^\star) - f(\alpha_t) \geqslant \int \langle \nabla\!\!\!\!\nabla f(\alpha_t)(x), T_t(x) - x\rangle\,\mathrm{d}\alpha_t(x)$. Since $v_t = -\nabla\!\!\!\!\nabla f(\alpha_t)$, this reads $f(\alpha_t) - f(\alpha^\star) \leqslant \int \langle v_t(x), T_t(x) - x\rangle\,\mathrm{d}\alpha_t(x)$. $\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2}\mathcal{W}_2^2(\alpha_t,\alpha^\star) = \int \langle x - T_t(x), v_t(x)\rangle\,\mathrm{d}\alpha_t(x)$, which proves the differential inequality. Integrating it from 0 to $t$ and using the monotonicity of $s \mapsto f(\alpha_s)$ gives

$$t\big(f(\alpha_t) - f(\alpha^\star)\big) \leqslant \int_0^t \big(f(\alpha_s) - f(\alpha^\star)\big)\,\mathrm{d}s \leqslant \frac{1}{2}\mathcal{W}_2^2(\alpha_0, \alpha^\star).$$

◻

**Convergence: the Wasserstein–PL viewpoint.** $|\partial f|^2(\alpha) \coloneqq \int \|\nabla\!\!\!\!\nabla f(\alpha)(x)\|^2\,\mathrm{d}\alpha(x)$

**Definition 14.17** (Wasserstein–Polyak–Łojasiewicz inequality)**.** Let $f_\star \coloneqq \inf_{\alpha\in\mathcal{P}_2(\mathbb{R}^d)} f(\alpha) > -\infty$. The functional $f$ satisfies the Wasserstein–PL inequality with constant $\kappa > 0$ if

$$|\partial f|^2(\alpha) \geqslant 2\kappa\big(f(\alpha) - f_\star\big) \tag{14.20}$$

for every $\alpha$ in the domain of $f$. In the smooth Otto notation this reads $\int \|\nabla\!\!\!\!\nabla f(\alpha)\|^2\,\mathrm{d}\alpha \geqslant 2\kappa\big(f(\alpha) - f_\star\big)$.

**Theorem 14.18** (Wasserstein–PL convergence)**.** *Assume that $f_\star > -\infty$, that $f$ satisfies the Wasserstein–PL inequality (14.20) with constant $\kappa > 0$, and that $(\alpha_t)_{t\geqslant 0}$ is a global Wasserstein gradient flow satisfying the full energy-dissipation identity*

$$\frac{\mathrm{d}}{\mathrm{d}t}f(\alpha_t) = -|\dot\alpha_t|^2 = -|\partial f|^2(\alpha_t) \qquad \textit{for a.e. } t > 0. \tag{14.21}$$

*Set $E(t) \coloneqq f(\alpha_t) - f_\star$. Then, for $0 \leqslant s \leqslant t$, $E(t) \leqslant e^{-2\kappa(t-s)}E(s)$, and the length of the flow segment satisfies $\mathrm{Length}_{\mathcal{W}_2}\big((\alpha_r)_{r\in[s,t]}\big) \leqslant \sqrt{\frac{2}{\kappa}}\big(\sqrt{E(s)} - \sqrt{E(t)}\big)$. If, in addition, $f$ is lower semicontinuous for $\mathcal{W}_2$-convergence, then $\alpha_t$ converges in $\mathcal{W}_2$ to a minimizer $\alpha_\infty \in \mathrm{Argmin}\, f$, and $\mathcal{W}_2(\alpha_t, \alpha_\infty) \leqslant \sqrt{\frac{2}{\kappa}}\sqrt{E(t)} \leqslant \sqrt{\frac{2}{\kappa}}\sqrt{E(0)}\,e^{-\kappa t}$. In particular, $\mathrm{dist}_{\mathcal{W}_2}(\alpha_t, \mathrm{Argmin}\, f) \leqslant \sqrt{\frac{2}{\kappa}}\sqrt{E(0)}\,e^{-\kappa t}$.*

*Proof.* The energy estimate is immediate from (14.21) and (14.20): $E'(t) = -|\partial f|^2(\alpha_t) \leqslant -2\kappa E(t)$ for almost every $t$. Gronwall's lemma gives $E(t) \leqslant e^{-2\kappa(t-s)}E(s)$.

The distance estimate uses the same two identities in a slightly different way. On the set where $E(r) > 0$, the PL inequality gives $|\partial f|(\alpha_r) \geqslant \sqrt{2\kappa E(r)}$. Using the full dissipation identity and $|\dot\alpha_r| = |\partial f|(\alpha_r)$,

$$\begin{aligned}
\mathrm{Length}_{\mathcal{W}_2}\big((\alpha_r)_{r\in[s,t]}\big) &= \int_s^t |\dot\alpha_r|\,\mathrm{d}r = \int_s^t |\partial f|(\alpha_r)\,\mathrm{d}r \\
&= \int_s^t \frac{|\partial f|^2(\alpha_r)}{|\partial f|(\alpha_r)}\,\mathrm{d}r \leqslant \frac{1}{\sqrt{2\kappa}}\int_s^t \frac{|\partial f|^2(\alpha_r)}{\sqrt{E(r)}}\,\mathrm{d}r \\
&= \frac{1}{\sqrt{2\kappa}}\int_s^t \frac{-E'(r)}{\sqrt{E(r)}}\,\mathrm{d}r = \sqrt{\frac{2}{\kappa}}\big(\sqrt{E(s)} - \sqrt{E(t)}\big).
\end{aligned}$$

If $E$ vanishes on a subinterval, then the flow is stationary there by (14.21), so the same estimate follows by applying the argument on the part where $E > 0$. Since Wasserstein distance is bounded by metric length, the same upper bound controls $\mathcal{W}_2(\alpha_s, \alpha_t)$. Letting $t \to +\infty$, the energy estimate gives $E(t) \to 0$, and the length estimate shows that the tail of $(\alpha_t)_t$ is Cauchy. Since $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$ is complete, there is a limit $\alpha_\infty$. Lower semicontinuity yields $f(\alpha_\infty) \leqslant \liminf_{t\to+\infty} f(\alpha_t) = f_\star$, hence $\alpha_\infty \in \operatorname{Argmin} f$. Sending the endpoint of the length estimate to $+\infty$ with $s = t$ fixed gives $\mathcal{W}_2(\alpha_t, \alpha_\infty) \leqslant \sqrt{\frac{2}{\kappa}}\sqrt{E(t)}$. The distance-to-the-set bound follows because $\alpha_\infty$ is one minimizer. □

**Geodesic strong convexity.**

**Proposition 14.19** (Strong geodesic convexity implies Wasserstein–PL)**.** *Assume that $f$ is $\lambda$-geodesically convex with $\lambda > 0$, that $f_\star$ is attained at some $\alpha^\star$, and that the Wasserstein slope is finite at $\alpha$. Then $|\partial f|^2(\alpha) \geqslant 2\lambda\big(f(\alpha) - f_\star\big)$. Thus positive Wasserstein strong convexity implies the Wasserstein–PL inequality with the same constant.*

*Proof.* Fix $\alpha$ and a minimizer $\alpha^\star$. Set $g = f(\alpha) - f_\star, \quad r = \mathcal{W}_2(\alpha, \alpha^\star), \quad a = |\partial f|(\alpha)$. If $r = 0$, then $\alpha = \alpha^\star$ and there is nothing to prove. Otherwise let $(\alpha_s)_{s\in[0,1]}$ be a constant-speed geodesic from $\alpha$ to $\alpha^\star$. By $\lambda$-geodesic convexity, $f(\alpha_s) \leqslant (1-s)f(\alpha) + sf_\star - \frac{\lambda}{2}s(1-s)r^2$. Therefore $f(\alpha) - f(\alpha_s) \geqslant sg + \frac{\lambda}{2}s(1-s)r^2$. Dividing by $\mathcal{W}_2(\alpha, \alpha_s) = sr$ and letting $s \downarrow 0$ in the definition of the descending metric slope yields $a \geqslant \frac{g}{r} + \frac{\lambda}{2}r$. Equivalently, $g \leqslant ar - \lambda r^2/2$. Maximizing the right-hand side over $r \geqslant 0$ gives $g \leqslant a^2/(2\lambda)$, which is the desired PL inequality. □

**Wasserstein Kurdyka–Łojasiewicz inequalities.**

**Definition 14.20** (Wasserstein Kurdyka–Łojasiewicz inequality)**.** Let $f_\star = \inf f$, let $E(\alpha) = f(\alpha) - f_\star$, and fix $c > 0$, $\theta \in [0, 1)$. We say that $f$ satisfies a power Wasserstein–KL inequality on a set $\mathcal{U} \subset \mathcal{P}_2(\mathbb{R}^d)$ if $|\partial f|(\alpha) \geqslant c\,E(\alpha)^\theta, \quad \alpha \in \mathcal{U}, \quad 0 < E(\alpha) < +\infty$. Equivalently, the desingularizing function $\psi(s) = \frac{s^{1-\theta}}{c(1-\theta)}$ satisfies $\psi'(E(\alpha))|\partial f|(\alpha) \geqslant 1$. The Wasserstein–PL inequality (14.20) is the special case $\theta = 1/2$ with $c = \sqrt{2\kappa}$. The regime $\theta > 1/2$ gives sublinear rates, while $\theta = 1/2$ is exactly the exponential PL regime.

**Theorem 14.21** (Sublinear convergence under Wasserstein–KL)**.** *Let $(\alpha_t)_{t\geqslant 0}$ be a Wasserstein gradient flow of $f$ satisfying the full energy-dissipation identity (14.21). Assume that $f_\star > -\infty$, that $E(t) = f(\alpha_t) - f_\star$ is finite, and that the trajectory stays in a region where the Wasserstein–KL inequality of Definition 14.20 holds with $\theta \in (1/2, 1)$. If $E(0) > 0$, then*

$$E(t) \leqslant \big(E(0)^{1-2\theta} + c^2(2\theta - 1)t\big)^{-1/(2\theta-1)}.$$

*Moreover, for every $t \geqslant 0$, $\mathrm{Length}_{\mathcal{W}_2}\big((\alpha_s)_{s\geqslant t}\big) \leqslant \frac{E(t)^{1-\theta}}{c(1-\theta)}$. If $f$ is lower semicontinuous on $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$, then $\alpha_t$ converges to some $\alpha_\infty \in \operatorname{Argmin} f$ and*

$$\mathcal{W}_2(\alpha_t, \alpha_\infty) \leqslant \frac{E(t)^{1-\theta}}{c(1-\theta)} = O\Big(t^{-(1-\theta)/(2\theta-1)}\Big).$$

*Proof.* If $E$ reaches zero at some time, then the flow is already at a global minimizer. Indeed, the integrated energy-dissipation identity forces both $|\dot\alpha_t|$ and $|\partial f|(\alpha_t)$ to vanish afterwards. We therefore argue on intervals where $E > 0$. For a.e. $t$, $-E'(t) = |\partial f|^2(\alpha_t) = |\dot\alpha_t|^2$. The Wasserstein–KL inequality yields $E'(t) \leqslant -c^2E(t)^{2\theta}$. Since $1-2\theta < 0$, $\frac{\mathrm{d}}{\mathrm{d}t}E(t)^{1-2\theta} = (1-2\theta)E(t)^{-2\theta}E'(t) \geqslant c^2(2\theta - 1)$, and integration from 0 to $t$ gives the announced polynomial bound.
For the length estimate, use again $|\dot\alpha_t| = |\partial f|(\alpha_t)$. For $T > t$,

$$\begin{aligned}\int_t^T |\dot\alpha_s|\,\mathrm{d}s = \int_t^T \frac{|\partial f|^2(\alpha_s)}{|\partial f|(\alpha_s)}\,\mathrm{d}s \leqslant \frac{1}{c}\int_t^T -E'(s)E(s)^{-\theta}\,\mathrm{d}s \\ = \frac{E(t)^{1-\theta} - E(T)^{1-\theta}}{c(1-\theta)} \leqslant \frac{E(t)^{1-\theta}}{c(1-\theta)}.\end{aligned}$$

Letting $T \to +\infty$ shows that the tail has finite length. Hence $(\alpha_t)_t$ is Cauchy in $(\mathcal{P}_2(\mathbb{R}^d), \mathcal{W}_2)$, which is complete, and converges to some $\alpha_\infty$. Since the energy bound gives $E(t) \to 0$, lower semicontinuity gives $f(\alpha_\infty) = f_\star$. The distance to $\alpha_\infty$ is bounded by the remaining tail length, which proves the last estimate. □

**Proposition 14.22** (Powers of PL energies are KL)**.** *Let $g \geqslant 0$ satisfy $\inf g = 0$ and the Wasserstein–PL inequality $|\partial g|^2(\alpha) \geqslant 2\kappa g(\alpha)$ on a region where $g > 0$. Fix $r \geqslant 1$ and set $f(\alpha) = g(\alpha)^r$. Whenever the metric-slope chain rule $|\partial(g^r)|(\alpha) = rg(\alpha)^{r-1}|\partial g|(\alpha)$ holds, $f$ satisfies the Wasserstein–KL inequality $|\partial f|(\alpha) \geqslant r\sqrt{2\kappa}\,f(\alpha)^{1-1/(2r)}$. Thus $f$ has KL exponent $\theta = 1 - 1/(2r)$. For $r = 1$ this is PL; for $r > 1$, Theorem 14.21 gives $f(\alpha_t) = O(t^{-r/(r-1)})$ along the $f$-gradient flow, under its hypotheses.*

*Proof.* By the chain rule and the PL inequality for $g$, $|\partial f| = rg^{r-1}|\partial g| \geqslant r\sqrt{2\kappa}\,g^{r-1/2} = r\sqrt{2\kappa}\,f^{1-1/(2r)}$. The exponent and rate follow by substituting $\theta = 1 - 1/(2r)$ into Theorem 14.21. □

**Convexity and curvature.** The same language is not restricted to subsets of $\mathbb{R}^d$. If $(\mathcal{X}, d, \mathfrak{m})$ is a geodesic metric-measure space, $\mathcal{W}_2$ geodesics can be defined by transporting each pair of endpoints along metric geodesics, or more intrinsically by dynamical optimal plans on path space, as discussed in Section **??**.

$$\mathrm{Ent}_{\mathfrak{m}}(\alpha) \coloneqq \begin{cases} \displaystyle\int_{\mathcal{X}} \rho\log\rho\,\mathrm{d}\mathfrak{m}, & \text{if } \alpha = \rho\,\mathfrak{m}, \\ +\infty, & \text{otherwise.}\end{cases}$$

**Theorem 14.23** (Ricci curvature and entropy convexity)**.** *Let $(M, g)$ be a smooth compact connected Riemannian manifold without boundary, and let $\mathfrak{m} = \mathrm{vol}_g$. For $\lambda \in \mathbb{R}$, the lower Ricci bound $\mathrm{Ric}_g \geqslant \lambda g$ holds if and only if $\mathrm{Ent}_{\mathfrak{m}}$ is $\lambda$-geodesically convex on $(\mathcal{P}_2(M), \mathcal{W}_2)$.*

## 14.3 Functional Inequalities via Optimal Transport

Optimal transport proves inequalities by turning comparison into geometry. One transports a density by a monotone or Brenier map, interpolates along the resulting rays, and converts concavity of a Jacobian determinant or convexity of an energy into an integral estimate.

**Brunn–Minkowski and isoperimetry.**

**Proposition 14.24** (Brunn–Minkowski and Euclidean isoperimetry)**.** *Let $A, B \subset \mathbb{R}^d$ be bounded Borel sets with positive Lebesgue measure. For $t \in [0,1]$, set $(1-t)A + tB := \{(1-t)x + ty \ : \ x \in A, \ y \in B\}$. Then* $\mathrm{Vol}((1-t)A + tB)^{1/d} \geqslant (1-t)\mathrm{Vol}(A)^{1/d} + t\,\mathrm{Vol}(B)^{1/d}$. $\mathrm{Per}(A) \geqslant d\,\omega_d^{1/d}\mathrm{Vol}(A)^{(d-1)/d}$. *The constant is sharp, with equality for Euclidean balls.*

*Proof.* We first give the argument for regular sets, the general Borel case following by approximation from inside and outside. Let $\alpha$ and $\beta$ be the uniform probability measures on $A$ and $B$, and let $T = \nabla\varphi$ be the Brenier map with $T_\sharp\alpha = \beta$. The interpolating map $T_t = (1-t)\mathrm{Id} + tT$ sends $\alpha$ to a measure $\alpha_t$ supported in $(1-t)A + tB$. At points where $T$ is differentiable, $\det DT_t = \det((1-t)\mathrm{Id} + tDT)$. Since $DT$ is symmetric positive semidefinite and $\det^{1/d}$ is concave on the positive semidefinite cone, $\det(DT_t)^{1/d} \geqslant (1-t) + t\det(DT)^{1/d}$. The change-of-variables identity for the uniform measures gives $\det(DT) = \mathrm{Vol}(B)/\mathrm{Vol}(A)$ almost everywhere. If $\rho_t$ denotes the density of $\alpha_t$, a second application of the same formula gives $\rho_t(T_t(x))^{-1/d} = \mathrm{Vol}(A)^{1/d}\det(DT_t(x))^{1/d} \geqslant c_t$, where $c_t = (1-t)\mathrm{Vol}(A)^{1/d} + t\mathrm{Vol}(B)^{1/d}$. Thus $\rho_t \leqslant c_t^{-d}$ almost everywhere. Since $\alpha_t$ has total mass one and is supported in $(1-t)A + tB$, this gives $\mathrm{Vol}((1-t)A + tB) \geqslant c_t^d$, which is the displayed Brunn–Minkowski inequality.
For isoperimetry, use the homogeneity of Brunn–Minkowski and apply it to $A$ and $\varepsilon B_1$. Writing $A + \varepsilon B_1$ for the parallel set gives $\mathrm{Vol}(A + \varepsilon B_1)^{1/d} \geqslant \mathrm{Vol}(A)^{1/d} + \varepsilon\omega_d^{1/d}$. Using $\mathrm{Vol}(A + \varepsilon B_1) = \mathrm{Vol}(A) + \varepsilon\,\mathrm{Per}(A) + o(\varepsilon)$ and differentiating at $\varepsilon = 0$ yields the claimed perimeter bound. Balls attain equality, so the constant is optimal. □

**Prékopa–Leindler.**

**Proposition 14.25** (Prékopa–Leindler inequality)**.** *Let $u, v, w : \mathbb{R}^d \to [0, +\infty)$ be integrable, and let $t \in [0,1]$. Assume that* $w((1-t)x + ty) \geqslant u(x)^{1-t}v(y)^t$ *for all $x, y \in \mathbb{R}^d$. Then*

$$\int_{\mathbb{R}^d} w \geqslant \left(\int_{\mathbb{R}^d} u\right)^{1-t}\left(\int_{\mathbb{R}^d} v\right)^t.$$

*Proof.* The endpoint cases are immediate, so fix $t \in (0,1)$. Set $U = \int u$ and $V = \int v$. If $U = 0$ or $V = 0$, the conclusion is trivial; hence assume $U, V > 0$. Let $\alpha = (u/U)\mathrm{d}x$ and $\beta = (v/V)\mathrm{d}x$, and let $T$ be the Brenier map from $\alpha$ to $\beta$. Put $T_t = (1-t)\mathrm{Id} + tT$. On a smooth positive regularization, $T_t$ is the gradient of a strictly convex function and is therefore one-to-one almost everywhere. The area formula gives $\int w \geqslant \int w(T_t(x))\det(DT_t(x))\mathrm{d}x$. For each eigenvalue $s \geqslant 0$ of $DT$, weighted arithmetic–geometric mean gives $(1-t) + ts \geqslant s^t$. Hence $\det((1-t)\mathrm{Id} + tDT) \geqslant \det(DT)^t$. $\int w \geqslant \int u(x)^{1-t}v(T(x))^t\det(DT(x))^t\mathrm{d}x$. Since $T_\sharp\alpha = \beta$, the Monge–Ampère identity gives $\det(DT(x)) = \frac{u(x)V}{v(T(x))U}$ almost everywhere. Substitution yields

$$\int w \geqslant \left(\frac{V}{U}\right)^t \int u(x)\mathrm{d}x = U^{1-t}V^t.$$

Approximation removes the smoothness and positivity assumptions. □

**Logarithmic Sobolev inequalities.**

**Definition 14.26** (Relative Fisher information and logarithmic Sobolev inequality)**.** Let $\beta \in \mathcal{P}(\mathbb{R}^d)$ have a positive density. For a probability measure $\alpha = h\beta$, define

$$\mathcal{I}(\alpha|\beta) := 4\int \|\nabla\sqrt{h}\|^2\,\mathrm{d}\beta = \int \|\nabla\log h\|^2\,\mathrm{d}\alpha, \tag{14.22}$$

whenever $\sqrt{h}$ has a weak gradient and the integral is finite, and set $\mathcal{I}(\alpha|\beta) = +\infty$ otherwise, including when $\alpha \not\ll \beta$. The measure $\beta$ satisfies a logarithmic Sobolev inequality with constant $\lambda > 0$ if

$$\mathrm{KL}(\alpha|\beta) \leqslant \frac{1}{2\lambda}\mathcal{I}(\alpha|\beta) \qquad \text{for every probability measure } \alpha. \tag{14.23}$$

**Proposition 14.27** (Curvature implies logarithmic Sobolev)**.** *Let $\beta = Z^{-1}e^{-V}\mathrm{d}x$ be a smooth probability measure on $\mathbb{R}^d$. If $\nabla^2 V \succeq \lambda\mathrm{Id}$ for some $\lambda > 0$, then $\beta$ satisfies (14.23) with constant $\lambda$.*

*Proof.* Apply the HWI inequality of Proposition 14.33. Under the curvature assumption, for every absolutely continuous $\alpha$, $\mathrm{KL}(\alpha|\beta) \leqslant \mathcal{W}_2(\alpha,\beta)\sqrt{\mathcal{I}(\alpha|\beta)} - \frac{\lambda}{2}\mathcal{W}_2^2(\alpha,\beta)$. For fixed $s = \sqrt{\mathcal{I}(\alpha|\beta)}$, maximize the right-hand side over $r = \mathcal{W}_2(\alpha,\beta) \geqslant 0$. The elementary inequality $rs - \lambda r^2/2 \leqslant s^2/(2\lambda)$ gives (14.23). Approximation yields the standard nonsmooth statement. This is the Otto–Villani transport route from the Bakry–Emery curvature bound to logarithmic Sobolev; the original $\Gamma_2$ calculus gives the same criterion directly. □

**Proposition 14.28** (Logarithmic Sobolev inequality implies entropy decay)**.** *Let $\beta = e^{-V}\mathrm{d}x/Z$ be a smooth probability measure on $\mathbb{R}^d$ satisfying the logarithmic Sobolev inequality (14.23) with constant $\lambda > 0$. Let $\alpha_t$ solve the Wasserstein gradient flow of $f(\alpha) = \mathrm{KL}(\alpha|\beta)$, namely $\partial_t\rho_t = \nabla\cdot(\rho_t\nabla V) + \Delta\rho_t, \qquad \alpha_t = \rho_t\mathrm{d}x$. If $\mathrm{KL}(\alpha_0|\beta) < +\infty$, then $\mathrm{KL}(\alpha_t|\beta) \leqslant e^{-2\lambda t}\,\mathrm{KL}(\alpha_0|\beta)$. Moreover, if $\alpha_0, \beta \in \mathcal{P}_2(\mathbb{R}^d)$, then $\mathcal{W}_2(\alpha_t,\beta) \leqslant \sqrt{\frac{2}{\lambda}\,\mathrm{KL}(\alpha_0|\beta)}\,e^{-\lambda t}$.*

*Proof.* The first variation of $f$ is $\log(\rho/\rho_\beta)$, up to an additive constant. Along the smooth Wasserstein gradient flow, $\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{KL}(\alpha_t|\beta) = -\int \|\nabla\log\frac{\mathrm{d}\alpha_t}{\mathrm{d}\beta}\|^2\,\mathrm{d}\alpha_t = -\mathcal{I}(\alpha_t|\beta)$. The logarithmic Sobolev inequality bounds the Fisher information below by $2\lambda\,\mathrm{KL}(\alpha_t|\beta)$. Hence $\frac{\mathrm{d}}{\mathrm{d}t}\mathrm{KL}(\alpha_t|\beta) \leqslant -2\lambda\,\mathrm{KL}(\alpha_t|\beta)$, and Gronwall's lemma gives the entropy estimate. The Wasserstein estimate follows from the length estimate in the Wasserstein–PL theorem: apply Theorem 14.18 to $f = \mathrm{KL}(\cdot|\beta)$, whose unique minimizer is $\beta$, and use the logarithmic Sobolev inequality as the corresponding Wasserstein–PL inequality. □

**Gaussian logarithmic Sobolev inequality.**

**Proposition 14.29** (Gaussian logarithmic Sobolev inequality)**.** *Let $\gamma_d$ be the standard Gaussian measure on $\mathbb{R}^d$. If $\rho > 0$ is smooth, has sufficient decay, and $\int \rho \,\mathrm{d}\gamma_d = 1$, then* $\mathrm{Ent}_{\gamma_d}(\rho) \coloneqq \int \rho \log \rho \,\mathrm{d}\gamma_d \leqslant \frac{1}{2}\int \frac{\|\nabla \rho\|^2}{\rho}\,\mathrm{d}\gamma_d$.

*Proof.* Let $\alpha = \rho\gamma_d$, and let $T = \nabla\varphi$ be the Brenier map with $T_\sharp\alpha = \gamma_d$. Write $T(x) = x + \xi(x)$. The Monge–Ampère equation reads $\rho(x)e^{-\|x\|^2/2} = e^{-\|T(x)\|^2/2}\det(DT(x))$. Thus $\log\rho(x) = -\langle x, \xi(x)\rangle - \frac{1}{2}\|\xi(x)\|^2 + \log\det(\mathrm{Id} + D\xi(x))$. Since $DT$ is positive semidefinite, $\log\det(\mathrm{Id} + D\xi) \leqslant \mathrm{tr}(D\xi) = \operatorname{div}\xi$. Multiplying by $\rho$ and integrating with respect to $\gamma_d$ gives $\mathrm{Ent}_{\gamma_d}(\rho) \leqslant \int \rho(\operatorname{div}\xi - \langle x, \xi\rangle)\,\mathrm{d}\gamma_d - \frac{1}{2}\int \rho\|\xi\|^2\,\mathrm{d}\gamma_d$. Gaussian integration by parts gives $\int \rho(\operatorname{div}\xi - \langle x, \xi\rangle)\,\mathrm{d}\gamma_d = -\int\langle\nabla\rho, \xi\rangle\,\mathrm{d}\gamma_d$. Completing the square pointwise, $-\langle\nabla\rho, \xi\rangle - \frac{1}{2}\rho\|\xi\|^2 \leqslant \frac{1}{2}\frac{\|\nabla\rho\|^2}{\rho}$. This proves the inequality. The general Sobolev-class statement follows by approximation. ☐

$$\rho_a(x) = \exp\left(\langle a, x\rangle - \frac{1}{2}\|a\|^2\right), \qquad a \in \mathbb{R}^d,$$

$\int \rho_a \log\rho_a \,\mathrm{d}\gamma_d = \frac{1}{2}\|a\|^2, \qquad \frac{1}{2}\int \frac{\|\nabla\rho_a\|^2}{\rho_a}\,\mathrm{d}\gamma_d = \frac{1}{2}\|a\|^2$. Thus equality is attained along translations, and the sharp logarithmic Sobolev constant of the standard Gaussian is $\lambda = 1$. Proposition 14.28 then gives the classical exponential convergence of the Ornstein–Uhlenbeck flow.

**Poincaré as a linearized Wasserstein–PL inequality.** $\beta_\varepsilon = (1 + \varepsilon\xi)\beta, \qquad \beta_0 = \beta.$

$$\mathsf{H}^f_\beta(\xi) \coloneqq \lim_{\varepsilon\to 0}\frac{2(f(\beta_\varepsilon) - f(\beta))}{\varepsilon^2}, \qquad \mathsf{D}^f_\beta(\xi) \coloneqq \lim_{\varepsilon\to 0}\frac{|\partial f|^2(\beta_\varepsilon)}{\varepsilon^2},$$

**Proposition 14.30** (Linearizing a Wasserstein–PL inequality)**.** *Let $\beta$ be a minimizer of an energy $f$, and assume that $f$ satisfies $|\partial f|^2(\alpha) \geqslant 2\kappa\big(f(\alpha) - f(\beta)\big)$ in a neighborhood of $\beta$. Let $\xi$ be a smooth bounded perturbation with $\int \xi\,\mathrm{d}\beta = 0$, and set $\beta_\varepsilon = (1+\varepsilon\xi)\beta$. If $\mathsf{H}^f_\beta(\xi)$ and $\mathsf{D}^f_\beta(\xi)$ exist, then $\kappa\,\mathsf{H}^f_\beta(\xi) \leqslant \mathsf{D}^f_\beta(\xi)$.*

*Proof.* Apply the PL inequality to $\beta_\varepsilon$: $|\partial f|^2(\beta_\varepsilon) \geqslant 2\kappa\big(f(\beta_\varepsilon) - f(\beta)\big)$. Divide by $\varepsilon^2$ and let $\varepsilon \to 0$. The definitions of $\mathsf{H}^f_\beta$ and $\mathsf{D}^f_\beta$ give the claim. ☐

For $f(\alpha) = \mathrm{KL}(\alpha|\beta)$, Proposition 14.30 says precisely that logarithmic Sobolev linearizes to Poincaré. Indeed, for $h_\varepsilon = 1 + \varepsilon\xi$,

$$\mathrm{KL}(h_\varepsilon\beta|\beta) = \frac{\varepsilon^2}{2}\int \xi^2\,\mathrm{d}\beta + o(\varepsilon^2), \qquad \mathcal{I}(h_\varepsilon\beta|\beta) = \varepsilon^2\int\|\nabla\xi\|^2\,\mathrm{d}\beta + o(\varepsilon^2).$$

**Proposition 14.31** (Poincaré from logarithmic Sobolev)**.** *Let $\beta$ be a smooth probability measure on $\mathbb{R}^d$. Assume that $\beta$ satisfies the logarithmic Sobolev inequality with constant $\lambda > 0$,*

$$\mathrm{KL}(\alpha|\beta) \leqslant \frac{1}{2\lambda}\mathcal{I}(\alpha|\beta), \qquad \mathcal{I}(\alpha|\beta) \coloneqq \int \|\nabla\log\frac{\mathrm{d}\alpha}{\mathrm{d}\beta}\|^2\,\mathrm{d}\alpha.$$

*Then, for every smooth $\xi$ with $\int\xi\,\mathrm{d}\beta = 0$, $\lambda\int\xi^2\,\mathrm{d}\beta \leqslant \int\|\nabla\xi\|^2\,\mathrm{d}\beta$. In particular, if $\mathrm{d}\beta = Z^{-1}e^{-V}\mathrm{d}x$ and $\nabla^2 V \succeq \lambda\mathrm{Id}$, so that $\beta$ is $\lambda$-strongly log-concave, then the conclusion holds by the Bakry–Emery criterion.*

*Proof.* For $h_\varepsilon = 1 + \varepsilon\xi$, with $|\varepsilon|$ small enough, one has $\beta_\varepsilon = h_\varepsilon\beta$ and $\mathrm{KL}(\beta_\varepsilon|\beta) = \int h_\varepsilon\log h_\varepsilon\,\mathrm{d}\beta = \frac{\varepsilon^2}{2}\int\xi^2\,\mathrm{d}\beta + o(\varepsilon^2)$, because $\int\xi\,\mathrm{d}\beta = 0$. The Fisher information expands as

$$\mathcal{I}(\beta_\varepsilon|\beta) = \int\|\nabla\log h_\varepsilon\|^2 h_\varepsilon\,\mathrm{d}\beta = \varepsilon^2\int\|\nabla\xi\|^2\,\mathrm{d}\beta + o(\varepsilon^2).$$

Inserting these expansions in the logarithmic Sobolev inequality and letting $\varepsilon \to 0$ gives the Poincaré inequality. The last assertion is the usual Bakry–Emery implication from $\lambda$-convexity of $V$ to the logarithmic Sobolev inequality. ☐

$L_\beta\xi \coloneqq \Delta\xi - \langle\nabla V, \nabla\xi\rangle = \rho_\beta^{-1}\nabla\cdot(\rho_\beta\nabla\xi)$. $\int\|\nabla\xi\|^2\,\mathrm{d}\beta = \langle\xi, -L_\beta\xi\rangle_{L^2(\beta)}$.

$$\|\dot h\|^2_{-1,\beta} \coloneqq \inf_v\left\{\int\|v\|^2\,\mathrm{d}\beta \;:\; \dot h = -\rho_\beta^{-1}\nabla\cdot(\rho_\beta v)\right\}.$$

**Talagrand's transport-entropy inequality.** The distance estimate in Theorem 14.18 already reveals the general principle: when $\mathrm{KL}(\cdot|\beta)$ satisfies logarithmic Sobolev with constant $\lambda$, the entropy flow yields the transport-entropy bound $\mathcal{W}_2^2(\alpha,\beta) \leqslant 2\,\mathrm{KL}(\alpha|\beta)/\lambda$.

**Proposition 14.32** (Gaussian $T_2$ inequality)**.** *For every $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, $\frac{1}{2}\mathcal{W}_2^2(\alpha,\gamma_d) \leqslant \mathrm{KL}(\alpha|\gamma_d)$, with the convention that the right-hand side is $+\infty$ when $\alpha \not\ll \gamma_d$.*

*Proof.* If $\mathrm{KL}(\alpha|\gamma_d) = +\infty$, the claim is immediate. Assume first that $\alpha = \rho\gamma_d$ with $\rho$ smooth and positive. Let $T = \nabla\varphi$ be the Brenier map with $T_\sharp\gamma_d = \alpha$, and write $T(x) = x + \xi(x)$. The change of variables gives $\rho(T(x))e^{-\|T(x)\|^2/2}\det(DT(x)) = e^{-\|x\|^2/2}$. Hence $\log\rho(T(x)) = \langle x, \xi(x)\rangle + \frac{1}{2}\|\xi(x)\|^2 - \log\det(\mathrm{Id} + D\xi(x))$. Using again $\log\det(\mathrm{Id} + D\xi) \leqslant \operatorname{div}\xi$ and integrating with respect to $\gamma_d$, $\mathrm{KL}(\alpha|\gamma_d) \geqslant \frac{1}{2}\int\|\xi\|^2\,\mathrm{d}\gamma_d + \int(\langle x, \xi\rangle - \operatorname{div}\xi)\,\mathrm{d}\gamma_d$. The last integral vanishes by Gaussian integration by parts. Since $T$ is optimal, $\int\|\xi\|^2\,\mathrm{d}\gamma_d = \mathcal{W}_2^2(\gamma_d,\alpha)$, which proves the claim. Approximation gives the general case. Equality holds for translations $\alpha = \mathcal{N}(a,\mathrm{Id})$, so the constant is sharp. ☐

**The HWI bridge.**

**Proposition 14.33** (Otto–Villani HWI inequality)**.** *Let $\beta = \rho_\beta \mathrm{d}x = e^{-V}\mathrm{d}x/Z$ be a smooth probability measure in $\mathcal{P}_2(\mathbb{R}^d)$, and assume that $\nabla^2 V \succeq \lambda \mathrm{Id}$ for some $\lambda \in \mathbb{R}$. For $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, define the relative Fisher information by $\mathcal{I}(\alpha|\beta) \coloneqq \int \|\nabla \log \frac{\rho_\alpha}{\rho_\beta}\|^2 \,\mathrm{d}\alpha$, when $\alpha = \rho_\alpha \mathrm{d}x$ and the expression is finite, and set $\mathcal{I}(\alpha|\beta) = +\infty$ otherwise. Then $\mathrm{KL}(\alpha|\beta) \leqslant \mathcal{W}_2(\alpha,\beta)\sqrt{\mathcal{I}(\alpha|\beta)} - \frac{\lambda}{2}\mathcal{W}_2^2(\alpha,\beta)$.*

*Proof.* If $\mathcal{I}(\alpha|\beta) = +\infty$, there is nothing to prove. Assume first that $\rho_\alpha$ is smooth and positive. Let $T$ be the Brenier map from $\alpha$ to $\beta$, set $v = T - \mathrm{Id}$, and let $\alpha_t = ((1-t)\mathrm{Id} + tT)_\sharp \alpha$. By Theorem 14.12 and Proposition 14.10, $f(\eta) = \mathrm{KL}(\eta|\beta)$ is $\lambda$-geodesically convex. Thus the one-dimensional function $e(t) = f(\alpha_t)$ satisfies $e(1) \geqslant e(0) + e'(0) + \frac{\lambda}{2}\mathcal{W}_2^2(\alpha,\beta)$. Since $e(1) = f(\beta) = 0$, it remains to compute the initial derivative. The continuity equation for the geodesic gives $\partial_t \alpha_t + \nabla\cdot(\alpha_t v_t) = 0$ and $v_0 = v$. Hence $e'(0) = \int \langle \nabla \log \frac{\rho_\alpha}{\rho_\beta},\, v\rangle \,\mathrm{d}\alpha$. Therefore $\mathrm{KL}(\alpha|\beta) \leqslant -\int \langle \nabla \log \frac{\rho_\alpha}{\rho_\beta},\, T - \mathrm{Id}\rangle \,\mathrm{d}\alpha - \frac{\lambda}{2}\mathcal{W}_2^2(\alpha,\beta)$. Cauchy–Schwarz and $\int \|T - \mathrm{Id}\|^2 \mathrm{d}\alpha = \mathcal{W}_2^2(\alpha,\beta)$ give the announced bound. The general case follows by standard approximation and lower semicontinuity of entropy, Fisher information, and $\mathcal{W}_2$. □

When $\lambda > 0$, the HWI inequality is a bridge from curvature to PL. It is not itself a PL inequality because it still contains the distance term $\mathcal{W}_2(\alpha,\beta)$. Optimizing the right-hand side, or simply using Young's inequality $rs - \lambda r^2/2 \leqslant s^2/(2\lambda)$, gives the logarithmic Sobolev estimate
$\mathrm{KL}(\alpha|\beta) \leqslant \frac{1}{2\lambda}\mathcal{I}(\alpha|\beta)$.

### 14.4 Training Two-Layer MLPs as Wasserstein Flows

**Wasserstein training of two-layer MLPs.** $x = (u,v) \in \mathbb{R}^d \times \mathbb{R}^{d'}$, where $u$ is the inner weight and $v$ is the outer vector weight. For a scalar nonlinearity $\sigma$, define the vector-valued feature
$\psi(x,z) = v\,\sigma(\langle u,\, z\rangle) \in \mathbb{R}^{d'}$. $G_X(z) = \frac{1}{n}\sum_{i=1}^n \psi(x_i,z)$, $\qquad G_\alpha(z) = \int \psi(x,z)\mathrm{d}\alpha(x)$, $\qquad \alpha = \frac{1}{n}\sum_i \delta_{x_i}$. Let $\zeta$ be a probability distribution on data-label pairs $(z,y) \in \mathbb{R}^d \times \mathbb{R}^{d'}$. The population risk is
$f(\alpha) = \int \ell(G_\alpha(z), y)\mathrm{d}\zeta(z,y)$,

$$\dot{x}_i = -n\nabla_{x_i} F(X), \qquad F(X) = f\left(\frac{1}{n}\sum_i \delta_{x_i}\right),$$

which is the gradient flow of $F(X) = f(\alpha_X)$ for this Wasserstein particle metric, equivalently Euclidean gradient descent with the time scale multiplied by $n$. It gives a particle discretization of (14.3).
Assume that $\ell$ is differentiable in its first variable. The first variation is

$$\delta f(\alpha)(x) = \int \langle \nabla_1 \ell(G_\alpha(z), y),\, \psi(x,z)\rangle \mathrm{d}\zeta(z,y), \tag{14.24}$$

$\mathbb{W} f(\alpha)(x) = \nabla_x \delta f(\alpha)(x) = \int [D_x\psi(x,z)]^\top \nabla_1 \ell(G_\alpha(z), y)\mathrm{d}\zeta(z,y)$. For the squared Euclidean loss $\ell(s,y) = \frac{1}{2}\|s-y\|^2$, the energy is the sum of a quadratic interaction and a linear potential:

$$f(\alpha) = \frac{1}{2}\iint k(x,x')\mathrm{d}\alpha(x)\mathrm{d}\alpha(x') + \int g(x)\mathrm{d}\alpha(x) + \frac{1}{2}\int \|y\|^2 \mathrm{d}\zeta(z,y), \tag{14.25}$$

$$k(x,x') = \int \langle \psi(x,z),\, \psi(x',z)\rangle \mathrm{d}\zeta(z,y), \qquad g(x) = -\int \langle y,\, \psi(x,z)\rangle \mathrm{d}\zeta(z,y). \tag{14.26}$$

$\delta f(\alpha)(x) = \int k(x,x')\mathrm{d}\alpha(x') + g(x)$, $\qquad \mathbb{W} f(\alpha)(x) = \int \nabla_x k(x,x')\mathrm{d}\alpha(x') + \nabla_x g(x)$.

**Classical convexity and stationarity.** $f(\alpha) = \frac{1}{2}\iint k(x,x')\mathrm{d}\alpha(x)\mathrm{d}\alpha(x') + \int V(x)\mathrm{d}\alpha(x) + C$, $Q((1-s)\alpha + s\beta) \leqslant (1-s)Q(\alpha) + sQ(\beta)$, $\qquad Q(\alpha) = \frac{1}{2}\iint k\,\mathrm{d}\alpha\mathrm{d}\alpha$.

**Proposition 14.34** (Affine convexity and stationary positive densities)**.** *Let $\Omega \subset \mathbb{R}^d$ be connected, let $f = Q + \int V\,\mathrm{d}\alpha + C$ be as above on $\mathcal{P}(\Omega)$, and assume that $Q$ is classically convex. Suppose that a Wasserstein gradient flow $\alpha_t$ of $f$ converges to $\alpha_\infty = \rho_\infty \mathrm{d}x$, where $\rho_\infty > 0$ almost everywhere on $\Omega$, and that $\inf_{t\geqslant 0} f(\alpha_t) > -\infty$. Assume that $\delta f(\alpha_\infty) \in C^1(\Omega)$ and that the squared Wasserstein slope is lower semicontinuous along the convergence, in the sense that*

$$\int_\Omega \|\nabla \delta f(\alpha_\infty)\|^2 \mathrm{d}\alpha_\infty \leqslant \liminf_{n\to\infty} \int_\Omega \|\nabla \delta f(\alpha_{t_n})\|^2 \mathrm{d}\alpha_{t_n}$$

*whenever $t_n \to \infty$. Then $\alpha_\infty$ is a global minimizer of $f$.*

*Proof.* The energy-dissipation identity and the lower bound on $f$ along the convergent trajectory imply $\int_0^\infty \int_\Omega \|\nabla \delta f(\alpha_t)\|^2 \mathrm{d}\alpha_t \mathrm{d}t < +\infty$. Hence there exists $t_n \to \infty$ for which the inner integral tends to zero. The assumed lower semicontinuity gives $\int_\Omega \|\nabla \delta f(\alpha_\infty)\|^2 \mathrm{d}\alpha_\infty = 0$. Because $\rho_\infty > 0$ almost everywhere and $\nabla \delta f(\alpha_\infty)$ is continuous, this gradient vanishes throughout $\Omega$. Connectedness then implies that $\delta f(\alpha_\infty)$ is constant. $\int_\Omega \delta f(\alpha_\infty)\mathrm{d}(\beta - \alpha_\infty) = 0$. Classical convexity of $f$ in the affine variable $\alpha$ gives the subgradient inequality $f(\beta) \geqslant f(\alpha_\infty) + \int \delta f(\alpha_\infty)\mathrm{d}(\beta - \alpha_\infty) = f(\alpha_\infty)$. Thus no competitor has smaller energy. Strict positivity is essential to this argument: without it, stationarity controls the first variation only on the support explored by the limiting measure. For square-loss two-layer mean-field models, (14.25) is exactly of this quadratic-plus-linear form, and positive semidefiniteness of the induced kernel $k$ is the classical convexity assumption. □

**Proposition 14.35** (Formal global optimality for two-homogeneous mean-field flows)**.** *Assume that the feature is positively two-homogeneous in the neuron variable, $\psi(\lambda x, z) = \lambda^2 \psi(x,z) \qquad (\lambda > 0)$, and that $f(\alpha) = J(G_\alpha)$ with $J$ convex and differentiable as a functional of the predictor. Let $\alpha$ be a smooth stationary point of the Wasserstein flow, so that $\nabla_x \delta f(\alpha)(x) = 0$ on $\mathrm{supp}(\alpha)$. Assume also full directional support: for every unit direction $\omega$, the support of $\alpha$ intersects the ray $\{\lambda\omega : \lambda > 0\}$. Then $\alpha$ is a global minimizer of $f$ over the mean-field model class.*

*Proof.* Choose the first-variation representative

$$h_\alpha(x) = \delta f(\alpha)(x) = \langle \nabla J(G_\alpha), \psi(x,\cdot)\rangle_\rho .$$

Additive constants in first variations do not affect the Wasserstein gradient or first-order inequalities between probability measures; this normalization is useful because it inherits the homogeneity of $\psi$. By two-homogeneity of $\psi$, one has $h_\alpha(\lambda x) = \lambda^2 h_\alpha(x), \qquad \lambda > 0$. Taking $x = 0$ and any $\lambda \neq 1$ in this identity gives $h_\alpha(0) = 0$. We first record that stationarity forces $h_\alpha$ to vanish on $\mathrm{supp}(\alpha)$. Indeed, if $x = r\omega \in \mathrm{supp}(\alpha)$ with $r > 0$ and $\|\omega\| = 1$, then $0 = \langle \nabla h_\alpha(r\omega), \omega\rangle = \frac{\mathrm{d}}{\mathrm{d}s} h_\alpha(s\omega)\Big|_{s=r} = 2r h_\alpha(\omega)$, so $h_\alpha(\omega) = 0$, and hence $h_\alpha(x) = r^2 h_\alpha(\omega) = 0$. The case $x = 0$ has already been covered, so $h_\alpha = 0$ on $\mathrm{supp}(\alpha)$ and $\int h_\alpha \,\mathrm{d}\alpha = 0$.
Suppose that a competitor $\beta$ satisfies $f(\beta) < f(\alpha)$. Since $G_\beta - G_\alpha = \int \psi(x,\cdot)\mathrm{d}(\beta-\alpha)(x)$, convexity of $J$ gives $f(\beta) - f(\alpha) \geqslant \int h_\alpha(x)\mathrm{d}(\beta-\alpha)(x)$. The left-hand side is negative, while $\int h_\alpha \,\mathrm{d}\alpha = 0$, so $\int h_\alpha \,\mathrm{d}\beta < 0$. Since $h_\alpha(0) = 0$, this strict negativity must occur away from the origin: there exists a nonzero point $x = r\omega$ with $r > 0$, $\|\omega\| = 1$, and $h_\alpha(x) < 0$. Equivalently $h_\alpha(\omega) = h_\alpha(x)/r^2 < 0$. By full directional support, there is $r_\omega > 0$ such that $r_\omega \omega \in \mathrm{supp}(\alpha)$. At this support point, $\langle \nabla h_\alpha(r_\omega\omega), \omega\rangle = 2r_\omega h_\alpha(\omega) < 0$, which contradicts the stationarity condition $\nabla h_\alpha = 0$ on $\mathrm{supp}(\alpha)$. Thus no competitor has smaller risk.
The rigorous Chizat–Bach theorem replaces the full directional support assumption by propagation and overparameterization hypotheses ensuring that any negative first-variation direction is represented by the evolving support. The same radial-derivative contradiction, applied after this support-propagation step, then rules out non-optimal stationary limits. □

### 14.5 Generalized Dynamic Wasserstein Flows

**Generalized Wasserstein flows.**

**Definition 14.36** (Generalized action gradient flow)**.** Let $d = \mathsf{D}_{\mathbb{A}}$ be the dynamic distance induced by the action $\mathbb{A}$ in Definition 13.8. Given $\alpha^{(0)} \in \mathcal{P}(\mathbb{R}^d)$ and $\tau > 0$, define

$$\alpha^{(\ell+1)} \in \operatorname*{argmin}_{\alpha \in \mathcal{P}(\mathbb{R}^d)} \left\{ \frac{1}{2\tau} d^2(\alpha^{(\ell)}, \alpha) + f(\alpha) \right\}. \tag{14.27}$$

Any continuous-time limit of its piecewise-constant interpolants is called a generalized action gradient flow of $f$ for $\mathbb{A}$.

$$\partial_t \alpha_t + \mathrm{div}(\alpha_t v_{\alpha_t}) = 0, \tag{14.28}$$

The metric ingredient needed below is the squared-speed action $\mathbb{A}(\alpha, w)$ introduced in Section 13.3. Its length distance is already defined in (13.15); the present section only uses the local cost in the infinitesimal variational step defined below.

$$\mathbb{A}_p(\alpha, w) = \left( \int \|w\|^p \mathrm{d}\alpha \right)^{2/p}.$$

$A_2(a, w) = a\|w\|^2, \qquad J_2(a, m) = \frac{\|m\|^2}{a}, \qquad \mathsf{D}_{\mathbb{A}} = \mathcal{W}_2 .$

**Definition 14.37** (Penalized minimization oracle)**.** Let $\mathbb{A}(\alpha, \cdot)$ be the local squared-speed action at $\alpha$. For a cotangent field $u$ for which the pairing below is finite, usually $u = \varpi f(\alpha)$, the penalized minimization oracle associated with this action is

$$\mathrm{PMO}_{\mathbb{A},\alpha}(u) \in \operatorname*{argmin}_{w} \left\{ \int \langle u(x), w(x)\rangle \mathrm{d}\alpha(x) + \frac{1}{2}\mathbb{A}(\alpha, w) \right\}, \tag{14.29}$$

where the minimization is over admissible finite-action velocity representatives for the chosen geometry. In Hilbertian examples $u \in L^2(\alpha; \mathbb{R}^d)$; for $\mathcal{W}_p$, the natural dual space is $L^q(\alpha; \mathbb{R}^d)$, with $q = p/(p-1)$.

For an energy $f$, the formal small-step expansion of (14.27), when the distance is generated by the squared-speed action $\mathbb{A}$ and the JKO displacement admits an admissible finite-action tangent representative, selects the PMO direction

$$v_\alpha = \mathrm{PMO}_{\mathbb{A},\alpha}(\varpi f(\alpha)). \tag{14.30}$$

$$\partial_t \alpha_t + \mathrm{div}(\alpha_t \; \mathrm{PMO}_{\mathbb{A},\alpha_t}(\varpi f(\alpha_t))) = 0. \tag{14.31}$$

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\alpha_t) = \langle \varpi f(\alpha_t), v_t\rangle_{\alpha_t} = -\mathbb{A}(\alpha_t, v_t), \qquad v_t = \mathrm{PMO}_{\mathbb{A},\alpha_t}(\varpi f(\alpha_t)).$$

For $\mathcal{W}_2$, $\mathbb{A}(\alpha, w) = \int \|w\|^2 \mathrm{d}\alpha$ and $\mathrm{PMO}_{\mathbb{A},\alpha}(u) = -u$. Hence (14.31) reduces to the standard Wasserstein gradient-flow equation

$$\partial_t \alpha_t = \mathrm{div}(\alpha_t \varpi f(\alpha_t)) ,$$

In the restricted Riemannian case of (13.16), where

$$\mathbb{A}(\alpha, w) = \langle Q_\alpha w, w\rangle_{L^2(\alpha)} ,$$

$$\mathrm{PMO}_{\mathbb{A},\alpha}(u) = -Q_\alpha^{-1} u, \qquad v_\alpha = -Q_\alpha^{-1} \varpi f(\alpha). \tag{14.32}$$

$$\mathbb{A}_p(\alpha, w) = \left( \int \|w\|^p \mathrm{d}\alpha \right)^{2/p}.$$

$$\mathrm{PMO}_{\mathbb{A}_p,\alpha}(u)(x) = -\,U^{2-q} \|u(x)\|^{q-2} u(x), \tag{14.33}$$

The examples below are organized by the same dictionary. Concave-mobility flows use $\mathbb{A}_{\theta,\lambda}(\alpha, v)$ from Section 13.3; spectral flows use $\mathbb{A}_\gamma(\alpha, v)$ from (13.23); kernelized Benamou–Brenier flows use $\mathbb{A}_k(\alpha, v) = \|v\|^2_{\mathcal{H}_k^d}$ with a restricted RKHS tangent space.

**General mobility flows.** Let $\lambda$ be Lebesgue measure, write $\alpha = \rho\lambda$, and let $u = \nabla(\delta f/\delta\rho)$. For the velocity action $\mathbb{A}_{\theta,\lambda}$, the PMO introduced in Definition 14.37 is the pointwise minimizer of
$\int \langle u,\, w\rangle\, \rho\, \mathrm{d}x + \frac{1}{2}\int \frac{\rho}{\theta(\rho)}\|w\|^2 \rho\, \mathrm{d}x$,

$$\mathrm{PMO}_{\mathbb{A}_{\theta,\lambda},\rho\lambda}(u) = -\frac{\theta(\rho)}{\rho}u. \tag{14.34}$$

$$\partial_t \rho_t = \nabla\cdot\left(\theta(\rho_t)\nabla\frac{\delta f}{\delta\rho}(\rho_t)\right). \tag{14.35}$$

$f(\rho) = \int U(\rho(x))\, \mathrm{d}x + \int V(x)\rho(x)\, \mathrm{d}x$,

$$\partial_t \rho = \nabla\cdot\big(\theta(\rho)\big(U''(\rho)\nabla\rho + \nabla V\big)\big).$$

**Dynamic spectral Wasserstein flows.** Specializing Definition 14.37 to the spectral action $\mathbb{A}_\gamma$, the descent direction associated with a field $g \in L^2(\alpha;\mathbb{R}^d)$ is simply $\mathrm{PMO}_{\mathbb{A}_\gamma,\alpha}(g)$. The field $g$ should be read as the Wasserstein gradient $g = \nabla\!\!\!\!\nabla f(\alpha) = \nabla\delta f(\alpha)$.

**Proposition 14.38** (Polar formula for the spectral direction)**.** *Let $S_\alpha(g) \coloneqq \int g(x)g(x)^\top \mathrm{d}\alpha(x)$. Assume that the inverse-trace problem admits a positive definite minimizer*

$$A_\alpha^\star \in \operatorname*{argmin}_{A\in\mathcal{B}_\gamma\cap\mathbb{S}^d_{++}} \operatorname{tr}\big(A^{-1}S_\alpha(g)\big).$$

*Then*
$$\mathrm{PMO}_{\mathbb{A}_\gamma,\alpha}(g)(x) = -(A_\alpha^\star)^{-1}g(x) \tag{14.36}$$

*is the spectral PMO direction. If the minimizer lies on the boundary of $\mathcal{B}_\gamma$, the same formula is understood by a limiting argument along positive definite minimizers, or equivalently by replacing the inverse by the corresponding inverse on the range of $S_\alpha(g)$.*

*Proof.* Using the polar formula $\gamma(M) = \sup_{A\in\mathcal{B}_\gamma}\operatorname{tr}(AM)$, the PMO objective (14.29) with $\mathbb{A} = \mathbb{A}_\gamma$ can be written as

$$\sup_{A\in\mathcal{B}_\gamma}\left\{\int\langle g,\, v\rangle \mathrm{d}\alpha + \frac{1}{2}\int v^\top A v \mathrm{d}\alpha\right\}.$$

For a fixed $A \succ 0$, the quadratic lower bound $\int\langle g,\, v\rangle\mathrm{d}\alpha + \frac{1}{2}\int v^\top Av\mathrm{d}\alpha$ is minimized by $v_A = -A^{-1}g$, with value $-\frac{1}{2}\operatorname{tr}(A^{-1}S_\alpha(g))$. Let $v_\star = -(A_\alpha^\star)^{-1}g$ and $M_\star = \int v_\star(x)v_\star(x)^\top\mathrm{d}\alpha(x) = (A_\alpha^\star)^{-1}S_\alpha(g)(A_\alpha^\star)^{-1}$. Since $A_\alpha^\star$ minimizes $A \mapsto \operatorname{tr}(A^{-1}S_\alpha(g))$ over the convex polar set, its first-order optimality condition gives $\operatorname{tr}(AM_\star) \leqslant \operatorname{tr}(A_\alpha^\star M_\star)$ for all $A \in \mathcal{B}_\gamma$. Thus $A_\alpha^\star$ is active in the polar representation of $\gamma(M_\star)$, and $\gamma(M_\star) = \operatorname{tr}(A_\alpha^\star M_\star) = \operatorname{tr}((A_\alpha^\star)^{-1}S_\alpha(g))$. For any $v$, the polar formula gives $\int\langle g,\, v\rangle\mathrm{d}\alpha + \frac{1}{2}\mathbb{A}_\gamma(\alpha, v) \geqslant \int\langle g,\, v\rangle\mathrm{d}\alpha + \frac{1}{2}\int v^\top A_\alpha^\star v\mathrm{d}\alpha$, and the right-hand side is minimized at $v_\star$. The activity identity above shows that equality holds at $v_\star$, hence $v_\star$ is the spectral PMO direction. The boundary case follows by approximation with positive definite matrices in the polar set. $\square$

For the trace gauge, $\mathcal{B}_\gamma = \{A\,;\, 0 \preceq A \preceq \mathrm{Id}\}$ and the minimizer is $A_\alpha^\star = \mathrm{Id}$. For the operator gauge $\gamma(M) = \lambda_{\max}(M)$, one has $\mathcal{B}_\gamma = \{A \succeq 0\,;\, \operatorname{tr}(A) \leqslant 1\}$; if $S_\alpha(g) \succ 0$, then

$$A_\alpha^\star = \frac{S_\alpha(g)^{1/2}}{\operatorname{tr}(S_\alpha(g)^{1/2})}, \qquad \mathrm{PMO}_{\mathbb{A}_\gamma,\alpha}(g)(x) = -\operatorname{tr}(S_\alpha(g)^{1/2})\, S_\alpha(g)^{-1/2}g(x).$$

Proposition 13.14 identifies the static spectral distance $\mathcal{W}_\gamma$ with the length distance generated by $\mathbb{A}_\gamma$. The infinitesimal descent direction is therefore defined directly by the general PMO problem (14.29) specialized to this action.

**Proposition 14.39** (Formal normalized spectral gradient flow)**.** *Assume that $f$ has a smooth first variation and that the spectral PMO is attained along a smooth curve. The formal gradient flow generated by the dynamic spectral action $\mathbb{A}_\gamma$ solves*

$$\partial_t\alpha_t + \operatorname{div}\big(\alpha_t\ \mathrm{PMO}_{\mathbb{A}_\gamma,\alpha_t}(\nabla\!\!\!\!\nabla f(\alpha_t))\big) = 0. \tag{14.37}$$

*Equivalently, if $A_t^\star$ solves the inverse-trace problem for $S_t = \int\nabla\!\!\!\!\nabla f(\alpha_t)(x)\nabla\!\!\!\!\nabla f(\alpha_t)(x)^\top\mathrm{d}\alpha_t(x)$, then the velocity is $v_t(x) = -(A_t^\star)^{-1}\nabla\!\!\!\!\nabla f(\alpha_t)(x)$. Moreover, along smooth solutions, $\frac{\mathrm{d}}{\mathrm{d}t}f(\alpha_t) = -\mathbb{A}_\gamma(\alpha_t, v_t)$. If the minimizing movements (14.27) for $d = \mathcal{W}_\gamma$ converge to a smooth curve with this minimal-action metric derivative, that limit satisfies the same equation.*

*Proof.* The action-based steepest-descent rule (14.30) gives $v_t = \mathrm{PMO}_{\mathbb{A}_\gamma,\alpha_t}(\nabla\!\!\!\!\nabla f(\alpha_t))$. Proposition 14.38 gives the explicit inverse-trace formula. Finally, the variation formula gives $\frac{\mathrm{d}}{\mathrm{d}t}f(\alpha_t) = \int\langle\nabla\!\!\!\!\nabla f(\alpha_t),\, v_t\rangle\mathrm{d}\alpha_t$. Since $v_t$ minimizes a 2-homogeneous penalized objective, the one-dimensional rescaling $s \mapsto sv_t$ has derivative zero at $s = 1$, hence $\int\langle\nabla\!\!\!\!\nabla f(\alpha_t),\, v_t\rangle\mathrm{d}\alpha_t + \mathbb{A}_\gamma(\alpha_t, v_t) = 0$, which proves the dissipation identity. $\square$

## 14.6 Nonlocal Wasserstein Flows

**Nonlocal Wasserstein flows and fractional PDEs.**

**Proposition 14.40** (Entropy gradient flow for reversible jump kernels)**.** *Under the regularity and irreducibility assumptions used in Proposition 13.19, the Markov semigroup generated by the closure of $L\psi(x) = \int\big(\psi(y) - \psi(x)\big)K(x, \mathrm{d}y)$ is the gradient flow, for the distance $\mathcal{W}_K$ defined in (13.29), of the relative entropy $\mathrm{Ent}_{\mathfrak{m}}(\alpha) = \int\rho\log\rho\,\mathrm{d}\mathfrak{m}$, $\alpha = \rho\mathfrak{m}$. Equivalently, in weak form, the entropy flow is*

$$\partial_t\rho_t = L\rho_t. \tag{14.38}$$

*When $K$ is singular, the integral defining $L$ is understood in the principal-value sense.*

*Proof idea.* The key identity is the logarithmic-mean relation $\theta(a, b)(\log a - \log b) = a - b$. The first variation of $\mathrm{Ent}_{\mathfrak{m}}$ is $\log\rho + 1$. With the sign convention of (13.28), the steepest descent velocity is $v(x, y) = -\bar{\nabla}(\log\rho + 1)(x, y) = \log\rho(x) - \log\rho(y)$. Hence $\theta(\rho(x), \rho(y))v(x, y) = \rho(x) - \rho(y)$. Substituting this in (13.28) and using the symmetry of $\mathsf{J}$ gives $\frac{\mathrm{d}}{\mathrm{d}t}\int\varphi\rho\,\mathrm{d}\mathfrak{m} = \int\varphi(x)\int(\rho(y) - \rho(x))K(x, \mathrm{d}y)\,\mathfrak{m}(\mathrm{d}x)$, which is the weak form of $\partial_t\rho = L\rho$. The metric properties needed to justify this computation as a gradient-flow identity are the distance and geodesic results recalled in Proposition 13.19. $\square$

$K_s(x, \mathrm{d}y) = c_{d,s} \frac{\mathrm{d}y}{|x-y|^{d+s}}$, $\qquad 0 < s < 2$. $L_s\psi(x) = \mathrm{p.v.} \int_{\mathbb{R}^d} \big(\psi(y) - \psi(x)\big) \frac{c_{d,s}}{|x-y|^{d+s}} \,\mathrm{d}y = -(-\Delta)^{s/2}\psi(x)$,

$$\partial_t \rho_t = -(-\Delta)^{s/2}\rho_t. \tag{14.39}$$

More generally, when a target-dependent jump kernel $K_\beta$ satisfies detailed balance with a desired invariant measure $\beta = \rho_\beta \mathfrak{m}$, the same construction can be applied to $r_t = \mathrm{d}\alpha_t/\mathrm{d}\beta$. The gradient flow of $\mathrm{KL}(\alpha|\beta)$ is then $\partial_t r_t = L_\beta r_t$.

**Discrete Wasserstein flows on Markov chains.** $\mathrm{Ent}_\pi(\rho) := \sum_i \pi_i \rho_i \log \rho_i$. $\frac{\rho^{(\ell+1)} - \rho^{(\ell)}}{\tau} = -\mathcal{K}_{\rho^{(\ell+1)}} \log \rho^{(\ell+1)} + o(1) = K\rho^{(\ell+1)} + o(1)$. Here $(K\rho)_i = \sum_j K_{i,j}(\rho_j - \rho_i)$. Under smoothness and strict positivity, $\rho^{(\ell+1)} = \rho^{(\ell)} + O(\tau)$, so the last expression also equals $K\rho^{(\ell)} + O(\tau)$.

**Proposition 14.41** (Entropy gradient flow of a reversible Markov chain)**.** *Let $K$ be reversible with invariant law $\pi$. The gradient flow of $\mathrm{Ent}_\pi$ for the metric $\mathcal{W}_K$ is the forward equation of the Markov chain,*

$$\dot\rho_i(t) = \sum_j K_{i,j}\big(\rho_j(t) - \rho_i(t)\big). \tag{14.40}$$

*Equivalently, for the masses $a_i(t) = \pi_i \rho_i(t)$, this is $\dot a_i(t) = \sum_j (a_j(t) K_{j,i} - a_i(t) K_{i,j})$.*

*Proof.* The first variation of the entropy, with respect to the weighted pairing $\sum_i \pi_i \xi_i \varphi_i$, is $\log \rho_i + 1$. Constants do not contribute to $\mathcal{K}_\rho$, hence the metric gradient-flow equation is $\dot\rho = -\mathcal{K}_\rho \log \rho$. Using the identity $\theta(a,b)(\log a - \log b) = a - b$, one obtains, componentwise, $\dot\rho_i = -\sum_j K_{i,j}\theta(\rho_i, \rho_j)(\log \rho_i - \log \rho_j) = \sum_j K_{i,j}(\rho_j - \rho_i)$, which is the density form of the Kolmogorov forward equation. Multiplying by $\pi_i$ and using detailed balance gives the equation for the probability masses. ◻

### 14.7 Dynamic Unbalanced OT and WFR Flows

The generalized dynamic Wasserstein flows of Section 14.5 are primarily mass-preserving: their tangent vectors are generated by continuity equations and are therefore velocity fields advecting the measure. Unbalanced transport changes the state space from probabilities to positive finite measures, where descent may both move and create or remove mass.

**Definition 14.42** (WFR minimizing movement)**.** Let $f : \mathcal{M}_+(\mathbb{R}^d) \to \mathbb{R} \cup \{+\infty\}$ be an energy on positive finite measures and let $\mathrm{WFR}_\kappa$ be the Wasserstein–Fisher–Rao distance with growth scale $\kappa > 0$, defined by the dynamic balance-equation formulation (13.37) and its static equivalence in Proposition 13.22. The unbalanced JKO step is

$$\alpha^{(\ell+1)} \in \operatorname*{argmin}_{\alpha \in \mathcal{M}_+(\mathbb{R}^d)} \frac{1}{2\tau} \mathrm{WFR}_\kappa^2(\alpha^{(\ell)}, \alpha) + f(\alpha). \tag{14.41}$$

The continuous-time limit, when it exists, is called the $\mathrm{WFR}_\kappa$ gradient flow of $f$.

$\partial_t \rho_t + \mathrm{div}(\rho_t v_t) = g_t \rho_t$,

**Proposition 14.43** (Formal WFR gradient-flow PDE)**.** *Assume that $f$ has a smooth first variation $\varphi_t = \delta f(\alpha_t)$ and that $\alpha_t = \rho_t \mathrm{d}x$ is smooth and positive. The formal $\mathrm{WFR}_\kappa$ gradient flow of $f$ is*

$$\partial_t \rho_t = \mathrm{div}(\rho_t \nabla \varphi_t) - \frac{1}{\kappa^2} \varphi_t \rho_t. \tag{14.42}$$

*Along such a smooth solution,*

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\alpha_t) = -\int \left( \|\nabla \varphi_t\|^2 + \frac{1}{\kappa^2}\varphi_t^2 \right) \rho_t \,\mathrm{d}x \leqslant 0. \tag{14.43}$$

*Proof.* For an infinitesimal tangent pair $(v, g)$, the density variation is $\zeta = -\mathrm{div}(\rho v) + g\rho$. The first variation of $f$ is $\int \varphi \zeta = \int \langle \nabla\varphi, v\rangle \rho \,\mathrm{d}x + \int \varphi g \rho \,\mathrm{d}x$, after integration by parts. For the inner product $\langle (v,g), (\tilde v, \tilde g)\rangle_\rho = \int \langle v, \tilde v\rangle \rho \,\mathrm{d}x + \kappa^2 \int g \tilde g \rho \,\mathrm{d}x$, the Riesz representative is $(\nabla\varphi, \varphi/\kappa^2)$. Steepest descent therefore uses $(v, g) = (-\nabla\varphi, -\varphi/\kappa^2)$, which gives (14.42). Substituting this pair in the first-variation formula gives

$$\frac{\mathrm{d}}{\mathrm{d}t} f(\alpha_t) = -\int \left( \|\nabla \varphi_t\|^2 + \frac{1}{\kappa^2}\varphi_t^2 \right) \rho_t \,\mathrm{d}x,$$

which is (14.43). ◻

### 14.8 Conditional Wasserstein Training of Infinite ResNets

**Finite-depth finite-width ResNets.**

$$z_{r+1} = z_r + \tau G_{r,n_r}(z_r), \qquad G_{r,n_r}(z) := \frac{1}{n_r} \sum_{i=1}^{n_r} \psi(\theta_{r,i}, z), \qquad z_0 = z, \qquad r = 0, \dots, L-1. \tag{14.44}$$

$f_{L,n}((\theta_{r,i})_{r,i}) = \int \ell(z_L^\theta(z), y) \mathrm{d}\zeta(z,y)$. $\alpha_r = \frac{1}{n_r}\sum_{i=1}^{n_r} \delta_{\theta_{r,i}}$, $G_{r,n_r}(z) = \int_\Theta \psi(\theta, z) \mathrm{d}\alpha_r(\theta)$. $\alpha^{L,n}(\mathrm{d}s, \mathrm{d}\theta) = \frac{1}{L}\sum_{r=0}^{L-1} \delta_{s_r}(\mathrm{d}s)\alpha_r(\mathrm{d}\theta)$, $\qquad s_r = r/L$,

**Finite-depth mean-field limit.** $z_{r+1} = z_r + \tau G_{\alpha_r}(z_r)$, $\qquad z_0 = z$, $\qquad G_{\alpha_r}(z) := \int_\Theta \psi(\theta, z) \mathrm{d}\alpha_r(\theta)$. $f_L(\alpha_0, \dots, \alpha_{L-1}) = \int \ell(z_L^\alpha(z), y) \mathrm{d}\zeta(z, y)$.

**Continuous-depth conditional geometry.** $\alpha(\mathrm{d}s, \mathrm{d}\theta) = \alpha_s(\mathrm{d}\theta)\mathrm{d}s$, or, equivalently, a probability measure on $[0,1] \times \Theta$ whose first marginal is Lebesgue measure. The general conditional Wasserstein distance is defined in Section 10.6.

$$\mathcal{W}_{2,\lambda}^2(\alpha, \beta) = \int_0^1 \mathcal{W}_2^2(\alpha_s, \beta_s) \mathrm{d}s. \tag{14.45}$$

**Infinite-depth mean-field ResNet.**

$$\partial_s z_s = G_{\alpha_s}(z_s) \coloneqq \int_\Theta \psi(\theta, z_s)\mathrm{d}\alpha_s(\theta), \qquad z_0 = z. \tag{14.46}$$

$f_{\mathrm{Res}}(\alpha) = \int \ell(z_1^\alpha(z), y)\mathrm{d}\zeta(z, y)$.

**Conditional Wasserstein gradient flow.**

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} f(\alpha + \varepsilon\xi)\Big|_{\varepsilon=0} = \int_S \int_\Theta \frac{\delta f}{\delta \alpha_s}(\alpha)(\theta)\mathrm{d}\xi_s(\theta)\mathrm{d}\lambda(s).$$

$$\partial_t \alpha_{t,s} + \mathrm{div}_\theta(\alpha_{t,s} v_{t,s}) = 0, \qquad v_{t,s}(\theta) = -\nabla_\theta \frac{\delta f}{\delta \alpha_s}(\alpha_t)(\theta), \tag{14.47}$$

for $\lambda$-a.e. condition $s$, whenever the first variation is differentiable in the parameter variable. In the ResNet case one takes $f = f_{\mathrm{Res}}$ and $S = [0, 1]$.

**Particle discretization and layerwise matching.** $\alpha_r = \frac{1}{n_r}\sum_{i=1}^{n_r} \delta_{\theta_{r,i}}$, $\quad \beta_r = \frac{1}{n_r}\sum_{i=1}^{n_r} \delta_{\theta'_{r,i}}$, $\mathcal{W}_{2,\lambda_L}^2(\alpha^{L,n}, \beta^{L,n}) = \frac{1}{L}\sum_{r=0}^{L-1} \mathcal{W}_2^2(\alpha_r, \beta_r)$. $\frac{1}{L}\sum_{r=0}^{L-1} \frac{1}{n_r}\sum_{i=1}^{n_r} \|\theta_{r,i} - \theta'_{r,i}\|^2$,

## 14.9 Second-Order Momentum Flows

**Finite-dimensional momentum.** Let $F : \mathbb{R}^N \to \mathbb{R}$ be a smooth finite-dimensional objective. With step size $h > 0$ and damping $\lambda_{\mathrm{fric}} > 0$, one convenient velocity form is

$$S^{(\ell+1)} = (1 - \lambda_{\mathrm{fric}} h)S^{(\ell)} - h\nabla F(X^{(\ell)}), \qquad X^{(\ell+1)} = X^{(\ell)} + hS^{(\ell+1)}. \tag{14.48}$$

If $S^{(\ell)}$ approximates $\dot{X}(\ell h)$, this is a semi-implicit, or symplectic-Euler, discretization of the first-order system: the velocity update is explicit, while the position uses the newly updated velocity. The limiting system is

$$\dot{X}(t) = S(t), \qquad \dot{S}(t) = -\lambda_{\mathrm{fric}} S(t) - \nabla F(X(t)), \tag{14.49}$$

$$\ddot{X}(t) + \lambda_{\mathrm{fric}} \dot{X}(t) + \nabla F(X(t)) = 0. \tag{14.50}$$

$$Y^{(\ell)} = X^{(\ell)} + \theta^{(\ell)}(X^{(\ell)} - X^{(\ell-1)}), \qquad X^{(\ell+1)} = Y^{(\ell)} - h\nabla F(Y^{(\ell)}), \tag{14.51}$$

$$\ddot{X}(t) + \frac{r}{t}\dot{X}(t) + \nabla F(X(t)) = 0. \tag{14.52}$$

**Empirical Wasserstein lift.**

$$\alpha_X \coloneqq \frac{1}{n}\sum_{i=1}^n \delta_{x_i}, \qquad F(X) \coloneqq f(\alpha_X). \tag{14.53}$$

$\|\dot{X}\|_n^2 \coloneqq \frac{1}{n}\sum_{i=1}^n \|\dot{x}_i\|^2$,

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon}\Big|_{\varepsilon=0} F(x_1, \ldots, x_i + \varepsilon\xi, \ldots, x_n) = \frac{1}{n}\langle \nabla\delta f(\alpha_X)(x_i), \xi\rangle.$$

Thus, if $\nabla^{(n)} F$ denotes the gradient for the metric $\|\cdot\|_n$, then

$$(\nabla^{(n)} F(X))_i = n\nabla_{x_i} F(X) = \nabla\delta f(\alpha_X)(x_i) = \nabla\!\!\!\!\nabla f(\alpha_X)(x_i). \tag{14.54}$$

$$v_\alpha(x) \coloneqq -\nabla\!\!\!\!\nabla f(\alpha)(x) = -\nabla\delta f(\alpha)(x). \tag{14.55}$$

$$\dot{x}_i(t) = s_i(t), \qquad \dot{s}_i(t) = -\lambda_{\mathrm{fric}} s_i(t) + v_{\alpha_t}(x_i(t)), \qquad \alpha_t = \frac{1}{n}\sum_{i=1}^n \delta_{x_i(t)}. \tag{14.56}$$

$\ddot{x}_i(t) + \lambda_{\mathrm{fric}}\dot{x}_i(t) = v_{\alpha_t}(x_i(t)) = -\nabla\!\!\!\!\nabla f(\alpha_t)(x_i(t))$.

**Proposition 14.44** (Discrete mechanical energy dissipation)**.** *Assume that $f$ and the particle solution are smooth enough to justify differentiating the identities above along (14.56). Define $\mathcal{H}_n(X, S) \coloneqq F(X) + \frac{1}{2n}\sum_{i=1}^n \|s_i\|^2$, $\quad S = (s_1, \ldots, s_n)$. Then along (14.56),*

$$\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{H}_n(X(t), S(t)) = -\frac{\lambda_{\mathrm{fric}}}{n}\sum_{i=1}^n \|s_i(t)\|^2 \leqslant 0. \tag{14.57}$$

*Proof.* By (14.54) and the definition $v_\alpha = -\nabla\!\!\!\!\nabla f(\alpha)$, one has $\nabla_{x_i} F(X) = -v_{\alpha_X}(x_i)/n$. Hence $\frac{\mathrm{d}}{\mathrm{d}t} F(X(t)) = \sum_{i=1}^n \langle \nabla_{x_i} F(X(t)), s_i(t)\rangle = -\frac{1}{n}\sum_{i=1}^n \langle v_{\alpha_t}(x_i(t)), s_i(t)\rangle$. On the other hand, $\frac{\mathrm{d}}{\mathrm{d}t}\frac{1}{2n}\sum_i \|s_i(t)\|^2 = \frac{1}{n}\sum_i \langle s_i(t), -\lambda_{\mathrm{fric}} s_i(t) + v_{\alpha_t}(x_i(t))\rangle$. The force terms cancel, leaving (14.57). $\square$

**Phase-space formulation.**

**Proposition 14.45** (Phase-space Liouville formulation)**.** *Let $v_\alpha$ be smooth enough that (14.56) has a classical solution. Define the empirical phase-space measure $\eta_t^n := \frac{1}{n}\sum_{i=1}^n \delta_{(x_i(t),s_i(t))} \in \mathcal{P}(\mathbb{R}_x^d \times \mathbb{R}_s^d)$, $\quad \alpha_t^n = (\pi_x)_\sharp \eta_t^n$. Then $\eta_t^n$ solves, in the sense of distributions,*

$$\partial_t \eta_t^n + \nabla_x \cdot (s\,\eta_t^n) + \nabla_s \cdot \big((-\lambda_{\text{fric}} s + v_{\alpha_t^n}(x))\,\eta_t^n\big) = 0. \tag{14.58}$$

*If, as $n \to \infty$, the empirical phase-space laws converge to a smooth density $\eta_t(x,s)$ and the nonlinear fields converge accordingly, the limiting kinetic equation is*

$$\partial_t \eta_t + \nabla_x \cdot (s\,\eta_t) + \nabla_s \cdot \big((-\lambda_{\text{fric}} s + v_{\alpha_t}(x))\,\eta_t\big) = 0, \qquad \alpha_t = (\pi_x)_\sharp \eta_t. \tag{14.59}$$

*Proof.* Let $\varphi \in C_c^\infty(\mathbb{R}_x^d \times \mathbb{R}_s^d)$. Along the particle system,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi\,\mathrm{d}\eta_t^n = \frac{1}{n}\sum_{i=1}^n \left[\langle \nabla_x \varphi(x_i,s_i),\, s_i\rangle + \langle \nabla_s \varphi(x_i,s_i),\, -\lambda_{\text{fric}} s_i + v_{\alpha_t^n}(x_i)\rangle\right].$$

Rewriting the right-hand side as an integral against $\eta_t^n$ and integrating by parts gives (14.58). Passing formally to the mean-field limit gives (14.59). $\square$

$k(x,y) = -\|x-y\|$, $\qquad f(\alpha) = \mathrm{MMD}_k^2(\alpha,\beta) = \iint k(x,y)\,\mathrm{d}(\alpha-\beta)(x)\mathrm{d}(\alpha-\beta)(y)$. $v_\alpha(x) = 2\int \frac{x-y}{\|x-y\|}\,\mathrm{d}(\alpha-\beta)(y)$.

**Entropy without an empirical energy.** For the numerical illustration, we add the quadratic confinement $V(x) = \|x\|^2/2$. In one dimension the overdamped equation becomes the Ornstein–Uhlenbeck Fokker–Planck equation $\partial_t \rho_t = \partial_{xx}\rho_t + \partial_x(x\rho_t)$, whose stationary law is the centered Gaussian $\mathcal{N}(0,1)$. The Newton lift is discretized directly in phase space: for a density $\eta_t(x,s)$ with spatial marginal $\rho_t(x) = \int \eta_t(x,s)\,\mathrm{d}s$, it reads $\partial_t \eta_t + \partial_x(s\eta_t) + \partial_s\big((-\partial_x \log \rho_t(x) - x)\eta_t\big) = 0$.

# 15 Generative Models via Transportation

## 15.1 Generative Models via Flow Matching

**Deterministic interpolants.** The word "stochastic" can hide two different levels of randomness. We first use the simpler one: after drawing a latent variable $U \sim \pi$, the path $t \mapsto I_t(U)$ is deterministic and differentiable.

**Definition 15.1** (Deterministic interpolant)**.** Let $\pi \in \mathcal{U}(\alpha_0,\alpha_1)$. A deterministic interpolant is a measurable family $I_t : \mathbb{R}^d \times \mathbb{R}^d \to \mathbb{R}^d$, differentiable in $t$, satisfying $I_0(x,y) = x$, $\qquad I_1(x,y) = y$. It induces $\alpha_t = (I_t)_\sharp \pi$ and has pathwise velocity $\partial_t I_t(x,y)$.

$$\forall t \in [0,1], \quad \alpha_t := (I_t)_\sharp \pi. \tag{15.1}$$

If $\pi = \alpha \otimes \beta$ and $\alpha = \frac{1}{n}\sum_i \delta_{x_i}$, $\beta = \frac{1}{m}\sum_j \delta_{y_j}$, then $\alpha_t$ consists of $n \times m$ Dirac masses $\alpha_t = \frac{1}{nm}\sum_{i,j} \delta_{I_t(x_i,y_j)}$. If $\pi = (\mathrm{Id},T)_\sharp \alpha$ is a Brenier-type coupling, then $\alpha_t = ((1-t)\mathrm{Id} + tT)_\sharp \alpha$ is the so-called McCann OT interpolation.

**Flow matching formula.**

**Proposition 15.2** (Flow matching vector field)**.** *For each fixed $t$, assume $\partial_t I_t \in L^2(\pi;\mathbb{R}^d)$. Consider the flow-matching problem over measurable fields $v_t : \mathbb{R}^d \to \mathbb{R}^d$*

$$\min_{v_t} \int_{\mathbb{R}^{d'}} \|v_t(I_t(u)) - [\partial_t I_t](u)\|^2\,\mathrm{d}\pi(u). \tag{15.2}$$

*Its minimizer is characterized $\alpha_t$-almost everywhere by the conditional expectation*

$$v_t(z) = \mathbb{E}_{u\sim\pi}\big([\partial_t I_t](u) \,\big|\, z = I_t(u)\big). \tag{15.3}$$

*Then the pair $(\alpha_t, v_t)$ satisfies the continuity equation (13.2).*

*Proof.* We first recall the two equivalent ways of writing the interpolated measure. Formally, one may write $\alpha_t(z) = \int_{\mathbb{R}^{d'}} \delta(z - I_t(u))\,\mathrm{d}\pi(u)$, while the rigorous meaning is that, for every smooth test function $\varphi$,

$$\int_{\mathbb{R}^d} \varphi(z)\,\mathrm{d}\alpha_t(z) = \int_{\mathbb{R}^{d'}} \varphi(I_t(u))\,\mathrm{d}\pi(u). \tag{15.4}$$

The minimizer in (15.2) is the orthogonal projection in $L^2(\pi;\mathbb{R}^d)$ of the latent velocity $\partial_t I_t(u)$ onto the closed subspace of functions that depend on $u$ only through $I_t(u)$. This projection is the conditional expectation (15.3). Formally, this can be read as $v_t(z) = \frac{1}{\alpha_t(z)}\int_{\mathbb{R}^{d'}} \delta(z - I_t(u))[\partial_t I_t](u)\,\mathrm{d}\pi(u)$, and rigorously it means that, for every smooth test vector field $m$,

$$\int \langle m(z),\, v_t(z)\rangle\,\mathrm{d}\alpha_t(z) = \int \langle m(I_t(u)),\, [\partial_t I_t](u)\rangle\,\mathrm{d}\pi(u). \tag{15.5}$$

The weak form of $\partial_t \alpha_t + \mathrm{div}(\alpha_t v_t) = 0$ is that, for every smooth scalar test function $\varphi$,

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(z)\,\mathrm{d}\alpha_t(z) - \int \langle v_t(z),\, \nabla\varphi(z)\rangle\,\mathrm{d}\alpha_t(z) = 0. \tag{15.6}$$

Using (15.4) and differentiating under the integral sign gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(z)\mathrm{d}\alpha_t(z) = \int \langle \nabla\varphi(I_t(u)),\, [\partial_t I_t](u)\rangle\mathrm{d}\pi(u). \tag{15.7}$$

On the other hand, applying (15.5) with $m = \nabla\varphi$ gives

$$\int \langle v_t(z),\, \nabla\varphi(z)\rangle \,\mathrm{d}\alpha_t(z) = \int \langle \nabla\varphi(I_t(u)),\, [\partial_t I_t](u)\rangle \mathrm{d}\pi(u). \tag{15.8}$$

Comparing (15.7) and (15.8) yields (15.6), which is the desired continuity equation. □

The conditional expectation in (15.3) has a simple measure-theoretic meaning. Let $\alpha_t = (I_t)_\sharp\pi$ and define the vector-valued flux measure $\omega_t$ on $\mathbb{R}^d$ by
$\int_{\mathbb{R}^d} \langle \psi(z),\, \mathrm{d}\omega_t(z)\rangle := \int_{\mathbb{R}^{d'}} \langle \psi(I_t(u)),\, [\partial_t I_t](u)\rangle \mathrm{d}\pi(u)$ $\mathrm{d}\omega_t(z) = v_t(z)\mathrm{d}\alpha_t(z)$, $\quad v_t = \frac{\mathrm{d}\omega_t}{\mathrm{d}\alpha_t}$. In the language of Lebesgue decomposition, $\omega_t$ has only an absolutely continuous part with respect to $\alpha_t$ and no singular part; the conditional expectation is precisely its density. This agrees with the flux notation used in the dynamic formulation of Section 13.2.
$v_t(z) = \int_{\{I_t(u)=z\}} [\partial_t I_t](u)\mathrm{d}\pi_{t,z}(u)$. Thus the solution of (15.2) is the conditional expectation of the velocities $\partial_t I_t$: intuitively, $v_t(z)$ is the average velocity of all trajectories passing through $z$.
For the exact field $v_t$, integrating the ODE $\dot{x} = v_t(x)$ defines a transport map $T_t$. If $v_t$ is regular enough, or more generally if the continuity equation has a unique solution for this velocity, then $(T_t)_\sharp\alpha_0 = \alpha_t$.
**Static-noise stochastic interpolants.**

**Definition 15.3** (Stochastic interpolant)**.** Let $(X_0, X_1) \sim \pi \in \mathcal{U}(\alpha_0, \alpha_1)$, and let $Z$ be an auxiliary random variable independent of $(X_0, X_1)$. A stochastic interpolant is $X_t = I_t(X_0, X_1, Z)$, where $t \mapsto I_t(x_0, x_1, z)$ is differentiable and $I_0(x_0, x_1, z) = x_0$, $\quad I_1(x_0, x_1, z) = x_1$. Its marginal curve is $\alpha_t = \mathrm{Law}(X_t)$.

$X_t = a(t)X_0 + b(t)X_1 + \gamma(t)Z$, $\quad \gamma(0) = \gamma(1) = 0$. The noise $Z$ is static: conditionally on $(X_0, X_1, Z)$, the path $t \mapsto X_t$ is differentiable. Thus this construction is exactly the previous push-forward framework with latent variable $u = (X_0, X_1, Z)$, law $\pi = \mathrm{Law}(X_0, X_1, Z)$, and interpolant given by the displayed $I_t$.
$v_t(x) = \mathbb{E}\big[\partial_t I_t(X_0, X_1, Z) \mid X_t = x\big]$,
**Connection with diffusion models.** In the special case where $I_t(x, y) = (1-t)x + ty$ is a linear interpolation and $\pi = \alpha \otimes \beta$, the curve $\alpha_t$ is a convolution of rescaled versions of $\alpha_0$ and $\alpha_1$. The flow-matching problem (15.2) becomes
$\min_{(v_t)_t} \int_{\mathbb{R}^d\times\mathbb{R}^d} \|v_t((1-t)x + ty) - (y - x)\|^2 \,\mathrm{d}\alpha_0(x)\mathrm{d}\alpha_1(y)$.
**Proposition 15.4** (Tweedie identity)**.** *Let $W$ be a random vector in $\mathbb{R}^d$ with law $\beta \in \mathcal{P}_1(\mathbb{R}^d)$. For $\sigma > 0$, observe $Z = W + \sigma\,\varepsilon$, where $\varepsilon \sim \mathcal{N}(0, \mathrm{Id})$ is independent of $W$. Denote by $\rho_\sigma$ the smooth positive density of $\beta_\sigma = \beta * \mathcal{N}\big(0, \sigma^2\mathrm{Id}\big)$, which is the law of $Z$. Then the conditional mean admits the following everywhere-defined version: $\mathbb{E}\big[W \mid Z = z\big] = z + \sigma^2\,\nabla\log\rho_\sigma(z)$ for all $z \in \mathbb{R}^d$.*

*Proof.* Let $\varphi_\sigma$ be the $\mathcal{N}(0, \sigma^2\mathrm{Id})$ density. Bayes' rule gives $\mathbb{E}[W \mid Z = z] = \frac{1}{\rho_\sigma(z)}\int_{\mathbb{R}^d} w\,\varphi_\sigma(z-w)\,\mathrm{d}\beta(w)$. Differentiating the Gaussian convolution under the integral sign and using $\nabla_z\varphi_\sigma(z-w) = -\sigma^{-2}(z-w)\,\varphi_\sigma(z-w)$ yields $\nabla\rho_\sigma(z) = \int \nabla_z\varphi_\sigma(z-w)\,\mathrm{d}\beta(w) = -\sigma^{-2}\Big(z - \mathbb{E}[W \mid Z = z]\Big)\,\rho_\sigma(z)$. Rearranging finishes the proof. □

**Proposition 15.5** (Gaussian-endpoint flow-matching field)**.** *Let $X \sim \alpha$ and $Y \sim \mathcal{N}(0, \mathrm{Id})$ be independent. For $t \in (0,1)$ set $Z_t = (1-t)\,X + t\,Y$, $\quad \alpha_t = \mathrm{Law}(Z_t) = \rho_t\,\mathrm{d}x$. The regression minimizer $v^\star : \mathbb{R}^d \times (0,1) \to \mathbb{R}^d$ of $\min_v \int_0^1\iint_{\mathbb{R}^d\times\mathbb{R}^d} \big|y - x - v\big((1-t)x + ty, t\big)\big|^2 \,\mathrm{d}\alpha(x)\,\mathrm{d}\mathcal{N}(y)\,\mathrm{d}t$ is $v^\star(x,t) = -\frac{1}{1-t}\,x - \frac{t}{1-t}\,\nabla\log\rho_t(x)$ $\quad (x \in \mathbb{R}^d,\ t \in (0,1))$. In particular, for each $t \in (0,1)$ this field is a gradient field,*

$$v^\star(\cdot, t) = -\nabla\left(\frac{\|\cdot\|^2}{2(1-t)} + \frac{t}{1-t}\log\rho_t\right).$$

*Proof.* Fix $t \in (0,1)$ and write $W = (1-t)X$, $\sigma = t$, so that $Z_t = W + \sigma\,Y$ matches the setting of Proposition 15.4. Conditional expectations satisfy $v^\star(z,t) = \mathbb{E}[Y - X \mid Z_t = z] = \frac{1}{t}\,\mathbb{E}[Z_t - W \mid Z_t = z] - \frac{1}{1-t}\,\mathbb{E}[W \mid Z_t = z]$. Applying Proposition 15.4 to $\mathbb{E}[W \mid Z_t = z]$ and noting $\mathbb{E}[Y \mid Z_t = z] = -\,t\,\nabla\log\rho_t(z)$ gives the claimed formula. □

**When is the induced map optimal?** Integrating the learned velocity gives a deterministic map from $\alpha_0$ to $\alpha_1$, but this map is not automatically the Brenier optimal map. It is optimal only in special cases where the accumulated flow remains the gradient of a convex potential.
**Proposition 15.6** (Gaussian flow matching and optimality)**.** *Let $\Sigma_0, \Sigma_1 \succ 0$ and let $X_0 \sim \mathcal{N}(0, \Sigma_0)$ and $X_1 \sim \mathcal{N}(0, \Sigma_1)$ be independent. Consider the linear flow-matching interpolation $Z_t = (1-t)X_0 + tX_1$, $\quad \alpha_t = \mathrm{Law}(Z_t) = \mathcal{N}(0, \Sigma_t)$,*

*where*
$$\Sigma_t = (1-t)^2\Sigma_0 + t^2\Sigma_1. \tag{15.9}$$

*Then the exact flow-matching velocity is affine, $v_t(z) = A_t z$, with*

$$A_t = \big(t\Sigma_1 - (1-t)\Sigma_0\big)\Sigma_t^{-1}. \tag{15.10}$$

*The induced flow map $T_t^{\mathrm{FM}}$ from $\alpha_0$ to $\alpha_t$ is*

$$T_t^{\mathrm{FM}} = \Sigma_0^{1/2}\Big((1-t)^2\mathrm{Id} + t^2\Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}\Big)^{1/2}\Sigma_0^{-1/2}. \tag{15.11}$$

*In particular,*
$$T_1^{\mathrm{FM}} = \Sigma_0^{1/2}\big(\Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}\big)^{1/2}\Sigma_0^{-1/2}. \tag{15.12}$$

*This terminal map coincides with the quadratic optimal transport map*

$$T^{\mathrm{OT}} = \Sigma_0^{-1/2}\big(\Sigma_0^{1/2}\Sigma_1\Sigma_0^{1/2}\big)^{1/2}\Sigma_0^{-1/2} \tag{15.13}$$

*if and only if $\Sigma_0\Sigma_1 = \Sigma_1\Sigma_0$.*

*Proof.* The conditional-expectation formula gives $v_t(z) = \mathbb{E}[X_1 - X_0 \mid Z_t = z]$. Since all variables are jointly Gaussian, this conditional expectation is linear and $v_t(z) = \mathrm{Cov}(X_1 - X_0, Z_t)\,\mathrm{Cov}(Z_t)^{-1} z = \big(t\Sigma_1 - (1-t)\Sigma_0\big)\Sigma_t^{-1} z$, which proves (15.10). To solve the characteristic equation, whiten the source by setting $C = \Sigma_0^{-1/2}\Sigma_1\Sigma_0^{-1/2}, \qquad \widetilde{Z}_t = \Sigma_0^{-1/2} Z_t$. In these coordinates the source covariance is Id and $\widetilde{\Sigma}_t = (1-t)^2\mathrm{Id} + t^2 C$. Because Id and $C$ commute, the affine flow map in whitened coordinates is simply $\widetilde{T}_t = \widetilde{\Sigma}_t^{1/2}$. Indeed, $\frac{\mathrm{d}}{\mathrm{d}t}\widetilde{\Sigma}_t^{1/2} = \big(tC - (1-t)\mathrm{Id}\big)\widetilde{\Sigma}_t^{-1/2}$, which is exactly the equation $\dot{\widetilde{T}}_t = \widetilde{A}_t \widetilde{T}_t$ with $\widetilde{T}_0 = \mathrm{Id}$. Returning to the original coordinates gives (15.11), and $t = 1$ gives (15.12).
Both $T_1^{\mathrm{FM}}$ and $T^{\mathrm{OT}}$ push $\mathcal{N}(0, \Sigma_0)$ to $\mathcal{N}(0, \Sigma_1)$. The Brenier map between nondegenerate Gaussians is the unique symmetric positive definite linear map with this property. Hence $T_1^{\mathrm{FM}} = T^{\mathrm{OT}}$ if and only if $T_1^{\mathrm{FM}}$ is symmetric positive definite. The map $T_1^{\mathrm{FM}}$ is similar to $C^{1/2}$, so if it is symmetric then it is automatically positive definite. Since $C^{1/2}$ is symmetric positive definite, $(T_1^{\mathrm{FM}})^\top = \Sigma_0^{-1/2} C^{1/2} \Sigma_0^{1/2}$. Thus symmetry of $T_1^{\mathrm{FM}}$ is equivalent to $\Sigma_0 C^{1/2} = C^{1/2}\Sigma_0$, hence to $\Sigma_0 C = C\Sigma_0$ by functional calculus. Multiplying this identity on the left and right by $\Sigma_0^{1/2}$ gives $\Sigma_0\Sigma_1 = \Sigma_1\Sigma_0$. Conversely, if $\Sigma_0$ and $\Sigma_1$ commute, they are orthogonally co-diagonalizable, and both (15.12) and (15.13) reduce in that basis to the diagonal map with entries $\sqrt{\lambda_{1,k}/\lambda_{0,k}}$. This proves the equivalence. □

**Variations on the interpolant.** $v_t(z) = \lambda'(t)\, v^{\mathrm{lin}}_{\lambda(t)}(z), \qquad v^{\mathrm{lin}}_r(z) = \mathbb{E}[Y - X \mid (1-r)X + rY = z]$. $Z_t = a_t X + b_t Y, \qquad Y \sim \mathcal{N}(0, \sigma^2\mathrm{Id})$, then both the mixture centers and the component variances are changed. Writing $p_t$ for the density of $Z_t$ and $s_t = \nabla \log p_t$, Tweedie's formula gives, away from times where $a_t = 0$,

$$v_t(z) = a_t'\,\mathbb{E}[X \mid Z_t = z] + b_t'\,\mathbb{E}[Y \mid Z_t = z] = \frac{a_t'}{a_t} z + \left(\frac{a_t' b_t^2}{a_t} - b_t' b_t\right)\sigma^2 s_t(z).$$

For the linear bridge, $a_t = 1 - t$ and $b_t = t$, this recovers the formula above. For the variance-preserving Ornstein–Uhlenbeck noising used in diffusion models,
$a_\tau = e^{-\tau}, \qquad b_\tau = \sqrt{1 - e^{-2\tau}}$, one obtains the forward probability-flow velocity $v_\tau(z) = -z - \sigma^2\nabla\log p_\tau(z)$. Sampling integrates this ODE backward in $\tau$.
$a_t = (1-t)(1-2t), \qquad b_t = t,$

### 15.2 One-Step Generative Models

**One-step models via parameter-domain discrepancy flows.** The most direct one-step strategy is to train the generator parameters themselves by descending a discrepancy between generated and data distributions. Let $\zeta$ be a simple latent distribution, let $f_\theta : \mathcal{Z} \to \mathcal{X}$ be a neural generator, and let $\beta$ be the data distribution.
$(f_\theta)_\sharp\zeta = \beta$, or, in practice, such that the generated law is close to $\beta$. Given a discrepancy $\mathcal{D}$ on probability measures, define

$$\mathcal{E}_\beta(\gamma) := \mathcal{D}(\gamma, \beta), \qquad H_\beta(\theta) := \mathcal{E}_\beta\big((f_\theta)_\sharp\zeta\big) = \mathcal{D}\big((f_\theta)_\sharp\zeta, \beta\big). \tag{15.14}$$

$$\dot{\theta}_t = -\nabla_\theta H_\beta(\theta_t). \tag{15.15}$$

$\alpha_t := (f_{\theta_t})_\sharp\zeta, \qquad X_t = f_{\theta_t}(Z), \qquad Z \sim \zeta$. If $t \mapsto \theta_t$ is smooth and $f_\theta$ is differentiable with respect to $\theta$, then
$\dot{X}_t = \partial_\theta f_{\theta_t}(Z)\dot{\theta}_t$. For $\alpha_t$-a.e. $x$, let $\eta_{t,x}$ be the disintegration of the latent law $\zeta$ with respect to the map $f_{\theta_t}$. Thus $\eta_{t,x}$ is supported on the fiber $\{z\,;\, f_{\theta_t}(z) = x\}$, and for every bounded measurable $\psi$,

$$\int \psi(z)\,\mathrm{d}\zeta(z) = \int_{\mathcal{X}} \left(\int \psi(z)\,\mathrm{d}\eta_{t,x}(z)\right)\mathrm{d}\alpha_t(x).$$

$$v_t(x) = \int \partial_\theta f_{\theta_t}(z)\dot{\theta}_t\,\mathrm{d}\eta_{t,x}(z) = -\int \partial_\theta f_{\theta_t}(z)\nabla_\theta H_\beta(\theta_t)\,\mathrm{d}\eta_{t,x}(z). \tag{15.16}$$

$\frac{\mathrm{d}}{\mathrm{d}t}\int \varphi(x)\,\mathrm{d}\alpha_t(x) = \int \langle \nabla\varphi(x),\, v_t(x)\rangle\,\mathrm{d}\alpha_t(x),$

$$\partial_t\alpha_t + \mathrm{div}(\alpha_t v_t) = 0. \tag{15.17}$$

The velocity (15.16) depends on the parameter value $\theta_t$ and on the latent disintegration, not only on the measure $\alpha_t$. If the parametrization is non-identifiable, two different parameters can generate the same measure while inducing different velocities.
$-\nabla\delta_\gamma\mathcal{E}_\beta(\gamma)\big|_{\gamma=\alpha_t}$. $\hat{\beta}_n = \frac{1}{n}\sum_{i=1}^n \delta_{Y_i}, \qquad Y_i \overset{\text{i.i.d.}}{\sim} \beta,$

$$\mathbb{E}\left[\nabla_\theta H_{\hat{\beta}_n}(\theta)\right] \neq \nabla_\theta H_\beta(\theta).$$

**One-step model using Wasserstein flow of discrepancy.** Let $\zeta$ be a simple latent distribution and let $\alpha_\theta = (G_\theta)_\sharp\zeta$ be the model distribution. Assume that the target data distribution is $\beta$. The Wasserstein-flow construction chooses a discrepancy $\mathcal{E}_\beta(\alpha)$,

$$\partial_t\alpha_t + \mathrm{div}(\alpha_t w_t) = 0, \qquad w_t(x) = -\nabla\delta_\alpha\mathcal{E}_\beta(\alpha_t)(x). \tag{15.18}$$

$$\min_\eta \int \|U_\eta(x) - w_t(x)\|^2\,\mathrm{d}\alpha_t(x). \tag{15.19}$$

$$\alpha_\theta^+ = (\mathrm{Id} + \tau U_{\eta_t})_\sharp\alpha_\theta, \qquad \text{or equivalently} \qquad G_\theta^+(z) = G_\theta(z) + \tau U_{\eta_t}(G_\theta(z)).$$

**Sliced-Wasserstein flow.** $\mathcal{E}_\beta(\alpha) = \frac{1}{2}\mathcal{SW}_2^2(\alpha, \beta)$, with $\mathcal{SW}_2$ defined in Section 10.2. With the continuity-equation convention $\partial_t\alpha_t + \mathrm{div}(\alpha_t w_t) = 0$, the descent velocity points from the current projected law toward the target projected law.

$$w_{\mathcal{SW}}[\alpha, \beta](x) = \int_{\mathbb{S}^{d-1}} \big(T_\theta(P_\theta x) - P_\theta x\big)\,\theta\,\mathrm{d}\sigma(\theta). \tag{15.20}$$

The sign follows from the one-dimensional identity: the first variation of $\frac{1}{2}\mathcal{W}_2^2(\alpha, \beta)$ has spatial derivative $s - T(s)$, so the descent velocity is $T(s) - s$, and composing with $P_\theta$ lifts it in the direction $\theta$. Thus empirical implementations only require one-dimensional sorting along sampled directions, followed by averaging the lifted projected displacements.

**Stein variational gradient descent.** For the formal density-level calculation, assume $\alpha = \rho_\alpha \,\mathrm{d}x$ and $\beta = \rho_\beta \,\mathrm{d}x$ have smooth positive densities, and let $v$ be a smooth compactly supported vector field. For the perturbation $\alpha_\varepsilon = (\mathrm{Id} + \varepsilon v)_\sharp \alpha$, integration by parts gives the Stein first variation

$$\frac{\mathrm{d}}{\mathrm{d}\varepsilon} \mathrm{KL}(\alpha_\varepsilon|\beta)\Big|_{\varepsilon=0} = -\int \big(\langle \nabla \log \rho_\beta(x),\, v(x)\rangle + \mathrm{div}\, v(x)\big)\mathrm{d}\alpha(x).$$

$$v_\alpha^{\mathrm{SVGD}}(x) = \int \Big(k(y,x)\nabla \log \rho_\beta(y) + \nabla_y k(y,x)\Big)\mathrm{d}\alpha(y). \tag{15.21}$$

$$\partial_t \alpha_t + \mathrm{div}\big(\alpha_t v_{\alpha_t}^{\mathrm{SVGD}}\big) = 0. \tag{15.22}$$

For particles $\alpha_n^\ell = n^{-1} \sum_i \delta_{X_i^\ell}$, this becomes the explicit update

$$X_i^{\ell+1} = X_i^\ell + \tau \frac{1}{n} \sum_{j=1}^n \Big(k(X_j^\ell, X_i^\ell)\nabla \log \rho_\beta(X_j^\ell) + \nabla_{X_j} k(X_j^\ell, X_i^\ell)\Big). \tag{15.23}$$

**Self-corrected drifting fields.**

$$B_\varepsilon[\nu](x) := \frac{\int (y-x) K_\varepsilon(x,y)\,\mathrm{d}\nu(y)}{\int K_\varepsilon(x,y)\,\mathrm{d}\nu(y)}. \tag{15.24}$$

For the Gaussian kernel $K_\varepsilon(x,y) = \exp(-\|x-y\|^2/(2\varepsilon))$, this normalized field is a score of a smoothed density:

$$B_\varepsilon[\nu](x) = \varepsilon \nabla \log\left(\int K_\varepsilon(x,y)\,\mathrm{d}\nu(y)\right). \tag{15.25}$$

$$u_t(x) = B_\varepsilon[\beta](x) - B_\varepsilon[\alpha_t](x) = \varepsilon \nabla \log \frac{\int K_\varepsilon(x,y)\,\mathrm{d}\beta(y)}{\int K_\varepsilon(x,y)\,\mathrm{d}\alpha_t(y)}. \tag{15.26}$$

The first term pulls samples toward data, while the second term corrects self-attraction and prevents all particles from collapsing onto the same high-density region. For a fixed reference measure, $B_\varepsilon[\nu]$ is precisely the Gaussian mean-shift displacement in (15.35): it moves $x$ toward the local kernel barycenter of $\nu$.

**Proposition 15.7** (Instantaneous gradient representation of drifting)**.** *Let $\alpha_t = \rho_t \,\mathrm{d}x$ be a smooth curve of probability measures with positive densities, and let $u_t = \nabla \varphi_t$ be a smooth time-dependent gradient field. Define the semi-relaxed functional*

$$\mathcal{R}_t(\alpha|\alpha_t) := -\int \varphi_t(x)\,\mathrm{d}\alpha(x) + \int \varphi_t(x)\,\mathrm{d}\alpha_t(x). \tag{15.27}$$

*Here $\alpha_t$ and $\varphi_t$ are frozen when taking the first variation with respect to the first argument $\alpha$. Then the continuity equation $\partial_t \alpha_t + \mathrm{div}(\alpha_t u_t) = 0$ is the formal Wasserstein gradient descent of the frozen time-dependent functional $\alpha \mapsto \mathcal{R}_t(\alpha|\alpha_t)$.*

*Proof.* Since $\alpha_t$ and $\varphi_t$ are fixed in the variation with respect to $\alpha$, the first variation is $\delta_\alpha \mathcal{R}_t(\alpha|\alpha_t)(x) = -\varphi_t(x)$. By Proposition 14.5, $\nabla\!\!\!\!\nabla \mathcal{R}_t(\alpha|\alpha_t) = \nabla \delta_\alpha \mathcal{R}_t(\alpha|\alpha_t) = -\nabla \varphi_t = -u_t$. The Wasserstein gradient-descent velocity is the negative of this gradient, namely $u_t$. Substituting this velocity in the continuity equation gives the claimed flow. □

## 15.3 Moment Measures

**Definition 15.8** (Moment measure of a convex function)**.** Let $u : \mathbb{R}^d \to \mathbb{R} \cup \{+\infty\}$ be a proper lower-semicontinuous convex function such that $Z_u := \int_{\mathbb{R}^d} e^{-u(x)}\,\mathrm{d}x < +\infty$. The normalized log-concave measure associated with $u$ is $\eta_u := Z_u^{-1} e^{-u(x)}\,\mathrm{d}x$. Whenever $\nabla u$ is defined $\eta_u$-almost everywhere, the moment measure of $u$ is $\mathfrak{M}(u) := (\nabla u)_\sharp \eta_u$.

$\int y \,\mathrm{d}\mathfrak{M}(u)(y) = Z_u^{-1} \int \nabla u(x) e^{-u(x)}\,\mathrm{d}x = -Z_u^{-1} \int \nabla(e^{-u(x)})\,\mathrm{d}x = 0$. Thus moment measures are necessarily centered. The nonsmooth theory uses essentially continuous convex functions: these are lower-semicontinuous convex functions whose set of discontinuity points has zero $\mathcal{H}^{d-1}$ measure.

**Theorem 15.9** (Cordero–Erausquin–Klartag)**.** *Let $\alpha \in \mathcal{P}_1(\mathbb{R}^d)$ be a probability measure. There exists an essentially continuous convex function $u$ such that $\alpha = \mathfrak{M}(u)$ if and only if $\int y\,\mathrm{d}\alpha(y) = 0$ and $\mathrm{supp}(\alpha)$ is not contained in a hyperplane. Under these conditions, $u$ is unique up to translations of the argument and additive constants.*

**Optimal-transport variational formulation.**

$$\mathcal{C}_\alpha(\eta) := \sup_{\pi \in \mathcal{U}(\eta,\alpha)} \int_{\mathbb{R}^d \times \mathbb{R}^d} x \cdot y \,\mathrm{d}\pi(x,y). \tag{15.28}$$

$$\mathcal{C}_\alpha(\eta) = \inf_{v \text{ convex}} \left\{ \int v(x)\,\mathrm{d}\eta(x) + \int v^*(y)\,\mathrm{d}\alpha(y) \right\}, \tag{15.29}$$

where the infimum is over convex functions for which both integrals are well defined and $v^*$ is the Legendre transform. If $\eta, \alpha \in \mathcal{P}_2(\mathbb{R}^d)$, then

$$\mathcal{C}_\alpha(\eta) = \frac{1}{2}\int \|x\|^2\,\mathrm{d}\eta(x) + \frac{1}{2}\int \|y\|^2\,\mathrm{d}\alpha(y) - \frac{1}{2}\mathcal{W}_2^2(\eta,\alpha). \tag{15.30}$$

$$\min_{\eta \in \mathcal{P}_1(\mathbb{R}^d)} \{\mathcal{H}(\eta) + \mathcal{C}_\alpha(\eta)\}, \qquad \mathcal{H}(r\,\mathrm{d}x) := \int r(x) \log r(x)\,\mathrm{d}x, \tag{15.31}$$

**Proposition 15.10** (Variational characterization of moment measures)**.** *Let $\alpha \in \mathcal{P}_1(\mathbb{R}^d)$ be centered and not supported on a hyperplane. Then the minimization problem (15.31) admits a solution, unique up to translations. Every minimizer is a probability measure with log-concave density of the form $\eta = Z_u^{-1}e^{-u}\,\mathrm{d}x$, where $u$ is convex, essentially continuous, and satisfies the moment-measure equation $\alpha = (\nabla u)_\sharp\eta$. Conversely, any essentially continuous convex $u$ satisfying this equation yields a global minimizer. If $\alpha \in \mathcal{P}_2(\mathbb{R}^d)$, the objective $\eta \mapsto \mathcal{H}(\eta) + \mathcal{C}_\alpha(\eta)$ is displacement convex along $\mathcal{W}_2$ geodesics of absolutely continuous measures in $\mathcal{P}_2(\mathbb{R}^d)$ whenever the terms are finite. When merely $\alpha \in \mathcal{P}_1(\mathbb{R}^d)$, the maximal-correlation term remains convex along such finite-second-moment geodesics by approximation; existence and uniqueness on the full $\mathcal{P}_1$ domain follow from the lower-semicontinuity argument of Santambrogio.*

*Proof.* The existence proof has two ingredients. First, since $\alpha$ is centered, translating $\eta$ does not change either $\mathcal{H}(\eta)$ or $\mathcal{C}_\alpha(\eta)$, so one can center a minimizing sequence. Second, the assumption that $\alpha$ is not supported on a hyperplane gives a coercive lower bound of the form $\mathcal{C}_\alpha(\eta) \geqslant c_\alpha \int \|x\|\,\mathrm{d}\eta(x)$ for centered absolutely continuous $\eta$, with $c_\alpha > 0$. Together with the lower-semicontinuity estimates for entropy and maximal correlation, this yields a minimizer.
Let $\eta = r\,\mathrm{d}x$ be a minimizer and let $u$ be a convex optimizer in the dual formula (15.29). Keeping $u$ fixed and varying $\eta$ in (15.31) gives the Euler equation $\log r(x)+1+u(x) = \text{constant} \quad \text{on } \{r>0\}$, so $\eta = Z_u^{-1}e^{-u}\mathrm{d}x$. The optimality condition for the scalar-product transport problem says that an optimal coupling is supported on $\{(x,y) : y \in \partial u(x)\}$. Since $\eta$ is absolutely continuous and $u$ is convex, $\partial u(x) = \{\nabla u(x)\}$ for $\eta$-almost every $x$, hence $\alpha = (\nabla u)_\sharp\eta$.
Conversely, assume $\eta = Z_u^{-1}e^{-u}\mathrm{d}x$ and $\alpha = (\nabla u)_\sharp\eta$. Let $\nu$ be a smooth compactly supported competitor, and let $T$ be the Brenier map from $\eta$ to $\nu$. Along the geodesic $\eta_t = ((1-t)\mathrm{Id} + tT)_\sharp\eta$, the right derivative of the entropy at $t=0$ is $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{H}(\eta_t)\Big|_{t=0^+} = -\int\langle T(x)-x,\, \nabla u(x)\rangle\,\mathrm{d}\eta(x)$, where the identity follows by differentiating the Jacobian formula and integrating by parts. The dual optimizer $u$ gives the upper directional bound $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{C}_\alpha(\eta_t)\Big|_{t=0^+} \leqslant \int\langle T(x)-x,\, \nabla u(x)\rangle\,\mathrm{d}\eta(x)$. Conversely, Santambrogio's derivative estimate gives $\frac{\mathrm{d}}{\mathrm{d}t}\mathcal{C}_\alpha(\eta_t)\Big|_{t=0^+} \geqslant \int\langle T(x)-x,\, \nabla u(x)\rangle\,\mathrm{d}\eta(x)$, because $(\mathrm{Id}, \nabla u)_\sharp\eta$ is optimal for the scalar-product problem. The two bounds coincide, so the first-order terms cancel. Hence the one-sided derivative of $\mathcal{H} + \mathcal{C}_\alpha$ at $\eta$ in every such direction is zero. Displacement convexity implies global minimality, and approximation removes the smooth compact-support restriction. Strict displacement convexity of entropy gives uniqueness, except for translations; translations do not change $\mathcal{C}_\alpha$ because $\alpha$ is centered. The entropy term is displacement convex by McCann's theorem, recalled in Theorem 14.12. If $\alpha$ has finite second moment, identity (15.30) writes $\mathcal{C}_\alpha$ as the sum of the 1-convex moment term $\eta \mapsto \frac{1}{2}\int\|x\|^2\,\mathrm{d}\eta$ and the $(-1)$-convex term $\eta \mapsto -\frac{1}{2}\mathcal{W}_2^2(\eta,\alpha)$, hence $\mathcal{C}_\alpha$ is displacement convex. For a target with only a finite first moment, Santambrogio obtains the same convexity along $\mathcal{P}_2$ geodesics by approximation and proves the full variational characterization by lower semicontinuity. □

**Conjugate moment measures for generation.**

$$\beta = (\nabla w^*)_\sharp \left(Z_w^{-1}e^{-w(z)}\,\mathrm{d}z\right). \tag{15.32}$$

## 15.4 Evolution in Depth of Transformers

**Attention as a context-dependent velocity.**

$$x_i \longmapsto x_i + \frac{1}{T}\,\mathrm{Att}_\theta^{(n)}(x_1,\dots,x_n)_i = x_i + \frac{1}{T}\sum_{j=1}^n \frac{e^{\langle Qx_i,\,Kx_j\rangle}Vx_j}{\sum_{\ell=1}^n e^{\langle Qx_i,\,Kx_\ell\rangle}}. \tag{15.33}$$

**Token measure evolution.** Let $x_i^{(\ell)}$ be token $i$ after layer $\ell$, and let $\alpha^{(\ell)} = n^{-1}\sum_i \delta_{x_i^{(\ell)}}$ be the corresponding empirical law. With $\tau = 1/T$, equation (15.33) is exactly

$$\alpha^{(\ell+1)} = \left(\mathrm{Id} + \tau\Gamma_{\theta^{(\ell)}}[\alpha^{(\ell)}]\right)_\sharp \alpha^{(\ell)}, \qquad \ell = 0,\dots,T-1. \tag{15.34}$$

$$\partial_t\alpha_t + \mathrm{div}(\alpha_t\Gamma_{\theta_t}[\alpha_t]) = 0.$$

**$L^2$ attention and mean shift.**

$$K_\varepsilon(x,y) = \exp(-\|x-y\|^2/(2\varepsilon)), \qquad \rho_\varepsilon[\alpha](x) = \int K_\varepsilon(x,y)\mathrm{d}\alpha(y), \qquad m_\varepsilon[\alpha](x) = \frac{\int yK_\varepsilon(x,y)\mathrm{d}\alpha(y)}{\rho_\varepsilon[\alpha](x)}.$$

$$M_\varepsilon[\alpha](x) \coloneqq m_\varepsilon[\alpha](x) - x = \frac{\int(y-x)K_\varepsilon(x,y)\mathrm{d}\alpha(y)}{\rho_\varepsilon[\alpha](x)} = \varepsilon\nabla\log\rho_\varepsilon[\alpha](x) \tag{15.35}$$

$$x_i^{(\ell+1)} = (1-\tau)x_i^{(\ell)} + \tau m_\varepsilon[\alpha^{(\ell)}](x_i^{(\ell)}) = x_i^{(\ell)} + \tau M_\varepsilon[\alpha^{(\ell)}](x_i^{(\ell)})$$

$\partial_t\alpha_t + \mathrm{div}\big(\alpha_t M_\varepsilon[\alpha_t]\big) = 0.$

**Consensus and Markov averaging.**

$$M_K[\alpha](x) \coloneqq \frac{\int(y-x)K(x,y)\mathrm{d}\alpha(y)}{\int K(x,y)\mathrm{d}\alpha(y)}. \tag{15.36}$$

For an empirical law $\alpha = \sum_{i=1}^n a_i\delta_{x_i}$, where $a_i > 0$ and $\sum_i a_i = 1$, let $X \in \mathbb{R}^{n\times d}$ have rows $x_i^\top$, set $\mathrm{a} = (a_i)_i$, and define

$$(\mathrm{K}_X)_{i,j} = K(x_i,x_j), \qquad \mathsf{P}_X \coloneqq \mathrm{diag}(\mathrm{K}_X\mathrm{a})^{-1}\mathrm{K}_X\,\mathrm{diag}(\mathrm{a}). \tag{15.37}$$

$$\dot{x}_i = \frac{\sum_j a_jK(x_i,x_j)(x_j-x_i)}{\sum_j a_jK(x_i,x_j)}, \qquad \text{or equivalently} \qquad \dot{X} = (\mathsf{P}_X - I_n)X. \tag{15.38}$$

**Dobrushin contraction from Hilbert geometry.** $d_{\mathcal{H}}(e^z, e^{z'}) = \|z - z'\|_V$. A row-stochastic matrix $\mathsf{P}$ is order preserving and satisfies $\mathsf{P}(z + s\mathbb{1}_n) = \mathsf{P}z + s\mathbb{1}_n$. It is therefore a linear topical map, so Proposition 8.8 gives nonexpansiveness on the additive quotient.

**Proposition 15.11** (Dobrushin contraction is the tangent form of Birkhoff contraction)**.** *Let $\mathsf{P} \in \mathbb{R}_+^{n\times n}$ be row-stochastic. Its Dobrushin coefficient is*

$$\delta(\mathsf{P}) := \frac{1}{2}\max_{i,\ell}\sum_j |\mathsf{P}_{i,j} - \mathsf{P}_{\ell,j}| = 1 - \min_{i,\ell}\sum_j \min\{\mathsf{P}_{i,j}, \mathsf{P}_{\ell,j}\}. \tag{15.39}$$

*It is the exact operator norm induced by $\|\cdot\|_V$ on $\mathbb{R}^n/\operatorname{Span}(\mathbb{1}_n)$:*

$$\delta(\mathsf{P}) = \sup_{z\notin \operatorname{Span}(\mathbb{1}_n)} \frac{\|\mathsf{P}z\|_V}{\|z\|_V}. \tag{15.40}$$

*Consequently,*

$$\|\mathsf{P}z\|_V \leqslant \delta(\mathsf{P})\|z\|_V \qquad (z \in \mathbb{R}^n). \tag{15.41}$$

*If $\mathsf{P} > 0$, then this exact tangent factor is bounded by the Birkhoff coefficient of Theorem 8.15:*

$$\delta(\mathsf{P}) \leqslant \lambda(\mathsf{P}) < 1. \tag{15.42}$$

*Proof.* Fix two rows $p = (\mathsf{P}_{i,j})_j$ and $p' = (\mathsf{P}_{\ell,j})_j$, and remove their common mass $r_j = \min\{p_j, p'_j\}$. The two residual nonnegative measures have the same mass $1 - \sum_j r_j = \frac{1}{2}\sum_j |p_j - p'_j|$. Their averages of $z$ can differ by at most this mass times $\|z\|_V$. Maximizing over the two output rows proves the upper bound in (15.40). Conversely, for a pair of rows attaining the maximum, choose $z_j = 1$ when $p_j \geqslant p'_j$ and $z_j = 0$ otherwise. Then $\|z\|_V = 1$ and $(\mathsf{P}z)_i - (\mathsf{P}z)_\ell = \frac{1}{2}\sum_j |p_j - p'_j|$, proving equality and hence (15.41).
Assume now that $\mathsf{P} > 0$. Apply Theorem 8.15 to $u_s = e^{sz}$ and $\mathbb{1}_n$. Since $\mathsf{P}\mathbb{1}_n = \mathbb{1}_n$, $d_{\mathcal{H}}(\mathsf{P}e^{sz}, \mathbb{1}_n) \leqslant \lambda(\mathsf{P}) d_{\mathcal{H}}(e^{sz}, \mathbb{1}_n) = \lambda(\mathsf{P})|s|\|z\|_V$. Moreover, $\mathsf{P}e^{sz} = \mathbb{1}_n + s\mathsf{P}z + O(s^2)$, hence $\log(\mathsf{P}e^{sz}) = s\mathsf{P}z + O(s^2)$. Divide by $|s|$ and let $s \to 0$ to obtain $\|\mathsf{P}z\|_V \leqslant \lambda(\mathsf{P})\|z\|_V$. Taking the exact quotient norm (15.40) yields (15.42). This Hilbert–Hopf–Dobrushin relation extends to Markov operators on general cones. ◻

$\|\mathsf{P}^\top \mathrm{p} - \mathsf{P}^\top \mathrm{p}'\|_{\ell^1} \leqslant \delta(\mathsf{P})\|\mathrm{p} - \mathrm{p}'\|_{\ell^1}$. Thus (15.41) contracts observables modulo constants, while the adjoint inequality contracts probability laws in total variation. The proposition also separates two useful constants.
The same coefficient applies to the measure-valued dynamics. For $\alpha \in \mathcal{P}(C)$, where $C \subset \mathbb{R}^d$ is compact, define the Markov operator

$$(\mathsf{P}_\alpha h)(x) := \int_C h(y)p_{\alpha,x}(y)\mathrm{d}\alpha(y), \qquad p_{\alpha,x}(y) := \frac{K(x,y)}{Z_\alpha(x)}, \qquad Z_\alpha(x) := \int_C K(x,z)\mathrm{d}\alpha(z). \tag{15.43}$$

$$\delta(\mathsf{P}_\alpha) := \frac{1}{2}\sup_{x,x'\in\operatorname{supp}\alpha}\int_C |p_{\alpha,x}(y) - p_{\alpha,x'}(y)|\,\mathrm{d}\alpha(y). \tag{15.44}$$

$\|\mathsf{P}_\alpha h\|_{V,\operatorname{supp}\alpha} \leqslant \delta(\mathsf{P}_\alpha)\|h\|_{V,\operatorname{supp}\alpha}$.

$$\eta_K(C) := \sup_{x,x',y,y'\in C}\frac{K(x,y)K(x',y')}{K(x',y)K(x,y')}, \qquad \lambda_K(C) := \frac{\sqrt{\eta_K(C)}-1}{\sqrt{\eta_K(C)}+1}. \tag{15.45}$$

$$\frac{(\mathsf{P}_X)_{i,j}(\mathsf{P}_X)_{\ell,r}}{(\mathsf{P}_X)_{i,r}(\mathsf{P}_X)_{\ell,j}} = \frac{K(x_i,x_j)K(x_\ell,x_r)}{K(x_i,x_r)K(x_\ell,x_j)}. \tag{15.46}$$

$$\delta(\mathsf{P}_X) \leqslant \lambda(\mathsf{P}_X) = \lambda(\mathrm{K}_X) \leqslant \lambda_K(C). \tag{15.47}$$

$$\bar{\delta}_K(C) := \sup_{\alpha\in\mathcal{P}(C)}\delta(\mathsf{P}_\alpha) \leqslant \lambda_K(C) < 1. \tag{15.48}$$

$$\lambda_K(C) \leqslant \frac{k_+ - k_-}{k_+ + k_-}. \tag{15.49}$$

**Theorem 15.12** (Dobrushin consensus for positive mean shift)**.** *Let $\alpha_0 \in \mathcal{P}(\mathbb{R}^d)$ have compact support, and set $C_0 = \operatorname{conv}(\operatorname{supp}\alpha_0)$, $\quad D_0 = \operatorname{diam}(\operatorname{supp}\alpha_0)$, $\quad \bar{\delta} = \bar{\delta}_K(C_0) < 1$. Assume that $K : C_0 \times C_0 \to (0,+\infty)$ is Lipschitz. For an empirical initial measure $\alpha_0 = \sum_i a_i\delta_{x_i^{(0)}}$, let $X^{(\ell)}$ follow the damped blurring iteration $X^{(\ell+1)} = \big((1-\tau)I_n + \tau\mathsf{P}_{X^{(\ell)}}\big)X^{(\ell)}$, $\quad 0 < \tau \leqslant 1$, define $\delta^{(\ell)} = \delta(\mathsf{P}_{X^{(\ell)}})$, $\quad q^{(\ell)} = 1 - \tau(1-\delta^{(\ell)})$, $\quad q_\tau = 1 - \tau(1-\bar{\delta}) < 1$.*

*Then*

$$D(X^{(\ell)}) := \max_{i,j}\|x_i^{(\ell)} - x_j^{(\ell)}\| \leqslant D_0\prod_{r=0}^{\ell-1}q^{(r)} \leqslant q_\tau^\ell D_0. \tag{15.50}$$

*All particles converge to a common $x_\infty^{\mathrm{d}} \in C_0$, and $\max_i\|x_i^{(\ell)} - x_\infty^{\mathrm{d}}\| \leqslant \frac{D_0}{1-\bar{\delta}}q_\tau^\ell$. The characteristic solution of*

$$\partial_t\alpha_t + \operatorname{div}\big(\alpha_t M_K[\alpha_t]\big) = 0, \qquad \alpha_{t=0} = \alpha_0, \tag{15.51}$$

*is globally defined and has nested convex hulls, $\operatorname{conv}(\operatorname{supp}\alpha_t) \subset \operatorname{conv}(\operatorname{supp}\alpha_s)$ $\quad (t \geqslant s \geqslant 0)$. Writing $D(t) = \operatorname{diam}(\operatorname{supp}\alpha_t)$, one has the adaptive estimate*

$$D(t) \leqslant D(s)\exp\Big(-\int_s^t\big[1 - \delta(\mathsf{P}_{\alpha_r})\big]\mathrm{d}r\Big) \leqslant e^{-(1-\bar{\delta})(t-s)}D(s) \qquad (t \geqslant s \geqslant 0). \tag{15.52}$$

$$\mathcal{W}_\infty(\alpha_t, \delta_{x_\infty^{\mathrm{c}}}) \leqslant \frac{D_0}{1-\bar{\delta}}e^{-(1-\bar{\delta})t}. \tag{15.53}$$

*Proof.* Because $K$ is positive and Lipschitz on the compact set $C_0 \times C_0$, its minimum is positive. Hence the denominator in (15.36) is uniformly bounded away from zero, and the characteristic vector field is uniformly Lipschitz in $x$ and Lipschitz in $\alpha$ for $\mathcal{W}_1$. Moreover, $x + M_K[\alpha](x) = \mathsf{P}_\alpha \mathrm{Id}(x)$ lies in $\mathrm{conv}(\operatorname{supp}\alpha)$. No characteristic can cross an outward supporting hyperplane of the current convex hull, so the convex hulls are nested and the dynamics is globally well posed.
For any compact set $S \subset \mathbb{R}^d$, the support-function identity

$$\operatorname{diam}(S) = \sup_{\theta \in \mathbb{S}^{d-1}} \left( \sup_{x\in S}\langle \theta, x\rangle - \inf_{x\in S}\langle \theta, x\rangle \right) \tag{15.54}$$

reduces Euclidean diameter contraction to scalar variation contraction. For a finite configuration, set $z_\theta = X\theta \in \mathbb{R}^n$. Proposition 15.11 gives

$$\begin{aligned} \|z_\theta^{(\ell+1)}\|_V &\leqslant (1-\tau)\|z_\theta^{(\ell)}\|_V + \tau \|\mathsf{P}_{X^{(\ell)}} z_\theta^{(\ell)}\|_V \\ &\leqslant \big[1 - \tau(1-\delta(\mathsf{P}_{X^{(\ell)}}))\big]\|z_\theta^{(\ell)}\|_V \leqslant q_\tau \|z_\theta^{(\ell)}\|_V. \end{aligned}$$

Taking the supremum over $\theta$ gives the one-step factor $q^{(\ell)}$; multiplying these factors and using $q^{(\ell)} \leqslant q_\tau$ proves (15.50). Since $\|x_i^{(\ell+1)} - x_i^{(\ell)}\| \leqslant \tau D(X^{(\ell)})$, every particle path is Cauchy; the vanishing diameter forces a common limit, and summing the uniform geometric tail yields the displayed pointwise estimate.
For the measure-valued flow, let $f_\theta(x) = \langle \theta, x\rangle$ and denote by $w_\theta(t) = \|f_\theta\|_{V,\operatorname{supp}\alpha_t}$ the width of the support in direction $\theta$. The upper Dini derivative of the extrema along characteristics satisfies

$$D^+ w_\theta(t) \leqslant \|\mathsf{P}_{\alpha_t} f_\theta\|_{V,\operatorname{supp}\alpha_t} - w_\theta(t) \leqslant -\big[1-\delta(\mathsf{P}_{\alpha_t})\big] w_\theta(t).$$

Gronwall's lemma, followed by the support-function identity (15.54), proves (15.52). Finally, every characteristic satisfies $\|\dot X_t\| \leqslant D(t)$. It is therefore Cauchy, all characteristic limits coincide because $D(t) \to 0$, and integrating the uniform exponential bound from $t$ to $+\infty$ gives (15.53). $\square$

For the Gaussian kernel $K_\varepsilon(x,y) = e^{-\|x-y\|^2/(2\varepsilon)}$, the cross-ratio has the closed form
$\log \frac{K_\varepsilon(x,y)K_\varepsilon(x',y')}{K_\varepsilon(x',y)K_\varepsilon(x,y')} = \frac{\langle x-x', y-y'\rangle}{\varepsilon}$. Hence, for a compact convex set $C$ of diameter $D$,

$$\eta_{K_\varepsilon}(C) = e^{D^2/\varepsilon}, \qquad \lambda_{K_\varepsilon}(C) = \tanh\left(\frac{D^2}{4\varepsilon}\right). \tag{15.55}$$

$$D(X^{(\ell+1)}) \leqslant \tanh\left(\frac{D(X^{(\ell)})^2}{4\varepsilon}\right) D(X^{(\ell)}) \leqslant \frac{D(X^{(\ell)})^3}{4\varepsilon},$$

$$D(t) \leqslant D(s)\exp\left(-\int_s^t \left[1 - \tanh\left(\frac{D(r)^2}{4\varepsilon}\right)\right]\mathrm{d}r\right). \tag{15.56}$$

**Scope of the consensus result.** The projective argument also clarifies what is not conserved and when clustering replaces consensus. Because row normalization makes the pairwise coefficients asymmetric, the consensus point need not equal the barycenter of $\alpha_0$; that barycenter is conserved for the constant kernel, but not in general.
**Gradient structure and limitations.** When the token space has dimension $d$ and the query/key space has dimension $r$, take $Q, K \in \mathbb{R}^{r\times d}$ and $V \in \mathbb{R}^{d\times d}$. If $V = Q^\top K$, the field $\Gamma_\theta[\alpha]$ is a gradient vector field in the token variable. Indeed, define the log-partition potential
$\Phi_\alpha(x) = \int \exp(\langle Qx, Ky\rangle)\mathrm{d}\alpha(y)$, $U_\alpha(x) = \log \Phi_\alpha(x)$. $\nabla_x U_\alpha(x) = \frac{\int Q^\top Ky \exp(\langle Qx, Ky\rangle)\mathrm{d}\alpha(y)}{\int \exp(\langle Qx, Kz\rangle)\mathrm{d}\alpha(z)} = \Gamma_\theta[\alpha](x)$. $f_{\mathrm{att}}(\alpha) = \int U_\alpha(x)\,\mathrm{d}\alpha(x)$ $\delta f_{\mathrm{att}}(\alpha)(z) = U_\alpha(z) + \int \frac{\exp(\langle Qx, Kz\rangle)}{\Phi_\alpha(x)}\,\mathrm{d}\alpha(x)$, up to an additive constant. The second term is the response of every query normalization to a perturbation of the key distribution, and its spatial gradient is absent from $\Gamma_\theta[\alpha]$.

## 15.5 Flows over the Gaussian Manifold

**Gaussianity preservation.**

**Proposition 15.13** (Affine velocities preserve Gaussianity)**.** *Let $\alpha_0 = \mathcal{N}(m_0, \Sigma_0)$, with $\Sigma_0 \succ 0$. Let $b_t \in \mathbb{R}^d$ and $A_t \in \mathbb{R}^{d\times d}$ be locally integrable on a time interval, and let $(m_t, \Sigma_t)$ solve $\dot m_t = b_t$, $\quad \dot\Sigma_t = A_t\Sigma_t + \Sigma_t A_t^\top$, $\quad (m_{t=0}, \Sigma_{t=0}) = (m_0, \Sigma_0)$. Then, as long as this matrix ODE is defined, $\Sigma_t \succ 0$ and $\alpha_t = \mathcal{N}(m_t, \Sigma_t)$ is the solution of the continuity equation $\partial_t\alpha_t + \operatorname{div}(\alpha_t v_t) = 0$, $\quad v_t(x) = b_t + A_t(x - m_t)$. In particular, if a smooth functional $f$ has a Wasserstein gradient on Gaussian measures of the affine form $\nabla\!\!\!\!\nabla f(\mathcal{N}(m,\Sigma))(x) = b_f(m,\Sigma) + A_f(m,\Sigma)(x-m)$, with $A_f(m,\Sigma)$ symmetric, then the Wasserstein gradient flow of $f$, initialized from any non-degenerate Gaussian, stays Gaussian and satisfies $\dot m_t = h_f(m_t,\Sigma_t)$, $\quad \dot\Sigma_t = H_f(m_t,\Sigma_t)$, where $h_f(m,\Sigma) = -b_f(m,\Sigma)$, $\quad H_f(m,\Sigma) = -\big(A_f(m,\Sigma)\Sigma + \Sigma A_f(m,\Sigma)\big)$. Conversely, fix a non-degenerate Gaussian $\mathcal{N}(m,\Sigma)$. Suppose that a finite-energy Wasserstein tangent field $V_{m,\Sigma}$ is represented by a gradient in $L^2(\mathcal{N}(m,\Sigma))$ and is tangent to the Gaussian manifold at this Gaussian. Then there exist $b(m,\Sigma)$ and a symmetric matrix $A(m,\Sigma)$ such that $V_{m,\Sigma}(x) = b(m,\Sigma) + A(m,\Sigma)(x-m)$ $\quad \mathcal{N}(m,\Sigma)$-a.e. Without the Wasserstein gradient representative, this converse is false because one may add velocity fields with zero $\mathcal{N}(m,\Sigma)$-weighted divergence. Finally, any smooth Gaussian curve with positive definite covariance can be generated by an affine velocity. If one wants the velocity to be a Wasserstein tangent gradient, one chooses the unique symmetric solution of the Lyapunov equation $A_t\Sigma_t + \Sigma_t A_t = \dot\Sigma_t$.*

*Proof.* Let $X_t$ follow the characteristic ODE $\dot X_t = b_t + A_t(X_t - m_t)$ with $X_0 \sim \mathcal{N}(m_0, \Sigma_0)$. Since $\dot m_t = b_t$, the centered variable $\tilde X_t = X_t - m_t$ solves the homogeneous linear ODE $\dot{\tilde X}_t = A_t \tilde X_t$. If $\Phi_t$ is the fundamental matrix $\dot\Phi_t = A_t\Phi_t$, $\Phi_{t=0} = \mathrm{Id}$, then $X_t = m_t + \Phi_t(X_0 - m_0)$, $\quad \Sigma_t = \Phi_t\Sigma_0\Phi_t^\top$. Hence $X_t$ is Gaussian and $\Sigma_t \succ 0$, and $\dot\Sigma_t = \frac{\mathrm{d}}{\mathrm{d}t}\mathbb{E}\big(\tilde X_t \tilde X_t^\top\big) = A_t\Sigma_t + \Sigma_t A_t^\top$. This proves Gaussian preservation and the moment ODE. The Wasserstein gradient-flow statement follows by inserting the descent velocity $v_t = -\nabla\!\!\!\!\nabla f(\alpha_t)$.
For the converse, fix $\alpha = \mathcal{N}(m,\Sigma)$ and denote its density by $\rho$. Tangency to the Gaussian manifold means that the density variation $-\operatorname{div}(\rho V)$ is generated by some moment variation $(\dot m, \dot\Sigma)$, with $\dot\Sigma$ symmetric. Set $b = \dot m$, and let $A = A^\top$ be the unique

solution of $A\Sigma+\Sigma A=\dot\Sigma$. By the first part of the proposition, the affine gradient field $V_0(x)=b+A(x-m)$ generates exactly the same infinitesimal Gaussian variation. Hence $\operatorname{div}\big(\rho(V-V_0)\big)=0$ in the distributional sense. Both $V$ and $V_0$ belong to the $L^2(\alpha)$ closure of gradient fields. The weighted Helmholtz decomposition therefore makes $V-V_0$ simultaneously a tangent gradient and orthogonal to every tangent gradient, so $V=V_0$ in $L^2(\alpha)$. Equivalently, for a smooth representative $V-V_0=\nabla\psi$, integration by parts gives $\int\|\nabla\psi\|^2\mathrm{d}\alpha=0$. This proves that the Wasserstein tangent representative is affine. The qualification is essential: without selecting the gradient representative, one can add a nonzero field with zero $\alpha$-weighted divergence without changing the Gaussian curve.

For a prescribed smooth Gaussian curve, set $b_t=\dot m_t$ and choose any matrix $A_t$ satisfying $A_t\Sigma_t+\Sigma_tA_t^\top=\dot\Sigma_t$. Since $\Sigma_t$ is positive definite, the Lyapunov map $A\mapsto A\Sigma_t+\Sigma_tA$ is invertible on symmetric matrices, which gives the unique symmetric choice when a gradient velocity is required. In that case $v_t$ is the gradient of the quadratic potential $x\mapsto\langle b_t,\,x\rangle+\langle A_t(x-m_t),\,x-m_t\rangle/2$. ☐

**Gaussian-preserving gradient flows.**

**Proposition 15.14** (Gaussian closure catalogue)**.** *Let $\gamma=\mathcal{N}(\bar m,\bar\Sigma)$ on $\mathbb{R}^d$, with $\bar\Sigma\succ0$, and let the initial condition be $\alpha_0=\mathcal{N}(m_0,\Sigma_0)$, with $\Sigma_0\succ0$. For each functional displayed below, let $\alpha_t$ be the usual Wasserstein gradient flow on $\mathcal{P}_2(\mathbb{R}^d)$, initialized at $\alpha_0$. Then the Gaussian family is invariant for these dynamics: as long as the solution exists and $\Sigma_t\succ0$, $\alpha_t=\mathcal{N}(m_t,\Sigma_t)$, and the mean and covariance satisfy $\dot m_t=h(m_t,\Sigma_t)$, $\quad\dot\Sigma_t=H(m_t,\Sigma_t)$. Write $\delta_m=m-\bar m$, $A=\bar\Sigma^{-1}$, and $M_{\Sigma,\bar\Sigma}:=\Sigma^{-1/2}\big(\Sigma^{1/2}\bar\Sigma\Sigma^{1/2}\big)^{1/2}\Sigma^{-1/2}$. With the normalizations displayed in the first column, the mean vector field $h$ and covariance vector field $H$ are listed in the following table. Gradients with respect to $\Sigma$ are symmetric Frobenius gradients on the cone of covariance matrices.*

***Gaussian-preserving Wasserstein gradient flows.***

*Here, in the MMD row, $R=\Sigma+m\,m^\top-\bar\Sigma-\bar m\,\bar m^\top$. The debiased Sinkhorn row uses the notation of Corollary 8.24: for Gaussian inputs, $\bar{\mathcal{L}}^{\varepsilon}_{\|\cdot-\cdot\|^2}(\alpha,\gamma)=\|\delta_m\|^2+\mathcal{B}_\varepsilon(\Sigma,\bar\Sigma)^2$, with the closed-form covariance gradient*

$$G_\varepsilon(\Sigma,\bar\Sigma)=\tau_\varepsilon(\Sigma)-\bar\Sigma^{1/2}\tau_\varepsilon\big(B^{1/2}_{\Sigma,\bar\Sigma}\big)B^{-1/2}_{\Sigma,\bar\Sigma}\bar\Sigma^{1/2},\qquad B_{\Sigma,\bar\Sigma}=\bar\Sigma^{1/2}\Sigma\bar\Sigma^{1/2}.$$

*Here $\tau_\varepsilon$ is the scalar function $\tau_\varepsilon(r):=\frac{\sqrt{\varepsilon^2+16r^2}-\varepsilon}{4r}$, $\qquad r>0$, applied to positive matrices by spectral calculus. Equivalently, for $M\succ0$, $\tau_\varepsilon(M)=\big(\sqrt{\varepsilon^2\mathrm{Id}+16M^2}-\varepsilon\mathrm{Id}\big)(4M)^{-1}$. With this convention, $G_\varepsilon=\nabla_\Sigma\mathcal{B}_\varepsilon(\Sigma,\bar\Sigma)^2$. In the sliced row, $\sigma$ is the normalized spherical measure on $\mathbb{S}^{d-1}$, and*

$$G_{\mathrm{sw}}(\Sigma,\bar\Sigma)=\int_{\mathbb{S}^{d-1}}\left(1-\sqrt{\frac{\theta^\top\bar\Sigma\theta}{\theta^\top\Sigma\theta}}\right)\theta\theta^\top\,\mathrm{d}\sigma(\theta).$$

*Here*

$$\mathcal{I}(\alpha|\gamma)=\int\left|\nabla\log\frac{\rho(x)}{\rho_\gamma(x)}\right|^2\rho(x)\,\mathrm{d}x=\int|\nabla\log\rho(x)+A(x-\bar m)|^2\,\rho(x)\,\mathrm{d}x\qquad(\alpha=\rho\,\mathrm{d}x),$$

*where $\rho_\gamma$ is the density of $\gamma$.*

*Proof.* For the moment-functional row, the formula is exact on the full Wasserstein space. Indeed, write $m_\alpha=\int x\,\mathrm{d}\alpha(x)$, $\qquad\Sigma_\alpha=\int(x-m_\alpha)(x-m_\alpha)^\top\mathrm{d}\alpha(x)$. If $\eta$ is a signed perturbation with zero total mass, then $\mathrm{d}m_\alpha[\eta]=\int x\,\mathrm{d}\eta(x)$, $\qquad\mathrm{d}\Sigma_\alpha[\eta]=\int(x-m_\alpha)(x-m_\alpha)^\top\mathrm{d}\eta(x)$. Thus, up to an irrelevant additive constant, the first variation is $\delta f(\alpha)(x)=\langle\nabla_m g,\,x\rangle+\langle\nabla_\Sigma g,\,(x-m_\alpha)(x-m_\alpha)^\top\rangle$, and its spatial gradient is the affine field $\nabla_x\delta f(\alpha)(x)=\nabla_m g+2\nabla_\Sigma g\,(x-m_\alpha)$.

For the quadratic potential row, the ambient first variation is $\delta f(\alpha)(x)=\frac12x^\top Bx+\langle\ell,\,x\rangle$, $\qquad\nabla_x\delta f(\alpha)(x)=Bm+\ell+B(x-m)$, which gives the displayed affine velocity. For the quadratic interaction row, the ambient first variation is $\delta f(\alpha)(x)=\frac12\int(x-y)^\top G(x-y)\mathrm{d}\alpha(y)$, $\qquad\nabla_x\delta f(\alpha)(x)=G(x-m_\alpha)$. These first variations are quadratic in $x$, hence the affine Gaussian flow coincides with the full Wasserstein gradient flow.

For the KL row, write $\alpha=\rho\,\mathrm{d}x$ and let $\rho_\gamma$ denote the density of $\gamma$. The ambient first variation is $\log(\rho/\rho_\gamma)$ up to an additive constant. If $\alpha=\mathcal{N}(m,\Sigma)$, then $\nabla\log(\rho/\rho_\gamma)(x)=-\Sigma^{-1}(x-m)+A(x-\bar m)$, so the descent velocity is $v(x)=-A\delta_m+(\Sigma^{-1}-A)(x-m)$. Proposition 15.13 gives $h(m,\Sigma)=-A\delta_m$ and $H(m,\Sigma)=2\mathrm{Id}-\Sigma A-A\Sigma$. For the Fisher row, writing $u=\log(\rho/\rho_\gamma)$, the ambient first variation of $\frac12\mathcal{I}(\alpha|\gamma)$ is, up to constants, $-\Delta u-\frac12|\nabla u|^2-\langle\nabla u,\,\nabla\log\rho_\gamma\rangle$. For Gaussian $\rho$ and Gaussian $\gamma$, this is quadratic in $x$. Multiplying by two for $\mathcal{I}(\alpha|\gamma)$ gives the descent velocity $v(x)=-2A^2\delta_m+2(\Sigma^{-2}-A^2)(x-m)$. Hence the Wasserstein velocity is affine and the Gaussian family is closed, with the displayed $h$ and $H$. Although the ambient Fisher flow is a fourth-order PDE for a generic density, it is therefore an exact finite-dimensional closure in this quadratic-reference case.

For $\mathcal{W}_2^2(\alpha,\gamma)$, the Brenier map from $\mathcal{N}(m,\Sigma)$ to $\gamma$ is $T(x)=\bar m+M_{\Sigma,\bar\Sigma}(x-m)$. The descent velocity for the unhalved squared distance is $2(T-\mathrm{Id})$, which gives the mean and covariance equations through Proposition 15.13. For the MMD row, $k(x,y)=\langle x,\,y\rangle^2$ identifies the kernel mean embedding with the raw second moment. Thus $\mathrm{MMD}_k^2(\alpha,\gamma)=\|\Sigma+m\,m^\top-\bar\Sigma-\bar m\,\bar m^\top\|_{\mathrm{F}}^2=\|R\|_{\mathrm{F}}^2$, and the ambient first variation is $2x^\top Rx$, whose descent velocity is $-4Rx$.

For the Sinkhorn row, Corollary 8.24 gives the closed Gaussian formula as a squared mean displacement plus the smoothed Bures term $\mathcal{B}_\varepsilon^2$. This is not only a formula on the Gaussian submanifold. The first variation of the entropic value with respect to the first marginal is a Sinkhorn dual potential, and for Gaussian marginals the cross and self dual potentials are quadratic. Their debiased combination is therefore quadratic, so its spatial gradient is affine. Since the restriction of the functional to Gaussians is $(m,\Sigma)\mapsto\|m-\bar m\|^2+\mathcal{B}_\varepsilon(\Sigma,\bar\Sigma)^2$, the quadratic coefficient is $G_\varepsilon=\nabla_\Sigma\mathcal{B}_\varepsilon(\Sigma,\bar\Sigma)^2$, and the moment-functional computation gives the displayed $h$ and $H$. To compute this gradient, set $B_{\Sigma,\bar\Sigma}=\bar\Sigma^{1/2}\Sigma\bar\Sigma^{1/2}$. Since $\psi_\varepsilon'(r)=-2\tau_\varepsilon(r)$, differentiating $\operatorname{tr}\psi_\varepsilon(B^{1/2}_{\Sigma,\bar\Sigma})$ gives $-\bar\Sigma^{1/2}\tau_\varepsilon\big(B^{1/2}_{\Sigma,\bar\Sigma}\big)B^{-1/2}_{\Sigma,\bar\Sigma}\bar\Sigma^{1/2}$, whereas differentiating the self term $-\frac12\operatorname{tr}\psi_\varepsilon(\Sigma)$ gives $\tau_\varepsilon(\Sigma)$. This proves the displayed closed form for $G_\varepsilon$. When $\bar\Sigma=\mathrm{Id}$, this covariance contribution is a spectral function of $\Sigma$. If $\lambda$ is an eigenvalue of $\Sigma$, the corresponding scalar velocity is $4\sqrt{\lambda+\varepsilon^2/16}-4\sqrt{\lambda^2+\varepsilon^2/16}$, which gives the displayed covariance-eigenvalue equation by functional calculus.

For sliced Wasserstein, the exact ambient Wasserstein gradient is obtained by averaging the one-dimensional gradients. Each projection is Gaussian:

$$(P_\theta)_\sharp\alpha=\mathcal{N}(\langle\theta,\,m\rangle,\theta^\top\Sigma\theta),\qquad(P_\theta)_\sharp\gamma=\mathcal{N}(\langle\theta,\,\bar m\rangle,\theta^\top\bar\Sigma\theta).$$

Hence
$$\mathcal{SW}_2^2(\alpha,\gamma) = \int_{\mathbb{S}^{d-1}} \left[\langle\theta,\,\delta_m\rangle^2 + \left(\sqrt{\theta^\top\Sigma\theta} - \sqrt{\theta^\top\bar\Sigma\theta}\right)^2\right] \mathrm{d}\sigma(\theta).$$
If $T_\theta$ is the monotone transport from $(P_\theta)_\sharp\alpha$ to $(P_\theta)_\sharp\gamma$, then the descent velocity for the unhalved sliced objective is $2\int_{\mathbb{S}^{d-1}}\big(T_\theta(P_\theta x) - P_\theta x\big)\theta\,\mathrm{d}\sigma(\theta)$. For Gaussian marginals, $T_\theta$ is affine. Thus this velocity is affine in $x$, and Proposition 15.13 applies. The spherical identity $\int \theta\theta^\top \mathrm{d}\sigma(\theta) = \mathrm{Id}/d$ gives the mean equation, and differentiating the covariance term gives $G_{\mathrm{sw}}$. ☐

**One-dimensional case.** Fix a Gaussian target $\gamma = \mathcal{N}(\bar m, \bar\sigma^2)$, with $\bar\sigma > 0$, and write $\alpha = \mathcal{N}(m,\sigma^2)$, with $\sigma > 0$. By Proposition 2.39, the parametrization $(m,\sigma)\mapsto\mathcal{N}(m,\sigma^2)$ identifies the one-dimensional Gaussian family with the open Euclidean half-plane.
$g_f(m,\sigma) \coloneqq f\big(\mathcal{N}(m,\sigma^2)\big)$. Set $\delta = m - \bar m$. For the Sinkhorn row below, with $\varepsilon > 0$, abbreviate
$$\psi_\varepsilon(r) \coloneqq -\frac{\sqrt{\varepsilon^2+16r^2}-\varepsilon}{2} - \frac{\varepsilon}{2}\log\left(\frac{2\varepsilon}{\sqrt{\varepsilon^2+16r^2}+\varepsilon}\right), \qquad r > 0.$$
**Constrained evolution on the Gaussian manifold.** $\mathcal{G} = \{\mathcal{N}(m,\Sigma) : m\in\mathbb{R}^d,\ \Sigma\succ 0\}$ be the Gaussian submanifold of $\mathcal{P}_2(\mathbb{R}^d)$. The Wasserstein gradient of a functional constrained to a smooth submanifold $\mathcal{M}\subset\mathcal{P}_2$ is defined as the Riesz representative of the differential restricted to tangent velocities of $\mathcal{M}$.
$\alpha^{(\ell+1)} \in \underset{\alpha\in\mathcal{M}}{\operatorname{argmin}}\ \frac{1}{2\tau}\mathcal{W}_2^2(\alpha,\alpha^{(\ell)}) + f(\alpha)$. For $\mathcal{M} = \mathcal{G}$, tangent velocities are affine gradient fields $v(x) = b + A(x-m)$ with $A = A^\top$.
The constrained gradient is therefore the $L^2(\mathcal{N}(m,\Sigma))$ projection of the ambient Wasserstein gradient onto this finite-dimensional affine space, whenever the ambient gradient exists.
**Proposition 15.15** (Gaussian-constrained Wasserstein gradients)**.** *Let $f$ be a smooth functional and assume that its restriction to nondegenerate Gaussian measures can be written as $f(\mathcal{N}(m,\Sigma)) = F(m,\Sigma)$. Then the Wasserstein gradient constrained to the Gaussian family is the affine vector field $v_F(x) = \nabla_m F(m,\Sigma) + 2\nabla_\Sigma F(m,\Sigma)(x-m)$, where $\nabla_\Sigma F$ denotes the symmetric matrix derivative. Equivalently, $v_F$ is the $L^2(\mathcal{N}(m,\Sigma))$ projection of the ambient Wasserstein gradient onto affine gradient fields, whenever the ambient gradient exists. Hence the gradient descent flow constrained to Gaussian measures satisfies*
$$\dot m_t = -\nabla_m F(m_t,\Sigma_t), \qquad \dot\Sigma_t = -2\big(\Sigma_t\nabla_\Sigma F(m_t,\Sigma_t) + \nabla_\Sigma F(m_t,\Sigma_t)\Sigma_t\big), \tag{15.57}$$
*and the descent velocity is affine.*

*Proof.* Test the functional along a Gaussian tangent vector, represented by an affine gradient field $v(x) = b + A(x-m)$ with $A$ symmetric. The induced first-order variations are $\dot m = b$ and $\dot\Sigma = A\Sigma + \Sigma A$. Therefore
$$\mathrm{d}F(m,\Sigma)[b, A\Sigma+\Sigma A] = \langle\nabla_m F,\, b\rangle + \mathrm{tr}(\nabla_\Sigma F(A\Sigma+\Sigma A))\,.$$
Since $A$, $\Sigma$ and $\nabla_\Sigma F$ are symmetric, the second term equals $2\,\mathrm{tr}(\nabla_\Sigma F\, A\Sigma) = \int\langle 2\nabla_\Sigma F(x-m),\, A(x-m)\rangle \mathrm{d}\mathcal{N}(m,\Sigma)(x)$. Together with the mean term, this gives $\mathrm{d}F(m,\Sigma)[\dot m,\dot\Sigma] = \int\langle v_F(x),\, v(x)\rangle \mathrm{d}\mathcal{N}(m,\Sigma)(x)$ for all affine gradient fields $v$. This identifies the constrained Wasserstein gradient in the induced $L^2(\alpha)$ metric, or equivalently the projection of the ambient gradient when it exists. Substituting the descent velocity $-v_F$ in Proposition 15.13 gives (15.57). ☐

**Non-variational Gaussian-preserving flows.**
**Contractive Gaussian projection.**

**Definition 15.16** (Gaussian projection)**.** For $\alpha\in\mathcal{P}_2(\mathbb{R}^d)$, its Gaussian projection is $\mathrm{Gauss}(\alpha) \coloneqq \mathcal{N}(m_\alpha,\Sigma_\alpha)$, the Gaussian with the same mean and covariance as $\alpha$.

**Theorem 15.17** (Gelbrich theorem)**.** *For every $\alpha,\beta\in\mathcal{P}_2(\mathbb{R}^d)$, $\mathcal{W}_2^2\big(\mathrm{Gauss}(\alpha),\mathrm{Gauss}(\beta)\big) = \|m_\alpha - m_\beta\|^2 + \mathcal{B}^2(\Sigma_\alpha,\Sigma_\beta) \leqslant \mathcal{W}_2^2(\alpha,\beta)$.*

*Proof.* Take any coupling $(X,Y)$ of $\alpha$ and $\beta$, center the variables, and write $C = \mathbb{E}[(X-m_\alpha)(Y-m_\beta)^\top]$. In the positive definite case, positivity of the block covariance matrix implies the factorization $C = \Sigma_\alpha^{1/2}K\Sigma_\beta^{1/2}$ with $\|K\|_{\mathrm{op}}\leqslant 1$, and therefore, by operator/nuclear norm duality,
$$\mathrm{tr}\,C \leqslant \mathrm{tr}\left((\Sigma_\alpha^{1/2}\Sigma_\beta\Sigma_\alpha^{1/2})^{1/2}\right).$$
The semidefinite case follows by adding $\eta\mathrm{Id}$ to both covariance matrices and letting $\eta\downarrow 0$. Expanding $\mathbb{E}\|X-Y\|^2$ gives the lower bound $\mathbb{E}\|X-Y\|^2 \geqslant \|m_\alpha - m_\beta\|^2 + \mathcal{B}^2(\Sigma_\alpha,\Sigma_\beta)$. Taking the infimum over couplings proves the inequality, while equality for Gaussian laws is Proposition 2.39. ☐

**Theorem 15.18** (Hugo Lavenant Gaussian-preservation criterion)**.** *Let $f:\mathcal{P}_2(\mathbb{R}^d)\to(-\infty,+\infty]$ satisfy $f\big(\mathrm{Gauss}(\alpha)\big)\leqslant f(\alpha)\quad\forall\alpha\in\mathcal{P}_2(\mathbb{R}^d)$, with* Gauss *defined in Theorem 15.17. If $\gamma$ is Gaussian and $\alpha^\star$ minimizes the JKO step $\eta\mapsto f(\eta) + \frac{1}{2\tau}\mathcal{W}_2^2(\gamma,\eta)$, then* $\mathrm{Gauss}(\alpha^\star)$ *is also a minimizer. If this JKO minimizer is unique, it is Gaussian.*

*Proof.* For the JKO claim, $\mathrm{Gauss}(\gamma) = \gamma$ because $\gamma$ is Gaussian. Hence, for any competitor $\eta$, $f\big(\mathrm{Gauss}(\eta)\big) + \frac{1}{2\tau}\mathcal{W}_2^2\big(\gamma,\mathrm{Gauss}(\eta)\big) \leqslant f(\eta) + \frac{1}{2\tau}\mathcal{W}_2^2(\gamma,\eta)$. Applying this to a minimizer $\eta = \alpha^\star$ shows that $\mathrm{Gauss}(\alpha^\star)$ is again a minimizer. Uniqueness forces $\alpha^\star = \mathrm{Gauss}(\alpha^\star)$. ☐

**Moment closure beyond Gaussianity.**

$$\dot{m}_t = -\nabla_m g(m_t, \Sigma_t), \qquad \dot{\Sigma}_t = -2\big(\Sigma_t \nabla_\Sigma g(m_t, \Sigma_t) + \nabla_\Sigma g(m_t, \Sigma_t)\Sigma_t\big). \tag{15.58}$$

For Gaussian initial data, this affine velocity also preserves Gaussianity; for general initial data, the law need not become Gaussian, but its mean and covariance still follow the same closed vector field $(h, H)$ listed in that proposition.

**Proposition 15.19** (Scalar moment closure and eikonal characterization)**.** *Let $\Omega \subset \mathbb{R}^d$ be connected and open, let $\varphi \in C^2(\Omega)$, and set $m_\varphi(\alpha) \coloneqq \int_\Omega \varphi(x)\mathrm{d}\alpha(x)$ for probability measures such that $\varphi$ and $\|\nabla\varphi\|^2$ are integrable. For every smooth $g : \mathbb{R} \to \mathbb{R}$, the Wasserstein gradient flow of $f_g(\alpha) = g(m_\varphi(\alpha))$ satisfies, whenever the differentiation is justified, $\dot{m}_t = -g'(m_t)\int_\Omega \|\nabla\varphi(x)\|^2\mathrm{d}\alpha_t(x), \qquad m_t = m_\varphi(\alpha_t)$. This equation is autonomous for every $g$, meaning that its right-hand side depends on $\alpha_t$ only through $m_t$, if and only if there are constants $a, b \in \mathbb{R}$ such that*

$$\|\nabla\varphi(x)\|^2 = a + b\varphi(x) \qquad (x \in \Omega). \tag{15.59}$$

*In that case, $\dot{m}_t = -(a + bm_t)g'(m_t)$. Moreover, on every connected open set $U \subset \{x : \nabla\varphi(x) \neq 0\}$, the restriction of (15.59) to $U$ is equivalent to the existence of an eikonal coordinate $r \in C^2(U)$ and constants $c, \kappa, \lambda \in \mathbb{R}$ such that*

$$\|\nabla r\| = 1, \qquad \varphi = c + \kappa r + \frac{\lambda}{2}r^2. \tag{15.60}$$

*Under (15.59), one may take $\lambda = b/2$.*

*Proof.* The first variation is $\delta f_g(\alpha)(x) = g'(m_\varphi(\alpha))\varphi(x)$, so the continuity equation gives the displayed evolution of $m_t$. Condition (15.59) then proves sufficiency.

For necessity, it is enough to take $g(s) = s$ and write $\psi = \|\nabla\varphi\|^2$. Autonomy for Dirac laws first shows that $\psi = q \circ \varphi$ for some function $q$ on $\varphi(\Omega)$. Because $\Omega$ is connected, $\varphi(\Omega)$ is an interval. Comparing a Dirac law with any two-point mixture having the same $\varphi$-moment shows that $q\big((1-\zeta)s_0 + \zeta s_1\big) = (1-\zeta)q(s_0) + \zeta q(s_1) \qquad (0 \leqslant \zeta \leqslant 1)$. Hence $q$ is affine, which is precisely (15.59). On a regular set $U$, the right-hand side of (15.59) is positive. Fix $s_0 \in \varphi(U)$ and define $r(x) = \int_{s_0}^{\varphi(x)} \frac{\mathrm{d}s}{\sqrt{a+bs}}$. Then $\|\nabla r\| = 1$. Inverting this change of variables gives (15.60) with $c = s_0$, $\kappa = \sqrt{a + bs_0}$, and $\lambda = b/2$. Conversely, (15.60) gives $\|\nabla\varphi\|^2 = \kappa^2 - 2\lambda c + 2\lambda\varphi$. □